

Realized Volatility

Estimation, Forecasting, and ML

A Learning Guide

Ryan Vincent

Last compiled: July 6, 2026

Contents

I	What Is Volatility and How Do You Measure It?	10
1	Returns, Variance, and Why Volatility Matters	11
1.1	What Are Returns?	11
1.1.1	Simple Returns	11
1.1.2	Log Returns	11
1.2	Variance and Standard Deviation	12
1.2.1	Annualizing Volatility	13
1.3	Why Volatility Matters	14
1.4	Stylized Facts of Financial Returns	15
1.4.1	Fact 1: Returns Are Approximately Uncorrelated	15
1.4.2	Fact 2: Volatility Clusters	15
1.4.3	Fact 3: Fat Tails	16
1.4.4	Fact 4: The Leverage Effect	17
1.4.5	Summary of Stylized Facts	18
1.5	Conditional vs. Unconditional Volatility	19
	Summary	20
2	Realized Volatility	22
2.1	The Theoretical Target: Integrated Variance	22
2.2	Realized Variance as an Estimator	23
2.2.1	Construction	23
2.2.2	Why It Works: Convergence to Quadratic Variation	24
2.2.3	Diagram: Building RV from a Price Path	27
2.3	How Frequently to Sample	28
2.3.1	The Theory: More Is Better	28
2.3.2	The Practice: Microstructure Noise	28
2.3.3	The Bias-Variance Tradeoff	29
2.3.4	The Volatility Signature Plot	30
2.4	5-Minute RV: The Practical Workhorse	31
2.5	Realized Volatility vs. Realized Variance	31
	Summary	32
3	Microstructure Noise and Robust Estimators	34
3.1	The Microstructure Noise Problem	34
3.1.1	Quantifying the Bias	36
3.2	The Volatility Signature Plot	37
3.3	Two-Scales Realized Volatility (TSRV)	38
3.3.1	Construction	38
3.3.2	Convergence Rate	39
3.4	Multi-Scale Realized Volatility (MSRV)	40

3.5	The Realized Kernel	40
3.5.1	Construction	41
3.5.2	Why It Works	41
3.5.3	Bandwidth Selection	42
3.6	Pre-Averaging	42
3.6.1	Construction	43
3.7	Other Estimators	44
3.8	Which Estimator to Use	44
3.8.1	Comparison Table	44
3.8.2	Decision Flowchart	44
	Summary	47
4	Jumps and Continuous Variation	49
4.1	Why Prices Jump	49
4.1.1	Diagram: Continuous vs. Jump Price Paths	50
4.2	Bipower Variation	50
4.2.1	Why the $\pi/2$ Scaling Factor	51
4.2.2	Diagram: How RV and BPV Respond to a Jump	52
4.3	The BNS Jump Test	53
4.4	Lee-Mykland: Detecting Individual Jumps	54
4.4.1	Diagram: Intraday Jump Detection	55
4.5	Aït-Sahalia-Jacod Test	56
4.6	Threshold and Truncation Methods	57
4.7	Why the Decomposition Matters for Forecasting	58
4.7.1	Diagram: The Decomposition Pipeline	58
	Summary	59
II	Forecasting Volatility with Classical Models	61
5	The GARCH Family	62
5.1	The ARCH Model	62
5.2	GARCH(1,1): The Workhorse Model	63
5.3	The Leverage Effect	65
5.4	GJR-GARCH: A Simple Asymmetric Extension	66
5.5	EGARCH: Log-Variance and Guaranteed Positivity	66
5.6	Long Memory in Volatility: FIGARCH	67
5.7	Realized GARCH: Bringing Intraday Data into GARCH	68
5.8	The HEAVY Model	69
5.9	Comparison: GARCH vs. Realized GARCH vs. HEAVY	70
5.10	Summary	71
6	The HAR Model and Its Extensions	73
6.1	The Heterogeneous Market Hypothesis	73
6.2	The HAR Model	74
6.2.1	Typical Coefficient Values	76
6.3	HAR-J and HAR-CJ: Adding Jumps	76
6.3.1	HAR-J	77
6.3.2	HAR-CJ	77
6.4	SHAR: Good Volatility, Bad Volatility	78
6.5	HARQ: Handling Measurement Error	79
6.6	HAR-X and Beyond: Adding Exogenous Predictors	81

6.7	Ridge and Elastic-Net HAR: Shrinking Collinear Components	82
6.7.1	The Ridge Objective	82
6.7.2	Closed-Form Solution	83
6.7.3	Why Ridge Damps the Noisy Collinear Directions	84
6.7.4	Bias, Variance, and the Hoerl–Kennard Guarantee	85
6.7.5	Ridge vs. Lasso on Correlated HAR Features	86
6.7.6	Why L_1 Produces Sparsity but L_2 Does Not	87
6.7.7	Elastic Net: Sparsity with Grouping	87
6.7.8	Tuning and Diagnostics	88
6.8	Why HAR Is Hard to Beat	89
	Summary	90
	Key Results	91
7	Rough Volatility	92
7.1	Why Path Shape Matters: A Hedging Motivation	92
7.2	What Is Roughness?	93
7.3	Volatility Is Rough	94
7.3.1	How to Estimate H : The Variogram Method	95
7.4	The RFSV Forecasting Formula	97
7.5	Rough Volatility for Pricing	98
7.6	Fact or Artefact?	98
7.7	The Universal LSTM Connection	100
	Summary	101
	Key Results	102
III	The Volatility Surface and Options-Implied Information	103
8	Options Basics and the Volatility Surface	104
8.1	What Is an Option?	104
8.1.1	Payoff at Expiry	105
8.1.2	Moneyness	105
8.2	Black-Scholes in One Page	105
8.2.1	The Greeks: Sensitivity to Inputs	108
8.3	Implied Volatility	109
8.4	The Implied Volatility Surface	111
8.4.1	The Volatility Smile and Skew	112
8.4.2	The Term Structure of Implied Volatility	112
8.4.3	The Butterfly Spread	113
8.4.4	The Full Surface	114
8.4.5	Fitting the Surface: The SVI Parametrization	114
8.5	PCA of the Volatility Surface	116
8.6	Local Volatility	117
8.7	Model-Free Implied Variance and the VIX	119
8.7.1	Model-Free Implied Variance	119
8.7.2	The VIX Index	120
8.8	Variance Swaps: Trading Realized Volatility	121
8.8.1	Definition and Payoff	121
8.8.2	Log-Contract Replication	122
8.9	Connecting the Vol Surface to Volatility Forecasting	123
8.10	Summary	123

9	The Variance Risk Premium	125
9.1	What Is the Variance Risk Premium?	125
9.2	Why VRP Exists	127
9.2.1	Risk Aversion and Downside Protection Demand	127
9.2.2	The Insurance Analogy, Formalized	127
9.2.3	Equilibrium Account: Long-Run Risk	127
9.3	VRP Predicts Returns	128
9.4	VRP Predicts Future Volatility	128
9.5	Decomposing VRP	129
9.5.1	Uncertainty vs. Risk Aversion	129
9.5.2	Normal-Times vs. Jump-Tail VRP	130
9.6	The Gamma P&L Formula: From Forecast to Money	130
9.6.1	Setup: Delta-Hedged Option P&L	130
9.6.2	Derivation from the Black–Scholes PDE	131
9.7	Vol-of-Vol	133
9.7.1	VVIX: The Volatility of VIX	133
9.7.2	Realized Vol-of-Vol	134
9.7.3	Jumps in VIX	135
9.7.4	Diagram: VIX vs. Realized Vol	135
9.8	ML Approaches to VRP	136
9.8.1	Improved $\mathbb{P}$ -Measure Forecasts	136
9.8.2	VRP as a Trading Signal	136
	Summary	137
IV	ML Methods for Volatility	139
10	Feature Engineering for Volatility	140
10.1	Feature Taxonomy	140
10.2	Triple Expansion: Level, Change, Z-Score	140
10.3	Lagged RV Transforms	142
10.4	Realized Quarticity and the HARQ Feature	144
10.5	Signed and Asymmetric Features	145
10.6	Higher Moments	147
10.7	Microstructure and Limit Order Book Features	147
10.7.1	VPIN and Kyle’s Lambda	149
10.7.2	Amihud Illiquidity Ratio	151
10.7.3	Microprice and Volume Features	152
10.7.4	Order Flow Imbalance and Depth Features	152
10.8	Options-Implied Features	154
10.9	Cross-Asset Features	157
10.10	Long-Memory Features	158
10.10.1	Vol-of-Vol and Regime Duration	159
10.11	Calendar and Event Features	160
10.11.1	Event-Driven Volatility: Beyond Binary Dummies	161
10.11.2	Calendar Proximity Measures	161
10.12	Sentiment and Text Features	163
10.13	Feature Importance and Selection	163
10.14	Proving a Feature Adds Value: The Incremental-Information Test	167
10.14.1	Two Nested Models	167
10.14.2	Three Verdicts: QLIKE, SHAP, and the Information Coefficient	168
10.14.3	Is the Improvement Real? The Diebold–Mariano Test	169

10.15	The Diminishing Returns Curve	170
10.16	Summary	172
11	Tree-Based Methods for Volatility	174
11.1	Why Trees for Volatility	174
11.2	Tree Foundations: From One Tree to a Forest	175
11.2.1	How a regression tree chooses its splits	175
11.2.2	Why one fully grown tree is too unreliable	176
11.2.3	Bagging: averaging away the variance	177
11.2.4	Random forests: decorrelating the trees	177
11.2.5	Out-of-bag error, and why it lies on time series	178
11.3	LightGBM and XGBoost	179
11.4	Hyperparameters for Volatility Data	181
11.5	Interpreting Tree Models: TreeSHAP and the SHAP Plot Toolkit	182
11.5.1	The exponential cost that TreeSHAP defeats	182
11.5.2	The four SHAP plots: a read-and-use taxonomy	183
11.5.3	SHAP vs. MDI vs. MDA: when to use which	184
11.5.4	Presenting SHAP to the desk	185
11.6	The Christensen–Siggaard–Veliyev Evidence	186
11.7	The Optiver Kaggle Evidence	187
11.8	The Honest Assessment	187
11.9	Ensemble with HAR	189
11.10	DART: Dropout Regularization for Boosted Trees	190
11.11	Summary	192
12	Rashomon Sets and Interpretable Trees	194
12.1	The Problem with Single-Model Explanations	194
12.1.1	Why SHAP Importance Is Unreliable with Near-Substitute Features	195
12.1.2	Importance Instability in Practice	195
12.2	The Rashomon Set	195
12.2.1	Formal Definition	196
12.2.2	How Large Is the Rashomon Set?	196
12.3	Optimal Sparse Decision Trees	197
12.3.1	Why Not Just Use CART?	197
12.3.2	Branch-and-Bound with Dynamic Programming	197
12.3.3	SPLIT: Greedy Where It Doesn’t Matter, Optimal Where It Does	198
12.3.4	STreeD: Piecewise-Linear Regression Trees	198
12.4	Enumerating the Rashomon Set	199
12.4.1	TreeFARMS: Exact Enumeration	199
12.4.2	RESPLIT: Fast Approximation via Greedy Leaves	200
12.4.3	LicketyRESPLIT: Polynomial-Time Approximation	201
12.5	What the Rashomon Set Reveals	202
12.5.1	Model Class Reliance (MCR)	202
12.5.2	Variable Importance Clouds (VIC)	202
12.5.3	Rashomon Importance Distributions (RID)	204
12.6	Rashomon Analysis for Volatility Forecasting	206
12.6.1	Regime-Stable Feature Selection	206
12.6.2	Prediction Multiplicity: Quantifying Model-Choice Uncertainty	206
12.6.3	Defensible Presentations	207
12.7	Summary and Connections	207
13	Deep Learning for Volatility	208

13.1	LSTMs and GRUs	208
13.1.1	LSTM Cell Architecture	208
13.1.2	LSTMs for Realized Volatility	210
13.2	Temporal Convolutional Networks	211
13.2.1	Dilated Causal Convolutions	211
13.2.2	DeepVol	212
13.3	DeepLOB: Learning from Limit Order Books	213
13.3.1	Architecture	213
13.4	Transformers and Attention	214
13.4.1	Graph Transformers for Cross-Asset Volatility	215
13.5	Modern Time-Series Architectures	215
13.6	Generative and Latent-Variable Approaches	216
13.6.1	Neural SDEs and CDEs	216
13.6.2	Autoencoders for the Implied Volatility Surface	217
13.6.3	Deep Stochastic Volatility	217
13.6.4	Normalizing Flows for RV Distribution	217
13.7	The Honest Bottom Line	218
13.8	Summary	219
14	Hybrid and Ensemble Models	221
14.1	Why This Chapter Matters	221
14.2	Why Hybrids Win	221
14.2.1	The Variance Budget Argument	221
14.2.2	Variance Decomposition	222
14.2.3	Architecture Diagram	222
14.3	HAR-SVR: Support Vector Regression on HAR Residuals	222
14.3.1	Intuition	222
14.3.2	Procedure	223
14.4	GARCH-Informed Neural Networks	224
14.4.1	Motivation	224
14.4.2	Architecture	224
14.4.3	Why This Works	224
14.5	NLP-Augmented Volatility Models	225
14.5.1	Motivation	225
14.5.2	The Rahimikia-Zohren-Poon Pipeline	225
14.5.3	What the Literature Finds	226
14.6	Ensemble: HAR + LightGBM	226
14.6.1	The Safest Performer	226
14.6.2	Architecture Diagram	227
14.7	Comparing Ensemble Architectures	227
14.7.1	Architecture A: Feature Stacking	227
14.7.2	Architecture B: Residual Stacking (Three-Stage Pipeline)	228
14.7.3	Architecture C: Prediction Blending	229
14.7.4	Architecture Comparison	230
14.8	Identifying Regimes: GMM, EM, and BIC	230
14.8.1	The GMM Density Model for Soft Regime Clustering	232
14.8.2	The EM Algorithm: Soft Labels, Then Updated Blobs	233
14.8.3	Choosing the Number of Regimes: BIC	235
14.8.4	The sklearn Workflow	236
14.8.5	Adding Persistence: Markov-Switching and HMMs	236
14.8.6	Regime Backtesting Hygiene	239
14.8.7	Static vs. Regime-Dependent Weights	239

14.9	When to Use Pure ML vs. Hybrid	240
14.9.1	Practical Checklist	240
14.9.2	Standalone vs. Hybrid: A Visual Comparison	241
14.10	Summary	241
V	Multivariate Volatility and Connectedness	244
15	Realized Covariance and Multivariate Forecasting	245
15.1	Realized Covariance	245
15.1.1	The Synchronicity Problem	245
15.1.2	Refresh-Time Sampling	246
15.1.3	Hayashi–Yoshida Estimator	246
15.2	Multivariate Realized Kernel	248
15.3	DCC-GARCH	248
15.4	Wishart Autoregressive (WAR) Model	249
15.5	HAR-DRD and Multivariate HARQ	250
15.6	Cholesky-HAR	252
15.7	Graph-Based Methods	252
15.8	CNN-RCOV	254
15.9	Geometric Deep Learning on the SPD Manifold	254
15.10	The PSD Constraint	256
15.11	Summary	256
16	Volatility Spillovers and Connectedness	258
16.1	Diebold–Yilmaz Spillover Indices	258
16.1.1	Setup: VAR on Realized Volatilities	258
16.1.2	Generalized Forecast Error Variance Decomposition	259
16.1.3	Spillover Measures	260
16.1.4	Spillover Network Diagram	260
16.1.5	Worked Example: 3-Asset VAR(1)	260
16.2	Time-Varying Spillovers	262
16.2.1	Rolling-Window Approach	262
16.2.2	TVP-VAR Connectedness	263
16.3	Network Visualization and Interpretation	263
16.3.1	Visual Encoding	264
16.3.2	Calm vs. Crisis Networks	264
16.4	Cross-Asset Universality	264
16.4.1	Pooled Panel Forecasting Across Instruments	265
16.5	Spillover Indices as Predictive Features	270
16.5.1	Three Feature Families	270
16.5.2	Practical Considerations	270
16.6	Summary	271
VI	Evaluation and Practice	272
17	Forecast Evaluation	273
17.1	Why Evaluation Methodology Matters	273
17.2	MSE and Its Limitations for Volatility	274
17.2.1	The MSE Formula	274
17.2.2	The Proxy Problem	274

17.2.3	Why MSE Is Still Not Enough	274
17.3	QLIKE: The Preferred Loss Function	274
17.3.1	Intuition	275
17.3.2	The QLIKE Formula	275
17.3.3	Worked Example: MSE vs. QLIKE Rankings	276
17.3.4	Retransformation Bias	278
17.4	Mincer–Zarnowitz Regressions	279
17.4.1	The Regression	279
17.4.2	Interpreting Deviations	281
17.5	The Diebold–Mariano Test	281
17.5.1	Setup	281
17.5.2	The Test Statistic	281
17.5.3	Worked Example	282
17.6	The Model Confidence Set	283
17.6.1	Intuition	283
17.6.2	The Procedure	283
17.6.3	Practical Use	283
17.7	Purged K-Fold Cross-Validation with Embargo	285
17.7.1	Why Standard K-Fold Fails	285
17.7.2	The Fix: Purging and Embargo	286
17.7.3	Worked Example	286
17.7.4	Cross-Sectional Leakage When Pooling Assets	287
17.7.5	The One-Standard-Error Rule for Hyperparameter Selection	288
17.8	Combinatorial Purged CV and the Distribution of OOS Performance	289
17.8.1	The Two Counting Formulas	289
17.9	Probability of Backtest Overfitting (PBO)	292
17.9.1	Combinatorially Symmetric Cross-Validation (CSCV)	292
17.10	The Deflated Sharpe Ratio	294
17.10.1	The Multiple Testing Problem	294
17.10.2	The DSR Formula	295
17.10.3	Worked Example	295
17.10.4	The Haircut Sharpe Ratio in Detail	297
17.11	What Doesn't Work	300
17.11.1	Lookahead Bias: A Taxonomy of Four Sources	301
17.12	Putting It All Together: An Evaluation Workflow	303
17.13	Summary	304
17.13.1	Key Results Recap	305
18	Practical Applications and Project Directions	307
18.1	Volatility Targeting: The Simplest Economic Value Test	307
18.1.1	The EWMA Baseline	307
18.1.2	The Volatility-Targeting Formula	307
18.1.3	Worked Example: Vol-Targeting the S&P 500	308
18.1.4	Connection to Time-Series Momentum	309
18.2	Trading Costs, Turnover, and Net Economic Value	309
18.2.1	Net P&L on the Vol-Targeting Weight	310
18.2.2	Turnover and the Sharpe Drag	311
18.2.3	The Sharpe-vs-Cost Curve and Breakeven Cost	312
18.2.4	What SPX Futures Actually Cost	313
18.2.5	Performance-Metric Reference	313
18.2.6	Fragility: Where Does the P&L Come From?	315
18.3	Dealer Gamma and Structured Products Feedback	315

18.3.1	How Dealer Hedging Amplifies or Suppresses Vol	315
18.3.2	Structured Products and the Short-Gamma Overhang	316
18.3.3	Pin Risk at Expiry	316
18.4	Communicating Volatility Forecasting Results	316
18.4.1	Frame It as a Hypothesis Test, Not a Fishing Expedition	317
18.4.2	One Slide Per Claim	317
18.4.3	Negative Results Are a Credibility Signal	318
18.4.4	The Written Report and Its Page Budget	318
18.4.5	Handling Desk Questions	319
19	From Forecast to P&L: A Realistic, Evaluable IV–RV Straddle	321
19.1	The Strategy in One Picture	321
19.2	The Signal and the Anti-Lookahead Protocol	323
19.3	The P&L Engine: The Gamma Identity	324
19.4	Where the Clean Identity Breaks	326
19.5	Option Transaction Costs	327
19.6	The Hedge Also Costs: Leland and Its Critique	329
19.7	The Variance the Hedge Injects	330
19.8	Calibrating the Kurtosis κ from 5-Minute Data	331
19.9	Assembling the Backtest	333
19.10	Evaluation: From QLIKE to a Deflated Sharpe	333
19.11	What to Compute on Our Data	335
19.12	Summary	336
20	Predicting Drawdowns of a Daily Variance-Swap Seller: The GSVIVS01 Index	337
20.1	The Strategy in One Picture	338
20.2	What a Variance Swap Pays	339
20.3	The Fair Strike: Model-Free Implied Variance	340
20.4	From Integral to Traded Strip: The CBOE Discrete Formula	342
20.5	How GSVIVS01 Trades It: 0DTE Replication	343
20.6	Delta Hedging and the Index	344
20.7	The Risk Profile: Short Gamma, Short Vega, Long Theta	345
20.8	The Drawdown Mechanism: When Realized Variance Beats the Strike	346
20.9	The Signal: A Variance-Risk-Premium Gap on the Forecast	348
20.10	Drawdown Prediction as Cost-Sensitive Classification	349
20.11	From Prediction to Position: The Overlay	351
20.12	Evaluation: The Project’s Metric	351
20.13	What to Compute on Our Data	352
20.14	Summary	353

Part I

What Is Volatility and How Do You Measure It?

Chapter 1

Returns, Variance, and Why Volatility Matters

Financial markets produce a stream of prices. To do anything quantitative with those prices, you first need to convert them into *returns*, measure how much those returns vary, and understand *why* that variation matters. This chapter covers that ground.

1.1 What Are Returns?

A stock price on its own tells you almost nothing about performance. Knowing that Apple closed at \$187 today is useless without knowing where it closed yesterday. *Returns* solve this: they express price changes as fractions (or percentages), making different assets and time periods comparable.

Natural Logarithm Properties

The natural logarithm $\ln(\cdot)$ is the inverse of the exponential function: $\ln(e^x) = x$. Three properties matter here:

- $\ln(A/B) = \ln A - \ln B$ (quotients become differences)
- $\ln(AB) = \ln A + \ln B$ (products become sums)
- For small x : $\ln(1 + x) \approx x$ (so log returns $\approx$ simple returns when returns are small)

1.1.1 Simple Returns

We want a single number that tells us “what fraction of my investment did I gain or lose in one period?” The simple (arithmetic) return does exactly this:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (1.1)$$

- R_t : simple return from period $t - 1$ to t
- P_t : price at the end of period t
- P_{t-1} : price at the end of the previous period

1.1.2 Log Returns

The simple return works fine in many contexts, but for volatility research we almost always use a different version: the log return. Instead of computing the fractional change directly, we take the natural logarithm of the price ratio:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln P_t - \ln P_{t-1} \quad (1.2)$$

- r_t : log return from period $t - 1$ to t
- $\ln(\cdot)$: natural logarithm

Why Log Returns?

Log returns have two properties that make them the default in volatility research:

1. **Time additivity.** The multi-period log return is the sum of single-period log returns: $r_{1:T} = r_1 + r_2 + \dots + r_T$.

Why this works: the 2-day log return is $\ln(P_2/P_0) = \ln((P_1/P_0) \cdot (P_2/P_1)) = \ln(P_1/P_0) + \ln(P_2/P_1) = r_1 + r_2$. The log property $\ln(AB) = \ln A + \ln B$ does all the work. Simple returns compound multiplicatively instead: the 2-day simple return is $(1 + R_1)(1 + R_2) - 1$, not $R_1 + R_2$, which makes variance formulas much more complicated.

2. **Approximate symmetry.** If you gain $+0.05$ log return and then lose -0.05 log return, the total is $\ln(P_{\text{final}}/P_{\text{start}}) = 0.05 + (-0.05) = 0$, meaning $P_{\text{final}} = P_{\text{start}}$ exactly. Simple returns don't cancel this way: $\$100 \times 1.05 \times 0.95 = \$99.75 \neq \$100$.

Throughout this guide, r_t always means the log return unless stated otherwise.

Worked Example:

Computing Returns A stock closes at \$100.00 on Monday, \$105.00 on Tuesday, and \$102.00 on Wednesday.

Tuesday (price rises from \$100 to \$105):

$$R_{\text{Tue}} = \frac{105 - 100}{100} = 0.0500 \quad (+5.00\%)$$

$$r_{\text{Tue}} = \ln\left(\frac{105}{100}\right) = \ln(1.05) = 0.0488 \quad (+4.88\%)$$

Wednesday (price falls from \$105 to \$102):

$$R_{\text{Wed}} = \frac{102 - 105}{105} = -0.0286 \quad (-2.86\%)$$

$$r_{\text{Wed}} = \ln\left(\frac{102}{105}\right) = \ln(0.9714) = -0.0290 \quad (-2.90\%)$$

Two-day log return (time additivity in action):

$$r_{\text{Mon} \rightarrow \text{Wed}} = r_{\text{Tue}} + r_{\text{Wed}} = 0.0488 + (-0.0290) = 0.0198$$

Check directly: $\ln(102/100) = \ln(1.02) = 0.0198$. ✓

The simple returns are close to the log returns because the magnitudes are small (under 5%). For daily equity returns, which are typically well under 1%, the two are nearly identical.

1.2 Variance and Standard Deviation

With returns in hand, the next question is: how spread out are they? Variance answers this by measuring the average squared deviation from the mean return.

Expected Value Notation

Throughout this guide, $\mathbb{E}[X]$ means the **expected value** of X : the long-run average you would get if you could observe X infinitely many times. For example, $\mathbb{E}[r_t]$ is the average return you'd observe over an infinite number of trading days. When you see $\mathbb{E}[\cdot]$ in a formula, read it as “the average of what’s inside the brackets.”

Sample Variance and Standard Deviation

Given T log returns $r_1, r_2, \dots, r_T$ with sample mean $\bar{r} = \frac{1}{T} \sum_{t=1}^T r_t$, we want a single number that captures “how spread out are these returns?” The sample variance answers this by measuring the average squared distance of each return from the mean:

$$\hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2 \quad (1.3)$$

- $\hat{\sigma}^2$: sample variance (the hat denotes an estimate)
- T : number of observations
- r_t : log return at time t
- $\bar{r}$: sample mean of returns
- $T-1$: Bessel’s correction (dividing by $T-1$ instead of T gives an unbiased estimate)

The sample standard deviation is $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$.

Why $T-1$?

Dividing by $T-1$ instead of T corrects for the fact that you estimated $\bar{r}$ from the same data. Using $\bar{r}$ instead of the true mean μ systematically shrinks the deviations, so dividing by the smaller number $T-1$ compensates. This is called Bessel’s correction. For large T the difference is negligible.

1.2.1 Annualizing Volatility

Raw daily standard deviations are tiny numbers (around 0.01 for a typical equity index), so practitioners report volatility on an annual scale. The key insight: if daily returns are independent, variances add. The n -day return is $r_1 + r_2 + \dots + r_n$ (time additivity), and for independent random variables, $\text{Var}(r_1 + r_2 + \dots + r_n) = \text{Var}(r_1) + \text{Var}(r_2) + \dots + \text{Var}(r_n) = n\sigma^2$. Taking the square root: the n -day standard deviation is $\sqrt{n}\sigma$.

$$\hat{\sigma}_{\text{annual}} = \hat{\sigma}_{\text{daily}} \times \sqrt{252} \quad (1.4)$$

- 252: approximate number of trading days per year in U.S. equity markets
- $\sqrt{252} \approx 15.87$

Why This Matters

Your model’s output (daily RV forecasts) will need to be converted to the annualized scale for comparison with implied volatility and industry benchmarks.

The Square-Root-of-Time Rule

The $\sqrt{252}$ scaling assumes returns are independent and identically distributed. When volatility clusters (Section 1.4), this assumption fails, and multi-day volatility may be

higher or lower than the square-root rule predicts. Chapters 5 and 6 develop models that handle this.

1.3 Why Volatility Matters

You now know how to compute returns and measure their spread. The next question: why does that spread matter so much? Variance and standard deviation sit at the center of four major problems in finance.

Finance Concepts Preview

The following four applications are covered briefly here to motivate the rest of the guide. You do not need to understand them fully yet; each connects to a later chapter.

Options pricing. An option gives the holder the right (not obligation) to buy or sell an asset at a fixed price by a future date. The value of this right depends critically on how much the underlying price might move, which is volatility. The Black-Scholes model (Black and Scholes, 1973) takes volatility as an *input* and produces an option price as output. If your volatility estimate is wrong, your price is wrong. Chapter 8 explores this connection.

Risk management. Value-at-Risk (VaR) estimates the worst-case loss over a holding period at a given confidence level. In the simplest version, VaR is directly proportional to volatility: if volatility doubles, your risk bound roughly doubles. For a 99% daily VaR, the worst expected loss is about 2.33 times the daily volatility (e.g., if daily vol is 1%, the worst-case daily loss at 99% confidence is roughly 2.33%).

Portfolio construction. Mean-variance optimization (Markowitz, 1952) and risk-parity strategies both require a covariance matrix as input. Volatility is the diagonal of that matrix. Better volatility forecasts produce better-diversified portfolios.

Trade execution. When a desk needs to buy a large position, it splits the order over time to reduce market impact. The participation rate (fraction of daily volume per time slice) depends on intraday volatility: higher volatility means wider price swings, which changes optimal execution speed.

What Makes Volatility Special

Volatility is one of the few quantities in finance that is (a) directly observable from intraday data, (b) economically central to pricing, hedging, and risk control, and (c) forecastable with meaningful accuracy. That combination makes it the right place to apply ML carefully.

Preview: What Is Realized Volatility?

The central concept of this guide is **realized volatility** (RV): the sum of squared intraday returns over one day, $RV_t = \sum_{i=1}^M r_{t,i}^2$, where M is the number of intraday observations (e.g., 78 five-minute returns per trading day). Instead of using a model to *estimate* what volatility was, RV directly *measures* it from high-frequency data. Chapter 2 develops this fully; for now, just know that RV gives you a daily number that tells you “how volatile was the market today,” and your project’s goal is to forecast tomorrow’s RV given today’s information.

1.4 Stylized Facts of Financial Returns

Before building models, you need to know what patterns actually appear in return data. [Cont \(2001\)](#) catalogued a set of “stylized facts”: statistical regularities observed across equities, foreign exchange, and commodities, across decades and geographies. Four of these facts matter most for volatility modeling.

1.4.1 Fact 1: Returns Are Approximately Uncorrelated

Autocorrelation

Autocorrelation at lag k measures how correlated a time series is with itself shifted k periods into the past. A value near $+1$ means “if it was high today, it will tend to be high k days from now”; near 0 means “no relationship”; near -1 means “high today predicts low k days later.” When we say “return autocorrelations are zero,” we mean knowing today’s return tells you nothing about tomorrow’s.

Daily return autocorrelations are statistically indistinguishable from zero for lags beyond a few minutes ([Cont, 2001](#)). In plain terms: knowing today’s return tells you almost nothing about tomorrow’s return. This is consistent with market efficiency; if returns were predictable, traders would exploit the pattern until it disappeared.

No Free Lunch in Means

If positive returns reliably followed positive returns, everyone would buy after an up day, driving the price up immediately and eliminating the pattern. Competition among traders keeps return autocorrelations near zero. Volatility, by contrast, is *not* a quantity you can directly trade on the same way, so its autocorrelation can persist.

1.4.2 Fact 2: Volatility Clusters

Although returns themselves are uncorrelated, *squared* returns and *absolute* returns show strong, slowly decaying positive autocorrelation ([Cont, 2001](#); [Mandelbrot, 1963](#)). Large moves tend to follow large moves, and calm periods tend to follow calm periods.

[Engle \(1982\)](#) formalized this observation with the ARCH model: the variance of next period’s return depends on recent squared returns. Chapter 5 develops this family of models in detail.

Cont (2001), Stylized Fact: Volatility Clustering

The autocorrelation of absolute returns remains significantly positive over lags of several weeks and decays slowly (approximately as a power law with exponent $\beta \in [0.2, 0.4]$), a pattern that is remarkably stable across asset classes and time periods.

What “Power Law Decay” Means

A **power law decay** means the autocorrelation drops as $\text{lag}^{-\beta}$, much more slowly than exponential decay. With $\beta \approx 0.3$, the autocorrelation at lag 100 is still about $100^{-0.3} \approx 0.25$: volatility 100 days ago is still informative about today’s volatility. This “long memory” is why simple models that only look at yesterday fail. The HAR model (Chapter 6) addresses this by explicitly including weekly and monthly vol averages as predictors.

In Figure 1.1, returns visibly alternate between calm and turbulent stretches.

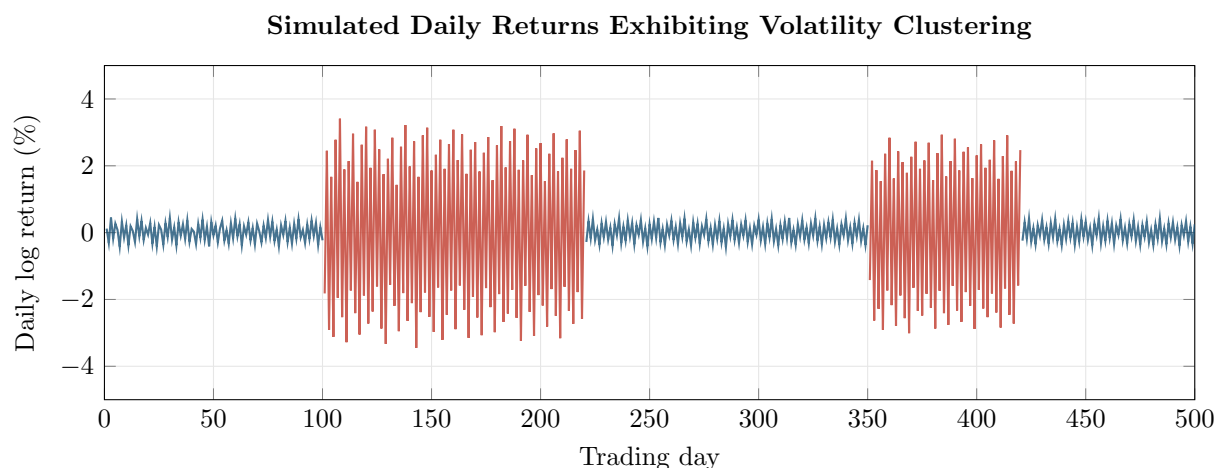


Figure 1.1: Simulated daily log returns exhibiting volatility clustering. Calm periods (blue, days 1–100, 221–350, 421–500) have returns within $\pm 0.5\%$; turbulent periods (red, days 101–220, 351–420) reach $\pm 3\%$. The clustering is visible by eye: large returns beget large returns.

Figure 1.2 quantifies this. Return autocorrelations hover near zero at all lags, but squared-return autocorrelations remain positive for many lags, confirming that volatility is persistent.

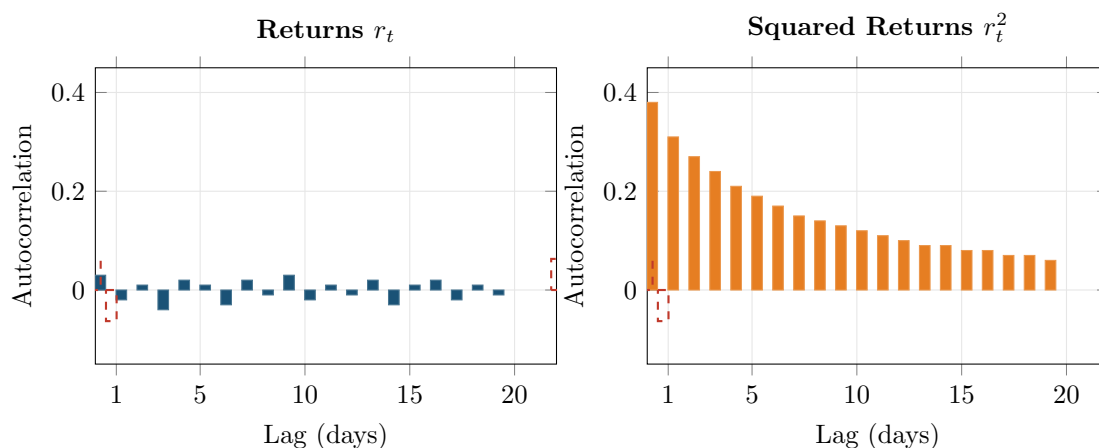


Figure 1.2: Autocorrelation functions for returns (left) and squared returns (right), based on representative daily equity index data. Returns show no significant autocorrelation at any lag (all bars within the dashed 95% significance bands). Squared returns show strong autocorrelation that decays slowly, reflecting volatility clustering.

1.4.3 Fact 3: Fat Tails

If returns were normally distributed, a daily move of 4 standard deviations would occur about once every 63 years. In practice, moves this large happen several times per year. The return distribution has “fat tails” (also called heavy tails): extreme returns are far more frequent than a Gaussian model predicts.

[Mandelbrot \(1963\)](#) first documented this for cotton prices, proposing that returns follow a stable Paretian distribution rather than a Gaussian. [Fama \(1965\)](#) confirmed the finding for stock returns, observing leptokurtic (heavy-tailed) distributions across U.S. equities.

Kurtosis

We need a way to measure “how fat are the tails?” compared to a normal distribution. Kurtosis does this by looking at the fourth power of deviations (which amplifies extreme values even more than squaring):

$$\kappa = \frac{\mathbb{E}[(r_t - \mu)^4]}{(\mathbb{E}[(r_t - \mu)^2])^2} \quad (1.5)$$

- κ : kurtosis
- $\mu = \mathbb{E}[r_t]$: population mean
- For a normal distribution, $\kappa = 3$
- *Excess kurtosis* $= \kappa - 3$; values above zero indicate fatter tails than normal

In Plain English

The 4th power is the key: if extreme returns are rare (thin tails), raising them to the 4th power doesn't add much. If extreme returns are common (fat tails), the 4th power blows them up and the ratio exceeds 3. The further above 3, the fatter the tails.

Why This Matters

Fat tails mean that extreme vol events (crashes, spikes) are far more likely than a normal model predicts. Your vol forecasting model must handle these outliers gracefully: if you assume normality, you will systematically underestimate the risk of large moves, and your model will appear to work in calm markets but fail catastrophically during crises.

Typical daily equity index returns have excess kurtosis in the range 5–10 (Cont, 2001).

Figure 1.3 illustrates fat tails with a **Q–Q (quantile-quantile) plot**. Here is how it works: sort your actual returns from smallest to largest, then ask “if these returns *were* normally distributed, what values would I expect at each position in the sorted list?” Plot the expected-if-normal values on the x -axis and the actual values on the y -axis. If the data is truly normal, expected equals actual at every point, so you get a straight diagonal line. When the dots curve away from the line at the extremes, it means the tails of your actual distribution are fatter than normal: extreme moves happen more often than the bell curve predicts.

Why Fat Tails Matter for Models

Any model that assumes normally distributed returns will underestimate the probability of large moves. This affects risk management (VaR breaches), option pricing (mispricing of out-of-the-money options), and backtesting (understating drawdowns). Chapters 4 and 5 introduce models that accommodate fat tails.

1.4.4 Fact 4: The Leverage Effect

Negative returns tend to increase future volatility more than positive returns of the same magnitude. Black (1976) first noted this asymmetry and proposed a mechanism: when a stock price falls, the firm's equity value shrinks relative to its debt, so leverage (debt/equity) rises, making the stock riskier and more volatile.

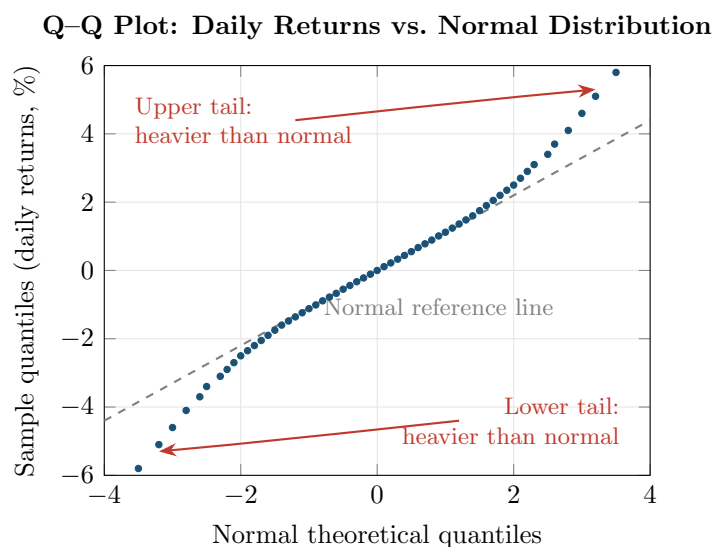


Figure 1.3: Q–Q plot of representative daily equity returns against a normal distribution. If returns were Gaussian, the dots would lie on the dashed diagonal. The S-shaped departure, first documented by [Fama \(1965\)](#), shows that both tails of the empirical distribution are heavier than normal: extreme moves (positive and negative) occur much more often than a Gaussian model predicts.

Leverage Effect

The leverage effect is the negative correlation between an asset’s return and changes in its subsequent volatility:

$$\text{Corr}(r_t, \sigma_{t+1}^2) < 0 \quad (1.6)$$

- r_t : return at time t
- σ_{t+1}^2 : conditional variance at time $t + 1$
- The negative sign means: price drops $\rightarrow$ volatility rises

Why This Matters

The leverage effect means your vol forecasting model should treat negative and positive returns differently. A symmetric model (basic GARCH) that treats +2% and –2% as equally informative about tomorrow’s vol will systematically underpredict vol after selloffs and overpredict after rallies. This is why GJR-GARCH and EGARCH (Chapter 5) add an asymmetry term, and why it often improves forecast accuracy.

In equity markets, the leverage effect is pronounced. In foreign exchange and commodity markets, the asymmetry is weaker or sometimes reversed ([Cont, 2001](#)). Chapter 5 introduces GJR-GARCH and EGARCH models that capture this asymmetry directly.

1.4.5 Summary of Stylized Facts

Four Facts That Shape Every Volatility Model

1. **Uncorrelated returns:** tomorrow's return direction is unpredictable from today's.
 2. **Volatility clustering:** large moves follow large moves, small follow small.
 3. **Fat tails:** extreme returns occur far more often than a Gaussian model predicts.
 4. **Leverage effect:** negative returns raise future volatility more than positive returns.
- Any useful volatility model must reproduce (at minimum) facts 2 and 3. A good model also captures fact 4.

1.5 Conditional vs. Unconditional Volatility

Everything so far has treated volatility as a single number: the standard deviation of the full return sample. That number is the *unconditional* volatility. It answers the question “what is the typical size of a daily return, averaging over all market conditions?”

Traders need a different answer: given what has happened up to today, how volatile will the market be *tomorrow*? That is the *conditional* volatility.

Unconditional vs. Conditional Variance

Unconditional variance:

$$\sigma^2 = \text{Var}(r_t) \quad (1.7)$$

A single number, constant over time. It is the long-run average variance.

Conditional variance:

$$\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1}) \quad (1.8)$$

- σ_t^2 : variance of the return at time t , conditional on past information
- $\mathcal{F}_{t-1}$: the **information set** available at time $t - 1$. Read this as “everything you could possibly know as of yesterday”: all past returns, prices, volumes, news, and any other observable data up through time $t - 1$. The fancy script- $\mathcal{F}$ notation comes from probability theory (where it is called a “filtration”); all it means in practice is “given what we knew yesterday.” This notation appears throughout the entire guide whenever a quantity is conditioned on past information.

The conditional variance changes from day to day as new information arrives.

Weather Analogy

Unconditional volatility is like the average annual rainfall for your city: a useful summary, but it does not tell you whether to carry an umbrella tomorrow. Conditional volatility is the weather forecast: it uses recent data (yesterday's pressure, today's cloud cover) to predict tomorrow's conditions. This guide is about building the forecast, not computing the average.

Worked Example:

Conditional vs. Unconditional in Practice Over 2024, the S&P 500's annualized volatility was about 16% (the unconditional estimate). But during a single turbulent week in August, realized daily vol spiked to 35% annualized; during a calm stretch in November, it dropped to 9%. The unconditional model always says “16%.” A conditional model notices the recent spike and says “tomorrow will be around 30%” (or notices the calm and

says “around 10%”).

The unconditional variance and conditional variance are connected by a standard probability result called the **law of total variance**. You do not need to derive it; just understand what it says about how the long-run average relates to the time-varying quantity:

$$\sigma^2 = \mathbb{E}[\sigma_t^2] + \text{Var}(\mathbb{E}[r_t | \mathcal{F}_{t-1}]) \quad (1.9)$$

- $\mathbb{E}[\sigma_t^2]$: average of the conditional variances over time
- $\text{Var}(\mathbb{E}[r_t | \mathcal{F}_{t-1}])$: variance of the conditional mean (small for equities, since expected returns are nearly constant at daily frequency)

Why This Matters

This equation tells you that the long-run average vol is a meaningful “anchor” for your forecasts. Many models (GARCH, HAR) build in mean-reversion: after a vol spike, forecasts should gradually drift back toward this long-run level. If your model’s forecasts don’t average out to something close to the unconditional variance over long horizons, something is likely wrong.

When the conditional mean is approximately constant (a reasonable assumption for daily equity returns), the unconditional variance is simply the time average of the conditional variances. The gap between the two is what makes volatility modeling interesting.

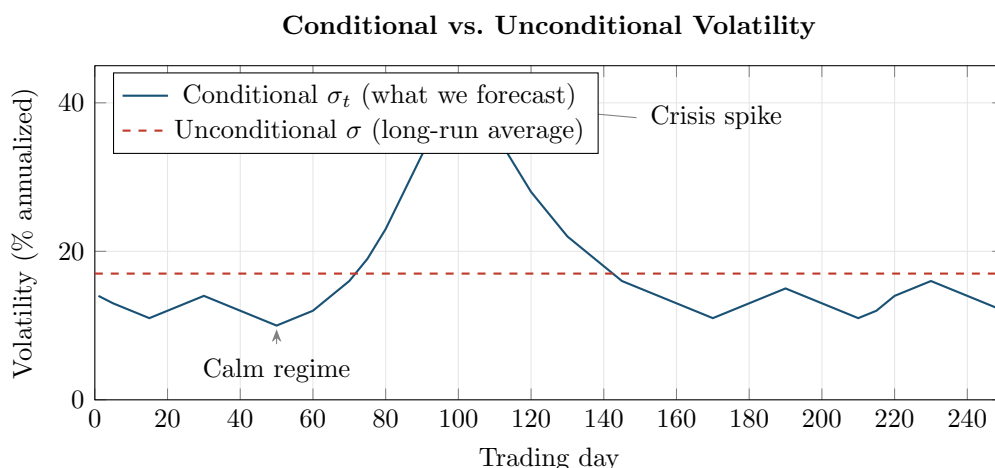


Figure 1.4: The unconditional volatility (dashed red) is a single long-run average: it tells you nothing about which regime you’re in today. The conditional volatility (solid blue) varies over time, spiking during crises and falling during calm periods. Your project’s goal is to forecast the blue line one step ahead; knowing the red line alone is not enough.

The Central Goal of This Guide

The goal of every model in Chapters 5–14 is to produce accurate estimates of the conditional variance σ_t^2 . Chapter 2 shows how to measure the realized conditional variance from high-frequency data. Chapters 5–14 show how to forecast it. Chapter 17 shows how to tell whether your forecasts are any good.

Summary

- **Returns** convert raw prices into comparable, analyzable quantities. Log returns ($r_t = \ln P_t - \ln P_{t-1}$) are preferred because they are additive over time.
- **Simple vs. log returns** are nearly identical for small magnitudes (daily equities); they diverge for large moves.
- **Sample variance** ($\hat{\sigma}^2$) measures dispersion around the mean return; standard deviation ($\hat{\sigma}$) is its square root.
- **Annualized volatility** scales daily volatility by $\sqrt{252}$, assuming independent returns.
- **Volatility matters** because it is a direct input to options pricing, risk management (VaR), portfolio optimization, and trade execution.
- **Stylized fact 1:** Returns are approximately uncorrelated (no predictability in the mean).
- **Stylized fact 2:** Squared and absolute returns are strongly autocorrelated (volatility clusters).
- **Stylized fact 3:** Return distributions have fat tails (excess kurtosis of 5–10 for daily equity returns).
- **Stylized fact 4:** The leverage effect makes volatility respond asymmetrically to negative vs. positive returns.
- The **unconditional** variance is a single long-run number; the **conditional** variance σ_t^2 varies over time and is what we want to forecast.
- The **central goal** of this guide is to estimate and forecast the conditional variance σ_t^2 using both classical and ML methods.
- [Cont \(2001\)](#) provides the canonical reference for stylized facts across asset classes.
- [Mandelbrot \(1963\)](#) and [Fama \(1965\)](#) established that financial returns have fat tails, not Gaussian distributions.
- [Engle \(1982\)](#) introduced ARCH, the first formal model of time-varying conditional variance, covered in Chapter 5.

Table 1.1: Key results referenced in this chapter.

Paper	Result	Relevance
Mandelbrot (1963)	Daily commodity returns follow a stable Paretian (fat-tailed) distribution, not Gaussian; sample variance does not converge.	First evidence that normal-distribution assumptions fail for financial data.
Fama (1965)	Confirmed fat tails (leptokurtosis) in daily U.S. stock returns via S-shaped Q–Q departures from normality.	Extended Mandelbrot’s finding to equities; supported the stable distribution hypothesis.
Black (1976)	Documented negative correlation between stock returns and subsequent volatility changes (the leverage effect).	Motivates asymmetric volatility models (GJR-GARCH, EGARCH) in Chapter 5.
Engle (1982)	Introduced ARCH: conditional variance is a function of past squared residuals; applied to UK inflation.	Founded the entire field of conditional volatility modeling (Chapter 5).
Cont (2001)	Canonical enumeration of stylized facts (absence of return autocorrelation, volatility clustering with slow power-law decay, fat tails, leverage effect) across equities, FX, and commodities.	Benchmark reference for the empirical properties any volatility model must match.

Chapter 2

Realized Volatility

Why This Chapter

Realized volatility is the target variable for every project direction in this guide. Chapters 3 and 4 refine the estimator (handling noise and jumps), Chapter 6 builds the HAR forecasting model around it, and Chapter 10 uses RV-derived features. You need to understand RV at an intuitive level (what it measures, how it is constructed, why 5-minute sampling) before any of that.

Chapter 1 introduced variance as a measure of how spread out returns are. That chapter computed variance from daily returns over weeks or months, producing a single number for the whole sample: unconditional volatility. This chapter moves to a different question: how volatile was the market *today*? Answering that requires looking inside the trading day at high-frequency (intraday) returns.

2.1 The Theoretical Target: Integrated Variance

Before building an estimator, you need to know what you are estimating. The theoretical quantity is called *integrated variance*.

Integrals as Accumulated Area

If $f(x)$ is a non-negative function, the integral $\int_a^b f(x) dx$ equals the area under the curve f between a and b . When f varies over time, the integral accumulates all of those local values into a single total. You can think of $\int_a^b f(x) dx$ as “add up all the tiny contributions of f across the interval $[a, b]$.”

Price Processes (Informal)

In continuous-time finance, the log price $p_t = \ln P_t$ evolves according to

$$dp_t = \mu_t dt + \sigma_t dW_t$$

You do not need stochastic calculus to follow this chapter. Here is what the equation says in plain English: over any tiny time interval, the change in the log price is made up of two pieces:

- $\mu_t dt$: a small predictable nudge (the drift, or expected return per unit time). At intraday horizons this is negligibly small and we ignore it.

- $\sigma_t dW_t$: a random shock whose size is controlled by σ_t (the instantaneous volatility). W_t is a Wiener process (continuous random walk), so dW_t is a tiny random push that is equally likely to be positive or negative.

The key takeaway: σ_t is the quantity that controls how much the price wiggles at each instant, and it can change from moment to moment within the day.

Time Notation Convention

Throughout this guide, t is an integer day counter. Day t runs from the market close on day $t - 1$ to the market close on day t . Inside day t , we use s (in integrals) or i (in sums) to index moments or sub-intervals within that day. So σ_s is the instantaneous volatility at some moment s during the day, while $r_{t,i}$ is the return over the i -th intraday interval on day t .

At every instant during the trading day, there is a local volatility level σ_t describing how rapidly the price fluctuates. This level is not constant; it changes as news arrives, liquidity shifts, and traders adjust positions. *Integrated variance* adds up all of these instantaneous variance levels across the day to produce a single summary of total price variation.

Think of it by analogy: if you drive a car at varying speeds throughout the day, the total distance driven is the integral of your speed over time. Integrated variance is the “total distance” of price fluctuation.

Integrated Variance

For a log-price process p_t with instantaneous variance σ_t^2 , the integrated variance over day t (from the close of day $t - 1$ to the close of day t) is:

$$IV_t = \int_{t-1}^t \sigma_s^2 ds \quad (2.1)$$

- IV_t : integrated variance over day t
- σ_s^2 : instantaneous variance at time s within the day
- ds : an infinitesimal increment of time
- The integral sums the instantaneous variances from the previous close ($t - 1$) to today's close (t)

Integrated variance is a *latent* quantity: you never observe σ_s^2 directly. The entire point of this chapter is to estimate IV_t from observable intraday price data.

2.2 Realized Variance as an Estimator

You now know the target: integrated variance IV_t . The question is how to estimate it from data you can actually observe. The answer, developed by [Andersen et al. \(2001\)](#) and [Barndorff-Nielsen and Shephard \(2002\)](#), is simple: sum up squared intraday returns.

2.2.1 Construction

Divide the trading day into n equal intervals. At the end of each interval, record the price. Compute the log return over each interval. Square each return and sum.

Realized Variance

Given n intraday log returns $r_{t,1}, r_{t,2}, \dots, r_{t,n}$ within day t , the realized variance is:

$$\text{RV}_t = \sum_{i=1}^n r_{t,i}^2 \quad (2.2)$$

- RV_t : realized variance for day t
- $r_{t,i}$: the i -th intraday log return on day t ; that is, $r_{t,i} = p_{t,i} - p_{t,i-1}$, where $p_{t,i}$ is the log price at the end of the i -th interval
- n : the number of intraday intervals (e.g., $n = 78$ for 5-minute intervals over a 6.5-hour U.S. equity trading day)
- No mean subtraction: the sample mean of intraday returns is so close to zero that omitting it improves finite-sample performance ([Andersen et al., 2003](#))

Why Squaring Works (Deeper)

From the price process equation above ($dp_t = \mu_t dt + \sigma_t dW_t$), each small intraday return is approximately the local volatility times a random draw: $r_{t,i} \approx \sigma_{t,i} \epsilon_i$, where $\sigma_{t,i}$ is the volatility during interval i and ϵ_i is a random variable with mean zero and variance one (coming from the Wiener process increments). Squaring gives $r_{t,i}^2 \approx \sigma_{t,i}^2 \epsilon_i^2$. Since ϵ_i has variance 1 and mean 0, $\mathbb{E}[\epsilon_i^2] = \text{Var}(\epsilon_i) + (\mathbb{E}[\epsilon_i])^2 = 1 + 0 = 1$. So on average, each squared return equals the local variance: $\mathbb{E}[r_{t,i}^2] \approx \sigma_{t,i}^2$. Summing across the day accumulates these local variance snapshots. As intervals shrink ($n \rightarrow \infty$), the random fluctuations in each ϵ_i^2 average out, and the sum converges to the integrated variance.

2.2.2 Why It Works: Convergence to Quadratic Variation

Why should summing squared (rather than absolute or cubed) returns give us anything meaningful? The answer comes from a deep result in probability theory: for processes like stock prices, the sum of squared increments converges to a specific quantity called the **quadratic variation**, and this quantity equals the integrated variance we're trying to measure. This subsection explains that result.

Quadratic Variation (Informal)

For a continuous-time stochastic process X_t , the quadratic variation $[X]_t$ measures the cumulative squared increments of the path up to time t . It is defined as the limit of the sum of squared increments as the partition becomes infinitely fine:

$$[X]_t = \lim_{n \rightarrow \infty} \sum_{i=1}^n (X_{t_i} - X_{t_{i-1}})^2$$

For a process with continuous paths (no jumps), the quadratic variation equals the integrated variance: $[X]_t = \int_0^t \sigma_s^2 ds$. If the process has jumps, the quadratic variation also captures the sum of squared jumps.

The chain to remember: RV (what you compute from data) $\rightarrow$ QV (the limit as sampling gets infinitely fine) = IV (the true volatility you want, if there are no jumps). Quadratic variation is the theoretical bridge connecting your finite-sample computation to the true quantity of interest.

The central result is this: as you sample more frequently (as $n \rightarrow \infty$ and the length of each

interval $\Delta \rightarrow 0$), the sum of squared returns converges to the quadratic variation of the log-price process (Andersen et al., 2001; Barndorff-Nielsen and Shephard, 2002):

$$RV_t \xrightarrow{p} [p]_t \quad \text{as } \Delta \rightarrow 0 \quad (2.3)$$

- $\xrightarrow{p}$: convergence in probability (meaning the estimate gets arbitrarily close to the true value as sampling gets finer)
- $[p]_t$: quadratic variation of the log-price process on day t
- $\Delta = 1/n$: the length of each sampling interval (as a fraction of the trading day)

In Plain English

Think of it like measuring the total wobble of a drunk person's walk: the more finely you measure their steps, the more zigzags you capture. As your measurement gets infinitely fine, you converge on the *true* total wobble. That true total wobble is the quadratic variation $[p]_t$.

Now, what exactly is this quadratic variation equal to? That depends on whether the price path had any sudden jumps during the day.

If the price path is continuous (no jumps), then the quadratic variation equals the integrated variance, exactly:

$$[p]_t = IV_t \quad (\text{no jumps}) \quad (2.4)$$

If the price path has jumps (sudden discontinuous moves, like a price gap after an earnings announcement), the quadratic variation picks up an extra component:

$$[p]_t = IV_t + \sum_{s \leq t} (J_s)^2 \quad (\text{with jumps}) \quad (2.5)$$

- J_s : the jump in log price at time s (zero if no jump occurs at s). Some papers write this as Δp_s ; we use J_s here to avoid confusion with Δ (sampling interval length) used elsewhere.
- The sum looks like it runs over infinitely many times, but in practice there are only a few jumps per day (or none), so the sum has at most a handful of nonzero terms.
- Separating jumps from continuous variation is the subject of Chapter 4.

Why This Matters

In practice, jumps are relatively rare but can be large. Including jump variation in your forecasting target adds noise: jumps are hard to predict (they're often driven by surprise events), so your model gets a noisier signal. Many of the best forecasting models (HAR-CJ, SHAR) separate the jump and continuous components and model them separately, because the continuous part is far more persistent and predictable.

Andersen et al. (2001), Barndorff-Nielsen and Shephard (2002): RV Consistency

Under mild regularity conditions, realized variance RV_t is a consistent estimator of quadratic variation $[p]_t$. In the absence of jumps, RV_t consistently estimates integrated variance IV_t . This result holds regardless of the specific form of σ_t ; the volatility process can be stochastic, path-dependent, or driven by latent factors.

Figure 2.1 summarizes the full chain of relationships between the quantities introduced so far.

Figure 2.2 illustrates the convergence idea: as you chop the day into more intervals, the sum of squared returns gets closer to the true integrated variance.

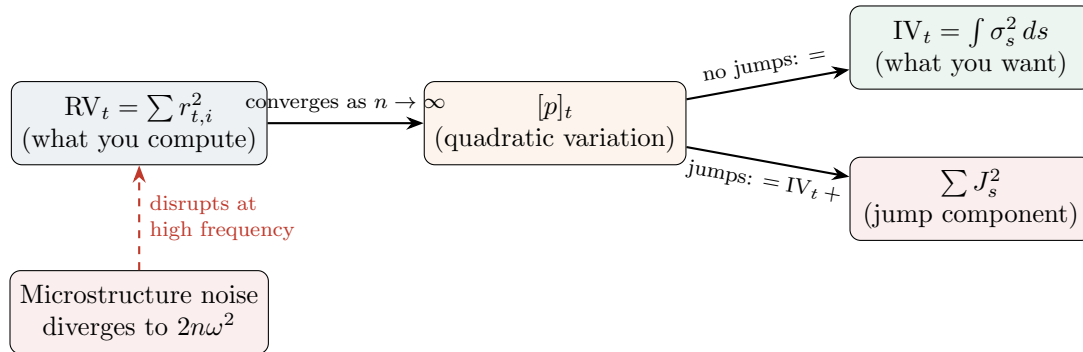


Figure 2.1: The conceptual chain connecting the three key quantities. **RV** (what you compute from data) converges to **QV** (a mathematical limit) as sampling gets finer. Without jumps, QV equals **IV** (the true volatility you want). With jumps, QV includes both IV and a jump component (Chapter 4 separates them). Microstructure noise disrupts the convergence at high sampling frequencies (Chapter 3 fixes this).

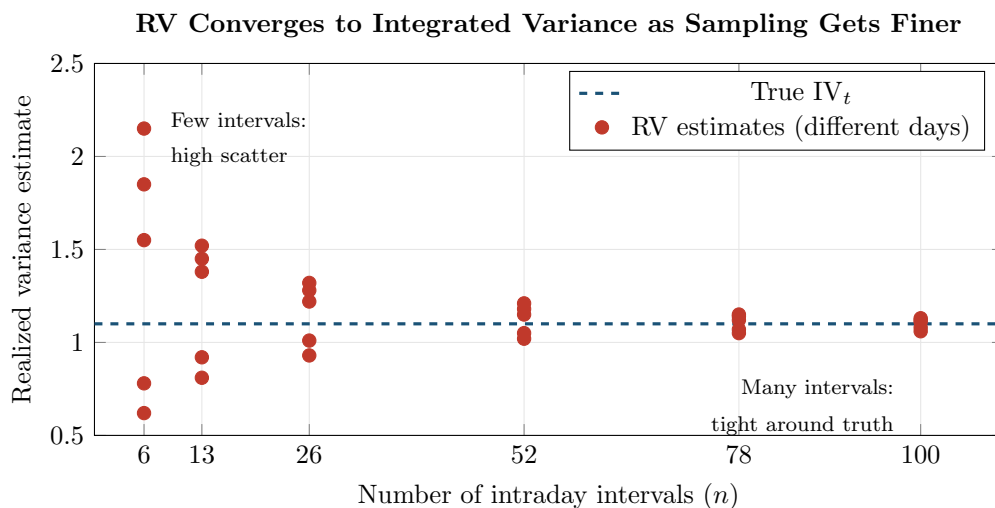


Figure 2.2: Convergence of RV to integrated variance. Each red dot is an RV estimate from a different simulated day. With only 6 intervals (hourly sampling), estimates are scattered widely around the true IV_t (dashed blue). With 78 intervals (5-minute sampling), estimates cluster tightly. This is the convergence result of Equation (2.3) in action. (In practice, microstructure noise prevents going much beyond 78 intervals.)

2.2.3 Diagram: Building RV from a Price Path

Figure 2.3 shows how RV is constructed from a single day's price data.

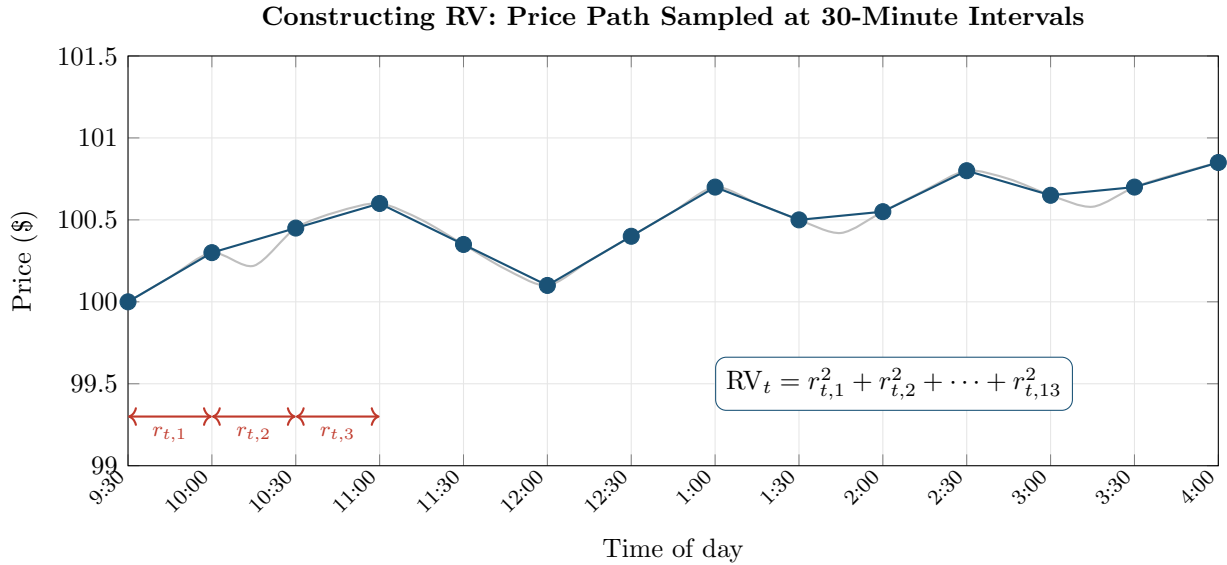


Figure 2.3: Constructing realized variance from a price path. Gray: the continuous intraday price. Blue dots: prices sampled at half-hourly intervals ($n = 13$ intervals). Each interval produces a log return $r_{t,i}$. RV is the sum of all squared returns. Sampling more frequently (e.g., every 5 minutes instead of 30) produces more terms in the sum and a more precise estimate.

Worked Example:

Computing RV from Half-Hourly Returns A stock opens at \$100.00 and you record its price every 30 minutes over a shortened trading day (6 intervals for simplicity). The prices and resulting log returns are:

Time	Price (\$)	Log return $r_{t,i}$	$r_{t,i}^2$
9:30 (open)	100.00		
10:00	100.30	$\ln(100.30/100.00) = +0.003000$	9.000×10^{-6}
10:30	100.10	$\ln(100.10/100.30) = -0.001998$	3.992×10^{-6}
11:00	100.50	$\ln(100.50/100.10) = +0.003992$	15.936×10^{-6}
11:30	100.20	$\ln(100.20/100.50) = -0.002992$	8.952×10^{-6}
12:00	100.60	$\ln(100.60/100.20) = +0.003988$	15.904×10^{-6}
12:30 (close)	100.45	$\ln(100.45/100.60) = -0.001491$	2.223×10^{-6}

Step 1: Sum the squared returns.

$$RV_t = (9.000 + 3.992 + 15.936 + 8.952 + 15.904 + 2.223) \times 10^{-6} = 56.007 \times 10^{-6}$$

Step 2: Convert to realized volatility.

$$\sqrt{RV_t} = \sqrt{56.007 \times 10^{-6}} = 0.00748 \quad (0.75\% \text{ daily})$$

Step 3: Annualize.

$$\text{Annualized realized volatility} = 0.00748 \times \sqrt{252} = 0.119 \quad (11.9\%)$$

For context, 11.9% annualized is a bit below the S&P 500's long-run average (roughly 15–20%). With only 6 intervals, this estimate is noisy. Sampling every 5 minutes would give 78 intervals over a 6.5-hour day, producing a much more precise estimate.

2.3 How Frequently to Sample

You now have an estimator (RV_t) and a convergence result (it approaches $[p]_t$ as $n \rightarrow \infty$). The obvious next step is to sample as frequently as possible: every second, every tick, every trade. In theory, this is optimal. In practice, it fails.

2.3.1 The Theory: More Is Better

The convergence result in Equation (2.3) says that RV becomes a better estimator as Δ shrinks. The convergence result tells you RV approaches IV_t as you sample more frequently, but how *fast*? How precise is your estimate with, say, 78 five-minute intervals versus 390 one-minute intervals? Barndorff-Nielsen and Shephard (2002) showed the estimation error follows a central limit theorem:

$$\sqrt{n}(RV_t - IV_t) \xrightarrow{d} \mathcal{N}\left(0, 2 \int_{t-1}^t \sigma_s^4 ds\right) \quad (2.6)$$

- $\xrightarrow{d}$: convergence in distribution (the error becomes approximately normally distributed)
- $\sqrt{n}$: the estimation error shrinks at rate $1/\sqrt{n}$
- $RV_t - IV_t$: the gap between your estimate and the truth
- $2 \int_{t-1}^t \sigma_s^4 ds$: the asymptotic variance of the error (it depends on the “quarticity,” the integral of σ^4 , which measures how variable the volatility itself was during the day)

In Plain English

This equation says: “the error in your RV estimate is approximately bell-shaped, centered on zero, and shrinks as $1/\sqrt{n}$.” Doubling the number of intervals (going from 5-minute to 2.5-minute sampling) cuts your standard error by a factor of $\sqrt{2} \approx 1.41$.

Why does σ^4 appear? Each squared return $r_{t,i}^2$ estimates the local variance $\sigma_{t,i}^2$, but with some error. The variance of that estimation error is proportional to $\sigma_{t,i}^4$ (because computing the variance of a squared quantity involves squaring something that is already squared). Summing these error variances across the day gives $\int \sigma_s^4 ds$, called the “quarticity.” On days when volatility was itself highly variable within the day, quarticity is large, and your RV estimate is less precise.

Why This Matters

The HARQ model (Chapter 6) explicitly accounts for this: it estimates each day’s RV precision from the CLT and downweights noisy observations. Days with few intraday observations (holiday-shortened sessions) or wildly variable intraday vol have noisier RV, and a smart model adjusts for this.

So with perfect data, you would sample every millisecond and get an essentially noise-free estimate of integrated variance.

2.3.2 The Practice: Microstructure Noise

Real transaction prices are not clean readings of a frictionless price process. They are contaminated by *microstructure noise*: the cumulative effect of bid-ask bounce, discrete price grids, order-processing delays, and other trading frictions.

Bid-Ask Bounce

When you buy a stock, you pay the *ask* price (slightly above the “true” value). When you sell, you receive the *bid* price (slightly below). The difference between bid and ask is the *spread*. Even if the true value is perfectly constant, alternating buys and sells cause the transaction price to bounce between bid and ask. This artificial oscillation creates spurious volatility in very high-frequency returns.

The standard model for microstructure noise (Andersen et al., 2001; Barndorff-Nielsen and Shephard, 2002) captures this with a simple additive structure: the price you actually observe is the “true” price plus some random noise from trading frictions.

$$p_{t,i}^* = p_{t,i} + \varepsilon_{t,i} \quad (2.7)$$

- $p_{t,i}^*$: observed (noisy) log price
- $p_{t,i}$: true (latent) efficient log price
- $\varepsilon_{t,i}$: microstructure noise, typically assumed i.i.d. with $\mathbb{E}[\varepsilon_{t,i}] = 0$ and $\text{Var}(\varepsilon_{t,i}) = \omega^2$

In Plain English

The noise variance ω^2 is a property of the market: liquid stocks with tight bid-ask spreads have small ω^2 ; illiquid stocks have larger ω^2 .

Why This Matters

When you download high-frequency price data for your project, you are downloading $p_{t,i}^*$, not $p_{t,i}$. This noise is baked into your data and cannot be removed by cleaning alone. The entire next chapter (Chapter 3) is about building RV estimators that work correctly despite this contamination.

When you compute returns from noisy prices, each observed return picks up the noise: $r_{t,i}^* = r_{t,i} + (\varepsilon_{t,i} - \varepsilon_{t,i-1})$. The noise part $(\varepsilon_{t,i} - \varepsilon_{t,i-1})$ has variance $\text{Var}(\varepsilon_{t,i}) + \text{Var}(\varepsilon_{t,i-1}) = 2\omega^2$ (since the two noise terms are independent). When you sample very frequently, the true returns $r_{t,i}$ become tiny (the price barely moves in a millisecond), but each noise contribution is still $2\omega^2$ regardless of the interval length. So the noise dominates, and RV diverges:

$$\text{RV}_t^{(\text{noisy})} \rightarrow 2n\omega^2 \quad \text{as } n \rightarrow \infty \quad (2.8)$$

In Plain English

Each noisy return contributes roughly $2\omega^2$ of garbage on average (from the noise term bouncing back and forth), and you’re summing n of these. As n grows, the garbage accumulates linearly while the true signal grows more slowly, so the noise wins. This is why sampling every tick or every second gives you a wildly inflated RV.

Sampling Too Fast Destroys the Estimate

With tick-by-tick or second-by-second data, the bid-ask bounce inflates realized variance far above the true integrated variance. Sampling more frequently makes the estimate *worse*, not better.

2.3.3 The Bias-Variance Tradeoff

The result is a tradeoff:

- **Sample too infrequently** (e.g., hourly): few observations, high estimation variance, but little noise contamination.
- **Sample too frequently** (e.g., every second): many observations, low estimation variance in theory, but massive upward bias from microstructure noise.

There is an optimal sampling frequency that balances these two forces. The concept is often visualized with a *volatility signature plot*.

2.3.4 The Volatility Signature Plot

Volatility Signature Plot

A volatility signature plot displays the average realized variance (or realized volatility), computed across many days, as a function of the sampling frequency. The x -axis shows the return interval (e.g., 1 second, 1 minute, 5 minutes, ...), and the y -axis shows the resulting average RV_t .

Figure 2.4 shows the typical shape.

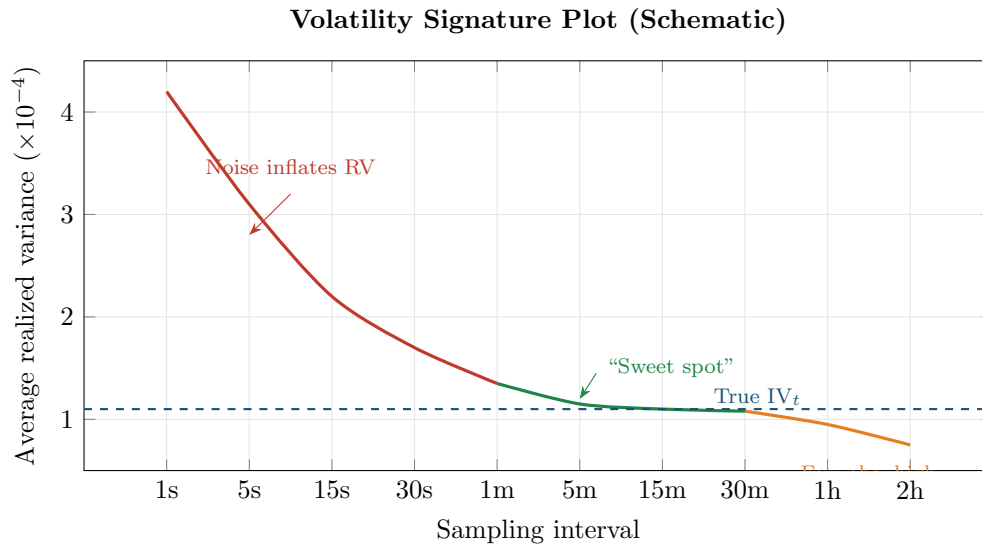


Figure 2.4: Schematic volatility signature plot. At very high frequencies (left, red), microstructure noise inflates realized variance. At very low frequencies (right, orange), too few observations make the estimate imprecise and biased downward. The green region (roughly 1–15 minutes) is the “sweet spot” where RV is closest to the true integrated variance (dashed blue line). Chapter 3 develops noise-robust estimators that work at higher frequencies.

The plot has three regions:

1. **High-frequency region** (left): RV rises sharply, contaminated by bid-ask bounce and microstructure effects.
2. **Intermediate region** (center): RV flattens near the true integrated variance. This is the practical sweet spot.
3. **Low-frequency region** (right): RV drifts downward and becomes noisy. The downward bias occurs because with fewer sample points, you miss many intraday price wiggles. Returns computed over long intervals smooth out the zigzags between your sparse sample points, producing smaller squared returns and hence a lower RV estimate.

The Fundamental Tradeoff

In theory, sample as fast as possible. In practice, noise wins beyond about 1-minute returns for most liquid assets. The volatility signature plot is a diagnostic tool: if average RV is still rising as you increase frequency, you are in the noise-contaminated region and need to back off (or use a noise-robust estimator from Chapter 3).

2.4 5-Minute RV: The Practical Workhorse

Given the tradeoff from the previous section, what sampling frequency should you use in practice? The answer, for most applications, is 5 minutes.

This number is an empirical finding rather than a derivation from first principles (the optimal frequency depends on the noise-to-signal ratio, which varies by asset and time period), and it has proved remarkably robust.

Liu et al. (2015): Does Anything Beat 5-Minute RV?

Liu et al. (2015) conducted a large-scale comparison of approximately 400 realized volatility estimators (including subsampled, kernel-based, pre-averaged, and multi-scale estimators) applied to 31 assets across 5 asset classes (equities, equity indices, exchange rates, bonds, and commodities). Their conclusion: for the purpose of *forecasting* future volatility, “it is difficult to significantly outperform” the simple 5-minute realized variance. More sophisticated estimators sometimes produce marginally more accurate daily estimates, but these gains do not reliably translate into better forecasts.

Why 5 minutes? Two practical reasons:

1. For liquid assets (major equity indices, G10 FX pairs, actively traded stocks), the volatility signature plot typically flattens by the 5-minute mark. At this frequency, microstructure noise is small relative to genuine price variation.
2. 5-minute sampling produces 78 intraday observations per day for U.S. equities (6.5 hours $\times$ 12 intervals per hour), which provides enough observations for the central limit theorem in Equation (2.6) to give a reasonable approximation.

5 Minutes Is Not Universal

For illiquid assets (small-cap stocks, emerging-market currencies, less-traded commodities), the noise-contaminated region can extend to 15 or even 30 minutes. Always check the volatility signature plot for your specific asset before committing to a sampling frequency.

2.5 Realized Volatility vs. Realized Variance

This chapter has used “realized variance” for $RV_t = \sum r_{t,i}^2$ and “realized volatility” for $\sqrt{RV_t}$. Not every paper follows this convention.

Variance vs. Volatility: A Factor-of-10 Error

The S&P 500’s typical daily realized variance is around 1×10^{-4} . Its typical daily realized volatility is $\sqrt{1 \times 10^{-4}} = 0.01$ (1%). Confusing the two is a factor-of-100 error in the daily number, which becomes roughly a factor of 10 when annualized ($0.01 \times \sqrt{252} \approx 0.16$ vs. $0.0001 \times \sqrt{252} \approx 0.0016$). Whenever you read a paper, check whether the authors report

RV_t or $\sqrt{RV_t}$, and note the units (decimal, percent, or basis points).

Conventions in This Guide

Throughout this guide:

- RV_t (and “RV”) always refers to *realized variance*: $RV_t = \sum_{i=1}^n r_{t,i}^2$
- Realized *volatility* refers to $\sqrt{RV_t}$, the square root of realized variance
- When annualizing, multiply $\sqrt{RV_t}$ by $\sqrt{252}$ (as in Section 1.2.1 of Chapter 1)

These conventions follow Andersen et al. (2003) and Barndorff-Nielsen and Shephard (2002).

Some papers (particularly in the HAR literature, Chapter 6) work with the natural logarithm of realized variance, $\ln(RV_t)$. The log transform has two practical benefits:

1. RV_t is right-skewed and bounded below by zero; $\ln(RV_t)$ is approximately Gaussian (Andersen et al., 2001, 2003), making standard regression assumptions more plausible.
2. Forecasting $\ln(RV_t)$ automatically ensures that predictions of the variance level are positive (since $e^x > 0$ for all x).

When you encounter $\ln(RV_t)$ in later chapters, this is why.

Summary

- **Integrated variance** $IV_t = \int_{t-1}^t \sigma_s^2 ds$ is the theoretical target: the total variance accumulated within a single day.
- **Realized variance** $RV_t = \sum_{i=1}^n r_{t,i}^2$ is a model-free estimator of integrated variance, constructed by summing squared intraday returns.
- **Convergence**: as the sampling frequency increases ($n \rightarrow \infty$), RV converges in probability to the quadratic variation of the log-price process (Andersen et al., 2001; Barndorff-Nielsen and Shephard, 2002).
- In the **absence of jumps**, quadratic variation equals integrated variance, so RV consistently estimates IV_t .
- With **jumps**, quadratic variation equals integrated variance plus the sum of squared jumps. Chapter 4 addresses separating the two components.
- **Microstructure noise** (bid-ask bounce, discrete prices) corrupts very high-frequency returns, causing RV to diverge as sampling frequency increases.
- The **volatility signature plot** (average RV vs. sampling interval) visualizes the bias-variance tradeoff: too fast means noise contamination, too slow means imprecise estimates.
- **5-minute RV** is the standard benchmark. Liu et al. (2015) showed it is hard to beat for forecasting across 31 assets and ~ 400 competing estimators.
- For **illiquid assets**, 5 minutes may still be in the noise-contaminated region; always check the volatility signature plot.
- **Convention**: this guide uses RV_t for realized variance and $\sqrt{RV_t}$ for realized volatility. Confusing the two is roughly a factor-of-10 annualized error.
- The **log transform** $\ln(RV_t)$ is approximately Gaussian and guarantees positive forecasts, which is why it appears frequently in later chapters.
- Chapter 3 develops noise-robust estimators that can safely use higher frequencies; Chapter 4 separates continuous and jump variation.

Table 2.1: Key results referenced in this chapter.

Paper	Result	Relevance
Andersen et al. (2001)	Realized variance converges to quadratic variation as sampling frequency increases; the distribution of $\ln(RV_t)$ is approximately Gaussian.	Foundational result establishing RV as a consistent, model-free volatility estimator.
Andersen et al. (2003)	Formal econometric theory for realized variance: consistency, asymptotic distribution, practical implementation with 5-minute FX returns, and the result that mean subtraction is unnecessary.	Provided the CLT for RV and demonstrated that dropping the intraday mean improves finite-sample accuracy.
Barndorff-Nielsen and Shephard (2002)	Parallel asymptotic theory for realized variance: consistency, CLT, and the additive noise model for microstructure contamination.	Second foundational contribution; BPV (introduced in later BNS papers) is the basis for jump detection in Chapter 4.
Liu et al. (2015)	Compared ~ 400 realized estimators across 31 assets in 5 asset classes; simple 5-minute RV is very hard to beat for volatility forecasting.	Justifies 5-minute RV as the default benchmark for all forecasting models in this guide.

Chapter 3

Microstructure Noise and Robust Estimators

Why This Chapter

Project 2 (LOB-based intraday) works directly in the high-frequency regime where noise is most severe.

Chapter 2 ended with a paradox. The theory says “sample as fast as possible”: more intraday returns means a more precise estimate of integrated variance. But the volatility signature plot showed that sampling faster than about 5 minutes makes the estimate *worse*, not better, because microstructure noise inflates the sum of squared returns. This chapter explains exactly where that noise comes from, quantifies the damage it does, and develops four estimators that fix the problem: TSRV, MSRV, the realized kernel, and pre-averaging.

3.1 The Microstructure Noise Problem

Chapter 2 introduced the noise model $p_{t,i}^* = p_{t,i} + \varepsilon_{t,i}$ (Equation 2.7) and showed that noise causes RV to diverge. This section digs into *where* the noise comes from, because understanding the source tells you when it matters and when it does not.

Bid, Ask, Mid, and Spread

At any moment, a stock has two prices: the *bid* (the highest price a buyer will pay) and the *ask* (the lowest price a seller will accept). The *mid price* is the average, $(P_{\text{bid}} + P_{\text{ask}})/2$. The *spread* is $P_{\text{ask}} - P_{\text{bid}}$. Transaction prices alternate between bid and ask depending on who initiates the trade (a market buy hits the ask; a market sell hits the bid).

Sampling Frequency Terminology

A 6.5-hour U.S. equity trading day has $6.5 \times 3,600 = 23,400$ one-second intervals.

There are three main sources of microstructure noise.

Source 1: Bid-ask bounce. Even if the true value of a stock is perfectly constant at \$50.00, alternating buys and sells produce transaction prices that bounce between \$49.99 (bid) and \$50.01 (ask). Computing returns from these prices generates a sequence of +0.02, −0.02, +0.02, ... that has nothing to do with actual volatility. Squaring and summing these fake returns

inflates RV. Bid-ask bounce is the dominant source of noise for most liquid assets (Hansen and Lunde, 2006).

Figure 3.1 illustrates how bid-ask bounce creates spurious volatility even when the true price is constant.

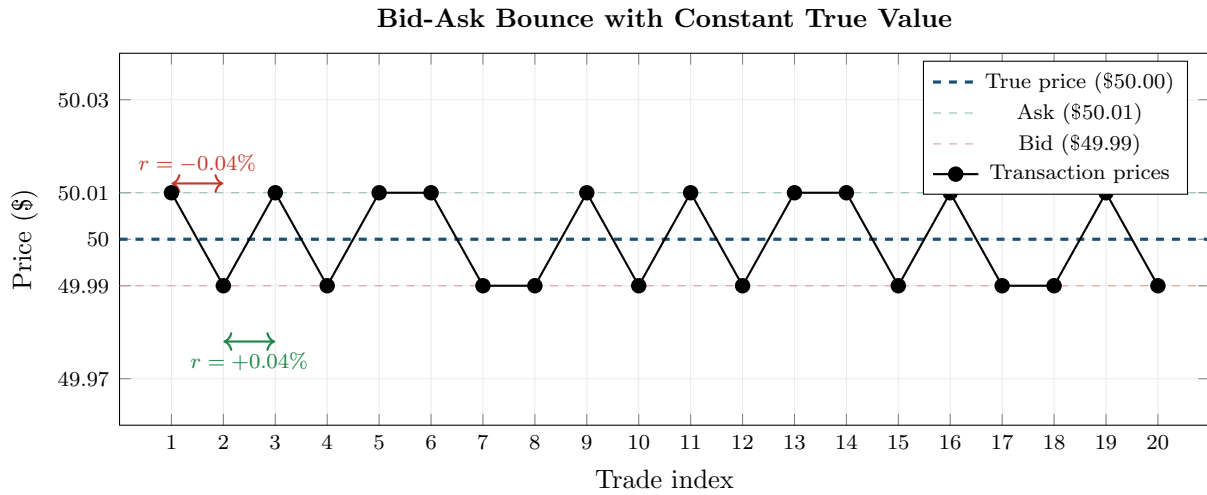


Figure 3.1: Bid-ask bounce creates spurious volatility. The true price (blue dashed) is constant at \$50.00, but transaction prices alternate between the bid (\$49.99) and ask (\$50.01). Each trade generates a return of $\pm 0.04\%$ that is pure noise. Squaring and summing these fake returns inflates RV far above zero.

Source 2: Discrete tick sizes. Prices on exchanges are quoted in discrete increments (one cent for U.S. equities). When the true efficient price is \$50.003, the observed price must round to either \$50.00 or \$50.01. This rounding adds a small, mean-zero error to every observation. At very high frequencies, these rounding errors accumulate into a non-trivial noise term.

Source 3: Price staleness. Illiquid stocks may not trade every second. If you sample the “last trade price” at regular intervals, you often repeat the same stale price. The resulting return is zero, which underestimates true volatility. When a new trade finally arrives, the return is artificially large. This creates spurious autocorrelation in returns (Aït-Sahalia et al., 2005).

Why Noise Kills High-Frequency RV

At very high frequencies, observed prices alternate between bid and ask. Squaring these back-and-forth moves inflates the sum far beyond the true volatility. The signal (real price moves) scales with the time interval Δ , but the noise (bid-ask bounce) has a roughly constant variance ω^2 per observation. As you shrink Δ , the noise-to-signal ratio explodes: each return is mostly noise, and you are summing $n = 1/\Delta$ of them.

Source 4: Adverse selection (information asymmetry). Beyond mechanical noise, spreads reflect a deeper economic force. Glosten and Milgrom (1985) showed that market makers widen spreads because some counterparties possess private information. The **bid-ask spread** must compensate for losses to informed traders:

$$s_t \propto \alpha \cdot \sigma_v$$

where α is the probability the counterparty is informed and σ_v is the volatility of the information signal. When volatility is high, information asymmetry increases (there is more to know), so

spreads widen mechanically. This creates a feedback loop: high vol leads to wider spreads, which creates more noise in transaction prices, which creates more biased RV estimates.

Beyond being a nuisance to filter, microstructure noise therefore carries information about market quality and **informed-trading intensity** that can itself predict future volatility (Chapter 10).

A related insight from Roll (1984)¹: the first-order autocovariance of price changes satisfies $\text{Cov}(\Delta p_t, \Delta p_{t+1}) = -s^2/4$. This connects the noise-robust estimators developed later in this chapter (which exploit autocovariance structure) to a liquidity interpretation.

3.1.1 Quantifying the Bias

The standard noise model assumes the observed log price is the true (efficient) price plus i.i.d. noise:

$$p_{t,i}^* = p_{t,i} + \varepsilon_{t,i}, \quad \varepsilon_{t,i} \stackrel{\text{iid}}{\sim} (0, \omega^2)$$

How far off is RV computed from noisy prices? Hansen and Lunde (2006) and Aït-Sahalia et al. (2005) showed that, under this model, the expected value of noisy RV decomposes into signal plus a noise term that scales with the number of observations:

$$\mathbb{E}[\text{RV}_t^{(\text{noisy})}] = \text{IV}_t + 2n\omega^2 \quad (3.1)$$

- $\text{RV}_t^{(\text{noisy})}$: realized variance computed from observed (noisy) prices
- IV_t : true integrated variance (the target)
- n : number of intraday returns
- ω^2 : variance of the microstructure noise per observation
- $2n\omega^2$: the noise-induced bias, linear in n

In Plain English

Using more returns makes the contamination worse, not better. This is the opposite of what you would expect from classical statistics, where more data means more precision.

Why This Matters

This equation is the reason your vol forecasting project uses 5-minute returns instead of tick-by-tick data. At 5-minute frequency ($n \approx 78$ for U.S. equities), the bias $2n\omega^2$ is small enough to ignore for most liquid stocks. If you ever move to higher-frequency data, you will need the noise-robust estimators developed later in this chapter.

The bias $2n\omega^2$ grows linearly with the number of observations. As $n \rightarrow \infty$ (tick-by-tick), the bias term dominates and $\text{RV}_t^{(\text{noisy})} \rightarrow \infty$. The IV_t term becomes a negligible fraction of the total.

The Bias Is Not Just Upward

The i.i.d. noise model predicts a purely upward bias. In practice, noise can also be negatively autocorrelated (alternating buys and sells) or positively autocorrelated (momentum in order flow), which can shift the bias in either direction. Hansen and Lunde (2006) found that the simple i.i.d. model is a useful first approximation for liquid stocks, but the actual

¹Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *The Journal of Finance*, 39(4), 1127–1139.

noise structure is more complex. Always inspect the volatility signature plot for your specific asset.

3.2 The Volatility Signature Plot

Chapter 2 introduced the volatility signature plot (Figure 2.4). The signature plot is a *diagnostic tool*: plot it before choosing an estimator or sampling frequency.

Volatility Signature Plot (Formal)

The volatility signature plot displays $\bar{RV}(\Delta)$, the average realized variance computed across T days, as a function of the sampling interval Δ :

$$\bar{RV}(\Delta) = \frac{1}{T} \sum_{t=1}^T RV_t(\Delta)$$

- Δ : sampling interval (e.g., 1 second, 1 minute, 5 minutes, ...)
- $RV_t(\Delta)$: realized variance for day t computed at interval Δ
- T : number of trading days in the sample

The x -axis shows Δ (often on a log scale), and the y -axis shows $\bar{RV}(\Delta)$.

Figure 3.2 illustrates the three regimes.

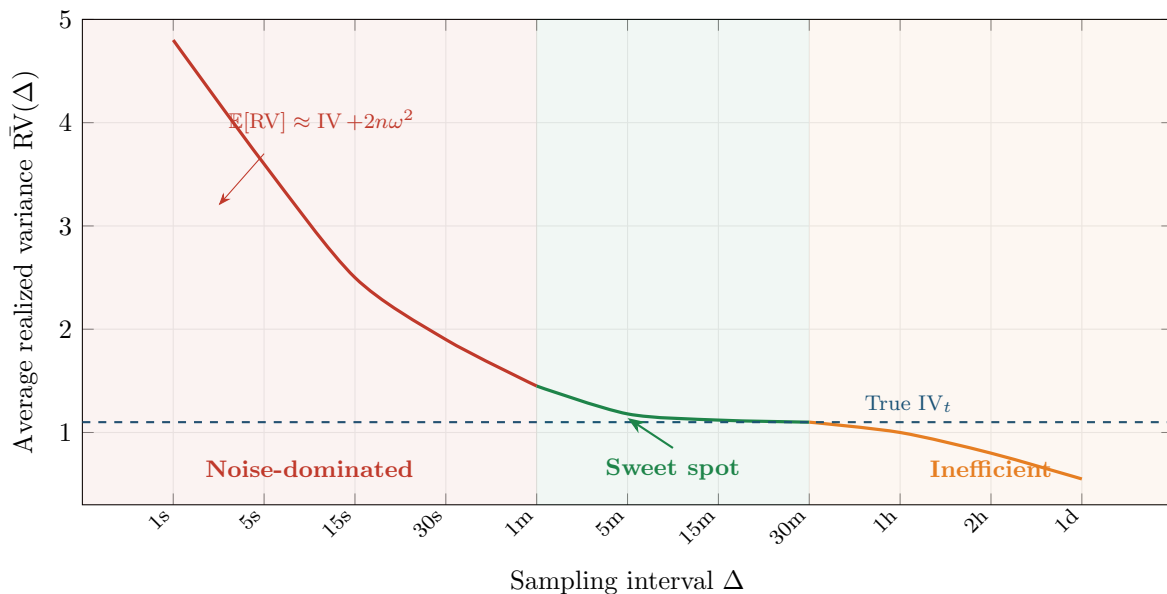


Figure 3.2: Volatility signature plot with three regimes. **Left (red)**: noise dominates; RV inflated by $2n\omega^2$. **Center (green)**: sweet spot where RV is close to true IV_t . **Right (orange)**: too few observations; high estimation variance. The dashed blue line is the true integrated variance. The curve flattens around 1–15 minutes for liquid U.S. equities.

The Volatility Signature Plot Is a Diagnostic, Not a Conclusion

The shape of the volatility signature plot tells you where noise begins to bite for your specific asset. Before computing RV for any analysis, plot $\bar{RV}(\Delta)$ for a range of frequencies. If the curve is still rising at your chosen frequency, you are in the noise-contaminated region

and must either reduce frequency or use a noise-robust estimator from this chapter.

The Sweet Spot Varies by Asset

The 5-minute rule of thumb holds for liquid large-cap equities and major FX pairs. For less liquid instruments (small-cap stocks, emerging-market currencies, corporate bonds), the noise-contaminated region can extend to 15 or even 30 minutes (Hansen and Lunde, 2006). For extremely liquid futures (e.g., E-mini S&P 500), the curve may flatten by 1 minute. Always check empirically.

3.3 Two-Scales Realized Volatility (TSRV)

The volatility signature plot shows that standard RV is badly biased at high frequencies. One approach (Chapter 2) is to avoid the problem by sampling at 5 minutes. A better approach is to *correct* for the noise and use all the high-frequency data. The Two-Scales Realized Volatility (TSRV) estimator, developed by Zhang et al. (2005), is the simplest noise-robust estimator.

The Two-Scales Idea

Compute RV at two different frequencies: a “fast” scale (e.g., every 1 minute) and a “slow” scale (e.g., every 30 minutes). The fast-scale RV picks up both signal and noise: $RV^{(\text{fast})} \approx IV_t + 2n_{\text{fast}}\omega^2$. The slow-scale RV also picks up signal and noise, but with far fewer observations: $RV^{(\text{slow})} \approx IV_t + 2n_{\text{slow}}\omega^2$. However, the *noise per observation* is the same at both scales. By taking a specific weighted difference of the two, you can cancel the noise term and isolate the signal.

3.3.1 Construction

TSRV works with subsampled realized variances. Instead of computing a single RV on a non-overlapping grid, you average over all possible starting points.

Subsampling

A “subsampled” RV at scale K takes the original n tick prices and computes K separate realized variances, each on a non-overlapping grid offset by one tick. For example, with $K = 3$, you compute RV using ticks $\{1, 4, 7, \dots\}$, then $\{2, 5, 8, \dots\}$, then $\{3, 6, 9, \dots\}$, and average the three. This “average RV” uses all the data while spacing returns K ticks apart.

Denote the subsampled (averaged) RV at scale K as $RV_t^{(\text{avg}, K)}$. TSRV uses a fast scale $K = 1$ (every tick) and a slow scale $K = K_{\text{slow}}$ (spaced-out returns):

Two-Scales Realized Volatility (TSRV)

$$\widehat{IV}_t^{\text{TSRV}} = RV_t^{(\text{avg}, K_{\text{slow}})} - \frac{\bar{n}_{K_{\text{slow}}}}{n} RV_t^{(\text{all})} \quad (3.2)$$

- $RV_t^{(\text{avg}, K_{\text{slow}})}$: the subsampled (averaged) RV computed with returns spaced K_{slow} ticks apart
- $RV_t^{(\text{all})}$: realized variance computed from all n tick-by-tick returns (the “fast scale”)
- $\bar{n}_{K_{\text{slow}}} = (n - K_{\text{slow}} + 1)/K_{\text{slow}}$: the average number of returns per subsample at the slow scale

- n : total number of tick-level returns
- The ratio $\bar{n}_{K_{\text{slow}}}/n$ is a small number; the second term removes the noise bias from the first

Why the Subtraction Works

Both $RV_t^{(\text{avg}, K_{\text{slow}})}$ and $RV_t^{(\text{all})}$ contain a noise bias proportional to $2\omega^2 \times (\text{number of returns})$. The subsampled slow-scale RV has bias $\approx 2\bar{n}_{K_{\text{slow}}}\omega^2$. The all-tick RV has bias $\approx 2n\omega^2$. Subtracting $(\bar{n}_{K_{\text{slow}}}/n) \times RV_t^{(\text{all})}$ removes exactly $2\bar{n}_{K_{\text{slow}}}\omega^2$ from the slow-scale estimate, canceling the noise.

Figure 3.3 illustrates the two scales on the same price path.

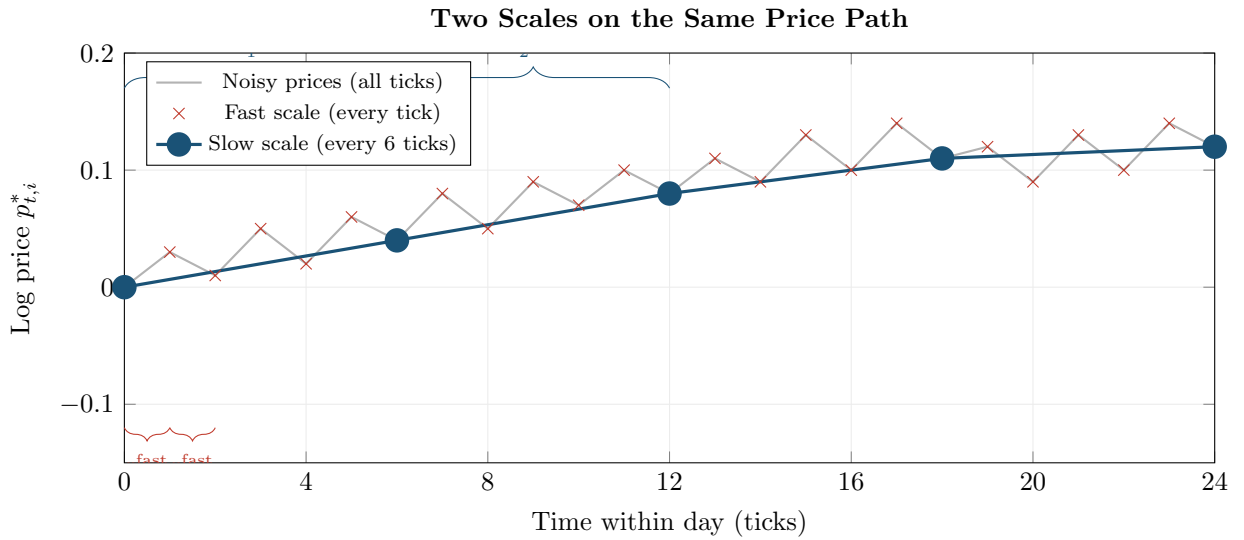


Figure 3.3: TSRV uses two scales on the same price path. **Red:** fast scale (every tick), picks up both signal and noise. **Blue:** slow scale (every 6 ticks), picks up signal with less noise but fewer observations. TSRV subtracts a noise correction derived from the fast scale to debias the slow-scale estimate.

3.3.2 Convergence Rate

Zhang et al. (2005): TSRV Convergence

Under the i.i.d. noise model, the optimal choice of K_{slow} yields a convergence rate of $n^{-1/6}$ for the TSRV estimator:

$$\widehat{IV}_t^{\text{TSRV}} - IV_t = O_p(n^{-1/6})$$

For comparison, the standard 5-minute RV (without noise correction) converges at rate $n^{-1/2}$ but only *if you ignore noise*. With noise, standard RV does not converge at all. TSRV is the first estimator that is consistent in the presence of noise.

The $n^{-1/6}$ rate is slower than the noise-free $n^{-1/2}$. This is the price you pay for not knowing the noise level ω^2 a priori. Subsequent estimators (MSRV, realized kernel) improve this rate.

Worked Example:

TSRV on a 6.5-Hour Trading Day For a liquid stock with $n = 23,400$ one-second ticks and a slow scale $K_{\text{slow}} = 390$, the fast-scale $\text{RV}_t^{(\text{all})} = 3.2 \times 10^{-4}$ (noise-inflated), the slow-scale $\text{RV}_t^{(\text{avg}, 390)} = 1.3 \times 10^{-4}$, and the correction factor $\bar{n}_{390}/n \approx 0.00252$, giving

$$\widehat{\text{IV}}_t^{\text{TSRV}} = 1.3 \times 10^{-4} - 0.00252 \times 3.2 \times 10^{-4} \approx 1.29 \times 10^{-4}.$$

The correction is small here because the slow scale already removes most noise. In cases where the slow scale is less aggressive (e.g., $K_{\text{slow}} = 30$), the correction is larger. The annualized volatility is $\sqrt{1.29 \times 10^{-4} \times \sqrt{252}} \approx 18.0\%$.

3.4 Multi-Scale Realized Volatility (MSRV)

TSRV uses two scales. A natural extension is to use *many* scales simultaneously. Zhang (2006) developed the Multi-Scale Realized Volatility (MSRV) estimator, which combines information from multiple time scales to achieve a faster convergence rate.

Multi-Scale Realized Volatility (MSRV)

MSRV takes a weighted combination of subsampled RVs at J different scales $K_1 < K_2 < \dots < K_J$:

$$\widehat{\text{IV}}_t^{\text{MSRV}} = \sum_{j=1}^J a_j \text{RV}_t^{(\text{avg}, K_j)} \quad (3.3)$$

- $\text{RV}_t^{(\text{avg}, K_j)}$: subsampled RV at scale K_j (same construction as in TSRV)
- a_j : weights chosen to (i) cancel the noise bias and (ii) minimize variance
- J : number of scales; Zhang (2006) showed that the optimal J grows with n
- The weights satisfy $\sum_j a_j = 1$ (so the estimator targets IV_t) and a second constraint that cancels the ω^2 bias

Zhang (2006): MSRV Achieves the Optimal Rate

The MSRV estimator achieves the convergence rate $n^{-1/4}$:

$$\widehat{\text{IV}}_t^{\text{MSRV}} - \text{IV}_t = O_p(n^{-1/4})$$

This is the best possible rate for any estimator under the i.i.d. noise model without knowing ω^2 . For context: with $n = 23,400$ one-second observations, $n^{-1/4} \approx 0.081$, compared to $n^{-1/6} \approx 0.188$ for TSRV. MSRV is roughly 2.3 times more precise than TSRV for the same data.

In practice, MSRV is straightforward to implement (it is a weighted sum of subsampled RVs), but the optimal weight selection requires choosing J and the scale grid $\{K_j\}$. Zhang (2006) provides explicit formulas. For most applied work, the realized kernel (next section) is preferred because it achieves the same $n^{-1/4}$ rate with a simpler tuning procedure.

3.5 The Realized Kernel

The realized kernel, developed by Barndorff-Nielsen et al. (2008), is the most widely used noise-robust estimator in the volatility literature. It achieves the optimal $n^{-1/4}$ convergence rate

(same as MSRV) with an intuitive construction.

Noise as Spurious Autocorrelation

Under the i.i.d. noise model, the *observed* returns $r_{t,i}^*$ have negative first-order autocorrelation: if the noise pushes the price up this tick, the next return is more likely to be negative (reverting the noise). This spurious autocorrelation shows up in the autocovariances of returns. The realized kernel idea: compute the autocovariance function of observed returns and use a kernel weighting to remove the noise-induced autocorrelation while keeping the genuine signal.

Autocovariance

The autocovariance at lag h of a sequence $\{x_1, \dots, x_n\}$ measures the linear association between observations h steps apart:

$$\hat{\gamma}_h = \frac{1}{n} \sum_{i=1}^{n-h} (x_i - \bar{x})(x_{i+h} - \bar{x})$$

Kernel Functions

A kernel function $k(x)$ is a smooth weighting function satisfying $k(0) = 1$ and $k(x) \rightarrow 0$ as $|x| \rightarrow \infty$. Common examples: the Bartlett kernel $k(x) = 1 - |x|$ for $|x| \leq 1$ (a triangle), and the Parzen kernel (a piecewise cubic that is smoother at the boundary).

3.5.1 Construction

The realized kernel takes the autocovariances of observed returns and weights them with a kernel function.

Realized Kernel

$$\hat{K}_t = \sum_{h=-H}^H k\left(\frac{h}{H+1}\right) \hat{\gamma}_h \quad (3.4)$$

- $\hat{\gamma}_h = \sum_{i=1}^{n-|h|} r_{t,i}^* r_{t,i+|h|}^*$: the h -th sample autocovariance of observed (noisy) returns; note $\hat{\gamma}_0 = \text{RV}_t^{(\text{noisy})}$
- $k(\cdot)$: a kernel function with $k(0) = 1$, typically the Parzen kernel
- H : bandwidth (number of lags included); controls the bias-variance tradeoff
- The sum runs from $-H$ to H , so both positive and negative lags are included
- $k(h/(H+1))$: the weight assigned to autocovariance at lag h ; lags near zero get high weight, distant lags get low weight

Why This Matters

If you extend your project beyond 5-minute RV (for example, to compute more accurate daily IV estimates or to work with tick-level LOB data), the realized kernel from the `highfrequency` R package is the standard tool.

The “flat-top” property is important: [Barndorff-Nielsen et al. \(2008\)](#) require $k'(0) = 0$ (the kernel is flat at the origin), which ensures that the noise bias is removed at the correct rate. The Parzen kernel satisfies this.

Figure 3.4 shows how the Parzen kernel assigns weights to different lags.

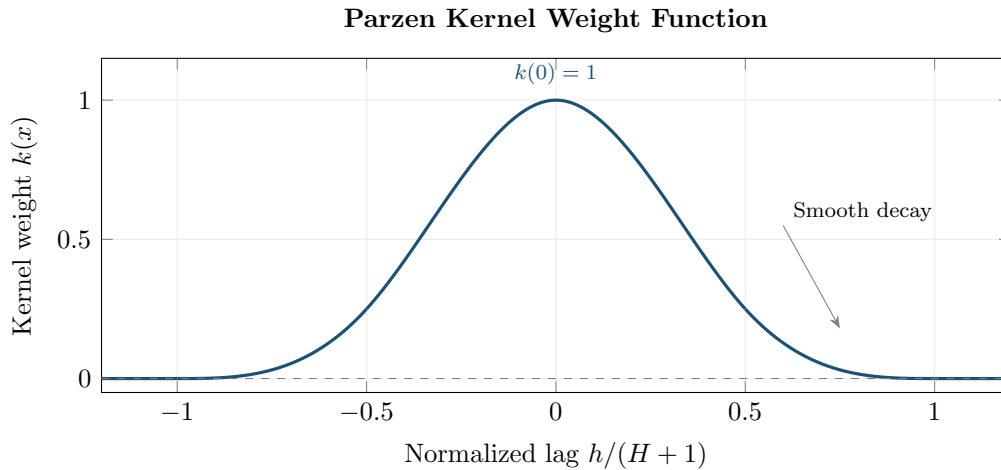


Figure 3.4: The Parzen kernel assigns full weight ($k(0) = 1$) to the zero-lag autocovariance (which is just RV) and smoothly downweights higher lags. Lags beyond H receive zero weight. The flat top ($k'(0) = 0$) ensures correct noise removal.

3.5.2 Why It Works

The logic is:

1. **Lag 0:** The zero-lag autocovariance $\hat{\gamma}_0 = \text{RV}_t^{(\text{noisy})}$ contains the signal plus noise bias.
2. **Lags 1, 2, ...:** The noise-induced autocorrelation appears at lag ± 1 (and possibly a few more lags for dependent noise). These autocovariances are negative on average, reflecting the bid-ask bounce reversal.
3. **Kernel weighting:** By including these negative autocovariances with appropriate weights, their sum partially cancels the positive noise bias in $\hat{\gamma}_0$.
4. **Higher lags:** Beyond a few lags, the noise-induced autocorrelation dies out, and genuine autocorrelation (if any) is captured.

3.5.3 Bandwidth Selection

The bandwidth H controls how many autocovariance lags are included.

- **Too small H :** not enough lags to fully remove the noise bias; the estimator is biased upward.
- **Too large H :** too many noisy autocovariance estimates are included; the estimator has high variance.

Barndorff-Nielsen et al. (2008) showed that the optimal bandwidth scales as $H^* \propto n^{3/5}$. For $n = 23,400$ one-second observations, this gives $H^* \approx 23,400^{3/5} \approx 418$. Data-driven bandwidth selection methods (analogous to those used in spectral density estimation) are available and recommended for applied work.

3.6 Pre-Averaging

Pre-averaging, developed by Jacod et al. (2009), takes a different approach to the noise problem. Instead of correcting the autocovariance structure (as the realized kernel does), it smooths the noise out of the price path *before* computing squared returns.

Smoothing Out the Bounce

If bid-ask bounce makes individual prices noisy, averaging a few adjacent prices should cancel some of the noise (positive and negative noise terms offset). Pre-averaging does exactly this: replace each price $p_{t,i}^*$ with a local average of nearby prices, then compute returns and sum their squares. The local averaging shrinks the noise while preserving the signal.

3.6.1 Construction

Pre-Averaged Realized Variance

Choose a block length L and a weight function $g : [0, 1] \rightarrow \mathbb{R}$ (typically $g(x) = \min(x, 1 - x)$). Define the pre-averaged price:

$$\bar{p}_{t,i}^* = \sum_{j=1}^{L-1} g\left(\frac{j}{L}\right) \Delta p_{t,i+j}^* \quad (3.5)$$

where $\Delta p_{t,i+j}^* = p_{t,i+j}^* - p_{t,i+j-1}^*$ is the tick-by-tick return. The pre-averaged realized variance is:

$$\widehat{\text{IV}}_t^{\text{PA}} = \frac{1}{L\psi_2} \sum_{i=0}^{n-L} (\bar{p}_{t,i}^*)^2 - \frac{\psi_1}{2L\psi_2} \text{RV}_t^{(\text{all})} \quad (3.6)$$

- $\bar{p}_{t,i}^*$: the pre-averaged return at position i (a weighted sum of $L - 1$ tick returns)
- $g(j/L)$: weight function; the standard choice $g(x) = \min(x, 1 - x)$ gives a triangular shape that ramps up then ramps down
- $\psi_1 = \int_0^1 [g'(x)]^2 dx$ and $\psi_2 = \int_0^1 [g(x)]^2 dx$: constants determined by the weight function
- $\text{RV}_t^{(\text{all})}$: tick-by-tick RV (used for a small bias correction, analogous to the TSRV correction)
- L : block length; the optimal choice is $L \propto n^{1/2}$

In Plain English

Step 1 replaces each noisy price with a local moving average of nearby prices (Equation 3.5). Step 2 computes a sum of squared “pre-averaged returns” (Equation 3.6), minus a small bias correction (the second term) that plays the same debiasing role as the fast-scale correction in TSRV.

Figure 3.5 compares raw and pre-averaged returns.

Jacod et al. (2009): Pre-Averaging Convergence

With the optimal block length $L \propto n^{1/2}$, the pre-averaged estimator achieves the optimal convergence rate $n^{-1/4}$, the same as MSRV and the realized kernel. The pre-averaging approach also provides a feasible central limit theorem, enabling confidence intervals for integrated variance.

3.7 Other Estimators

The four estimators above (TSRV, MSRV, realized kernel, pre-averaging) are the most cited. Several other approaches exist and are worth brief mention.

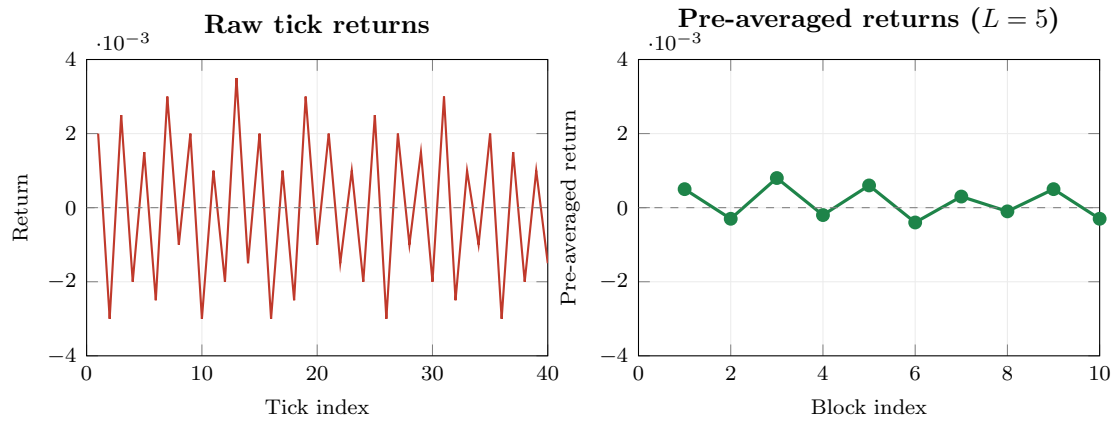


Figure 3.5: Raw tick returns (left, red) versus pre-averaged returns (right, green) with block length $L = 5$. The raw returns show large, alternating moves (bid-ask bounce). Pre-averaging smooths out the noise, producing smaller, more stable returns that better reflect true price variation.

Subsampled/Averaged RV. Before TSRV was developed, practitioners used averaged RV (computing multiple RVs on offset grids and averaging them) to reduce noise. This is the slow-scale component of TSRV without the bias correction. It reduces noise variance but does not eliminate the bias. It remains a useful quick fix when the noise level is low.

Fourier Estimator. [Malliavin and Mancino \(2002, 2009\)](#) proposed estimating integrated variance via the Fourier coefficients of the price process. The idea: the variance is related to the squared magnitude of low-frequency Fourier coefficients, while noise concentrates at high frequencies. By truncating high-frequency components, you filter out the noise. The Fourier estimator is less widely used in practice, but it has attractive theoretical properties and does not require equally spaced observations.

Quasi-Maximum Likelihood Estimator (QMLE). [Xiu \(2010\)](#) proposed treating the noisy price observations as a state-space model: the true price is the latent state, and the observed price adds Gaussian noise. Applying the Kalman filter produces a quasi-maximum likelihood estimate of integrated variance and the noise variance ω^2 jointly. The QMLE achieves the optimal $n^{-1/4}$ rate and provides a natural estimate of ω^2 as a byproduct.

All Roads Lead to $n^{-1/4}$

Under the standard i.i.d. noise model, the best possible convergence rate is $n^{-1/4}$. All of the modern estimators (MSRV, realized kernel, pre-averaging, QMLE) achieve this rate. They differ in implementation details, robustness to dependent noise, and ease of tuning, but their asymptotic efficiency is the same.

3.8 Which Estimator to Use

Which estimator you use depends on your goal.

Figure 3.6 shows how the different estimators behave on the volatility signature plot. Noise-robust estimators produce a flat line across frequencies, while naive RV diverges.

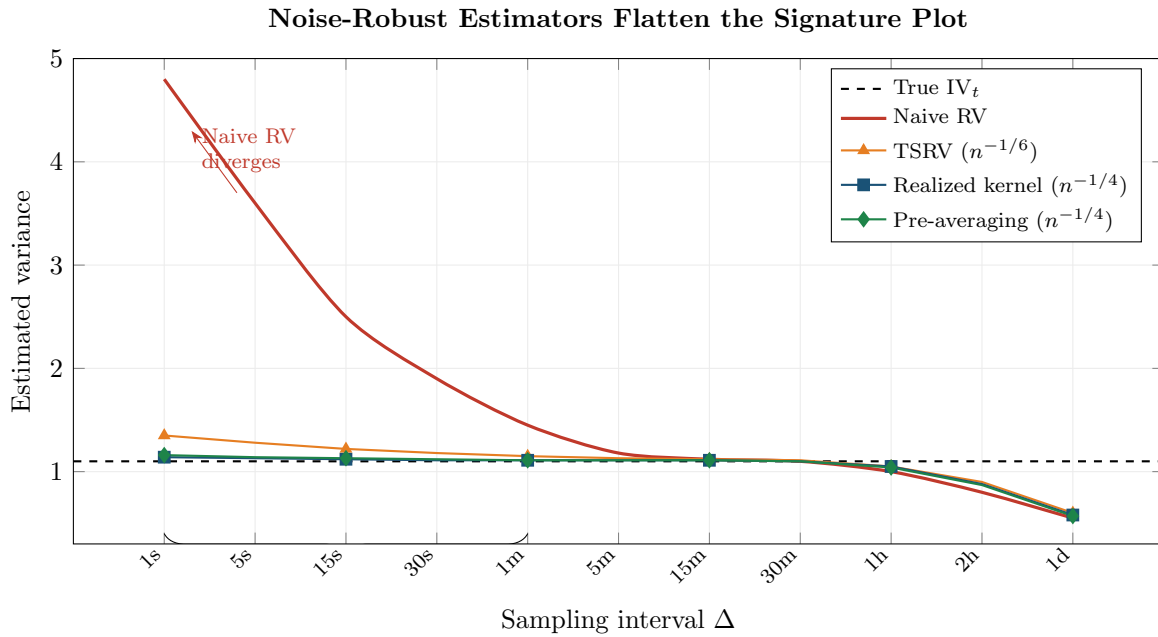


Figure 3.6: Volatility signature plot comparing estimators. Naive RV (red) diverges at high frequencies due to noise. TSRV (orange) is consistent but has a slower convergence rate, producing slight upward bias at the highest frequencies. The realized kernel (blue) and pre-averaging (green) achieve the optimal $n^{-1/4}$ rate and remain close to the true IV_t across all frequencies. At very low frequencies (right side), all estimators lose precision due to too few observations.

3.8.1 Comparison Table

Table 3.1 summarizes the key properties.

3.8.2 Decision Flowchart

Figure 3.7 provides a practical decision tree.

Practical Guidance

For daily RV forecasting (Chapters 6 and 10), simple 5-minute RV suffices. Liu et al. (2015) showed that noise-robust estimators rarely produce better *forecasts* of future volatility, even though they produce more accurate *estimates* of today's integrated variance. For estimation accuracy, intraday analysis, or when working with tick data in Project 2, use the realized kernel as the default.

Why This Matters

For your HAR-based vol forecasting project, the *difference* between noise-robust RV (e.g., realized kernel) and 5-minute RV on the same day is itself a signal: it measures how much microstructure noise is present, which proxies for liquidity and can be used as an ML feature.

Summary

- **Three noise sources:** bid-ask bounce (dominant for liquid assets), discrete tick sizes, and price staleness each corrupt high-frequency prices.

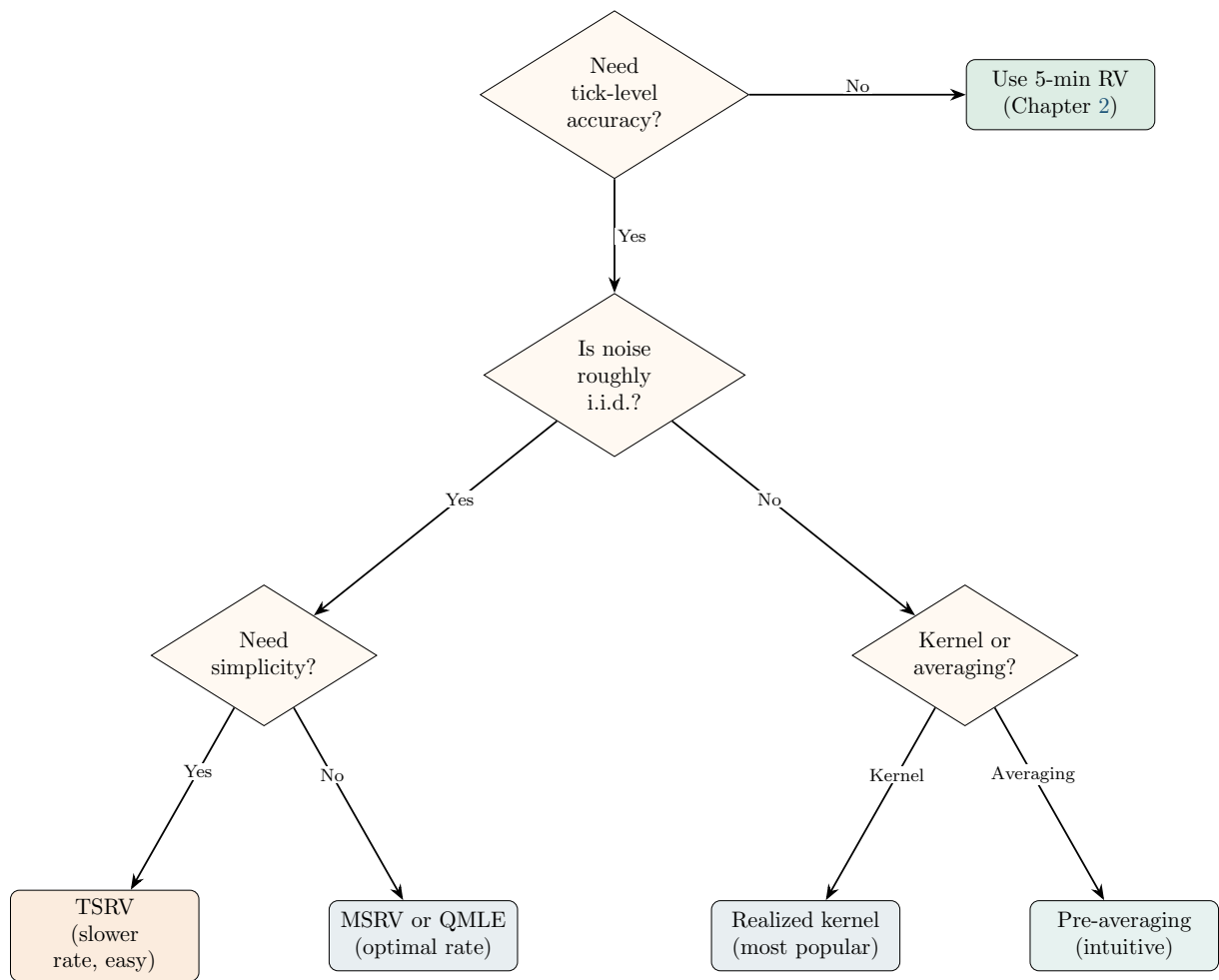


Figure 3.7: Decision tree for choosing a noise-robust estimator. For daily volatility forecasting, 5-min RV is the starting point. For intraday analysis or when estimation accuracy matters, use the realized kernel (default) or pre-averaging. TSRV is the simplest noise-robust estimator and a good starting point for learning.

Table 3.1: Comparison of noise-robust realized variance estimators.

Estimator	Rate	Noise assumption	Practical notes
5-min RV	$n^{-1/2}$ (no noise)	Avoids noise	Simple; hard to beat for forecasting (Liu et al., 2015)
TSRV	$n^{-1/6}$	i.i.d.	First consistent estimator with noise; easy to implement
MSRV	$n^{-1/4}$	i.i.d.	Optimal rate; weight selection requires tuning
Realized kernel	$n^{-1/4}$	Dependent OK	Most widely used; Parzen kernel standard; bandwidth data-driven
Pre-averaging	$n^{-1/4}$	Dependent OK	Intuitive; provides feasible CLT; block length $L \propto n^{1/2}$
QMLE	$n^{-1/4}$	Gaussian noise	Joint estimate of IV and ω^2 ; state-space framework

- **Noise model:** observed log price $p^* = p + \varepsilon$, where ε is mean-zero noise with variance ω^2 . Under this model, noisy RV has bias $2n\omega^2$, which diverges as $n \rightarrow \infty$.
- **Volatility signature plot:** plot average RV vs. sampling frequency. If the curve is still rising at your chosen frequency, you are in the noise-contaminated region. Always check this diagnostic before computing RV.
- **TSRV** (Zhang et al., 2005): combines a fast-scale and slow-scale RV to cancel noise. First consistent estimator with noise. Convergence rate $n^{-1/6}$.
- **MSRV** (Zhang, 2006): extends TSRV to multiple scales, achieving the optimal convergence rate $n^{-1/4}$.
- **Realized kernel** (Barndorff-Nielsen et al., 2008): weights the autocovariances of observed returns with a flat-top kernel (e.g., Parzen). Handles dependent noise. Rate $n^{-1/4}$. The most widely used noise-robust estimator.
- **Pre-averaging** (Jacod et al., 2009): smooths prices over local blocks before computing squared returns. Rate $n^{-1/4}$. Provides a feasible CLT for inference.
- **Other approaches:** subsampled/averaged RV (simple but biased), Fourier estimator (Malliavin and Mancino, 2002), QMLE (Xiu, 2010).
- **All optimal estimators converge at $n^{-1/4}$** under i.i.d. noise. They differ in robustness to dependent noise, implementation complexity, and tuning requirements.
- **For forecasting**, 5-minute RV is hard to beat (Liu et al., 2015). Noise-robust estimators improve *estimation* accuracy but these gains rarely translate into forecast improvements.
- **For estimation and intraday work**, the realized kernel is the default choice. Pre-averaging is a close second.
- Chapter 4 addresses a separate contamination: jump variation mixed into the continuous component.
- Chapter 10 uses both 5-minute RV and noise-robust estimators as ML features.

Table 3.2: Key results referenced in this chapter.

Paper	Result	Relevance
Hansen and Lunde (2006)	Characterized the effect of microstructure noise on realized variance; showed noise causes divergence as sampling frequency increases; documented that the i.i.d. noise model is a useful first approximation.	Foundational empirical analysis of the noise problem; motivates all robust estimators.
Aït-Sahalia et al. (2005)	Showed that optimal sampling frequency balances bias (from noise) and variance (from fewer observations); derived the optimal Δ^* under the i.i.d. noise model.	Theoretical foundation for the volatility signature plot and the bias-variance tradeoff.
Zhang et al. (2005)	Introduced TSRV: first estimator consistent for integrated variance in the presence of noise. Convergence rate $n^{-1/6}$.	The simplest noise-robust estimator; foundational for MSRV and other extensions.
Zhang (2006)	Extended TSRV to multiple scales (MSRV), achieving the optimal convergence rate $n^{-1/4}$.	Proved the $n^{-1/4}$ efficiency bound for noise-contaminated estimation.
Barndorff-Nielsen et al. (2008)	Developed the realized kernel: flat-top kernel weighting of autocovariances. Rate $n^{-1/4}$, handles dependent noise.	The most widely used noise-robust estimator in practice.
Jacod et al. (2009)	Introduced pre-averaging: smooth prices locally before computing RV. Rate $n^{-1/4}$, feasible CLT.	Intuitive alternative to the realized kernel.
Xiu (2010)	Proposed QMLE via Kalman filter on a state-space model. Rate $n^{-1/4}$, jointly estimates IV and ω^2 .	Useful when you also need a noise variance estimate.
Liu et al. (2015)	Compared ~ 400 estimators across 31 assets: noise-robust estimators rarely improve <i>forecasts</i> over 5-min RV.	Key practical finding: estimation accuracy $\neq$ forecast accuracy.

Chapter 4

Jumps and Continuous Variation

Why This Chapter

Separating jump variation from continuous variation is critical for forecasting. The HAR-J and HAR-CJ extensions (Chapter 6) split the RV signal into components with different persistence and predictability. Signed jump features appear in Chapter 10. Projects 1 and 4 use jump-decomposed features.

Chapter 2 showed that realized variance RV_t converges to the quadratic variation of the log-price process. When the price path is continuous (no jumps), quadratic variation equals integrated variance, and RV_t is a clean estimator of IV_t . But prices do jump. Earnings surprises, central bank announcements, and flash crashes all produce sudden, large price moves that do not come from the smooth diffusion process. When jumps are present, RV_t picks up *both* continuous and jump variation (Equation 2.5 in Chapter 2). This chapter develops the tools to separate them.

4.1 Why Prices Jump

Prices move for two fundamentally different reasons. Most of the time, they drift and fluctuate continuously as traders gradually digest information, adjust positions, and provide liquidity. This smooth fluctuation is the *diffusion* component. Occasionally, a piece of news is so large or so sudden that the price discontinuously leaps to a new level. This is a *jump*.

Concrete examples of jump triggers:

- **Earnings announcements:** A company reports revenue 15% above consensus after the close. The stock gaps up 8% at the next open.
- **Macro releases:** Non-farm payrolls (NFP) come in at +350k vs. a +180k forecast. Treasury yields spike 15 basis points in seconds.
- **Central bank decisions:** The Fed surprises with a 50bp rate cut when 25bp was priced. Equity indices jump 2% in one tick.
- **Flash crashes:** The May 6, 2010 flash crash sent the Dow down nearly 1,000 points (roughly 9%) in minutes before recovering.
- **Geopolitical shocks:** The Swiss National Bank abandoned its EUR/CHF floor on January 15, 2015. EUR/CHF dropped roughly 30% in seconds.

The Jump-Diffusion Price Process

Chapter 2 introduced the pure-diffusion log-price process: $dp_t = \mu_t dt + \sigma_t dW_t$. Adding jumps extends this to:

$$dp_t = \mu_t dt + \sigma_t dW_t + J_t dN_t$$

where N_t is a counting process (it increases by 1 each time a jump occurs) and J_t is the jump size (a random variable, possibly negative). You do not need to work with this process directly. The key takeaway: the price path now has two sources of variation, and the quadratic variation captures both:

$$[p]_t = \underbrace{\int_{t-1}^t \sigma_s^2 ds}_{\text{continuous}} + \underbrace{\sum_{s \in (t-1, t]} J_s^2}_{\text{jumps}}$$

This chapter is about estimating each piece separately.

4.1.1 Diagram: Continuous vs. Jump Price Paths

Figure 4.1 shows two price paths over the same day. The left path evolves smoothly (pure diffusion). The right path is identical except for a single large jump at 2:15pm.

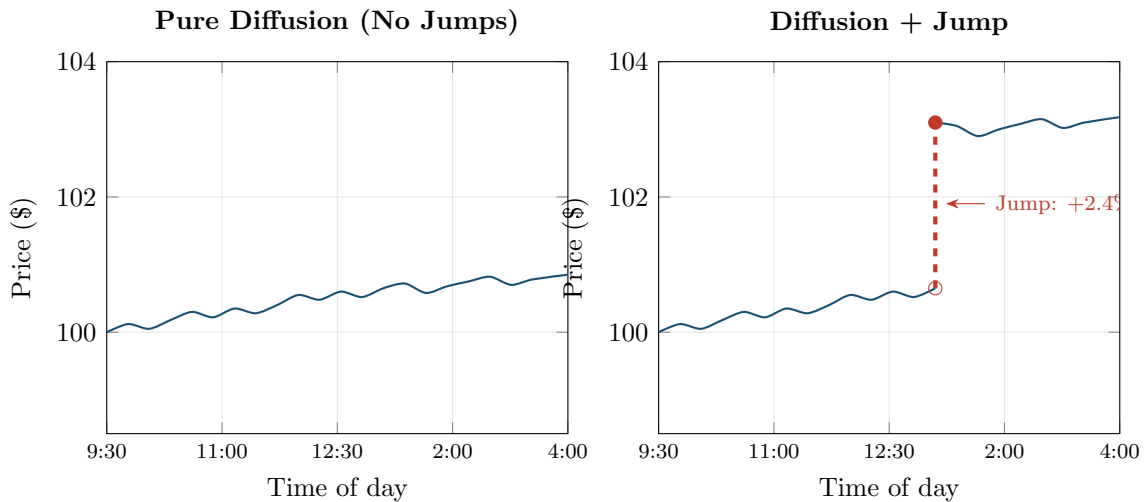


Figure 4.1: Two price paths over the same trading day. **Left:** pure diffusion with no jumps. The path is continuous and smooth. **Right:** diffusion plus a single jump at 2:00pm (red dashed line). The open circle marks the pre-jump price; the filled circle marks the post-jump price. Both paths have similar diffusion volatility, but the right path has much higher quadratic variation due to the jump.

4.2 Bipower Variation

You now know that RV_t captures both continuous and jump variation. The next step is to build an estimator that captures *only* the continuous part. The key idea, due to [Barndorff-Nielsen and Shephard \(2004\)](#), is bipower variation.

Why Multiplying Neighbors Kills Jumps

RV_t squares each return, so a single jump return of +2% contributes $(0.02)^2 = 4 \times 10^{-4}$ to the sum. Bipower variation instead multiplies each return's absolute value by its neighbor's absolute value. A jump return of +2% is large, but its neighbor (one interval before or after) is typically a normal-sized diffusion return, say 0.05%. The product $|0.02| \times |0.0005| = 1 \times 10^{-5}$ is far smaller than the squared jump 4×10^{-4} . The jump's contribution is “diluted” by the normal-sized neighbor. This is the core mechanism: jumps are isolated events, so a product of consecutive returns dampens them.

Bipower Variation

$$BPV_t = \frac{\pi}{2} \sum_{i=2}^n |r_{t,i}| |r_{t,i-1}| \quad (4.1)$$

- BPV_t : bipower variation for day t
- $|r_{t,i}|$: absolute value of the i -th intraday log return
- $|r_{t,i-1}|$: absolute value of the adjacent (previous) intraday log return
- The sum runs from $i = 2$ to n (you need pairs of consecutive returns, so you lose one observation)
- $\frac{\pi}{2} \approx 1.5708$: a scaling constant that corrects for the fact that $\mathbb{E}[|Z|] = \sqrt{2/\pi}$ when $Z \sim \mathcal{N}(0, 1)$

4.2.1 Why the $\pi/2$ Scaling Factor

Mean Absolute Value of a Standard Normal

If $Z \sim \mathcal{N}(0, 1)$, then $\mathbb{E}[|Z|] = \sqrt{2/\pi} \approx 0.7979$. This follows from integrating the folded normal density. The important consequence: $\mathbb{E}[|Z|]^2 = 2/\pi$, which is *not* equal to $\mathbb{E}[Z^2] = 1$. Taking absolute values and then multiplying introduces a bias relative to squaring, and the scaling factor corrects for it.

Consider two consecutive diffusion returns with no jumps. Each return is approximately $r_{t,i} \approx \sigma_{t,i} \epsilon_i$, where ϵ_i is standard normal. The product of absolute values is:

$$|r_{t,i}| |r_{t,i-1}| \approx \sigma_{t,i} \sigma_{t,i-1} |\epsilon_i| |\epsilon_{i-1}|$$

Taking expectations:

$$\mathbb{E}[|r_{t,i}| |r_{t,i-1}|] \approx \sigma_{t,i} \sigma_{t,i-1} (\mathbb{E}[|\epsilon|])^2 = \sigma_{t,i} \sigma_{t,i-1} \cdot \frac{2}{\pi}$$

To make the sum converge to $\int \sigma_s^2 ds$ (integrated variance), you need to undo the $2/\pi$ factor. Multiplying by $\pi/2$ does exactly that.

Barndorff-Nielsen and Shephard (2004): BPV Consistency

Under mild regularity conditions, bipower variation converges in probability to integrated variance even in the presence of finite-activity jumps:

$$BPV_t \xrightarrow{p} IV_t = \int_{t-1}^t \sigma_s^2 ds \quad \text{as } n \rightarrow \infty$$

This result holds regardless of whether jumps occurred during day t . In contrast, RV_t

converges to integrated variance *plus* the sum of squared jumps.

4.2.2 Diagram: How RV and BPV Respond to a Jump

Figure 4.2 shows the key mechanism. A sequence of six 5-minute returns includes one jump return (r_4). The left panel shows each return's contribution to RV_t (its squared value). The right panel shows each return's contribution to BPV_t (its absolute value times the previous absolute value, scaled by $\pi/2$).

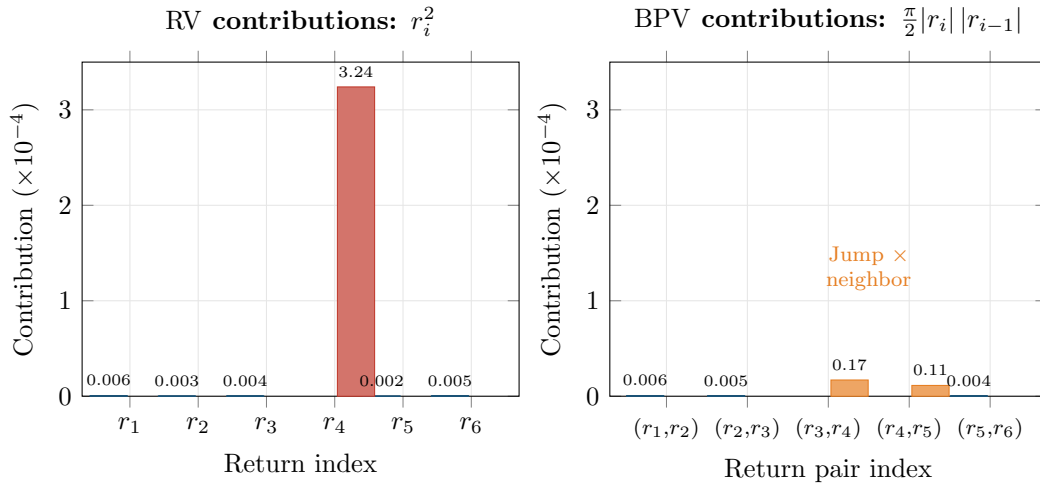


Figure 4.2: How a jump affects RV vs. BPV. **Left:** the jump return r_4 contributes 3.24×10^{-4} to RV (via r_4^2), dominating all other terms. **Right:** in BPV, the jump is multiplied by its normal-sized neighbors. The two terms involving r_4 contribute only 0.17 and 0.11×10^{-4} , roughly $50\times$ smaller than the squared jump. BPV “sees through” the jump and recovers the continuous variation.

Worked Example:

Computing RV and BPV for a Day with a Jump A stock is sampled every 5 minutes. You observe the following six returns (a shortened day for illustration):

Index	Return r_i	r_i^2	$ r_i r_{i-1} $	$\frac{\pi}{2} r_i r_{i-1} $
1	+0.0008	0.64×10^{-6}	—	—
2	−0.0005	0.25×10^{-6}	4.00×10^{-7}	6.28×10^{-7}
3	+0.0006	0.36×10^{-6}	3.00×10^{-7}	4.71×10^{-7}
4	+0.0180	324.0×10^{-6}	10.80×10^{-6}	16.96×10^{-6}
5	−0.0004	0.16×10^{-6}	7.20×10^{-6}	11.31×10^{-6}
6	+0.0007	0.49×10^{-6}	2.80×10^{-7}	4.40×10^{-7}

Return $r_4 = +0.0180$ (a 1.8% move in a single 5-minute bar) is the jump. The other returns are typical: 4–8 basis points each.

Step 1: Compute RV_t .

$$RV_t = \sum_{i=1}^6 r_i^2 = (0.64 + 0.25 + 0.36 + 324.0 + 0.16 + 0.49) \times 10^{-6} = 325.90 \times 10^{-6}$$

The jump return contributes $324.0/325.9 = 99.4\%$ of total RV.

Step 2: Compute BPV_t .

$$\text{BPV}_t = \frac{\pi}{2} \sum_{i=2}^6 |r_i| |r_{i-1}| = (6.28 + 4.71 + 16,960 + 11,310 + 4.40) \times 10^{-7} \times \frac{1}{10}$$

Summing the $\frac{\pi}{2} |r_i| |r_{i-1}|$ column:

$$\text{BPV}_t = (0.628 + 0.471 + 16.96 + 11.31 + 0.440) \times 10^{-6} = 29.81 \times 10^{-6}$$

Step 3: Estimate the jump component.

$$J_t^2 = \text{RV}_t - \text{BPV}_t = 325.90 \times 10^{-6} - 29.81 \times 10^{-6} = 296.09 \times 10^{-6}$$

Interpretation. RV_t is inflated by the jump: $\sqrt{325.9 \times 10^{-6}} = 1.81\%$ daily vol. BPV_t strips the jump and recovers the diffusion component: $\sqrt{29.81 \times 10^{-6}} = 0.55\%$ daily vol. The jump component $J_t^2 = 296 \times 10^{-6}$ accounts for 91% of the quadratic variation, consistent with a single large jump in an otherwise quiet day.

Note: BPV_t is not zero in the jump terms because the jump (r_4) is still multiplied by its (normal) neighbors. But those products are far smaller than r_4^2 . With 78 returns instead of 6, the effect would be even more diluted.

4.3 The BNS Jump Test

BPV_t gives you an estimate of continuous variation, and $\text{RV}_t - \text{BPV}_t$ gives you an estimate of jump variation. But both are estimates, subject to sampling noise. On any given day, $\text{RV}_t - \text{BPV}_t$ is slightly positive even with no jump, due to estimation error. You need a formal statistical test to determine whether the difference is large enough to declare a jump day.

Barndorff-Nielsen and Shephard (2006) proposed the following test.

BNS Jump Test Statistic

$$Z_{\text{BNS},t} = \frac{\text{RV}_t - \text{BPV}_t}{\sqrt{\vartheta \max\left(\frac{1}{n} \text{RQ}_t, 10^{-10}\right)}} \quad (4.2)$$

- $Z_{\text{BNS},t}$: the test statistic for day t
- $\text{RV}_t - \text{BPV}_t$: the estimated jump component (the difference between realized variance and bipower variation)
- $\vartheta = (\pi^2/4) + \pi - 5 \approx 0.6090$: a constant from the asymptotic theory
- $\text{RQ}_t = \frac{n}{3} \sum_{i=1}^n r_{t,i}^4$: realized quarticity, a consistent estimator of integrated quarticity $\int \sigma_s^4 ds$ under the null of no jumps, needed to scale the variance of the numerator. A jump-robust alternative is the tripower quarticity $\text{TPQ}_t = n \mu_{4/3}^{-3} \sum_{i=3}^n |r_{t,i}|^{4/3} |r_{t,i-1}|^{4/3} |r_{t,i-2}|^{4/3}$, where $\mu_{4/3} = \mathbb{E}[|Z|^{4/3}]$; in practice either estimator is used
- n : number of intraday returns
- $\max(\cdot, 10^{-10})$: a numerical safeguard to prevent division by zero
- Under the null hypothesis of no jumps, $Z_{\text{BNS},t} \xrightarrow{d} \mathcal{N}(0, 1)$ as $n \rightarrow \infty$

In Plain English

This is a signal-to-noise ratio. The numerator measures how much jump variation you think occurred (the gap between RV and BPV). The denominator measures how noisy that estimate is (derived from realized quarticity, which captures the variability of the variability). If the ratio exceeds a standard normal critical value, the jump is real, not a statistical artifact.

Why This Matters

The BNS test gives you a daily binary jump indicator: did a statistically significant jump occur on day t ? In your HAR-J model, this indicator determines whether you include the jump component $J_t^2 = \max(RV_t - BPV_t, 0)$ as a separate regressor. Days without significant jumps get $J_t^2 = 0$, keeping the jump feature clean and preventing estimation noise from leaking in.

How to Use the BNS Test

Compare $Z_{\text{BNS},t}$ to standard normal critical values. At the 5% level (one-sided), declare a jump day if $Z_{\text{BNS},t} > 1.645$. At the 1% level, use 2.326. If $Z_{\text{BNS},t} \leq 1.645$, you cannot reject the null of no jumps, so you treat RV_t as pure continuous variation for that day.

Quarticity

Just as variance is the integral of σ^2 , *quarticity* is the integral of σ^4 :

$$\int_{t-1}^t \sigma_s^4 ds$$

This quantity controls the variance of the $RV_t - BPV_t$ difference. You never need to compute it by hand. The realized quarticity estimator $RQ_t = \frac{n}{3} \sum r_{t,i}^4$ does it for you using fourth powers of returns. A jump-robust alternative (tripower quarticity) uses products of three consecutive absolute returns raised to the $4/3$ power, analogous to how BPV uses products of two absolute returns.

BNS Test Size Distortion in Finite Samples

The BNS test has known size distortions in finite samples: with 78 returns per day, it can over-reject the null (detect jumps that are not there) or under-reject (miss small jumps). [Huang and Tauchen \(2005\)](#) documented these problems and proposed a ratio form of the test statistic that has better finite-sample properties. For production use, consider the ratio variant or the corrected versions in [Barndorff-Nielsen and Shephard \(2006\)](#).

4.4 Lee-Mykland: Detecting Individual Jumps

The BNS test answers: “Did a jump occur at any point during day t ?” It does not tell you *when* the jump happened or how large it was. If you want intraday jump timing and sizes, you need the [Lee and Mykland \(2008\)](#) test.

From Daily to Intraday Jump Detection

The BNS test is like checking whether a patient had a fever at some point during the day. The Lee-Mykland test is like checking the patient's temperature every hour and flagging the specific times when fever occurred. It tests each individual return against a local volatility estimate to see if that return is “too large” to be diffusion.

The Lee-Mykland test works as follows. For each intraday return $r_{t,i}$, compute a local volatility estimate from a window of recent returns (excluding the return being tested). Then standardize the return by this local volatility. If the standardized return exceeds a threshold derived from extreme-value theory, flag it as a jump.

Lee-Mykland Test Statistic

$$\mathcal{L}_{t,i} = \frac{|r_{t,i}|}{\hat{\sigma}_{t,i}} \quad (4.3)$$

- $\mathcal{L}_{t,i}$: the test statistic for the i -th return on day t
- $|r_{t,i}|$: absolute value of the return being tested
- $\hat{\sigma}_{t,i}$: local volatility estimate from a window of K returns centered around (but excluding) return i , typically computed as $\hat{\sigma}_{t,i} = \sqrt{\text{BPV}_K / K}$, where BPV_K is the bipower variation over the K -return window

A return is classified as a jump if the statistic exceeds a critical value from the Gumbel distribution (extreme-value theory), after centering and scaling by C_n and S_n (constants that depend on the number of observations n):

$$\frac{\mathcal{L}_{t,i} - C_n}{S_n} > \beta_\alpha \quad (4.4)$$

where β_α is the $(1 - \alpha)$ quantile of the standard Gumbel distribution.

In Plain English

The Lee-Mykland statistic asks: “How many local standard deviations is this return?” A normal diffusion return might be 1 or 2 local standard deviations. A jump will be 5, 10, or more. The Gumbel critical value accounts for the fact that with many returns per day, even under pure diffusion you expect a few moderately large ones by chance. Only returns that exceed this multiple-testing-adjusted threshold are flagged as jumps.

Why This Matters

Lee-Mykland gives you jump *times and sizes*, whereas the BNS test gives only a daily indicator. This is the basis for signed jump features: you can separately measure “bad jumps” (large negative returns) and “good jumps” (large positive returns) within the day. Signed jump variation feeds directly into the SHAR model and the asymmetric jump features in Chapter 10, where negative jumps predict higher future volatility than positive jumps of the same magnitude.

4.4.1 Diagram: Intraday Jump Detection

Figure 4.3 shows an intraday return path with detected jump returns highlighted.

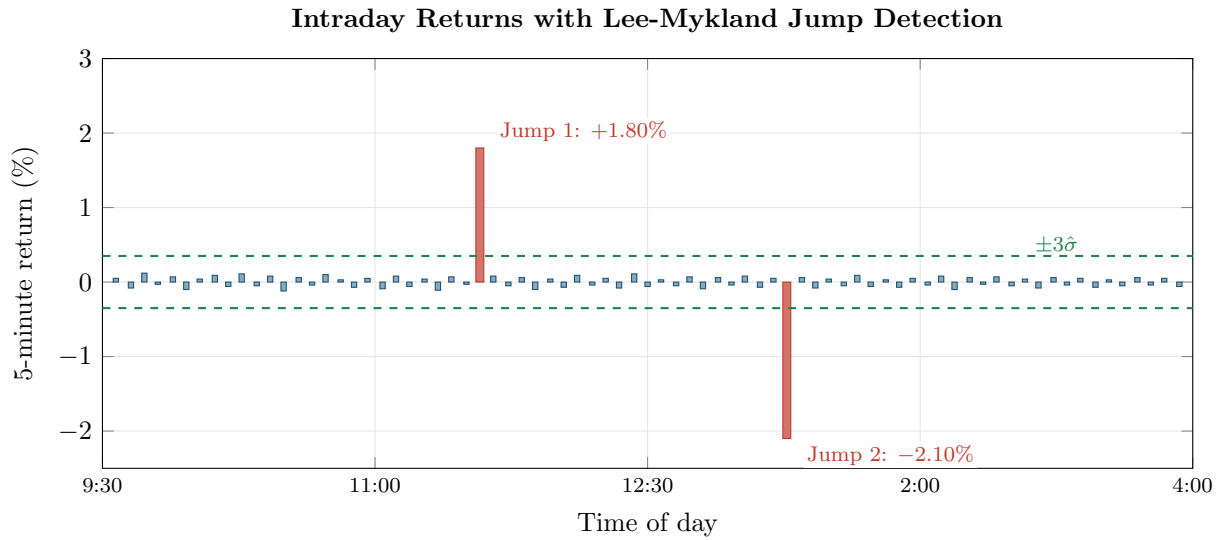


Figure 4.3: Intraday 5-minute returns with Lee-Mykland jump detection. Blue bars: normal diffusion returns (typically $\pm 0.1\%$). Red bars: detected jump returns (far exceeding the local $\pm 3\sigma$ band, shown as green dashed lines). Jump 1 at 12:15pm (+1.80%) might correspond to a macro release. Jump 2 at 2:00pm (−2.10%) is a large negative jump. The Lee-Mykland test identifies both the times and sizes of jumps, information that the BNS test (which only flags the day) cannot provide.

BNS vs. Lee-Mykland: When to Use Which

Use the BNS test when you need a daily-level binary indicator: “did a jump occur today?” This is sufficient for most forecasting applications (e.g., HAR-J in Chapter 6). Use Lee-Mykland when you need intraday jump timing (e.g., studying how volatility behaves in the minutes after a jump) or when you want to construct signed jump features (Chapter 10).

4.5 Aït-Sahalia-Jacod Test

Both the BNS test and the Lee-Mykland test assume that microstructure noise is negligible at the sampling frequency you use. As Chapter 2 discussed, this is a reasonable assumption at 5-minute frequency for liquid assets, but it fails at higher frequencies. Aït-Sahalia and Jacod (2009) proposed an alternative that is robust to microstructure noise.

Power Variation at Two Scales

The idea is to compare realized power variation computed at two different sampling frequencies. If no jumps are present, the ratio of these two quantities converges to a known constant (that depends only on the power used, not on the volatility level). If jumps are present, the ratio shifts. The test detects this shift.

Power Variation

Instead of always squaring returns, we can raise absolute returns to any power p . This single parameter p controls whether the estimator “listens to” jumps or ignores them.

$$V_t^{(p)} = \sum_{i=1}^n |r_{t,i}|^p \quad (4.5)$$

- $p = 2$: this reduces to RV_t (realized variance)
- $p = 1$: sum of absolute returns
- The key property: for $p < 2$, the power variation is dominated by diffusion (jumps contribute negligibly). For $p = 2$, jumps contribute fully. For $p > 2$, jumps dominate.

In Plain English

Think of the power p as a volume knob for jump sensitivity. Small returns (diffusion) and large returns (jumps) are both present in the data. When p is low (say $p = 1$), raising everything to p compresses the big returns more than the small ones, so jumps fade into the background. When p is high (say $p = 4$), the large returns get amplified disproportionately, so jumps scream. At $p = 2$ (standard RV), you get an honest count of both.

The Aït-Sahalia and Jacod (2009) test statistic compares $V_t^{(p)}$ computed at two sampling frequencies, Δ and $k\Delta$ (e.g., 5-minute and 15-minute). Under the null of no jumps, the ratio converges to a constant $k^{p/2-1}$. Under the alternative (jumps present), the ratio is different.

Aït-Sahalia and Jacod (2009): Power-Variation-Ratio Test

The test statistic

$$S_t = \frac{V_t^{(p)}(\Delta)}{V_t^{(p)}(k\Delta)}$$

converges to $k^{p/2-1}$ under the null of no jumps. Significant deviation from this value indicates jump activity. Because the test uses ratios of power variations at different frequencies, microstructure noise affects numerator and denominator similarly, making the test more robust than BNS in noisy settings.

The practical advantage of this test is that you can apply it at higher sampling frequencies (1-minute or even 30-second) without the noise distortions that affect BNS. The cost is additional complexity and the need to choose k and p . For a first pass, the BNS test at 5-minute frequency is usually sufficient.

4.6 Threshold and Truncation Methods

The BNS approach estimates continuous variation indirectly: compute BPV_t , and the jump component is the residual $RV_t - BPV_t$. Threshold methods take the opposite approach: directly classify each return as either a jump or a diffusion return, then compute realized variance using only the diffusion returns.

Threshold Realized Variance

Instead of dampening jumps implicitly (as BPV does), we can remove them explicitly: classify each return as “jump” or “not jump” based on a size cutoff, then sum squared returns only for the non-jump returns.

$$TRV_t = \sum_{i=1}^n r_{t,i}^2 \cdot \mathbf{1}\{|r_{t,i}| \leq \vartheta_i\} \quad (4.6)$$

- TRV_t : threshold realized variance for day t
- $r_{t,i}^2$: squared i -th intraday return

- $\mathbf{1}\{\cdot\}$: indicator function (equals 1 if the condition is true, 0 otherwise)
- ϑ_i : threshold for return i , typically set as $\vartheta_i = c \hat{\sigma}_i \Delta^\varpi$, where c is a constant (e.g., $c = 3$), $\hat{\sigma}_i$ is a local volatility estimate, and $\varpi \in (0, 1/2)$ controls how the threshold scales with sampling frequency

Why This Matters

The threshold approach is the basis of the HAR-CJ model (Corsi et al., 2010), one of the strongest baselines for your vol forecasting project. HAR-CJ enters $C_t = \text{TRV}_t$ and $J_t = \text{RV}_t - \text{TRV}_t$ as separate features, letting the model learn that continuous variation is persistent (high coefficient) while jump variation is transient (low coefficient). If your ML model cannot beat HAR-CJ, the decomposition itself is doing most of the heavy lifting.

Mancini (2009) established the theoretical foundation for threshold estimators. Corsi et al. (2010) combined the threshold approach with the HAR forecasting model to produce the HAR-CJ decomposition. Their key contribution was showing that the threshold-based continuous and jump components improve forecasting performance when entered separately into the HAR model, because they have different persistence.

Corsi et al. (2010): Threshold HAR-CJ

Splitting RV_t into a continuous component ($C_t = \text{TRV}_t$) and a jump component ($J_t = \text{RV}_t - \text{TRV}_t$) using the threshold approach, and entering them separately into the HAR model, produces significantly better volatility forecasts than the baseline HAR. The improvement comes from allowing the model to assign different persistence parameters to continuous and jump variation.

4.7 Why the Decomposition Matters for Forecasting

You have now seen four ways to separate jumps from continuous variation: bipower variation, the BNS test, Lee-Mykland, and threshold truncation. Why does this separation matter? The answer is forecasting: continuous and jump variation behave very differently over time.

Different Persistence

Continuous volatility is highly persistent. If a stock had high diffusion volatility today, it will very likely have high diffusion volatility tomorrow, and next week, and even next month. This is the long-memory property that the HAR model (Chapter 6) exploits. Jump variation is the opposite. A jump today tells you almost nothing about whether a jump will occur tomorrow. Earnings announcements, FOMC decisions, and geopolitical shocks do not cluster in the same way that diffusion volatility does. The jump component is nearly unpredictable.

This difference has a direct implication for forecasting models. If you feed raw RV_t into a model, you are mixing a highly persistent signal (continuous variation) with transient noise (jump variation). The model has to learn a single set of persistence parameters that compromises between the two. Separating them lets the model learn different dynamics for each component.

4.7.1 Diagram: The Decomposition Pipeline

Figure 4.4 summarizes the full decomposition pipeline from raw intraday returns to the separated components that feed into forecasting models.

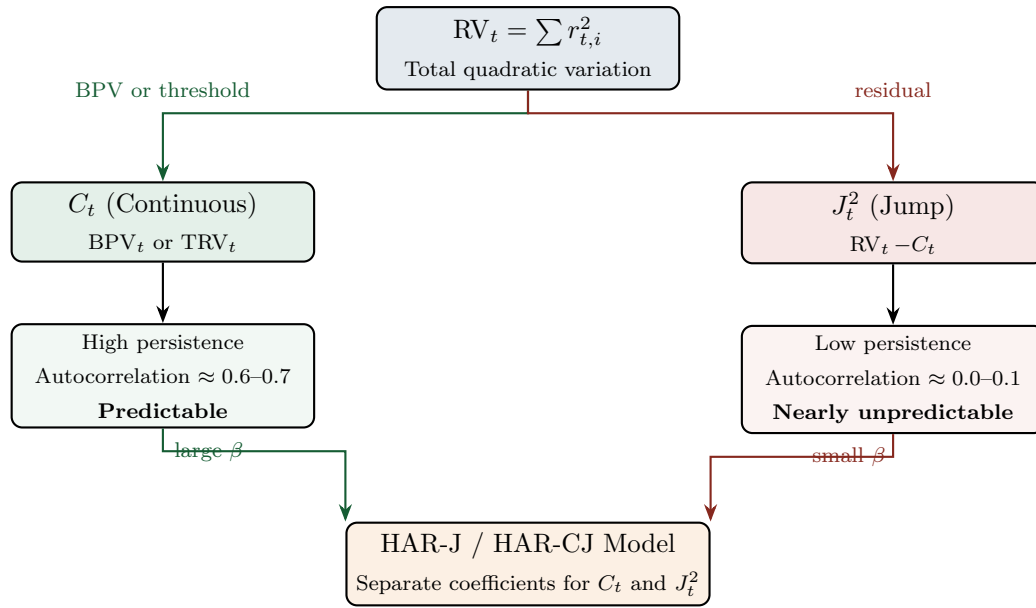


Figure 4.4: The decomposition pipeline. RV_t is split into a continuous component C_t (estimated via BPV or threshold truncation) and a jump component J_t^2 (the residual). The continuous component is highly persistent and drives forecasting accuracy. The jump component is nearly unpredictable but still included to prevent it from contaminating the continuous signal. In the HAR-J and HAR-CJ models, each component receives its own coefficient, letting the model exploit the different persistence.

There is an additional asymmetry. [Bollerslev et al. \(2009a\)](#) and [Patton and Sheppard \(2015a\)](#) found that “bad jumps” (large negative returns) are more informative for future volatility than “good jumps” (large positive returns). A 5% daily drop signals higher volatility going forward; a 5% daily rally does not, to the same degree. This connects to the signed volatility decompositions (SHAR) covered in Chapter 6 and the signed jump features in Chapter 10.

Do Not Treat Jump and Continuous Components as Interchangeable

The continuous and jump components of RV_t have different statistical properties: different persistence, different distributional shapes, and different predictive content. Lumping them together (as raw RV_t does) throws away information. Treating the jump component as if it were as persistent as continuous variation will lead to overstated volatility forecasts after jump days. Always decompose before forecasting.

Summary

- **Jumps** are sudden, large price moves caused by discrete news events (earnings, macro releases, geopolitical shocks). They are distinct from the smooth diffusion that drives normal price variation.
- **Quadratic variation** equals integrated variance plus the sum of squared jumps: $[p]_t = IV_t + \sum J_s^2$. Standard RV_t captures both; this chapter develops tools to separate them.
- **Bipower variation** $BPV_t = \frac{\pi}{2} \sum_{i=2}^n |r_{t,i}| |r_{t,i-1}|$ multiplies consecutive absolute returns, dampening the contribution of isolated jumps ([Barndorff-Nielsen and Shephard, 2004](#)).
- **BPV consistency**: BPV_t converges to IV_t even in the presence of finite-activity jumps. $RV_t - BPV_t$ consistently estimates the jump component.
- The $\frac{\pi}{2}$ scaling factor corrects for the fact that $\mathbb{E}[|Z|]^2 = 2/\pi$ when $Z \sim \mathcal{N}(0, 1)$.

- The **BNS jump test** formalizes this: $Z_{\text{BNS},t} = (RV_t - BPV_t) / \sqrt{\vartheta \cdot RQ_t / n}$ is asymptotically standard normal under the null of no jumps (Barndorff-Nielsen and Shephard, 2006). Compare to 1.645 (5%) or 2.326 (1%).
- The BNS test has **finite-sample size distortions** (it can over-reject). The ratio variant or Huang-Tauchen correction improves performance (Huang and Tauchen, 2005).
- The **Lee-Mykland test** detects individual jumps within the day, providing jump timing and sizes rather than only a daily binary indicator (Lee and Mykland, 2008).
- The **Aït-Sahalia-Jacod test** uses power-variation ratios at two frequencies and is more robust to microstructure noise than the BNS test (Aït-Sahalia and Jacod, 2009).
- **Threshold methods** classify each return as jump or diffusion based on a size cut-off. Threshold realized variance sums squared returns only for below-threshold returns (Mancini, 2009; Corsi et al., 2010).
- **Continuous variation is persistent**; jump variation is transient. Separating them improves forecasting by allowing models to assign different persistence to each component.
- **Bad jumps** (negative) are more informative for future volatility than good jumps (positive), motivating signed jump features in Chapter 10 (Bollerslev et al., 2009a; Patton and Sheppard, 2015a).
- The HAR-J and HAR-CJ models (Chapter 6) exploit this decomposition directly; Projects 1 and 4 use jump-decomposed features.

Table 4.1: Key results referenced in this chapter.

Paper	Result	Relevance
Barndorff-Nielsen and Shephard (2004)	Bipower variation BPV_t converges to integrated variance IV_t even in the presence of finite-activity jumps.	Foundational estimator for separating continuous and jump variation.
Barndorff-Nielsen and Shephard (2006)	BNS jump test: $Z_{\text{BNS},t}$ is asymptotically $\mathcal{N}(0, 1)$ under no jumps; formal test for jump days.	Standard daily jump test used in HAR-J models.
Lee and Mykland (2008)	Intraday jump detection: identifies individual jump times and sizes using local volatility and extreme-value theory.	Enables signed jump features and intraday jump analysis.
Aït-Sahalia and Jacod (2009)	Power-variation-ratio test for jump activity, robust to microstructure noise.	Alternative to BNS for noisy high-frequency data.
Corsi et al. (2010)	Threshold-based HAR-CJ decomposition improves volatility forecasts by separating persistent continuous and transient jump components.	Direct input to HAR-CJ forecasting model (Chapter 6).
Andersen et al. (2007)	Jump component of RV has near-zero persistence; forecasting improves when continuous and jump components are modeled separately.	Motivates the jump-continuous decomposition for all forecasting in this guide.

Part II

Forecasting Volatility with Classical Models

Chapter 5

The GARCH Family

Why This Chapter?

GARCH models forecast volatility using only daily returns, without intraday data. Project 1 (HARQ-X) uses GARCH as a comparison baseline, and Realized GARCH and the HEAVY model bridge the daily-return and realized-volatility worlds.

Chapters 2–4 built volatility estimators from intraday data. This chapter steps back in time: before high-frequency data was widely available, how did people forecast volatility? The answer is the GARCH family, a set of models that extract volatility dynamics entirely from daily returns.

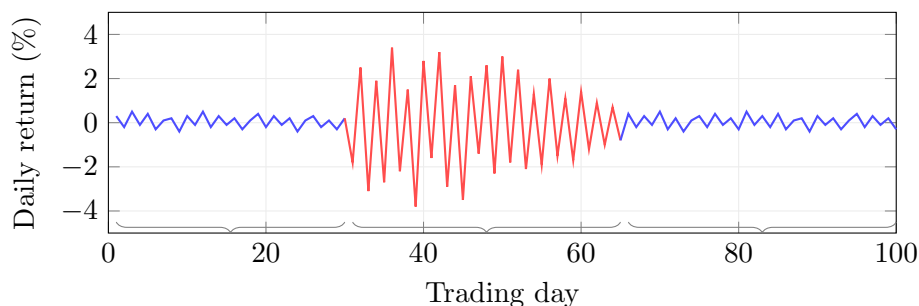
5.1 The ARCH Model

We start with the simplest idea: yesterday’s return tells you something about today’s volatility.

Conditional vs. Unconditional Variance

The *unconditional* variance of a return series is a single number computed over the full sample: $\text{Var}(r_t)$. The *conditional* variance $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1})$ is the variance you would forecast for period t , given everything you know up through period $t - 1$ (the information set $\mathcal{F}_{t-1}$). GARCH-family models are models for this conditional variance.

Volatility clustering: calm periods alternate with turbulent periods



Engle (1982) proposed the ARCH(q) model. In its simplest form, ARCH(1):

$$\sigma_t^2 = \omega + \alpha_1 r_{t-1}^2 \quad (5.1)$$

- σ_t^2 : conditional variance for period t (the quantity we forecast)
- $\omega > 0$: baseline variance level, ensuring σ_t^2 stays positive even when $r_{t-1} = 0$

- $\alpha_1 \geq 0$: sensitivity to the most recent squared return
- r_{t-1}^2 : squared return from the previous period, a noisy proxy for realized variance

The general ARCH(q) model adds more lags:

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i r_{t-i}^2 \quad (5.2)$$

- q : number of lags included
- $\alpha_i \geq 0$: weight on the squared return from i days ago

ARCH's Parameter Problem

Volatility is persistent: a shock today still affects volatility weeks later. Capturing this with ARCH requires many lags ($q = 10$ or more), each with its own parameter. This makes estimation fragile and overfitting likely. GARCH solves this.

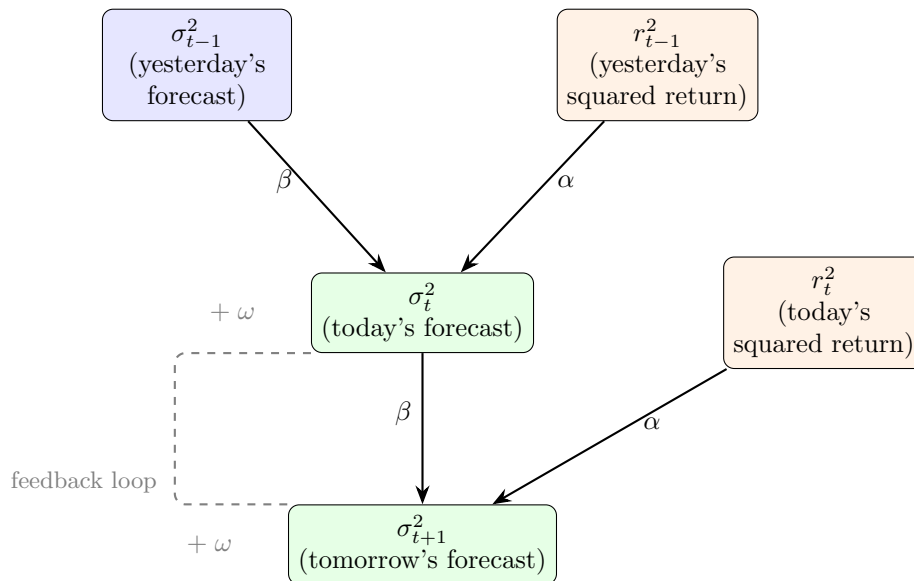
5.2 GARCH(1,1): The Workhorse Model

ARCH needs many parameters to capture persistence. The key insight of GARCH is to add a single feedback term: let yesterday's *conditional variance* help predict today's.

GARCH as Exponential Smoothing

Think of a weather forecast. Tomorrow's temperature forecast depends on two things: (1) what actually happened today (the "news"), and (2) what you predicted yesterday. GARCH works the same way: the new volatility forecast blends today's squared return (news) with yesterday's forecast (memory). A single memory term replaces an infinite history of squared returns.

Today's variance forecast feeds into tomorrow's, creating a self-reinforcing loop that captures persistence (see diagram).



Bollerslev (1986) introduced the GARCH(1,1) model:

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (5.3)$$

- σ_t^2 : conditional variance for period t

- $\omega > 0$: intercept, related to the long-run average variance
- $\alpha \geq 0$: reaction coefficient; how strongly the model reacts to new information (yesterday's squared return)
- $\beta \geq 0$: persistence coefficient; how much of yesterday's variance forecast carries forward
- r_{t-1}^2 : yesterday's squared return (the “shock” or “news”)
- σ_{t-1}^2 : yesterday's conditional variance forecast (the “memory”)

Why This Matters

GARCH(1,1) is your simplest conditional-variance baseline. In a model horse race, any ML model or HAR variant that cannot beat GARCH(1,1) out of sample is not worth the added complexity. Typical $\alpha + \beta \approx 0.98$ for equity indices, meaning volatility shocks are extremely persistent, the same stylized fact that HAR captures with its weekly and monthly RV components.

Stationarity Condition for GARCH(1,1)

The process is covariance-stationary if and only if $\alpha + \beta < 1$. When this holds, the unconditional (long-run) variance is:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta} \quad (5.4)$$

- $\bar{\sigma}^2$: the unconditional variance, i.e., the level to which σ_t^2 mean-reverts over time
- $\alpha + \beta$: total persistence; values close to 1 mean slow mean-reversion (volatility shocks are long-lived)

If $\alpha + \beta = 1$, the model becomes IGARCH (Integrated GARCH): shocks never decay, and the unconditional variance is undefined (Engle and Bollerslev, 1986).

In Plain English

The unconditional variance is the “gravity” level that the conditional variance always pulls toward. This is mean reversion in volatility, and the speed of that reversion is $1 - \alpha - \beta$.

Why This Matters

When you forecast RV at the 22-day horizon, mean-reversion speed matters enormously. A GARCH model with $\alpha + \beta = 0.98$ will still forecast elevated volatility three weeks after a shock, while HAR's monthly component captures the same persistence more flexibly. Comparing the implied half-lives of GARCH versus HAR forecasts is a simple diagnostic for your project.

Worked Example:

GARCH(1,1) One-Step Forecast Suppose you have estimated parameters $\omega = 0.00001$, $\alpha = 0.08$, $\beta = 0.90$. Yesterday's return was $r_{t-1} = -0.03$ (a 3% loss), and yesterday's conditional variance was $\sigma_{t-1}^2 = 0.0004$ (implying $\sigma_{t-1} = 0.02$, or 2% daily vol).

Step 1: Compute the forecast. Summing the baseline, news, and memory components gives $\sigma_t^2 = 0.00001 + 0.000072 + 0.00036 = 0.000442$, so $\sigma_t = \sqrt{0.000442} \approx 2.10\%$.

Most of the forecast (81%) comes from the memory term $\beta\sigma_{t-1}^2$, reflecting the high persistence typical of financial volatility.

Step 2: Check long-run variance.

$$\bar{\sigma}^2 = \frac{0.00001}{1 - 0.08 - 0.90} = \frac{0.00001}{0.02} = 0.0005 \Rightarrow \bar{\sigma} \approx 2.24\%$$

With $\alpha + \beta = 0.98$, mean-reversion is very slow: it takes roughly $\ln(0.5)/\ln(0.98) \approx 34$ days for the conditional variance to close half the gap to $\bar{\sigma}^2$.

GARCH(1,1) Is Usually Enough

Hansen and Lunde (2005) compared 330 GARCH-type models on exchange rate and equity data. For exchange rates, no model significantly outperformed GARCH(1,1). For equities, models with a leverage effect (Section 5.3) did better, but higher-order specifications like GARCH(2,1) or GARCH(1,2) provided negligible improvement. In practice, GARCH(1,1) captures the bulk of conditional variance dynamics. The gains from asymmetric extensions come from modeling leverage, not from adding lags.

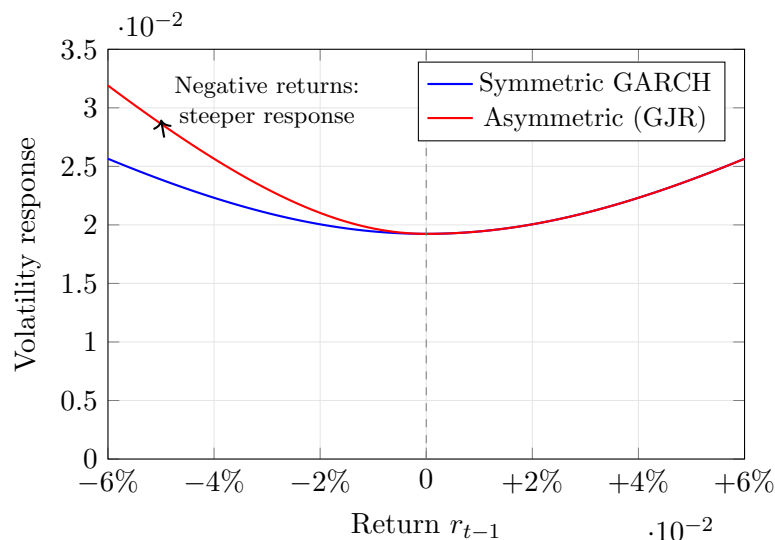
5.3 The Leverage Effect

GARCH(1,1) treats positive and negative returns symmetrically: a +3% return and a −3% return produce the same $r_{t-1}^2 = 0.0009$ and thus the same volatility forecast. In real equity markets, this is wrong.

Why Negative Returns Increase Volatility More

Black (1976) documented this asymmetry and proposed an explanation. When a stock drops, the firm's equity value falls while its debt stays fixed. The firm becomes more leveraged (higher debt-to-equity ratio), making its equity riskier, so volatility rises. A positive return has the opposite effect but to a smaller degree. This “leverage effect” is one of the most robust stylized facts in equity markets.

For the same magnitude of return, negative returns produce a larger volatility response than positive returns (diagram).



Standard GARCH(1,1) cannot capture this asymmetry because it depends on r_{t-1}^2 , which discards the sign of the return. The next two sections present models that fix this.

5.4 GJR-GARCH: A Simple Asymmetric Extension

The most direct way to capture leverage is to add an indicator function that activates only for negative returns.

Glosten et al. (1993) proposed:

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \gamma \mathbf{1}_{\{r_{t-1} < 0\}} r_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (5.5)$$

- ω, α, β : same roles as in GARCH(1,1) (Equation 5.3)
- $\gamma \geq 0$: the asymmetry (leverage) parameter
- $\mathbf{1}_{\{r_{t-1} < 0\}}$: indicator function equal to 1 when $r_{t-1} < 0$, and 0 otherwise

The effect is straightforward:

- After a *positive* return: the effective coefficient on r_{t-1}^2 is just α .
- After a *negative* return: the effective coefficient is $\alpha + \gamma$, giving a stronger volatility response.

Why This Matters

The leverage effect is one of the strongest predictable patterns in equity volatility. Adding an asymmetry term (whether in GJR-GARCH or as a signed-return feature in your ML model) typically improves QLIKE by 5–15%. If your HAR baseline does not include a negative-return indicator, GJR-GARCH shows exactly why it should.

5.5 EGARCH: Log-Variance and Guaranteed Positivity

GJR-GARCH has a practical annoyance: you need parameter constraints ($\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$) to ensure $\sigma_t^2 > 0$. Unconstrained optimization sometimes produces estimates that violate these bounds.

Nelson (1991) proposed EGARCH, which models the *log* of variance. Since $\exp(\cdot)$ is always positive, the variance is automatically positive regardless of parameter signs.

$$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \alpha \left(\frac{|r_{t-1}|}{\sigma_{t-1}} - \sqrt{\frac{2}{\pi}} \right) + \gamma \frac{r_{t-1}}{\sigma_{t-1}} \quad (5.6)$$

- $\ln \sigma_t^2$: log conditional variance (the model's target)
- ω : intercept on the log-variance scale (can be any real number)
- β : persistence of log-variance; $|\beta| < 1$ for stationarity
- $|r_{t-1}|/\sigma_{t-1}$: the absolute value of the standardized return, measuring shock magnitude
- $\sqrt{2/\pi}$: the expected value of $|z|$ when $z \sim \mathcal{N}(0, 1)$; subtracting it centers the magnitude term at zero
- α : the size effect; controls how strongly large shocks (of either sign) increase variance
- r_{t-1}/σ_{t-1} : the signed standardized return
- γ : the sign effect (asymmetry parameter); $\gamma < 0$ means negative returns increase log-variance

How the Sign Effect Works

The γ term adds $\gamma \cdot z_{t-1}$ to log-variance, where $z_{t-1} = r_{t-1}/\sigma_{t-1}$ is the standardized return. If $\gamma = -0.10$ and the standardized return is $z = -2$ (a two-standard-deviation loss), the contribution is $(-0.10)(-2) = +0.20$: log-variance rises. If $z = +2$ (a two-standard-

deviation gain), the contribution is $(-0.10)(+2) = -0.20$: log-variance falls. The same shock magnitude produces opposite effects depending on sign.

Why This Matters

Many ML models for realized volatility also work in log space ($\log RV_t$), just as EGARCH does. If you include EGARCH as a baseline, its sign-effect coefficient γ gives you a direct comparison point for how well your ML model captures asymmetry.

EGARCH Forecasting Complication

Multi-step forecasts from EGARCH require computing $\mathbb{E}[\exp(\cdot)]$, which does not simplify as cleanly as for linear GARCH. In practice, simulation-based forecasts are used for horizons beyond one step.

5.6 Long Memory in Volatility: FIGARCH

Volatility autocorrelations in financial data decay very slowly. If you compute the autocorrelation of daily squared returns (or absolute returns) at lags 1, 5, 20, 100, you find that the autocorrelation is still positive even at lag 100. Standard GARCH(1,1) implies *exponential* decay of autocorrelations, which is too fast.

Long Memory and Fractional Integration

A time series has *long memory* if its autocorrelations decay at a hyperbolic (power-law) rate rather than an exponential rate. In the context of ARIMA, a non-integer differencing parameter $d \in (0, 0.5)$ produces this behavior: the series is neither stationary ($d = 0$) nor unit-root ($d = 1$), but something in between. The same idea can be applied to the variance equation of a GARCH model.

Baillie et al. (1996) introduced FIGARCH, which replaces the integer-differencing implicit in GARCH with fractional differencing.

FIGARCH Differencing Parameter

The FIGARCH model introduces a parameter $d \in (0, 1)$ governing the rate at which past shocks influence current variance:

- $d = 0$: equivalent to GARCH; shock impact decays exponentially (short memory)
- $d = 1$: equivalent to IGARCH; shock impact never decays (unit root in variance)
- $0 < d < 1$: shock impact decays hyperbolically (long memory); past shocks matter, but their influence eventually fades

The full FIGARCH(1,d,1) specification is written using the lag operator L :

$$(1 - \beta L)\sigma_t^2 = \omega + [1 - \beta L - (1 - \phi L)(1 - L)^d]r_t^2 \quad (5.7)$$

- L : lag operator ($Lx_t = x_{t-1}$)
- $(1 - L)^d$: fractional differencing operator with parameter d
- ϕ : an additional ARCH-side parameter
- β : persistence parameter, same role as in GARCH

The algebra matters less than the implication: FIGARCH matches the slow decay of volatility autocorrelations far better than GARCH.

Why This Matters

Long memory in volatility is precisely why HAR works: its weekly and monthly components act as a discrete approximation to the slow-decaying memory that FIGARCH models continuously. If your ML residual model finds that long-lag features improve forecasts, FIGARCH's d parameter gives you a theoretical explanation for why.

FIGARCH vs. Rough Volatility

Both FIGARCH and rough volatility models (Chapter 7) address the same empirical fact: volatility has long memory. FIGARCH does so within the discrete-time GARCH framework by adding the parameter d . Rough volatility uses continuous-time fractional Brownian motion with Hurst parameter $H \approx 0.1$. The two approaches are complementary views of the same phenomenon.

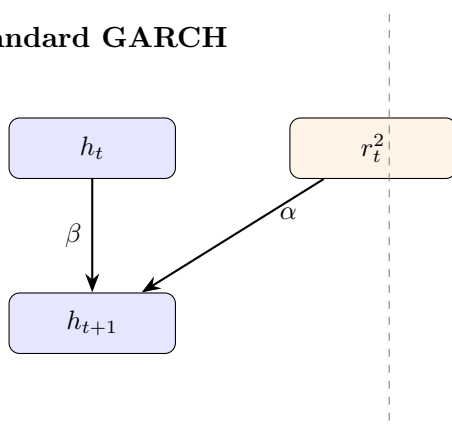
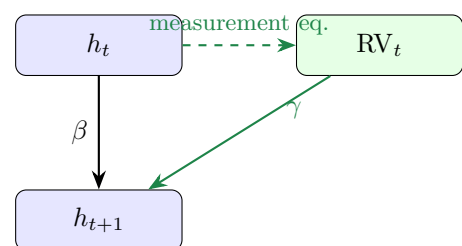
5.7 Realized GARCH: Bringing Intraday Data into GARCH

Standard GARCH uses only daily returns to infer volatility. But if you have access to intraday data, you can compute realized volatility (RV_t , as defined in Chapter 2), a far more precise measure of what actually happened. Realized GARCH incorporates RV_t directly into the GARCH framework.

The Measurement Problem

In GARCH, the conditional variance σ_t^2 is a latent (unobserved) variable. The model infers it from the noisy signal r_t^2 (the squared daily return). But r_t^2 is an extremely noisy estimator of daily variance: on average, the noise-to-signal ratio exceeds 5. Realized volatility RV_t is computed from many intraday returns and is orders of magnitude more precise. Realized GARCH exploits this better signal.

What Realized GARCH adds is a measurement equation linking RV_t to the latent conditional variance h_t (see diagram).

Standard GARCH**Realized GARCH**

Hansen et al. (2012) specify three equations. The return equation is standard:

$$r_t = \sqrt{h_t} z_t, \quad z_t \sim \mathcal{N}(0, 1) \quad (5.8)$$

- r_t : daily return
- h_t : conditional variance (the latent variable we model)
- z_t : standardized innovation, assumed i.i.d. standard normal

This is the standard return-generating process: the daily return is volatility times a random shock. It is shared with standard GARCH and is not where Realized GARCH innovates.

The measurement equation links the observed RV_t to the latent h_t :

$$\log RV_t = \xi + \delta \log h_t + \tau(z_t) + u_t \quad (5.9)$$

- $\log RV_t$: log realized volatility (observed from intraday data)
- ξ : intercept allowing for a level difference between RV_t and h_t
- δ : loading of $\log h_t$ on $\log RV_t$; typically close to 1
- $\tau(z_t)$: a leverage function of the standardized return, capturing the asymmetry between positive and negative returns (often specified as $\tau_1 z_t + \tau_2(z_t^2 - 1)$)
- u_t : measurement noise, independent of z_t

The GARCH equation uses RV_t instead of r_t^2 :

$$\log h_{t+1} = \omega + \beta \log h_t + \gamma \log RV_t \quad (5.10)$$

- ω : intercept on the log-variance scale
- β : persistence of the conditional variance
- γ : sensitivity to the realized measure; this is where intraday information enters the model

In Plain English

Realized GARCH is a three-part system. (1) Returns are driven by latent variance, as usual. (2) The measurement equation says “realized volatility is a noisy, possibly biased reading of that latent variance, with an asymmetry correction.” (3) The GARCH equation says “tomorrow’s latent variance depends on today’s latent variance (persistence) plus today’s realized volatility (new information).” The key upgrade over standard GARCH is in step (3): instead of feeding in the noisy squared return r_t^2 , you feed in RV_t , which is orders of magnitude more precise.

Why This Matters

Realized GARCH is the bridge between the GARCH world and the RV world your project lives in. It uses RV_t directly as an input, just as your HAR and ML models do. Including Realized GARCH as a baseline lets you ask: “Does my ML model add value beyond what a well-specified parametric model can extract from the same RV_t data?” If your ML model only matches Realized GARCH, the complexity may not be justified.

Realized GARCH Outperforms Standard GARCH

[Hansen et al. \(2012\)](#) find that Realized GARCH with log-linear specification substantially outperforms standard GARCH(1,1) in both in-sample fit and out-of-sample forecasting on Dow Jones Industrial Average stocks. Squared returns become statistically insignificant once a realized measure is included. The leverage function $\tau(z_t)$ is highly significant, confirming the importance of asymmetry.

5.8 The HEAVY Model

Realized GARCH is not the only way to combine daily and intraday information. [Shephard and Sheppard \(2010\)](#) proposed the HEAVY (High-frEQuency-bAsed VolatilitY) model, which takes a different but related approach: instead of treating RV_t as a noisy measurement of a latent h_t ,

HEAVY models both the conditional variance of returns and the conditional expectation of RV_t as a joint system.

The HEAVY system has two equations. The first governs the conditional variance of daily returns:

$$\sigma_{R,t}^2 = \omega_R + \alpha_R RV_{t-1} + \beta_R \sigma_{R,t-1}^2 \quad (5.11)$$

- $\sigma_{R,t}^2$: conditional variance of daily returns
- RV_{t-1} : yesterday's realized variance (replaces r_{t-1}^2 from GARCH)
- α_R : sensitivity to the realized measure
- β_R : persistence of the return variance

The second governs the conditional expectation of realized variance:

$$\mu_{M,t} = \omega_M + \alpha_M RV_{t-1} + \beta_M \mu_{M,t-1} \quad (5.12)$$

- $\mu_{M,t}$: conditional expectation of RV_t (what you expect today's realized variance to be)
- α_M, β_M : analogous to GARCH parameters but for the realized measure

In Plain English

HEAVY runs two GARCH-like equations side by side. The first forecasts the variance of daily returns, and the second forecasts realized variance itself. Both use yesterday's RV as input instead of r^2 . Think of it as two parallel trackers: one for “what will the daily return variance be?” and one for “what will intraday volatility look like?”

Why This Matters

The second HEAVY equation (Equation 5.12) is essentially forecasting RV_t using an AR(1) in RV with GARCH-style persistence, making it a close relative of the HAR model's daily component. Comparing HEAVY's $\mu_{M,t}$ forecasts against your HAR forecasts at the one-day horizon reveals how much the multi-horizon structure of HAR adds beyond a simple autoregressive specification.

HEAVY Model Performance

Shephard and Sheppard (2010) find that the HEAVY model outperforms GARCH both in-sample and out-of-sample across a range of asset classes. Forecast gains are most pronounced at short horizons (one to five days). The model adjusts quickly to structural breaks in the volatility level because RV_{t-1} responds immediately to intraday price variation, while GARCH must wait for the slow accumulation of squared daily returns.

5.9 Comparison: GARCH vs. Realized GARCH vs. HEAVY

The main distinction across the three approaches is the data input: standard GARCH uses only daily returns, while Realized GARCH and HEAVY incorporate intraday information through RV_t (see table).

	GARCH(1,1)	Realized GARCH	HEAVY
Data input	Daily returns only	Daily returns + RV_t	Daily returns + RV_t
Volatility driver	r_{t-1}^2	$\log RV_t$	RV_{t-1}
Latent variance	Yes (σ_t^2)	Yes (h_t), linked to RV_t via measurement eq.	Two: $\sigma_{R,t}^2$ (returns) and $\mu_{M,t}$ (RV)
Leverage effect	Not in basic form; needs GJR or EGARCH	Built in via $\tau(z_t)$	Not in basic form; extensions exist
Structural breaks	Slow adjustment	Faster (uses RV_t)	Fastest (direct RV input)
Estimation	MLE on daily returns	Joint MLE	Equation-by-equation or joint MLE
Use when	No intraday data available	Intraday data available; want a single integrated framework	Intraday data available; want fast response to regime changes

GARCH Is Not a Straw Man

When intraday data is unavailable (many asset classes, historical periods, or emerging markets), GARCH remains the best option. Even with intraday data, GARCH serves as a useful baseline: any more complex model should demonstrably outperform it. [Andersen and Bollerslev \(1998\)](#) showed that GARCH forecasts appear poor only when evaluated against noisy proxies like r_t^2 ; evaluated against RV_t , they perform respectably.

5.10 Summary

- ARCH ([Engle, 1982](#)) models conditional variance as a function of past squared returns but requires many lags to capture persistence.
- GARCH(1,1) ([Bollerslev, 1986](#)) adds a single feedback term ($\beta\sigma_{t-1}^2$), capturing persistence with just three parameters (ω, α, β).
- The stationarity condition is $\alpha + \beta < 1$; the unconditional variance is $\omega/(1 - \alpha - \beta)$.
- Higher-order GARCH(p, q) rarely improves on GARCH(1,1) in practice ([Hansen and Lunde, 2005](#)).
- The leverage effect ([Black, 1976](#)): negative returns increase volatility more than positive returns of the same magnitude.
- GJR-GARCH ([Glosten et al., 1993](#)) adds an indicator term $\gamma \mathbf{1}_{\{r < 0\}} r_{t-1}^2$ to capture leverage simply.
- EGARCH ([Nelson, 1991](#)) models log-variance, guaranteeing positivity without parameter constraints and capturing both sign and size effects.
- FIGARCH ([Baillie et al., 1996](#)) introduces fractional integration ($d \in (0, 1)$) to match the slow, hyperbolic decay of volatility autocorrelations.
- Realized GARCH ([Hansen et al., 2012](#)) links RV_t to the latent conditional variance via a measurement equation, substantially outperforming standard GARCH.

- The HEAVY model (Shephard and Sheppard, 2010) builds a joint system for daily return variance and RV_t , adjusting quickly to structural breaks.
- GARCH remains the right baseline when intraday data is unavailable.
- The HAR model (Chapter 6) offers a simpler, non-parametric alternative for RV forecasting that often matches or beats Realized GARCH.

Key Results

Result	Source	Finding
ARCH model	Engle (1982)	Conditional variance depends on past squared returns; Nobel Prize-winning framework
GARCH(1,1)	Bollerslev (1986)	Single memory term replaces many ARCH lags; three parameters capture the bulk of variance dynamics
Leverage effect	Black (1976)	Negative returns increase volatility more than positive returns of the same magnitude
GJR-GARCH	Glosten et al. (1993)	Indicator-based asymmetry; effective coefficient on negative shocks is $\alpha + \gamma$
EGARCH	Nelson (1991)	Log-variance specification; no positivity constraints; separates sign and size effects
GARCH(1,1) benchmark	Hansen and Lunde (2005)	Among 330 models, no significant improvement from higher-order lags; leverage matters for equities
FIGARCH	Baillie et al. (1996)	Fractional integration parameter d captures hyperbolic decay of volatility autocorrelations
Realized GARCH	Hansen et al. (2012)	Incorporating RV_t via measurement equation substantially outperforms daily-return-only GARCH
HEAVY model	Shephard and Sheppard (2010)	Joint return/RV system; fast adjustment to regime changes; strongest gains at short horizons

Chapter 6

The HAR Model and Its Extensions

Why This Chapter

HAR is THE benchmark for volatility forecasting. Projects 1, 4, and 5 all use HAR variants as their primary baseline.

Chapter 5 forecast volatility using only daily returns. That approach works when intraday data is unavailable, but it uses a noisy proxy (r_t^2) for what actually happened during the day. Chapter 2 showed that realized variance $RV_t = \sum r_{t,i}^2$, computed from intraday returns, is a far more precise measure of daily volatility. The natural next question: can you forecast tomorrow's RV directly from past RV values, without the latent-variable machinery of GARCH?

The answer is yes, and the model that does it is remarkably simple: three OLS coefficients.

6.1 The Heterogeneous Market Hypothesis

Mean Reversion

A quantity *mean-reverts* if it tends to return to a long-run average after being pushed away. If today's value is unusually high, mean reversion says tomorrow's value is more likely to be lower (and vice versa). The speed of mean reversion determines how long departures persist.

Start with an observation about who trades in financial markets. Not everyone operates at the same frequency. Day traders react to what happened in the last few hours. Portfolio managers at mutual funds or pension funds rebalance weekly. Large institutional allocators (sovereign wealth funds, endowments) adjust positions monthly or quarterly.

Each type of participant pays attention to volatility at their own characteristic horizon. A day trader cares about yesterday's realized volatility. A weekly rebalancer looks at the average volatility over the past week. A monthly allocator responds to the volatility level over the past month.

Müller et al. (1993) formalized this as the *Heterogeneous Market Hypothesis* (HMH): the market is a superposition of participants operating at different time scales, and these participants interact. When monthly allocators adjust positions, it creates volatility that daily traders react to, and vice versa.

Three Clocks Running Simultaneously

Imagine three clocks ticking at different speeds. Clock 1 (daily) ticks every day and captures fast-moving volatility reactions: earnings surprises, Fed announcements, flash crashes. Clock 2 (weekly) ticks every week and captures medium-term dynamics: multi-day volatility clusters, the weekly options expiration cycle. Clock 3 (monthly) ticks every month and captures the slow-moving background volatility level: business-cycle shifts, prolonged risk-on/risk-off regimes. Observed volatility is a mixture of all three clocks. The HAR model puts one predictor per clock.

Figure 6.1 illustrates how the three participant types feed into observed market volatility.

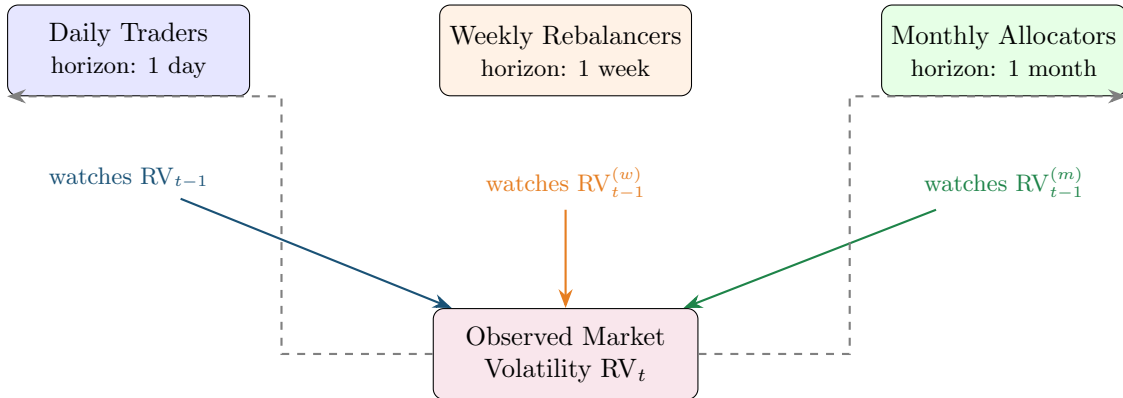


Figure 6.1: The Heterogeneous Market Hypothesis. Three types of market participants operate at different horizons, each responding to volatility averaged over their characteristic time scale. Their collective actions produce the observed realized volatility. Dashed arrows indicate feedback: today's volatility feeds back into each participant's future decisions.

Why This Matters for Forecasting

If the market were homogeneous (all participants on the same clock), a simple AR(1) model would suffice. The heterogeneity means that volatility dynamics involve multiple time scales simultaneously.

6.2 The HAR Model

OLS Regression

Ordinary Least Squares (OLS) fits a linear equation $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \varepsilon$ by choosing $\beta_0, \dots, \beta_k$ to minimize the sum of squared residuals $\sum \varepsilon_t^2$. The resulting coefficients are the best linear unbiased estimators under standard assumptions (Gauss–Markov theorem). If you can set up a forecasting problem as a linear regression, OLS gives you the answer.

The insight from Section 6.1 suggests three predictors: yesterday's RV, the average RV over the past week, and the average RV over the past month. [Corsi \(2009\)](#) turned this directly into a regression. The key move is defining the weekly and monthly averages.

Weekly and Monthly Realized Variance

The weekly average realized variance on day t is:

$$RV_t^{(w)} = \frac{1}{5} \sum_{i=0}^4 RV_{t-i} \quad (6.1)$$

The monthly average realized variance on day t is:

$$RV_t^{(m)} = \frac{1}{22} \sum_{i=0}^{21} RV_{t-i} \quad (6.2)$$

- $RV_t^{(w)}$: the arithmetic mean of the five most recent daily realized variances (including today)
- $RV_t^{(m)}$: the arithmetic mean of the 22 most recent daily realized variances (including today)
- 5 trading days = 1 week; 22 trading days $\approx$ 1 month
- These are backward-looking averages, not forecasts

The model regresses tomorrow's realized variance on yesterday's daily, weekly, and monthly values.

The HAR-RV Model (,)

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1} \quad (6.3)$$

- RV_{t+1} : tomorrow's realized variance (the target you are forecasting)
- β_0 : intercept, related to the long-run average level of RV
- β_d : coefficient on yesterday's daily RV; captures the short-term reaction
- RV_t : yesterday's realized variance
- β_w : coefficient on the weekly average; captures medium-term persistence
- $RV_t^{(w)}$: average of the five most recent daily RV values (Equation 6.1)
- β_m : coefficient on the monthly average; captures long-term persistence
- $RV_t^{(m)}$: average of the 22 most recent daily RV values (Equation 6.2)
- ε_{t+1} : forecast error

Estimation: OLS. No maximum likelihood, no iterative optimization, no latent variables.

HAR Mimics Long Memory with Three Coefficients

Volatility autocorrelations decay slowly (Chapter 5, Section 5.6). A pure AR model would need many lags to capture this. HAR achieves the same effect with a trick: the weekly average $RV^{(w)}$ implicitly includes lags 1 through 5, and the monthly average $RV^{(m)}$ implicitly includes lags 1 through 22. Three coefficients, but through the averages, information from 22 past days enters the forecast. The result is autocorrelation decay that closely approximates a long-memory process, with none of the estimation complexity of FIGARCH (Corsi, 2009).

Log or Level?

Many papers estimate HAR on $\ln(RV)$ rather than RV in levels. As noted in Chapter 2 (Section 2.5), $\ln(RV_t)$ is approximately Gaussian, which makes OLS residuals better be-

haved and guarantees positive forecasts (since $e^x > 0$). The log specification generally forecasts better. Throughout this chapter, all equations are written in levels for clarity; in practice, use logs unless you have a specific reason not to.

6.2.1 Typical Coefficient Values

What do the coefficients look like on real data? Corsi (2009) calibrates the HAR simulation with $\beta_d = 0.36$, $\beta_w = 0.28$, $\beta_m = 0.28$, values chosen to produce realistic dynamics; the empirical estimates on S&P 500 futures (1990–2007) are $\beta_d \approx 0.37$, $\beta_w \approx 0.34$, $\beta_m \approx 0.22$. All three components contribute meaningfully. The monthly term's coefficient is similar to the daily term's, confirming that long-horizon persistence matters.

The in-sample R^2 is typically 0.40–0.60 for daily-horizon forecasts on equity index RV. This is high for a financial time series, and it comes from just three predictors.

Figure 6.2 summarizes the HAR cascade: how the three components at different time scales feed into a single forecast.

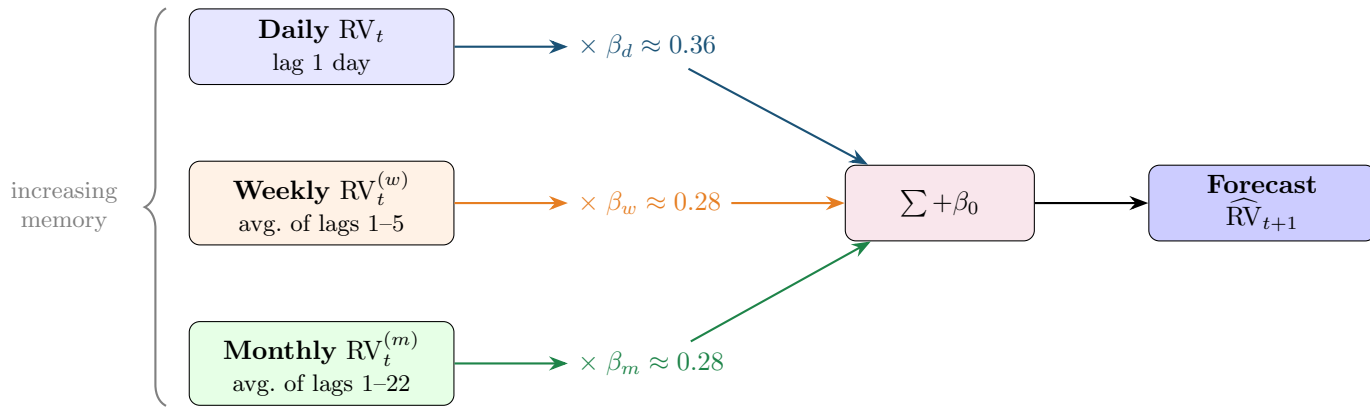


Figure 6.2: The HAR cascade. Three components at daily, weekly, and monthly time scales are weighted by OLS coefficients and summed to produce the forecast. Through the weekly and monthly averages, information from 22 past days enters the model with only three coefficients, approximating long-memory decay. Typical coefficient values are from Corsi (2009) on S&P 500 data.

6.3 HAR-J and HAR-CJ: Adding Jumps

The basic HAR uses total realized variance RV_t as the predictor. But as discussed in Chapter 2 (Equation 2.5), RV captures both the continuous component of price variation and any jumps that occurred during the day. A natural question: do jumps and continuous variation have different forecasting power?

Bipower Variation

Bipower variation (BPV) is an estimator of integrated variance that is robust to jumps. It is defined as:

$$BPV_t = \frac{\pi}{2} \sum_{i=2}^n |r_{t,i}| |r_{t,i-1}| \quad (6.4)$$

- $|r_{t,i}|$: absolute value of the i -th intraday return
- The product of consecutive absolute returns averages out jump contributions because

it is unlikely that two consecutive intervals both contain a jump

- $\pi/2$: a scaling constant that ensures BPV_t converges to integrated variance IV_t (not quadratic variation) as sampling frequency increases

The key property: $\text{BPV}_t \rightarrow \text{IV}_t$ even when jumps are present, whereas $\text{RV}_t \rightarrow \text{IV}_t + \sum(\text{jumps})^2$. The difference $\text{RV}_t - \text{BPV}_t$ therefore isolates the jump component. BPV was introduced by [Barndorff-Nielsen and Shephard \(2004\)](#) and is covered in detail in Chapter 4.

6.3.1 HAR-J

[Andersen et al. \(2007\)](#) proposed the HAR-J model, which adds a jump component as an additional predictor:

$$\text{RV}_{t+1} = \beta_0 + \beta_d \text{RV}_t + \beta_w \text{RV}_t^{(w)} + \beta_m \text{RV}_t^{(m)} + \beta_J J_t + \varepsilon_{t+1} \quad (6.5)$$

- $J_t = \max(\text{RV}_t - \text{BPV}_t, 0)$: the estimated jump component on day t ; the max ensures non-negativity, since measurement error can make $\text{RV}_t - \text{BPV}_t$ slightly negative even on jumpless days
- β_J : coefficient on jumps; typically estimated as negative and small, meaning jumps have a weak or transient effect on future volatility
- All other terms are the same as in Equation (6.3)

[Andersen et al. \(2007\)](#): Jumps Matter Less Than You Expect

[Andersen et al. \(2007\)](#) find that the jump component J_t is statistically significant but economically small in forecasting next-day RV. Most of the predictive power comes from the continuous component. Jumps are largely transient: a jump on day t does not meaningfully raise volatility on day $t + 1$.

6.3.2 HAR-CJ

[Corsi et al. \(2010\)](#) take the separation further with the HAR-CJ model, which replaces total RV with the continuous and jump components at all three horizons:

$$\text{RV}_{t+1} = \beta_0 + \beta_d^C C_t + \beta_w^C C_t^{(w)} + \beta_m^C C_t^{(m)} + \beta_d^J J_t + \beta_w^J J_t^{(w)} + \beta_m^J J_t^{(m)} + \varepsilon_{t+1} \quad (6.6)$$

- $C_t = \text{BPV}_t$: the continuous component of day t 's variation
- $J_t = \max(\text{RV}_t - \text{BPV}_t, 0)$: the jump component of day t
- $C_t^{(w)}, C_t^{(m)}$: weekly and monthly averages of the continuous component, constructed like $\text{RV}_t^{(w)}$ and $\text{RV}_t^{(m)}$
- $J_t^{(w)}, J_t^{(m)}$: weekly and monthly averages of the jump component

The HAR-CJ model has six slope coefficients instead of three. The empirical finding is consistent with HAR-J: continuous coefficients (β^C) are large and significant; jump coefficients (β^J) are small and often insignificant at weekly and monthly horizons ([Corsi et al., 2010](#)).

In Plain English

HAR-CJ goes further than HAR-J by building separate daily, weekly, and monthly averages for the continuous part and the jump part. Instead of asking “how volatile was the market?” at each horizon, it asks two questions: “how volatile was the smooth trading activity?” and “how big were the jumps?”

Figure 6.3 shows how HAR-CJ decomposes realized variance before forecasting, compared to the standard HAR approach.

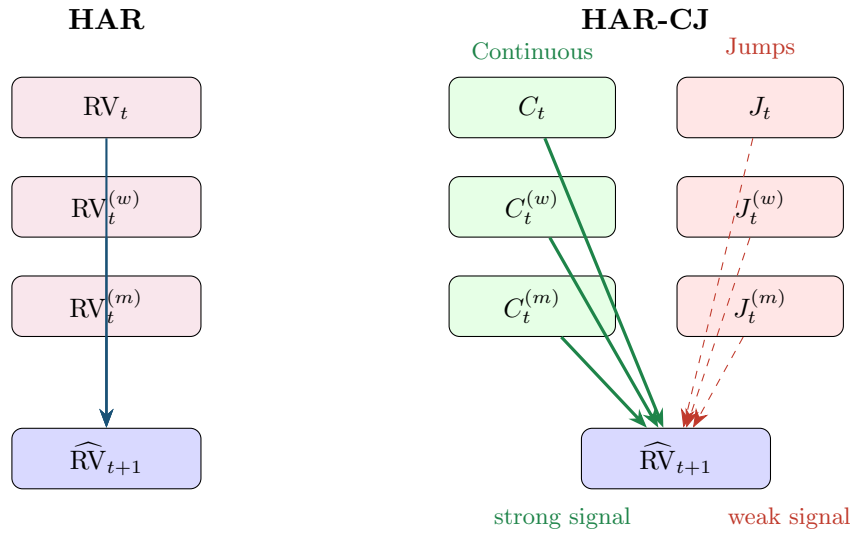


Figure 6.3: HAR vs. HAR-CJ. HAR feeds total RV at three horizons into the forecast (left). HAR-CJ first decomposes RV into continuous and jump components at each horizon (right). Thick green arrows indicate strong forecasting power from continuous components; thin dashed red arrows indicate the weak contribution of jumps.

6.4 SHAR: Good Volatility, Bad Volatility

HAR and its jump extensions treat all returns symmetrically: a large up move and a large down move both increase RV by the same amount (both r^2 terms are positive). But Chapter 5 documented the leverage effect (Section 5.3): negative returns increase volatility more than positive returns of the same magnitude. Can you bring this asymmetry into the HAR framework?

Realized Semivariance

Realized semivariance decomposes realized variance into contributions from positive and negative intraday returns:

$$RS_t^+ = \sum_{i=1}^n r_{t,i}^2 \mathbf{1}_{\{r_{t,i} > 0\}}, \quad RS_t^- = \sum_{i=1}^n r_{t,i}^2 \mathbf{1}_{\{r_{t,i} < 0\}} \quad (6.7)$$

- RS_t^+ : positive semivariance, the sum of squared positive intraday returns
- RS_t^- : negative semivariance, the sum of squared negative intraday returns
- $\mathbf{1}_{\{r_{t,i} > 0\}}$: indicator function, equal to 1 if $r_{t,i} > 0$
- $RS_t^+ + RS_t^- = RV_t$: the two halves sum to total realized variance

Patton and Sheppard (2015a): Bad Volatility Is More Persistent

Decomposing realized variance into positive semivariance RS^+ and negative semivariance RS^- substantially improves forecasts. Bad volatility (RS^- , from downward price moves) is significantly more persistent than good volatility (RS^+ , from upward moves).

The SHAR (Semivariance HAR) model replaces the daily RV term with RS^+ and RS^- :

$$RV_{t+1} = \beta_0 + \beta_d^+ RS_t^+ + \beta_d^- RS_t^- + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1} \quad (6.8)$$

- RS_t^+ : yesterday's positive semivariance ("good" volatility)
- RS_t^- : yesterday's negative semivariance ("bad" volatility)
- $\beta_d^- > \beta_d^+$ is the typical finding: negative semivariance has a larger coefficient, meaning bad volatility predicts more future volatility
- The weekly and monthly components remain as total RV averages

Why Downward Moves Are More Informative

Consider two days with identical $RV = 2 \times 10^{-4}$. On Day A, most of the variation came from upward moves ($RS^+ = 1.6$, $RS^- = 0.4$): a rally day with choppy upward movement. On Day B, most came from downward moves ($RS^+ = 0.4$, $RS^- = 1.6$): a selloff with persistent downward pressure.

Day B predicts higher future volatility. Why? Selloffs trigger margin calls, portfolio insurance rebalancing, stop-loss orders, and panic selling, all of which generate additional volatility. Rallies of the same magnitude do not create the same cascading effects. This asymmetry is the leverage effect operating at the intraday level.

Patton and Sheppard (2015a) show that SHAR improves forecast accuracy relative to HAR across multiple asset classes, with the largest gains on equity indices where the leverage effect is strongest.

6.5 HARQ: Handling Measurement Error

All the models so far treat every day's RV_t as equally reliable. But it is not. Chapter 2 showed that RV is an *estimate* of integrated variance, subject to estimation error that depends on how volatile volatility was during the day.

Realized Quarticity

The precision of realized variance as an estimator of integrated variance depends on the *integrated quarticity*, the integral of σ_s^4 over the day (see Equation 2.6 in Chapter 2). Realized quarticity (RQ_t) is a consistent estimator of this quantity:

$$RQ_t = \frac{n}{3} \sum_{i=1}^n r_{t,i}^4 \quad (6.9)$$

- $r_{t,i}^4$: the fourth power of the i -th intraday return
- $n/3$: a scaling constant that ensures consistency
- High RQ_t means that volatility itself was volatile during the day, making RV_t a noisy estimate of IV_t
- Low RQ_t means volatility was stable, making RV_t precise

The idea behind HARQ is simple: when yesterday's RV is noisy (high RQ), the model should rely less on it. When it is precise (low RQ), the model should rely more on it.

Bollerslev et al. (2016a): HARQ Adjusts for Measurement Error

Bollerslev et al. (2016a) allow the daily coefficient in the HAR model to vary with realized quarticity RQ_t . On noisy days (high RQ), the daily RV coefficient shrinks toward zero, down-weighting the unreliable estimate. On precise days (low RQ), the coefficient is

large, fully exploiting the signal. This simple modification produces statistically significant forecast improvements.

$$RV_{t+1} = \beta_0 + (\beta_d + \beta_{dQ}\sqrt{RQ_t}) RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1} \quad (6.10)$$

- β_d : baseline daily coefficient (the coefficient when $RQ_t = 0$, i.e., the hypothetical noise-free case)
- β_{dQ} : the adjustment coefficient; typically estimated as negative
- $\sqrt{RQ_t}$: square root of realized quarticity, a measure of how noisy RV_t is as an estimate
- $\beta_d + \beta_{dQ}\sqrt{RQ_t}$: the effective daily coefficient, which varies from day to day
- When RQ_t is large (noisy day), $\beta_{dQ}\sqrt{RQ_t}$ is a large negative number, shrinking the effective coefficient toward zero
- When RQ_t is small (precise day), the adjustment is negligible and the coefficient stays near β_d

Why This Matters

HARQ is THE paper your project builds on. The idea that measurement quality should modulate how much weight the model gives to each input is exactly the kind of structure an ML model can generalize. Your project explores whether tree ensembles or neural networks can learn this adaptive weighting from data, potentially extending it to weekly and monthly coefficients or to other noise proxies beyond RQ .

The following diagram shows how the effective daily coefficient changes with measurement noise.

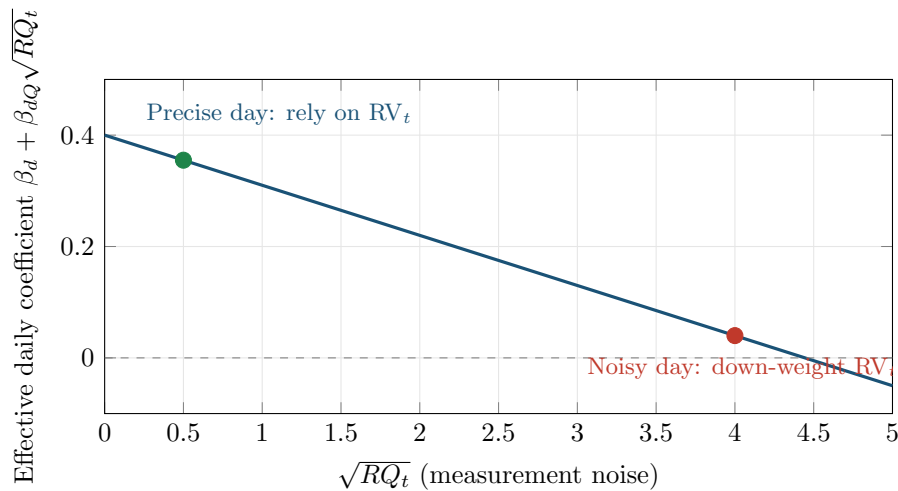


Figure 6.4: HARQ's adaptive coefficient. On days when realized quarticity is low (left), the daily coefficient is large and the model trusts RV_t . On days when quarticity is high (right), the coefficient shrinks toward zero, shifting weight to the more stable weekly and monthly averages.

HARQ Is the Strongest Univariate RV Forecast

Among models that use only past RV and its measurement-error statistics, HARQ is the strongest in the literature (Bollerslev et al., 2016a). It is the bar that any ML model must clear. If your tree ensemble or neural network cannot beat HARQ on out-of-sample QLIKE (Chapter 17), you have not learned useful nonlinear structure; you have likely overfit.

HARQ Needs Realized Quaticity

HARQ requires computing RQ_t from intraday returns. If your data source provides only daily RV without the underlying intraday returns, you cannot construct RQ_t and HARQ is not available. In that case, fall back to SHAR or plain HAR as your benchmark.

6.6 HAR-X and Beyond: Adding Exogenous Predictors

All HAR variants so far use only past RV (and its decompositions) as predictors. But other variables contain information about future volatility: the VIX, lagged signed returns, macroeconomic indicators, trading volume, options-implied information. The HAR-X framework adds these as extra regressors.

HAR-X

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \sum_{j=1}^p \gamma_j X_{j,t} + \varepsilon_{t+1} \quad (6.11)$$

- $X_{j,t}$: the j -th exogenous predictor observed on day t
- γ_j : coefficient on the j -th predictor
- p : number of exogenous predictors
- Common choices for $X_{j,t}$: VIX level, VIX innovations (ΔVIX), lagged daily return r_t (signed, to capture leverage), overnight return, trading volume, implied–realized volatility spread

Why This Matters

HAR-X is the linear version of what your ML model will do. Any feature you feed to a random forest or neural network, you should first test in a HAR-X regression to see whether it helps linearly. If it does, the ML question becomes: does the feature interact nonlinearly with other inputs? If it does not help even linearly, adding it to an ML model is unlikely to help and may cause overfitting.

[Bollerslev et al. \(2018a\)](#) take this to a large scale in their “Risk Everywhere” paper, constructing a large cross-section of HAR-X type models with many macro and market predictors.

When p is large, you face a classic overfitting problem: many predictors, limited data. [Audrino and Knaus \(2016\)](#) propose the “Lassoing the HAR” approach, applying the Lasso (L1 regularization) to the HAR-X framework:

$$\min_{\beta, \gamma} \sum_t \left(RV_{t+1} - \beta_0 - \beta_d RV_t - \beta_w RV_t^{(w)} - \beta_m RV_t^{(m)} - \sum_j \gamma_j X_{j,t} \right)^2 + \lambda \sum_j |\gamma_j| \quad (6.12)$$

- λ : regularization penalty; larger λ shrinks more coefficients to exactly zero
- The penalty applies to the exogenous coefficients γ_j , not the core HAR terms
- The Lasso selects which exogenous variables matter while keeping the core HAR structure intact

Lasso Regularization

The Lasso (Least Absolute Shrinkage and Selection Operator) adds an L_1 penalty ($\lambda \sum |\gamma_j|$) to the OLS objective. This has two effects: (1) it shrinks coefficients toward zero, reducing overfitting; (2) it sets some coefficients to exactly zero, performing automatic variable selection. The tuning parameter λ controls the strength of the penalty and is chosen by cross-validation.

HAR-X Is the Bridge to ML

HAR-X with many predictors is effectively a linear ML model. Adding Lasso regularization makes it a penalized linear model with variable selection. Chapters 11–14 ask the next question: can *nonlinear* models (tree ensembles, neural networks) improve on this, or does the extra flexibility just lead to overfitting? The answer depends critically on the feature set and forecast horizon.

6.7 Ridge and Elastic-Net HAR: Shrinking Collinear Components

The Collinearity Problem

Two predictors are **collinear** when one is nearly a linear function of the other, so the data carry little independent information about their separate coefficients. The HAR design matrix is a textbook case: the weekly average $RV_t^{(w)}$ contains RV_t as one of its five terms, and the monthly average $RV_t^{(m)}$ contains both as part of its 22 terms (Definition 6.2). When columns of the design matrix $\mathbf{X}$ are nearly linearly dependent, the Gram matrix $\mathbf{X}^\top \mathbf{X}$ (a summary grid of how the predictors overlap) has a near-zero **eigenvalue** (a single number that here signals a direction in the data carrying almost no information), OLS coefficients swing wildly from sample to sample, and adding even one exogenous predictor can flip a sign.

The Lasso of Section 6.6 answers “which extra predictors matter?” by zeroing out the weak ones. But zeroing is exactly the wrong move for the *core* HAR terms. When RV_t , $RV_t^{(w)}$, and $RV_t^{(m)}$ are strongly correlated (pairwise correlations routinely exceed 0.8 on equity index data), the Lasso keeps one and drops the rest essentially at random, a disaster for a model whose entire premise (Section 6.1) is that all three horizons carry signal.

The principled fix is **ridge regression**, which adds an L_2 penalty instead of an L_1 penalty. Ridge shrinks correlated coefficients *together* rather than forcing a choice between them, so the daily, weekly, and monthly components all survive in damped form. **Elastic net** then blends the two penalties to get sparsity over the exogenous predictors *and* ridge-style grouping over the collinear HAR core.

6.7.1 The Ridge Objective

Start from the penalized least-squares problem. Write the HAR-X regression of Equation (6.11) in matrix form, stacking the daily, weekly, monthly, and any exogenous columns into a single design matrix $\mathbf{X}$ with coefficient vector β . Ridge regression (Hoerl and Kennard, 1970) adds a squared-coefficient penalty to the residual sum of squares:

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \underbrace{\|\mathbf{y} - \mathbf{X}\beta\|_2^2}_{\text{fit to past RV}} + \underbrace{\lambda \|\beta\|_2^2}_{\text{shrinkage penalty}} \quad (6.13)$$

Reading it left to right: $\arg \min_{\beta}$ means “find the particular coefficient list β that makes everything to its right as small as possible,” and the double bars $\|\mathbf{v}\|_2^2$ are shorthand for “take every number in the list $\mathbf{v}$, square it, and add them all up.” So the first term $\|\mathbf{y} - \mathbf{X}\beta\|_2^2$ is just the total squared forecast error written compactly, and the second term penalizes the total squared coefficient size.

- $\mathbf{y} \in \mathbb{R}^n$: the vector of realized-variance targets RV_{t+1} across the n training days
- $\mathbf{X} \in \mathbb{R}^{n \times p}$: the design matrix whose columns are RV_t , $\text{RV}_t^{(w)}$, $\text{RV}_t^{(m)}$, and any exogenous predictors $X_{j,t}$
- $\beta \in \mathbb{R}^p$: the HAR coefficient vector $(\beta_d, \beta_w, \beta_m, \gamma_1, \dots)$
- $\|\mathbf{y} - \mathbf{X}\beta\|_2^2 = \sum_t (\text{RV}_{t+1} - \mathbf{x}_t^\top \beta)^2$: the ordinary HAR sum of squared forecast errors; here $\mathbf{x}_t$ is the one row of $\mathbf{X}$ for day t (that day’s feature values), and $\mathbf{x}_t^\top \beta$ multiplies each feature by its coefficient and adds them up, i.e. it is the model’s forecast for day t
- $\|\beta\|_2^2 = \sum_j \beta_j^2$: the squared L_2 norm of the coefficients; penalizes large coefficients
- $\lambda \geq 0$: the regularization strength. At $\lambda = 0$ this is plain OLS-HAR; as $\lambda \rightarrow \infty$ all coefficients shrink toward zero

In Plain English

Ridge-HAR fits the same daily/weekly/monthly regression as ordinary HAR, but it charges a fee for every unit of coefficient size. Because the fee is on the *squared* size, it punishes one big swollen coefficient far more than two moderate ones that add up to the same forecast. So when the daily and weekly terms are nearly interchangeable, ridge prefers to split the weight between them rather than dump it all on one. The result is a HAR whose three components stay sensible even when the data cannot cleanly tell them apart.

6.7.2 Closed-Form Solution

Unlike the Lasso, ridge has a closed form. Expand the objective in Equation (6.13) and set the gradient to zero:

$$\begin{aligned} \mathcal{L}(\beta) &= (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) + \lambda \beta^\top \beta \\ \nabla_{\beta} \mathcal{L} &= -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\beta) + 2\lambda \beta = \mathbf{0} \end{aligned} \quad (6.14)$$

This step is algebra you can take on faith; the next equation is what you actually use. In words: the triangle symbol ∇ (“nabla”) means the slope of the error surface, and at the bottom of a valley the slope is flat (zero), so setting $\nabla_{\beta} \mathcal{L} = \mathbf{0}$ locates the best β . The superscript $\top$ is the transpose (flipping a grid so its rows become columns), which is the bookkeeping that lets these grids multiply together.

Solving for β gives the ridge estimator:

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y} \quad (6.15)$$

- $\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{p \times p}$: the Gram matrix; its eigenvalues (each a single number measuring how strongly the data vary along one direction) measure how much the data spread along each direction. For collinear HAR columns, the smallest eigenvalue is near zero, meaning the data barely move that way; this is exactly the collinear case

- $\mathbf{I}$: the identity matrix, the grid-version of the number 1 (ones down the diagonal, zeros elsewhere), so $\lambda\mathbf{I}$ just means “add the amount λ along the diagonal”; this adds λ to every eigenvalue of $\mathbf{X}^\top\mathbf{X}$, lifting the smallest one away from zero and guaranteeing the inverse exists
- the superscript -1 on $(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})$ is the matrix inverse, the grid-version of dividing, which undoes a multiplication
- $\mathbf{X}^\top\mathbf{y} \in \mathbb{R}^p$: the cross-covariance between each lagged RV feature and tomorrow’s RV
- Setting $\lambda = 0$ recovers the OLS-HAR estimator $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ behind Equation (6.3)

In Plain English

OLS-HAR has to invert $\mathbf{X}^\top\mathbf{X}$, and when the daily/weekly/monthly columns nearly overlap that matrix is almost singular, so the inversion magnifies tiny data wiggles into huge coefficient swings. Ridge slides λ down the diagonal first, which is like propping up the near-flat direction so the inversion no longer blows up. The bigger λ is, the more the wobbly directions get held in place, and the more stable the three HAR coefficients become from one sample to the next.

6.7.3 Why Ridge Damps the Noisy Collinear Directions

To see *which* part of the HAR signal ridge shrinks, look through the singular value decomposition (SVD). Write $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ with singular values $d_1 \geq d_2 \geq \dots \geq d_p$. You can think of a **singular value** d_j as a single number measuring how much the data stretch along one underlying direction: a big d_j means lots of variation (trustworthy), a tiny d_j means the data barely move that way (the fragile, collinear direction). The decomposition $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ is just the standard recipe that extracts those directions ($\mathbf{U}$ and $\mathbf{V}$ hold the direction grids, $\mathbf{D}$ holds the d_j); you do not need its details to follow the argument. The squared singular values d_j^2 are exactly the eigenvalues of the Gram matrix $\mathbf{X}^\top\mathbf{X}$ from the previous subsection, which is the link the worked example below relies on. The directions with large d_j are well-determined (the data move a lot along them); the directions with small d_j are the noisy, near-collinear combinations of RV_t , $\text{RV}_t^{(w)}$, $\text{RV}_t^{(m)}$ where the data barely move. Along each SVD direction, ridge replaces the OLS amplification factor d_j^{-1} with a damped factor:

$$\underbrace{\frac{1}{d_j}}_{\text{OLS amplifies}} \longrightarrow \underbrace{\frac{d_j}{d_j^2 + \lambda}}_{\text{ridge damps}} \quad (6.16)$$

- d_j : the j -th singular value of $\mathbf{X}$; large for well-determined directions, small for collinear ones
- d_j^{-1} : the OLS weight along direction j , which explodes when d_j is tiny
- $d_j/(d_j^2 + \lambda)$: the ridge weight, which $\rightarrow d_j^{-1}$ when $d_j^2 \gg \lambda$ (well-determined directions untouched) and $\rightarrow 0$ when $d_j^2 \ll \lambda$ (noisy directions damped out)

In Plain English

OLS-HAR puts the most faith in exactly the directions where it has the least evidence: when the three RV averages nearly coincide, the leftover direction that distinguishes them has almost no variation, so OLS multiplies it by a giant number and reads pure noise as signal. Ridge does the opposite: it leaves the strong, well-measured directions alone and quietly turns down the volume on the weak, collinear ones. That is the entire point of ridge-HAR: rather than shrinking the three coefficients equally, it selectively mutes the

unreliable combination of them.

Why This Matters

The noisy collinear direction in a HAR regression is precisely where measurement error in RV does the most damage to coefficients (the same theme HARQ tackles in Section 6.5). Ridge’s selective damping is a model-free cousin of HARQ’s *RQ*-weighting: instead of down-weighting noisy *days*, ridge down-weights the noisy *direction* in coefficient space. Both push forecasts away from over-trusting fragile information.

6.7.4 Bias, Variance, and the Hoerl–Kennard Guarantee

Why accept any shrinkage at all, given that ridge biases the coefficients? First, three terms in plain English: **bias** means the forecast is systematically off-target on average; **variance** means it jumps around wildly from one data sample to the next; **mean squared error** (MSE) bundles both into a single score, bias-squared plus variance. As λ grows from zero, the squared bias of $\hat{\beta}_{\text{ridge}}$ rises smoothly from zero (the OLS estimator is unbiased) while the estimator variance falls. The mean squared error is their sum, and it traces a U-shape in λ (Figure 6.5): a wrong-but-stable estimate beats a right-on-average-but-wild one. The foundational result makes this precise.

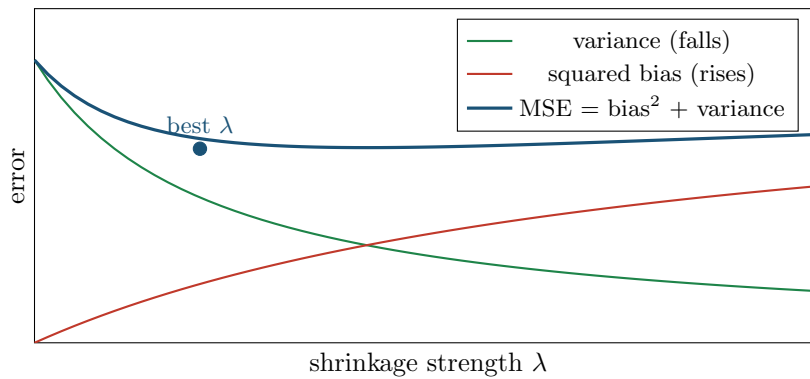


Figure 6.5: The bias–variance trade-off behind ridge. As shrinkage λ rises from zero, the estimator variance (green) falls while the squared bias (red) rises from zero. Their sum, the mean squared error (blue), traces a U-shape; the minimum sits at a strictly positive λ , so some shrinkage always beats OLS.

Hoerl–Kennard Theorem (Hoerl and Kennard, 1970)

This is the formal statement; if the notation is unfamiliar, skip to the plain-English box below. A quick legend: ε is the random noise in volatility (the error term); $\mathbb{E}[\dots]$ means “on average”; σ^2 is the size of that noise; and d_j^{-2} means 1 divided by the singular value squared, which is what explodes when a direction is collinear. For any linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \varepsilon$ with $\mathbb{E}[\varepsilon] = \mathbf{0}$ (the noise is zero on average) and $\text{Cov}(\varepsilon) = \sigma^2 \mathbf{I}$ (every day’s noise has the same size σ^2 and is uncorrelated across days), there *always* exists a $\lambda > 0$ for which the ridge estimator has strictly lower mean squared error than OLS:

$$\text{MSE}(\hat{\boldsymbol{\beta}}_{\text{ridge}}) < \text{MSE}(\hat{\boldsymbol{\beta}}_{\text{OLS}}).$$

The intuition: OLS variance is proportional to $\sigma^2 \sum_j d_j^{-2}$ (recall the eigenvalues of $\mathbf{X}^\top \mathbf{X}$ are $\lambda_j = d_j^2$), which becomes enormous when any singular value d_j is small, exactly

the collinear HAR case. A small dose of bias buys a large variance reduction. The MSE-optimal λ depends on the unknown β and σ^2 , so in practice it is chosen by cross-validation.

Worked Example: Ridge Splits Weight Across RV_d and RV_w

Take the two most collinear HAR predictors: yesterday's daily realized variance RV_t (call it RV_d) and the weekly average $RV_t^{(w)}$ (call it RV_w), which literally contains RV_d as one of its five terms. On equity index data their correlation is around $\rho = 0.95$. Suppose, after standardizing, the two SVD directions of this 2-column block have squared singular values (eigenvalues of $\mathbf{X}^\top \mathbf{X}$) of roughly $d_1^2 \approx 195$ (the common "level" direction both features share) and $d_2^2 \approx 5$ (the fragile direction that tells them apart).

OLS-HAR. Since $d_2^2 \approx 5$, the singular value itself is $d_2 \approx \sqrt{5} \approx 2.24$, so OLS amplifies the fragile direction by $1/d_2 \approx 0.45$ in coefficient space, magnifying noise. A single volatile week can hand all the credit to RV_d and a negative coefficient to RV_w (illustrative values, not computed here): $\hat{\beta}_d \approx 3.2$, $\hat{\beta}_w \approx -3.4$, a nonsensical split that flips sign in the next sample.

Ridge-HAR at $\lambda = 10$. The shrinkage factor along each direction is $d_j^2/(d_j^2 + \lambda)$:

$$\text{level direction: } \frac{195}{195 + 10} = 0.95, \quad \text{fragile direction: } \frac{5}{5 + 10} = 0.33.$$

Ridge barely touches the well-measured level the two features agree on (factor 0.95) but damps the fragile distinguishing direction by two-thirds (factor 0.33). Projecting back to the coefficients, that damped fragile direction is what turns the wild OLS split into a stable one (again illustrative): $\hat{\beta}_d \approx 1.3$, $\hat{\beta}_w \approx 1.2$, both positive, both contributing.

	OLS-HAR	Ridge-HAR ($\lambda = 10$)
$\hat{\beta}_d$	+3.2	+1.3
$\hat{\beta}_w$	-3.4	+1.2
Stable across resamples?	no (flips sign)	yes

Takeaway. Ridge does not shrink β_d and β_w equally. It leaves the shared volatility level (what both terms agree on) almost untouched and damps only the noisy direction where the data cannot reliably separate "yesterday" from "this week" (the Lasso would instead zero one of the two; see Table 6.1).

6.7.5 Ridge vs. Lasso on Correlated HAR Features

The two penalties behave very differently precisely where it matters for HAR: correlated predictors. Table 6.1 summarizes the contrast.

Do Not Lasso the Core HAR Terms

The Lasso is the right tool for selecting among the *exogenous* HAR-X predictors (Section 6.6), where most candidates are genuinely irrelevant. It is the wrong tool for the three core RV components. Because $RV_t^{(w)}$ and $RV_t^{(m)}$ are built from RV_t , they are mechanically collinear, and the Lasso will drop one whole time scale to satisfy its sparsity preference, choosing essentially at random which one survives across resamples. That destroys the multi-horizon structure that is the entire reason HAR works (Section 6.1). Keep the core HAR terms unpenalized or ridge-penalized; reserve L_1 for the exogenous block.

Table 6.1: Ridge (L_2) versus Lasso (L_1) on a HAR regression. The correlated-features row is the decisive one: the daily, weekly, and monthly RV terms are strongly collinear by construction, so the Lasso's tendency to keep one and drop the rest is a liability, while ridge's group shrinkage preserves all three time scales.

Property	Ridge (L_2)	Lasso (L_1)
Penalty	$\lambda \sum_j \beta_j^2$	$\lambda \sum_j \beta_j $
Sparsity	No (all $\beta_j \neq 0$)	Yes (many $\beta_j = 0$)
Closed form	Yes (Eq. 6.15)	No (coordinate descent)
Correlated HAR terms	Shrinks RV_d, RV_w, RV_m together	Keeps one, zeros the others (arbitrary)
Constraint geometry	Ball (circle)	Diamond
Variable selection	None	Built in
Best when	Many correlated, all useful	Few strong, mostly irrelevant

6.7.6 Why L_1 Produces Sparsity but L_2 Does Not

The difference in Table 6.1 comes down to geometry. Both penalties can be written as a constraint: minimize the HAR sum of squared errors subject to a budget t on the total coefficient size, $\|\beta\|_2^2 \leq t$ (ridge) or $\|\beta\|_1 \leq t$ (Lasso). The OLS-HAR loss has elliptical *contours*, like the rings on a contour map: each ring marks a level of equal forecast error, and the centre is the best (lowest-error) point. These rings are elongated and tilted because the daily and weekly terms are correlated. The regularized solution is where the smallest such ellipse first touches the constraint region. Figure 6.6 shows why the shape of that region decides whether a coefficient lands on exactly zero.

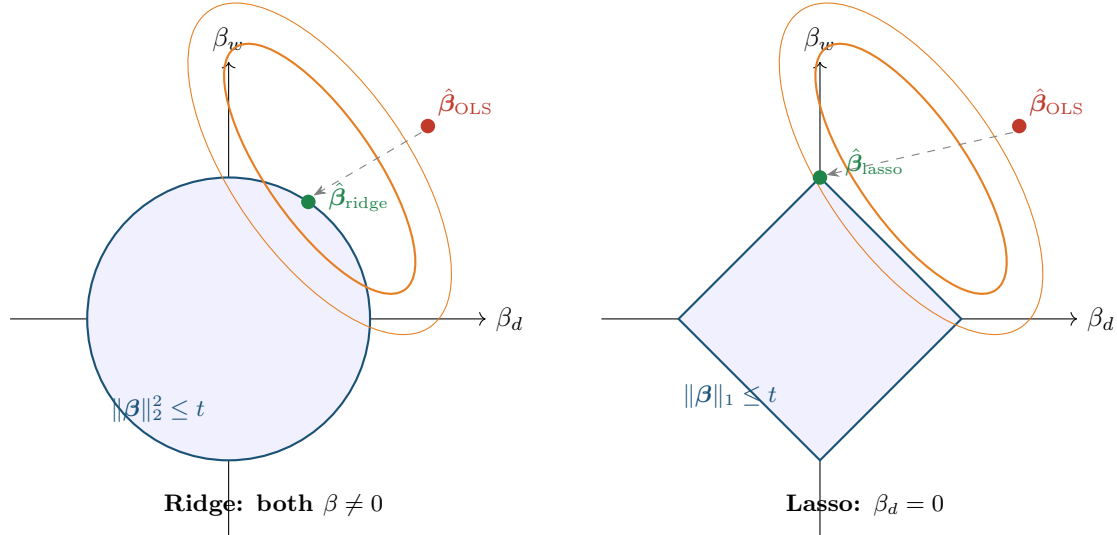


Figure 6.6: Why L_1 produces sparsity and L_2 does not. The tilted orange ellipses are contours of the HAR sum-of-squares loss; the tilt reflects the strong correlation between RV_d and RV_w . The regularized estimate is where the loss contour first meets the constraint region. **Left (ridge):** the L_2 ball is round, so the tangency point is generically off-axis, leaving both β_d and β_w nonzero, shrunk together. **Right (Lasso):** the L_1 diamond has sharp corners on the axes, and an elongated ellipse almost always touches a corner first, setting $\beta_d = 0$. For collinear HAR components, that corner means dropping a whole time scale.

In Plain English

Picture inflating the blue constraint region until it just kisses the orange loss contours. A round ball gets kissed on its smooth side, so both coefficients keep some value. A diamond gets kissed on a pointy corner, and the corners sit exactly on the axes where one coefficient is zero. The kink in the L_1 penalty is what makes those corners, and the corners are what produce sparsity. Ridge has no kink, no corners, and so no exact zeros.

6.7.7 Elastic Net: Sparsity with Grouping

Ridge keeps every coefficient; the Lasso zeros many but mishandles correlated groups. The **elastic net** (Zou and Hastie, 2005) combines both penalties so you can select among exogenous HAR-X predictors *and* keep correlated terms grouped:

$$\hat{\beta}_{\text{enet}} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \left[\underbrace{\alpha \|\beta\|_1}_{\text{selection}} + \underbrace{\frac{1-\alpha}{2} \|\beta\|_2^2}_{\text{grouping}} \right] \quad (6.17)$$

- $\alpha \in [0, 1]$: the mixing parameter. At $\alpha = 1$ elastic net is pure Lasso; at $\alpha = 0$ it is pure ridge
- $\lambda \geq 0$: the overall regularization strength, as in Equation (6.13)
- $\alpha \|\beta\|_1$: the L_1 component that zeros out weak exogenous predictors. The L_1 norm $\|\beta\|_1 = \sum_j |\beta_j|$ adds up the absolute sizes of the coefficients (ignoring sign), whereas the L_2 norm $\|\beta\|_2^2$ adds up their squares; the squaring is what makes L_2 punish one big coefficient far more
- $\frac{1-\alpha}{2} \|\beta\|_2^2$: the L_2 component that shrinks correlated coefficients toward each other; the factor $\frac{1}{2}$ is a conventional scaling that keeps the algebra tidy and does not change the behaviour

The Grouping Effect

Zou and Hastie (2005) prove the **grouping effect**: if two features are highly correlated, elastic net assigns them nearly equal coefficients, rather than the Lasso's all-or-nothing pick. The strength of the grouping is controlled by the L_2 weight $(1 - \alpha)$: more L_2 means tighter grouping. For HAR this is exactly right: the daily, weekly, and monthly RV terms form a natural correlated group that elastic net keeps intact while pruning useless exogenous regressors; this is the principled middle ground between ridge-HAR (no selection) and Lasso-HAR (group-blind selection).

6.7.8 Tuning and Diagnostics

Ridge, Lasso, and elastic net all leave you with a tuning parameter λ (and, for elastic net, the mix α). How you choose them matters most: the wrong validation scheme silently leaks future volatility into the past.

Tune λ with Purged CV, Not i.i.d. K-Fold

Standard K -fold CV assumes independent observations; realized variance violates this badly: it is highly persistent (Section 6.1), and the HAR features $\text{RV}_t^{(w)}, \text{RV}_t^{(m)}$ are overlapping moving averages, so adjacent days share information by construction. If day t is in the training fold and day $t + 1$ in the validation fold, the model has effectively already seen the answer, and the chosen λ will be far too small (too little shrinkage), inflating in-sample fit and collapsing out-of-sample. Tune λ (and α) with **purged K -fold CV with**

embargo (Chapter 17, Section 17.7), and size the embargo to cover the 22-day reach of the monthly average. Generalized cross-validation (GCV) gives ridge a fast leave-one-out shortcut, but it inherits the i.i.d. assumption, so treat it only as a rough first pass and confirm with purged CV before trusting any reported improvement.

Practical λ Selection and Path Diagnostics

Four habits make regularized HAR robust:

1. **Log-spaced λ grid.** Search λ on a geometric grid. Log-spaced means the candidate λ values grow by a constant multiple ($\dots, 0.01, 0.1, 1, 10, \dots$) rather than by constant steps, because λ matters on a multiplicative scale; `np.logspace(-4, 4, 50)` is just “50 values from 0.0001 to 10000” in code. Refine around the minimum, and extend the grid if the optimum hits a boundary.
2. **The one-standard-error rule.** Instead of the λ that minimizes purged-CV error, pick the *largest* λ whose CV error is within one standard error of the minimum. This yields a more heavily shrunk, simpler HAR that forecasts about as well and is far less prone to overfitting the validation noise.
3. **The regularization path as a feature-importance diagnostic.** Plotting each coefficient as λ increases from 0 to ∞ shows which HAR features persist longest (most informative) and which shrink to zero first (noise). For Lasso/elastic-net-HAR the order in which exogenous predictors drop out is a free importance ranking.
4. **Effective degrees of freedom.** Report model complexity as a continuous quantity, $df(\lambda) = \sum_{j=1}^p d_j^2 / (d_j^2 + \lambda)$ (the big $\sum$ means “add the following up across all p directions”), which runs from p at $\lambda = 0$ (full OLS-HAR) down to 0 as $\lambda \rightarrow \infty$. Notice this is just the sum of the per-direction shrinkage factors $d_j^2 / (d_j^2 + \lambda)$ from Equation (6.16): each well-measured direction contributes nearly a whole knob, each damped collinear direction only a fraction, and adding them up gives the effective knob-count. It is the regularized analog of “number of parameters” and lets you compare HAR variants on equal footing.

Implementation Checklist for Ridge/Elastic-Net HAR

1. **Standardize before penalizing.** The L_2 and L_1 penalties are scale-dependent: a feature in raw RV units and one in percentage points are penalized unequally. Standardize every column to zero mean and unit variance *inside each training fold* (never on the full sample) before fitting.
2. **Leave the intercept unpenalized.** You do not want to shrink the long-run mean of RV toward zero. Fit β_0 without penalty; standard libraries do this by default.
3. **Use the CV wrappers, but supply your own splits.** `RidgeCV`, `LassoCV`, and `ElasticNetCV` automate the λ (and α) search, but their default i.i.d. folds are wrong for RV. Pass a purged-CV splitter (Chapter 17) so the tuning respects the time ordering.
4. **Fit in logs.** As in Section 6.2, estimating on $\ln RV$ keeps residuals near-Gaussian and forecasts positive; the penalty and standardization arguments are unchanged.

6.8 Why HAR Is Hard to Beat

If HAR is so simple (three coefficients, OLS), why is it the benchmark rather than a stepping stone that ML quickly surpasses? Three reasons.

Reason 1: Volatility is highly persistent and approximately linear. The autocorrelation

function of RV_t decays slowly and smoothly. A model that captures the right decay rate with the right mixture of time scales gets most of the forecastable variation. HAR does exactly this. The residual nonlinearity is real but small relative to the linear component.

Reason 2: The signal-to-noise ratio is low. Even though volatility is more predictable than returns, the day-to-day fluctuations in RV are large. A model that tries to fit these fluctuations more precisely (e.g., by adding interaction terms, polynomial features, or deep layers) tends to fit noise rather than signal, especially with the relatively short samples typical in finance (10–20 years of daily data is 2,500–5,000 observations).

Reason 3: Daily horizon, RV-only features. When you restrict the feature set to past RV values and forecast one day ahead, there is little nonlinear structure for ML to exploit. The gains from ML come from two sources that HAR cannot access:

1. **Richer features:** options-implied volatility, cross-asset information, order flow, macroeconomic data, text/news sentiment (Chapter 10).
2. **Longer horizons:** at weekly or monthly forecast horizons, nonlinear regime dynamics and mean-reversion patterns become more pronounced, giving tree-based models and neural networks more to work with.

The HAR Litmus Test

If your ML model does not beat HAR on the same features (past RV only) and the same forecast horizon (one day), you have not learned nonlinear structure; you have overfit noise. Always report HAR alongside your ML results. If you beat HAR, the follow-up question is: does the improvement come from richer features (good) or from fitting noise in RV (bad)? The answer determines whether the ML model has real economic value.

Why This Matters

This table is your strategic roadmap. If you use RV-only features at the daily horizon, HAR will probably tie or beat your ML model, and that is expected. Your project should aim for the bottom-right cell: richer features (e.g., VIX, signed returns, cross-asset vol) at weekly or monthly horizons, where both the feature advantage and the nonlinear structure advantage compound. Design your experiments accordingly.

The following table summarizes where ML has and has not consistently beaten HAR in the literature.

Setting	HAR vs. ML	Why
Daily horizon, RV-only features	HAR wins or ties	Little nonlinear signal
Daily horizon, rich features	ML often wins	Extra features carry new info
Weekly/monthly horizon, RV-only	ML sometimes wins	Nonlinear regime effects
Weekly/monthly horizon, rich features	ML usually wins	Both advantages compound

Publication Bias

Papers that fail to beat HAR are less likely to be published. The literature therefore overstates the frequency with which ML improves on HAR. Be skeptical of reported improvements below 5–10% in out-of-sample QLIKE, and always check whether the improvement is statistically significant via a Diebold–Mariano test (Chapter 17).

Summary

- The **Heterogeneous Market Hypothesis** (Müller et al., 1993) posits that markets are driven by participants operating at daily, weekly, and monthly horizons, whose interactions produce the observed volatility dynamics.
- The **HAR model** (Corsi, 2009) translates this directly into a regression: $RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}$, estimated by OLS.
- HAR **mimics long memory** with only three coefficients by embedding lags 1–22 through the weekly and monthly averages.
- **Typical** R^2 for daily-horizon HAR on equity index RV: 0.40–0.60.
- **HAR-J** (Andersen et al., 2007) adds a jump component $J_t = \max(RV_t - BPV_t, 0)$. Jumps are statistically significant but economically small predictors.
- **HAR-CJ** (Corsi et al., 2010) separates continuous and jump components at all three horizons. Continuous variation dominates the forecast; jumps are largely transient.
- **SHAR** (Patton and Sheppard, 2015a) decomposes daily RV into positive (RS^+) and negative (RS^-) semivariance. Bad volatility (RS^-) is significantly more persistent, reflecting the leverage effect at the intraday level.
- **HARQ** (Bollerslev et al., 2016a) allows the daily coefficient to vary with realized quarticity RQ_t , down-weighting RV_t on noisy days. It is the strongest univariate RV forecast in the literature.
- **HAR-X** adds exogenous predictors (VIX, returns, macro). With many predictors, Lasso regularization (Audrino and Knaus, 2016) prevents overfitting while preserving the core HAR structure.
- HAR is **extremely competitive** at the daily horizon with RV-only features. ML gains come from richer feature sets and longer horizons.
- The **HAR litmus test**: if your ML model cannot beat HAR on the same features and horizon, it has overfit noise, not learned nonlinear structure.
- All HAR variants are estimated by **OLS** (or penalized OLS), require no iterative optimization, and produce interpretable coefficients. This simplicity is a feature, not a limitation.
- Throughout this guide, HAR (or HARQ) is the **mandatory baseline** for every forecasting model.

Key Results

Result	Source	Finding
Heterogeneous Market Hypothesis	Müller et al. (1993)	Markets are driven by participants at multiple horizons; volatility dynamics are a superposition of time scales
HAR model	Corsi (2009)	Three OLS coefficients (daily, weekly, monthly RV) approximate long-memory dynamics; $R^2 \approx 0.40\text{--}0.60$ for daily equity index forecasts
HAR-J (jumps)	Andersen et al. (2007)	Adding jump component improves fit marginally; jumps are largely transient and do not persist into future volatility
HAR-CJ	Corsi et al. (2010)	Full continuous/jump decomposition at all horizons; continuous variation dominates forecasting power
SHAR (semivariance)	Patton and Sheppard (2015a)	Negative semivariance is more persistent than positive; capturing the leverage effect improves forecasts
HARQ (measurement error)	Bollerslev et al. (2016a)	Varying the daily coefficient with $\sqrt{RQ_t}$ down-weights noisy estimates; strongest univariate RV forecast
Lassoing the HAR	Audrino and Knaus (2016)	Lasso selects relevant exogenous predictors while preserving core HAR structure; prevents overfitting with many regressors
Risk Everywhere	Bollerslev et al. (2018a)	Large-scale HAR-X with macro/market predictors; confirms predictability from multiple exogenous sources

Chapter 7

Rough Volatility

Why This Chapter

Rough volatility provides an alternative explanation for the slow decay in volatility autocorrelation that HAR (Chapter 6) captures with three components and FIGARCH (Chapter 5) captures with fractional integration. The RFSV model is a parsimonious forecaster competitive with HAR and LSTM. The Cont–Das counterargument warns against taking roughness as ground truth. Project 4 (rough vol vs. deep learning) directly tests RFSV against universal LSTM.

Chapter 6 showed that HAR’s three-component structure (daily, weekly, monthly) approximates the slow power-law decay of volatility autocorrelations. Chapter 5 showed that FIGARCH (Section 5.6) captures the same slow decay with a fractional differencing parameter d . Both approaches are empirically motivated heuristics: they match the shape of the autocorrelation function, but neither explains *why* the decay is so slow.

This chapter presents a different starting point. Instead of asking “how should I model the autocorrelation structure?” it asks: “what kind of stochastic process would *generate* the autocorrelation structure we observe?” The answer, proposed by Gatheral et al. (2018), is that the log of realized volatility behaves like a *fractional Brownian motion* with a very low Hurst exponent, around $H \approx 0.1$. This makes the volatility path much rougher (more jagged) than standard Brownian motion ($H = 0.5$).

7.1 Why Path Shape Matters: A Hedging Motivation

Consider a practical puzzle that motivates the chapter.

An options trader buys a 30-day ATM straddle and delta-hedges daily. Over the month, realized vol comes in at exactly 20%. But the trader’s P&L depends on *when* the moves happened, not just their aggregate size.

Two Paths, Same RV, Different P&L

Path A: the stock drifts quietly for 25 days then has 5 days of extreme moves. Path B: moves are spread evenly across all 30 days. Both paths have identical 30-day $RV = 20\%$. But the trader’s cumulative gamma P&L (Section 9.6) differs because:

- Gamma varies with moneyness: it is highest when the stock is near the strike.
- On Path A, most of the vol occurs after the stock has moved away from the strike

(gamma is low), so the trader captures less.

- On Path B, moves happen while gamma is still high, so the trader captures more.

The conclusion: forecasting *average* realized vol is necessary but not sufficient. The **path texture**, how vol is distributed across time and price levels, also determines economic outcomes.

This is exactly what rough volatility ($H \ll 0.5$) captures: rough paths have more fine-grained variation at short time scales, meaning the vol arrives in frequent small bursts rather than rare large moves. For a delta-hedger, rough paths produce more predictable P&L (the vol arrives continuously rather than in lumps). [Rosenbaum and Zhang \(2022\)](#) showed that both the universal LSTM and the parametric rough-vol model converge on this same characterization of the volatility process: one from data, one from theory.

7.2 What Is Roughness?

The central concept is the *Hurst exponent*, a single number that describes how jagged or smooth a random path looks.

Brownian Motion (Standard)

A standard Brownian motion W_t is the canonical “random walk in continuous time” (continuous paths, Gaussian independent increments of variance h , nowhere differentiable). Its key property here is *self-similarity* with scaling factor $H = 1/2$: rescaling time by a factor c rescales the path by $c^{1/2}$, so a zoomed-in portion looks statistically identical to the original.

Fractional Brownian Motion (fBM)

Fractional Brownian motion B_t^H , introduced by [Mandelbrot and Van Ness \(1968\)](#), generalizes standard Brownian motion by allowing the self-similarity exponent H to take any value in $(0, 1)$. Its key properties:

- $B_0^H = 0$, and increments are Gaussian with mean zero.
- $\text{Var}(B_{t+h}^H - B_t^H) = h^{2H}$.
- The path is self-similar with exponent H : rescaling time by c rescales the path by c^H .
- When $H = 1/2$, fBM reduces to standard Brownian motion.
- When $H \neq 1/2$, increments are *not* independent. They are negatively correlated if $H < 1/2$ and positively correlated if $H > 1/2$.

The parameter H is called the *Hurst exponent* (after the hydrologist Harold Edwin Hurst, who studied long-range dependence in Nile river levels).

The Hurst exponent controls two things simultaneously: the roughness of the path and the correlation structure of increments.

Hurst Exponent and Path Regularity

For a fractional Brownian motion B_t^H with Hurst exponent $H \in (0, 1)$:

$$\text{Var}(B_{t+h}^H - B_t^H) = h^{2H} \quad (7.1)$$

- H : the Hurst exponent, controlling both path roughness and increment correlation
- h : the time lag

- h^{2H} : the variance of the increment over a window of length h

Figure 7.1 shows sample paths at four values of H .

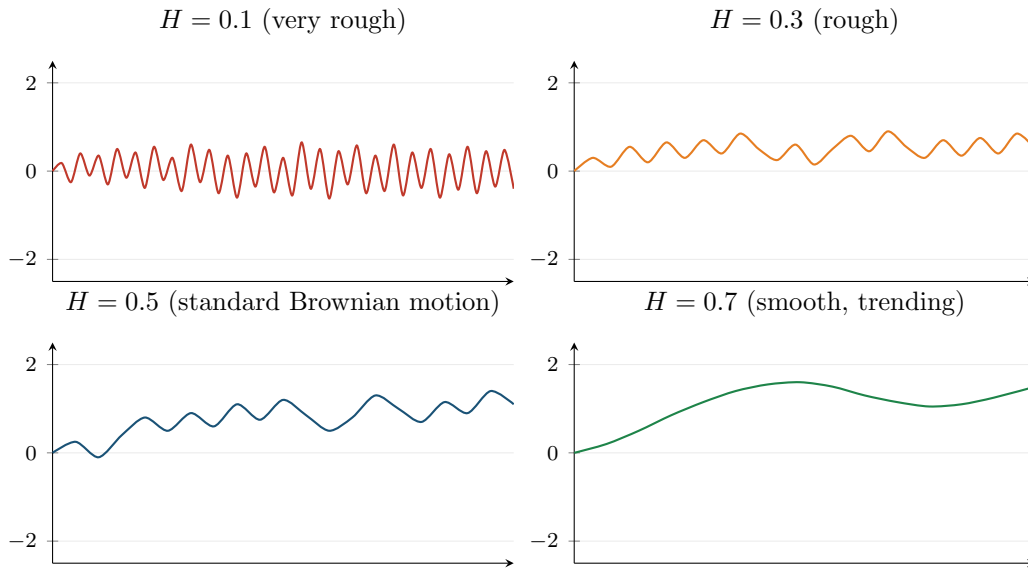


Figure 7.1: Sample paths of fractional Brownian motion at four values of the Hurst exponent H . Top left ($H = 0.1$, red): extremely rough, rapidly oscillating. Top right ($H = 0.3$, orange): rough but with visible short-range structure. Bottom left ($H = 0.5$, blue): standard Brownian motion, the familiar random walk. Bottom right ($H = 0.7$, green): smooth, trending trajectory. Empirically, log RV behaves like the top-left panel.

Figure 7.2 provides a compact reference for how different H values map to path behavior and real-world examples.

7.3 Volatility Is Rough

The central empirical claim of the rough volatility literature can now be stated precisely.

Gatheral et al. (2018) studied the time series of $\log RV_t$ (5-minute realized variance, as defined in Chapter 2) across a range of equity indices and bond futures, including the DAX, Bund, S&P 500, and NASDAQ, as well as roughly twenty other indices from the Oxford-Man realized library. For each asset, they estimated the Hurst exponent H of the log RV series.

Gatheral et al. (2018): Volatility Is Rough

Across equity indices and bond futures, the Hurst exponent of $\log RV_t$ is consistently around $H \approx 0.1$ (ranging from roughly 0.06 to 0.2 depending on the asset), far below the $H = 0.5$ of standard Brownian motion. This means that log-volatility paths are much rougher (more jagged, more rapidly oscillating) than any standard diffusion model would predict.

To put $H = 0.1$ in context: standard stochastic volatility models (Heston, SABR) assume that the volatility process is driven by a standard Brownian motion, which has $H = 0.5$. An H of 0.1 means the volatility path is far more jagged than a standard diffusion: it reverses direction much more frequently, creating the rapidly oscillating pattern visible in the top-left panel of Figure 7.1.

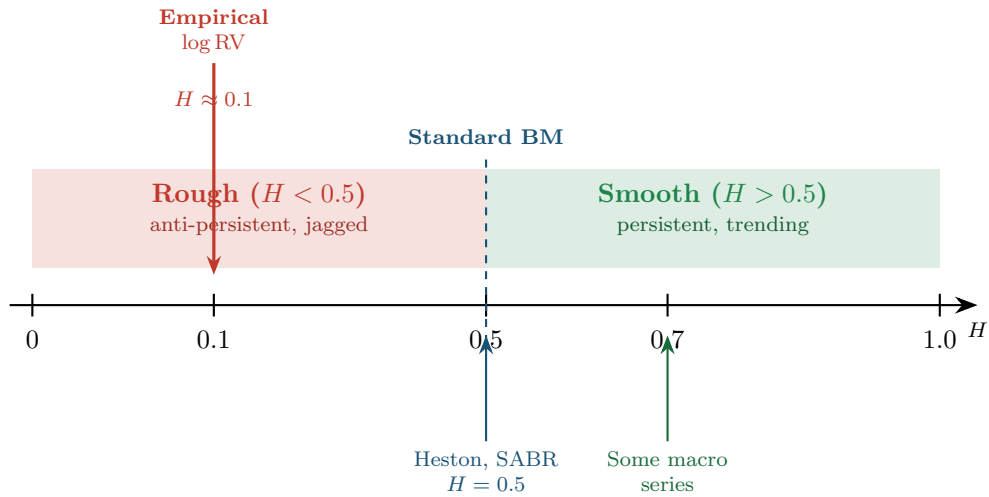


Figure 7.2: The Hurst exponent number line. Empirical log-volatility sits at $H \approx 0.1$, deep in the rough regime. Standard stochastic volatility models (Heston, SABR) assume $H = 0.5$. The gap between 0.1 and 0.5 is the motivation for the entire rough volatility program.

7.3.1 How to Estimate H : The Variogram Method

The estimation approach in Gatheral et al. (2018) uses the scaling relationship in Equation (7.1). If X_t is an fBM with exponent H , then the variance of its increments scales as h^{2H} . In log-log coordinates, this becomes a straight line.

Variogram Estimator of H

For a time series $X_1, X_2, \dots, X_T$, define the empirical q -th moment of increments at lag h :

$$m(q, h) = \frac{1}{T-h} \sum_{t=1}^{T-h} |X_{t+h} - X_t|^q \quad (7.2)$$

- X_t : the time series (here, $\log RV_t$)
- h : the lag (number of days)
- q : the moment order (typically $q = 2$ for the variance, or $q = 1$ for the mean absolute increment)
- T : the number of observations

If X_t behaves like fBM with exponent H , then $m(q, h) \propto h^{qH}$. Taking logs: $\log m(q, h) = qH \cdot \log h + \text{const}$. Regressing $\log m(q, h)$ on $\log h$ across multiple lags gives a slope of qH , from which H is recovered.

In Plain English

The variogram asks how fast the change in log-volatility grows as you widen the time window. For a standard random walk, doubling the window roughly doubles the variance of the change. For rough volatility, doubling the window barely increases the variance at all, because the anti-persistent path keeps reversing direction and staying near its starting point.

Why This Matters

The variogram is your primary diagnostic tool for checking whether your realized volatility series exhibits the rough-vol property. Before fitting any model (HAR, RFSV, or LSTM), estimating H from the data tells you whether the long-memory structure that these models exploit is actually present in your specific asset. If $\hat{H}$ comes back near 0.1, you have confirmation that multi-scale models like HAR are appropriate; if it comes back near 0.5, simpler AR(1)-type models may suffice.

Worked Example:

Estimating H from a Variogram You have $T = 1,000$ daily observations of $\log RV_t$ for the S&P 500. Using $q = 2$, you compute the average squared increment at lags $h = 1, 2, 5, 10, 20$ days and pass to log-log coordinates:

Lag h (days)	$m(2, h)$	$\log_{10} h$	$\log_{10} m(2, h)$
1	0.050	0.000	-1.301
2	0.059	0.301	-1.229
5	0.074	0.699	-1.131
10	0.088	1.000	-1.056
20	0.104	1.301	-0.983

The OLS slope of $\log_{10} m(2, h)$ on $\log_{10} h$ is approximately 0.245. Since slope = $qH = 2H$, we recover $H = 0.245/2 = 0.12$. For comparison, a standard Brownian motion would give a slope of $2 \times 0.5 = 1.0$.

This is close to the $H \approx 0.1$ reported by [Gatheral et al. \(2018\)](#). The slow increase of $m(2, h)$ with lag (i.e., the low slope) reflects the anti-persistent, rapidly oscillating nature of log-volatility.

Figure 7.3 shows the variogram in log-log coordinates, illustrating how the slope distinguishes rough from smooth processes.

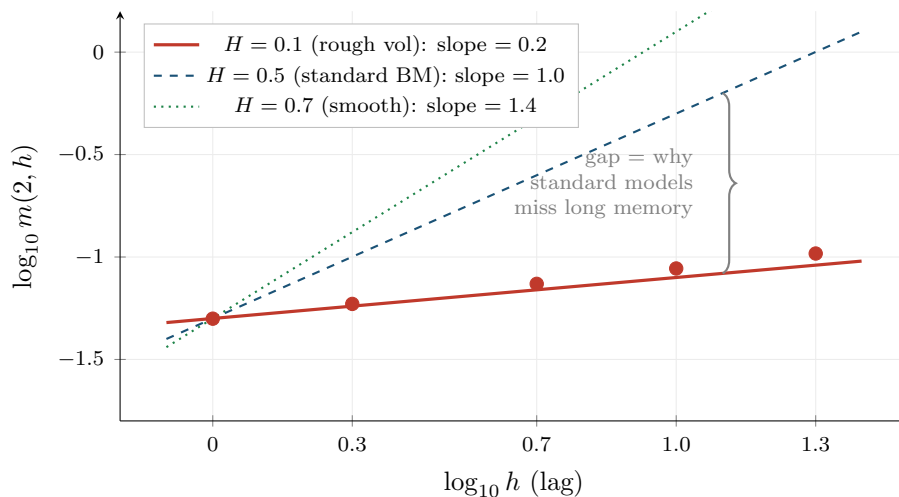


Figure 7.3: Variogram in log-log coordinates for three Hurst exponent values. The red line and data points ($H = 0.1$) show the flat slope characteristic of rough volatility: variance of increments grows very slowly with lag. The blue dashed line ($H = 0.5$) shows standard Brownian motion with a steep slope of 1.0. The gap between the two slopes quantifies how much standard diffusion models underestimate the persistence of volatility.

Bennedsen et al. (2022): Cross-Asset Universality

Bennedsen et al. (2022) extend the analysis of Gatheral et al. (2018) to a broad cross-section of asset classes (equities, equity indices, FX, fixed income, commodities) and confirm that $H \approx 0.1$ is universal. The Hurst exponent does not vary meaningfully across asset classes, geographies, or time periods. This universality suggests that $H \approx 0.1$ reflects a deep structural property of volatility dynamics, not a peculiarity of any particular market.

7.4 The RFSV Forecasting Formula

The observation that log RV behaves like fBM with $H \approx 0.1$ has a direct forecasting payoff. If you know the process is fBM, the optimal linear forecast is known in closed form. This gives a volatility forecasting model called RFSV (Rough Fractional Stochastic Volatility).

The intuition is simple. For fBM, the best forecast of the future value is a weighted average of all past observations, where the weights are determined by H . When H is small (rough regime), the weights decay slowly, meaning distant past observations still carry information. This slow weight decay is what produces the long-memory behavior that HAR approximates with three components.

RFSV Forecast

Model $\log \text{RV}_t$ as fractional Brownian motion with Hurst exponent H and a constant mean μ :

$$\log \text{RV}_t = \mu + \sigma_v B_t^H \quad (7.3)$$

- $\log \text{RV}_t$: the natural logarithm of day t 's realized variance
- μ : the unconditional mean of log-volatility
- σ_v : the volatility-of-volatility (a scaling constant)
- B_t^H : fractional Brownian motion with exponent H

The optimal one-step-ahead forecast, given observations through day T , is:

$$\widehat{\log \text{RV}}_{T+1} = \sum_{k=0}^{T-1} w_k \log \text{RV}_{T-k} \quad (7.4)$$

- w_k : the weight on the observation k days in the past
- The weights $\{w_k\}$ are determined by the covariance structure of fBM (i.e., by H)
- They sum to 1 and decay as a power law: $w_k \propto k^{H-3/2}$ for large k
- When $H = 0.1$, the decay is $k^{-1.4}$, which is slow enough that observations from weeks ago still contribute meaningfully

One Parameter Does the Work of Three

RFSV achieves HAR-level forecasting accuracy with essentially one free parameter (H), because the Hurst exponent $H \approx 0.1$ encodes the entire autocorrelation structure. HAR uses three coefficients ($\beta_d, \beta_w, \beta_m$) to approximate the same slow decay. RFSV derives the weights from first principles given H . The fact that both approaches perform similarly is evidence that the slow power-law decay really is the dominant structure in volatility dynamics.

Why RFSV Weights Decay Slowly

Consider two forecasting extremes. If $H = 0.5$ (standard BM), increments are independent and only the most recent observation matters; the weights drop to zero immediately. If $H = 0.01$ (extremely rough), the process is so anti-persistent that every past reversal carries information about the current level; the weights decay very slowly. At $H = 0.1$, you are close to the second extreme. The unnormalized weight on an observation from 22 days ago is $22^{-1.4} \approx 1.3\%$ of the weight on yesterday's observation, but because the weights are normalized to sum to one and lag 1 dominates, the lag-22 group (lags 6–22) collectively receives roughly a quarter of the total weight. This is why HAR's monthly component ($RV^{(m)}$) carries a large coefficient in Chapter 6: it is picking up the long tail of the RFSV weight function.

Why This Matters

RFSV is both a benchmark and a diagnostic for your vol forecasting project. As a benchmark, it tells you what a single-parameter model can achieve: if your ML model (LSTM, transformer) cannot beat RFSV out of sample, the added complexity is not justified. As a diagnostic, the RFSV weight function shows you exactly which lags carry information. If your ML model's learned attention pattern or feature importances do not roughly match the $k^{-1.4}$ decay shape, either the model has found genuinely new structure or it is overfitting.

7.5 Rough Volatility for Pricing

This guide focuses on realized volatility estimation and forecasting, not on derivatives pricing. But the rough volatility framework has had its greatest quantitative-finance impact in the pricing domain, briefly summarized here.

The Volatility Smile Problem

Standard stochastic volatility models (e.g., Heston) cannot simultaneously fit the steep short-maturity smile in S&P 500 options and the VIX implied volatility ("VVIX") smile.

Bayer et al. (2016) introduced the *rough Bergomi* model, the first pricing model built on rough volatility. In the rough Bergomi model, the variance process is driven by fractional Brownian motion with $H \approx 0.07$, rather than the standard Brownian motion used in models like Heston.

Two key results from the pricing literature:

1. **Rough Bergomi** (Bayer et al., 2016): the short-maturity behavior of the implied volatility smile is controlled by the Hurst exponent H . With $H \approx 0.07$, the model generates steep short-dated smiles that match market data far better than Heston. Simulation is required for pricing (no closed-form solution), but the model is parsimonious.
2. **Quadratic rough Heston**: a tractable variant that jointly fits SPX option smiles and VIX option smiles, a combination that no standard model can achieve. The roughness parameter is again $H \approx 0.05$ – 0.1 .

7.6 Fact or Artefact?

The rough volatility paradigm rests on an empirical observation: $\log RV_t$ has $H \approx 0.1$. But RV_t is not the true spot volatility σ_t^2 . It is a noisy estimate of it, contaminated by the microstructure

noise discussed in Chapter 3 (Section 2.3). Could the apparent roughness be an artefact of this noise?

Cont and Das (2024) argue that the answer is yes, at least in part.

Cont and Das (2024): Roughness as a Noise Artefact

Observed roughness of realized volatility estimates is partly a microstructure-noise artefact. When i.i.d. noise contaminates the high-frequency returns used to compute RV, the resulting RV series appears rougher than the true spot volatility process. Even if the true σ_t^2 follows a standard semimartingale with $H = 0.5$, the estimated $\log RV_t$ can exhibit $H \approx 0.1$ due to the noise channel.

The mechanism is illustrated in Figure 7.4.

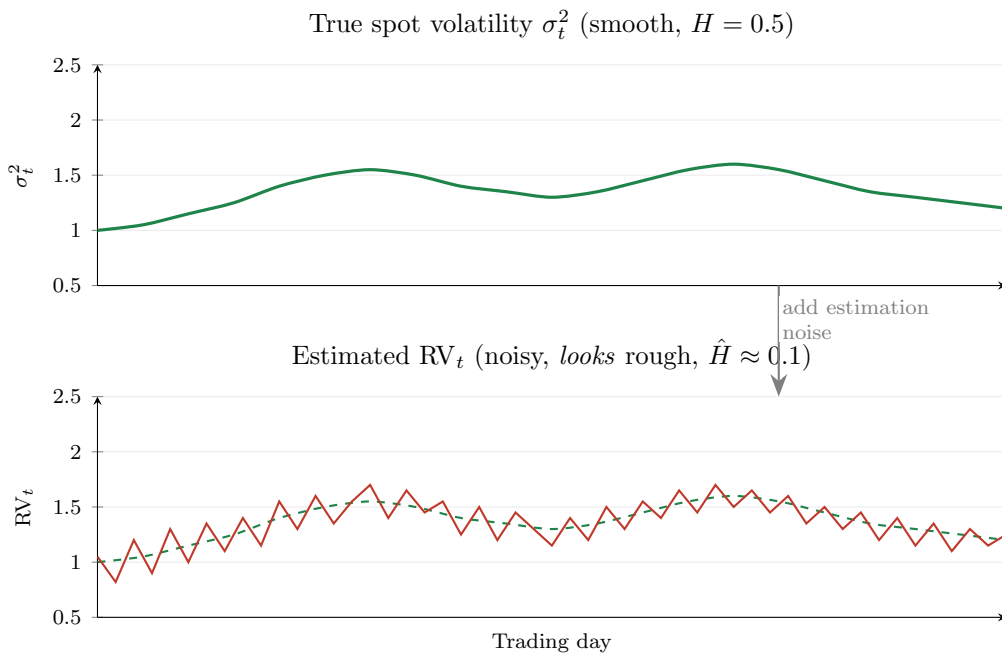


Figure 7.4: How microstructure noise creates apparent roughness. Top: the true spot volatility σ_t^2 is a smooth process ($H = 0.5$). Bottom: the estimated RV_t (red) adds i.i.d. estimation noise around the true level (dashed green). The noisy RV_t series looks rough (anti-persistent, rapidly oscillating) even though the underlying process is smooth. Applying the variogram estimator to the red series yields $\hat{H} \approx 0.1$, but this reflects noise contamination, not true roughness of σ_t^2 .

The Cont and Das (2024) argument proceeds in three steps:

1. **Noise adds estimation errors.** Each day's RV_t differs from the true IV_t by an estimation error η_t (Chapter 2, Equation 2.6). Under certain models (e.g., the Ornstein–Uhlenbeck stochastic volatility model), these log-estimation errors are approximately i.i.d. Gaussian across days.
2. **Independent errors create anti-persistence.** If you add i.i.d. noise to a smooth signal, the increments of the noisy series are negatively autocorrelated. An unusually high noise draw today will be followed (on average) by a smaller one tomorrow, creating artificial “reversals” in the series.
3. **Anti-persistence lowers $\hat{H}$.** The variogram estimator interprets these reversals as roughness and produces $\hat{H} \ll 0.5$.

Do Not Equate $\hat{H} = 0.1$ with True Roughness

The observed $H \approx 0.1$ from $\log RV_t$ data does not establish that the true spot volatility process σ_t^2 is rough. The noise channel documented by [Cont and Das \(2024\)](#) is a plausible alternative explanation. The truth may lie in between: some genuine roughness in σ_t^2 , amplified by estimation noise in RV_t . For forecasting, this distinction does not matter much (see Section 7.7). For model calibration and theoretical claims about the nature of volatility, it matters a great deal.

The Coin-Flip Analogy

Suppose you flip a fair coin 100 times and record the running total of heads minus tails. The resulting path is a standard random walk ($H = 0.5$). Now suppose someone records your totals but makes small random errors (e.g., occasionally miscounting by ± 1). If you estimate H from their error-contaminated records, you will get $\hat{H} < 0.5$, because the errors introduce artificial reversals. The path *looks* rougher than it is. Cont–Das argue that RV_t is the error-contaminated record and σ_t^2 is the true total.

7.7 The Universal LSTM Connection

[Rosenbaum and Zhang \(2022\)](#) train a single LSTM (Long Short-Term Memory neural network) on hundreds of individual stocks simultaneously. The trained network, which they call the “universal LSTM,” consistently outperforms asset-specific RFSV models. However, a combined RFSV + Quadratic Rough Heston parametric forecaster with fixed (non-asset-specific) parameters matches the LSTM’s performance, suggesting both capture the same underlying regularity.

This convergence is suggestive. Two very different approaches (a parametric fBM model with one parameter and a nonparametric neural network with thousands of parameters) arrive at the same forecast. The natural interpretation is that both are learning the same underlying statistical regularity in RV data.

Why This Matters

The universal LSTM result sets a clear success criterion for your project. To justify an LSTM or transformer, you need to demonstrate statistically significant improvement via Diebold-Mariano tests, not just lower in-sample QLIKE. The universality property also suggests that training on a broad cross-section of assets (rather than a single stock) may improve generalization.

[Rosenbaum and Zhang \(2022\)](#) also document a “universality” property: the LSTM trained on US equities transfers well to European equities it has never seen, without degradation in performance. This is consistent with the cross-asset universality of $H \approx 0.1$ documented by [Bennedsen et al. \(2022\)](#).

[Rosenbaum and Zhang \(2022\)](#): Universal LSTM

A single LSTM trained on hundreds of stocks consistently outperforms asset-specific RFSV models. A combined RFSV + QRH parametric forecaster with fixed parameters matches the LSTM, and both outperform HAR. The universal LSTM also transfers from US equities to European equities without degradation, suggesting that the learned kernel is market-independent.

The connection to the rest of this guide is direct. Chapter 13 develops LSTM and transformer-based forecasting models in full detail. When you reach that chapter, the key question will be: does your deep learning model learn something *beyond* the rough-vol kernel, or is it just a complicated way to recover the same $H \approx 0.1$ decay structure? If the latter, RFSV with one parameter is preferable on parsimony grounds. If the former, you need to demonstrate the improvement with a proper out-of-sample test (Chapter 17).

Summary

- The **Hurst exponent** H of a fractional Brownian motion controls path roughness. $H = 0.5$ is standard Brownian motion; $H < 0.5$ is rougher (more jagged); $H > 0.5$ is smoother.
- **Fractional Brownian motion** (fBM) generalizes standard BM by allowing correlated increments: negatively correlated when $H < 0.5$ (anti-persistent), positively correlated when $H > 0.5$ (persistent).
- Gatheral et al. (2018) showed that $\log RV_t$ behaves like fBM with $H \approx 0.1$ across equity indices and bond futures. This is the “volatility is rough” result.
- H can be estimated from data using the **variogram method**: regress $\log m(q, h)$ on $\log h$ and divide the slope by q .
- Bennedsen et al. (2022) confirmed the **cross-asset universality** of $H \approx 0.1$ across equities, FX, fixed income, and commodities.
- The **RFSV model** forecasts $\log RV_{t+1}$ as a weighted average of past $\log RV$ values, with weights derived from the fBM covariance structure. It achieves HAR-level accuracy with essentially one parameter (H).
- RFSV weights decay as $k^{H-3/2}$. At $H = 0.1$, the decay is slow ($k^{-1.4}$), explaining why HAR’s monthly component carries a large coefficient.
- **Rough Bergomi** (Bayer et al., 2016) and quadratic rough Heston use $H \approx 0.07$ – 0.1 for option pricing. They fit short-maturity smiles and VIX smiles better than standard models. This guide focuses on forecasting, not pricing.
- Cont and Das (2024) argue that observed roughness ($H \approx 0.1$) is **partly a microstructure-noise artefact**. Estimation noise in RV_t creates artificial anti-persistence that the variogram interprets as roughness.
- The practical implication: $H \approx 0.1$ from $\log RV_t$ is a useful forecasting input but does not prove that the true spot volatility σ_t^2 is rough.
- Rosenbaum and Zhang (2022) show that a **universal LSTM** trained on hundreds of stocks outperforms asset-specific RFSV models and transfers to unseen equities in other markets. A combined RFSV + QRH parametric forecaster matches the LSTM, suggesting both capture the same statistical kernel.
- Whether spot vol is truly rough or the roughness is a noise artefact, the **forecasting value is identical**. The distinction matters for pricing-model calibration and theory, not for forecasting accuracy.
- RFSV connects to FIGARCH (Chapter 5, Section 5.6) and HAR (Chapter 6) as three different parameterizations of the same long-memory phenomenon. It connects forward to deep learning (Chapter 13) as a benchmark that any LSTM must beat to justify its complexity.

Key Results

Result	Source	Finding
Volatility is rough	Gatheral et al. (2018)	$\log RV_t$ behaves like fBM with $H \approx 0.1$ across equity indices and bond futures; far below $H = 0.5$ of standard models
Cross-asset universality	Bennedsen et al. (2022)	$H \approx 0.1$ is universal across asset classes, geographies, and time periods
Rough Bergomi pricing	Bayer et al. (2016)	First rough-vol pricing model; $H \approx 0.07$ generates steep short-maturity implied volatility smiles
Roughness as artefact	Cont and Das (2024)	Microstructure noise in RV estimates creates apparent roughness; $\hat{H} \approx 0.1$ may not reflect true spot-vol dynamics
Universal LSTM	Rosenbaum and Zhang (2022)	Single LSTM trained on hundreds of stocks outperforms RFSV; RFSV + QRH parametric combination matches LSTM performance

Part III

The Volatility Surface and Options-Implied Information

Chapter 8

Options Basics and the Volatility Surface

Why This Chapter Matters

Options-implied information is a rich source of features for volatility forecasting (Chapter 10). The variance risk premium (Chapter 9) is defined as the gap between implied and realized volatility, both of which you need this chapter to understand. Project 5 (VRP ML trader) directly trades that gap. Even for non-options projects, understanding the volatility surface is necessary because VIX and IV-derived features appear in virtually every competitive feature set for realized volatility forecasting (Gu et al., 2020a).

This chapter introduces options from scratch, builds up to the Black-Scholes formula (just enough to define implied volatility), and then shows how the entire volatility surface encodes the market's fears and expectations. You will finish with the model-free VIX, which connects options prices directly to expected future variance.

8.1 What Is an Option?

Background for This Chapter

You need comfort with:

- Expected values and probability distributions (any intro probability course).
- The standard normal CDF, $\Phi(z) = P(Z \leq z)$ for $Z \sim \mathcal{N}(0, 1)$.
- Logarithms, exponentials, and basic calculus (derivatives, integrals).
- The concept of variance and standard deviation from Chapter 1.

No prior knowledge of options, derivatives, or financial markets is assumed.

Option Contract

An **option** is a contract that gives the holder the *right, but not the obligation*, to buy or sell an **underlying asset** (the stock, index, or commodity the option is written on) at a pre-agreed price. Two parameters define any option:

- **Strike price** K : the pre-agreed transaction price.
- **Expiry date** T : the deadline by which the right must be exercised.

There are two types:

- A **call option** gives the right to *buy* the underlying at price K by time T .

- A **put option** gives the right to *sell* the underlying at price K by time T .

The buyer pays an upfront price (the **premium**) for this right. The seller (“writer”) collects the premium but takes on the obligation.

8.1.1 Payoff at Expiry

At expiry, the option is worth exactly what you would gain by exercising it (or zero if exercising would lose money). Let S_T denote the price of the underlying asset at expiry.

Option Payoffs at Expiry

$$\text{Call payoff} = \max(S_T - K, 0) \quad (8.1)$$

$$\text{Put payoff} = \max(K - S_T, 0) \quad (8.2)$$

where:

- S_T = price of the underlying asset at expiry.
- K = strike price (pre-agreed transaction price).
- The $\max(\cdot, 0)$ reflects that you never exercise at a loss; you simply let the option expire.

Why “Right but Not Obligation” Matters

Suppose you hold a call option with strike $K = 100$. If $S_T = 120$, you exercise: buy at 100, the asset is worth 120, you gain 20. If $S_T = 80$, you do nothing: buying at 100 when the market price is 80 would be foolish. Your payoff is never negative. This asymmetry is the entire reason options have value, and why someone must be paid a premium to write them.

Figure 8.1 shows the payoff profiles. The kink at K is the signature of an option; linear instruments (stocks, futures) have straight-line payoffs.

8.1.2 Moneyness

Traders classify options by how the current spot price S compares to the strike K :

Moneyness

For a **call** option:

- **In-the-money (ITM)**: $S > K$ (exercising now would be profitable).
- **At-the-money (ATM)**: $S \approx K$.
- **Out-of-the-money (OTM)**: $S < K$ (exercising now would not be profitable).

For a **put**, the directions reverse: ITM when $S < K$, OTM when $S > K$. A common quantitative measure of moneyness is $m = K/S$ (or its log, $\ln(K/S)$).

8.2 Black-Scholes in One Page

Now that you know what options pay, the natural question is: what should you pay *today* for an option that expires in the future? This section presents the answer that [Black and Scholes \(1973\)](#) derived, which earned a Nobel Prize and remains the bedrock of options pricing.

The core idea: an option’s value today equals the expected payoff at expiry, discounted to the present, under a probability measure that accounts for risk. [Black and Scholes \(1973\)](#) showed

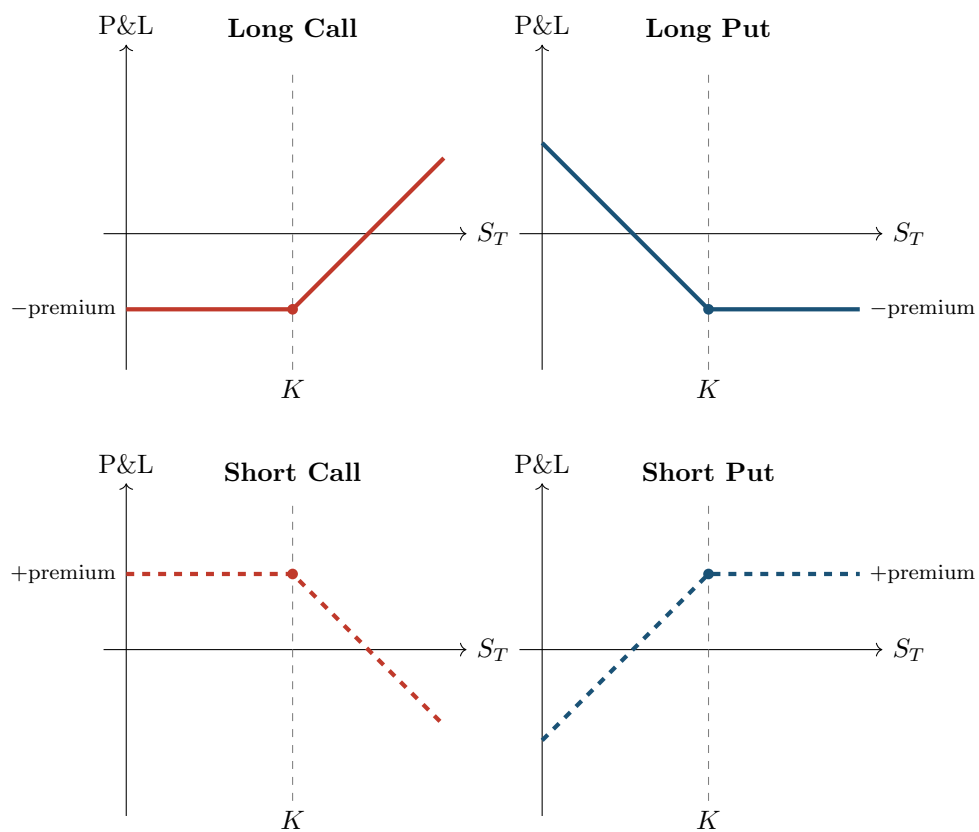


Figure 8.1: Option payoff diagrams (profit/loss including the premium paid or received). Long positions (top row) have limited downside and potentially large upside. Short positions (bottom row) are the mirror image: limited upside, potentially large downside. The kink at the strike K is the defining feature of an option.

that under a specific set of assumptions, this expected value has a closed-form solution.

The Black-Scholes Assumptions

The formula assumes:

1. The underlying price follows geometric Brownian motion (log-normal returns).
2. Volatility σ is **constant** over the life of the option.
3. There are no jumps in the price.
4. Trading is continuous and frictionless (no transaction costs, unlimited short selling).
5. The risk-free interest rate r is constant and known.
6. No dividends.

Every one of these assumptions is wrong in practice, and the volatility surface is what that wrongness looks like.

The Black-Scholes Formula for a European Call

^aA **European** option can only be exercised at expiry, not before. An **American** option can be exercised at any time up to expiry. For the purpose of defining implied volatility, the European version is standard.

$$C = S \Phi(d_1) - K e^{-rT} \Phi(d_2) \quad (8.3)$$

where

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T} \quad (8.4)$$

and:

- C = price of the call option today.
- S = current spot price of the underlying asset.
- K = strike price.
- r = continuously compounded risk-free interest rate (annualized).
- T = time to expiry in years (e.g., 3 months = 0.25).
- σ = annualized volatility of the underlying's log returns.
- $\Phi(\cdot)$ = CDF of the standard normal distribution.
- d_1 = a standardized measure of how far in-the-money the option is, adjusted for drift and time.
- $d_2 = d_1$ minus a volatility-time adjustment; $\Phi(d_2)$ is the risk-neutral probability of exercise.

Reading the Formula

Break Equation (8.3) into two pieces:

- $S \Phi(d_1)$: what you expect to receive (the asset value, weighted by the probability-adjusted chance it ends up above K).
- $K e^{-rT} \Phi(d_2)$: what you expect to pay (the strike, discounted to today, weighted by the risk-neutral probability of actually exercising).

The call price is the difference: expected receipt minus expected cost.

Worked Example:

Black-Scholes Call Price Suppose the S&P 500 is at $S = 100$, and you want to price a 3-month call with strike $K = 105$. The risk-free rate is $r = 5\%$ per year and volatility is $\sigma = 20\%$ per year.

Step 1: Compute d_1 and d_2 .

$$d_1 = \frac{\ln(100/105) + (0.05 + 0.04/2)(0.25)}{0.20\sqrt{0.25}} = \frac{-0.04879 + 0.0175}{0.10} = \frac{-0.03129}{0.10} = -0.3129$$

$$d_2 = -0.3129 - 0.20\sqrt{0.25} = -0.3129 - 0.10 = -0.4129$$

Step 2: Look up the normal CDF values.

$$\Phi(-0.3129) = 0.3772, \quad \Phi(-0.4129) = 0.3398$$

Step 3: Plug into the formula.

$$\begin{aligned} C &= 100 \times 0.3772 - 105 e^{-0.05 \times 0.25} \times 0.3398 \\ &= 37.72 - 105 \times 0.9876 \times 0.3398 \\ &= 37.72 - 35.24 \\ &= \$2.48 \end{aligned}$$

The call is out-of-the-money ($S < K$), so the premium is modest: \$2.48 on a \$100 underlying. You are paying for the *possibility* that the index rises above 105 within three months.

Why This Matters

Its key role for you is as the *inverse function* that maps observed option prices to implied volatilities (Section 8.3), which then become features and benchmarks for your ML model.

Black-Scholes Is Wrong, but Universal

Black-Scholes is wrong in nearly all its assumptions. Returns have fat tails (Chapter 1), volatility is not constant (Chapters 5–7), and prices jump (Chapter 4). But the formula remains the market’s common language for quoting option prices. When a trader says “this option is trading at 25 vol,” they mean the Black-Scholes implied volatility is 25%. The formula is not a belief about reality; it is a translation device.

8.2.1 The Greeks: Sensitivity to Inputs

The Black-Scholes formula expresses the option price as a function of five inputs. The partial derivatives of the price with respect to each input are called the **Greeks**, and they tell you how sensitive the option’s value is to small changes in market conditions.

Key Greeks

For a European call option priced by Black-Scholes:

- **Delta** ($\Delta = \partial C / \partial S$): sensitivity to the underlying price. A delta of 0.60 means the call gains roughly \$0.60 for each \$1 rise in the stock.
- **Gamma** ($\Gamma = \partial^2 C / \partial S^2$): rate of change of delta. High gamma means delta shifts rapidly with the stock price, making hedging more difficult.
- **Theta** ($\Theta = \partial C / \partial T$): time decay. Options lose value as expiry approaches (all else equal), and theta quantifies that bleed.
- **Vega** ($\mathcal{V} = \partial C / \partial \sigma$): sensitivity to volatility. This is the Greek that matters most for your project.
- **Rho** ($\rho = \partial C / \partial r$): sensitivity to the risk-free rate. Usually small for short-dated

options.

In Plain English

Vega is the most important Greek for volatility trading because it tells you how many dollars you make or lose per percentage-point change in implied vol. A high-vega position is a direct bet on volatility; a zero-vega portfolio is insulated from vol changes.

Figure 8.2 shows how vega varies with moneyness and time to expiry.

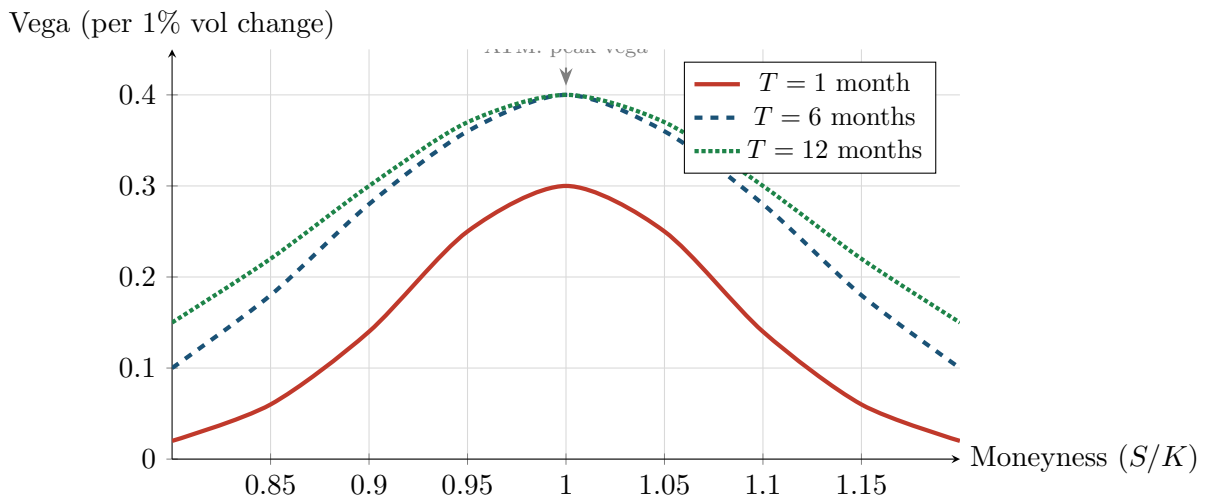


Figure 8.2: Vega profiles across moneyness for different maturities. Vega peaks at-the-money and falls off for deep ITM or OTM options. Longer-dated options have higher vega (more time for vol to matter) and a wider peak (more strikes are sensitive to vol changes). Short-dated ATM options have concentrated vega, making them efficient instruments for short-horizon vol trades.

8.3 Implied Volatility

Reversing the Black-Scholes formula extracts the market's view of volatility.

Every input to the Black-Scholes formula (S , K , r , T) is directly observable from market data, except one: σ . If you also observe the market price of the option C_{mkt} , you can back out the volatility the market is implicitly using.

Implied Volatility

The **implied volatility** σ_{imp} is the value of σ that makes the Black-Scholes formula match the observed market price:

$$C_{\text{mkt}} = C_{\text{BS}}(S, K, r, T, \sigma_{\text{imp}}) \quad (8.5)$$

where:

- C_{mkt} = the option price observed in the market.
- $C_{\text{BS}}(\cdot)$ = the Black-Scholes pricing function (Equation 8.3).
- σ_{imp} = the implied volatility (the unknown being solved for).

There is no closed-form solution for σ_{imp} ; it must be found numerically (e.g., Newton-Raphson or bisection).

The “Wrong Number in the Wrong Formula”

Implied volatility is famously described as “the wrong number to put in the wrong formula to get the right price” (Rebonato, 2004). It is *not* a forecast of future volatility. It is the market’s consensus price of uncertainty, contaminated by:

- **Risk premia:** sellers of options demand compensation for bearing tail risk, inflating IV above expected realized vol.
- **Supply and demand:** heavy demand for downside protection (put buying) pushes up IV for low-strike options.
- **Model error:** since Black-Scholes is misspecified, IV absorbs all the model’s errors into a single number.

Despite all this, IV is extremely useful. It is a real-time, forward-looking, market-priced measure of uncertainty.

Why This Matters

Implied volatility is the market’s consensus forecast of future vol. Comparing it to your RV forecast gives you the **variance risk premium**: $VRP \approx IV^2 - \widehat{RV}^2$. If your ML model produces a more accurate RV forecast than the market’s implied estimate, the VRP becomes a tradeable signal (Chapter 9). IV is also one of the strongest single features for predicting future realized vol (Gu et al., 2020a), so it will appear directly in your feature set (Chapter 10).

Computing IV: The Newton-Raphson Method

The definition above says IV must be found “numerically.” The standard method is **Newton-Raphson iteration**, which exploits the fact that vega (the derivative of option price with respect to σ) is available in closed form.

Starting from an initial guess σ_0 , the iteration refines the estimate:

$$\sigma_{n+1} = \sigma_n - \frac{C_{BS}(\sigma_n) - C_{mkt}}{\mathcal{V}(\sigma_n)}, \quad (8.6)$$

where:

- $C_{BS}(\sigma_n)$: the Black-Scholes price evaluated at the current volatility guess σ_n .
- C_{mkt} : the observed market price of the option.
- $\mathcal{V}(\sigma_n) = S\sqrt{T}N'(d_1)$: the Black-Scholes vega, with $N'(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ the standard normal density.

In Plain English

Newton-Raphson asks: “Given my current guess σ_n , the BS formula gives price $C_{BS}(\sigma_n)$. The market price is C_{mkt} . How much should I adjust σ to close the gap?” The answer is to divide the price error by the sensitivity of the price to σ (vega). If vega is 12 cents per vol point and the price error is 60 cents, adjust σ by $60/12 = 5$ vol points. The method converges quadratically – each iteration roughly doubles the number of correct digits – so 3–5 iterations typically suffice.

Initial guess. Brenner and Subrahmanyam (1988) provide a simple closed-form approximation for ATM options:

$$\sigma_0 \approx \sqrt{\frac{2\pi}{T}} \cdot \frac{C_{mkt}}{S}. \quad (8.7)$$

This approximation exploits the fact that the ATM Black-Scholes call price is approximately $C \approx S\sigma\sqrt{T/(2\pi)}$. For options far from the money, use $\sigma_0 = 0.20$ (20%) as a reliable starting point.

When Newton-Raphson Fails

Vega approaches zero for deep in-the-money and deep out-of-the-money options (where the option price is insensitive to volatility changes). Dividing by near-zero vega produces enormous, erratic steps. Two safeguards: (1) bound each step: $|\sigma_{n+1} - \sigma_n| \leq 0.25$; (2) if vega drops below a threshold (e.g., 10^{-8}), fall back to bisection, which is slower (linear convergence) but unconditionally stable. In practice, a hybrid Newton-Raphson/bisection algorithm handles the entire strike range reliably.

Why This Matters

If your project uses options data (SPX IV surface from Marquee), you will not receive pre-computed implied volatilities for every strike and maturity. You will often need to invert the Black-Scholes formula yourself to convert option prices into IV. Implement Newton-Raphson with the Brenner-Subrahmanyam initial guess and bisection fallback. The entire IV surface construction pipeline – from raw option prices to smooth SVI-fitted IV – starts with this inversion at each individual (K, T) point.

Implied volatility is *not* a single number for a given underlying. Every (strike, maturity) pair has its own IV. This leads directly to the next section.

8.4 The Implied Volatility Surface

The implied volatility extracted from options depends on which option you look at, giving rise to a two-dimensional surface.

Implied Volatility Surface

The **implied volatility surface** is the function

$$\sigma_{\text{imp}}(K, T) \tag{8.8}$$

mapping each (strike, maturity) pair to the implied volatility extracted from the corresponding option price. In practice, moneyness $m = K/S$ (or $\ln(K/S)$) replaces the raw strike to make the surface comparable across different underlying price levels.

- K (or m) = strike price (or moneyness).
- T = time to expiry.
- σ_{imp} = implied volatility at that (strike, maturity) point.

What a Flat Surface Would Mean

If Black-Scholes were correct (constant volatility, log-normal returns, no jumps), every option on the same underlying would produce the *same* implied volatility regardless of strike or maturity. The surface would be perfectly flat. The fact that the surface is *not* flat is direct evidence that Black-Scholes is wrong, and the shape of the surface tells you *how* it is wrong.

8.4.1 The Volatility Smile and Skew

The cross-section of IV at a fixed maturity reveals two patterns, shown in Figure 8.3.

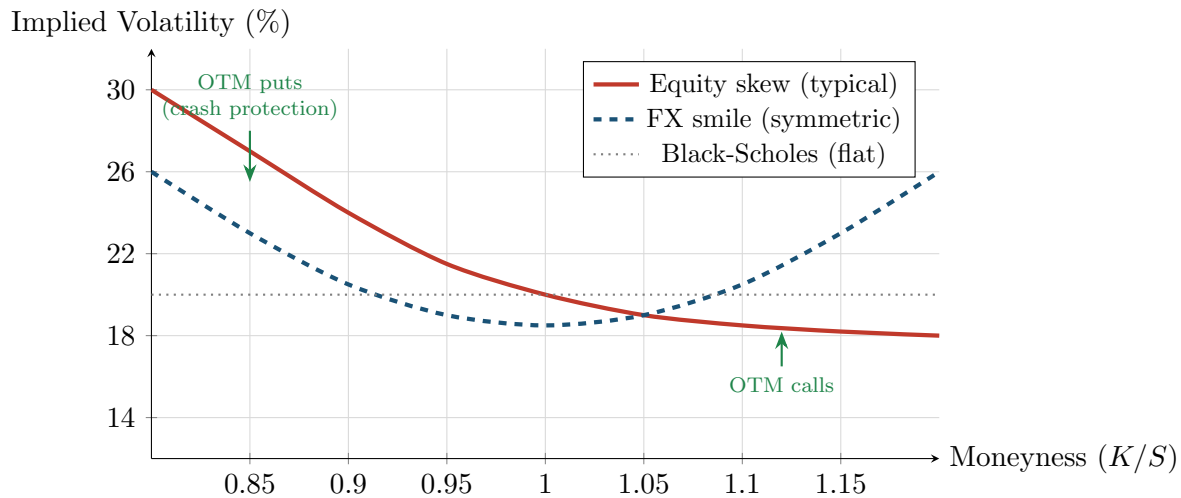


Figure 8.3: Implied volatility vs. moneyness at a fixed maturity. The red solid line shows the typical equity skew: IV increases sharply for low strikes (OTM puts) due to crash-risk demand. The blue dashed line shows a symmetric “smile” common in FX markets, where both tails carry extra uncertainty. The gray dotted line is the flat surface that Black-Scholes would predict.

The skew (red curve): for equity index options, IV is much higher for low strikes ($K/S < 1$, OTM puts) than for high strikes ($K/S > 1$, OTM calls). This pattern emerged after the 1987 crash and has persisted ever since. It reflects the market’s pricing of left-tail (crash) risk: investors pay a premium for downside protection.

The smile (blue curve): for foreign exchange options, IV is elevated for both deep OTM puts and deep OTM calls, creating a U-shape. This reflects the market’s view that large moves in either direction are more likely than the log-normal model assumes (fat tails on both sides).

Both patterns are inconsistent with constant-volatility Black-Scholes, which would produce the flat gray line.

8.4.2 The Term Structure of Implied Volatility

The second dimension is maturity. At-the-money IV is not the same at 1 month as at 1 year.

IV Term Structure

- **Normal conditions:** the term structure slopes upward (long-dated IV > short-dated IV), reflecting the greater uncertainty over longer horizons and mean-reversion in volatility.
- **Crisis periods:** the term structure inverts (short-dated IV > long-dated IV), because near-term fear spikes while the market expects volatility to eventually normalize.
- Short-dated IV is more volatile than long-dated IV, consistent with the mean-reverting nature of volatility established in Chapters 5 and 6.

Figure 8.4 contrasts these two regimes.

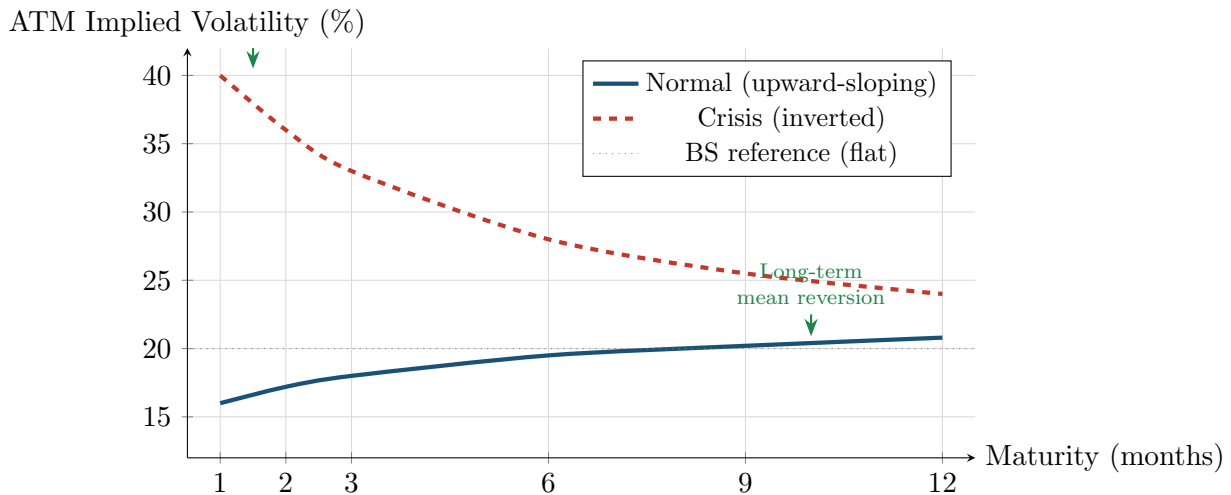


Figure 8.4: ATM implied volatility term structure under two regimes. In normal conditions (blue), the term structure slopes upward as longer horizons carry greater uncertainty. During crises (red dashed), the term structure inverts: near-term IV spikes while long-term IV reflects the expectation that volatility will eventually normalize. The spread between short-dated and long-dated ATM IV is itself a useful feature for forecasting models.

8.4.3 The Butterfly Spread

The skew tells you about the *asymmetry* of the market's fear – how much more expensive downside protection is relative to upside. But it says nothing about whether *both* tails are being bid up simultaneously. The **butterfly spread** fills that gap.

When portfolio managers buy both OTM puts (crash protection) and OTM calls (upside participation or short-squeeze hedging) at the same time, IV rises on both wings relative to ATM. This symmetric fattening of the tails is invisible to a skew measure, which only captures the *difference* between the two wings. The butterfly captures the *average* elevation of both wings above the center.

The butterfly spread is constructed from three implied volatilities at a fixed maturity: the 25-delta put, the 25-delta call, and the ATM (50-delta) volatility.

Butterfly Spread

The **butterfly spread** (or **butterfly**) is defined as:

$$BF_t = \frac{1}{2}(\sigma_{25\Delta P} + \sigma_{25\Delta C}) - \sigma_{ATM} \quad (8.9)$$

where:

- $\sigma_{25\Delta P}$ = implied volatility of the 25-delta put (an OTM put whose Black-Scholes delta is -0.25).
- $\sigma_{25\Delta C}$ = implied volatility of the 25-delta call (an OTM call whose Black-Scholes delta is $+0.25$).
- σ_{ATM} = at-the-money implied volatility (50-delta).
- BF_t = the butterfly spread on day t , measured in volatility points.

The butterfly is always non-negative in a no-arbitrage market (a negative value would imply the smile is concave, which violates convexity constraints on the implied volatility curve).

In Plain English

The butterfly measures how much higher the average wing volatility is compared to the center of the smile. Think of it as the “curvature” or “U-shape” of the smile at a fixed maturity. A large butterfly means traders are paying up for protection on *both* sides of the distribution – they expect fat tails, regardless of direction. A small butterfly means the smile is relatively flat and the market sees tail risk as modest.

To see why the butterfly and skew are complementary rather than redundant, note that they decompose the two wing volatilities into orthogonal components. The **risk reversal** (skew) is the difference between the wings:

$$RR_t = \sigma_{25\Delta C} - \sigma_{25\Delta P}$$

while the butterfly is their average elevation above ATM. Together, the two reconstruct the full wing structure:

$$\sigma_{25\Delta P} = \sigma_{ATM} + BF_t - \frac{1}{2} RR_t \quad (8.10)$$

$$\sigma_{25\Delta C} = \sigma_{ATM} + BF_t + \frac{1}{2} RR_t \quad (8.11)$$

The risk reversal captures **directional asymmetry** (skewness demand); the butterfly captures **symmetric tail thickness** (kurtosis demand). One measures the tilt of the smile; the other measures its curvature.

Crisis Behavior

During crises, the butterfly typically spikes alongside VIX, but with a distinctive pattern. Portfolio insurance programs bid up OTM puts (the dominant effect in the skew), and at the same time, short-covering and tail-risk hedging bid up OTM calls. When *both* wings are elevated, the butterfly rises sharply.

A rising butterfly with a steepening skew is the signature of a broad-based panic: the market is pricing extreme moves in both directions, not just a directional crash. In contrast, a rising skew with a *stable* butterfly signals targeted downside hedging without generalized tail fear.

Symmetric tail pricing (high butterfly) tends to precede periods of elevated but two-sided chopiness, while pure skew spikes often precede sharp, directional sell-offs that resolve more quickly.

Why This Matters

The butterfly spread is one of nine options-implied features in the project’s feature pipeline (Chapter 10). It provides information that the risk reversal (skew) cannot: whether the market prices *symmetric* tail risk rather than directional fear alone. Empirically, the butterfly is most useful as a crisis-detection signal. Spikes in the butterfly predict periods of elevated realized volatility at weekly and monthly horizons, especially when combined nonlinearly with VIX and the term structure slope in tree-based models.

8.4.4 The Full Surface

Figure 8.5 combines both dimensions into a three-dimensional view.

8.4.5 Fitting the Surface: The SVI Parametrization

The IV surface is a collection of discrete points (one per traded option). To use it as a continuous object – for interpolation, extrapolation, or computing smooth Greeks – you need a

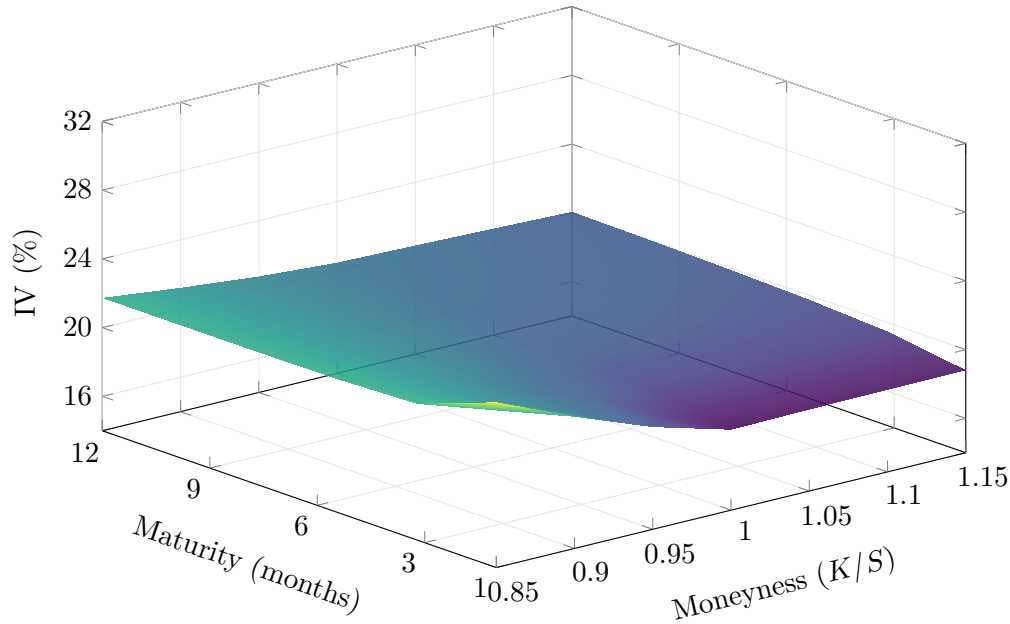


Figure 8.5: The implied volatility surface for a typical equity index. Left side (low moneyness, OTM puts): IV is elevated, especially at short maturities, reflecting concentrated crash-risk demand. Right side (high moneyness, OTM calls): IV is lower. Short maturities (front of the surface): the skew is steeper and IV levels are more extreme. Long maturities (back): the surface flattens as mean-reversion dampens both the level and the skew.

parametric model that fits through those points. The **Stochastic Volatility Inspired (SVI)** parametrization (Gatheral and Jacquier, 2014) is the industry standard for equity index surfaces.

Raw SVI Parametrization

For log-moneyness $k = \ln(K/F)$ (where F is the forward price), the **total implied variance** $w(k) = \sigma_{\text{imp}}^2(k) \cdot T$ of a single maturity slice is:

$$w(k) = a + b \left\{ \rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right\}, \quad (8.12)$$

with five parameters $\{a, b, \rho, m, \sigma\}$ satisfying:

- $a \in \mathbb{R}$: vertical level of the smile (shifts total variance up or down).
- $b \geq 0$: controls the slope of both wings (larger b = steeper wings).
- $|\rho| < 1$: controls the tilt (negative ρ steepens the left/put wing, flattens the right/call wing).
- $m \in \mathbb{R}$: horizontal translation (shifts the smile's center left or right along log-moneyness).
- $\sigma > 0$: controls the curvature at the money (smaller σ = sharper ATM bend).

The non-negativity constraint $a + b\sigma\sqrt{1 - \rho^2} \geq 0$ ensures $w(k) \geq 0$ for all k .

In Plain English

The SVI formula says: total implied variance is a shifted, tilted hyperbola in log-moneyness. The square root $\sqrt{(k - m)^2 + \sigma^2}$ creates the “smile” shape – it curves upward for strikes far from the center m . The $\rho(k - m)$ term tilts the smile: for equities, $\rho < 0$, making the left wing (OTM puts) steeper than the right wing (OTM calls). The parameter a shifts the whole curve vertically, and b scales the wings. The five parameters

give enough flexibility to fit real market smiles accurately while producing smooth, well-behaved extrapolations into the tails.

Why SVI? Two properties make SVI theoretically well-motivated rather than merely a convenient curve fit (Gatheral and Jacquier, 2014):

1. As $|k| \rightarrow \infty$, $w(k)$ becomes linear in k , matching **Roger Lee's moment formula** – the theoretical constraint that total variance must grow at most linearly in extreme log-moneyness.
2. The large-maturity limit of the **Heston** stochastic volatility model's implied variance is exactly SVI. The parametrization is not arbitrary.

What each parameter controls. When calibrating to market data, you can think of the five parameters as knobs:

- Raise a : the entire smile shifts up (higher overall variance).
- Raise b : both wings get steeper (more extreme strike vol increases).
- Lower ρ (more negative): the put wing steepens, the call wing flattens (stronger skew).
- Shift m right: the smile's center moves to higher moneyness.
- Lower σ : the ATM region becomes more “V-shaped” (sharper curvature at the money).

Why This Matters

SVI is how practitioners interpolate and extrapolate the IV surface. For your project, SVI matters in two ways: (1) if you compute surface-derived features (skew, butterfly, term structure slope) from raw option prices, SVI gives you a smooth surface to read those features from, avoiding the noise of individual option quotes; (2) the SVI parameters themselves – especially b (wing steepness) and ρ (skew) – can serve as features for vol forecasting, since they summarize the entire smile shape in five numbers. Fit SVI slice-by-slice (one set of parameters per maturity), then track how ρ and b change over time.

No-Arbitrage Requires Care

A naive SVI fit can produce **calendar spread arbitrage**: total variance that decreases with maturity at some strikes. Gatheral and Jacquier (2014) provide sufficient conditions for arbitrage-free SVI surfaces. For a surface (not just a single slice), use the **SSVI** extension, which parametrizes the entire surface jointly and guarantees no static arbitrage under simple parameter constraints. Always verify $\partial_T w(k, T) \geq 0$ for all k after calibration.

8.5 PCA of the Volatility Surface

The IV surface is a high-dimensional object (potentially hundreds of strike-maturity points), but its daily movements are driven by a small number of factors.

Cont and da Fonseca (2002) applied principal component analysis (PCA) to the daily changes of S&P 500 implied volatility surfaces and found a remarkably low-dimensional structure.

PCA in One Paragraph

PCA finds orthogonal directions of maximum variance in a dataset. The first principal component (PC1) captures the most variance, PC2 captures the most remaining variance orthogonal to PC1, and so on. If you have studied linear algebra, PCA extracts the eigenvectors of the covariance matrix, sorted by eigenvalue magnitude.

Three Factors Drive the IV Surface (Cont and da Fonseca, 2002)

PCA on daily changes in the implied volatility surface yields three dominant factors:

1. **Level** (PC1, ~70% of variance): a parallel shift; the entire surface moves up or down together. This is the “VIX factor” for equity indices.
2. **Slope** (PC2, ~15% of variance): the skew steepens or flattens; OTM put IV moves relative to OTM call IV.
3. **Curvature** (PC3, ~10% of variance): the smile becomes more or less convex; both tails move relative to ATM.

Together, these three factors explain roughly 95% of daily surface variation.

Figure 8.6 illustrates the three factor loadings.

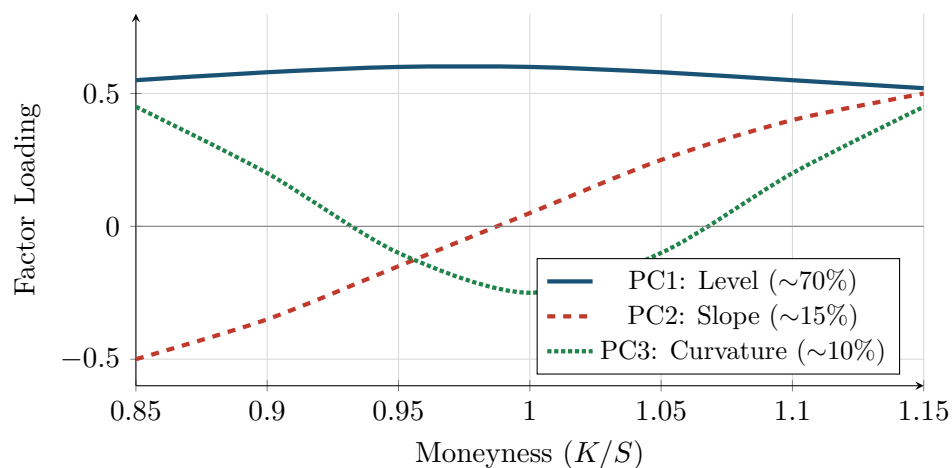


Figure 8.6: Stylized PCA factor loadings for the implied volatility surface at a fixed maturity, following Cont and da Fonseca (2002). PC1 (level) loads uniformly across strikes. PC2 (slope) loads negatively on low strikes and positively on high strikes, capturing skew rotation. PC3 (curvature) loads positively at both extremes and negatively at ATM, capturing smile convexity changes.

Practical Use: Surface Compression

Represent each day’s surface (or surface change) by just its PC1, PC2, and PC3 scores – a three-feature compression for feature engineering (Chapter 10).

8.6 Local Volatility

Requirements for This Section

You should be comfortable with the Black-Scholes framework (Section 8.2), partial derivatives (including second-order), and the concept of the implied volatility surface from the preceding sections.

The implied volatility surface gives us a snapshot of how the market prices options across strikes and maturities, but it does not tell us *how to interpolate* between quoted points or *how to extrapolate* beyond them without introducing arbitrage. A naive interpolation (e.g., linear in strike) can easily violate calendar-spread or butterfly-spread no-arbitrage conditions. **Local volatility** solves this problem: it provides the unique continuous diffusion model consistent with all observed option prices.

The key result is due to [Dupire \(1994\)](#). Given a continuum of European call prices $C(K, T)$ observed across strikes K and maturities T , the **Dupire formula** extracts the local volatility:

$$\sigma_{\text{loc}}^2(K, T) = \frac{\frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K}}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}} \quad (8.13)$$

The numerator combines two effects: $\partial C / \partial T$ captures the time decay of the option (how much value accrues as the horizon extends), while $rK \partial C / \partial K$ is a cost-of-carry correction arising from discounting. The denominator, $\frac{1}{2} K^2 \partial^2 C / \partial K^2$, is proportional to the **risk-neutral density** of the underlying at strike K and maturity T – it measures how much probability the market assigns to the stock landing near K . For the formula to be well-defined, the denominator must be strictly positive (a condition equivalent to the absence of butterfly arbitrage).

Local Vol as Market-Implied Instantaneous Vol

Local vol is the unique diffusion coefficient $\sigma(S, t)$ such that a one-factor model

$$\frac{dS}{S} = \mu dt + \sigma(S, t) dW$$

reproduces *all* observed option prices simultaneously. It is the market's pricing of instantaneous volatility conditional on the stock being at price K at time T . Unlike implied volatility (which averages over the entire path to expiry), local vol is a pointwise, path-independent quantity – the “microscopic” volatility the market embeds at each node of the (K, T) grid.

Worked Example:

Computing Local Vol from a Call Surface Setup. Consider a stock at $S = 100$ with risk-free rate $r = 2\%$. We observe European call prices at three strikes and two maturities:

Strike K	$C(K, 0.25)$	$C(K, 0.50)$
90	12.85	14.60
100	5.60	7.95
110	1.80	4.10

We compute $\sigma_{\text{loc}}^2(100, 0.25)$ using finite differences.

Step 1: Estimate $\partial C / \partial T$ at $K = 100$, $T = 0.25$.

$$\frac{\partial C}{\partial T} \approx \frac{C(100, 0.50) - C(100, 0.25)}{0.50 - 0.25} = \frac{7.95 - 5.60}{0.25} = 9.40$$

Step 2: Estimate $\partial C / \partial K$ at $K = 100$, $T = 0.25$. Using a central difference with $\Delta K = 10$:

$$\frac{\partial C}{\partial K} \approx \frac{C(110, 0.25) - C(90, 0.25)}{2 \times 10} = \frac{1.80 - 12.85}{20} = -0.5525$$

Step 3: Estimate $\partial^2 C / \partial K^2$ at $K = 100$, $T = 0.25$.

$$\frac{\partial^2 C}{\partial K^2} \approx \frac{C(110, 0.25) - 2C(100, 0.25) + C(90, 0.25)}{10^2} = \frac{1.80 - 11.20 + 12.85}{100} = 0.0345$$

Step 4: Plug into the Dupire formula.

$$\text{Numerator} = \frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K} = 9.40 + 0.02 \times 100 \times (-0.5525) = 9.40 - 1.105 = 8.295$$

$$\text{Denominator} = \frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2} = \frac{1}{2} (100)^2 \times 0.0345 = 172.5$$

$$\sigma_{\text{loc}}^2(100, 0.25) = \frac{8.295}{172.5} = 0.0481$$

Therefore $\sigma_{\text{loc}}(100, 0.25) = \sqrt{0.0481} \approx 21.9\%$.

Local Vol Is Not a Forecast

Local vol perfectly fits today's option prices but has no predictive power for tomorrow. It generates flat forward volatility smiles (the smile does not move), which contradicts observed behavior. Use it as an interpolation and arbitrage-checking tool, not as a model of reality. For forecasting, use the realized measures and ML models from later chapters.

Why This Matters

The ratio of local vol at 90% moneyness to ATM local vol is a cleaner skew measure than raw IV skew, potentially giving your model a more informative tail-risk signal.

8.7 Model-Free Implied Variance and the VIX

The previous sections define implied volatility through the lens of Black-Scholes, which means the IV number carries the model's errors. This section presents a method that extracts expected future variance from option prices without assuming any pricing model.

8.7.1 Model-Free Implied Variance

[Britten-Jones and Neuberger \(2000\)](#) proved a striking result: under very general conditions, the expected integrated variance of the underlying (under the risk-neutral probability measure) can be read directly from option prices across all strikes, with no parametric model needed.

Risk-Neutral Measure

In derivatives pricing, the “risk-neutral” (or $\mathbb{Q}$) measure is a probability weighting under which all assets earn the risk-free rate on average. It is *not* the real-world probability of outcomes; it is a mathematical device that makes pricing consistent with no-arbitrage. Expectations under $\mathbb{Q}$ incorporate risk premia: they overweight bad outcomes relative to real-world probabilities because investors demand compensation for bearing risk.

Model-Free Implied Variance ([Britten-Jones and Neuberger, 2000](#))

The risk-neutral expected integrated variance over horizon $[0, T]$ equals:

$$\mathbb{E}^{\mathbb{Q}} \left[\int_0^T \sigma_t^2 dt \right] = \frac{2}{T} \left[\int_0^F \frac{P(K, T)}{K^2} dK + \int_F^\infty \frac{C(K, T)}{K^2} dK \right] \quad (8.14)$$

where:

- $\mathbb{E}^{\mathbb{Q}}[\cdot]$ = expectation under the risk-neutral probability measure.

- σ_t^2 = instantaneous variance of the underlying at time t .
- $\int_0^T \sigma_t^2 dt$ = integrated variance over the period (the object that realized volatility estimates; see Chapter 2).
- F = forward price of the underlying.
- $P(K, T)$ = price of a European put with strike K and maturity T (used for $K < F$, i.e., OTM puts).
- $C(K, T)$ = price of a European call with strike K and maturity T (used for $K > F$, i.e., OTM calls).
- The $1/K^2$ weighting gives proportionally more weight to lower-strike options.

Integrating option prices across all strikes cancels out the dependence on any particular volatility model.

Why It Works Without a Model

A portfolio of options across all strikes effectively replicates a contract on realized variance itself. Each option at strike K provides local information about the probability of the price reaching K . By weighting them by $1/K^2$ and integrating, you reconstruct the full distribution and extract its second moment (variance) without ever assuming what that distribution looks like.

8.7.2 The VIX Index

The CBOE Volatility Index (VIX) is the market's most-watched "fear gauge." It implements the model-free approach of [Britten-Jones and Neuberger \(2000\)](#) on S&P 500 options with a 30-day horizon.

VIX Construction (,)

The VIX is defined as:

$$\text{VIX}^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2 \quad (8.15)$$

and $\text{VIX} = 100 \times \sqrt{\text{VIX}^2/100^2}$, reported in annualized percentage points.

- T = time to expiry (30 calendar days, $T \approx 30/365$).
- K_i = strike price of the i -th OTM option.
- ΔK_i = spacing between adjacent strikes $((K_{i+1} - K_{i-1})/2)$.
- $Q(K_i)$ = midpoint of the bid-ask spread for the OTM option at strike K_i (puts for $K_i < F$, calls for $K_i > F$).
- F = forward index level derived from put-call parity.
- K_0 = the first strike below F .
- r = risk-free rate.
- The second term is a small correction for the difference between the forward and the first-below-forward strike.

In practice, the CBOE interpolates between the two nearest-to-30-day maturities to target exactly 30 days.

Equation (8.15) is the discrete approximation of the continuous integral in Equation (8.14): the sum over OTM options across all available strikes replaces the integral, and the $\Delta K_i/K_i^2$ weighting mirrors the $1/K^2$ in the continuous formula.

Interpreting VIX Levels

VIX is quoted in annualized percentage points. Typical values for the S&P 500:

- **VIX** $\approx$ **12–15**: calm markets, low uncertainty.
- **VIX** $\approx$ **20–25**: elevated uncertainty, moderate fear.
- **VIX** $>$ **30**: high fear; historically associated with market sell-offs.
- **VIX** $>$ **50**: extreme panic (2008 financial crisis peak: ~ 80 ; March 2020: ~ 82).

To convert VIX to expected 30-day realized vol: $\sigma_{30d} \approx \text{VIX}/100$. For example, $\text{VIX} = 20$ implies the options market prices roughly 20% annualized vol over the next month.

VIX Is Not a Volatility Forecast

VIX is the risk-neutral expected volatility, *not* a forecast of realized volatility under real-world probabilities. VIX systematically overstates future realized volatility. The average gap between VIX-squared and subsequent 30-day realized variance is the **variance risk premium** (Chapter 9). Historically, VIX exceeds subsequent realized vol roughly 85% of the time (Carr and Wu, 2009a). Using VIX as a direct volatility forecast is a common and costly mistake: it confuses the risk-neutral measure $\mathbb{Q}$ with the real-world measure $\mathbb{P}$.

8.8 Variance Swaps: Trading Realized Volatility

Background for This Section

This section requires:

- The model-free implied variance integral (Section 8.7.1).
- The VIX construction and interpretation (Section 8.7).
- Realized variance and its estimation (Chapter 2).

VIX tells you the fair price of future variance. But how do you actually *trade* it? The **variance swap** is the instrument that lets you take a direct position on realized volatility without managing Greeks day-to-day.

8.8.1 Definition and Payoff

Variance Swap

A **variance swap** is an over-the-counter derivative whose payoff at expiry is:

$$\text{Payoff} = N_{\text{var}} \times (\text{RV}^2 - K_{\text{var}}) \quad (8.16)$$

where:

- N_{var} = the **variance notional** (dollars per variance point squared).
- K_{var} = the **variance strike** (the agreed-upon fair variance level, set at inception so the swap has zero initial value).
- RV^2 = the annualized realized variance of the underlying over the contract period, typically computed from daily log returns.

The buyer of the swap profits when realized variance exceeds the strike; the seller profits when it falls below.

Traders often think in terms of **vega notional** rather than variance notional, because it is more intuitive. The vega notional approximates the dollar P&L per volatility point (not variance

point):

$$N_{\text{vega}} = N_{\text{var}} \times 2\sqrt{K_{\text{var}}} \quad (8.17)$$

For example, if $K_{\text{var}} = 0.04$ (i.e., 20% vol) and you want \$100,000 per vol point, then $N_{\text{var}} = N_{\text{vega}}/(2 \times 0.20) = \$250,000$ per variance point.

8.8.2 Log-Contract Replication

The theoretical foundation: Britten-Jones and Neuberger (2000) and Carr and Wu (2009a) showed that continuously delta-hedging a portfolio of OTM options weighted by $1/K^2$ replicates the payoff $-2\log(S_T/S_0)$. The realized P&L from this hedge equals the realized variance minus the cost of the portfolio. Therefore, the cost of this portfolio – which is exactly the model-free implied variance integral from Section 8.7.1 – equals the fair variance swap strike K_{var} .

In other words, the VIX formula (Equation 8.15) computes the fair strike of a 30-day variance swap on the S&P 500. $\text{VIX}^2/10,000$ is the annualized K_{var} for that maturity.

Worked Example:

Discretizing the Variance Swap Strike Setup. Consider an underlying at $S = 100$ with $r = 0$. We observe 5 OTM puts and 5 OTM calls with the following mid-market prices. Using uniform strike spacing $\Delta K = 5$ for simplicity:

Step 1: Compute weights. For each option, the weight from the model-free integral discretization is:

$$w_i = \frac{2 \Delta K_i}{K_i^2}$$

Step 2: Compute contributions. Each option's contribution to K_{var} is $w_i \times Q_i$, where Q_i is the option price.

K	Type	Price Q_i	ΔK	Weight w_i	Contribution $w_i \times Q_i$
80	Put	0.22	5	$2 \times 5/80^2 = 0.001563$	0.000344
85	Put	0.45	5	$2 \times 5/85^2 = 0.001384$	0.000623
90	Put	0.95	5	$2 \times 5/90^2 = 0.001235$	0.001173
95	Put	2.10	5	$2 \times 5/95^2 = 0.001108$	0.002327
98	Put	3.40	5	$2 \times 5/98^2 = 0.001041$	0.003539
102	Call	3.20	5	$2 \times 5/102^2 = 0.000961$	0.003076
105	Call	1.85	5	$2 \times 5/105^2 = 0.000907$	0.001678
110	Call	0.80	5	$2 \times 5/110^2 = 0.000826$	0.000661
115	Call	0.30	5	$2 \times 5/115^2 = 0.000756$	0.000227
120	Call	0.10	5	$2 \times 5/120^2 = 0.000694$	0.000069

Step 3: Sum all contributions.

$$\begin{aligned}
 K_{\text{var}} &= \sum_i w_i \times Q_i = 0.000344 + 0.000623 + 0.001173 + 0.002327 + 0.003539 \\
 &\quad + 0.003076 + 0.001678 + 0.000661 + 0.000227 + 0.000069 = 0.01372
 \end{aligned}$$

Annualizing over $T = 30/365 \approx 0.08219$ years:

$$K_{\text{var}}^{\text{ann}} = \frac{K_{\text{var}}}{T} = \frac{0.01372}{0.08219} \approx 0.1669$$

Step 4: Convert to implied vol.

$$\sqrt{K_{\text{var}}^{\text{ann}}} = \sqrt{0.1669} \approx 40.9\%$$

This high implied vol reflects the elevated option prices in our stylized example. For context, if the S&P 500 traded with a vol surface consistent with $VIX = 40$, then $VIX^2/10,000 = 0.1600$, close to our 0.1669. The small gap is truncation error from the limited strike range (80–120); additional OTM options below 80 and above 120 would add positive increments to the sum.

Trading Your Vol Forecast Directly

A variance swap lets you monetize your RV forecast without managing delta, gamma, or theta. If you believe next-month RV will be 22% but the var swap strike is K_{var} corresponding to 19%, you buy the swap. At expiry your P&L is simply $N_{\text{var}} \times (0.22^2 - 0.19^2) = N_{\text{var}} \times 0.0123$. No path dependence before expiry settlement, no Greeks to manage – just a pure bet on realized variance.

Convexity and Mark-to-Market

A **volatility swap** (payoff = $\sigma_{\text{realized}} - K_{\text{vol}}$) is NOT the same as a variance swap. Because $\mathbb{E}[\sigma] < \sqrt{\mathbb{E}[\sigma^2]}$ (Jensen’s inequality), the vol swap strike is lower than $\sqrt{K_{\text{var}}}$. The gap depends on the volatility of volatility (vol-of-vol, Chapter 9 Section 9.7). Also: before expiry, variance swaps have substantial mark-to-market risk as implied vol moves. A position can be deeply underwater mid-life even if it ultimately settles in your favor.

Why This Matters

If your ML model forecasts RV more accurately than the market-implied K_{var} , you can systematically harvest the mispricing – the core mechanism behind Project 5 (VRP ML Trader) and the economic-value test for any of the other project directions.

8.9 Connecting the Vol Surface to Volatility Forecasting

IV Surface Features for ML Models

The volatility surface provides a rich set of features for realized volatility forecasting models (Chapters 10–14):

- **VIX level:** the single most common implied-vol feature. Often the strongest univariate predictor of future realized vol (Gu et al., 2020a).
- **VVIX** (the “vol of vol”): implied volatility of VIX options; captures uncertainty about uncertainty.
- **Skew slope:** the difference in IV between OTM puts and ATM options; a proxy for tail-risk pricing.
- **Butterfly spread** (Section 8.4.3): average wing IV minus ATM IV; captures symmetric tail thickness (kurtosis demand), orthogonal to skew.
- **Term spread:** the difference between long-dated and short-dated ATM IV; signals mean-reversion expectations.
- **PCA scores** (Section 8.5): three-number summary of the entire surface.
- **Variance risk premium:** VIX^2 minus recent realized variance; the subject of Chapter 9.

8.10 Summary

- An option gives the right (not obligation) to buy (call) or sell (put) at a fixed strike K by expiry T .
- Call payoff: $\max(S_T - K, 0)$; put payoff: $\max(K - S_T, 0)$.
- Black-Scholes (Black and Scholes, 1973) provides a closed-form call price: $C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2)$. The formula assumes constant volatility, log-normal returns, no jumps, and frictionless markets.
- Implied volatility σ_{imp} is the σ that makes Black-Scholes match the market price. It is not a forecast; it is the market's price of uncertainty, contaminated by risk premia and model error.
- The IV surface $\sigma_{\text{imp}}(K, T)$ varies across strikes and maturities. Skew (higher IV for low strikes) reflects crash-risk pricing; smile (U-shape) reflects fat-tail pricing.
- The butterfly spread $\text{BF}_t = \frac{1}{2}(\sigma_{25\Delta P} + \sigma_{25\Delta C}) - \sigma_{\text{ATM}}$ measures symmetric tail thickness (kurtosis demand), complementing the skew's measure of directional asymmetry. It spikes during crises when both wings are bid up.
- PCA of the IV surface (Cont and da Fonseca, 2002) yields three dominant factors (level, slope, curvature) explaining $\sim 95\%$ of daily variation.
- Model-free implied variance (Britten-Jones and Neuberger, 2000) extracts expected future variance from the cross-section of option prices without assuming any pricing model.
- VIX implements model-free implied variance for S&P 500 options over 30 days. VIX systematically overestimates future realized vol; the gap is the variance risk premium (Chapter 9).
- VIX, skew, butterfly, term structure, PCA scores, and VRP are all feature candidates for ML volatility models (Chapter 10).
- The IV surface is observed daily, forward-looking, and market-priced, making it a uniquely valuable complement to backward-looking realized volatility measures.

Table 8.1: Key Results from Chapter 8

Concept	Key Formula / Result	Significance
Black-Scholes	$C = Ke^{-rT} \Phi(d_2)$ — $S\Phi(d_1)$	Defines the common language for quoting option prices as volatilities
Implied volatility	Solve $C_{\text{mkt}} = C_{\text{BS}}(\sigma_{\text{imp}})$	Market-priced, forward-looking measure of uncertainty
IV surface shape	Skew, smile, term structure	Reveals crash-risk pricing, fat-tail beliefs, and mean-reversion expectations
Butterfly spread	$\text{BF}_t = \frac{1}{2}(\sigma_{25\Delta P} + \sigma_{25\Delta C}) - \sigma_{\text{ATM}}$	Symmetric tail thickness; orthogonal to skew; crisis-detection signal
PCA of IV surface	3 factors $\approx 95\%$ of variance	Level, slope, curvature; compress the surface for feature engineering
Model-free variance	$\frac{2}{T} \int \frac{O(K)}{K^2} dK$	Extracts expected variance without model assumptions
VIX	CBOE implementation, 30-day	Not a vol forecast; systematically overestimates realized vol (includes risk premium)

Chapter 9

The Variance Risk Premium

Why This Chapter Matters

The variance risk premium links implied volatility (Chapter 8) to realized volatility (Chapter 2). It predicts equity returns (Bollerslev et al., 2009b), forecasts future realized vol through mean reversion, and is a direct trading signal (Project 5: VRP ML trader). VRP features appear in virtually every competitive feature set (Chapter 10).

9.1 What Is the Variance Risk Premium?

Chapter 8 showed that VIX systematically overstates future realized volatility: the options market charges more for volatility exposure than what actually materializes.

Imagine you own a house and buy fire insurance. You pay \$1,200 per year in premiums, but the expected fire loss (probability of fire times damage) is only \$400. The \$800 gap is the insurance premium: compensation the insurer earns for bearing your tail risk.

The variance risk premium is the same concept applied to volatility. Options sellers are the “insurers” of the stock market. They sell downside protection (puts) and volatility exposure (straddles) to hedgers and speculators. In return, they earn a premium: the gap between what the options market prices in and what actually happens.

Risk-Neutral ($\mathbb{Q}$) vs. Physical ($\mathbb{P}$) Probability Measures

Two probability measures appear throughout this chapter:

- The **physical measure** $\mathbb{P}$ describes the actual frequencies of outcomes in the real world. If you simulate the S&P 500 forward under $\mathbb{P}$, your simulated returns match the historical distribution (including its mean, fat tails, and skew).
- The **risk-neutral measure** $\mathbb{Q}$ is a mathematical reweighting of $\mathbb{P}$ that makes asset pricing consistent with no-arbitrage. Under $\mathbb{Q}$, bad outcomes (crashes, spikes in volatility) receive higher weight than under $\mathbb{P}$ because investors demand compensation for bearing those risks.

Option prices reflect $\mathbb{Q}$ -expectations, not $\mathbb{P}$ -expectations. VIX is a $\mathbb{Q}$ -measure of expected variance. Realized volatility is observed under $\mathbb{P}$. The VRP is the difference between the two.

The variance risk premium at time t is the difference between the risk-neutral expected variance (what the options market prices) and the physical expected variance (what actually tends to happen) over a horizon h (typically 30 calendar days):

Variance Risk Premium

$$\text{VRP}_t = \mathbb{E}_t^{\mathbb{Q}}[\text{RV}_{t,t+h}] - \mathbb{E}_t^{\mathbb{P}}[\text{RV}_{t,t+h}] \quad (9.1)$$

- VRP_t : the variance risk premium at time t .
- $\mathbb{E}_t^{\mathbb{Q}}[\cdot]$: the risk-neutral (options-market-implied) expectation, conditional on information at time t .
- $\mathbb{E}_t^{\mathbb{P}}[\cdot]$: the physical (real-world) expectation, conditional on information at time t .
- $\text{RV}_{t,t+h}$: realized variance over the period from t to $t+h$ (defined in Chapter 2).
- h : the forecast horizon, typically 30 calendar days (~ 22 trading days) to match VIX.

In practice, neither expectation is directly observed. You need proxies for each:

Operationalized VRP

The standard operational measure of VRP is:

$$\widehat{\text{VRP}}_t = \underbrace{\left(\frac{\text{VIX}_t}{100}\right)^2}_{\text{proxy for } \mathbb{E}_t^{\mathbb{Q}}[\text{RV}_{t,t+h}]} - \underbrace{\hat{\text{RV}}_{t,t+h}^{\mathbb{P}}}_{\text{proxy for } \mathbb{E}_t^{\mathbb{P}}[\text{RV}_{t,t+h}]} \quad (9.2)$$

- $(\text{VIX}_t/100)^2$: VIX squared (converted from percentage points to decimal), which equals the model-free risk-neutral expected variance over 30 days (Section 8.7 of Chapter 8).
- $\hat{\text{RV}}_{t,t+h}^{\mathbb{P}}$: a physical-measure forecast of future realized variance. Common choices:
 - Backward-looking RV: use $\text{RV}_{t-h,t}$ (the past 30 days' realized variance) as a naive forecast.
 - HAR forecast: use the HAR model (Chapter 6) to produce $\hat{\text{RV}}_{t+h}$, a more sophisticated forecast.
- $\widehat{\text{VRP}}_t > 0$: implied variance exceeds expected realized variance (normal; VRP is positive on average).
- $\widehat{\text{VRP}}_t < 0$: rare, but occurs when realized vol spikes above what was priced in (e.g., crash events).

Why This Matters

This is the equation you will compute every day in a VRP trading strategy. Your ML model replaces the naive backward-looking RV with a more accurate $\hat{\text{RV}}_{t+h}^{\mathbb{P}}$, producing a cleaner VRP signal. A HAR or XGBoost forecast that improves QLIKE by 5–10% sharpens the VRP estimate that drives every downstream trading decision.

Ex-Ante vs. Ex-Post VRP

Two variants appear in the literature and they measure different things:

- **Ex-ante VRP**: $\text{VIX}_t^2 - \hat{\text{RV}}_{t,t+h}^{\mathbb{P}}$, where $\hat{\text{RV}}$ is a *forecast* of future variance (e.g., from HAR). This is available in real time and can be used as a trading signal.
- **Ex-post VRP**: $\text{VIX}_t^2 - \text{RV}_{t,t+h}$, where $\text{RV}_{t,t+h}$ is the *actual* realized variance over the next 30 days. This is only known after the fact and cannot be used for trading.

Bollerslev et al. (2009b) use the ex-ante version with backward-looking RV. Bekaert and Hoerova (2014) show that the choice of $\mathbb{P}$ -measure proxy matters for return predictability. Always state which version you are using.

9.2 Why VRP Exists

The previous section showed that VRP is positive on average: VIX-squared systematically exceeds realized variance. Why? If markets were risk-neutral (if investors did not care about risk), then $\mathbb{Q} = \mathbb{P}$ and VRP would be zero. But investors are risk-averse, and that risk aversion creates the premium.

Three forces contribute:

9.2.1 Risk Aversion and Downside Protection Demand

Risk-averse investors are willing to overpay for insurance against large losses. Pension funds, endowments, and portfolio managers routinely buy put options and volatility protection even though these instruments lose money on average. They accept this negative expected return because the protection pays off precisely when their portfolios suffer most.

This is not irrational. A 40% drawdown can trigger margin calls, force liquidation, or violate regulatory capital requirements. The cost of ruin is asymmetric: a dollar lost in a crash is more painful than a dollar gained in calm markets. Paying a small insurance premium to avoid catastrophic outcomes is rational for constrained investors, even if the premium exceeds the actuarial cost.

9.2.2 The Insurance Analogy, Formalized

Return to the fire insurance example. The homeowner pays \$1,200/year for a policy with an expected loss of \$400. The insurer earns the \$800 difference. But the insurer also bears the risk of correlated claims (a wildfire that burns an entire neighborhood). The premium compensates for this systematic, non-diversifiable risk.

In the options market:

- Hedgers (pension funds, portfolio managers) are the homeowners buying protection.
- Options sellers (market makers, hedge funds) are the insurers collecting the premium.
- The VRP is the \$800 gap: compensation for bearing volatility risk, especially in its most extreme (crash) form.

9.2.3 Equilibrium Account: Long-Run Risk

[Drechsler and Yaron \(2011\)](#) provide a formal equilibrium explanation. In their model, the representative agent has *Epstein-Zin preferences* (a generalization of standard utility that separates risk aversion from the willingness to substitute consumption over time). Uncertainty about future economic growth fluctuates over time: there are periods when the economy's growth rate is predictable and periods when it is not. This "volatility of volatility" in the real economy generates a variance risk premium in asset markets.

Why VRP Is Positive

VRP exists because:

1. **Risk aversion:** investors overpay for crash protection relative to its actuarial value.
2. **Non-diversifiable risk:** volatility spikes coincide with market crashes, making volatility risk systematic (not diversifiable away).
3. **Uncertainty about uncertainty:** in equilibrium models like [Drechsler and Yaron \(2011\)](#), time-varying economic uncertainty generates a positive VRP.

The premium is not a market inefficiency but rational compensation for bearing a risk that hurts most when it materializes.

9.3 VRP Predicts Returns

Options sellers collect the VRP as an insurance premium, but the premium also carries information: it predicts future equity returns. This is the central empirical finding of [Bollerslev et al. \(2009b\)](#).

The logic: when VRP is high, the market is demanding steep compensation for bearing volatility risk. This signals unusually high risk aversion or uncertainty. Assets priced under high risk aversion offer higher expected returns as compensation. When VRP is low, the market is complacent, and expected returns are lower.

Bollerslev et al. (2009b): VRP Predicts Quarterly Equity Returns

[Bollerslev et al. \(2009b\)](#) regress future quarterly S&P 500 excess returns on the ex-ante VRP and find:

- The VRP coefficient is positive and statistically significant: higher VRP today predicts higher equity returns over the next quarter.
- A one-standard-deviation increase in VRP predicts higher quarterly excess returns.
- Using simple backward-looking RV, the R^2 for quarterly return prediction is about 4%; using a HAR-based expected variance proxy, R^2 exceeds 15%, substantially exceeding the dividend yield and other popular predictors at the quarterly horizon.
- The predictability is concentrated at the quarterly (3–6 month) horizon and fades at longer horizons.

They operationalize VRP in two ways: a simple version using $VIX_t^2 - RV_{t-22,t}$ (backward-looking monthly RV), and an expected variance premium using a HAR-RV forecast as the $\mathbb{P}$ -measure proxy.

Worked Example:

VRP as a Return Predictor In March 2020, VIX hit 82.7 while trailing realized vol was around 30%. VRP was roughly $(0.827)^2 - (0.30)^2 = 0.684 - 0.090 = 0.594$, an extreme level. The BTZ framework would predict very high future returns, and indeed the S&P 500 rallied over 40% in the subsequent two quarters.

VRP Is Not a Timing Tool

VRP predicts returns in a statistical, population-average sense. Even the best specification ($R^2 \approx 15\%$) leaves 85% of quarterly return variation unexplained. A high VRP does *not* guarantee positive returns in any single quarter. VRP is highest during crises, precisely when portfolio constraints and drawdown pain are most severe. Trading on VRP requires the ability and willingness to take risk when it feels worst.

9.4 VRP Predicts Future Volatility

VRP also contains information about future realized volatility. The mechanism is mean reversion.

When VRP is large (implied far above realized), two things tend to happen:

1. Realized volatility rises toward implied. The options market is pricing in higher future vol for a reason: upcoming risk events, deteriorating conditions, or high uncertainty. Realized vol tends to catch up.

2. Implied volatility falls toward realized. As the risk event passes or uncertainty resolves, the fear premium deflates.

The net effect is convergence: the gap closes from both sides, but with realized vol doing more of the adjustment.

The mean-reversion channel also explains a practical finding from volatility forecasting: including VIX (or VIX-squared) alongside lagged RV in a HAR-type model (Chapter 6) improves out-of-sample forecasts. VIX carries forward-looking information that backward-looking RV alone does not capture. The VRP quantifies how much extra forward-looking information VIX contributes beyond what lagged RV already tells you.

9.5 Decomposing VRP

The headline VRP is a single number, but it mixes together different sources of risk. Two decompositions from the literature isolate the components.

9.5.1 Uncertainty vs. Risk Aversion

Bekaert and Hoerova (2014) decompose the VIX-squared into two pieces:

$$\text{VIX}_t^2 = \underbrace{\mathbb{E}_t^{\mathbb{P}}[\text{RV}_{t,t+h}]}_{\text{expected variance}} + \underbrace{\text{VRP}_t}_{\text{variance risk premium}} \quad (9.3)$$

Rearranging Equation (9.1); the insight is in how each component behaves:

- $\mathbb{E}_t^{\mathbb{P}}[\text{RV}_{t,t+h}]$: the **expected variance** component, which captures physical uncertainty about future returns. When economic conditions deteriorate, this rises.
- VRP_t : the **risk premium** component, which captures the market's risk aversion and demand for protection. When fear rises faster than objective uncertainty, this rises.

In Plain English

When VIX is high, this decomposition asks: is VIX high because the world is genuinely risky right now (the expected variance piece), or because investors are unusually scared and demanding extra compensation for bearing that risk (the VRP piece)? The same VIX level can mean very different things depending on which component is driving it. Separating the two tells you whether the market is pricing objective danger or subjective fear.

Why This Matters

Do not feed raw VIX-squared as a single feature in your RV forecasting model: it mixes a signal that reflects genuine future risk (useful for forecasting RV) with one that reflects risk aversion (useful for forecasting returns, not RV). Instead, split it via your HAR or ML $\mathbb{P}$ -measure proxy into expected-variance and VRP components and use them as distinct features.

Bekaert and Hoerova (2014): Return Predictability Lives in the VRP Component

Bekaert and Hoerova (2014) find that:

- The VRP component significantly predicts future equity returns (consistent with

Bollerslev et al. 2009b).

- The expected-variance component does *not* predict returns. High objective uncertainty alone does not imply high future returns; it is the risk-aversion markup that carries the signal.
- The expected-variance component predicts future realized volatility (naturally, since it is a forecast of RV).
- Different $\mathbb{P}$ -measure proxies (backward-looking RV, GARCH forecasts, HAR forecasts) yield materially different VRP estimates and different predictive power. The choice of proxy is not innocuous.

9.5.2 Normal-Times vs. Jump-Tail VRP

Bollerslev and Todorov (2011) decompose the VRP along a different dimension: the portion attributable to “normal” continuous fluctuations vs. the portion attributable to rare, large jumps (tail events).

Normal vs. Tail VRP (,)

$$\text{VRP}_t = \underbrace{\text{VRP}_t^{\text{diffusive}}}_{\text{normal-times premium}} + \underbrace{\text{VRP}_t^{\text{tail}}}_{\text{jump-tail premium}} \quad (9.4)$$

- $\text{VRP}_t^{\text{diffusive}}$: the premium for bearing day-to-day, continuous variance risk. This component is relatively stable and modest.
- $\text{VRP}_t^{\text{tail}}$: the premium for bearing rare, large-jump risk (crashes). This component is highly time-varying and spikes during stress.

Bollerslev and Todorov (2011) find that the tail component drives most of the time-variation in the aggregate VRP. During calm markets, the tail premium is small and the overall VRP is moderate. During crises, the tail premium explodes, reflecting the market’s intense fear of further crashes.

Why This Matters

This decomposition connects directly to jump detection from Chapter 4. The bipower variation and jump test statistics you computed there separate realized variance into continuous and jump components. The same separation applied to the risk premium side tells you whether the market is overpricing jump risk or diffusive risk. If your ML model can forecast the jump component of RV better than the continuous component (or vice versa), you know which piece of the VRP you are best positioned to harvest. A model that excels at predicting jump arrivals would target the tail VRP; a model that excels at forecasting the smooth diffusive component would target the normal-times VRP.

9.6 The Gamma P&L Formula: From Forecast to Money

Required Background

This section requires delta and gamma from Chapter 8 (the Black–Scholes section), variance swap strike mechanics (Section 8.8), and the VRP definition (Section 9.1 earlier in this chapter).

You have built an ML model that forecasts RV more accurately than the options market implies.

How much money does that improved forecast generate? This section derives the exact formula linking forecast accuracy to trading profit.

9.6.1 Setup: Delta-Hedged Option P&L

Consider a trader who buys an option at implied vol σ_i and delta-hedges continuously. Each day, the option's value changes due to two effects: (a) the delta-hedged P&L from the stock's realized move, and (b) time decay (theta). Because the trader maintains a delta-neutral position, the first-order stock exposure cancels. What remains is the second-order (gamma) exposure to realized moves versus the cost of carrying that exposure (theta).

9.6.2 Derivation from the Black–Scholes PDE

The starting point is the Black–Scholes PDE, which describes the equilibrium between time decay and gamma exposure when volatility equals the implied level. This equation governs the “fair cost” of holding a delta-hedged option:

$$\Theta + \frac{1}{2}\Gamma S^2\sigma_i^2 = rV \quad (9.5)$$

- Θ : theta, the option's time decay (dollars lost per day from the passage of time).
- Γ : gamma, the option's second-order sensitivity to the stock price.
- S : current stock price.
- σ_i : implied volatility (the market's priced-in vol).
- r : risk-free rate; V : option value.

For a delta-hedged book, the stock component nets out, leaving theta as the “cost of gamma.”

In reality, the stock moves with realized volatility σ_r , not implied volatility σ_i . The actual P&L per infinitesimal time step for a delta-hedged long option position is:

$$d(\text{P\&L}) = \frac{1}{2}\Gamma S^2(r_t^2 - \sigma_i^2) dt \quad (9.6)$$

- r_t : the log-return over the interval dt .
- r_t^2 : the realized variance contribution for that period.
- $\sigma_i^2 dt$: the implied variance “charged” by the market over the same period.

Over the full life of the option, the cumulative hedging P&L is:

$$\text{Total P\&L} = \sum_{t=1}^N \frac{1}{2}\Gamma_t S_t^2(\sigma_{r,t}^2 - \sigma_i^2) \Delta t \quad (9.7)$$

- N : total number of hedging intervals over the option's life.
- Γ_t, S_t : gamma and stock price at each rebalance, which change as the stock moves.
- $\sigma_{r,t}^2$: realized variance in interval t ; σ_i^2 : the constant implied variance you paid for.

In Plain English

The Black–Scholes PDE tells you what an option is “worth” if the stock moves exactly at implied vol. Equations (9.6) and (9.7) capture what happens when reality disagrees with the market's assumption. Each day, you earn the difference between what the stock actually did (r_t^2) and what the option charged you for ($\sigma_i^2 dt$), scaled by your gamma exposure. Over the life of the option, these daily differences accumulate. If realized vol consistently exceeds implied, the long-gamma trader profits; if realized vol falls short, the short-gamma trader (vol seller) profits.

For an at-the-money option where gamma remains roughly constant over the period, this simplifies to:

$$\text{P\&L} \approx \frac{1}{2} \Gamma S^2 (RV^2 - IV^2) T \quad (9.8)$$

where RV^2 and IV^2 are the annualized realized and implied variances, and T is the time period in years.

The Gamma P&L Formula

The daily hedging P&L for a delta-hedged option position is:

$$\text{Daily Hedging P\&L} = \frac{1}{2} \Gamma S^2 (\sigma_{\text{realized}}^2 - \sigma_{\text{implied}}^2) \Delta t \quad (9.9)$$

Positive gamma (long options) profits when realized vol exceeds implied vol. **Negative gamma** (short options) profits when realized vol is below implied vol. The VRP ($IV > RV$ on average) means selling gamma is systematically profitable: you collect theta that, on average, exceeds the realized moves you pay out.

Figure 9.1 shows the P&L of a delta-hedged long option position as a function of realized volatility relative to implied volatility.

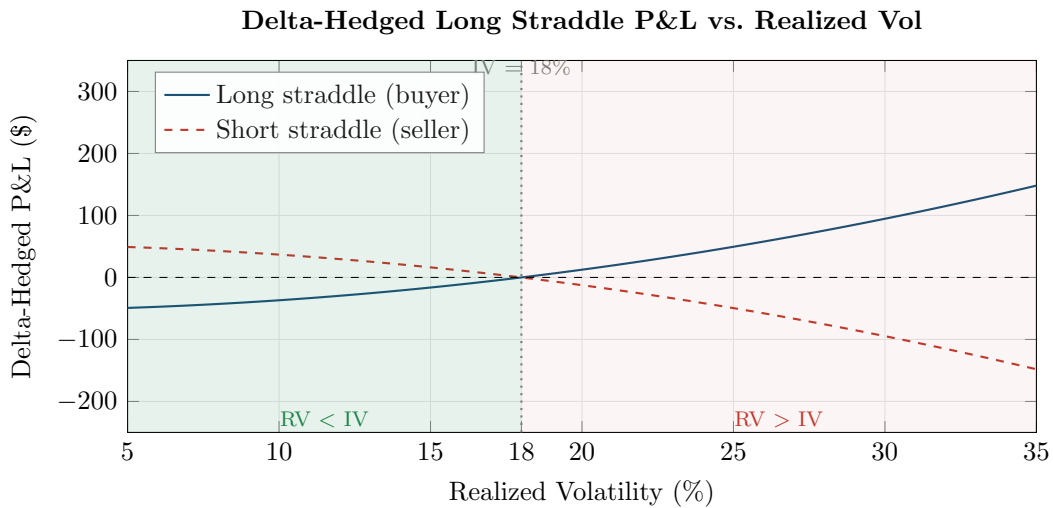


Figure 9.1: P&L of a delta-hedged straddle as a function of realized volatility, assuming $IV = 18\%$, $\Gamma = 0.04$, $S = 100$, $T = 30$ days. The seller (dashed red) profits in the green-shaded region where realized vol falls below implied vol, which is the typical VRP regime. The buyer (solid blue) profits only when realized vol exceeds implied. Since VRP is positive roughly 85% of the time, the seller has a statistical edge.

Worked Example: Delta-Hedged Straddle: Five Days of Gamma P&L

Setup. Buy a 30-day ATM straddle on a stock at $S = 100$, with implied volatility $IV = 18\%$. The approximate ATM gamma for this option is $\Gamma = 0.04$. Delta-hedge daily. Over 5 days, the stock follows the path: 100, 101.2, 99.8, 100.5, 102.1, 101.0.

Parameters. The daily implied variance charge is:

$$\sigma_i^2 \cdot \Delta t = \frac{(0.18)^2}{252} = \frac{0.0324}{252} = 0.0001286$$

Computation. Each day, compute the log-return $r_t = \ln(S_t/S_{t-1})$, the squared return

r_t^2 (realized variance for that day), and the gamma P&L:

$$\text{P\&L}_t = \frac{1}{2} \times 0.04 \times S_t^2 \times (r_t^2 - 0.0001286)$$

Day	S_t	r_t (%)	$r_t^2 (\times 10^{-4})$	$\sigma_i^2 \Delta t (\times 10^{-4})$	Gamma P&L (\$)
1	101.2	+1.19	1.423	1.286	+0.003
2	99.8	-1.39	1.941	1.286	+0.013
3	100.5	+0.70	0.489	1.286	-0.016
4	102.1	+1.58	2.495	1.286	+0.025
5	101.0	-1.08	1.173	1.286	-0.002
Cumulative (5 days)					+0.023

Interpretation. The annualized realized volatility over these 5 days was approximately 19.5%, above the 18% implied vol we paid. The cumulative gamma P&L is positive (\$0.023 per share, or \$2.30 per standard 100-share contract). On days where the squared return exceeded $\sigma_i^2 \Delta t$ (days 1, 2, 4), we profited; on quiet days (days 3, 5), we lost to theta.

Path Dependence: Gamma Is Not Constant

Gamma is *not* constant. As the stock moves away from the strike, gamma falls. Two stock paths with identical 30-day realized vol can produce very different hedging P&L because gamma was high during the low-vol days and low during the high-vol days. This is why forecasting *when* vol occurs (intraday patterns, jump timing) matters, not just the average level. A model that predicts the same total RV but correctly identifies which days will be volatile is worth more than a model that gets only the level right.

Why This Matters

For your internship evaluation: a 5% QLIKE improvement in your RV forecast means you can identify days when the market misprices vol by more. In a vol-trading context, this translates to approximately 2–5 bps per unit of vega exposure per day (order of magnitude). The exact number depends on your gamma profile and hedging frequency. Section 18.1 quantifies this via a simpler mechanism (vol-targeting Sharpe ratio improvement).

9.7 Vol-of-Vol

VIX measures the expected volatility of the S&P 500. But VIX itself is volatile. The “volatility of VIX” captures a distinct risk factor: uncertainty about the level of future uncertainty.

9.7.1 VVIX: The Volatility of VIX

VIX as an Underlying

Chapter 8 defined VIX as the model-free implied volatility of the S&P 500 over 30 days. VIX itself is a traded index: there are options written on VIX (VIX options, traded at Cboe). You can apply the same model-free variance extraction method from Chapter 8 to VIX options to get the implied volatility of VIX. This is VVIX.

VVIX

VVIX is the model-free implied volatility of VIX, computed from VIX options using the same methodology as VIX itself (Cboe Exchange, 2019).

- VVIX is quoted in annualized percentage points (like VIX).
- Typical range: 80–120 in calm markets, spiking to 150+ during stress.
- Interpretation: VVIX = 100 means the options market prices VIX’s annualized volatility at 100%.

Concretely, “same methodology as VIX” means the following. Recall from Chapter 8 that the VIX formula extracts model-free implied variance from a strip of out-of-the-money options. VVIX applies this identical formula to *VIX options* rather than S&P 500 options.

To compute VVIX, you need two expiries of VIX options that bracket a 30-day horizon. For each expiry $j \in \{1, 2\}$, compute the single-term implied variance:

$$\sigma_j^2 = \frac{2}{T_j} \sum_i \frac{\Delta K_i}{K_i^2} e^{R_j T_j} Q_j(K_i) - \frac{1}{T_j} \left(\frac{F_j}{K_{0,j}} - 1 \right)^2, \quad (9.10)$$

where:

- T_j : time to expiration of the j -th VIX option series (in years).
- K_i : strike price of the i -th out-of-the-money VIX option (calls for $K_i > K_{0,j}$, puts for $K_i < K_{0,j}$, both at $K_{0,j}$).
- ΔK_i : half the distance between the strikes on either side of K_i .
- $Q_j(K_i)$: midpoint of the bid-ask spread of the VIX option at strike K_i .
- F_j : forward VIX level implied by put-call parity: $F_j = K_{\text{ATM}} + e^{R_j T_j} (C_{\text{ATM}} - P_{\text{ATM}})$.
- $K_{0,j}$: first strike at or below F_j .
- R_j : risk-free rate to expiry j .

Then interpolate to a constant 30-day maturity:

$$\text{VVIX} = 100 \times \sqrt{\left[T_1 \sigma_1^2 \frac{N_{T_2} - N_{30}}{N_{T_2} - N_{T_1}} + T_2 \sigma_2^2 \frac{N_{30} - N_{T_1}}{N_{T_2} - N_{T_1}} \right] \times \frac{N_{365}}{N_{30}}}, \quad (9.11)$$

where N_{T_j} is the number of minutes to expiry j , $N_{30} = 43,200$ (minutes in 30 days), and $N_{365} = 525,600$ (minutes in 365 days).

In Plain English

The interpolation in Equation (9.11) blends the near-term and next-term results to produce a constant 30-day measure, ensuring VVIX is always comparable across dates regardless of the options expiry calendar.

Why This Matters

You do not need to compute VVIX yourself; Cboe publishes it daily. But understanding the formula matters because it reveals what VVIX captures: the cost of insuring against VIX moves, aggregated across all strikes. When VVIX is high, the market expects VIX to swing violently, which predicts fatter tails in realized volatility changes. Include VVIX as a feature column for RV forecasting at horizons of 1–5 days, where its predictive power is strongest.

9.7.2 Realized Vol-of-Vol

You can also compute a backward-looking measure: the realized volatility of the VIX time series itself. This is constructed exactly like RV from Chapter 2, but applied to VIX returns rather than S&P 500 returns:

$$RV_t^{\text{VIX}} = \sum_{i=1}^n (\Delta \ln \text{VIX}_{t,i})^2 \quad (9.12)$$

- $\Delta \ln \text{VIX}_{t,i}$: the i -th intraday log return of VIX on day t .
- This measures how much VIX itself fluctuated within the day.

9.7.3 Jumps in VIX

VIX is known for sudden spikes: it can double in a single day during a market crash. These VIX jumps represent a distinct risk factor beyond the level of VIX. A market with VIX at 20 that is stable is very different from a market with VIX at 20 that just fell from 40 (or is about to spike to 40). The vol-of-vol measures capture this instability.

9.7.4 Diagram: VIX vs. Realized Vol

Figure 9.2 shows the persistent gap between VIX and subsequent realized volatility over a long sample.

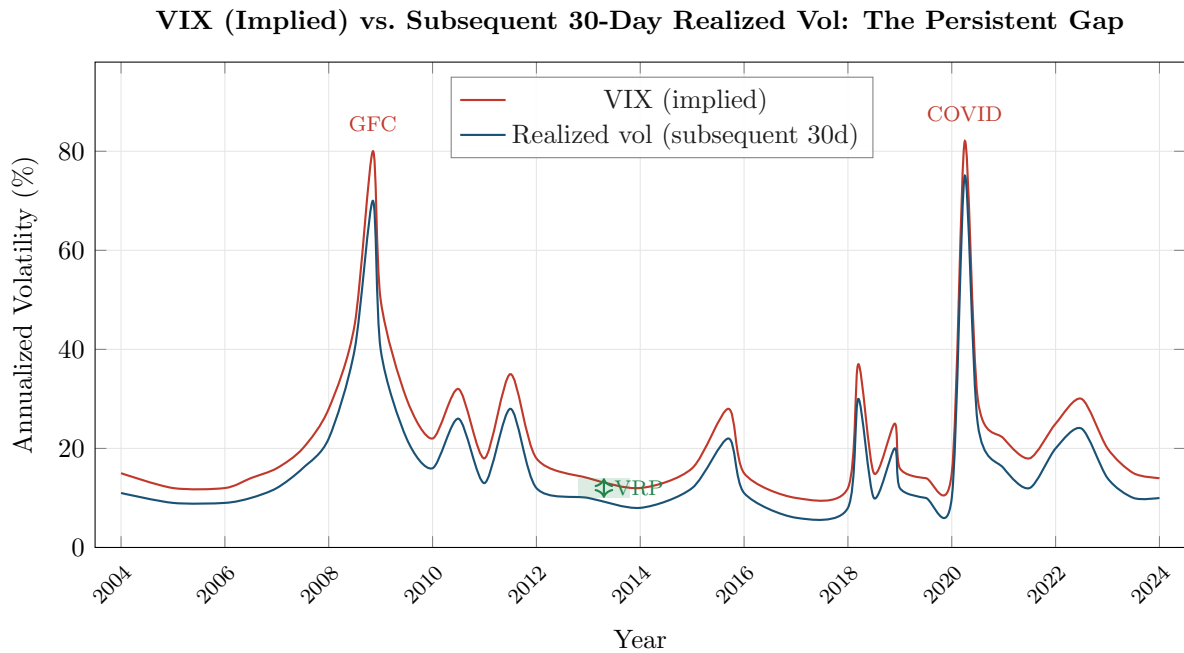


Figure 9.2: VIX (red) vs. subsequent 30-day realized volatility (blue), 2004–2024 (schematic). The persistent gap between the two lines is the variance risk premium. VIX almost always exceeds subsequent realized vol. The gap widens during crises (2008, 2020) when the demand for protection surges. The green shaded region illustrates the VRP during a calm period.

The key visual pattern: the red line (VIX) sits above the blue line (realized vol) nearly everywhere. The gap is the VRP. It widens dramatically during crises and narrows during calm markets, but it almost never flips negative for extended periods.

Carr and Wu (2009a) document that VIX exceeds subsequent realized vol roughly 85% of the

time. The 15% of months where realized vol exceeds VIX are concentrated in sudden-onset crises, when realized vol spikes before the options market fully adjusts.

9.8 ML Approaches to VRP

The traditional VRP literature uses simple proxies (backward-looking RV, GARCH forecasts) for the $\mathbb{P}$ -measure expectation. ML methods can improve this proxy, and the VRP itself can serve as a feature or trading signal in ML pipelines.

9.8.1 Improved $\mathbb{P}$ -Measure Forecasts

Fouhy (2024) proposes a hierarchical XGBoost approach to the VRP pipeline:

1. **Stage 1:** Train an XGBoost model to forecast realized variance RV_{t+h} using lagged RV, HAR-style features, and additional predictors (macro indicators, sentiment, past VIX). This produces a high-quality $\hat{RV}_{t+h}^{\mathbb{P}}$.
2. **Stage 2:** Compute VRP as $VIX_t^2 - \hat{RV}_{t+h}^{\mathbb{P}}$.
3. **Stage 3:** Use the VRP (and its components) as input features for a second-stage model that predicts returns or constructs a trading signal.

Figure 9.3 illustrates the full pipeline from RV forecast to trading decision.

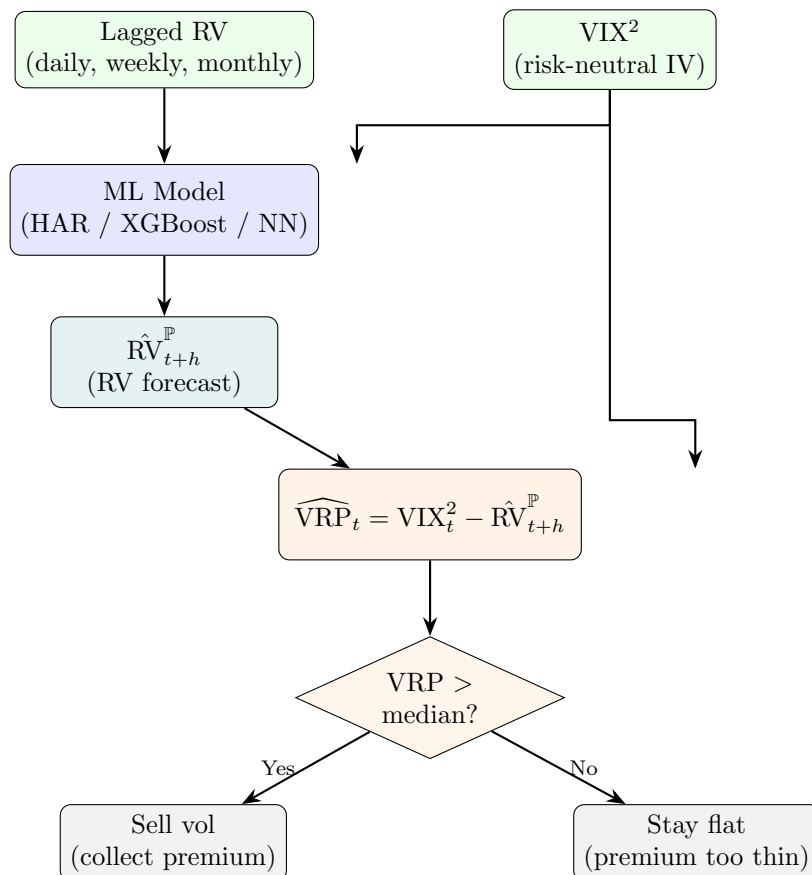


Figure 9.3: The VRP trading pipeline. Your ML model forecasts realized variance (left path), which is compared to the options market’s implied variance (VIX-squared) to compute the VRP signal. The signal drives the trading decision: sell volatility when the premium is rich, stay flat when it is thin.

9.8.2 VRP as a Trading Signal

The most direct use of VRP is as a signal for a delta-hedged volatility strategy. The idea is simple:

- When VRP is high (implied $\gg$ realized): *sell* volatility (sell options, collect the premium). The options market is overpricing future vol, so you profit as realized vol comes in below implied.
- When VRP is low or negative (implied $\approx$ or $<$ realized): *buy* volatility or stay flat. The premium is thin or inverted, and the risk-reward of selling options is poor.

A concrete implementation: sell a 30-day delta-hedged straddle on the S&P 500 when the ex-ante VRP exceeds its trailing median, and flatten the position otherwise. Delta-hedging removes directional exposure, so the trade profits or loses based purely on realized vs. implied variance. This is Project 5 in Chapter 18.

Short Volatility Carries Tail Risk

Selling volatility is the most reliable way to harvest the VRP, but it is inherently a strategy with negative skewness: small, frequent gains and rare, large losses. VRP harvesting strategies lost 20–40% in the October 2008 crash and the March 2020 COVID sell-off. ML-based regime detection (Chapter 10) and position sizing are essential to manage this tail risk.

Summary

- The **variance risk premium** (VRP) is the difference between risk-neutral expected variance ($\mathbb{Q}$, measured by VIX-squared) and physical expected variance ($\mathbb{P}$, measured by realized variance or a model-based forecast).
- VRP is **positive on average**: VIX overstates subsequent realized vol about 85% of the time (Carr and Wu, 2009a). This gap is compensation for bearing volatility risk.
- The standard operational measure is $\widehat{\text{VRP}}_t = (\text{VIX}_t/100)^2 - \hat{\text{RV}}_{t,t+h}^{\mathbb{P}}$, where $\hat{\text{RV}}$ is either backward-looking RV or a model forecast (e.g., HAR).
- **VRP exists** because of risk aversion, demand for crash protection, and equilibrium pricing of uncertainty about future growth (Drechsler and Yaron, 2011).
- **VRP predicts returns**: Bollerslev et al. (2009b) show that VRP predicts quarterly equity excess returns with R^2 of 4–15% (depending on the $\mathbb{P}$ -measure proxy), beating the dividend yield as a return predictor.
- **VRP predicts volatility**: high VRP signals that realized vol will likely rise toward implied vol through mean reversion.
- **Decomposition 1** (Bekaert and Hoerova, 2014): $\text{VIX}^2 = \text{expected variance} + \text{VRP}$. Return predictability lives in the VRP component, not the expected-variance component.
- **Decomposition 2** (Bollerslev and Todorov, 2011): $\text{VRP} = \text{normal-times premium} + \text{jump-tail premium}$. Most time-variation in VRP comes from the tail component (crash fear).
- **VVIX** measures the implied volatility of VIX itself (vol-of-vol). It captures uncertainty about the future level of uncertainty, a distinct risk factor.
- **Realized vol-of-vol** is computed by applying the RV estimator from Chapter 2 to VIX returns. VIX jumps are a distinct risk factor beyond the VIX level.
- The persistent gap between VIX and realized vol (Figure 9.2) is the visual signature of the VRP. It widens in crises and narrows in calm markets but rarely flips negative.
- **ML approaches** (Fouhy, 2024): tree ensembles can improve the $\mathbb{P}$ -measure forecast, yielding a cleaner VRP estimate. VRP is also a powerful feature for downstream ML

models (Chapter 10).

- **VRP as a trading signal:** sell volatility when VRP is high, flatten when it is low. This is a direct, implementable strategy (Project 5, Chapter 18), but it carries significant tail risk.
- Ex-ante VRP (using a forecast for $\mathbb{P}$) is available in real time and can be traded. Ex-post VRP (using actual future RV) is only known after the fact. Always state which version you are using.
- The choice of $\mathbb{P}$ -measure proxy (backward-looking RV, GARCH, HAR, XGBoost) materially affects VRP estimates and their predictive power (Bekaert and Hoerova, 2014).

Table 9.1: Key results referenced in this chapter.

Paper	Result	Relevance
Bollerslev et al. (2009b)	VRP predicts quarterly equity excess returns with R^2 of 4–15% (depending on $\mathbb{P}$ -proxy), beating the dividend yield.	Establishes VRP as a top return predictor; motivates its use as an ML feature.
Drechsler and Yaron (2011)	Long-run risk model with Epstein-Zin preferences generates a positive VRP in equilibrium through time-varying economic uncertainty.	Provides the theoretical foundation for why VRP exists.
Bekaert and Hoerova (2014)	Decompose VIX^2 into expected variance and VRP; return predictability resides in the VRP component, not in expected variance.	Clarifies that risk aversion, not objective uncertainty, drives return predictability.
Bollerslev and Todorov (2011)	Decompose VRP into normal-times and jump-tail components; the tail premium drives most time-variation.	Most of VRP is crash-fear compensation, not day-to-day variance risk.
Carr and Wu (2009a)	Document that VIX exceeds subsequent realized vol $\sim 85\%$ of the time; provide a model-free framework for variance risk premia across assets.	Quantifies the frequency and magnitude of the VRP.
Fouhy (2024)	Hierarchical XGBoost for VIX-to-RV-to-VRP pipeline; ML-based $\mathbb{P}$ -measure forecasts improve VRP estimation.	Connects VRP to ML methods used in Chapters 10–18.

Part IV

ML Methods for Volatility

Chapter 10

Feature Engineering for Volatility

Application

This chapter is the bridge between volatility theory (Chapters 1–9) and ML modeling (Chapters 11–14). Every feature described here is a column in the input matrix that tree models, neural networks, and hybrid methods will consume. The feature set you choose matters more than the model you choose. Projects 1–5 all draw their inputs from this catalog.

We have spent nine chapters building a toolkit of volatility estimators, noise corrections, jump decompositions, parametric models, long-memory specifications, options-implied measures, and risk premia. Now we pivot from *measuring* volatility to *predicting* it with machine learning. The raw material for any ML model is its feature matrix $\mathbf{X} \in \mathbb{R}^{T \times p}$, where each row is a date and each column is a predictor. This chapter catalogs the features that the literature and Kaggle competitions have found useful, organizes them into families, and flags the pitfalls that make or break a forecasting pipeline.

10.1 Feature Taxonomy

Figure 10.1 groups every feature family into five branches. You do not need all of them for every project; the tree is a menu, not a checklist.

Figure 10.2 shows how raw market data flows through the feature engineering pipeline into the model. Every feature must pass through the temporal alignment gate: it can only use data available at or before time t to predict the target at time $t + 1$.

10.2 Triple Expansion: Level, Change, Z-Score

A systematic expansion technique applies to every continuous feature in the catalog. For any base quantity x_t , construct three variants:

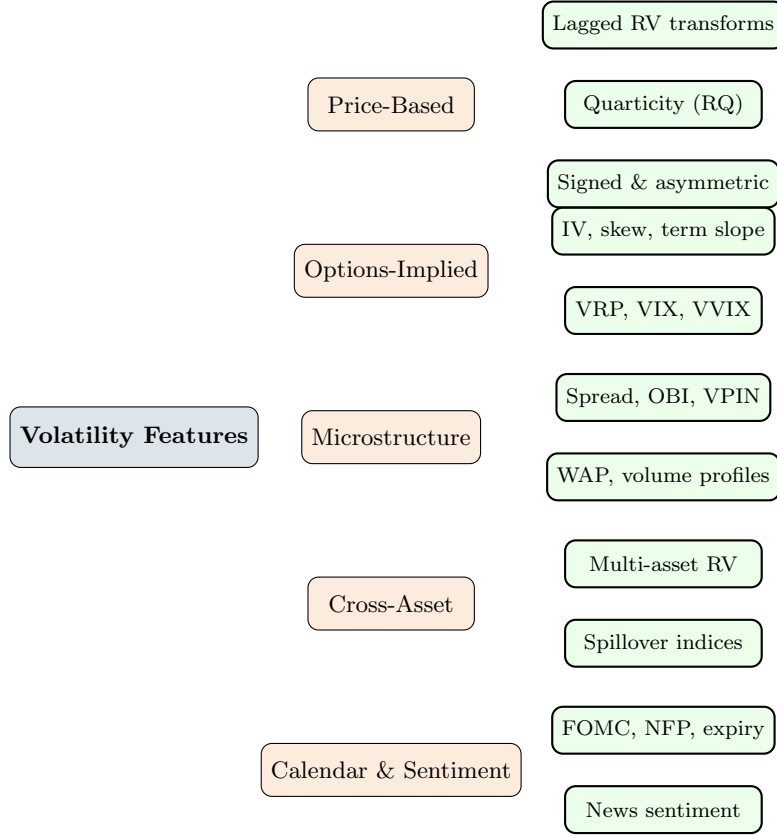


Figure 10.1: Feature taxonomy for volatility forecasting. Price-based features (left) are available for any asset with intraday data. Options-implied features require a liquid derivatives market. Microstructure features need tick-level LOB data. Cross-asset and calendar/sentiment features supplement any core set.

Triple Expansion

Given a continuous feature x_t and a lookback window w (typically 22 trading days), the **triple expansion** produces:

$$x_t^{\text{level}} = x_t, \quad (10.1)$$

$$x_t^{\text{change}} = x_t - x_{t-k}, \quad k \in \{1, 5\}, \quad (10.2)$$

$$x_t^{\text{z-score}} = \frac{x_t - \bar{x}_{t,w}}{\hat{\sigma}_{x,t,w}}, \quad (10.3)$$

where $\bar{x}_{t,w} = \frac{1}{w} \sum_{i=0}^{w-1} x_{t-i}$ is the trailing mean and $\hat{\sigma}_{x,t,w}$ is the trailing standard deviation over the same window.

- x_t^{level} : the raw value of the feature, capturing the current **state**.
- x_t^{change} : the first difference over k periods, capturing **momentum** (is the feature rising or falling?).
- $x_t^{\text{z-score}}$: the standardized deviation from the recent mean, capturing **anomaly** (is the current value unusual?).

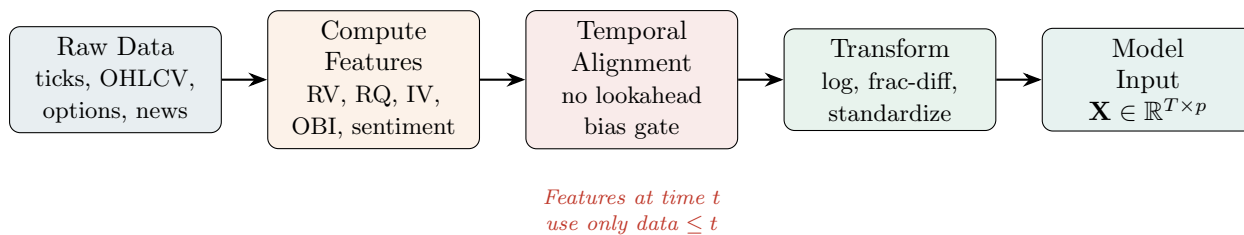


Figure 10.2: Feature engineering pipeline. Raw market data is processed into features, passed through a temporal alignment gate to prevent lookahead bias, transformed for stationarity, and assembled into the model input matrix $\mathbf{X}$.

In Plain English

Consider the bid-ask spread. Its level tells you how wide the spread is right now. Its change tells you whether the spread is widening (deteriorating liquidity) or narrowing (improving liquidity). Its z-score tells you whether today's spread is abnormally wide relative to its own recent history. A spread of 5 cents means very different things for a stock that typically trades at 3 cents (z-score ≈ 2 , alarm) versus one that typically trades at 6 cents (z-score ≈ -1 , calm). Each variant captures genuinely different information.

Which features get expanded. The triple expansion applies to *continuous* features: RV, RQ, bid-ask spread, OBI, implied volatility, VRP, volume ratios, and any numeric quantity with a meaningful scale. It does *not* apply to categorical or binary features such as FOMC dummies, day-of-week indicators, or earnings announcement flags. These are already discrete and cannot be meaningfully z-scored.

Multicollinearity: a non-issue for trees. In a linear regression, including level, change, and z-score of the same base feature would introduce severe multicollinearity (the three are highly correlated). This would inflate standard errors and produce unstable coefficient estimates. For tree-based models (random forests, gradient boosting), multicollinearity is harmless: trees select the most informative split at each node and are unaffected by correlated alternatives. Neural networks are also largely robust to correlated inputs, especially with regularization. The triple expansion is therefore a “free” way to enrich the feature set for non-linear models.

Why This Matters

In the vol-project-ref feature composition (Chapter 8), the triple expansion is identified as a core engineering principle: it is the primary reason feature counts grow from ~ 20 base features to ~ 80 model inputs.

10.3 Lagged RV Transforms

The single most predictive feature for tomorrow's realized volatility is yesterday's realized volatility. The HAR model (Chapter 6) already exploits this by using three horizons that smooth RV over different lookback windows:

$$RV_t^{(d)} = RV_t, \quad (10.4)$$

$$RV_t^{(w)} = \frac{1}{5} \sum_{i=0}^4 RV_{t-i}, \quad (10.5)$$

$$RV_t^{(m)} = \frac{1}{22} \sum_{i=0}^{21} RV_{t-i}. \quad (10.6)$$

- $RV_t^{(d)}$: daily RV, the single most recent observation.
- $RV_t^{(w)}$: weekly RV, the average of the past 5 trading days.
- $RV_t^{(m)}$: monthly RV, the average of the past 22 trading days.

Why This Matters

These three features are the columns your HAR baseline consumes. Every ML model you build should include $\log RV_t^{(d)}$, $\log RV_t^{(w)}$, and $\log RV_t^{(m)}$ as core inputs. The goal of fancier feature engineering is to add value on top of these, not to replace them. If an ML model cannot beat HAR using only these three features, the extra complexity is not justified.

These three variables alone explain roughly 40–60% of the variation in next-day log-RV for equity indices. For ML models, you should include all three, but also consider the following transforms.

Log-RV. Working with $\log RV_t$ rather than RV_t has two benefits:

- It stabilizes the variance of the target (variance of variance is highly skewed).
- It makes the distribution closer to Gaussian, which helps linear baselines and loss-function calibration.

Most HAR papers estimate the model in logs; you should do the same with your feature columns.

Square-root RV. An intermediate transform: $\sqrt{RV_t}$ approximates realized *volatility* (standard deviation), which is more interpretable than variance and less compressed than log.

Signed-magnitude transform. The square-root transform $\sqrt{x}$ is only defined for non-negative quantities such as RV_t itself. But many of the most informative features in this chapter are *signed*: they can be strongly negative as well as positive. The simplest example you have already met is a 5-day return, which can be +3% or −3%; the cumulative signed return over the past 5 days is what we will call **signed momentum**. Several features we will meet later in this chapter are signed too (signed jumps J_t^- and J_t^+ from Equation 10.11, realized skewness $RSkew_t$ from Equation 10.12, and order flow imbalance OFI_k from Equation 10.23). For all of these we want a transform that compresses large magnitudes the way $\sqrt{\cdot}$ does, yet preserves the sign that carries the leverage-effect information. The **signed-magnitude transform** (also called the signed square-root) does exactly this:

$$x_t^{\text{sgn}} = \text{sign}(x_t) \underbrace{|x_t|^{1/2}}_{\text{compressed magnitude}}, \quad \text{sign}(x_t) = \begin{cases} +1 & x_t > 0 \\ 0 & x_t = 0 \\ -1 & x_t < 0. \end{cases} \quad (10.7)$$

- where x_t is any signed feature (e.g. a signed jump difference $J_t^+ - J_t^-$, a 5-day signed return, realized skewness, or OFI), $|x_t|^{1/2}$ compresses large magnitudes, and $\text{sign}(x_t)$ re-attaches the direction.

In Plain English

Think of the signed-magnitude transform as “square root, but it remembers which way the move went.” A raw signed feature such as the signed-jump difference is heavy-tailed: a single crash-day value can be ten times larger than a typical day, and that one observation will dominate any linear baseline or distance-based model. Taking $\sqrt{|x|}$ pulls that extreme value back toward the pack, so the feature stops being a near-binary “crash / no crash” switch and becomes a smooth, graded signal. The power $1/2$ pulls big numbers down hard ($\sqrt{100} = 10$) while barely touching small ones ($\sqrt{1} = 1$), which is exactly the compression we want; plain $|x|$ would not compress at all. A value of exactly zero stays zero, so a feature that is usually zero (like a signed jump on a calm day) is left untouched. Re-attaching the sign keeps the asymmetry intact: a compressed -1.4% down-jump still reads as bad volatility (the downside kind that risk managers fear) and a compressed $+1.4\%$ up-jump still reads as good volatility (we formalize good vs. bad volatility in Section 10.5 later in this chapter).

Why This Matters

Apply the signed-magnitude transform to every signed feature before it enters a linear or ridge-HAR baseline, where heavy tails inflate coefficient variance and a handful of crisis days can flip the sign of an estimate. The signed jump difference $J_t^+ - J_t^-$, signed 5-day momentum, and realized skewness are the prime candidates: each is dominated by a few extreme days, and the leverage effect (Chapter 5) means the *sign* is precisely the part you must preserve. Tree models are scale-invariant (they only care about the *order* of values, not their size, so shrinking outliers does not change which way a split goes) and do not strictly need it, but it still helps gradient boosting place cleaner split points away from the tail. As with every continuous feature, you can then layer the triple expansion (Section 10.2) on top of x_t^{sgn} .

Ratio features. The ratio $\text{RV}_t^{(d)} / \text{RV}_t^{(m)}$ captures how today’s volatility compares to its recent average. Values above 1 indicate a local spike; values well below 1 indicate calm. This is a simple but effective regime indicator.

Worked Example: Building HAR-style features

Suppose the last 22 daily RV values (most recent first) are:

$$\text{RV}_t = 1.8 \times 10^{-4}, \quad \text{RV}_{t-1} = 1.5 \times 10^{-4}, \quad \dots$$

and you compute $\text{RV}_t^{(w)} = 1.6 \times 10^{-4}$, $\text{RV}_t^{(m)} = 1.2 \times 10^{-4}$.

Your feature row for date t contains:

- $\log \text{RV}_t^{(d)} = \log(1.8 \times 10^{-4}) \approx -8.62$
- $\log \text{RV}_t^{(w)} = \log(1.6 \times 10^{-4}) \approx -8.74$
- $\log \text{RV}_t^{(m)} = \log(1.2 \times 10^{-4}) \approx -9.03$
- Ratio: $\text{RV}_t^{(d)} / \text{RV}_t^{(m)} = 1.50$ (volatility currently 50% above its monthly average)

The ratio signals an elevated-volatility regime.

10.4 Realized Quarticity and the HARQ Feature

You learned in Chapter 2 that realized variance is a noisy estimator: its sampling error depends on the fourth moment of returns. The realized quarticity (RQ) quantifies exactly this noise.

Realized Quarticity

Given n intraday returns $r_{t,i}$ on day t ,

$$\text{RQ}_t = \frac{n}{3} \sum_{i=1}^n r_{t,i}^4. \quad (10.8)$$

Under standard diffusion assumptions, $\text{RQ}_t \xrightarrow{p} \int_0^1 \sigma_s^4 ds$ as $n \rightarrow \infty$.

- n is the number of intraday returns (e.g., 78 for 5-minute bars in a 6.5-hour session).
- $r_{t,i}^4$ is the fourth power of each intraday return; this heavily up-weights large moves.
- The factor $n/3$ provides the correct asymptotic scaling.

In Plain English

Realized quarticity measures how “wild” the intraday price path was. By raising each return to the fourth power, extreme moves dominate the sum. A day with one large intraday swing followed by calm will have high RQ even if overall RV is moderate. In short, RQ tells you how much you should trust today’s RV number: high RQ means the RV estimate is noisy and unreliable.

Why RQ matters for feature engineering. The asymptotic variance of RV_t is proportional to RQ_t . When RQ is high, today’s RV number is unreliable. [Bollerslev et al. \(2016b\)](#) exploit this insight in the HARQ model: they interact the daily RV component with $\sqrt{\text{RQ}_t}$, allowing the model to down-weight noisy days.

The HARQ Feature

In the HARQ model (Chapter 6), the coefficient on $\text{RV}_t^{(d)}$ is made state-dependent:

$$\beta_{d,t} = \beta_d + \beta_{dQ} \sqrt{\text{RQ}_t}.$$

The product $\text{RV}_t^{(d)} \cdot \sqrt{\text{RQ}_t}$ is itself a feature. When RQ_t is large, the interaction term pulls the effective weight on daily RV toward zero, letting the more stable weekly and monthly averages dominate the forecast. For ML models, include both $\text{RV}_t^{(d)}$ and $\sqrt{\text{RQ}_t}$ as separate columns; the tree or network can learn the interaction without you specifying it.

Why This Matters

The HARQ interaction feature is the single most important extension beyond baseline HAR for your project. [Bollerslev et al. \(2016b\)](#) show it improves QLIKE by 5–15% across equity indices.

10.5 Signed and Asymmetric Features

Volatility is not symmetric. Negative returns drive future volatility up more than positive returns of the same magnitude (the leverage effect you met in Chapter 5). Several features capture this asymmetry directly from high-frequency data.

Realized semivariances. To separate the contribution of upward and downward price moves

to total volatility, we split intraday returns by sign:

$$RV_t^+ = \sum_{i=1}^n r_{t,i}^2 \mathbf{1}(r_{t,i} > 0), \quad (10.9)$$

$$RV_t^- = \sum_{i=1}^n r_{t,i}^2 \mathbf{1}(r_{t,i} < 0). \quad (10.10)$$

- RV_t^+ : realized semivariance from positive returns (“good volatility”).
- RV_t^- : realized semivariance from negative returns (“bad volatility”).
- $\mathbf{1}(\cdot)$: indicator function selecting returns of the specified sign.

In Plain English

Total RV treats a 2% up-move and a 2% down-move identically. Semivariances break this symmetry. RV^- isolates the volatility coming from declines, which is the part that matters most for risk management and future vol prediction. RV^+ captures the “calm” upside moves. Keeping them separate lets a model learn that a day dominated by selling pressure has different forecasting implications than a day dominated by buying.

Why This Matters

Include RV_t^- and RV_t^+ (and their weekly/monthly averages) as separate feature columns. If you must prune features, keep RV_t^- and drop RV_t^+ before dropping other features.

By construction, $RV_t = RV_t^+ + RV_t^-$ (ignoring zero returns). The SHAR model in Chapter 6 uses RV_t^+ and RV_t^- as separate regressors, allowing different persistence for up-moves and down-moves.

Patton and Sheppard 2015b

Replacing total RV with the signed pair (RV_t^+, RV_t^-) in the HAR framework reduces out-of-sample QLIKE loss by 3–8% for equity index volatility. The coefficient on RV_t^- is roughly twice that on RV_t^+ , confirming the leverage effect at intraday frequency.

Signed jumps. From Chapter 4, the jump component is $J_t = \max(RV_t - BPV_t, 0)$. You can further decompose:

$$J_t^+ = \sum_i r_{t,i}^2 \mathbf{1}(r_{t,i} > 0, |r_{t,i}| > \theta_t), \quad J_t^- = \sum_i r_{t,i}^2 \mathbf{1}(r_{t,i} < 0, |r_{t,i}| > \theta_t), \quad (10.11)$$

where θ_t is an intraday jump threshold (e.g., from Lee-Mykland, Chapter 4).

- J_t^+ : squared returns from large positive moves exceeding the jump threshold.
- J_t^- : squared returns from large negative moves exceeding the jump threshold.
- θ_t : intraday threshold distinguishing jumps from continuous price variation.

In Plain English

Signed jumps isolate the extreme tail events and ask: did the big intraday moves come from sudden rallies or sudden crashes? A day dominated by negative jumps (a flash crash, a surprise rate hike) behaves very differently from a day dominated by positive jumps (a short squeeze, a surprise earnings beat). Splitting jumps by sign gives the model a finer-grained view of tail risk than total jump variation alone.

Why This Matters

In HAR-CJ extensions, including J_t^- separately improves QLIKE by 1–3%.

Realized semicovariances. Bollerslev et al. (2020) extend semivariances to the multivariate case, computing covariances conditional on the sign of the market return. For single-asset forecasting, the key insight is that the downside component of any cross-asset covariance feature (Chapter 15) carries more information than the upside component.

Why negatives matter more

When prices fall, leveraged investors face margin calls, hedging demand spikes, and correlations rise. The mechanism is mechanical, not behavioral: a short gamma position that loses money forces the holder to sell more, amplifying volatility. Features that isolate this downside channel capture a structural driver, not a statistical accident.

10.6 Higher Moments

Realized skewness and kurtosis measure the shape of the intraday return distribution. They are computed from higher powers of intraday returns:

$$\text{RSkew}_t = \frac{\frac{1}{n} \sum_{i=1}^n r_{t,i}^3}{\left(\frac{1}{n} \sum_{i=1}^n r_{t,i}^2\right)^{3/2}}, \quad (10.12)$$

$$\text{RKurt}_t = \frac{\frac{1}{n} \sum_{i=1}^n r_{t,i}^4}{\left(\frac{1}{n} \sum_{i=1}^n r_{t,i}^2\right)^2}. \quad (10.13)$$

- RSkew_t : realized skewness, measuring asymmetry of intraday returns (negative = more large down-moves).
- RKurt_t : realized kurtosis, measuring tail heaviness (higher = more extreme moves relative to typical).
- Both are normalized by powers of RV so they are scale-free.

Modest Predictive Power

Realized skewness and kurtosis are noisy estimators and have shown only modest incremental forecasting power for next-day RV in most studies. Include them in your initial feature set, but do not be surprised if tree-based importance scores rank them low. Their main use is as conditioning variables: when skewness is very negative, the volatility process may behave differently (regime-dependent dynamics).

10.7 Microstructure and Limit Order Book Features

If you have tick-level or LOB snapshot data (common in Kaggle competitions like Optiver’s Realized Volatility Prediction), you unlock a family of features that daily or 5-minute data cannot provide.

Bid-ask spread. The quoted spread $s_t = p_t^{\text{ask}} - p_t^{\text{bid}}$ is a proxy for liquidity and informed-trading intensity. Wider spreads predict higher near-term volatility. Use the time-weighted average spread across the observation window.

Order book imbalance (OBI). The balance of resting orders on each side of the book reveals directional pressure. OBI formalizes this:

$$\text{OBI}_t = \frac{V_t^{\text{bid}} - V_t^{\text{ask}}}{V_t^{\text{bid}} + V_t^{\text{ask}}}, \quad (10.14)$$

- V_t^{bid} : total visible volume resting at the best bid price.
- V_t^{ask} : total visible volume resting at the best ask price.
- OBI ranges from -1 (all volume on the ask side) to $+1$ (all on the bid side).

In Plain English

OBI measures who is more eager to trade. Extreme imbalance in either direction signals one-sided pressure that tends to resolve through a price move, and price moves create volatility.

Why This Matters

The absolute value $|\text{OBI}_t|$ is often more useful than signed OBI for volatility forecasting, since extreme imbalance in either direction predicts elevated vol.

Weighted average price (WAP) log returns. The WAP blends bid and ask prices, weighting each side by the opposite side's depth to reflect where the “true” price likely sits:

$$\text{WAP}_t = \frac{p_t^{\text{bid}} \cdot V_t^{\text{ask}} + p_t^{\text{ask}} \cdot V_t^{\text{bid}}}{V_t^{\text{bid}} + V_t^{\text{ask}}}. \quad (10.15)$$

- $p_t^{\text{bid}}, p_t^{\text{ask}}$: best bid and ask prices.
- $V_t^{\text{bid}}, V_t^{\text{ask}}$: volume at the best bid and ask.
- The cross-weighting pulls WAP toward the side with less volume (where the next fill is more likely).

In Plain English

The simple midprice treats both sides of the book equally, but if there are 10,000 shares on the bid and only 100 on the ask, the next trade is far more likely to happen at the ask. WAP accounts for this asymmetry by weighting each price by the opposite side's volume, pulling the estimated fair value toward the thinner side of the book. Returns computed from WAP are less contaminated by bid-ask bounce.

Why This Matters

When computing RV from LOB data, use WAP-based log returns rather than midprice returns as your raw input. This reduces microstructure noise at the source, before any noise-robust estimator is applied, giving you a cleaner RV target and cleaner lagged-RV features. Top Optiver competition solutions all used WAP returns for exactly this reason.

Price acceleration. Beyond first-order returns, the second difference of log-prices captures changes in momentum within the observation window:

$$a_{t,i} = \Delta \log(\text{WAP}_{t,i}) - \Delta \log(\text{WAP}_{t,i-1}). \quad (10.16)$$

- $\Delta \log(\text{WAP}_{t,i})$: the log return between consecutive WAP observations at times $i - 1$ and i .
- $a_{t,i}$: the change in the log return, analogous to acceleration in physics.

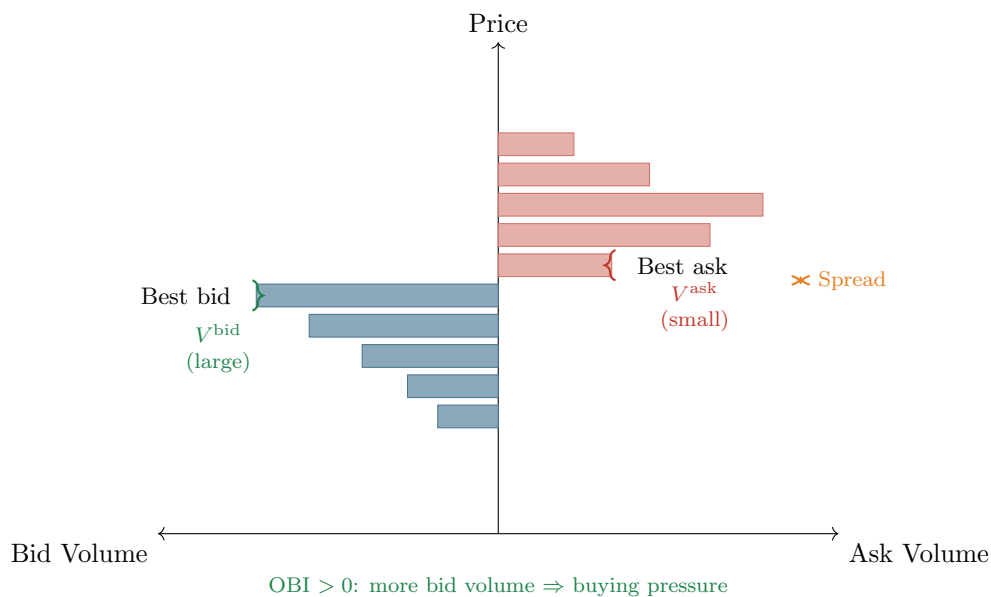


Figure 10.3: Limit order book snapshot with feature annotations. The spread (orange) is the gap between best bid and best ask. OBI compares bid volume (blue, left) to ask volume (red, right). In this snapshot, OBI is positive (more resting bid volume), suggesting net buying pressure.

In Plain English

If the price was falling and then starts falling faster, that is negative acceleration. If the price was falling and then slows down, that is positive acceleration (deceleration). Aggregating the magnitude of these acceleration values over a window captures how “jerky” the price path is. A smooth trending market has low acceleration; a choppy market with rapid reversals has high acceleration. This jerkiness is a leading indicator of near-term volatility.

Why This Matters

This was one of the highest-importance features in top Optiver competition solutions. It captures information orthogonal to RV (which uses only first differences), providing the model with a signal about momentum instability that raw RV misses.

Volume profiles. Aggregate volume into time buckets (e.g., 2-minute bins within a 10-minute window). The ratio of volume in the last bucket to the first captures whether activity is accelerating or decelerating. Volume acceleration tends to precede volatility spikes.

10.7.1 VPIN and Kyle’s Lambda

Two microstructure features capture the intensity of *informed trading* rather than just the state of the order book. Both originate from market microstructure theory and measure how much private information is currently flowing into the market.

VPIN (Volume-Synchronized Probability of Informed Trading). Standard time-based sampling mixes high-activity and low-activity periods. [Easley et al. \(2012\)](#) propose an alternative: partition the trading day into **volume buckets**, each containing the same total volume V_{bucket} . Within each bucket, classify trades as buyer- or seller-initiated (typically using the tick rule or bulk volume classification), and compute the absolute imbalance.

The VPIN algorithm proceeds in three steps. First, partition trades into n equal-volume buckets, where each bucket contains exactly $V_{\text{bucket}} = V_{\text{day}}/n$ shares:

$$\text{VPIN}_t = \frac{1}{n} \sum_{\tau=1}^n \frac{|V_{\tau}^B - V_{\tau}^S|}{V_{\text{bucket}}}, \quad (10.17)$$

- V_{τ}^B : volume classified as buyer-initiated in bucket τ .
- V_{τ}^S : volume classified as seller-initiated in bucket τ ($V_{\tau}^B + V_{\tau}^S = V_{\text{bucket}}$).
- n : number of volume buckets per estimation window (typically 50).
- VPIN ranges from 0 (perfectly balanced flow) to 1 (completely one-sided flow).

In Plain English

VPIN asks: “In each chunk of trading volume, how one-sided was the flow?” When informed traders are active, they trade aggressively on one side, creating persistent imbalances. Uninformed traders, by contrast, are roughly equally likely to buy or sell, so their flow approximately cancels out. High VPIN therefore signals that informed traders are dominating the flow, which tends to precede large price moves and elevated volatility. The key innovation is measuring in *volume time* rather than clock time: a bucket fills quickly during high-activity periods and slowly during calm periods, automatically adapting the sampling rate to market conditions.

Why This Matters

VPIN is most valuable at the intraday and 1-day horizons, where it captures the arrival of informed order flow before it fully resolves into price moves. At longer horizons, VPIN’s signal decays because the information has already been incorporated into prices.

Kyle’s lambda (λ): price impact per unit of order flow. Kyle (1985) shows that in a market with a single informed trader and competitive market makers, the equilibrium pricing rule is linear in aggregate order flow. The slope of this relationship, **Kyle’s lambda**, measures the market’s **price impact** per unit of signed volume:

$$\Delta p_t = \alpha + \underbrace{\lambda}_{\text{price impact}} \cdot \underbrace{(\text{signed volume})_t}_{\text{buy vol} - \text{sell vol}} + \varepsilon_t. \quad (10.18)$$

- Δp_t : change in the midprice over interval t .
- λ : Kyle’s lambda, the slope coefficient. A higher λ means each unit of net order flow moves the price more (the market is less liquid or more informationally sensitive).
- Signed volume: the difference between buyer-initiated and seller-initiated volume (positive = net buying).

In practice, estimate λ by regressing midprice changes on signed volume over a rolling window (e.g., 60 five-minute intervals):

$$\Delta p_{t,i} = \hat{\alpha} + \hat{\lambda}_t \cdot (V_{t,i}^B - V_{t,i}^S) + \hat{\varepsilon}_{t,i}, \quad i = 1, \dots, m, \quad (10.19)$$

where m is the number of intraday intervals in the rolling window.

In Plain English

Kyle’s lambda answers: “How many basis points does the price move for every million dollars of net order flow?” When λ is high, even modest buying or selling pressure moves

prices substantially – the market is “thin” or carrying a lot of private information. When λ is low, the market can absorb large orders without significant price impact – it is deep and liquid. Rising λ predicts higher near-term volatility because it signals that order flow is becoming more informative (or the market is becoming less resilient), both of which lead to larger price swings.

Why This Matters

Estimate $\hat{\lambda}_t$ on a rolling 60-interval window using 5-minute data and include it as a feature column. $\hat{\lambda}_t$ captures a different dimension of market quality than the bid-ask spread: the spread measures the cost of a small trade, while λ measures the cost of a *large* trade. During stress periods, λ spikes before the spread widens, making it a leading indicator.

Microstructure features are asset-specific

Features like OBI and spread depend heavily on the market’s microstructure (tick size, maker/taker fees, minimum lot sizes). A feature that works for liquid US equities may be meaningless for an illiquid commodity future. Always recalibrate when switching asset classes.

10.7.2 Amihud Illiquidity Ratio

The features above (OBI, spread, WAP) require tick-level or LOB data. The **Amihud illiquidity ratio** (Amihud, 2002) provides a liquidity measure that requires only daily returns and dollar volume – data available for any traded asset.

Amihud Illiquidity

For stock i on day d , the daily illiquidity ratio is the absolute return per dollar of trading volume:

$$\text{ILLIQ}_{i,t} = \frac{1}{D_t} \sum_{d=1}^{D_t} \frac{|r_{i,d}|}{\text{DVOL}_{i,d}} \times 10^5, \quad (10.20)$$

where:

- $r_{i,d}$: the return of stock i on day d .
- $\text{DVOL}_{i,d}$: the dollar trading volume on day d (price $\times$ shares traded).
- D_t : the number of trading days in the averaging window (e.g., 22 for monthly).
- The 10^5 scaling factor prevents the ratio from being extremely small (returns of order 10^{-2} divided by volume of order 10^6 – 10^9).

In Plain English

The Amihud ratio answers a simple question: “How much does the price move per dollar of trading activity?” This is exactly Kyle’s concept of market depth, but measured with data available for any publicly traded asset – no tick data, no LOB snapshots, just daily returns and volume. When ILLIQ spikes, the market is having trouble absorbing order flow, which reliably precedes periods of elevated realized volatility.

Why This Matters

Include $\log(1 + \text{ILLIQ}_{i,t})$ as a feature column for single-stock vol forecasting. The log transform compresses the heavy right tail (ILLIQ is highly skewed). For the E-mini,

ILLIQ is less informative since the contract is extremely liquid, but for the 30 mega-cap equities in your panel, cross-sectional variation in ILLIQ captures stock-specific liquidity regimes that predict idiosyncratic vol. Apply the triple expansion (level, change, z-score) as with any continuous feature. A rising Amihud ratio (increasing illiquidity) is a leading indicator of higher near-term volatility, especially for mid-cap stocks where liquidity can deteriorate rapidly.

10.7.3 Microprice and Volume Features

The simple midprice $(P_{\text{bid}} + P_{\text{ask}})/2$ assigns equal weight to both sides of the book regardless of depth. The **microprice** (Cartea et al., 2015) corrects for order-book imbalance by shifting the estimated fair value toward the thinner side of the book:

$$S_t^* = \frac{V_t^{\text{ask}} \cdot P_t^{\text{bid}} + V_t^{\text{bid}} \cdot P_t^{\text{ask}}}{V_t^{\text{bid}} + V_t^{\text{ask}}}. \quad (10.21)$$

- $P_t^{\text{bid}}, P_t^{\text{ask}}$: best bid and ask prices.
- $V_t^{\text{bid}}, V_t^{\text{ask}}$: total visible volume at the best bid and ask.
- When bid volume dominates, the microprice shifts toward the ask (fair value is higher).

Why This Matters

For microstructure-level projects, the deviation $|S^* - S_{\text{mid}}|$ between microprice and mid-price is itself a useful feature: it measures how asymmetric the book is. Large deviations indicate one-sided pressure that often precedes short-term volatility. Use microprice returns rather than midprice returns as the basis for computing short-horizon RV targets.

Volume features. Daily trading volume is highly correlated with daily volatility. More usefully for forecasting, volume is *persistent*: high-volume days tend to cluster, just like high-vol days. This means volume features can serve as leading indicators:

- $\log(\text{volume}/\text{MA}_{20}\text{volume})$: normalized volume; values above zero indicate above-average activity.
- Volume acceleration: change in log-volume over the past 3 days.
- Microprice–midprice deviation: $|S^* - S_{\text{mid}}|$ as a measure of book imbalance pressure.

Volume Predicts Vol

Abnormal volume today predicts elevated RV tomorrow, even after controlling for lagged RV. The intuition: volume spikes reflect new information arriving (earnings whispers, large block trades, portfolio rebalancing) that has not yet fully resolved into price moves. Include at least one volume feature in any feature set.

10.7.4 Order Flow Imbalance and Depth Features

The features above (OBI, spread, VPIN) treat the order book as a static snapshot. **Order flow imbalance (OFI)** (Cont et al., 2014) instead tracks how the book *changes* over time, capturing the dynamic pressure of order arrivals and cancellations.

Event-level contribution. Between any two consecutive order book observations $n - 1$ and n , exactly one thing happens: demand increases, demand decreases, supply increases, or supply decreases. The contribution of event n to order flow is:

$$e_n = \underbrace{\mathbf{1}_{P_n^B \geq P_{n-1}^B} q_n^B - \mathbf{1}_{P_n^B \leq P_{n-1}^B} q_{n-1}^B}_{\text{bid-side change}} - \underbrace{\mathbf{1}_{P_n^A \leq P_{n-1}^A} q_n^A + \mathbf{1}_{P_n^A \geq P_{n-1}^A} q_{n-1}^A}_{\text{ask-side change (negated)}}, \quad (10.22)$$

where:

- P_n^B, P_n^A : best bid and ask prices at observation n .
- q_n^B, q_n^A : queue sizes (shares) at the best bid and ask.
- $\mathbf{1}_{\{\cdot\}}$: indicator function.
- A market sell and a cancel-buy of the same size produce the same e_n , because both reduce the bid queue identically.

Interval-level OFI. Aggregate the event contributions over a time interval $[t_{k-1}, t_k]$:

$$\text{OFI}_k = \sum_{n \in [t_{k-1}, t_k]} e_n. \quad (10.23)$$

The price impact regression. Cont et al. (2014) show that mid-price changes are linear in OFI:

$$\Delta P_k = \beta \cdot \text{OFI}_k + \varepsilon_k, \quad (10.24)$$

where β is the **price impact coefficient** (basis points per unit of order flow) and ε_k is residual noise from deeper book levels. Estimated on 10-second intervals for 50 US equities, this simple regression achieves $R^2 \approx 65\%$ – far higher than trade-based measures ($R^2 \approx 32\%$ for signed trade imbalance). When both OFI and trade imbalance are included, trade imbalance becomes insignificant: OFI subsumes its information.

In Plain English

OBI (Section 10.7) asks “what does the book look like right now?” OFI asks “how is the book changing?” The distinction matters: a balanced book that is rapidly losing bid volume (market sells arriving, or limit buys being cancelled) has low OBI but strongly negative OFI. OFI captures the *flow* of information into the market, not just the *state* of the book, which is why it explains 65% of short-term price variation.

Why This Matters

For the E-mini L2 data, compute OFI over 10-second or 1-minute intervals and aggregate to your forecasting window (e.g., standard deviation or sum of squared OFI values over the observation window). The aggregated OFI variability is a volatility feature: days with large swings in order flow produce higher near-term RV. The price impact coefficient $\hat{\beta}$ itself is also a feature – it is inversely proportional to market depth, so rising $\hat{\beta}$ signals deteriorating liquidity and predicts higher volatility.

Depth ratio. When multi-level LOB data is available (as with E-mini L2), you can look beyond the best quotes. The **depth ratio** aggregates volume across multiple price levels:

$$\text{DR}_t^{(L)} = \frac{\sum_{\ell=1}^L V_t^{\text{bid}, \ell}}{\sum_{\ell=1}^L V_t^{\text{ask}, \ell}}, \quad (10.25)$$

where $V_t^{\text{bid}, \ell}$ and $V_t^{\text{ask}, \ell}$ are the resting volumes at the ℓ -th best bid and ask levels, and L is the number of levels included (typically 5 or 10).

Unlike OBI (which uses only the best quote), the depth ratio captures **structural imbalance** deeper in the book. A depth ratio far from 1 at levels 2–5 can signal informed positioning that has not yet reached the best quote, providing lead time over L1-only features.

Market urgency. The **market urgency** composite combines spread and imbalance into a single feature:

$$\text{Urgency}_t = s_t \times |\text{OBI}_t|, \quad (10.26)$$

where s_t is the bid-ask spread and $|\text{OBI}_t|$ is the absolute order book imbalance.

In Plain English

Market urgency captures when *both* conditions for a large price move are present simultaneously: a wide spread (the market is thin and vulnerable) and a strong imbalance (one side is dominating). Either condition alone is weaker: a wide spread with balanced flow just means low liquidity, and strong imbalance with a tight spread means the market can absorb the pressure. When both align, the next price move is likely to be large and fast.

Why This Matters

Include all three features – OFI variability, depth ratio, and market urgency – in your microstructure feature set. For the E-mini L2 data, use $L = 5$ for the depth ratio. For the 30 equities (L1 only), you can compute OBI and market urgency but not the multi-level depth ratio.

10.8 Options-Implied Features

Options prices embed the market's forward-looking expectation of volatility (Chapter 8) and the variance risk premium the market charges for bearing that risk (Chapter 9). This makes options-implied quantities natural predictors.

ATM implied volatility. The at-the-money IV for a given maturity is the most direct options-based feature. For forecasting 1-month RV, use the 30-day ATM IV. For daily RV, short-dated (weekly) options are more informative.

Skew (25-delta risk reversal). The risk reversal measures the difference in implied volatility between out-of-the-money calls and puts at matched delta:

$$\text{RR}_{25} = \text{IV}_{25\Delta\text{call}} - \text{IV}_{25\Delta\text{put}} . \quad (10.27)$$

- $\text{IV}_{25\Delta\text{call}}$: implied volatility of the 25-delta call (out-of-the-money upside).
- $\text{IV}_{25\Delta\text{put}}$: implied volatility of the 25-delta put (out-of-the-money downside).

In Plain English

The risk reversal tells you which side of the distribution the options market is more worried about. For equities, puts are almost always more expensive than calls (negative RR), reflecting the market's willingness to pay a premium for downside protection. When the skew becomes more negative than usual, the market is pricing in elevated crash risk. This forward-looking fear signal often precedes actual volatility increases.

Term structure slope. The slope of the IV term structure reveals whether the market expects volatility to increase or decrease over time:

$$\text{TS}_t = \text{IV}_t^{(3m)} - \text{IV}_t^{(1m)} . \quad (10.28)$$

- $\text{IV}_t^{(3m)}$: 3-month at-the-money implied volatility.
- $\text{IV}_t^{(1m)}$: 1-month at-the-money implied volatility.

In Plain English

Normally, longer-dated options cost more in vol terms because there is more time for uncertainty to accumulate (contango, $TS > 0$). When the term structure inverts ($TS < 0$), it means near-term IV exceeds longer-term IV, signaling that the market sees a current crisis that it expects to resolve. An inverted term structure is analogous to an inverted yield curve: it reflects stress in the near term that the market believes is temporary.

Why This Matters

Consider the term structure slope's interaction with lagged RV. When TS is negative (backwardation), the market has already priced in a vol spike, so your model should dampen its vol-up forecasts. When TS is positive and rising, the market expects calm to continue.

VRP proxy. The variance risk premium measures the gap between what the options market expects and what actually materializes. From Chapter 9:

$$VRP_t = \frac{VIX_t^2}{10,000} - \mathbb{E}_t[RV_{t+30}], \quad (10.29)$$

- $VIX_t^2/10,000$: implied variance from VIX, annualized and scaled to match RV units.
- $\mathbb{E}_t[RV_{t+30}]$: expected realized variance over the next 30 days, approximated by trailing 30-day RV.

In Plain English

VRP is the “insurance premium” the market charges for bearing volatility risk. When VRP is high, implied vol far exceeds realized vol, meaning options are expensive relative to what actually happens. This premium tends to mean-revert: when it is abnormally high, future implied vol is likely to fall (or realized vol is likely to rise) to close the gap. VRP captures information that neither lagged RV nor current IV alone provides.

Why This Matters

Include VRP as a feature column using the simple proxy $VIX_t^2/10,000 - RV_t^{(m)}$. Be careful about temporal alignment: the “expected” component must use only backward-looking data. Using a forward-looking RV estimate in the VRP proxy introduces lookahead bias. For Project 5 (VRP ML trader), this feature becomes the primary signal rather than just an input.

VIX and VVIX. The VIX itself is a feature for equity volatility. The VVIX (vol-of-vol, Chapter 9) captures uncertainty about volatility. High VVIX predicts fatter tails in the distribution of future RV changes.

VIX futures term structure. VIX futures trade at multiple maturities. The ratio of a far-dated VIX future to spot VIX is a powerful regime indicator:

$$VTS_t = \frac{F_t^{(3m)}(VIX)}{VIX_t}, \quad (10.30)$$

where $F_t^{(3m)}(VIX)$ is the 3-month VIX future and VIX_t is the spot VIX.

When $VTS_t > 1$ (**contango**), far-dated futures trade above spot VIX. This is the normal state: volatility mean-reverts, so the market anchors the far future closer to the long-run average than

the current (lower) spot level. When $VTS_t < 1$ (**backwardation**), spot VIX exceeds the far future. This signals a crisis: the market has spiked in the near term but expects the spike to be temporary.

The economic mechanism is mean reversion. As [Bennett \(2014\)](#) documents, the front-month VIX future has roughly 90% sensitivity (delta) to spot VIX, while the 6-month future has only 55%. When VIX spikes above 40, the 6-month future barely moves because the market does not expect the high-volatility environment to persist that long.

Forward implied volatility. The *additive variance rule* lets you extract the market's expectation for volatility between two future dates. Given ATM implied volatilities σ_1, σ_2 at maturities $T_1 < T_2$, the **forward volatility** from T_1 to T_2 is:

$$\sigma_{T_1 \rightarrow T_2} = \sqrt{\frac{\sigma_2^2 T_2 - \sigma_1^2 T_1}{T_2 - T_1}}. \quad (10.31)$$

- σ_1, σ_2 : ATM implied volatilities for maturities T_1 and T_2 .
- $\sigma_1^2 T_1$ and $\sigma_2^2 T_2$: total implied variances (accumulated variance to each maturity).
- $\sigma_{T_1 \rightarrow T_2}$: the implied volatility for the *forward period* between T_1 and T_2 only.

In Plain English

Total implied variance accumulates like distance: the total variance to 6 months equals the variance to 3 months plus the variance from 3 to 6 months. The forward vol formula “subtracts out” the near-term variance to isolate what the market expects for the later period. If 3-month IV is much higher than forward 3-to-6-month vol, the market is pricing a near-term event (FOMC, earnings) that will resolve before the forward period begins. This is a more refined feature than the raw term structure slope because it directly answers: “What does the market expect for each specific future window?”

Why This Matters

Include VTS (the VIX futures ratio) as a regime indicator feature. Values below 0.9 strongly predict elevated near-term RV, while values above 1.1 signal calm. Also compute forward volatility at matched horizons (e.g., 1-to-3 month forward vol for predicting 1-month-ahead RV) and include it alongside the raw term structure slope.

Options features are forward-looking

Every price-based feature (RV, RQ, signed jumps) is backward-looking: it summarizes what has already happened. Options-implied features are the only family that reflects the market's consensus about the future. When both are available, combining them almost always improves forecasts. The gains are largest at horizons beyond one week, where the informational advantage of options over recent realized quantities is greatest.

Why This Matters

For the options-implied feature family (ATM IV, RR_{25} , TS slope, VRP, VVIX), the gain over HAR-family baselines varies sharply by horizon. At the 1-day horizon, the gain is modest (1–3% QLIKE). At the 1-week and 1-month horizons, the gain can reach 5–10% because options encode forward-looking information that lagged RV cannot.

10.9 Cross-Asset Features

Volatility does not live in isolation. A shock in crude oil can spill over to energy equities, then to credit spreads, then to broad equity vol. Cross-asset features capture these transmission channels.

Multi-asset RV. For an equity single-name model, include sector-level RV and index-level RV as additional features. For an FX pair, include RV of correlated pairs and interest-rate volatility.

Spillover indices. Diebold and Yilmaz (2012) propose a variance decomposition of a VAR on a panel of volatilities, producing a total spillover index and directional “to” and “from” connectedness for each asset. A rising spillover index means that shocks are propagating more readily across the system.

Diebold-Yilmaz Spillover Index

Fit a VAR(p) to a vector of N volatility series. Compute the H -step forecast error variance decomposition θ_{ij}^H , normalized so each row sums to 1. The total spillover index is:

$$S^H = \frac{\sum_{i \neq j} \theta_{ij}^H}{\sum_{i,j} \theta_{ij}^H} \times 100. \quad (10.32)$$

- θ_{ij}^H is the fraction of the H -step forecast error variance of series i attributable to shocks from series j .
- When $i = j$, the contribution is “own”; when $i \neq j$, it is “cross.”
- S^H ranges from 0 (no spillovers) to 100 (all variance driven by cross-shocks).

In Plain English

The spillover index answers a simple question: what fraction of each asset’s volatility is coming from other assets rather than from its own dynamics? When the index is high, markets are tightly coupled and a shock anywhere propagates everywhere. When the index is low, each asset’s volatility is driven mainly by its own news. The index rises during crises (2008, COVID) and falls during calm, asset-specific regimes.

Why This Matters

For Project 3 (Multivariate RC with GNNs), the spillover index is a natural global feature. For single-asset forecasting, include the rolling spillover index and its 5-day change as feature columns. Rising connectedness warns that your single-asset model may face larger forecast errors because external shocks are more likely to contaminate your target asset’s volatility. This signal helps a tree model adjust its forecasts during contagion regimes.

In practice, you compute the spillover index on a rolling window (200 trading days is typical) and include its level and change as features. Rising connectedness forecasts higher broad-market volatility.

Curse of dimensionality

Adding 20 cross-asset RV features is tempting but dangerous when your sample is 1,000–2,000 daily observations. Either use PCA to reduce the cross-asset panel to 3–5 factors, or let a tree model’s built-in feature selection handle the pruning.

10.10 Long-Memory Features

Volatility has long memory: the autocorrelation of log-RV decays hyperbolically, not exponentially (Chapter 7). Standard differencing (Δ^1) removes long memory but also destroys the very persistence that predicts future levels. Fractional differencing provides a middle ground.

Fractional differencing. Standard first-differencing ($d = 1$) achieves stationarity but destroys the long memory that makes volatility predictable. Fractional differencing lets you dial the differencing order to a non-integer value, preserving some memory while achieving stationarity. Following [Lopez de Prado \(2018\)](#) (Chapter 5), define the fractional difference operator:

$$(1 - L)^d x_t = \sum_{k=0}^{\infty} \binom{d}{k} (-1)^k x_{t-k}, \quad (10.33)$$

- L : the lag operator ($Lx_t = x_{t-1}$).
- $d \in (0, 1)$: the fractional differencing order (a continuous knob between “no differencing” and “full differencing”).
- $\binom{d}{k}$: generalized binomial coefficients that produce exponentially decaying weights on past values.

In Plain English

Fractional differencing is a weighted average of the current value and all past values, where the weights decay slowly for small d and quickly for large d . At $d = 0$, you keep the raw series (full memory, possibly non-stationary). At $d = 1$, you get the standard first difference (stationary but memoryless). The trick is to find the smallest d that makes the series stationary, because that value preserves the maximum amount of predictive long-range dependence while satisfying the stationarity assumptions that many ML models require.

Why This Matters

Apply fractional differencing to $\log RV_t$ with $d \approx 0.35$ – 0.45 and include the result as a feature column. This is especially important for linear models and neural networks that assume stationary inputs. Tree models are less sensitive to non-stationarity, but fractionally differenced log-RV can still help by making the signal cleaner.

The binomial coefficients $\binom{d}{k}$ for non-integer d are:

$$\binom{d}{k} = \frac{d(d-1)(d-2) \cdots (d-k+1)}{k!}.$$

In practice, you truncate the infinite sum at a lag k^* where $|\binom{d}{k^*}|$ falls below a threshold (e.g., 10^{-4}).

- At $d = 0$: no differencing; the series retains all memory but may be non-stationary.
- At $d = 1$: standard first differencing; stationary but nearly all long-range dependence is destroyed.
- At $d \approx 0.35$ – 0.45 : the series passes the ADF test for stationarity while retaining most of its autocorrelation structure.

To find the right d in practice: grid-search over $d \in \{0.05, 0.10, \dots, 1.00\}$, apply fractional differencing to $\log RV_t$, and pick the smallest d for which the ADF test rejects at the 5% level.

Rolling Hurst exponent. From Chapter 7, the Hurst exponent H of log-RV can be estimated via the variogram. A rolling 60-day estimate of H captures time-varying roughness. When H

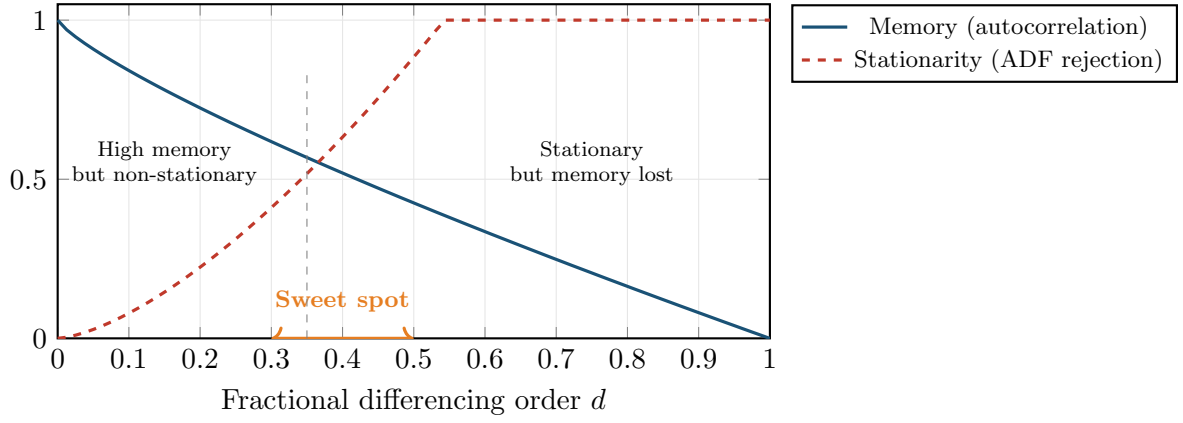


Figure 10.4: The fractional differencing tradeoff. As d increases from 0 to 1, the series becomes more stationary (dashed red, ADF rejection rate rises) but loses memory (solid blue, autocorrelation decays). The sweet spot around $d \in [0.3, 0.5]$ achieves stationarity while preserving enough memory for prediction.

drops below 0.1, the vol-of-vol process is particularly rough, and mean-reversion is fast. When H rises toward 0.3–0.4, persistence is higher and trends in volatility last longer.

Worked Example: Fractional differencing of log-RV

Let $d = 0.4$ and truncate at $k^* = 10$. The weights are:

$$w_0 = 1, \quad w_1 = -0.4, \quad w_2 = \frac{-0.4 \times (-1.4)}{2} = 0.28, \quad w_3 = \frac{0.28 \times (-2.4)}{3} = -0.224, \quad \dots$$

The fractionally differenced series is:

$$\tilde{x}_t = \sum_{k=0}^{10} w_k x_{t-k} = x_t - 0.4 x_{t-1} + 0.28 x_{t-2} - 0.224 x_{t-3} + \dots$$

Unlike $\Delta^1 x_t = x_t - x_{t-1}$, this weighted sum retains some of the level information from x_t while achieving stationarity.

10.10.1 Vol-of-Vol and Regime Duration

Fractional differencing captures long memory through the autocorrelation structure of the RV series itself. Two complementary features capture different aspects of memory: the *stability* of the volatility process and the *time elapsed* since the last regime change.

Vol-of-vol. The **volatility of volatility** measures how much the volatility process itself is fluctuating. It is computed as the rolling standard deviation of daily RV over a 22-day window:

$$\text{VoV}_t = \sqrt{\frac{1}{21} \sum_{i=0}^{21} (\text{RV}_{t-i} - \bar{\text{RV}}_{t,22})^2}, \quad (10.34)$$

- $\bar{\text{RV}}_{t,22} = \frac{1}{22} \sum_{i=0}^{21} \text{RV}_{t-i}$: the trailing 22-day mean of daily RV.
- VoV_t : standard deviation of daily RV over the past month, capturing how “unstable” the volatility regime is.

In Plain English

Vol-of-vol tells you whether volatility has been roughly constant over the past month or swinging wildly. During the 2020 COVID crash, both RV and vol-of-vol were extremely high. But in some regimes, RV is elevated but stable (a sustained high-vol period like mid-2022), meaning vol-of-vol is moderate even though RV is high. The distinction matters for forecasting: when vol-of-vol is high, your model should have wider prediction intervals because the volatility process is itself unpredictable. When vol-of-vol is low, volatility is clustered tightly around its current level, and the forecast is more reliable.

Why This Matters

Tree models can use VoV_t as a split variable to identify regimes where the standard HAR-style forecast (based on mean persistence) is less reliable. If the VVIX index is available (it is for SPX), it provides a forward-looking analog of VoV; when both are available, include both and let the model choose.

Regime duration. The **regime duration** feature counts the number of trading days since the last “volatility event,” defined as a day where RV exceeded its trailing mean by more than 2 standard deviations:

$$D_t = t - \max\{\tau \leq t : \text{RV}_\tau > \bar{\text{RV}}_{\tau,66} + 2\hat{\sigma}_{\text{RV},\tau,66}\}, \quad (10.35)$$

- $\bar{\text{RV}}_{\tau,66}$: the trailing 66-day (3-month) mean of daily RV at time τ .
- $\hat{\sigma}_{\text{RV},\tau,66}$: the trailing 66-day standard deviation of daily RV.
- D_t : number of trading days since the last 2-sigma RV spike.

In Plain English

Regime duration acts as a **mean-reversion clock**. After a volatility spike, there is typically a decay period where RV gradually returns to its long-run mean. The duration feature tells the model how far along this decay process we are. When D_t is small (say, 3 days since the last spike), mean reversion is still in progress and the forecast should remain elevated. When D_t is large (say, 45 days since the last spike), the market has been calm for a while, and the probability of a new spike is rising (vol clustering means calm periods do not last forever). Trees can learn this non-monotonic relationship: short durations predict high vol (mean-reversion tail), and very long durations predict moderately elevated vol (the calm-before-the-storm effect).

10.11 Calendar and Event Features

Certain days are systematically different.

Macro announcements. FOMC rate decisions, non-farm payroll (NFP) releases, and CPI prints are associated with elevated realized volatility. Encode these as binary dummies: $\mathbf{1}(\text{FOMC}_t)$, $\mathbf{1}(\text{NFP}_t)$, $\mathbf{1}(\text{CPI}_t)$. Also include the day before the event (anticipation effect) and the day after (digestion effect).

Earnings and corporate events. For single-stock volatility, earnings announcement dates dominate all other calendar effects. A binary earnings dummy plus the number of days until the next earnings date (“earnings countdown”) can be powerful.

Options expiry and quarter-end. Monthly options expiry (third Friday) and quarterly triplewitching dates show elevated intraday volume and can distort RV measurements. Quarter-end

rebalancing by institutional investors also creates temporary volatility.

Day-of-week effects. Mondays historically show higher volatility (three calendar days of news compressed into one trading day), but this effect has weakened over time.

Calendar features are supplements, not drivers

Each individual calendar dummy is weak. They rarely rank in the top 10 features by importance in a tree model. Their value is incremental: they help the model explain residual variation after the heavy-lifting features (lagged RV, options-implied measures) have done their work. Do not build a model primarily on calendar features.

10.11.1 Event-Driven Volatility: Beyond Binary Dummies

The basic calendar dummies above capture whether an event occurs. But events affect volatility in richer ways that better feature engineering can exploit.

Term structure kinks. Before a known event (e.g., FOMC on Wednesday), the IV term structure shows a characteristic kink: the option expiring just after the event has elevated IV (it straddles the uncertainty), while the option expiring just before has lower IV (the event is not included). The magnitude of this kink, the **event-implied vol**, quantifies how much extra vol the market expects from the event. Extract it as:

$$\sigma_{\text{event}} = \sqrt{\frac{T_2 \sigma_2^2 - T_1 \sigma_1^2}{T_2 - T_1}} \quad (10.36)$$

- T_1, T_2 : expiry times of two options that bracket the event ($T_1 < t_{\text{event}} < T_2$).
- σ_1, σ_2 : ATM implied volatilities for those expiries.
- σ_{event} : the implied vol attributable to the event alone.

In Plain English

This formula “subtracts out” the normal day-to-day volatility to isolate the extra uncertainty the market assigns to the event itself. By comparing two options that differ only in whether they include the event day, you can back out how much additional variance the event contributes.

Historical event-day RV ratios. Compute the ratio $\text{RV}(\text{event day})/\text{RV}(\text{surrounding days})$ historically for each event type. FOMC days typically show $1.5\text{--}2\times$ normal RV; NFP days show $1.3\text{--}1.5\times$. Use these ratios as multiplicative adjustments to your baseline forecast on event days.

Days-to-next-event features. Rather than a binary “is today an event,” encode continuous distance: `days_to_FOMC`, `days_since_FOMC`, `days_to_earnings`. This lets the model learn the anticipation buildup and post-event decay.

10.11.2 Calendar Proximity Measures

The binary dummies above answer a yes/no question: “Is today an FOMC day?” But volatility does not jump from zero to one on event day. It ramps up gradually in the days before and decays gradually in the days after. **Continuous proximity measures** capture this ramp by encoding the distance to the next event as a real-valued feature.

Calendar Proximity Feature

For a scheduled event of type e (FOMC, NFP, earnings, options expiry), define the **proximity measure**:

$$\text{prox}_{e,t} = \max(0, W_e - |t - t_e^{\text{next}}|), \quad (10.37)$$

where t_e^{next} is the date of the next occurrence of event e , and W_e is a window parameter (e.g., $W_e = 5$ trading days for FOMC, $W_e = 3$ for NFP).

- t_e^{next} : the next scheduled occurrence of event e after date t .
- W_e : the anticipation window in trading days.
- $\text{prox}_{e,t}$: a triangular function that ramps from 0 to W_e as the event approaches and falls back to 0 afterward.

In Plain English

A proximity measure says “FOMC is 3 days away” – and the model can learn that volatility starts compressing 2–3 days before FOMC (the “pre-FOMC drift”) and then expands sharply on announcement day. This anticipation ramp is invisible to binary dummies: 5 days before FOMC looks identical to 50 days before FOMC in a binary encoding, but the market’s behavior is measurably different. The continuous encoding lets tree models learn the full shape of the event response, not just the on/off state.

FOMC compression and expansion. Equity index volatility shows a well-documented pattern around FOMC meetings: RV is suppressed 1–3 days before the announcement as traders reduce risk ahead of the decision, then spikes on the announcement day and the following session. A proximity feature centered on the FOMC date captures both the compression phase (positive proximity, negative vol effect) and the expansion phase (event day, positive vol effect). Tree models can split on $\text{prox}_{\text{FOMC},t}$ to learn this asymmetric pattern.

Options expiry and gamma unwind. Monthly options expiry (third Friday) and quarterly triple-witching dates create mechanical volatility through **gamma exposure unwind**. As options approach expiry, delta-hedging activity by market makers intensifies, compressing intraday volatility when gamma is positive and amplifying it when gamma is negative. The proximity-to-expiry feature lets the model learn that the 2–3 days before expiry have distinct vol dynamics driven by hedging flows rather than information arrival.

Earnings proximity (single names). For individual stocks, earnings announcements dominate all other calendar effects. The proximity feature $\text{prox}_{\text{earn},t}$ captures the well-known pre-earnings volatility buildup, which begins roughly 5 trading days before the announcement as implied vol rises and option volume spikes. Post-earnings, the “volatility crush” typically resolves within 1–2 days.

Why This Matters

Replace raw binary event dummies with continuous proximity measures in your feature pipeline. For each event type, compute $\text{prox}_{e,t}$ with an appropriate window: $W = 5$ for FOMC, $W = 3$ for NFP/CPI, $W = 5$ for earnings (single names), $W = 3$ for options expiry. Also include the raw `days_to_event` as a separate feature (without the triangular window) so the model can learn longer-range anticipation effects. Proximity features provide 1–2% incremental QLIKE improvement over binary dummies alone at the 1-day horizon, with the gain concentrated around event days where the baseline forecast has the largest errors.

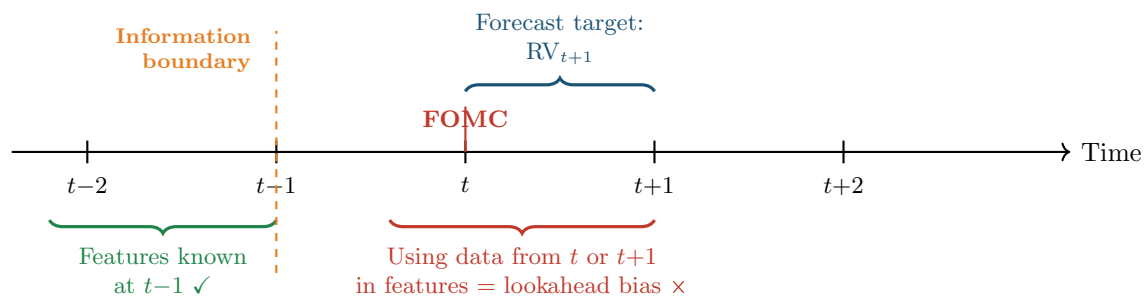


Figure 10.5: Temporal alignment for feature engineering. Features used to predict RV_{t+1} must be computed from data available at or before time $t-1$ (the close of trading on the day before the forecast is made). Using same-day or future data introduces lookahead bias. The orange dashed line marks the information boundary.

Look-Ahead Risk with Events

Earnings dates are announced approximately 2–4 weeks before the event. FOMC dates are published annually. These are safe to use. But some events (emergency Fed meetings, unscheduled news) cannot be known in advance. Never include an event indicator for a date that was not knowable at time $t-1$. For earnings, use the *announced* date, not the ex-post actual date (companies occasionally reschedule).

10.12 Sentiment and Text Features

News and social media sentiment contain forward-looking information that market prices may not yet fully reflect.

Audrino et al. (2020) construct daily sentiment indices from financial news articles and show that adding a negative-sentiment variable to the HAR model improves volatility forecasts, particularly during crisis periods. The key finding: negative sentiment matters more than positive sentiment, echoing the asymmetry we saw in signed semivariances (Section 10.5).

Rahimikia et al. (2021a) apply transformer-based NLP models (FinBERT) to extract sentiment from financial text and demonstrate incremental predictive power for equity volatility at daily and weekly horizons.

Sentiment for Volatility

Both Audrino et al. (2020) and Rahimikia et al. (2021a) find that text-derived sentiment adds 1–3% improvement in QLIKE loss over HAR-family baselines, with gains concentrated in high-volatility periods. The effect is modest but robust across different text sources and NLP methods.

If you have access to a news or social media feed, construct:

- A daily sentiment score (average polarity of articles mentioning the asset).
- A volume-of-news feature (number of articles; more attention predicts higher vol).
- A disagreement feature (standard deviation of sentiment across articles).

For most academic and internship projects, news data is hard to obtain. Treat sentiment features as a “nice to have” rather than a core requirement.

10.13 Feature Importance and Selection

With dozens of candidate features, you need a principled way to assess which ones actually help.

Mean Decrease in Accuracy (MDA). Permute one feature column at a time and measure how much out-of-sample loss increases. The feature whose permutation hurts most is the most important. MDA is model-agnostic and properly accounts for nonlinear interactions.

Mean Decrease in Impurity (MDI). For tree-based models, sum the impurity reduction (e.g., variance reduction for regression trees) across all splits that use a given feature. MDI is fast but biased: it favors high-cardinality features and features correlated with other important variables.

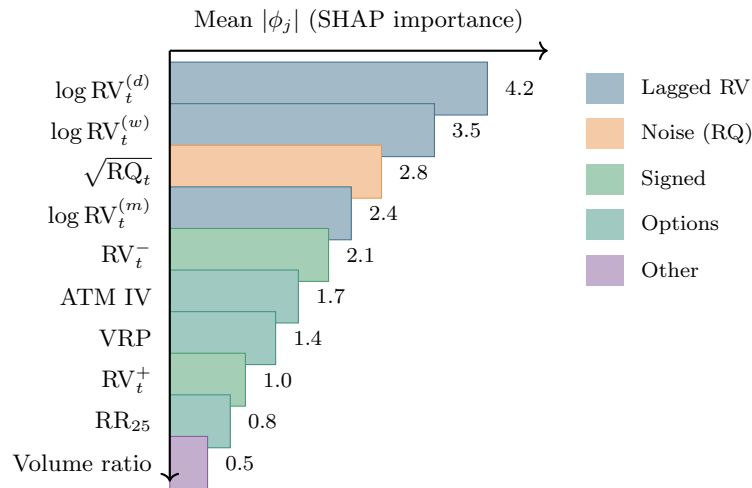


Figure 10.6: Illustrative SHAP feature importance ranking for a gradient-boosted tree model forecasting next-day log-RV. Lagged RV features (blue) dominate, but noise-awareness ($\sqrt{RQ}$, orange) and asymmetry (RV^- , green) provide substantial incremental value. Options-implied features (teal) add forward-looking information. Exact values are dataset-dependent; the ranking pattern is robust.

SHAP (SHapley Additive exPlanations). SHAP values decompose each prediction into additive contributions from each feature, grounded in cooperative game theory. For a single prediction $\hat{y}_i$, SHAP decomposes it as:

$$\hat{y}_i = \phi_0 + \sum_{j=1}^p \phi_j^{(i)}, \quad (10.38)$$

- ϕ_0 : the base value (mean prediction across the training set).
- $\phi_j^{(i)}$: feature j 's contribution to prediction i (can be positive or negative).
- The sum of all SHAP values plus the base value exactly equals the model's prediction.

In Plain English

SHAP answers the question: “For this particular prediction, how much did each feature push the forecast up or down relative to the average?” It borrows from game theory the idea of fairly splitting credit among players in a cooperative game. Each feature is a “player,” the prediction is the “payout,” and SHAP computes each feature’s fair share by considering all possible orderings in which features could be added to the model.

Why This Matters

Use SHAP values to explain your model's predictions in your internship presentation. A SHAP summary plot showing that $\log \text{RV}_t^{(d)}$ and $\sqrt{\text{RQ}_t}$ are the top two features validates that your ML model has learned economically sensible patterns. SHAP also helps debug: if a calendar dummy ranks in the top 3, something is likely wrong (possible lookahead bias or data leakage). Always compute SHAP across multiple purged CV folds to assess stability.

SHAP instability for correlated features

When two features are highly correlated (e.g., $\text{RV}_t^{(d)}$ and $\log \text{RV}_t^{(d)}$), SHAP values become unstable: small perturbations in the training data can swap their importance rankings. This does not mean the model is wrong; it means the importance is genuinely shared and cannot be cleanly attributed. Always check stability by computing SHAP across multiple train/test splits. If two features swap ranks across folds, treat them as interchangeable rather than debating which is “truly” more important.

Accumulated Local Effects (ALE) plots. Christensen et al. (2023) advocate ALE plots over partial dependence plots (PDPs) for volatility models. ALE plots avoid the extrapolation problem of PDPs by computing effects only within the observed feature distribution.

ALE Plot

For feature x_j , partition its range into K bins. In each bin $[z_{k-1}, z_k]$, compute the average change in the model's prediction as x_j moves across the bin, holding other features at their observed values:

$$\widehat{\text{ALE}}_j(x) = \sum_{k=1}^{k_x} \frac{1}{n_k} \sum_{i: x_j^{(i)} \in [z_{k-1}, z_k]} \left[\hat{f}(z_k, \mathbf{x}_{-j}^{(i)}) - \hat{f}(z_{k-1}, \mathbf{x}_{-j}^{(i)}) \right], \quad (10.39)$$

where:

- k_x is the bin containing x ,
- n_k is the number of observations in bin k ,
- $\hat{f}(z_k, \mathbf{x}_{-j}^{(i)})$ evaluates the model at the bin boundary z_k with all other features at their observed values for observation i .

In Plain English

Unlike partial dependence plots, which can be misleading when features are correlated (they evaluate the model at feature combinations that never occur in practice), ALE plots only compute effects within the observed data range. The result is a curve showing the marginal effect of one feature on the prediction, holding everything else at its actual (not hypothetical) value.

Why This Matters

Use ALE plots to sanity-check your trained model. The ALE plot for $\log \text{RV}_t^{(d)}$ should show a roughly increasing, concave relationship: higher lagged RV predicts higher future RV, but with diminishing effect at extreme levels (mean reversion). If the ALE plot shows a non-monotonic or economically implausible pattern, your model may be overfitting noise. Include ALE plots for the top 3–5 features in your internship report as evidence that the

model has learned meaningful structure.

ALE plots tell you the *shape* of the relationship: is the effect of lagged RV on predicted volatility linear, concave, or threshold-like? This is valuable for sanity-checking whether the model has learned economically sensible patterns.

Importance Stability Protocol

Before reporting which features matter, turn the qualitative “is the ranking consistent?” question into a single number. Run MDA (and, separately, mean-|SHAP|) across $K = 5$ purged CV folds (splitting the timeline into chunks, training on some and testing on the rest, with a gap to prevent leakage; see Chapter 17, Section 17.7), rank the features within each fold, and compute the **average pairwise Spearman rank correlation** (a score from -1 to $+1$: $+1$ means two lists put the features in exactly the same order, 0 means unrelated orderings) of those fold-level rankings:

$$\rho_{\text{stab}} = \frac{2}{K(K-1)} \sum_{k=1}^{K-1} \sum_{l=k+1}^K \underbrace{\rho_S(r^{(k)}, r^{(l)})}_{\text{agreement of folds } k \text{ and } l}, \quad (10.40)$$

where $r^{(k)}$ is the ordered list of features for fold k (rank 1 = most important, rank 2 = next, and so on) and ρ_S is the Spearman rank correlation. In words: for every *pair* of the 5 folds, score how similarly they ranked the features (that is the ρ_S term); then take the plain average over all 10 pairs (the prefactor $\frac{2}{K(K-1)}$ is just 1 over the number of pairs, and l starting at $k+1$ counts each pair once). A score near 1 means all folds agree on which features matter. Read the number against fixed thresholds:

- $\rho_{\text{stab}} > 0.8$ (**stable**): the same features dominate across folds; proceed with confidence.
- $0.5 \leq \rho_{\text{stab}} \leq 0.8$ (**caution**): the top 2–3 features are consistent but mid-ranked features shuffle; report the stable ones and flag the rest.
- $\rho_{\text{stab}} < 0.5$ (**red flag**): rankings are essentially random across folds, a classic symptom of fitting noise on a short sample.

Compute ρ_{stab} for *both* the MDA rankings and the mean-|SHAP| rankings. When the two agree (both high or both low) you have a clean verdict. When they *disagree*, the feature’s importance is concentrated in a handful of extreme observations; use the disagreement as a diagnostic, not a tie-break, and inspect the high-leverage dates with an ALE or SHAP dependence plot before trusting the feature (see the worked example below).

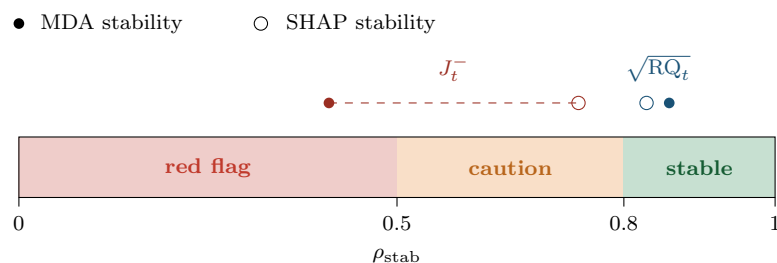


Figure 10.7: The stability scale with its three threshold bands. $\sqrt{RQ_t}$ lands in the green “stable” band for both MDA (filled) and SHAP (hollow) and is trustworthy. J_t^- straddles the red “red flag” band on MDA and the amber “caution” band on SHAP (dashed connector); that wide split is the diagnostic that its importance is concentrated in a few turbulent dates.

Worked Example: Reading a stability table for two candidate features

You run 5-fold purged CV and rank features within each fold by both MDA and mean- $|\text{SHAP}|$. For $\sqrt{\text{RQ}_t}$, the two metrics agree: $\rho_{\text{stab}}^{\text{MDA}} = 0.86$ and $\rho_{\text{stab}}^{\text{SHAP}} = 0.83$. Both clear the 0.8 threshold, and they agree: $\sqrt{\text{RQ}_t}$ is a genuinely stable, trustworthy feature, consistent with the HARQ insight (Bollerslev et al., 2016b).

For the signed-jump feature J_t^- , the picture splits: $\rho_{\text{stab}}^{\text{MDA}} = 0.41$ (red flag) but $\rho_{\text{stab}}^{\text{SHAP}} = 0.74$ (caution). The disagreement is the signal. MDA works by shuffling a feature's column to destroy its information and seeing how much worse the forecast gets; if the column is zero on most days, shuffling those zeros changes almost nothing, so MDA rates the feature low on quiet folds. J_t^- is mostly zero and occasionally huge, so a fold that happens to contain a crash day attributes large SHAP credit to it, while permutation MDA, which measures average accuracy loss, sees little change on the many quiet days and ranks it inconsistently. The honest reading is not “MDA says drop it, SHAP says keep it” but “ J_t^- 's importance is concentrated in a few turbulent dates; verify on those dates before deciding.” This is precisely the kind of feature for which the signed-magnitude transform (Equation 10.7) tames the tail and often stabilizes the ranking.

10.14 Proving a Feature Adds Value: The Incremental-Information Test

Omitted-Variable Bias

If a candidate predictor x and the target y are both driven by a third variable z that you left out of the model, the apparent predictive relationship between x and y is partly *spurious*: x looks useful only because it is a noisy stand-in for z . Including z as a control and asking whether x *still* helps is the standard cure. In volatility forecasting the usual omitted variable is the overall level of market fear, and the cleanest observable proxy for it is the VIX. The question of this section is therefore: *does my new feature add information beyond lagged RV and VIX, or is it just a noisy repackaging of what they already say?*

You have just learned how to rank features by importance (Section 10.13). But a high importance score inside one model does not prove a feature *adds value* relative to a sensible baseline. The decisive question for every candidate feature is: does the forecast get better when I add it, and worse when I take it away? This is the **incremental-information test**, and it is the gate every feature must pass before it earns a column in your production matrix.

10.14.1 Two Nested Models

The test compares two models, one *nested* inside the other (the bigger model contains every feature of the smaller one, plus exactly one more).

- **Model A (the baseline)**: the HAR core plus VIX. Features are $\log \text{RV}_t^{(d)}$, $\log \text{RV}_t^{(w)}$, $\log \text{RV}_t^{(m)}$ (Equations 10.4–10.6) and $\log \text{VIX}_t$.
- **Model B (the challenger)**: everything in Model A *plus* the single candidate feature x_t you are evaluating.

Because Model A is *nested* inside Model B (B contains every column A has, and one more), any out-of-sample improvement from A to B is attributable to x_t alone. Including VIX in the baseline is the central design choice: VIX is the omitted variable that confounds nearly every volatility feature, so a candidate that beats *HAR core + VIX* has demonstrated information the market's own fear gauge does not already contain.

Why HAR Core Alone Is Too Weak a Baseline

A candidate feature will almost always beat plain HAR core, because almost everything correlates with the level of volatility. Skew, VRP, semivariances, and a noisy VIX proxy all clear that low bar. The interesting and decision-relevant question is whether the feature beats a baseline that *already includes the obvious forward-looking control*. If your candidate only helps when VIX is absent, you have built a VIX proxy, not a new signal; that is exactly the omitted-variable trap. Always put VIX (or, if options data is unavailable, $RV_t^{(m)}$ as a level control) in Model A.

10.14.2 Three Verdicts: QLIKE, SHAP, and the Information Coefficient

We judge the candidate on three complementary axes. Each answers a slightly different question, and a feature you trust should pass all three.

(1) Out-of-sample QLIKE improvement. Re-estimate both models with purged K -fold cross-validation (splitting the timeline into chunks, training on some and testing on the rest, with a gap to prevent leakage; Chapter 17, Section 17.7) and compare their out-of-sample QLIKE loss. The incremental value of the feature is the loss reduction:

$$\Delta \text{QLIKE} = \underbrace{\overline{\text{QLIKE}}_A}_{\text{HAR core} + \text{VIX}} - \underbrace{\overline{\text{QLIKE}}_B}_{+ \text{candidate } x_t}, \quad (10.41)$$

- $\overline{\text{QLIKE}}_A$: average out-of-sample QLIKE of the baseline across the CV test folds.
- $\overline{\text{QLIKE}}_B$: average out-of-sample QLIKE of the challenger across the same folds.
- $\Delta \text{QLIKE} > 0$ means the candidate *lowered* loss (improvement); $\Delta \text{QLIKE} \leq 0$ means it added nothing or hurt.

Why This Matters

This is the single most important feature-vetting computation in your project. Your QLIKE improvement target of 30–80 bps (Chapter 17) is the *cumulative* budget; the incremental test tells you how much of that budget each feature is actually spending.

(2) SHAP attribution in Model B. Refit SHAP (Equation 10.38, Figure 10.6) on Model B and read off the mean absolute SHAP value of the candidate, $|\phi_x|$, the average over the test-fold observations of $|\phi_x^{(i)}|$, the candidate's per-prediction SHAP contribution from Equation 10.38. A feature with a real ΔQLIKE but near-zero SHAP is a contradiction worth investigating; a feature with large SHAP but zero ΔQLIKE is importance shared with a correlated baseline column, because SHAP can hand a feature credit that really belongs to a near-identical baseline column it travels with. Removing the candidate then changes nothing (zero ΔQLIKE) since the baseline column still carries the signal (the SHAP-instability warning of Section 10.13). Agreement between the two is the reassuring case.

(3) Information coefficient (IC). The third lens, borrowed from active portfolio management, measures how well the feature *by itself* lines up with the target. We introduce it here as a named metric you will use throughout the project.

Information Coefficient

The **information coefficient** of a feature x_t for a target y_{t+1} (here $y_{t+1} = \log RV_{t+1}$) is the rank correlation between the feature and the realized target over the out-of-sample period:

$$\text{IC} = \rho_S(x_t, y_{t+1}) = \text{corr}(\text{rank}(x_t), \text{rank}(y_{t+1})), \quad (10.42)$$

- ρ_S : the Spearman (rank) correlation coefficient, robust to the heavy tails endemic to volatility data.
- x_t : the candidate feature observed at time t (strictly before the target).
- y_{t+1} : the realized target one step ahead, e.g. next-day log RV.
- $IC \in [-1, 1]$; the magnitude measures predictive strength and the sign measures direction.

In Plain English

The information coefficient asks: “On the days this feature was high, was tomorrow’s volatility also high?” It is a rank correlation, so it does not care about the exact scale of the feature, only whether the ordering of feature values matches the ordering of realized outcomes. It is the *size* (distance from zero) that measures strength, so an IC of -0.18 is just as strong as $+0.18$; the sign only tells you the direction. An IC of 0.05 sounds tiny, but in volatility forecasting a stable IC even in the 0.03 – 0.10 range for a feature that is *not* already spanned by HAR core is meaningful here (weak but useful at the low end, strong toward 0.10). Forecasting one-step-ahead volatility is hard, and a small edge applied every day adds up over a long backtest. The danger is the opposite of intuition: a large *raw* IC often just means the feature tracks the level of RV, which HAR core already supplies. That is why IC is read alongside, not instead of, the nested Δ QLIKE.

Why This Matters

Report IC for every candidate feature in your feature audit. Compute it two ways: the *raw* IC of x_t against $\log RV_{t+1}$, and the *residual* IC of x_t against the residual of $\log RV_{t+1}$ after regressing out HAR core and VIX. *Regressing out* HAR core and VIX means: first predict the candidate feature using only HAR core and VIX, then keep only the leftover part (the *residual*) that those baseline features could *not* explain. The residual IC then asks whether that leftover part still lines up with tomorrow’s volatility; if it does not, the feature was just echoing VIX. The gap between the two ICs is the omitted-variable story made quantitative: if the raw IC is large but the residual IC collapses to near zero, the feature is a level proxy with no incremental content. A feature whose residual IC survives is exactly the kind that will also produce a positive Δ QLIKE in the nested test.

10.14.3 Is the Improvement Real? The Diebold–Mariano Test

A positive Δ QLIKE from Equation 10.41 is not proof: with a finite sample it could be luck. The **Diebold–Mariano (DM) test** (Chapter 17, Section 17.5; Diebold and Mariano, 1995) is the formal instrument for deciding whether the loss difference between two forecasts is statistically distinguishable from zero. Form the per-period loss differential $d_t = \text{QLIKE}_t^A - \text{QLIKE}_t^B$ and test $\mathbb{E}[d_t] = 0$ (the symbol $\mathbb{E}[\cdot]$ denotes the expected, i.e. long-run average, value) using the DM statistic (Chapter 17, Equation 17.7). The candidate adds value only if the DM test *rejects* equal predictive accuracy in favour of Model B. As a rule of thumb, a DM statistic bigger than about 1.96 in magnitude means the improvement is unlikely (less than a 5% chance) to be a fluke; anything well below that leaves us unable to rule out luck.

The Same Caveat the Diminishing-Returns Curve Will Repeat

Many candidate features produce a positive but *insignificant* Δ QLIKE: the point estimate improves, but the DM test cannot reject equality. This is the data telling you the feature lives on the flat part of the diminishing-returns curve (Section 10.15). A long string of

individually insignificant features can collectively overfit your validation folds and inflate the model's apparent edge.

Table 10.1: The incremental-information test summarized. Each candidate is judged against the same baseline (HAR core + VIX) on three axes, with the Diebold–Mariano test as the significance gate. “Keep” requires a positive, significant ΔQLIKE and a surviving residual IC. “Rejects equal accuracy” means the test concludes Model B is genuinely better, not just better by luck.

Axis	What it measures	Computed on	Keep if
ΔQLIKE (Eq. 10.41)	Loss reduction A→B	Purged-CV test folds	> 0 and DM-significant
DM test	Significance of ΔQLIKE	Per-period loss diff	Rejects equal accuracy
Mean SHAP in B	Attributed contribution	Model B test folds	Non-negligible, stable
Residual IC (Eq. 10.42)	Edge beyond HAR+VIX	Out-of-sample ranks	Still correlates with target after removing

Worked Example: A skew feature that fails the test it looks like it should pass

Suppose you are evaluating the 25-delta risk reversal RR_{25} as a candidate. In isolation it looks excellent: its raw IC against next-day log RV is -0.18 (steep negative skew precedes high vol), comfortably the largest raw IC among your options features, and it ranks third by SHAP when added to plain HAR core. The naive conclusion is “keep it, it is a top-3 feature.”

Now run the proper test. Model A is HAR core + $\log VIX_t$; Model B adds RR_{25} .

- **Residual IC:** after regressing out HAR core and VIX, the residual IC of RR_{25} collapses from -0.18 to -0.04 . Most of the raw signal was VIX in disguise: when fear spikes, both VIX and the put skew move together.
- **ΔQLIKE :** the out-of-sample loss falls by only 0.004 ($\overline{\text{QLIKE}}_A = 0.318$ vs. $\overline{\text{QLIKE}}_B = 0.314$).
- **DM test:** the DM statistic is 1.1, far short of the 1.96 needed to reject equal accuracy at the 5% level. The improvement is not significant.

The verdict is *do not keep RR_{25} as a standalone column at the one-day horizon*. Its apparent power was almost entirely shared with VIX. This is exactly the omitted-variable trap, and it is why the residual IC and the nested DM test, not the raw IC or the SHAP rank, are the deciding evidence. (The same feature can still pass at longer horizons, where Table 10.2 shows options features gain a genuine edge over VIX.)

The Incremental-Information Test in One Line

A feature earns its column only if, relative to HAR core + VIX, it delivers a *positive and Diebold–Mariano-significant* out-of-sample ΔQLIKE and a *residual* information coefficient that survives controlling for the baseline.

10.15 The Diminishing Returns Curve

You have now seen seven layers of features: lagged RV transforms, noise-awareness features (RQ), signed and asymmetric measures, options-implied quantities, microstructure signals, cross-asset spillovers, calendar/event features, and long-memory/regime features. *How much does each layer actually contribute?*

The answer follows a characteristic **diminishing returns curve**: the first few feature families provide the vast majority of forecasting accuracy, and each subsequent layer adds progressively

less.

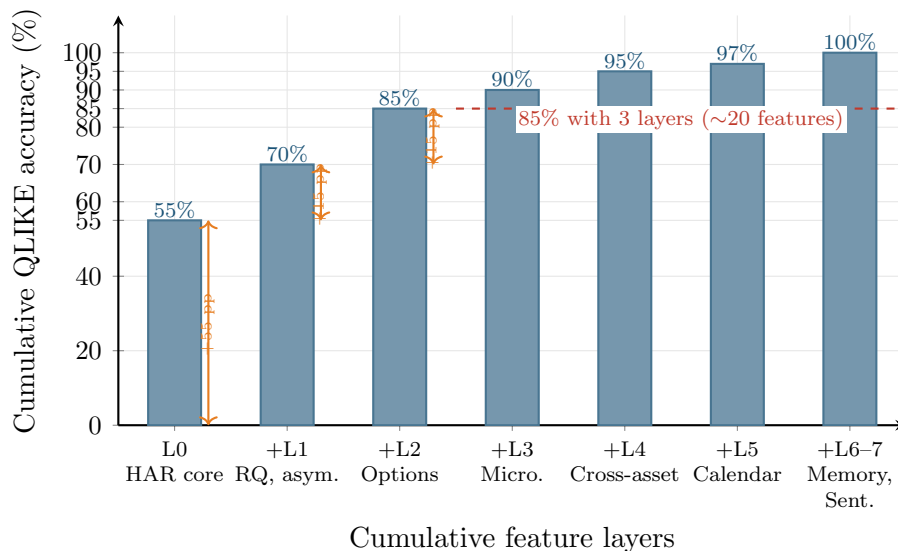


Figure 10.8: Diminishing returns staircase for volatility forecasting features. Each bar shows the cumulative QLIKE accuracy (normalized so the full feature set achieves 100%) when adding one more feature layer. The first three layers (HAR core, noise/asymmetry, options) achieve 85% of attainable accuracy with roughly 20 features. The remaining four layers add the final 15% with 60–100 additional features. Exact percentages are illustrative and based on findings in Christensen et al. (2023) and the vol-project-ref feature composition analysis.

Interpreting the staircase. The staircase in Figure 10.8 shows approximate cumulative accuracy contributions, normalized so that the full feature set achieves 100%.

- **Layer 0 (HAR core):** $RV_t^{(d)}$, $RV_t^{(w)}$, $RV_t^{(m)}$ alone achieve roughly 55% of attainable accuracy. This is the single most important result in volatility forecasting: three simple features explain more than half the variation.
- **Layer 1 (Noise + Asymmetry):** Adding $\sqrt{RQ_t}$, RV_t^- , RV_t^+ , and jump components pushes accuracy to roughly 70%. The HARQ and SHAR improvements are captured here.
- **Layer 2 (Options):** ATM IV, skew, VRP, and term structure slope bring the total to roughly 85%. These forward-looking features represent the single largest marginal gain beyond the core RV features.
- **Layers 3–7 (Microstructure, Cross-asset, Calendar, Memory, Sentiment):** The remaining five layers collectively contribute the final 15%, with each individual layer adding 2–5 percentage points.

The curve shifts with forecast horizon. The diminishing returns curve is not fixed; it depends heavily on the forecast horizon. Table 10.2 summarizes which features dominate at each horizon.

Table 10.2: Feature dominance by forecast horizon. The “dominant” column lists the feature families that contribute the most marginal accuracy at each horizon. Based on findings in Christensen et al. (2023) and Bollerslev et al. (2024).

Horizon	Dominant features	ML vs. linear gap	Key insight
$h = 1$ day	Lagged RV, RQ, RV^-	Small ($\sim 5\%$)	HARQ nearly optimal
$h = 5$ days	Options (VRP, skew) + lagged RV	Moderate ($\sim 10\%$)	VRP begins to matter
$h = 22$ days	VRP, term structure, Hurst	Large ($\sim 15\%$)	Options have max advantage

At the 1-day horizon, the HAR core dominates and ML adds relatively little. [Bollerslev et al. \(2024\)](#) demonstrate that a properly fitted HAR model with daily re-estimation and a 630-day training window is “hard to beat” even with gradient-boosted trees and neural networks, when the feature set is restricted to lagged RV and VIX. The implication is stark: if your ML model does not beat a well-tuned HAR at $h = 1$, the issue is likely the baseline, not the model.

At the 1-week and 1-month horizons, the picture changes. [Christensen et al. \(2023\)](#) show that ML models gain the most from additional features at longer horizons, where the informational advantage of options-implied and macroeconomic variables over lagged RV is greatest. At $h = 22$, ML models with the full feature set ($\mathcal{M}_{\text{ALL}}$) consistently outperform all HAR variants, and the Diebold-Mariano test frequently rejects equal predictive accuracy.

Perfect L0–L2 Before Chasing Marginal Features

The diminishing returns curve delivers a clear engineering message: invest your effort in getting Layers 0–2 right before adding Layers 3–7. Specifically:

1. Get the HAR baseline correct: daily re-estimation, 500–800 day training window, log-RV target ([Bollerslev et al., 2024](#)).
2. Add $\sqrt{\text{RQ}_t}$, RV^- , RV^+ , and jump components. This gives you HARQ/SHAR-level performance.
3. Add options features (ATM IV, VRP, skew, term structure slope) if available. This is the largest single-layer gain.
4. Only then consider microstructure, cross-asset, calendar proximity, and sentiment features. Each adds 2–5% marginal improvement.

Chasing Layer 5–7 features when your Layer 0 baseline is poorly calibrated (wrong training window, wrong re-estimation frequency, wrong target transform) is optimizing the wrong thing.

10.16 Summary

- **Lagged RV transforms** ($\text{RV}^{(d)}$, $\text{RV}^{(w)}$, $\text{RV}^{(m)}$, log, ratios) are the single strongest feature family, explaining 40–60% of next-day RV variation.
- **Realized quarticity** (RQ) measures the noise in today’s RV estimate. The HARQ interaction feature down-weights noisy days ([Bollerslev et al., 2016b](#)).
- **Signed semivariances** (RV^+ , RV^-) capture the leverage effect. The downside component carries roughly twice the predictive weight of the upside component ([Patton and Sheppard, 2015b](#)).
- **Signed jumps** (J^+ , J^-) from Chapter 4 provide additional asymmetric information; negative jumps are substantially more informative.
- **Higher moments** (realized skewness, kurtosis) have modest stand-alone power but serve as regime indicators.
- **Microstructure features** (spread, OBI, WAP returns, volume profiles, VPIN) are powerful for short-horizon forecasting with tick data but are asset-specific.
- **Options-implied features** (ATM IV, skew, term structure slope, VRP, VVIX) are uniquely forward-looking and provide the largest gains at horizons beyond one week.
- **Cross-asset features** (multi-asset RV, Diebold-Yilmaz spillover indices) capture contagion channels but require care to avoid curse-of-dimensionality problems ([Diebold and Yilmaz, 2012](#)).

- **Fractional differencing** $((1 - L)^d$ with $d \approx 0.35\text{--}0.45$) preserves long memory while achieving stationarity (Lopez de Prado, 2018).
- **Rolling Hurst exponent** from Chapter 7 captures time-varying roughness of the volatility process.
- **Calendar dummies** (FOMC, NFP, earnings, expiry) are weak individually but provide incremental improvement.
- **Sentiment features** add 1–3% QLIKE improvement, concentrated in crisis periods (Audrino et al., 2020; Rahimikia et al., 2021a).
- **Feature importance tools** (MDA, SHAP, ALE) should be evaluated across multiple CV folds to ensure stability; correlated features naturally produce unstable attribution (Christensen et al., 2023).

Chapter 10 Key Results

Feature Family	Key Contribution	Best Horizon
Lagged RV (d/w/m)	Baseline predictive power, 40–60% R^2	1–5 days
Realized quarticity	Noise-aware weighting (HARQ)	1 day
Signed semivariances	Leverage effect, $RV^- \approx 2 \times$ weight of RV^+	1–5 days
Signed jumps	Asymmetric tail events	1 day
Higher moments	Regime conditioning	Auxiliary
Microstructure/LOB	High-frequency liquidity signals	Intraday–1 day
Options-implied	Forward-looking; VRP, skew, VVIX	1 week–1 month
Cross-asset RV	Spillovers, contagion	1–5 days
Frac. differencing	Stationarity with memory preservation	All
Calendar dummies	Event-day adjustment	Event-specific
Sentiment	Crisis-period improvement	1 day–1 week

Chapter 11

Tree-Based Methods for Volatility

Application

This chapter is where ML meets volatility forecasting for the first time. Tree-based methods (LightGBM, XGBoost) are the workhorse models for tabular volatility data, just as they dominate Kaggle competitions and quantitative finance pipelines. Projects 1, 2, and 5 all use tree ensembles as their primary model. The central question: when do trees beat HAR (Chapter 6), and when don't they?

11.1 Why Trees for Volatility

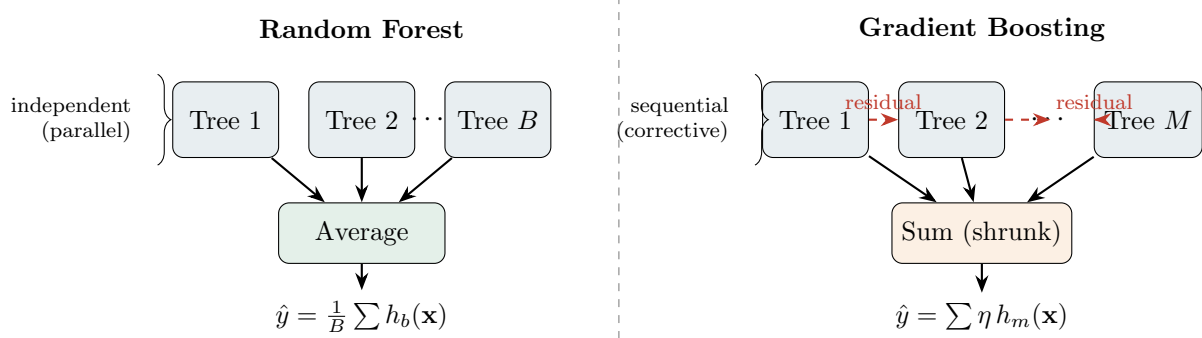
Chapter 10 built a rich feature matrix: lagged RV transforms, realized quarticity, signed semi-variances, jumps, options-implied measures, cross-asset signals, and calendar dummies. All of these live in a flat table where each row is a date and each column is a number. No pixels, no word tokens, no sequential structure that demands recurrence. This is *tabular data*, and tree-based ensembles are the best off-the-shelf learners for tabular data (Gu et al., 2020b).

Why? Four reasons, each directly relevant to volatility:

1. **Automatic nonlinear interactions.** HAR (Chapter 6) is linear in its three RV averages. If the effect of yesterday's RV depends on whether the VIX is above 25, you must hand-craft that interaction term for HAR. A tree discovers it automatically: one split on VIX, another on lagged RV, and the interaction is encoded in the tree structure.
2. **Threshold effects.** Volatility regimes change sharply. A tree split at $RV_{t-1} = 0.0004$ captures a regime boundary that a linear model can only approximate with polynomial terms.
3. **Fast training.** A LightGBM model with 500 trees trains in seconds on 1,250 rows. This speed enables hundreds of purged cross-validation iterations (Chapter 17), which is essential for honest evaluation on small samples.
4. **Built-in feature importance.** Split-count importance and SHAP values tell you *which* features drive predictions. This matters for interpretability, and it connects back to the feature selection discussion in Chapter 10.

The diagram below contrasts the two main tree ensemble strategies. A **random forest** trains many independent trees on bootstrapped samples and averages their predictions (variance reduction through decorrelation). **Gradient boosting** trains trees sequentially, each one correcting the errors of the ensemble so far (bias reduction through iterative refinement). Both are used

in volatility forecasting, but gradient boosting (LightGBM, XGBoost) typically wins on tabular data.



11.2 Tree Foundations: From One Tree to a Forest

What a Tree Actually Is

The last section claimed trees “discover threshold effects and interactions automatically.” This section opens the box. You need only one idea from Chapter 6: a forecast of tomorrow’s RV is a function of today’s feature vector $\mathbf{x}_t$ (lagged RV averages, realized quarticity, VIX, jumps). We ask: how does a tree turn that vector into a number, and why do we never use a single tree to do it?

The depth-2 tree drawn later in Section 11.3 (the one that splits on $\text{RV}_{t-1} = 0.0004$) is a finished object; the next three subsections show how such a tree is built, why one tree is too unreliable to forecast with, and how an ensemble of many trees fixes that.

11.2.1 How a regression tree chooses its splits

A **regression tree** carves the feature space $\mathbb{R}^p$ (the space of all possible feature vectors with p features) into J non-overlapping rectangular boxes $R_1, \dots, R_J$, called **leaves**, and predicts a single constant inside each box (Breiman et al., 1984). The constant is just the average target of the training observations that landed in that box:

$$\hat{f}(\mathbf{x}) = \sum_{j=1}^J \underbrace{c_j}_{\text{leaf mean}} \underbrace{\mathbf{1}(\mathbf{x} \in R_j)}_{\text{which box?}}, \quad c_j = \frac{1}{|R_j|} \sum_{t: \mathbf{x}_t \in R_j} \text{RV}_t, \quad (11.1)$$

where:

- R_j is the j -th rectangular leaf region,
- c_j is the **leaf value**: the mean realized volatility of the training days that fall in R_j . Here $|R_j|$ is the count of training days inside box R_j (the vertical bars mean “how many,” not absolute value), and the subscript $t: \mathbf{x}_t \in R_j$ means “take every day t whose features land in this box,”
- $\mathbf{1}(\mathbf{x} \in R_j)$ is the indicator, an on/off switch that equals 1 when the day’s feature vector $\mathbf{x}$ lands inside box R_j and 0 otherwise (the symbol $\in$ means “is inside”), so exactly one leaf fires per observation and only one term in the sum survives,
- $\sum_{j=1}^J$ means add up across all J leaves, and the hat on $\hat{f}$ marks it as the model’s estimate,
- J is the number of leaves.

In Plain English

A tree is a lookup table with smart boundaries. To forecast tomorrow's RV, you walk down the tree answering yes/no questions about today's features ("is RV_{t-1} below 0.0004?"), arrive at one leaf, and read off the average RV of all the historical days that ended up in that same leaf. The prediction is piecewise constant: every day inside a given box gets the identical forecast, no matter where in the box it sits. In symbols, that whole equation just says: find your one box, read off its stored average c_j .

The boxes are not handed to the tree; it builds them *greedily* (it grabs the single best split available right now and never goes back to reconsider it) by **recursive binary splitting** (it then repeats the very same procedure inside each new child group). Start with all training days in one root node. Consider every feature j and every candidate threshold s , and pick the single split (j, s) that most reduces the sum of squared residuals (SSR) across the two children that the split would create. A **residual** is how far a day's actual RV sits from the box's predicted average (actual minus predicted):

$$\min_{j, s} \left[\underbrace{\sum_{t: x_{tj} \leq s} (RV_t - c_L)^2}_{\text{left-child SSR}} + \underbrace{\sum_{t: x_{tj} > s} (RV_t - c_R)^2}_{\text{right-child SSR}} \right], \quad (11.2)$$

where:

- x_{tj} is the value of feature j on day t (e.g. RV_{t-1}),
- s is the split threshold being tested,
- c_L, c_R are the means of the left and right children that the split produces.

The $\min_{j, s}[\dots]$ means: try every feature j and every cut-point s , and pick the single pair that makes the total spread in the bracket as small as possible. The winning split becomes a node; the algorithm then recurses into each child and repeats, stopping when a child is too small (`min_child_samples`), too deep (`max_depth`), or no split reduces SSR enough. Minimising the SSR is identical to maximising the **variance reduction** of the target, which is why this rule is often called the variance-reduction (or MSE-reduction) criterion. (You can skip the next sentence: volatility forecasting is regression, so SSR is the criterion throughout this chapter.) For a classification target the same machinery swaps SSR for an impurity measure such as **Gini impurity** $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$ or **entropy** $H = -\sum_k \hat{p}_k \log_2 \hat{p}_k$, where k indexes the classes and $\hat{p}_k$ is the estimated share of a node's days in class k .

11.2.2 Why one fully grown tree is too unreliable

Let the tree grow with no stopping rule and it will keep splitting until each leaf holds a single training day. Training error is then exactly zero: the tree has memorised the sample. But the structure it learned is a fragile accident of *this* particular sample. A single tree is a **high-variance estimator** (its output swings wildly if the training data changes even a little, an instability unrelated to the volatility we are forecasting): perturb the training data slightly (drop a week, add a month) and the chosen thresholds, and even which feature splits at the root, can change completely, producing a different forecast function.

A Single Tree Will Memorise Volatility Noise

Volatility data is short ($\sim 1,250$ daily rows), heavy-tailed, and highly autocorrelated (Section 11.4). An unconstrained tree will carve a dedicated leaf around a single crisis week

and predict that week's RV perfectly in-sample while generalising terribly out-of-sample. This is why nobody forecasts RV with one tree, and why every result in this chapter uses an *ensemble* of trees.

There are two ways to tame this variance: average many independent trees (bagging and random forests, below) or add small trees sequentially (gradient boosting, Section 11.3).

11.2.3 Bagging: averaging away the variance

Bagging (bootstrap aggregating) reduces variance by training many trees on resampled copies of the data and averaging them. A **bootstrap sample** draws n rows from the n training days *with replacement*, so some days appear several times and others not at all. The fraction of *unique* original days that land in a given bootstrap sample converges to

$$1 - \underbrace{\left(1 - \frac{1}{n}\right)^n}_{\substack{\text{prob. a row is} \\ \text{never drawn}}} \xrightarrow{n \rightarrow \infty} 1 - e^{-1} \approx 0.632, \quad (11.3)$$

where:

- n is the number of training observations,
- $\left(1 - \frac{1}{n}\right)^n$ is the probability that a specific day is missed by all n draws,
- the arrow with $n \rightarrow \infty$ underneath means “as the dataset grows, this fraction settles at the value on the right”; e is Euler's constant (≈ 2.718), so $e^{-1} \approx 0.368$,
- 0.632 is the limiting **in-bag** fraction; the remaining $\approx 36.8\%$ are the **out-of-bag** (OOB) days for that tree, which become the free validation set used in the OOB subsection below.

Bagging then fits a tree h_b to each of B bootstrap samples and averages: $\hat{f}_{\text{bag}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B h_b(\mathbf{x})$, i.e. just take the plain average of the forecasts of all B trees, where $h_b(\mathbf{x})$ is tree b 's forecast.

In Plain English

Drawing n rows with replacement is like building each tree from a slightly different reshuffling of history: roughly two-thirds of the real days appear, one-third sit out. Each tree is still high-variance on its own, but their errors are partly independent, so averaging them cancels much of the noise, the same reason an average of noisy measurements is more precise than any single reading.

The catch: averaging only helps to the extent the trees disagree. If one feature dominates, every bagged tree splits on it first and the trees end up nearly identical, so the variance barely falls.

11.2.4 Random forests: decorrelating the trees

A **random forest** adds one decisive twist to bagging: at *each* split, the tree may only consider a random subset of m of the p features as split candidates. This forces different trees to build around different features, breaking the correlation that limits bagging. The variance of the forest average makes the payoff precise:

Random Feature Subsets Decorrelate the Forest

For a forest of B trees, each with single-tree variance σ^2 and average pairwise correlation ρ between trees, the variance of the averaged forecast is

$$\text{Var}(\hat{f}_{\text{RF}}) = \underbrace{\rho \sigma^2}_{\text{irreducible floor}} + \underbrace{\frac{1-\rho}{B} \sigma^2}_{\text{vanishes as } B \rightarrow \infty}. \quad (11.4)$$

- $\text{Var}(\dots)$: how much the forecast wobbles from sample to sample,
- σ^2 : that wobble for a single tree (this σ is the spread of tree forecasts, *not* the volatility we are forecasting),
- ρ : how similar the trees' answers are (0 = totally different, 1 = identical copies),
- B : the number of trees in the forest.

Adding trees ($B \rightarrow \infty$) kills the second term but never the first: the variance floor is $\rho\sigma^2$. The only way below that floor is to lower ρ itself, which is exactly what the random m -feature restriction does. Smaller m means more decorrelated trees (lower ρ) at the price of slightly higher individual-tree bias. The standard defaults are $m = \lfloor p/3 \rfloor$ for regression and $m = \lfloor \sqrt{p} \rfloor$ for classification, where the $\lfloor \cdot \rfloor$ brackets mean “round down to a whole number” (so with $p = 45$ features you try about 15 at each split).

Why This Matters

In RV forecasting the dominant predictor is overwhelmingly the lagged daily RV (Chapter 6), so an unrestricted forest would split on it in nearly every tree and decorrelate poorly. The random-feature trick is what lets the other feature families from Chapter 10 (RQ, signed semivariances, VIX, jumps) actually enter the ensemble. The same logic reappears as `colsample_bytree` for gradient boosting in Section 11.4: feature subsampling there decorrelates boosted trees for precisely this reason.

11.2.5 Out-of-bag error, and why it lies on time series

Because each tree omits its $\approx 36.8\%$ out-of-bag days (Equation 11.3), a forest comes with a free, built-in validation set: score each day using only the trees that did *not* see it, and the resulting **out-of-bag (OOB) error** approximates leave-one-out cross-validation (test on each day in turn while training on all the others) with no separate holdout needed. That is genuinely useful for cross-sectional, exchangeable data.

OOB Error Is Invalid for Volatility Time Series

OOB error assumes the observations are **exchangeable** (i.i.d., independent and identically distributed; the order of the rows does not matter). Volatility is not: if day $t+1$ is in a tree's bootstrap sample while day t is out-of-bag, scoring day t on that tree lets it “predict the past from the future,” because adjacent RV values are nearly identical (Section 11.4). OOB error therefore looks far better than true out-of-sample performance. Use it only as a quick development sanity check, never as your selection metric. For honest model selection, use purged K -fold cross-validation with an embargo (Section 17.7 of Chapter 17), which removes the leakage that breaks OOB.

11.3 LightGBM and XGBoost

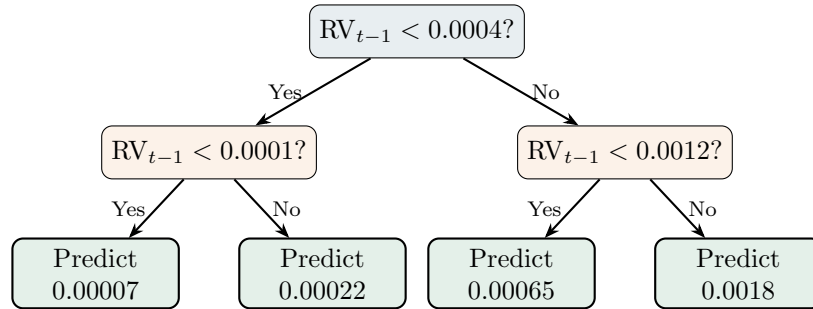
For the full algorithmic treatment (histogram binning, leaf-wise vs. level-wise growth, GOSS sampling), see a dedicated ML reference; here we focus on the choices that matter for forecasting RV.

Gradient Boosting in One Paragraph

You start with a constant prediction (the training-set mean of RV). You compute the residuals. You fit a shallow decision tree to those residuals. You add a shrunk version of that tree's predictions to the running forecast. You repeat. Each new tree corrects the mistakes of the ensemble so far. The “gradient” part: the residuals are the negative gradient of the loss function with respect to the current prediction, so this procedure is gradient descent in function space.

A single tree on lagged RV

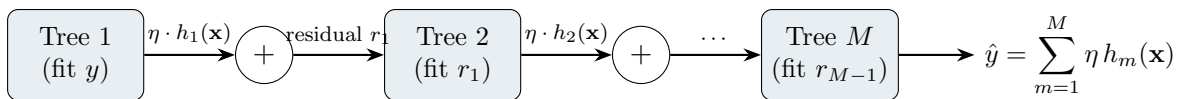
Before ensembles, consider one tree. The diagram below shows a depth-2 tree predicting tomorrow's RV from yesterday's value.



The tree partitions the feature space into four leaves, each returning a constant prediction. This is piecewise-constant approximation. A single shallow tree is a weak learner; boosting combines hundreds of them.

The boosting ensemble

The diagram below shows how gradient boosting builds its forecast sequentially. Each tree fits the residual errors of the ensemble so far.



The ensemble prediction is:

$$\hat{y}_t = \sum_{m=1}^M \eta h_m(\mathbf{x}_t), \quad (11.5)$$

where:

- $h_m(\mathbf{x}_t)$ is the prediction of the m -th tree given feature vector $\mathbf{x}_t$,
- $\eta \in (0, 1]$ is the learning rate (shrinkage),
- M is the number of boosting iterations (trees).

Loss function choice

The standard loss for regression is MSE: $\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_t (\text{RV}_t - \hat{y}_t)^2$. But Chapter 17 showed that QLIKE is the preferred loss for volatility forecasting (Audrino and Knaus, 2016):

$$\mathcal{L}_{\text{QLIKE}} = \frac{1}{N} \sum_{t=1}^N \left(\frac{\text{RV}_t}{\hat{y}_t} - \ln \frac{\text{RV}_t}{\hat{y}_t} - 1 \right). \quad (11.6)$$

- RV_t : the realized volatility actually observed on day t ,
- $\hat{y}_t$: the model's forecast for day t ,
- the summand equals zero when $\hat{y}_t = \text{RV}_t$ and is strictly positive otherwise.

In Plain English

QLIKE penalizes you more harshly for under-predicting volatility than for over-predicting it by the same amount. If realized vol is 20% and you forecast 10%, the penalty is much larger than if you forecast 30%. This asymmetry matches real risk management needs: underestimating volatility is more dangerous than overestimating it. MSE, by contrast, penalizes both directions equally.

Why This Matters

Audrino and Knaus (2016) show that QLIKE-optimized trees outperform MSE-optimized trees for realized volatility, so this custom loss is a core part of the pipeline.

Neither LightGBM nor XGBoost provides QLIKE natively, but both accept custom objective functions. You supply the gradient and Hessian:

$$g_t = \frac{\partial \mathcal{L}_{\text{QLIKE}}}{\partial \hat{y}_t} = -\frac{\text{RV}_t}{\hat{y}_t^2} + \frac{1}{\hat{y}_t}, \quad (11.7)$$

$$h_t = \frac{\partial^2 \mathcal{L}_{\text{QLIKE}}}{\partial \hat{y}_t^2} = \frac{2 \text{RV}_t}{\hat{y}_t^3} - \frac{1}{\hat{y}_t^2}. \quad (11.8)$$

- g_t : the gradient tells each tree which direction to adjust predictions,
- h_t : the Hessian tells each tree how aggressively to adjust (curvature information).

Why This Matters

Implementing these two formulas as a custom objective function in LightGBM or XGBoost is a 10-line code change, but it is the most effective modification you can make to the default pipeline.

Enforce Positive Predictions

QLIKE requires $\hat{y}_t > 0$. Clip predictions to a small positive floor (e.g., 10^{-8}) inside the custom objective, or the gradient explodes. Also apply the floor when evaluating the metric, not just during training.

Monotone constraints

Finance intuition says “higher recent volatility predicts higher future volatility.” You can encode this as a monotone constraint on lagged-RV features: the prediction must be non-decreasing in RV_{t-1} . Both LightGBM and XGBoost support per-feature monotone constraints. Use them sparingly (only for features where the monotone relationship is theoretically unambiguous), but when appropriate, they reduce overfitting and improve interpretability.

11.4 Hyperparameters for Volatility Data

Volatility data has three properties that make default hyperparameters dangerous:

1. **Small samples.** Five years of daily data gives you roughly 1,250 observations. This is orders of magnitude smaller than the datasets LightGBM and XGBoost were optimized for.
2. **High autocorrelation.** RV is strongly persistent (Chapter 6). Nearby observations are nearly identical, so effective sample size is smaller than the row count suggests.
3. **Heavy tails.** Volatility spikes create influential observations. A single crisis week can dominate the loss function.

Default Settings Will Overfit

Default LightGBM/XGBoost settings (`max_depth=6+`, `min_child_samples=20`) were designed for datasets with 100K+ rows. With 1,250 daily observations, defaults will memorize noise. The table below gives recommended ranges calibrated for volatility forecasting.

Parameter	Default	Vol Range	Rationale
<code>max_depth</code>	6–8	3–5	Shallow trees limit memorization
<code>min_child_samples</code>	20	50–200	Forces each leaf to generalize
<code>subsample</code>	1.0	0.6–0.8	Row subsampling adds regularization
<code>colsample_bytree</code>	1.0	0.6–0.8	Feature subsampling decorrelates trees
<code>learning_rate</code>	0.1	0.01–0.05	Slow learning + early stopping
<code>num_iterations</code>	100	500–2000	More trees at lower rate; early stop
<code>reg_lambda</code>	0	1–10	L2 leaf regularization

Worked Example: Tuning LightGBM on 5 Years of SPY RV

You have 1,258 daily observations of RV for SPY, with 45 features from Chapter 10. You reserve the last 6 months (126 days) as a holdout (never touched until final evaluation). The remaining 1,132 days go into purged 5-fold CV with a 5-day embargo (Chapter 17). **Step 1: Baseline.** Fit HAR on the same 1,132 training days. QLIKE = 0.142 averaged across folds.

Step 2: Defaults. Fit LightGBM with default settings. QLIKE = 0.158. The model overfits: training QLIKE = 0.041. The gap between training and validation loss is a red flag.

Step 3: Constrained. Set `max_depth=4`, `min_child_samples=100`, `subsample=0.7`, `colsample_bytree=0.7`, `learning_rate=0.02`, `num_iterations=1500` with early stopping (patience 50 rounds on purged validation fold). QLIKE = 0.131. Now the model improves over HAR by 7.7%, and training QLIKE = 0.098, a much healthier gap.

Step 4: Grid search. Sweep `max_depth` $\in \{3, 4, 5\}$, `min_child_samples` $\in \{50, 100, 200\}$ via purged CV. Best: `max_depth=4`, `min_child_samples=100` (confirming Step 3). This is a 12-trial search; record all trials in the experiment log for Deflated Sharpe Ratio correction (Chapter 17).

Early Stopping on Purged Validation

Early stopping is the single most important regularization technique for tree ensembles on small volatility data. But the validation set used for early stopping must respect the

purged CV structure: no overlap between training and validation dates, with an embargo gap. If you use a random validation split, early stopping will stop too late because the validation set is contaminated by lookahead.

11.5 Interpreting Tree Models: TreeSHAP and the SHAP Plot Toolkit

SHAP, MDA, and MDI from Chapter 10

Section 10.13 introduced three feature-importance ideas you will reuse here: **SHAP** values, which decompose a single prediction additively into per-feature contributions (Equation 10.38); **MDA** (mean decrease in accuracy), the permutation-based global importance; and **MDI** (mean decrease in impurity), the fast-but-biased split-based importance. It also introduced **ALE** plots (Equation 10.39). This section answers the question that Section left open: *how* do you compute SHAP for a tree ensemble cheaply, and *which* plot do you reach for when?

You have a tuned LightGBM forecasting next-day log-RV on the feature matrix from Chapter 10 (lagged RV transforms, realized quarticity, VIX, jumps). It beats HAR by 7.7% in QLIKE (Section 11.4). Your sponsor’s first question will be “*why* did the model forecast a vol spike for next Tuesday?”, not “what is the QLIKE?” Equation 10.38 promises an exact additive answer per day, but computing it honestly looks hopeless.

11.5.1 The exponential cost that TreeSHAP defeats

The SHAP value of feature i (recall from Chapter 10: feature i ’s fair share of the forecast) averages its *marginal contribution* (how much adding i changes the prediction) over *every* subset of the other features. With p features that is 2^p **coalitions** to evaluate, *for every observation* (“coalition” and “subset of features” mean the same thing here). The notation 2^p means 2 multiplied by itself p times, so every extra feature *doubles* the number of subsets. On the Chapter 10 matrix with, say, $p = 20$ features, that already gives $2^{20} \approx 10^6$ (about a million) model evaluations per day, times $\sim 1,250$ days, which is far too slow for a generic model. This exponential wall is exactly why [Lundberg and Lee \(2017\)](#) introduced model-specific shortcuts.

TreeSHAP ([Lundberg et al., 2020](#)) exploits the structure of decision trees to compute the *exact* same Shapley values in polynomial time. In the cost formula below, $O(\dots)$ is shorthand for how fast the computation time grows as the inputs grow: bigger inside the parentheses means slower.

$$\underbrace{O(T L D^2)}_{\text{TreeSHAP}} \quad \text{instead of} \quad \underbrace{O(T L 2^p)}_{\text{naive Shapley}}, \quad (11.9)$$

where:

- T is the number of trees in the ensemble,
- L is the maximum number of leaves per tree,
- D is the maximum tree depth,
- p is the number of features.

In plain terms: TreeSHAP’s cost grows with trees $\times$ leaves $\times$ depth-squared (manageable), while the naive method’s cost grows with 2^p , which doubles every time you add one feature (hopeless), so TreeSHAP is fast enough to run and the naive method is not.

In Plain English

The naive method asks “what happens to the prediction under all 2^p ways of hiding features?” TreeSHAP notices that a depth- D tree only ever splits on at most D features along any root-to-leaf path, so the only subsets that can possibly change a leaf’s reachability number $O(2^D)$, not $O(2^p)$. It pushes probability mass down the tree once and reads off the exact attributions, turning an exponential count over *features* into a polynomial count over *tree depth*. Because of the additivity axiom (a guarantee that the per-tree contributions add up cleanly), the ensemble’s SHAP value for a feature is simply the sum of its values across all trees.

Why This Matters

Because TreeSHAP is exact, it removes a confound when you compare importance across purged-CV folds for stability: any ranking changes you see are real instability, not Monte-Carlo noise from the explainer.

TreeSHAP: Interventional vs. Path-Dependent

TreeSHAP ships in two modes. The default **path-dependent** mode is faster but uses the tree’s internal node-coverage counts to marginalise missing features, which can assign small nonzero SHAP values to features the model never split on. The **interventional** mode marginalises against a background dataset, which is theoretically cleaner but slower. Both are still TreeSHAP (exact for their respective games) and differ from **KernelSHAP**, a model-agnostic *sampling* approximation usable on any model but noisy and orders of magnitude slower. For RV work: use path-dependent mode for fast exploration, switch to interventional with an RV-feature background sample for the numbers you put on a slide.

11.5.2 The four SHAP plots: a read-and-use taxonomy

The `shap` library produces four core plot types. Each answers a different question about your RV forecaster; the table is your lookup for which to reach for.

Plot	Scope		What it shows	Use it to...
Beeswarm (summary)	Global, days	all	One dot per day per feature; x = SHAP value, colour = feature value (red high, blue low); features sorted by mean $ \phi_j $	See which features drive forecasts overall and in which direction (does high lagged RV push the forecast up?)
Dependence	Per- observation scatter		One feature's value (x) vs. its SHAP value (y), coloured by the most-interacting feature	Reveal nonlinearity/thresholds in a feature's effect and two-way interactions (e.g. lagged RV $\times$ implied vol)
Waterfall	Single day		Bars walking from the base value $\mathbb{E}[f(\mathbf{x})]$ to the day's forecast, red up / blue down	Explain <i>one</i> prediction as an additive story; the presentation chart
Force	Single day		The same decomposition as the waterfall, in a compact horizontal bar	Embed a one-line explanation in a dashboard or monitor

Two symbols in the table: $\mathbb{E}[f(\mathbf{x})]$ is the **base value**, the model's average prediction over all days, i.e. its starting guess before any feature nudges it up or down; and ϕ_j is the SHAP contribution of feature j (from Chapter 10), so $|\phi_j|$ is its size ignoring whether it pushed the forecast up or down.

Dependence Plot vs. ALE: Two Views of the Same Feature

The SHAP **dependence** plot and the **ALE** (Accumulated Local Effects) plot (Equation 10.39, Chapter 10) answer related but distinct questions about, say, lagged RV. ALE gives the *marginal shape* (the average effect of this feature once you average over all the other features): a single smooth curve summarising “as lagged RV rises, does the forecast rise, fall, or flatten?”, averaged within the observed data range. The dependence plot gives the *per-observation scatter*: every day is a dot, exposing the spread around that average shape, and its colour axis surfaces the dominant *interaction* (e.g. the lagged-RV effect is amplified on high-VIX days). Read ALE for the functional form; read the dependence plot for heterogeneity and interactions. They are complements, not substitutes.

11.5.3 SHAP vs. MDI vs. MDA: when to use which

All three measures answer “which features matter?” but at different granularity, and they occasionally disagree. The reconciliation rule below is what you actually apply during RV model development.

Measure	Granularity	Strength / weakness	When to use
SHAP	Local (per day), sums to the prediction	Richest information; exact via TreeSHAP; can be unstable across correlated features (Section 10.13)	Presentation charts, debugging single forecasts, detecting nonlinearity
MDA	Global, in units of QLIKE/MSE drop	Model-agnostic, unbiased under independence (gives the right ranking when features are not correlated with each other); costs p extra passes; <i>must</i> use held-out data	Primary aggregate importance and cross-fold stability checks
MDI	Global, free byproduct of training	Biased toward high-cardinality features (those with many distinct values, hence many possible split points – a continuous RV vs. a yes/no jump flag) (Strobl et al., 2007)	Quick sanity check only; never a reported result

In practice you compute all three on your RV model and reconcile:

- **All three agree** on the top features (typically lagged RV then $\sqrt{RQ}$): robust evidence, proceed.
- **SHAP and MDA agree but MDI disagrees**: trust SHAP/MDA and ignore MDI; this is the MDI cardinality bias (see the warning below).
- **SHAP and MDA disagree**: the importance is likely driven by a few extreme days (vol spikes) that SHAP attributes per-observation but MDA averages away; inspect a SHAP dependence plot to locate them.

Do Not Report MDI as a Final Importance Result

MDI systematically over-credits features with many possible split points. A continuous feature such as lagged RV or VIX has hundreds of candidate thresholds, so the tree has more chances to split on it by luck alone, while a binary jump-day dummy has one. Strobl et al. (2007) show this rigorously: in simulations MDI assigned nonzero importance to continuous *noise* features. Use MDI to glance during development; use SHAP or MDA for anything you write down.

11.5.4 Presenting SHAP to the desk

The interpretation work only pays off if a non-technical audience believes it. Three rules, all of which trade rigor for being understood:

1. **Read the bars, not the math.** Say “this chart breaks the forecast into the contribution of each input; red bars pushed the predicted vol up, blue bars pushed it down, and the biggest bar is what mattered most that day.” Do not say “Shapley” unless asked, then explain the fair-credit game-theory idea (Equation 10.38, Chapter 10).
2. **Show exactly three waterfalls**, each illustrating a different mechanism, never twenty.

3. **Keep the global beeswarm in an appendix.** The summary plot is methodology, not a result; lead with the three stories and have the beeswarm as backup.

The three waterfalls to pick for an RV forecaster:

1. **A correct vol-spike forecast.** “The model predicted next-day RV roughly double its average. The drivers were a high lagged daily RV (+), a detected jump the prior session (+), and an elevated VIX (+). Realized vol spiked as forecast.”
2. **A missed forecast from an unseen event.** “The model predicted a calm day from low lagged RV and a flat VIX, but an unscheduled macro announcement triggered a spike. SHAP shows the forecast rested entirely on backward-looking RV features; the model cannot see a surprise that is not in its inputs.”
3. **A tree-vs-HAR disagreement.** “HAR forecast a mild rise; the tree forecast a sharp one. SHAP shows the gap came from an interaction between lagged RV and implied volatility, the HARQ-type effect (Chapter 6) the tree captures but linear HAR cannot, motivating the HAR+tree blend in Section 11.9.”

Why This Matters

The waterfall is the most persuasive chart in a volatility-model review because it mirrors the additive P&L attribution risk committees already use. “Lagged RV contributed +X, the jump term +Y, VIX +Z” is a sentence a portfolio manager can challenge on economic grounds, which is exactly the scrutiny that separates a real signal from an overfit one.

11.6 The Christensen–Siggaard–Veliyev Evidence

The most comprehensive academic horse-race of ML methods for realized volatility forecasting is Christensen et al. (2023). Their setup: 29 DJIA stocks, daily, weekly, and monthly RV forecasting, with a feature set spanning RV lags, implied volatility, VIX, momentum, volume, earnings announcements, and macroeconomic indicators.

Christensen, Siggaard, and Veliyev (2023)

Gradient-boosted trees are among the top-performing models for daily RV forecasting across 29 DJIA stocks. Three findings matter most:

1. **Rich features amplify the ML advantage.** When the feature set includes only RV lags (what HAR uses), trees offer minimal improvement. When the feature set expands to include implied volatility, VIX, volume, and macroeconomic indicators, trees pull ahead by 4–10% in MSE at the daily horizon, and by substantially more at longer horizons.
2. **Longer horizons favor ML.** The gap between tree models and HAR is larger for weekly and monthly RV prediction than for daily. At longer horizons, nonlinear feature interactions matter more because the direct autoregressive signal decays.
3. **Accumulated Local Effects (ALE) plots reveal drivers.** ALE plots (a model-agnostic alternative to partial dependence) show that the tree models primarily exploit the interaction between lagged RV and implied volatility, exactly the HARQ-type interaction from Chapter 6, but estimated nonparametrically rather than imposed by hand.

The study uses a static 70/10/20 train-validation-test split for the ML models, while the non-regularized HAR benchmarks use a rolling-window scheme. The out-of-sample test period spans several years, covering both calm and crisis regimes.

11.7 The Optiver Kaggle Evidence

Academic papers evaluate models carefully but on relatively simple feature sets. The Optiver Realized Volatility Prediction competition (Kaggle, 2021) provides the complementary experiment: thousands of teams competing to predict 10-minute-ahead realized volatility from limit order book data across over 100 stocks.

Optiver Kaggle Competition (2021)

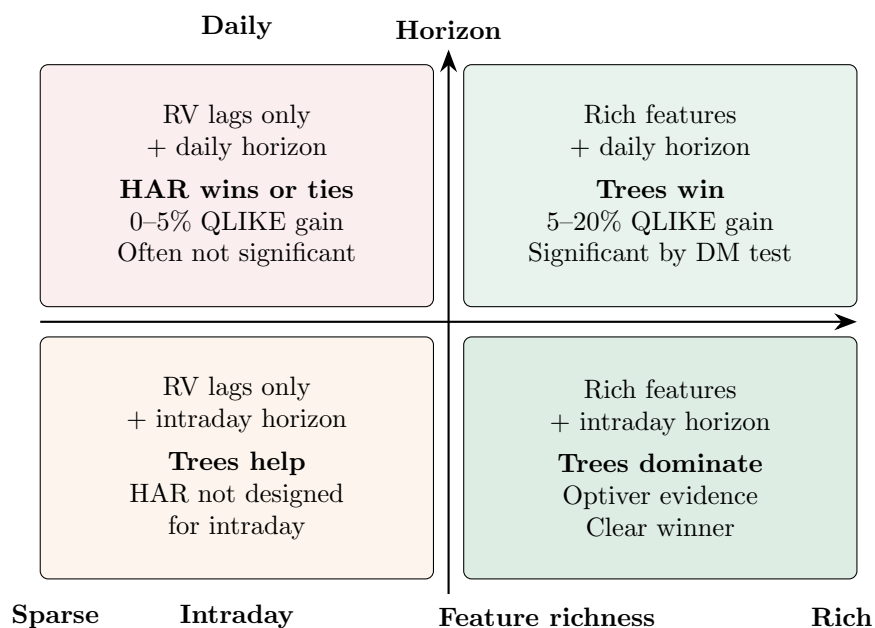
LightGBM ensembles dominated the leaderboard. The top solutions shared three characteristics:

1. **Feature engineering dominated.** WAP (weighted average price) returns, price acceleration, volume imbalance profiles, bid-ask spread dynamics, trade-flow toxicity (Chapter 10). Winners spent 80% of their effort on features and 20% on modeling.
2. **Trees beat deep learning.** Transformer and LSTM models were tried extensively. They did not beat well-tuned LightGBM on the public or private leaderboard.
3. **Blending helped, but only modestly.** Top solutions blended 3–5 LightGBM models (different feature subsets or random seeds). Gains from blending were 1–3%, far smaller than gains from better features.

The Optiver setting differs from academic setups in two ways. First, the prediction horizon is 10 minutes, not one day, so HAR is not a natural benchmark (it was designed for daily or longer horizons). Second, the feature space includes tick-level order book data, which is far richer than what most academic studies use. The lesson about feature engineering transfers directly to the daily setting, but the absolute performance numbers do not.

11.8 The Honest Assessment

The evidence above might suggest that trees always win. They do not. Here is a regime-by-regime summary of when ML adds value, grounded in the literature.



Daily horizon, RV-only features: HAR is extremely competitive

When you give a tree ensemble the same three features HAR uses (RV_{t-1} , $RV_{t-1}^{(w)}$, $RV_{t-1}^{(m)}$), the improvement over HAR is 0–5% in QLIKE, and it is often not statistically significant by the Diebold–Mariano test (Chapter 17). HAR already captures the dominant autoregressive structure. Trees can only add nonlinear kinks, which are small and unstable on 1,250 observations.

Bollerslev et al. (2024) demonstrate that a rolling-window HAR with properly selected window length matches or beats off-the-shelf ML models. The key insight: HAR’s advantage comes from its parsimonious structure (3 parameters), which is well-suited to small, noisy, autocorrelated data.

Daily horizon, rich features: trees win

When you add the full feature set from Chapter 10 (implied volatility, jumps, signed semivariances, cross-asset, sentiment), trees pull ahead by 5–20% in QLIKE. The reason: these features contain nonlinear interactions that HAR cannot exploit without hand-crafting interaction terms. The tree ensemble discovers these interactions automatically.

Christensen et al. (2023) confirm this pattern: the ML advantage grows with feature-set richness.

Intraday horizons: trees are necessary

HAR was designed for daily (or longer) horizons. For 10-minute or 1-hour ahead prediction from order book data, HAR has no natural formulation. Trees operate on any feature matrix, regardless of the time scale, and the Optiver evidence shows they dominate at short horizons.

Stress regimes: ML stumbles

Here is the uncomfortable finding. ML models, including trees, tend to underperform HAR during extreme events (VIX spikes, flash crashes, pandemic onset). The reason: tree ensembles trained predominantly on calm-regime data have never seen the patterns that emerge during crises. They extrapolate poorly because trees are piecewise-constant; predictions in extreme leaves are based on very few training observations.

Rahimikia and Poon (2020) document this explicitly: their ML model beats HAR on 90% of out-of-sample days, but fails catastrophically on the remaining 10%, which are precisely the days that matter most for risk management. Section 11.9 addresses this.

Does anything beat linear models?

Branco et al. (2024) ask this question directly and find that the answer is “often no, when the comparison is fair.” Their main point: many published ML-beats-HAR results use default hyperparameters for HAR (fixed window, OLS) while giving the ML model a full tuning budget. When both models receive equal care (rolling windows, feature selection, proper tuning), the gap shrinks or disappears for daily RV with standard features.

Where the ML Gains Come From

The gains from ML come from richer features and longer horizons, not from nonlinear modeling of the same three RV lags that HAR uses. If your tree model does not beat HAR on the same features, you have overfit noise (Chapter 6). If it does beat HAR, check whether the gain survives the Diebold–Mariano test and Deflated Sharpe Ratio (Chapter 17).

11.9 Ensemble with HAR

Section 11.8 revealed a tension: trees win most days but fail in stress. HAR is robust in stress but misses nonlinear patterns in calm periods. The natural solution: combine them.

Rahimikia and Poon (2020) propose exactly this. Their ML model (a random forest, but the logic extends to any tree ensemble) beats HAR on roughly 90% of out-of-sample days. On the remaining 10% (concentrated during stress), HAR wins decisively. The combined forecast outperforms both standalone models.

Simple weighted average

The simplest combination:

$$\hat{\sigma}_{\text{combo},t}^2 = w \cdot \hat{\sigma}_{\text{HAR},t}^2 + (1 - w) \cdot \hat{\sigma}_{\text{tree},t}^2, \quad (11.10)$$

where $w \in [0, 1]$ is estimated by minimizing QLIKE on a purged validation set. Typical values: $w \in [0.2, 0.4]$, giving the tree model majority weight but retaining HAR’s stabilizing influence.

- $\hat{\sigma}_{\text{HAR},t}^2$ is the HAR forecast from Chapter 6.
- $\hat{\sigma}_{\text{tree},t}^2$ is the LightGBM or XGBoost forecast.
- w is the HAR weight, estimated on the validation set.

Regime-switching combination

A more sophisticated version: let w depend on the current regime. Define a “stress indicator” s_t (e.g., $s_t = \mathbf{1}[\text{VIX}_t > 30]$ or a GMM-based regime probability). Then:

$$\hat{\sigma}_{\text{combo},t}^2 = w(s_t) \cdot \hat{\sigma}_{\text{HAR},t}^2 + [1 - w(s_t)] \cdot \hat{\sigma}_{\text{tree},t}^2, \quad (11.11)$$

with $w(s_t)$ increasing during stress (rely more on HAR when volatility is elevated). This connects directly to the regime overlay in Chapter 14.

Why This Matters

If your holdout period includes a volatility spike (e.g., a VIX jump above 30), this adaptive weighting can recover QLIKE losses that a fixed-weight blend would miss.

Worked Example: HAR + LightGBM Ensemble

Continuing the SPY example from Section 11.4. On the 126-day holdout:

Model	QLIKE	Hit Rate (vs. HAR)
HAR alone	0.148	—
LightGBM alone	0.133	72% of days
Combo ($w = 0.3$)	0.126	78% of days

The combination achieves a further 5.3% QLIKE improvement over the standalone tree model. Inspecting the days where LightGBM alone underperforms, they cluster around the three largest VIX spikes in the holdout period. HAR’s contribution on those days pulls the ensemble back toward the realized value.

Diebold–Mariano test, combo vs. HAR: $p = 0.02$ (significant at 5%). Combo vs. LightGBM alone: $p = 0.11$ (suggestive but not significant at 5%, reflecting the small holdout size).

Why Combining Works

HAR and tree ensembles make different types of errors. HAR's errors are small and unbiased in calm periods, large but mean-reverting in stress. Tree errors are small in calm periods but can be persistently biased during regimes unseen in training. Averaging diversifies the error, the same principle behind portfolio diversification.

11.10 DART: Dropout Regularization for Boosted Trees

Section 11.4 covered the standard regularization toolkit for gradient boosting: shallow trees, subsampling, and slow learning rates. There is one more technique worth understanding, and it addresses a subtle failure mode that the standard controls do not.

The over-specialization problem

Recall from Equation 11.5 that standard gradient boosting shrinks every tree's contribution by the same learning rate η . Early trees see large residuals (the raw signal), so they learn the dominant pattern: the strong autoregressive relationship between lagged RV and future RV. Later trees see only the leftover residuals after those early trees have already captured the bulk of the variance. Shrinkage treats every tree identically, so the early trees permanently dominate the ensemble's output.

This creates a problem called **over-specialization**: later trees become highly specialized to the narrow residual signal left by the first few trees. If those early trees overfit to noise in the training data, every subsequent tree inherits that bias. The ensemble's predictions become overly dependent on a small number of early trees.

In Plain English

Imagine a group project where one person does most of the work in the first week. Everyone else spends the remaining weeks patching small gaps in that person's draft. If the first person made a fundamental error, the entire project is built on a flawed foundation, and no amount of patching fixes it. Standard shrinkage is like asking everyone to speak more quietly; it does not change the fact that the first speaker set the direction.

Dropout applied to trees

DART (Dropouts meet Multiple Additive Regression Trees) solves over-specialization by borrowing the dropout idea from neural networks (Vinayak and Gilad-Bachrach, 2015). In each boosting round, instead of computing residuals from the full ensemble, DART randomly **drops** (removes) a subset of the previously built trees. The new tree is then trained on the residuals of the *reduced* ensemble, the one with the dropped trees removed.

Concretely, let $\mathcal{D} \subset \{1, \dots, m-1\}$ denote the set of tree indices dropped at round m , chosen by including each tree independently with **drop rate** p . The prediction used to compute residuals becomes:

$$\hat{y}_t^{(\text{drop})} = \sum_{k \notin \mathcal{D}} h_k(\mathbf{x}_t), \quad (11.12)$$

where:

- $h_k(\mathbf{x}_t)$ is the prediction of tree k for observation t (already including its learned weight),
- $\mathcal{D}$ is the random dropout set for this boosting round,

- $\hat{y}_t^{(\text{drop})}$ is the reduced ensemble prediction (with dropped trees excluded).

The new tree h_m is fitted to the residuals $r_t = y_t - \hat{y}_t^{(\text{drop})}$. Because some trees were dropped, these residuals are larger than they would be under the full ensemble, so the new tree must learn to compensate for the missing trees. This forces it to capture general patterns rather than narrow residual corrections.

In Plain English

After training, all trees are brought back for the final prediction. The result is an ensemble where each tree carries more independent information.

Prediction normalization

After fitting the new tree h_m , DART must normalize the ensemble so that adding the new tree does not inflate predictions. The dropped trees and the new tree are rescaled so the ensemble stays calibrated:

$$\hat{y}_t = \sum_{k \notin \mathcal{D}} h_k(\mathbf{x}_t) + \frac{|\mathcal{D}|}{|\mathcal{D}|+1} \sum_{k \in \mathcal{D}} h_k(\mathbf{x}_t) + \frac{1}{|\mathcal{D}|+1} h_m(\mathbf{x}_t), \quad (11.13)$$

where:

- $|\mathcal{D}|$ is the number of dropped trees,
- the factor $\frac{|\mathcal{D}|}{|\mathcal{D}|+1}$ rescales the dropped trees down to make room for the new tree,
- the factor $\frac{1}{|\mathcal{D}|+1}$ rescales the new tree so it contributes an “equal share” alongside the dropped trees.

In Plain English

Without normalization, the new tree would be trained against a gap left by the dropped trees (large residuals), so its raw predictions would be too large. The normalization splits the “budget” of the dropped trees evenly between the returning dropped trees and the new tree.

DART versus standard shrinkage

The key difference from the learning rate η in standard boosting:

	Shrinkage (η)	DART (drop rate p)
Mechanism	Scales <i>every</i> tree by the same constant η	Randomly removes <i>entire trees</i> each round
Effect on early trees	Reduced proportionally but still dominant	May be dropped, forcing later trees to learn independently
Diversity	All trees see residuals from the same full ensemble	Each tree sees residuals from a different random subset
Analogy	Turning down everyone’s volume equally	Randomly muting some speakers so others must step up

Shrinkage and DART are not mutually exclusive. In LightGBM, you can set `boosting_type='dart'` and still specify a learning rate, though in practice the learning rate is often set higher with DART (e.g., 0.05–0.1) because the dropout itself provides regularization.

DART for Volatility Forecasting

Over-specialization is particularly relevant for RV forecasting. The dominant signal in volatility data is the autoregressive persistence captured by HAR's three lagged averages (Chapter 6). In standard boosting, the first few trees learn this persistence, and all subsequent trees become narrowly specialized to small residual patterns that may not generalize. DART forces later trees to occasionally reconstruct the autoregressive signal on their own, building redundancy into the ensemble and reducing dependence on any single tree's overfitting.

In LightGBM, enable DART by setting `boosting_type='dart'`. The key hyperparameter is the drop rate: start with `drop_rate` $\in [0.05, 0.15]$ and tune via purged CV (Section 11.4). Higher drop rates increase diversity but slow convergence; lower rates approach standard GBDT behavior. DART disables early stopping (because the loss is non-monotone due to random dropping), so you must set `num.iterations` explicitly rather than relying on patience-based stopping.

DART Disables Early Stopping

With standard GBDT, you monitor validation loss and stop when it plateaus. DART's random dropout makes the validation loss noisy and non-monotone from round to round, so early stopping triggers prematurely. When using DART, fix the number of boosting rounds via cross-validation rather than relying on early stopping. This increases tuning cost, but the regularization benefit of dropout often compensates.

11.11 Summary

1. Tree-based ensembles (LightGBM, XGBoost) are the default ML model for tabular volatility data because they automatically discover nonlinear interactions and threshold effects.
2. Gradient boosting builds the forecast sequentially: each tree corrects the residual errors of the ensemble so far (Equation 11.5).
3. For volatility forecasting, use a custom QLIKE loss (Equations 11.6–11.8), not the default MSE loss.
4. Default hyperparameters overfit on small volatility samples. Use shallow trees (`max_depth` 3–5), large minimum leaf sizes (50–200), aggressive subsampling (0.6–0.8), and slow learning rates (0.01–0.05) with early stopping on a purged validation set.
5. Christensen et al. (2023): trees are among the best for daily RV forecasting, with gains amplified by richer feature sets and longer horizons.
6. The Optiver Kaggle competition confirms that feature engineering matters far more than model architecture: trees plus good features beat deep learning plus raw data.
7. With RV-only features at daily horizons, HAR is extremely competitive. Trees offer 0–5% QLIKE improvement, often not significant.
8. With rich features (implied vol, jumps, cross-asset) at daily horizons, trees win by 5–20% in QLIKE.
9. At intraday horizons, trees are clearly necessary; HAR was not designed for this regime.
10. During extreme stress, tree models tend to underperform HAR because they extrapolate poorly from calm-period training data.

11. Combining HAR and tree forecasts (Equation 11.10) captures the best of both: the tree's nonlinear skill in calm markets and HAR's robustness in stress (Rahimikia and Poon, 2020).
12. Branco et al. (2024) and Bollerslev et al. (2024) caution that many ML-beats-HAR claims reflect unfair comparisons. When both models are properly tuned, the gap is smaller than headline numbers suggest.
13. DART (Vinayak and Gilad-Bachrach, 2015) applies dropout to boosted trees: randomly dropping previous trees during each boosting round prevents over-specialization, where later trees become overly dependent on early trees that captured the dominant autoregressive signal. Use `drop_rate` $\in [0.05, 0.15]$; DART disables early stopping.
14. The gains from ML come from richer features and longer horizons, not from nonlinear modeling of the same three RV lags. Chapter 13 explores whether deep learning changes this conclusion.

Paper	Key Result	Relevance
Christensen et al. (2023)	Trees among best for daily RV; gains grow with feature richness and horizon length	Primary evidence for Sections 11.6 and 11.8
Branco et al. (2024)	Linear models competitive when comparison is fair	Calibrates expectations in Section 11.8
Bollerslev et al. (2024)	Rolling-window HAR with proper window matches off-the-shelf ML	Strongest HAR defense
Rahimikia and Poon (2020)	ML beats HAR 90% of days, fails in stress; ensemble solves it	Motivates Section 11.9
Audrino and Knaus (2016)	QLIKE-optimized trees outperform MSE-optimized trees for RV	Justifies custom loss in Section 11.3
Gu et al. (2020b)	Trees and neural nets dominate linear models in cross-sectional return prediction with rich features	Canonical ML horse-race; context for tree methods
Vinayak and Gilad-Bachrach (2015)	Dropout for boosted trees reduces over-specialization; each tree learns more independently	Regularization technique in Section 11.10

Chapter 12

Rashomon Sets and Interpretable Trees

You have built a volatility model, run SHAP, and found that VIX is the most important feature. You refit on a slightly different training window and now ATM implied volatility takes over. A third refit: lagged RV_{t-1} wins. Which feature is *actually* important? The unsettling answer is that any single model’s feature importance is a sample of size one from a large population of equally-good models. Drawing conclusions from that single sample is exactly the kind of reasoning we would reject in any other statistical context.

This chapter introduces a principled alternative. Instead of asking “what does *my* model say is important?” we ask “what do *all* near-optimal models agree is important?” The set of all such models is called the **Rashomon set**, and the tools for analyzing it give us something SHAP cannot: provably stable feature importance that does not change when you perturb the training window by a few days.

What You Need Before This Chapter

- **Chapter 11.** Decision tree fundamentals: how splits are chosen, how CART grows a tree, what makes a tree “optimal.” The distinction between greedy (top-down) and globally optimal (branch-and-bound) tree construction.
- **SHAP basics.** The idea that SHAP assigns each feature a contribution to each individual prediction, and that global SHAP importance is obtained by averaging $|\phi_i|$ across predictions. You should know what a SHAP summary plot looks like, but you do *not* need to understand the Shapley value derivation in detail.
- **Feature importance concepts.** Permutation importance (shuffle a feature, measure how much loss increases) and split-count importance (how often a feature appears in tree splits). Both are covered in Chapter 11.

12.1 The Problem with Single-Model Explanations

Feature importance from a single model is not a fact about the data. It is a fact about *one model fit to the data*. If there are many models that fit the data almost equally well, each may assign very different importances, and the one we happen to report is determined by arbitrary choices: the random seed, the exact train/test split boundary, the hyperparameter grid, even the order in which features are processed.

12.1.1 Why SHAP Importance Is Unreliable with Near-Substitute Features

Consider two features that are highly correlated: VIX (the CBOE volatility index) and the ATM implied volatility from the SPX options surface. Both carry similar information about the market’s forward-looking volatility expectation. A tree model must choose one or the other at each split. If the model happens to split on VIX first, VIX gets credit for the variance it explains, and ATM IV gets little credit because the residual information it adds is small. Reverse the split order and ATM IV gets the credit.

SHAP values (Lundberg and Lee, 2017) partially address this by considering all feature orderings. For a model f and a prediction at input x , the SHAP value for feature i is the unique additive attribution satisfying local accuracy, missingness, and consistency:

$$\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)], \quad (12.1)$$

where:

- F is the full feature set, $|F| = M$
- S is a subset of features *excluding* feature i
- $f_x(S) = \mathbb{E}[f(z) \mid z_S = x_S]$ is the expected prediction when only features in S are observed

Why SHAP Doesn’t Fully Solve the Problem

SHAP computes importances for all feature orderings, but it does so for a *single fixed model* f . The problem is upstream: the model f itself is just one of many near-optimal models. A different near-optimal model f' with the same predictive accuracy may produce entirely different SHAP values, especially for correlated features. Averaging over orderings within one model does not average over the space of *models*.

12.1.2 Importance Instability in Practice

The instability is not hypothetical. In volatility forecasting, features like VIX, VVIX, ATM IV, the 25-delta skew, and lagged RV_{t-1} are all moderately to highly correlated. A gradient-boosted tree fit on 2017–2021 data might rank VIX as the top feature, while the same model fit on 2017–2022 data (just one extra year) might rank RV_{t-1} first and demote VIX to third. The model’s accuracy barely changes between these two fits, but the “feature importance story” changes completely.

This creates a serious practical problem. In an internship setting, you will present your model’s feature importance to a desk or to a portfolio manager. If the ranking shuffles every time you retrain, your explanation is not credible. What you need is a statement like: “Across *all* models within 2% of optimal accuracy, VIX and ATM IV are interchangeable substitutes, but lagged RV is always important.” That statement requires looking at all good models, not just one.

12.2 The Rashomon Set

The idea of many equally-good models was articulated by Leo Breiman, who called it the **Rashomon effect**, a reference to the Akutagawa story (and Kurosawa film) in which multiple witnesses give contradictory but individually plausible accounts of the same event. The **Rashomon set** formalizes this: it is the collection of all models whose loss is close to the best achievable loss.

12.2.1 Formal Definition

We need three ingredients: a model class $\mathcal{F}$ (e.g., all decision trees of depth $\leq d$), a dataset $(\mathbf{X}, \mathbf{y})$, and a reference model f^* that achieves the best loss within $\mathcal{F}$.

ϵ -Rashomon Set (,)

Given a model class $\mathcal{F}$, a benchmark model $f^* \in \mathcal{F}$, and a tolerance $\epsilon > 0$, the ϵ -Rashomon set is

$$\mathcal{R}(\epsilon, f^*, \mathcal{F}) = \{f \in \mathcal{F} \mid L(f) \leq (1 + \epsilon) L(f^*)\}, \quad (12.2)$$

where $L(f)$ is the loss (e.g., misclassification rate, squared error) of model f evaluated on the training data.

The symbols:

- ϵ : the tolerance parameter. Setting $\epsilon = 0.02$ means we accept models within 2% of the best loss. Typical values in the literature range from 0.01 to 0.10 (Xin et al., 2022).
- f^* : the reference model, usually the empirical risk minimizer within $\mathcal{F}$ (e.g., found by GOSDT for optimal sparse trees).
- $L(f^*)$: the best achievable loss in the class.

For decision trees specifically, Xin et al. (2022) define the objective function as the misclassification loss plus a sparsity penalty:

$$\text{Obj}(t, \mathbf{X}, \mathbf{y}) = \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbb{I}[\hat{y}_i \neq y_i]}_{\text{misclassification loss}} + \underbrace{\lambda H_t}_{\text{sparsity penalty}}, \quad (12.3)$$

where:

- $\hat{y}_i$ is the prediction of tree t for observation i
- H_t is the number of leaves in tree t
- $\lambda > 0$ is a regularization parameter penalizing complexity

The Rashomon set threshold then becomes $\theta_\epsilon = (1 + \epsilon) \times \text{Obj}(t_{\text{ref}}, \mathbf{X}, \mathbf{y})$, and any tree t with $\text{Obj}(t, \mathbf{X}, \mathbf{y}) \leq \theta_\epsilon$ is in the set (Xin et al., 2022).

Rashomon Sets for Regression

Our volatility project uses squared-error loss (or QLIKE), not misclassification. The Rashomon set concept applies identically: replace $L(f)$ with QLIKE or MSE and everything carries through. The computational algorithms (TreeFARMS, SPLIT) currently target classification trees, but the conceptual framework (and the VIC/RID analysis tools) is loss-function agnostic.

12.2.2 How Large Is the Rashomon Set?

The hypothesis space of sparse trees is enormous. For trees of depth at most 4 with only 10 binary features, the number of possible trees exceeds 9.3×10^{20} (Xin et al., 2022). The Rashomon set, fortunately, is usually a tiny fraction of this space, but “tiny fraction” can still be a very large number in absolute terms.

Xin et al. (2022) report that on the COMPAS dataset with $\lambda = 0.005$ and a 15% Rashomon threshold, the set contains approximately 10^{12} trees. On smaller datasets (Monk2, Bar), sets of 10^5 to 10^8 trees are typical. The key insight from their experiments is that *natural baselines* (BART, Random Forest, CART + sampling) find at best a tiny sliver of the Rashomon set: they recover only hundreds to thousands of trees when the true set contains millions or more.

Worked Example:

Two Trees, Different Features, Same Accuracy Consider a toy dataset with 100 observations, 3 binary features (X_1, X_2, X_3), and a binary label Y . Suppose the optimal tree t^* has 4 leaves, splits on X_1 then X_2 , and achieves $\text{Obj}(t^*, \mathbf{X}, \mathbf{y}) = 0.08 + 0.01 \times 4 = 0.12$. With $\epsilon = 0.05$, the Rashomon threshold is $\theta_{0.05} = 1.05 \times 0.12 = 0.126$. Now consider tree t' with 3 leaves that splits on X_3 first, then X_1 . Suppose it achieves $\text{Obj}(t', \mathbf{X}, \mathbf{y}) = 0.09 + 0.01 \times 3 = 0.12$.

Tree	Splits on	Leaves	Error	Objective
t^*	X_1, X_2	4	0.08	0.12
t'	X_3, X_1	3	0.09	0.12

Both trees are in the Rashomon set. Tree t^* relies heavily on X_2 ; tree t' does not use X_2 at all but uses X_3 instead. If we report SHAP importance for t^* alone, we conclude X_2 is important. The Rashomon set reveals that X_2 and X_3 are substitutes: the data supports models using either one.

12.3 Optimal Sparse Decision Trees

Before we can enumerate *all* near-optimal trees, we must be able to find *one* provably optimal tree.

12.3.1 Why Not Just Use CART?

CART (Breiman et al., 1984) and other greedy algorithms build trees top-down, choosing the best split at each node independently. This is fast ($O(npd)$ for n samples, p features, depth d), but greedy splits can be suboptimal. Babbar et al. (2025) show that greedy methods exhibit an average gap of 1–2 percentage points from the optimum, and on some datasets (e.g., COMPAS) the gap can reach 10 percentage points.

The problem with greedy construction is that a split that looks best at the root may set up poor options downstream. Global optimization avoids this by searching the *entire* space of trees up to a given depth.

12.3.2 Branch-and-Bound with Dynamic Programming

Modern optimal tree algorithms (GOSDT, MurTree) solve the optimization problem exactly:

$$\mathcal{L}^*(D, d, \lambda) = \min_{T \in \mathcal{T}} L(T, D, \lambda) \quad \text{s.t.} \quad \text{depth}(T) \leq d, \quad (12.4)$$

where $L(T, D, \lambda) = \frac{1}{N} \sum_{i=1}^N \ell(T(\mathbf{x}_i), y_i) + \lambda S(T)$ is the regularized loss and $S(T)$ is the number of leaves (Babbar et al., 2025).

The key insight behind branch-and-bound is that the optimal solution for a dataset D at depth d' depends on the solutions for subsets $D(f)$ and $D(\bar{f})$ at depth $d' - 1$, where f is the splitting feature. Starting from the root, the algorithm considers all candidate features, tracking upper

and lower bounds on the objective at each split. When the lower bound of a subtree exceeds the current best upper bound, that entire subtree is pruned.

12.3.3 SPLIT: Greedy Where It Doesn't Matter, Optimal Where It Does

A key empirical observation by Babbar et al. (2025) is that greedy splits near the *leaves* are almost always optimal, while greedy splits near the *root* often deviate from the optimum. This makes sense: near the leaves, there are only a few possible splits left, so the greedy choice among them is unlikely to be far from the best. Near the root, a bad split propagates errors through the entire tree.

SPLIT (Sparse Lookahead for Interpretable Trees) exploits this observation. It takes a **lookahead depth** parameter $d_l < d$ and performs full branch-and-bound optimization for splits up to depth d_l , then switches to greedy splitting for the remaining $d - d_l$ levels.

SPLIT Runtime

For a dataset with n samples, k binary features, depth budget d , and lookahead depth d_l , SPLIT (Algorithm 2 in Babbar et al. (2025)) has runtime

$$\mathcal{O}(n(d - d_l) k^{d_l+1} + n k^{d-d_l}).$$

The polynomial-time variant, LicketySPLIT, achieves $\mathcal{O}(nk^2d^2)$, comfortably polynomial and dramatically faster than the $\mathcal{O}((2k)^d)$ worst case of fully optimal methods.

Interpretable Trees for Vol Forecasting

For our volatility project, a depth-5 tree with 10–15 features is a realistic interpretable baseline. An optimal tree of this size can be found in under a second with LicketySPLIT.

12.3.4 STreeD: Piecewise-Linear Regression Trees

The optimal tree methods discussed so far assume each leaf predicts a *constant*: the mean of the training labels that fall into that leaf. This creates a staircase-shaped prediction function: the forecast jumps from one flat value to another at each split boundary. van den Bos et al. (2024) introduce **STreeD** (STreeD Regression Trees), a dynamic-programming framework that extends optimal regression trees to **piecewise-linear** leaf models. They develop three methods of increasing expressiveness:

1. **SRT-C** (piecewise-constant): An improved DP algorithm for constant-leaf regression trees with a specialized depth-two solver that achieves orders-of-magnitude speedups over previous optimal methods (e.g., $18\times$ faster than OSRT on average).
2. **SRT-SL** (simple linear regression): The *first optimal method* for piecewise simple linear regression trees. Each leaf fits a one-variable linear model $y = \hat{\beta}_0 + \hat{\beta}_j x_j$, selecting the single best feature j for that leaf. Ridge regularization prevents overfitting in small leaves.
3. **SRT-L** (multiple linear regression): The *first optimal method* for piecewise multiple linear regression trees. Each leaf fits a full linear model with elastic net regularization ($\ell_1 + \ell_2$ penalty), solved via coordinate descent.

The key idea behind SRT-SL is that each leaf selects *one* continuous feature and fits a simple linear regression with ridge regularization. The leaf-level objective for a leaf containing data subset $\mathcal{D}$ is:

$$\min_{j, \hat{\beta}_0, \hat{\beta}_j} \sum_{(x, y) \in \mathcal{D}} (y - \hat{\beta}_0 - x_j \hat{\beta}_j)^2 + \gamma \hat{\beta}_j^2, \quad (12.5)$$

where:

- j is the index of the continuous feature selected for this leaf
- $\hat{\beta}_0$ is the intercept, $\hat{\beta}_j$ is the slope on feature x_j
- $\gamma > 0$ is the ridge regularization parameter
- The leaf selects whichever feature j yields the smallest regularized SSE

The closed-form solutions for the optimal coefficients are:

$$\hat{\beta}_j = \frac{n \sum x_j y - \sum y \sum x_j}{n \sum x_j^2 - (\sum x_j)^2 + n\gamma}, \quad (12.6)$$

$$\hat{\beta}_0 = \frac{\sum y}{n} - \hat{\beta}_j \frac{\sum x_j}{n}, \quad (12.7)$$

where all sums run over the instances in the leaf and $n = |\mathcal{D}|$.

Why Piecewise-Linear Leaves Matter

A constant-leaf tree predicts $\bar{y}$ in each partition, creating a staircase function. A linear-leaf tree fits a slope within each partition, so it captures *local trends*. For volatility, this means a leaf can express “RV increases linearly with VIX within this regime” rather than just “RV is high in this regime.” The tree handles the nonlinear regime boundaries (splits), and the linear models handle the smooth within-regime relationships.

Scalability. The depth-two algorithm from van den Bos et al. (2024) is the key to STreeD’s performance advantage. By precomputing per-instance costs (the sums $\sum y$, $\sum y^2$, $\sum x_j$, $\sum x_j^2$, and $\sum x_j y$ for every feature j and every data subset defined by pairs of binary splits), the depth-two solver avoids redundant traversals of the data. Remarkably, fitting a simple linear regression model per leaf (SRT-SL) costs almost nothing extra compared to fitting a constant (SRT-C), because the additional statistics ($\sum x_j$, $\sum x_j^2$, $\sum x_j y$) can be accumulated in the same pass.

Interpretability. Every SRT-SL prediction decomposes into two transparent components: (1) a root-to-leaf path of binary splits that determines which regime the input falls into, and (2) a one-variable linear formula in the leaf that produces the forecast. This is what the interpretability literature calls a **short linear formula**: the entire model is human-readable and can be written on a single page.

STreeD for Volatility Forecasting

SRT-SL is particularly appealing for our project. Imagine a depth-3 tree that splits on lagged-RV quintile at the root, then on a VIX threshold, creating 4–8 leaves. Within each leaf, a simple linear regression on one feature (e.g., RV_{t-1} or the VIX level) captures the local relationship. The result is an interpretable model that can express statements like: “In the high-vol, backwardated-VIX regime, tomorrow’s RV increases by 0.15 for each unit increase in today’s RV.” This is far more informative than a constant prediction per regime, and every coefficient has a clear economic interpretation.

12.4 Enumerating the Rashomon Set

Finding one optimal tree is step one. The real power comes from finding *all* near-optimal trees, the complete Rashomon set. This section covers three algorithms of increasing scalability.

12.4.1 TreeFARMS: Exact Enumeration

TreeFARMS (Trees FAST RashoMon Sets) by [Xin et al. \(2022\)](#) was the first algorithm to *completely* enumerate the Rashomon set for sparse decision trees. It builds on the GOSDT branch-and-bound framework and modifies it in two key ways:

1. **Rashomon pruning.** Instead of pruning subproblems whose lower bound exceeds the optimal objective (as in standard GOSDT), TreeFARMS prunes those whose lower bound exceeds the *Rashomon threshold* θ_ϵ . This retains more of the search space: exactly the near-optimal region.
2. **Return all models.** Instead of returning only the single best tree, TreeFARMS returns every tree in the Rashomon set, stored in a compact **Model Set** data structure.

The Model Set representation is critical for scalability. A **Model Set Instance (MSI)** represents a subproblem paired with its objective value. Many trees share identical subtrees, and the Model Set exploits this by storing shared components once. The loss function for decision trees takes on a discrete set of values (approximately n distinct values for n training samples), while the number of trees in the Rashomon set can be orders of magnitude larger. By grouping trees with the same objective, TreeFARMS avoids massive data duplication.

TreeFARMS: From Optimization to Feasibility

Standard ML algorithms solve an *optimization* problem: find the single best model. TreeFARMS solves a *feasibility* problem: find all models within ϵ of the best. This reframing is the core contribution. Perhaps the tiny sacrifice in empirical risk makes the difference between a model that can be used (interpretable, fair, aligned with domain knowledge) and one that cannot.

Limitations. TreeFARMS provides exact enumeration but its runtime and memory scale exponentially with tree depth and the number of features. On the Bike dataset ($n \approx 17,000$, $k = 60$ binary features, depth 5), TreeFARMS requires approximately 700 seconds and over 50 GB of memory ([Heile et al., 2025](#)). On larger datasets, it runs out of memory entirely.

12.4.2 RESPLIT: Fast Approximation via Greedy Leaves

[Babbar et al. \(2025\)](#) extend SPLIT to Rashomon set computation with the RESPLIT algorithm. The idea is simple: use SPLIT to find a set of “prefix trees” (partial trees optimized to the lookahead depth), then call TreeFARMS on each prefix to enumerate the near-optimal completions.

Because SPLIT uses greedy splits near the leaves, RESPLIT does not exhaustively search the full tree space. This makes it an *approximation*: it may miss some trees in the true Rashomon set. However, [Babbar et al. \(2025\)](#) show that the approximation is remarkably accurate. On six benchmark datasets, the Pearson correlation between variable importances computed from the RESPLIT-approximated Rashomon set and the full Rashomon set is nearly 1.0:

Dataset	Full (s)	RESPLIT (s)	Speedup	τ (correlation)
COMPAS	152	18	8×	1.000
Spambase	2,659	154	17×	0.930
Netherlands	4,255	216	20×	0.932
HELOC	5,564	337	17×	0.979
HIV	9,273	388	24×	0.959
Bike	14,330	194	74×	0.999

Source: Table 1 in Babbar et al. (2025). Parameters: 10 bootstrapped datasets, $\lambda = 0.02$, $\epsilon = 0.01$, depth 5, lookahead depth 2.

12.4.3 LicketyRESPLIT: Polynomial-Time Approximation

Heile et al. (2025) push the scalability further with LicketyRESPLIT, which replaces TreeFARMS entirely with a polynomial-time enumeration strategy. At each node, the algorithm considers all features and prunes splits whose LicketySPLIT-completed cost would exceed the Rashomon budget $B = (1+\epsilon) \cdot \text{LicketySPLIT}(D, \lambda, d)$. It then recursively enumerates left and right subtrees, filtering by the remaining budget.

The key theoretical results are:

LicketyRESPLIT Complexity (Heile et al. 2025)

Given a dataset D of size n with k features and max depth d :

- **Runtime:** $\mathcal{O}(|R| n k^3 d^3)$, where $|R|$ is the size of the recovered Rashomon set. This is polynomial in n , k , and d , and linear in $|R|$.
- **Memory:** $\mathcal{O}(nk + \sum_{f \in R} S(f))$, proportional to the input size plus the output size.

In practice, LicketyRESPLIT achieves order-of-magnitude improvements over both TreeFARMS and RESPLIT:

Dataset	LicketyRESPLIT Time / RAM	TreeFARMS Time / RAM	RESPLIT Time / RAM
Bike	18.8 s / 438 MB	685 s / 51 GB	184 s / 528 MB
Bank	123 s / 776 MB	OOM	238 s / 2 GB
Covertypes	507 s / 1.8 GB	1819 s / 68 GB	1295 s / 3 GB
Student	1.7 s / 370 MB	351 s / 4.7 GB	4.0 s / 382 MB

Source: Table 1 in Heile et al. (2025). Parameters: $\lambda = 0.01$, $\epsilon_{mult} = 0.01$, max depth = 5.

Despite being an approximation, LicketyRESPLIT achieves near-perfect precision and recall relative to the true Rashomon set. On six benchmark datasets, precision is ≥ 0.91 and recall is ≥ 0.90 across all tested configurations (Heile et al., 2025).

Exact vs. Approximate Enumeration

TreeFARMS gives you the *exact* Rashomon set: every tree in the set, and no tree outside it. RESPLIT and LicketyRESPLIT are approximations: they may miss a few trees (recall < 1) and occasionally include a tree slightly outside the boundary (precision < 1). For downstream variable importance analysis, this distinction rarely matters: the approximate sets produce virtually identical importance rankings. But if you need a formal guarantee (“no tree in the Rashomon set uses feature X_7 ”), only exact enumeration suffices.

Rashomon set algorithms are developing rapidly. Figure 12.1 summarizes the three approaches and their trade-offs.



Figure 12.1: Evolution of Rashomon set enumeration algorithms for sparse decision trees. Each generation trades a small amount of exactness for dramatic improvements in scalability.

12.5 What the Rashomon Set Reveals

Two complementary tools turn the raw set of models into concrete conclusions about feature importance.

12.5.1 Model Class Reliance (MCR)

The simplest analysis is to ask: for each feature, what is the *range* of its importance across all models in the Rashomon set? This is **Model Class Reliance (MCR)**, introduced by Dong and Rudin (2020) (building on the concept from Fisher et al., 2019).

For a single model f , the **model reliance** of feature j is defined as the ratio of the loss when feature j is randomly permuted to the original loss:

$$mr_j^{\text{ratio}}(f) = \frac{L(f; [X_{\setminus j}, \bar{X}_j], Y)}{L(f; X, Y)}, \quad (12.8)$$

where:

- $X_{\setminus j}$ denotes all features except feature j
- $\bar{X}_j$ is an independent copy of X_j (i.e., feature j is reshuffled)
- A reliance of 1.0 means the feature contributes nothing; larger values indicate greater importance

MCR defines two key quantities for each feature j :

$$\text{MCR}_-(j) = \min_{f \in \mathcal{R}} mr_j(f), \quad (12.9)$$

$$\text{MCR}_+(j) = \max_{f \in \mathcal{R}} mr_j(f). \quad (12.10)$$

The interpretation is clean:

- A feature with large MCR_- is important in *every* well-performing model. It is robustly important.
- A feature with small MCR_+ is unimportant in every well-performing model. It can be safely excluded.
- A feature with small MCR_- but large MCR_+ is a *substitute*: important in some models, not in others. It can be swapped for a correlated alternative without loss of accuracy.

Xin et al. (2022) showed that TreeFARMS enables *exact* MCR computation for decision trees by directly calculating variable importance for every tree in the set and then finding the min and max.

12.5.2 Variable Importance Clouds (VIC)

MCR gives a one-dimensional summary (min and max importance) for each feature independently. **Variable Importance Clouds (VIC)** (Dong and Rudin, 2020) capture the full joint structure.

Variable Importance Cloud (,)

The **model reliance function** $MR : \mathcal{F} \rightarrow \mathbb{R}^p$ maps each model to a vector of its reliances on all p features:

$$MR(f) = (mr_1(f), mr_2(f), \dots, mr_p(f)).$$

The **Variable Importance Cloud** of the Rashomon set $\mathcal{R}$ is the set of all such vectors:

$$VIC(\mathcal{R}) = \{MR(f) : f \in \mathcal{R}\}. \quad (12.11)$$

The VIC lives in p -dimensional space, where each axis represents the importance of one feature. To visualize it, Dong and Rudin (2020) project the VIC onto all pairs of features, producing **Variable Importance Diagrams (VIDs)**, 2D scatter plots that reveal substitution patterns:

- **Non-overlapping projections** along one axis: the two features have robustly distinct importance levels. One is always more important than the other, regardless of model choice.
- **Overlapping projections**: the features are substitutes. Some near-optimal models rely on one, some on the other.
- **Negative slope** in the 2D projection: the features are direct substitutes; as one's importance increases, the other's decreases. This is the signature of correlated features competing for the same splits.

Worked Example:

VIX, VVIX, and ATM IV as Near-Substitutes Imagine constructing the Rashomon set for a volatility classification task (“will tomorrow’s RV exceed its 90th percentile?”) using depth-4 trees with the following features: lagged RV_{t-1} , RV_{t-5} , VIX, VVIX, and ATM IV.

After enumeration, we compute $MR(f)$ for each tree f in the set and project onto two pairs:

Pair 1: VIX vs. ATM IV. The 2D cloud shows a strong negative slope: when a tree relies heavily on VIX, it does not rely on ATM IV, and vice versa. The projections onto each axis overlap substantially. These features are *substitutes*: the data supports using either one, but not both simultaneously.

Pair 2: Lagged RV_{t-1} vs. VIX. The cloud is a compact blob with no clear slope. The projection of RV_{t-1} onto its axis is consistently high (all models rely on it), while VIX varies. Lagged RV is *robustly important*; VIX is a useful but substitutable addition.

Conclusion for feature selection: We must include RV_{t-1} . We may include either VIX or ATM IV (but including both adds redundancy, not information). VVIX may or may not add value depending on what else is in the model.

VIC vs. Bootstrapped SHAP

A natural reaction is: “I could just bootstrap the data, refit SHAP each time, and get a distribution of importances.” This is fundamentally different from VIC. Bootstrapped SHAP resamples the *data* but always uses the same model class and always reports the importance of the single best model within that class. VIC holds the data fixed and varies the *model*: it considers every near-optimal model, not just the best one.

The distinction matters because bootstrap variability conflates data uncertainty with model uncertainty. VIC isolates model uncertainty: “given this exact dataset, how much does the importance depend on which near-optimal model we happened to pick?” This is the right question when your concern is the stability of your explanations, not the variability of your data.

12.5.3 Rashomon Importance Distributions (RID)

MCR and VIC answer the question “what is the range of importances across all near-optimal models *for this dataset?*” But this answer is fragile: the Rashomon set itself changes when the dataset is perturbed. [Donnelly et al. \(2023\)](#) show that both the Rashomon set size and the MCR range can vary wildly across bootstrap resamples of the *same* data, making MCR and VIC unstable summaries of feature importance.

The **Rashomon Importance Distribution (RID)** proposed by [Donnelly et al. \(2023\)](#) addresses this by combining bootstrap resampling *with* Rashomon set analysis in a two-level procedure that captures both sources of uncertainty simultaneously.

The five-step RID pipeline (illustrated in Figure 2 of [Donnelly et al. 2023](#)):

1. **Bootstrap.** Draw B bootstrap datasets $\mathcal{D}_1^{(n)}, \dots, \mathcal{D}_B^{(n)}$ from the original data $\mathcal{D}^{(n)}$ (sampling n observations with replacement).
2. **Find Rashomon set.** For each bootstrap dataset $\mathcal{D}_b^{(n)}$, compute the Rashomon set $\mathcal{R}_{\mathcal{D}_b}^\varepsilon$: all models in $\mathcal{F}$ whose loss is within ε of the best model *on that bootstrap sample*.
3. **Find importances.** For each model f in each Rashomon set, compute the variable importance $\phi_j(f, \mathcal{D}_b^{(n)})$ using any importance metric (permutation importance, SHAP, model reliance, etc.).
4. **Find CDF.** For each bootstrap b , compute the empirical CDF of importances across the Rashomon set for that bootstrap.
5. **Find PDF.** Average the CDFs across all B bootstraps to obtain the RID, then differentiate to get the marginal density.

Formally, RID is defined as a cumulative distribution function. For feature j , the RID at threshold k measures the expected fraction of near-optimal models whose importance for feature j is at most k , where the expectation is over the data distribution:

$$\text{RID}_j(k) = \mathbb{E}_{\mathcal{D}_b^{(n)} \sim \mathcal{P}_n} \left[\frac{|\{f \in \mathcal{R}_{\mathcal{D}_b}^\varepsilon : \phi_j(f, \mathcal{D}_b^{(n)}) \leq k\}|}{|\mathcal{R}_{\mathcal{D}_b}^\varepsilon|} \right], \quad (12.12)$$

where:

- $\mathcal{P}_n$ is the empirical distribution of the data (i.e., the bootstrap distribution)
- $\mathcal{R}_{\mathcal{D}_b}^\varepsilon$ is the Rashomon set for bootstrap sample $\mathcal{D}_b^{(n)}$

- $\phi_j(f, \mathcal{D}_b^{(n)})$ is the importance of feature j for model f evaluated on $\mathcal{D}_b^{(n)}$
- k ranges over $[\phi_{\min}, \phi_{\max}]$, the support of the importance metric

In practice, we estimate RID by replacing the expectation with a sample average over B bootstrap replicates:

$$\widehat{\text{RID}}_j(k) = \frac{1}{B} \sum_{b=1}^B \frac{|\{f \in \mathcal{R}_{\mathcal{D}_b}^\varepsilon : \phi_j(f, \mathcal{D}_b^{(n)}) \leq k\}|}{|\mathcal{R}_{\mathcal{D}_b}^\varepsilon|}. \quad (12.13)$$

In Plain English

RID asks: “If I drew a new dataset from the same population, computed all near-optimal models on it, and measured feature j ’s importance in each of those models, what distribution of importances would I see?” The bootstrap handles the “new dataset” part; the Rashomon set handles the “all near-optimal models” part. The result is a CDF that captures *both* data uncertainty and model-choice uncertainty in a single object.

Additive vs. Multiplicative Rashomon Threshold

Donnelly et al. (2023) define the Rashomon set using an **additive** threshold: $\mathcal{R}^\varepsilon = \{f \in \mathcal{F} : \ell(f) \leq \min_{f'} \ell(f') + \varepsilon\}$. This contrasts with the **multiplicative** threshold used by Xin et al. (2022) earlier in this chapter: $L(f) \leq (1 + \epsilon)L(f^*)$. The two formulations are equivalent when rescaled appropriately, but be careful not to mix them: an additive $\varepsilon = 0.05$ and a multiplicative $\epsilon = 0.05$ define different sets.

Why not just bootstrap? A natural alternative is naive bootstrap importance: draw B bootstrap samples, fit the best model on each, and collect the importances. This captures data uncertainty but uses only *one* model per resample (the best one), ignoring the many near-optimal alternatives. RID considers *all* near-optimal models per resample, capturing model-choice uncertainty that naive bootstrapping misses entirely.

Why not just MCR/VIC? MCR and VIC analyze the Rashomon set of a *single* dataset. They capture model-choice uncertainty but are blind to data uncertainty. Donnelly et al. (2023) demonstrate that MCR ranges are unstable across bootstrap resamples: for one variable on the Monk 3 dataset, the MCR range is $[-0.1, 0.33]$ on one resample and $[0.33, 0.36]$ on another: contradictory conclusions from the same underlying data.

RID = Bootstrap $\times$ Rashomon

RID is the only method that captures *both* sources of uncertainty simultaneously:

Method	Data uncertainty	Model uncertainty	Stable?
Single-model SHAP	No	No	No
Bootstrap importance	Yes	No	Partially
MCR / VIC	No	Yes	No
RID	Yes	Yes	Yes

Finite-sample guarantees. Donnelly et al. (2023) prove (Theorem 2) that the estimated $\widehat{\text{RID}}_j$ converges to the true RID_j at a rate controlled by the number of bootstraps: with probability at least $1 - \delta$,

$$|\widehat{\text{RID}}_j(k) - \text{RID}_j(k)| \leq t \quad \text{for all } k, \quad \text{when } B \geq \frac{1}{2t^2} \ln \frac{2}{\delta}.$$

For example, $B = 471$ bootstraps guarantees that the estimated RID is within $t = 0.075$ of the true value with 90% confidence. This is a practical guarantee: a few hundred bootstraps suffice for reliable results.

Metric-agnostic. RID works with any variable importance metric ϕ_j : permutation importance, SHAP, model reliance, conditional model reliance, or any other metric with a bounded range. The framework treats ϕ_j as a black box, making it immediately compatible with existing importance tools.

Empirical stability. On four synthetic data-generating processes, [Donnelly et al. \(2023\)](#) find that RID achieves a median Jaccard similarity of 0.69 across independently generated datasets, compared to below 0.55 for both MCR and VIC. RID is the only method whose importance intervals consistently overlap across independent datasets drawn from the same distribution.

RID for Volatility Feature Selection

For our project, RID would answer questions that neither SHAP nor MCR can: “Across bootstrap resamples of the training data *and* across all near-optimal models within each resample, what is the distribution of VIX’s importance?” If the RID CDF for VIX rises steeply near zero, VIX is unimportant in most models across most resamples, and we can drop it. If the CDF rises steeply above 0.2, VIX is robustly important. If the CDF rises gradually from 0 to 0.4, VIX is a substitute: sometimes important, sometimes not, depending on both the data sample and the model chosen. Unlike MCR, this conclusion is *stable*: it will not flip if we add one more month of data.

12.6 Rashomon Analysis for Volatility Forecasting

No published work has applied Rashomon set analysis to financial time-series forecasting. This is an open frontier, and it is directly relevant to our project. This section describes how the tools from Sections 12.4–12.5 could be deployed for realized volatility forecasting.

12.6.1 Regime-Stable Feature Selection

Volatility features that matter in a low-vol regime (2017–2019) may not matter in a crisis regime (March 2020) or a post-crisis recovery (2021). Single-model SHAP run on the full sample averages over these regimes, potentially concluding that a feature is “moderately important” when it is actually critical in crises and irrelevant otherwise.

A Rashomon-based approach to regime-stable feature selection:

1. **Rolling-window Rashomon sets.** For each rolling window (e.g., 252-day training sets advanced by one month), compute the Rashomon set and the MCR for each feature.
2. **Intersection across regimes.** A feature that has $\text{MCR}_- > 1$ (i.e., minimum importance above the baseline) in *every* window is regime-stable. A feature whose MCR range includes 1.0 in some windows is regime-dependent.
3. **Feature selection rule.** Include only regime-stable features in the final model. For regime-dependent features, consider regime-conditional inclusion (e.g., add VIX term structure only when $\text{VIX} > 20$).

12.6.2 Prediction Multiplicity: Quantifying Model-Choice Uncertainty

The Rashomon set also reveals **prediction multiplicity**: the range of forecasts that different near-optimal models produce for the same input. For a given feature vector $\mathbf{x}_t$ (today’s features),

we can compute $\{f(\mathbf{x}_t) : f \in \mathcal{R}\}$ and report the min, max, and spread.

If all near-optimal models agree that tomorrow's volatility will be high, that is a strong signal. If some predict high and others predict low, the forecast is sensitive to model choice, and we should report wider confidence intervals or flag the day as uncertain.

This is conceptually similar to the disagreement among models in an ensemble, but with one important difference: ensemble members are not guaranteed to be near-optimal, while Rashomon set members are.

12.6.3 Defensible Presentations

In a Goldman Sachs internship, you will present model results to people who will ask hard questions: “Why did your model use VIX and not ATM IV?” or “Your SHAP plot from last month looked completely different.”

Rashomon analysis gives you defensible answers:

- “I examined all 14,000 decision trees within 2% of optimal accuracy. Lagged RV is the top feature in every single one of them. VIX and ATM IV are interchangeable; the data supports either, so I chose VIX for simplicity.”
- “The feature importance is *provably stable*: no near-optimal model exists that does not rely on lagged RV.”
- “The prediction range across all near-optimal models for tomorrow is [0.00012, 0.00018]. The spread tells you how much of the forecast uncertainty comes from model choice rather than data noise.”

12.7 Summary and Connections

This chapter introduced a fundamentally different approach to model interpretability. Rather than explaining one model after the fact (SHAP), we examine the entire space of near-optimal models before committing to one.

Aspect	Single-Model (SHAP)	Rashomon Set
What is explained	One model's predictions	All near-optimal models
Feature importance	Point estimate for one model	Range $[\text{MCR}_-, \text{MCR}_+]$
Stability	Changes with refit	Provably stable within ϵ
Substitute detection	Not possible	Overlapping VIC projections
Model-choice uncertainty	Not quantified	Prediction multiplicity
Computational cost	Fast (one model)	Hours (enumeration needed)

Looking ahead. Chapter 13 will introduce deep learning methods for volatility, which are powerful but even harder to interpret than tree ensembles. The Rashomon perspective suggests a hybrid strategy: use deep models for raw predictive power, but use interpretable trees (and their Rashomon sets) to understand *which features matter and why*. The two approaches complement each other.

Chapter 13

Deep Learning for Volatility

Where This Chapter Fits

Project 2 (Intraday RV from LOB) and Project 4 (Rough Vol vs Deep Learning) use architectures from this chapter.

What You Need

- Backpropagation, gradient descent, and the idea of a loss function.
- Matrix multiplication at the level of $\mathbf{W}\mathbf{x} + \mathbf{b}$.
- Sigmoid $\sigma(\cdot)$ and ReLU($\cdot$) activations.
- The HAR model from Chapter 6 and the tree methods from Chapter 11.

13.1 LSTMs and GRUs

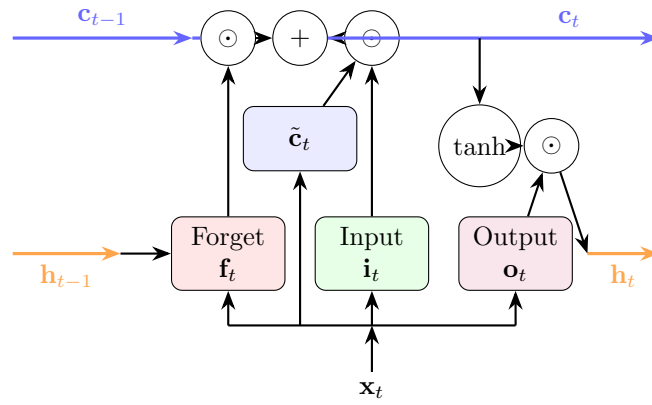
Chapter 11 established that gradient-boosted trees dominate *tabular* volatility forecasting. But volatility data is often sequential: a stream of 5-minute returns, a sequence of LOB snapshots, a panel of daily risk measures across assets. Recurrent neural networks, specifically the Long Short-Term Memory (LSTM) and the Gated Recurrent Unit (GRU), are purpose-built for sequences.

Why Sequences Need Special Architecture

A standard feedforward network treats its inputs as an unordered bag of numbers. If you feed it $(RV_{t-1}, RV_{t-2}, \dots, RV_{t-22})$, it does not know that RV_{t-1} is yesterday and RV_{t-22} is a month ago. A recurrent network processes inputs one step at a time, maintaining a hidden state $\mathbf{h}_t$ that summarizes everything it has seen so far. This hidden state is the network's "memory."

13.1.1 LSTM Cell Architecture

The LSTM cell solves the vanishing gradient problem that cripples simple recurrent networks. It uses three gates (forget, input, output) and a cell state $\mathbf{c}_t$ that acts as a conveyor belt for information.



LSTM Cell Equations

At each time step t , given input $\mathbf{x}_t$ and previous hidden state $\mathbf{h}_{t-1}$:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_f) \quad (\text{forget gate}) \quad (13.1)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_i) \quad (\text{input gate}) \quad (13.2)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_c) \quad (\text{candidate cell state}) \quad (13.3)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (\text{cell state update}) \quad (13.4)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_o) \quad (\text{output gate}) \quad (13.5)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (\text{hidden state}) \quad (13.6)$$

where:

- $[\mathbf{h}_{t-1}; \mathbf{x}_t]$ is the concatenation of previous hidden state and current input.
- $\mathbf{W}_f, \mathbf{W}_i, \mathbf{W}_c, \mathbf{W}_o$ are learned weight matrices.
- $\mathbf{b}_f, \mathbf{b}_i, \mathbf{b}_c, \mathbf{b}_o$ are bias vectors.
- $\odot$ denotes element-wise (Hadamard) multiplication.
- $\sigma(\cdot)$ squashes values to $[0, 1]$; each gate is a “soft switch.”

The Three Gates in Plain English

- **Forget gate $\mathbf{f}_t$:** “How much of the old memory should I keep?” Values near 0 erase; near 1 retain.
- **Input gate $\mathbf{i}_t$:** “How much of the new candidate should I write to memory?”
- **Output gate $\mathbf{o}_t$:** “How much of the current memory should I expose as my prediction?”

The cell state $\mathbf{c}_t$ flows through the network with only linear interactions (multiply and add), so gradients flow back through many time steps without vanishing. This is why LSTMs can learn long-range dependencies that simple RNNs cannot.

Why This Matters

An LSTM can learn the HAR structure automatically (daily, weekly, monthly lags map to different hidden-state components), but it can also discover more complex nonlinear patterns that HAR misses, such as asymmetric responses to positive and negative shocks. Because neural nets accept arbitrary loss functions, you can train the LSTM directly on QLIKE rather than MSE, which is straightforward in PyTorch or TensorFlow.

The GRU simplifies the LSTM by merging the forget and input gates into a single “update gate” and eliminating the separate cell state. GRUs have fewer parameters and train faster; in

practice, performance differences between LSTM and GRU are small for volatility tasks.

13.1.2 LSTMs for Realized Volatility

Bucci (2020) provides an early comparison. He tests LSTM and NARX (nonlinear autoregressive with exogenous inputs) networks against ARFIMA and other econometric models for monthly RV forecasting of the S&P 500.

Bucci (2020): LSTM for Monthly RV

Recurrent neural networks (LSTM, NARX) outperform traditional long-memory econometric models (ARFIMA, LSTAR) for monthly RV forecasting, particularly in terms of robust accuracy measures. The LSTM’s advantage is clearest during volatile periods (the 2007–2009 crisis subsample), when the relationship between past and future RV is non-linear.

This result is consistent with the lesson from Chapter 11: on daily RV with only lagged RV as inputs, simple models are hard to beat. The LSTM becomes genuinely useful when you change what you feed it.

Sirignano and Cont (2019) demonstrate a powerful idea: *pooling* data across assets. Instead of training one LSTM per stock, they train a single “universal” LSTM on all stocks simultaneously.

Sirignano–Cont (2019): Universal Features via Pooling

A single LSTM trained on pooled data across 1,000+ stocks learns universal features of price formation. Pooling works because volatility dynamics are similar across assets (the same “rough” kernel from Chapter 7). The pooled model outperforms asset-specific models, especially for assets with short histories.

Rosenbaum and Zhang (2022) connect LSTMs directly to rough volatility. They show that a universal LSTM, trained to forecast volatility, learns a kernel that matches the fractional kernel of the RFSV model from Chapter 7.

Rosenbaum–Zhang (2022): LSTM Rediscovered Roughness

An LSTM trained on raw volatility data learns the same power-law kernel $K(t) \propto t^{H-1/2}$ with $H \approx 0.1$ that defines the rough fractional stochastic volatility (RFSV) model. The LSTM and the RFSV forecast are nearly identical, suggesting both learn the same underlying structure.

The LSTM was given no prior knowledge of rough volatility, fractional Brownian motion, or Hurst exponents; it discovered roughness from data alone.

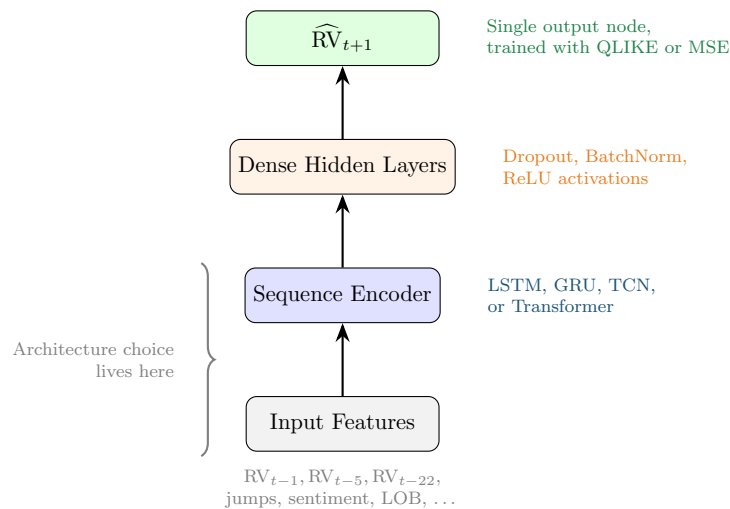
Rahimikia and Poon (2020) push the LSTM further by adding LOB features and news sentiment as inputs alongside standard RV lags.

Rahimikia–Poon (2020): LSTM with LOB + Sentiment

An LSTM incorporating limit order book features and news sentiment beats HAR on approximately 90% of trading days. However, the model fails during extreme stress events, precisely when accurate forecasts matter most.

Stress-Period Failure

Deep learning models trained on normal-regime data can fail catastrophically during crises. The LSTM in [Rahimikia and Poon \(2020\)](#) underperforms HAR during high-volatility episodes because the training data contains few such events. This is not specific to LSTMs; it affects all flexible models. Always evaluate forecast performance conditional on volatility regime (see Chapter 17).

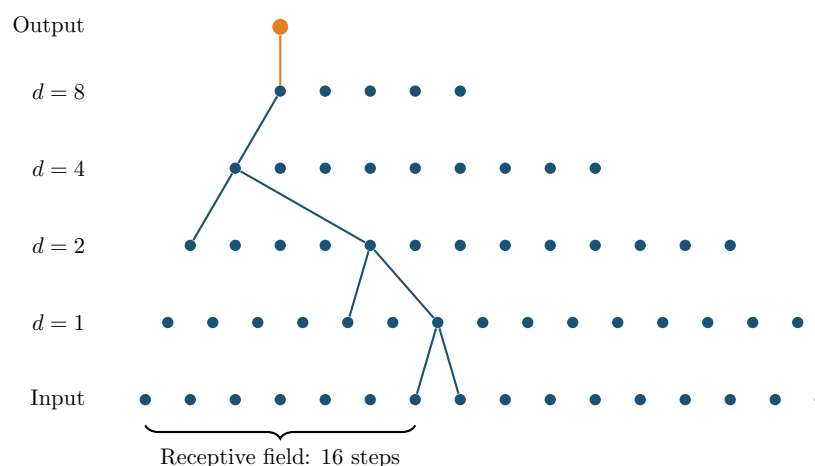


13.2 Temporal Convolutional Networks

LSTMs process sequences one step at a time. This is conceptually clean but creates a computational bottleneck: you cannot parallelize across time steps because each step depends on the previous hidden state. Temporal Convolutional Networks (TCNs) solve this by using *causal convolutions*: filters that only look backward in time, applied in parallel across the entire sequence.

13.2.1 Dilated Causal Convolutions

The key innovation is *dilation*. A standard 1D convolution with kernel size k looks at k consecutive time steps. A dilated convolution with dilation factor d skips $d - 1$ steps between inputs, so a kernel of size k covers a receptive field of $k + (k - 1)(d - 1)$ steps. By doubling the dilation factor at each layer ($d = 1, 2, 4, 8, \dots$), the receptive field grows exponentially while the number of parameters grows only linearly.



Dilated Causal Convolution

For a 1D input sequence $(x_0, x_1, \dots, x_T)$, a dilated causal convolution with kernel $\mathbf{w} = (w_0, \dots, w_{k-1})$ and dilation factor d computes:

$$y_t = \sum_{j=0}^{k-1} w_j \cdot x_{t-j \cdot d} \quad (13.7)$$

where:

- k is the kernel size (typically 2 or 3).
- d is the dilation factor; layer ℓ uses $d = 2^{\ell-1}$.
- **Causal:** y_t depends only on x_s for $s \leq t$. No future information leaks.
- With L layers and kernel size $k = 2$, the receptive field is 2^L time steps.

Why Dilation Works

Think of each layer as a zoom level. Layer 1 (dilation 1) sees adjacent time steps: tick-by-tick patterns. Layer 2 (dilation 2) sees every other step: short-term trends. Layer 4 (dilation 8) sees steps 8 apart: the overall shape of the day. With just 8 layers and kernel size 2, you cover $2^8 = 256$ time steps. That is an entire trading day of 5-minute bars (78 bars) with room to spare.

13.2.2 DeepVol

Moreno-Pino and Zohren (2022) build **DeepVol**, a TCN that forecasts daily RV directly from raw intraday returns. Instead of computing RV_t from 5-minute returns and then modeling RV_{t+1} as HAR does, DeepVol feeds the raw 5-minute returns into a dilated causal convolution stack and predicts RV_{t+1} end-to-end.

DeepVol: TCN for Daily RV from Raw Returns

DeepVol achieves state-of-the-art daily RV forecasting by processing raw 5-minute returns directly, bypassing the RV aggregation step entirely. The dilated causal convolution architecture provides two advantages over LSTMs:

1. **Parallelizable training:** all time steps are processed simultaneously.
2. **Interpretable receptive field:** you know exactly how many past time steps influence each prediction.

Worked Example:

DeepVol Receptive Field Calculation Suppose DeepVol uses $L = 7$ layers with kernel size $k = 2$ and dilation factors $d = 1, 2, 4, 8, 16, 32, 64$.

Receptive field per layer: Each layer with dilation d and kernel size 2 adds d new time steps to the receptive field. Total receptive field:

$$\text{RF} = 1 + \sum_{\ell=0}^{L-1} (k-1) \cdot 2^\ell = 1 + \sum_{\ell=0}^6 2^\ell = 1 + 127 = 128 \text{ steps.}$$

In trading time: At 5-minute sampling, one trading day has 78 bars (6.5 hours $\times$ 12 bars/hour). A receptive field of 128 steps covers $128/78 \approx 1.6$ trading days. This means each prediction uses roughly today's and yesterday's intraday data.

To cover one trading week (5 days, 390 bars), you need:

$$L = \lceil \log_2(390) \rceil = 9 \text{ layers} \quad (\text{RF} = 512 \text{ steps}).$$

TCN vs. LSTM for Volatility

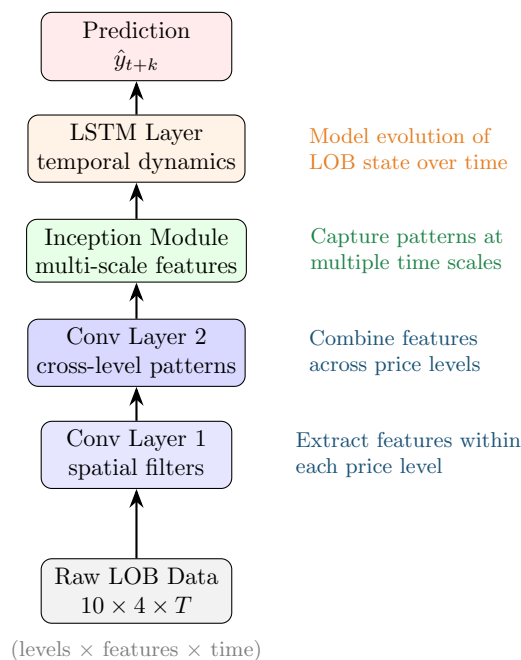
- TCN processes the entire input sequence in one forward pass; LSTM must step through sequentially.
- TCN's receptive field is explicit (2^L with kernel size 2); LSTM's effective memory is learned and opaque.
- TCN has no vanishing gradient problem by construction (parallel paths through residual connections).
- LSTM is more flexible for variable-length sequences; TCN requires fixed-length input (or padding).
- For fixed-length volatility sequences (e.g., one day of 5-min returns), TCN is generally preferred.

13.3 DeepLOB: Learning from Limit Order Books

The limit order book (LOB) is the richest source of short-horizon information in modern markets. At each moment, the LOB shows the queue of buy and sell orders at every price level. Chapter 10 discussed hand-crafted LOB features (order imbalance, depth ratios, spread). Zhang et al. (2019) ask: can a neural network learn better features directly from the raw LOB?

13.3.1 Architecture

DeepLOB combines CNNs (for spatial features across price levels) with LSTMs (for temporal dynamics across snapshots).



DeepLOB Input Representation

The input to DeepLOB at time t is a tensor $\mathbf{X}_t \in \mathbb{R}^{T \times 10 \times 4}$:

- **T time steps:** a rolling window of LOB snapshots (e.g., $T = 100$).
- **10 price levels:** the 5 best bid and 5 best ask levels.
- **4 features per level:** price, volume, price difference from mid, volume difference from mid (on each side).

The CNN layers convolve across the 10 price levels (spatial dimension), extracting patterns like order imbalance and depth concentration. The LSTM then processes the sequence of CNN-extracted features over time.

DeepLOB: End-to-End LOB Prediction

DeepLOB outperforms hand-crafted LOB features for short-horizon ($k = 10, 20, 50$ tick) mid-price movement prediction. The convolutional layers learn features that resemble (but improve upon) classical order imbalance measures. The architecture generalizes across stocks without retraining, consistent with the universal-features finding of [Sirignano and Cont \(2019\)](#).

DeepLOB for Volatility

DeepLOB was designed for mid-price direction prediction, but the same architecture predicts short-horizon RV from LOB data. Project 2 (Intraday RV from LOB) adapts DeepLOB by replacing the classification output with a regression head predicting RV over the next k ticks. The key insight: LOB state (depth imbalance, spread dynamics, queue lengths) contains predictive information about short-term volatility that is not captured by price-based features alone.

Worked Example:

DeepLOB Input Dimensions Consider S&P 500 E-mini futures (ES) with LOB snapshots every 100 milliseconds.

Input window: $T = 100$ snapshots (10 seconds of data).

Tensor shape: $100 \times 10 \times 4 = 4,000$ input values per sample.

Contrast with tabular features: A tree-based model using hand-crafted LOB features might have 20–30 features computed from the same raw data. DeepLOB uses 4,000 raw values and lets the CNN learn the right features. This is why deep learning wins on raw sequential data: the feature space is too rich for manual engineering.

13.4 Transformers and Attention

Transformers replaced LSTMs as the dominant sequence model in NLP (GPT, BERT) and are now being applied to financial time series. The core mechanism is *self-attention*: instead of processing the sequence step-by-step, the transformer computes a weighted sum over all positions, where the weights are learned from the data.

Scaled Dot-Product Attention

Given queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ (all derived from the input via learned linear projections):

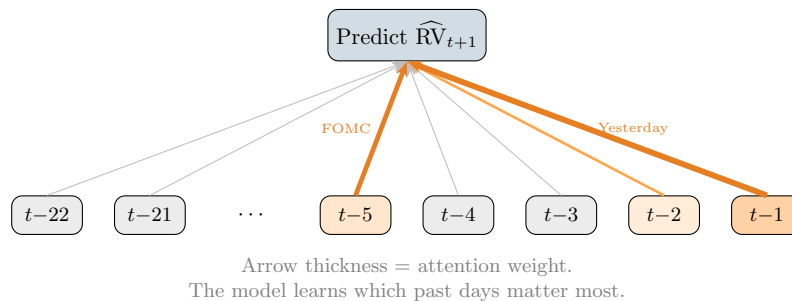
$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (13.8)$$

where:

- $\mathbf{Q} \in \mathbb{R}^{T \times d_k}$: queries (“what am I looking for?”).
- $\mathbf{K} \in \mathbb{R}^{T \times d_k}$: keys (“what do I contain?”).
- $\mathbf{V} \in \mathbb{R}^{T \times d_v}$: values (“what information do I carry?”).
- $\sqrt{d_k}$: scaling factor to prevent dot products from growing large.
- The softmax produces a $T \times T$ attention weight matrix; entry (i, j) is how much position i attends to position j .

Attention for Volatility

In an LSTM, the influence of a past observation on the current prediction decays as it recedes into the hidden state. With attention, every past observation can directly influence the current prediction with a learned weight. For volatility, this means the model can learn that, say, last Tuesday’s intraday pattern is highly relevant to today’s forecast (perhaps both are FOMC days) without relying on the hidden state to carry that information across the intervening days.



13.4.1 Graph Transformers for Cross-Asset Volatility

Chen and Roberts (2022) extend the transformer with a graph structure over assets. Instead of treating each asset’s time series independently, the attention mechanism defines a *learned graph*: the attention weight between asset i and asset j captures how much asset j ’s history informs the forecast for asset i .

Attention Weights as a Volatility Network

The attention weight matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ across N assets connects directly to the spillover analysis in Chapter 16 and the multivariate models in Chapter 15, but learns the network structure from data rather than imposing it via a VAR or DCC.

Transformer variants have also been applied to LOB data (TLOB), replacing the LSTM component of DeepLOB with multi-head self-attention. Early results are promising but not yet conclusive.

Transformer Evidence on Volatility Is Thin

Transformer results on volatility are preliminary. Most published results use short evaluation periods or small asset universes. Be skeptical of claims that lack Diebold-Mariano tests or Model Confidence Set analysis (Chapter 17). The core problem: transformers have many parameters and volatility datasets are small. A daily RV series for one asset gives you ~ 252 observations per year. Even 20 years is only 5,000 observations, far fewer than the millions of sentences used to train language models. Pooling across assets helps

(Section 13.1), but the evidence base is still thin compared to LSTMs and TCNs.

13.5 Modern Time-Series Architectures

The deep-learning-for-time-series field has produced a rapid succession of architectures: N-BEATS, N-HiTS, TiDE, TSMixer, PatchTST, and others. These were designed for general time-series forecasting (energy demand, weather, retail sales) and tested primarily on benchmarks like M4, M5, and the Monash archive. Their application to realized volatility is limited.

N-BEATS (Neural Basis Expansion Analysis) uses a stack of fully-connected blocks with residual connections. Each block produces a “backcast” (reconstruction of the input) and a “forecast,” and the blocks are organized into stacks that can be given interpretable basis functions (trend, seasonality).

N-HiTS extends N-BEATS with hierarchical interpolation, allowing different blocks to operate at different temporal resolutions (similar in spirit to the dilated convolutions of TCN).

PatchTST (Patch Time Series Transformer) divides the input time series into non-overlapping patches (subsequences) and treats each patch as a token for a transformer. This is conceptually similar to treating multi-day windows of 5-minute returns as input tokens.

TiDE (Time-series Dense Encoder) and **TSMixer** use MLP-based architectures that are simpler and faster than transformers while achieving competitive performance on standard benchmarks.

Modern Architectures: Solutions Looking for a Problem?

These architectures are well-engineered for general forecasting, but their evidence on RV specifically is thin. Most have not been tested with proper volatility evaluation (QLIKE loss, Diebold-Mariano tests against HAR, Model Confidence Set). PatchTST is the most promising for volatility because its patching mechanism naturally handles the multi-scale structure of intraday data (5-minute patches within daily windows within weekly patterns). Test them if you have the engineering bandwidth, but do not expect them to dominate purpose-built approaches like DeepVol or the well-tuned HAR.

13.6 Generative and Latent-Variable Approaches

The methods above produce point forecasts: a single number $\widehat{RV}_{t+1}$. Generative models instead learn the *full distribution* of future volatility, or learn compressed representations of complex objects like the implied volatility surface. These are frontier methods with thin evidence.

13.6.1 Neural SDEs and CDEs

Kidger (2021) develops the mathematical framework for embedding stochastic differential equations into neural networks. A neural SDE replaces the hand-specified drift and diffusion of classical models (Heston, SABR, rough Bergomi) with neural networks that learn these functions from data:

$$dX_t = f_{\theta}(X_t, t) dt + g_{\theta}(X_t, t) dW_t \quad (13.9)$$

where:

- $f_{\theta}(\cdot)$: drift function, parameterized by a neural network with weights θ .
- $g_{\theta}(\cdot)$: diffusion function, also a neural network.
- W_t : standard Brownian motion.

In Plain English

A classical stochastic volatility model says “volatility evolves according to this specific formula” (e.g., Heston’s square-root process). A neural SDE says “volatility evolves according to *whatever function* a neural network learns from the data.” The drift f_{θ} captures the predictable part of how volatility moves (mean reversion, trends), and the diffusion g_{θ} captures the randomness (vol-of-vol, jump intensity). Both are flexible enough to approximate any continuous function, so you are not locked into a particular parametric assumption.

This is appealing for volatility because the ground truth *is* an SDE (Chapter 2). The neural SDE learns the drift and diffusion from data without committing to a specific parametric form (Heston, SABR, rough Bergomi). The controlled differential equation (CDE) variant handles irregularly-spaced observations, which is natural for tick data.

Neural SDEs Are Data-Hungry

Training neural SDEs requires backpropagation through the SDE solver (adjoint method), which is computationally expensive. The diffusion function g_{θ} is especially hard to learn because it governs the noise, not the signal. With typical volatility dataset sizes (a few thousand daily observations), neural SDEs are prone to overfitting. They are most promising when combined with cross-asset pooling and intraday data, where sample sizes are orders of magnitude larger.

13.6.2 Autoencoders for the Implied Volatility Surface

The implied volatility (IV) surface from Chapter 8 is high-dimensional: IV values at dozens of strikes and maturities, updated continuously. Ding et al. (2025) use autoencoders to compress this surface into a low-dimensional latent space.

Autoencoder for IV Surface

An autoencoder consists of:

- **Encoder** $f_{\theta} : \mathbb{R}^{K \times M} \rightarrow \mathbb{R}^d$: maps the IV surface (K strikes $\times$ M maturities) to a latent vector $\mathbf{z} \in \mathbb{R}^d$ with $d \ll K \times M$.
- **Decoder** $g_{\theta} : \mathbb{R}^d \rightarrow \mathbb{R}^{K \times M}$: reconstructs the surface from the latent vector.
- **Training objective**: minimize reconstruction error $\|\text{surface} - g_{\theta}(f_{\theta}(\text{surface}))\|^2$.

The latent vector $\mathbf{z}$ is a compressed summary of the entire IV surface. Typical latent dimensions are $d = 3$ to 8, meaning the IV surface (perhaps $20 \times 10 = 200$ values) is compressed to fewer than 10 numbers.

13.6.3 Deep Stochastic Volatility

Xu and Chen (2021) propose a deep stochastic volatility model that combines the structure of classical stochastic vol (latent variance process driving observed returns) with the flexibility of neural networks. The latent variance follows a neural-network-parameterized transition, and inference uses variational methods (a VAE-like architecture). This produces a full posterior distribution over future volatility, not just a point forecast.

13.6.4 Normalizing Flows for RV Distribution

Du et al. (2023) use normalizing flows co-trained with a VAE to model the full distribution of future RV.

Normalizing Flow (Sketch)

A normalizing flow transforms a simple base distribution $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ through a sequence of invertible, differentiable transformations $T_1, T_2, \dots, T_K$:

$$\mathbf{x} = T_K \circ T_{K-1} \circ \dots \circ T_1(\mathbf{z}) \quad (13.10)$$

The log-density of the output is computed via the change-of-variables formula:

$$\log p(\mathbf{x}) = \log p(\mathbf{z}) - \sum_{k=1}^K \log \left| \det \frac{\partial T_k}{\partial \mathbf{x}_{k-1}} \right| \quad (13.11)$$

where:

- Each T_k is designed to be invertible with a tractable Jacobian determinant.
- The composition of simple transformations can model complex, non-Gaussian distributions.
- For RV forecasting, the flow learns to map a Gaussian to the empirical distribution of future RV conditional on past information.

In Plain English

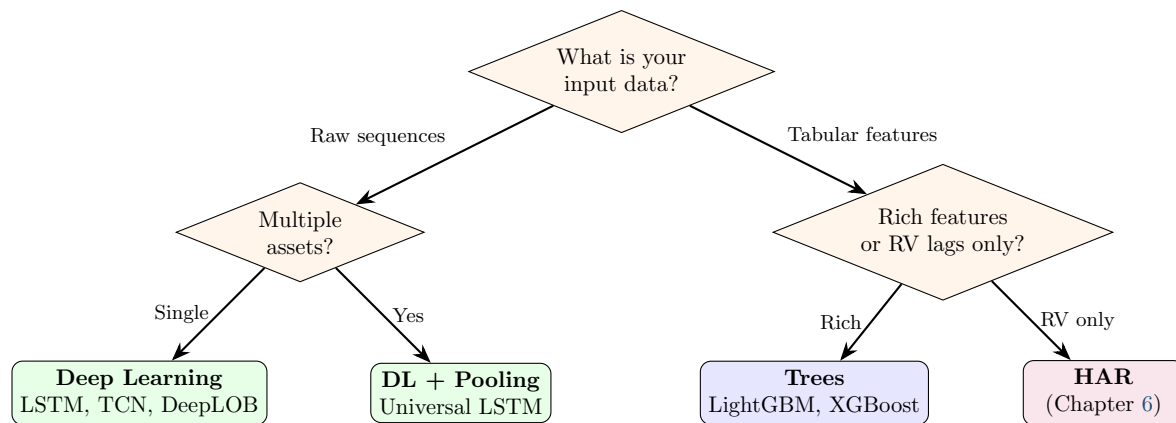
Start with a blob of points drawn from a simple Gaussian (a bell curve). Now warp that blob through a series of smooth, reversible stretches and squishes. After enough transformations, the warped blob can match any complicated distribution you want, such as the right-skewed, heavy-tailed distribution of realized volatility. The key constraint is that every transformation must be invertible, so you can always go backward from data to the Gaussian and compute exact likelihoods.

Why Distributional Forecasts Matter for Volatility

A point forecast $\widehat{\text{RV}}_{t+1} = 15\%$ tells you the expected level. A distributional forecast tells you “15% is the median, but there is a 10% probability that RV exceeds 30%.” This matters for risk management (you care about the tail, not the mean) and for the variance risk premium (Chapter 9), where the entire distribution of future RV determines the fair price of variance swaps. Normalizing flows and VAEs produce these distributional forecasts naturally.

13.7 The Honest Bottom Line

Deep learning is neither a silver bullet nor useless for volatility forecasting. The evidence, taken honestly, points to a clear division of labor.



Here is the evidence, scenario by scenario:

1. **Daily horizon, RV lags only.** HAR often matches deep learning (Christensen et al., 2023). Trees are slightly better than DL due to less overfitting on small datasets (Chapter 11). Do not use a 50,000-parameter LSTM when four parameters suffice.
2. **Daily horizon, rich tabular features.** Trees still usually win. The “tabular data advantage” of gradient-boosted trees over neural networks is well-documented: trees handle heterogeneous features, missing values, and small samples more gracefully.
3. **Raw sequential data (intraday returns, LOB snapshots).** Deep learning wins clearly. DeepVol (Moreno-Pino and Zohren, 2022) beats HAR and trees by processing raw 5-minute returns. DeepLOB (Zhang et al., 2019) beats hand-crafted LOB features. This is where DL adds genuine value: learning representations from raw, high-dimensional, sequential inputs.
4. **Cross-asset pooling.** Deep learning enables pooled training across assets (Sirignano and Cont, 2019). Trees cannot easily share learned representations between assets. If volatility dynamics are universal (Chapter 7), pooling helps, and neural networks make it natural.
5. **Distributional forecasts.** Normalizing flows, VAEs, and neural SDEs produce full predictive distributions. Trees and HAR produce point forecasts (or require separate quantile models). If you need the full distribution (for VRP pricing or tail risk), generative models have an architectural advantage.
6. **Longer horizons (weekly, monthly).** Evidence is mixed. DL can exploit richer temporal patterns at longer horizons, but the sample size shrinks (52 weekly observations per year), which favors simpler models.

The Decision Rule

Use deep learning when your data is sequential and raw, or when you want to pool across assets. Use trees when your data is tabular features. Use HAR when you have nothing but RV lags. If in doubt, start with HAR, add trees if you have features, and only reach for DL if you have raw sequential data or need cross-asset pooling.

Overfitting Is the Central Risk

Deep learning models have orders of magnitude more parameters than HAR or trees. A 2-layer LSTM with 64 hidden units has ~50,000 parameters. A HAR model has 4. A LightGBM with max_depth=4 and 200 trees has ~3,000 leaf values. On a daily RV series with 2,500 observations (10 years), the LSTM has 20 parameters per observation. Regularization (dropout, weight decay, early stopping) is not optional. Always compare against the HAR baseline on identical data.

13.8 Summary

- LSTMs and GRUs are the default sequential architecture for volatility. They use gates (forget, input, output) and a cell state to maintain long-range memory.
- On monthly S&P 500 RV, LSTMs and NARX networks outperform traditional long-memory models, especially during crises (Bucci, 2020). The value of LSTMs comes from feeding them raw sequences, not pre-computed features.
- Cross-asset pooling is a genuine advantage of neural networks. A universal LSTM trained on many assets outperforms asset-specific models (Sirignano and Cont, 2019) because volatility dynamics are similar across assets.
- LSTMs trained on raw volatility data rediscover the rough-volatility kernel ($H \approx 0.1$) without any prior knowledge of fractional processes (Rosenbaum and Zhang, 2022).
- Temporal Convolutional Networks (TCNs) use dilated causal convolutions to grow the receptive field exponentially while maintaining parallelizable training. DeepVol (Moreno-Pino and Zohren, 2022) achieves state-of-the-art daily RV forecasting from raw 5-minute returns.
- DeepLOB (Zhang et al., 2019) combines CNNs (spatial features across LOB levels) with LSTMs (temporal dynamics) and outperforms hand-crafted LOB features for short-horizon prediction.
- Transformers and attention mechanisms can capture long-range dependencies and learn cross-asset networks via attention weights (Chen and Roberts, 2022), but evidence on volatility tasks remains thin. Be skeptical of results without proper statistical testing.
- Modern time-series architectures (N-BEATS, PatchTST, TiDE, TSMixer) are well-engineered but have limited evidence on RV. PatchTST is the most promising due to its natural handling of multi-scale temporal structure.
- Generative methods (neural SDEs, autoencoders, normalizing flows) produce distributional forecasts or compressed representations of complex surfaces, but require large datasets and careful regularization.
- **The honest bottom line:** use DL for raw sequential data and cross-asset pooling; use trees for tabular features; use HAR for RV-only lags. Always compare against HAR as a baseline.
- Overfitting is the central risk. A 2-layer LSTM has $\sim 50,000$ parameters; HAR has 4. Regularization (dropout, early stopping, weight decay) is mandatory.
- LSTMs can fail during extreme stress (Rahimikia and Poon, 2020). Always evaluate performance conditional on volatility regime.

Paper	Architecture	Key Result	Relevance
Bucci (2020)	LSTM, NARX	Outperforms ARFIMA; gains in crises	Monthly RV
Sirignano and Cont (2019)	Universal LSTM	Pooling across assets improves forecasts	Cross-asset
Rosenbaum and Zhang (2022)	Universal LSTM	Learns rough-vol kernel ($H \approx 0.1$)	Theory valid
Rahimikia and Poon (2020)	LSTM + LOB + sentiment	Beats HAR 90% of days; fails in stress	Feature rich
Moreno-Pino and Zohren (2022)	TCN (DeepVol)	SOTA daily RV from raw 5-min returns	Raw sequence
Zhang et al. (2019)	CNN + LSTM (DeepLOB)	Beats hand-crafted LOB features	LOB prediction
Chen and Roberts (2022)	Graph Transformer	Learned cross-asset attention network	Multi-asset
Kidger (2021)	Neural SDE/CDE	Principled SDE inside neural network	Distributional
Ding et al. (2025)	Autoencoder	IV surface compressed to $d \leq 8$ factors	Feature extraction
Xu and Chen (2021)	Deep stochastic vol	Full posterior over future volatility	Uncertainty
Du et al. (2023)	Normalizing flow + VAE	Full predictive distribution of RV	Distributional

Next: Chapter 14 combines the strengths of HAR, trees, and deep learning into hybrid and ensemble models.

Chapter 14

Hybrid and Ensemble Models

14.1 Why This Chapter Matters

From Components to Combinations

Hybrid models let HAR handle the linear part and train ML only on the residual, and they are the safest bet in the volatility forecasting literature ([Rahimikia and Poon, 2020](#)). Project 1 (HARQ-X with ML Residual Augmentation) is a hybrid by design.

You already have two powerful toolkits: the econometric models of Chapters 5–6 and the machine-learning models of Chapters 11–13. Each has blind spots. HAR is parsimonious but linear; gradient-boosted trees are flexible but hungry for signal. This chapter shows you how to combine them so that each component operates where it is strongest.

The core idea: *decompose*, then *specialize*. Let a well-understood econometric model absorb the predictable structure, then aim the full power of ML at whatever remains. The resulting hybrid almost always outperforms either component alone, and it does so with lower variance and greater interpretability than a pure ML approach ([Rahimikia and Poon, 2020](#)).

14.2 Why Hybrids Win

14.2.1 The Variance Budget Argument

Consider a forecast target RV_{t+1} . A fitted HAR model typically explains 40–60% of next-day realized-volatility variation with just three coefficients ([Rahimikia and Poon, 2020](#)). That leaves 40–60% in the residual $e_t = RV_{t+1} - \widehat{HAR}_t$. If you train an ML model directly on RV_{t+1} , it must rediscover the linear structure that HAR already captures perfectly, wasting model capacity and inviting overfitting. If instead you train on e_t , three things change for the better:

1. **HAR never overfits.** Three coefficients on lagged averages cannot memorize noise, so the linear component is rock-solid out of sample.
2. **Residual targets have lower variance.** The signal-to-noise ratio for the ML component improves because you have removed the dominant low-frequency trend.
3. **Ensemble diversification.** Even if the ML model adds only modest accuracy, combining two weakly correlated forecasters reduces overall forecast variance by the standard diversification identity ([Rahimikia and Poon, 2020](#)).

Let HAR Do the Heavy Lifting

If HAR explains 60% of next-day RV with 3 coefficients, let it. Train ML on the residual. You get the reliability of HAR plus the flexibility of ML, without asking either model to do the other's job.

14.2.2 Variance Decomposition

Figure 14.1 illustrates the variance budget argument visually. The total variance of next-day RV is split into what HAR explains, what ML can capture from the residual, and irreducible noise.

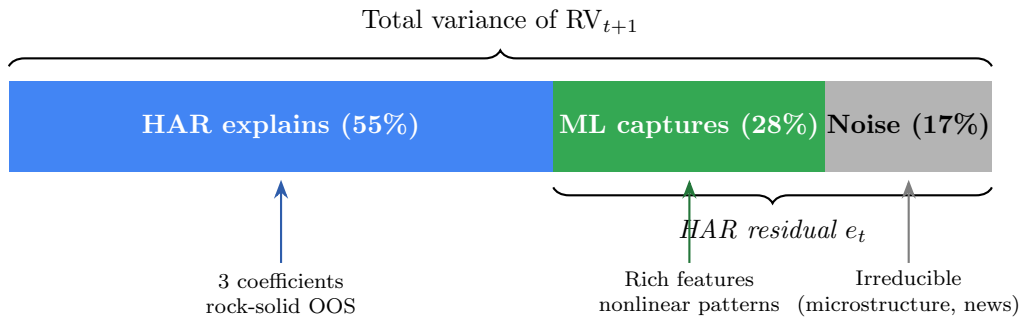


Figure 14.1: Variance budget for next-day RV forecasting. HAR absorbs the dominant linear signal cheaply; ML targets only the residual variance, where the signal-to-noise ratio is lower but nonlinear patterns exist. Noise is irreducible regardless of model complexity.

14.2.3 Architecture Diagram

Figure 14.2 illustrates the two-stage hybrid pipeline that recurs throughout this chapter. The key insight is that the ML model never sees the raw target; it only sees the residual that HAR could not explain.

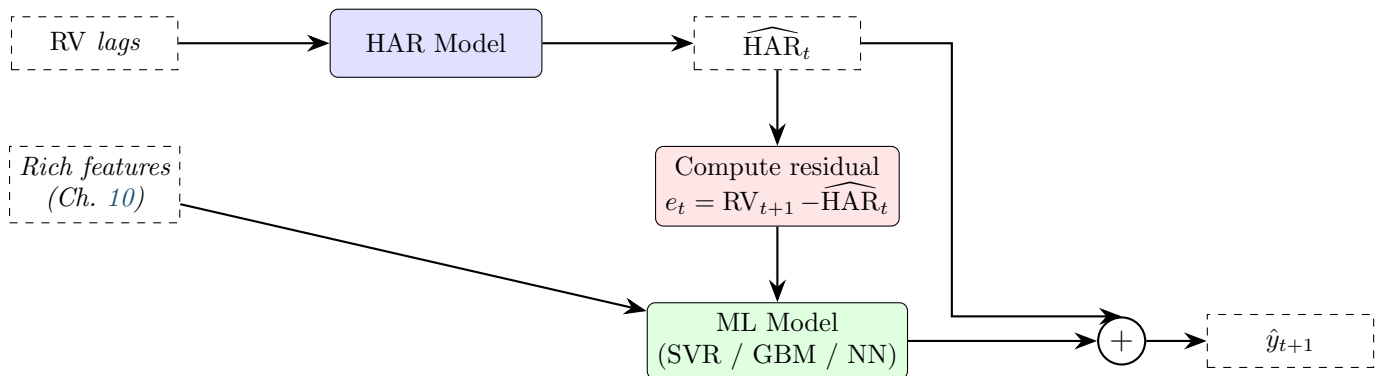


Figure 14.2: Two-stage hybrid pipeline. HAR absorbs the linear trend; ML targets only the residual. The final forecast sums both components.

14.3 HAR-SVR: Support Vector Regression on HAR Residuals

14.3.1 Intuition

Support vector regression (SVR) is a natural first choice for the residual stage because its ε -insensitive loss function automatically ignores small residuals. If the HAR fit is already

good for most days, many residuals will be near zero. SVR treats those as “correct enough” and focuses capacity on the days where HAR fails most, such as post-jump or regime-change dates (Rahimikia and Poon, 2020).

Why SVR Suits Residuals

SVR’s ε -tube acts as a built-in noise filter. Days where HAR is close enough (residual inside the tube) contribute zero loss. SVR concentrates its support vectors on the hard cases, which is exactly what you want when most of the signal has already been removed.

14.3.2 Procedure

1. Fit HAR on the training set:

$$\widehat{\text{HAR}}_t = \hat{\beta}_0 + \hat{\beta}_d \text{RV}_t + \hat{\beta}_w \text{RV}_t^{(w)} + \hat{\beta}_m \text{RV}_t^{(m)}. \quad (14.1)$$

2. Compute training residuals:

$$e_t = \text{RV}_{t+1} - \widehat{\text{HAR}}_t. \quad (14.2)$$

3. Construct a feature matrix $\mathbf{X}_t$ from the feature engineering toolkit of Chapter 10 (jump indicators, leverage, signed volatility, VIX basis, microstructure noise proxies).

4. Fit SVR with radial basis function (RBF) kernel on $(e_t, \mathbf{X}_t)$:

$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum_{t=1}^T \max(0, |e_t - f(\mathbf{X}_t)| - \varepsilon), \quad (14.3)$$

where each term is:

- $\frac{1}{2} \|w\|^2$: the regularization penalty that controls model complexity,
- C : the trade-off parameter between margin width and training error,
- ε : the tube width; residuals smaller than ε in absolute value incur zero loss,
- $f(\mathbf{X}_t) = w^\top \phi(\mathbf{X}_t) + b$: the SVR prediction in the kernel-induced feature space.

5. Produce the combined forecast:

$$\hat{y}_{t+1} = \widehat{\text{HAR}}_t + \widehat{\text{SVR}}(\mathbf{X}_t). \quad (14.4)$$

Why This Matters

This additive structure means you can always decompose your final forecast into “what HAR said” and “what ML corrected.” That decomposition is critical for interpretability at Goldman Sachs: you can explain the linear component with HAR coefficients and use SHAP values to explain the ML correction, giving the desk a complete audit trail.

Tune ε Carefully

Setting ε too large makes the SVR ignore all residuals. Setting it too small turns SVR into standard squared-loss regression and removes the noise-filtering benefit. Cross-validate ε on QLIKE or MSE, not on the number of support vectors.

14.4 GARCH-Informed Neural Networks

14.4.1 Motivation

The models in Sections 14.2–14.3 use a two-stage pipeline: fit an econometric model, then correct its errors. GARCH-Informed Neural Networks (GINN) take a different approach. They hard-wire the GARCH recursion directly into the neural network architecture so that the network learns corrections to the GARCH parameters rather than learning the entire volatility dynamics from scratch (Cuchiero et al., 2024).

GINN = Residual Learning for GARCH

GINN is to GARCH what a residual network is to a feedforward net. Instead of asking the network to learn σ_{t+1}^2 from raw data, you give it the GARCH update equation as a scaffold and let the network learn only the deviations. This dramatically reduces the function space the network must search.

14.4.2 Architecture

The standard GARCH(1,1) update is

$$\sigma_{t+1}^2 = \omega + \alpha \epsilon_t^2 + \beta \sigma_t^2. \quad (14.5)$$

GINN replaces the fixed parameters (ω, α, β) with time-varying outputs of a neural network g_{θ} :

$$\sigma_{t+1}^2 = \omega_t + \alpha_t \epsilon_t^2 + \beta_t \sigma_t^2, \quad (\omega_t, \alpha_t, \beta_t) = g_{\theta}(\mathbf{X}_t), \quad (14.6)$$

where each term is:

- $g_{\theta}(\mathbf{X}_t)$: a small feedforward network (typically 2 hidden layers, 32–64 units each) that maps auxiliary features $\mathbf{X}_t$ to time-varying GARCH parameters,
- $\omega_t, \alpha_t, \beta_t$: the GARCH coefficients at time t , constrained to satisfy $\omega_t > 0$, $\alpha_t \geq 0$, $\beta_t \geq 0$, and $\alpha_t + \beta_t < 1$ via softmax and sigmoid output activations (Cuchiero et al., 2024),
- ϵ_t^2 : the squared innovation (return shock),
- σ_t^2 : the previous conditional variance, fed back recurrently.

In Plain English

During calm markets the network can set β high (strong persistence); after a jump it can raise α (more reactive to shocks).

Why This Matters

GINN’s time-varying parameters are conceptually parallel to HARQ’s measurement-error-weighted coefficients. Both models ask: “What if the parameters themselves depend on the current regime?” If you extend HARQ with a neural network that modulates β_d , β_w , β_m based on jump intensity or microstructure noise, you are building a HAR analogue of GINN, and the GINN literature provides the architectural blueprint.

14.4.3 Why This Works

The key advantage is that GINN preserves the inductive bias of GARCH: volatility clusters, shocks decay exponentially, and the unconditional variance is finite. The neural network modulates these dynamics based on auxiliary information (sentiment, jump indicators, cross-asset

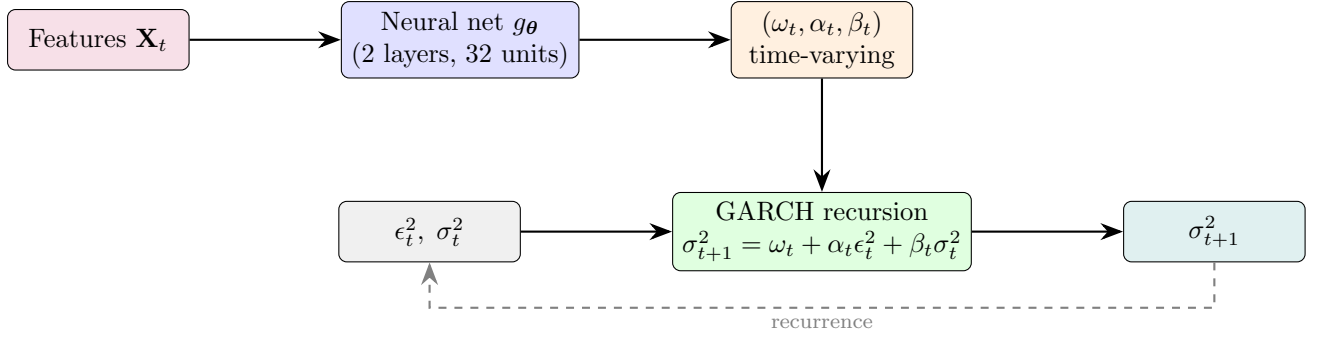


Figure 14.3: GINN architecture. A neural network produces time-varying GARCH parameters; the GARCH recursion itself is hard-wired, not learned.

signals) without having to discover the clustering structure itself. [Cuchiero et al. \(2024\)](#) show that GINN matches or outperforms both standard GARCH and unconstrained LSTMs on equity index volatility, with substantially fewer parameters than the LSTM.

14.5 NLP-Augmented Volatility Models

14.5.1 Motivation

News moves volatility, and the effect is asymmetric: negative news amplifies RV far more than positive news calms it. [Rahimikia et al. \(2021b\)](#) ask whether a machine-readable news signal can improve HAR forecasts. The answer is yes, but modestly and conditionally.

14.5.2 The Rahimikia-Zohren-Poon Pipeline

The procedure has three stages:

1. **Text embedding.** Collect financial news headlines (e.g., from Dow Jones Newswires) for each trading day. Train a Word2Vec skip-gram model on a financial corpus to obtain 300-dimensional word vectors (*FinText*). Arrange each day's tokens into a 500×300 sentence matrix (padding shorter sequences), then feed it through a convolutional neural network (CNN) with multiple filter sizes that learns to map the text directly to a volatility forecast ([Rahimikia et al., 2021b](#)).
2. **Augmented HAR (simplified view).** The simplest way to incorporate a text signal is to append a summary sentiment score s_t to the standard HAR regression:

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \gamma s_t + \varepsilon_{t+1}, \quad (14.7)$$

where:

- s_t : the daily text-derived signal (negative values indicate bearish tone),
- γ : the text loading; if news carries incremental information beyond lagged RV, then $\gamma \neq 0$.

In practice, [Rahimikia et al. \(2021b\)](#) go further: their NLP-ML model feeds the 500×300 sentence matrix directly into the CNN, bypassing the sentiment-score bottleneck. The linear augmented-HAR equation above is a useful pedagogical simplification that captures the same core idea: text provides an exogenous signal beyond lagged RV.

Why This Matters

At Goldman Sachs you may have access to proprietary news feeds or internal research sentiment. Public NLP signals show modest and conditional improvements, primarily on volatility jump days, but a curated internal signal could do better.

3. **Evaluation.** Compare QLIKE loss of HAR vs. HAR+NLP across the full sample and conditional on high-volatility periods.

14.5.3 What the Literature Finds

Rahimikia et al. (2021b) find that NLP-ML models strongly outperform HAR-family models on *volatility jump days* (roughly the top 10% of RV observations) but underperform on normal volatility days. During calm markets, the text signal adds negligible information because lagged RV already captures the low-volatility regime; the gains are concentrated in high-volatility episodes where news carries incremental content.

NLP Gains Are Small and Fragile

A 1–3% QLIKE improvement is economically meaningful only if you trade on volatility forecasts at scale. The improvement is sample-dependent, sensitive to the choice of news source, and does not survive aggressive transaction costs in short-horizon strategies (Rahimikia et al., 2021b). Do not overfit to the headline number.

Rahimikia and Poon (2020) provide additional evidence that hybrid models combining econometric baselines with text-derived features outperform pure text-based approaches, reinforcing the “let the baseline work first” principle of this chapter.

14.6 Ensemble: HAR + LightGBM

14.6.1 The Safest Performer

If you must choose one approach from this chapter for a production volatility forecast, choose a weighted average of HAR and LightGBM. Two combination strategies dominate practice.

Strategy A: Fixed Weighted Average

Choose a fixed weight $w \in [0, 1]$ and combine:

$$\hat{y}_{t+1} = w \widehat{\text{HAR}}_t + (1 - w) \widehat{\text{GBM}}_t, \quad (14.8)$$

where:

- w : the HAR weight, typically calibrated on a validation set or set to $w = 0.7$ as a robust default,
- $\widehat{\text{HAR}}_t$: the HAR forecast from Equation (14.1),
- $\widehat{\text{GBM}}_t$: the LightGBM forecast trained on the full feature set of Chapter 10.

The 70/30 Rule

A 70/30 HAR/LightGBM blend is remarkably hard to beat. It inherits HAR’s stability in calm markets and LightGBM’s ability to capture nonlinear effects during crises. You can optimize w on validation data, but the gain over a fixed 70/30 split is usually small.

Strategy B: Stacking with Ridge Meta-Learner

Instead of fixing w , learn the combination weights via ridge regression on out-of-sample predictions:

1. Generate OOS predictions from HAR and LightGBM using time-series cross-validation (expanding or rolling window).
2. Stack these predictions as features and fit a ridge regression:

$$\hat{y}_{t+1} = \alpha_0 + \alpha_1 \widehat{\text{HAR}}_t^{\text{OOS}} + \alpha_2 \widehat{\text{GBM}}_t^{\text{OOS}}, \quad (14.9)$$

where:

- $\widehat{\text{HAR}}_t^{\text{OOS}}$ and $\widehat{\text{GBM}}_t^{\text{OOS}}$ are out-of-sample predictions from the base models,
- $\alpha_0, \alpha_1, \alpha_2$ are learned by ridge regression with penalty λ chosen by cross-validation,
- the ridge penalty prevents the meta-learner from overfitting to the small differences between base models.

Why This Matters

The meta-learner can be trained by minimizing QLIKE rather than MSE, ensuring the combination weights are aligned with the project's primary loss function.

14.6.2 Architecture Diagram

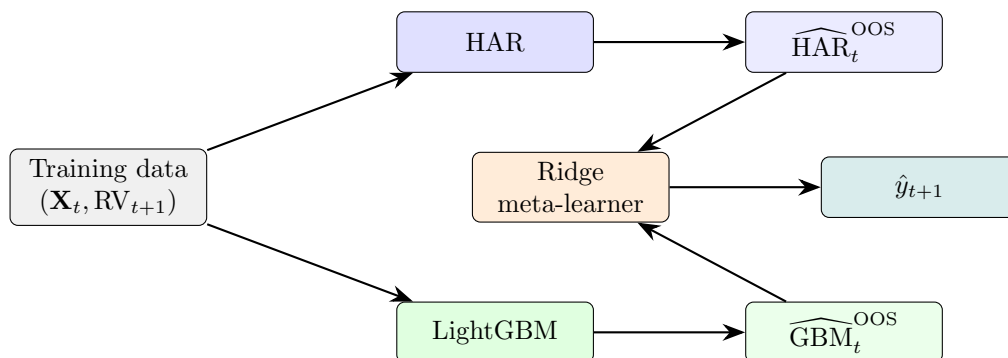


Figure 14.4: Stacking architecture. Base models produce OOS predictions; a ridge meta-learner learns the optimal combination.

14.7 Comparing Ensemble Architectures

So far this chapter has presented two combination strategies for HAR and LightGBM: a fixed weighted average (Equation (14.8)) and a ridge meta-learner on OOS predictions (Equation (14.9)). Both are instances of a broader design choice: how do you wire multiple models together? Three architectures dominate the ensemble literature, and each answers that question differently.

14.7.1 Architecture A: Feature Stacking

Feature stacking concatenates the internal representation of one model with the input features of another, training a single downstream model on the combined feature set. In the volatility context, this typically means training an LSTM on intraday sequences (e.g., 5-minute E-mini

bars) to produce a k -dimensional **embedding vector**, then appending that embedding to the tabular feature matrix before training LightGBM.

The combined feature vector fed to LightGBM becomes:

$$\tilde{\mathbf{X}}_t = [\mathbf{X}_t^{\text{tab}} \parallel \mathbf{h}_t^{\text{LSTM}}], \quad (14.10)$$

where each term is:

- $\mathbf{X}_t^{\text{tab}}$: the tabular feature matrix (RV lags, jump indicators, leverage, microstructure proxies, cross-asset signals) from Chapter 10,
- $\mathbf{h}_t^{\text{LSTM}}$: the k -dimensional embedding vector extracted from the LSTM's final hidden state after processing the intraday sequence,
- $\parallel$: concatenation along the feature axis.

In Plain English

The LSTM compresses a day of 5-minute bars into a summary vector that LightGBM treats as extra columns, hoping to capture intraday dynamics (order flow imbalances, microstructure noise patterns) that tabular statistics miss.

The Gradient Isolation Problem

Feature stacking has a fundamental flaw: **gradient isolation**. LightGBM is a tree-based model, so it cannot compute gradients with respect to its inputs. This means the LSTM embedding $\mathbf{h}_t^{\text{LSTM}}$ is never optimized for the tree's QLIKE objective. The LSTM learns representations that minimize its own loss function (typically MSE on log-RV), and the tree uses that representation as-is, whether or not it is what the tree needs.

Contrast this with an end-to-end neural architecture where you could backpropagate from the final loss through both components. In feature stacking, the two models live in separate optimization worlds:

1. The LSTM is trained to minimize $\mathcal{L}_{\text{LSTM}}$ on intraday sequences.
2. The embedding $\mathbf{h}_t$ is frozen and appended to $\mathbf{X}_t$.
3. LightGBM is trained to minimize $\mathcal{L}_{\text{GBM}}$ on $\tilde{\mathbf{X}}_t$, treating $\mathbf{h}_t$ as fixed columns.

If the embedding encodes information in a format that tree splits cannot exploit efficiently (e.g., information spread across multiple embedding dimensions in a way that requires linear combinations), LightGBM will underuse it. Worse, you cannot tell from the final forecast error whether a bad prediction came from a bad embedding or bad tree splits.

Gradient Isolation Breaks Joint Optimization

Because LightGBM cannot backpropagate into the LSTM, the embedding is optimized for the wrong objective. The LSTM minimizes its own loss; the tree minimizes a different loss on fixed embeddings. There is no mechanism to align the two. This is not a theoretical concern: it means the LSTM may learn to encode information that is useful for its own predictions but redundant or inaccessible to the tree.

14.7.2 Architecture B: Residual Stacking (Three-Stage Pipeline)

Section 14.3 introduced the two-stage residual hybrid: HAR first, then SVR on the residuals. **Residual stacking** generalizes this to three or more stages, where each model trains on the

residuals left by all prior stages. The canonical three-stage pipeline for volatility forecasting is:

1. **Stage 1: HAR (linear baseline).** Fit HAR on the training set exactly as in Equation (14.1). HAR captures the dominant multi-scale autoregressive structure of realized volatility. Compute residuals: $e_t^{(1)} = RV_{t+1} - \widehat{\text{HAR}}_t$.
2. **Stage 2: LightGBM (nonlinear interactions).** Train LightGBM on Stage 1 residuals $e_t^{(1)}$ using the full tabular feature set. The tree captures nonlinear patterns that HAR misses: regime interactions, jump-asymmetry effects, cross-asset spillovers. Compute residuals: $e_t^{(2)} = e_t^{(1)} - \widehat{\text{GBM}}(e_t^{(1)}, \mathbf{X}_t)$.
3. **Stage 3: LSTM (sequential dynamics).** Train an LSTM on Stage 2 residuals $e_t^{(2)}$ using intraday sequences. If any temporal structure remains after the tree has operated, the recurrent network can capture it. This stage is optional: if Stage 2 residuals are indistinguishable from white noise, the LSTM adds nothing and should be dropped.

The final forecast sums all stage contributions:

$$\hat{y}_{t+1} = \widehat{\text{HAR}}_t + \widehat{\text{GBM}}(\mathbf{X}_t) + \widehat{\text{LSTM}}(\mathbf{X}_t^{\text{seq}}), \quad (14.11)$$

where each term is:

- $\widehat{\text{HAR}}_t$: the HAR baseline forecast (Equation (14.1)),
- $\widehat{\text{GBM}}(\mathbf{X}_t)$: the LightGBM correction trained on HAR residuals with tabular features,
- $\widehat{\text{LSTM}}(\mathbf{X}_t^{\text{seq}})$: the LSTM correction trained on Stage 2 residuals with intraday sequences (set to zero if Stage 3 is dropped).

Residual Stacking Is Your Default Architecture

This three-stage pipeline maps directly onto the HARQ-X + ML residual direction. HARQ-X replaces plain HAR in Stage 1, giving even cleaner residuals because measurement-error adaptation removes a source of variation before the tree ever sees the data. Stage 2 (LightGBM on residuals) is where most of the incremental QLIKE improvement will come from. Stage 3 (LSTM on intraday sequences) is the optional upgrade path: add it only if you have evidence that Stage 2 residuals contain exploitable temporal structure.

Why does residual stacking avoid the gradient isolation problem? Because each model trains directly on a well-defined target (the residuals from the prior stage), using a loss function that is directly aligned with that target. There are no frozen embeddings, no cross-model feature coupling. If Stage 2 performs poorly, you know it is because LightGBM cannot explain the HAR residuals, not because some upstream embedding was misaligned.

14.7.3 Architecture C: Prediction Blending

Prediction blending is the simplest ensemble architecture: train each model independently, then combine their final predictions with a weighted average. This is what Section 14.6 already covers with the fixed 70/30 blend and the ridge meta-learner. The key distinction from the other architectures is that models never share information during training. Each model sees the original target RV_{t+1} (not a residual) and the features best suited to its architecture.

The general blending formula for K models is:

$$\hat{y}_{t+1} = \sum_{k=1}^K w_k \hat{y}_{t+1}^{(k)}, \quad \sum_{k=1}^K w_k = 1, \quad (14.12)$$

where each term is:

- $\hat{y}_{t+1}^{(k)}$: the independent forecast from model k ,
- w_k : the blend weight for model k , constrained to sum to one (though a ridge meta-learner as in Equation (14.9) relaxes this constraint).

Weight Schemes

Two approaches to setting blend weights:

Static weights. Choose weights once on a validation set and hold them fixed. The simplest principled choice is **inverse-QLIKE weighting**:

$$w_k = \frac{\text{QLIKE}_k^{-1}}{\sum_{j=1}^K \text{QLIKE}_j^{-1}}, \quad (14.13)$$

where each term is:

- QLIKE_k : the QLIKE loss of model k on the validation set,
- QLIKE_k^{-1} : the inverse loss; models with lower (better) QLIKE receive higher weight.

Regime-dependent weights. Use different blend weights in different volatility regimes. For example, give more weight to LightGBM during high-volatility periods (where nonlinear effects dominate) and more weight to HAR during calm periods (where the linear structure is sufficient). Section 14.8.7 discusses when this helps and when it does not.

Competition Evidence

Top-performing solutions in the Optiver Realized Volatility competition (Optiver, 2021) consistently chose prediction blending over feature stacking. Winners trained LightGBM and neural network branches independently, then combined outputs with simple weighted averages. The rationale: prediction blending is easier to debug (each branch can be evaluated in isolation), easier to iterate on (swap one branch without retraining others), and provides a natural fallback strategy (if one branch degrades, drop it and reweight).

Prediction Blending: Simplest and Often Best

Prediction blending requires no cross-model coupling, no shared training, and no sequential dependencies. Each model can be developed, validated, and debugged in isolation. Despite its simplicity, competition evidence from Optiver (Optiver, 2021) shows it matches or outperforms more complex architectures. It should be the default ensemble strategy unless you have specific evidence that residual stacking improves QLIKE.

14.7.4 Architecture Comparison

Table 14.1 summarizes the tradeoffs across the three architectures on five dimensions that matter for a production volatility pipeline.

14.8 Identifying Regimes: GMM, EM, and BIC

The regime-dependent weighting scheme of the next subsection, and the regime-switching combination of Chapter 11 (Equation (11.11)), both rely on a single ingredient we have so far left

Table 14.1: Three-way ensemble architecture comparison for volatility forecasting.

Dimension	Feature Stacking	Residual	Stack- ing	Prediction	Blend- ing
Complexity	High (joint training, embedding pipeline)	Moderate (sequential stages)		Low (independent models)	
Gradient flow	Broken (tree cannot backprop into LSTM)	Clean (each stage has its own target)		N/A (no cross-model coupling)	
Interpretability	Opaque (embedding dimensions lack semantics)	Clear (each stage's contribution is additive and measurable)		Clear (individual model forecasts are directly comparable)	
Fallback strategy	Must retrain tree without embedding	Drop later stages; keep HAR + Light-GBM		Drop one model; reweight the rest	
Literature support	Weak (no RV paper demonstrates gains)	Strong (HARQ-X residual literature)		Strong (Optiver, 2021); Kaggle competition evidence	

undefined: a *regime indicator*. Until now we have used ad-hoc thresholds ($RV_t^{(w)}$ above its 75th percentile, or $IV_t > 30$ for the VIX) to declare a market “stressed.” Those rules are easy to state but arbitrary: *why 75th and not 80th?* This section replaces the hand-picked threshold with a probabilistic model that lets the data decide both how many regimes exist and which one each day belongs to: the “GMM-based regime probability” that Chapter 11 promised when it defined the stress indicator s_t .

Multivariate Gaussian and Mixture Models

A d -dimensional Gaussian density is

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right),$$

where $\boldsymbol{\mu} \in \mathbb{R}^d$ is the mean and $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ is the covariance matrix (symmetric positive-definite: a square table recording each feature’s spread on its diagonal and how every pair of features co-moves off it). A **mixture model** says the data come from several such Gaussians, each chosen with some probability. If this is unfamiliar, review Bishop (2006), Ch. 2 and 9.

You do not need to understand every symbol above. The key idea is *distance from the center*: this is the ordinary bell curve generalized to d features at once. The $\exp(\dots)$ term is large when $\mathbf{x}$ sits close to the center $\boldsymbol{\mu}$ and shrinks as $\mathbf{x}$ moves away; the quadratic $(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$ measures that distance while accounting for how the features spread and co-move. The fraction in front is just a normalizing constant that makes the total probability sum to one; you never compute it by hand. Notationally, $|\boldsymbol{\Sigma}|$ is the **determinant** (a single number summarizing the overall spread), the superscript $^\top$ is the **transpose** (turn a column into a row so the multiplication works), and $\boldsymbol{\Sigma}^{-1}$ is the **matrix inverse**.

14.8.1 The GMM Density Model for Soft Regime Clustering

The idea is to treat each trading day as a point in a low-dimensional space of volatility descriptors and ask: *which cluster of historical days does today most resemble?* A **Gaussian Mixture Model (GMM)** formalizes “cluster” as a Gaussian blob and “which one” as a probability. We first state what the model is, then how to fit it.

Gaussian Mixture Model

A K -component Gaussian Mixture Model models the density of observations $\mathbf{x} \in \mathbb{R}^d$ as

$$p(\mathbf{x}) = \sum_{k=1}^K \underbrace{\pi_k}_{\text{how common regime } k \text{ is}} \underbrace{\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}_{\text{regime-}k \text{ shape}}, \quad (14.14)$$

where each term is:

- K : the number of mixture components, interpreted here as the number of **volatility regimes**,
- $\pi_k \geq 0$: the **mixing weight** (prior probability) of regime k , with $\sum_{k=1}^K \pi_k = 1$; the long-run fraction of days spent in regime k ,
- $\boldsymbol{\mu}_k \in \mathbb{R}^d$: the **mean** feature vector of regime k (e.g., the typical (RV, jump, IV) profile of a crisis),
- $\boldsymbol{\Sigma}_k \in \mathbb{R}^{d \times d}$: the **covariance** of regime k , describing how tightly days cluster around $\boldsymbol{\mu}_k$ and how the features co-move within the regime.

In Plain English

Equation (14.14) says: “Pretend volatility days are drawn from K different bell-shaped clouds. Each cloud has its own center (typical RV, jump size, VIX) and its own spread. A calm cloud sits at low RV and low VIX; a crisis cloud sits at high RV, large jumps, and a blown-out VIX.” Rather than slicing the data with a single hard line at the 75th percentile, the GMM draws soft, possibly overlapping blobs and lets a day belong partly to several of them.

Why This Matters

The regime descriptors we feed the GMM are precisely the quantities this guide has spent chapters building: the HAR components RV_t , $\text{RV}_t^{(w)}$, $\text{RV}_t^{(m)}$ (Chapter 6), the signed jump variation that separates good from bad volatility (Chapter 4), and the option-implied IV_t with its variance risk premium VRP_t (Chapter 9). A fitted GMM turns these into a soft regime label per day, which becomes the stress-indicator s_t that drives the regime-dependent blend weights in Section 14.8.7, replacing the arbitrary “ $\text{RV}^{(w)} > 75\text{th percentile}$ ” rule with a data-driven one.

What goes into $\mathbf{x}_t$. The regime model takes *volatility-state descriptors*, not your full prediction feature set. A compact, defensible choice for daily RV forecasting is

$$\mathbf{x}_t = (\text{RV}_t, \text{RV}_t^{(w)}, \text{RV}_t^{(m)}, J_t^+, J_t^-, \text{IV}_t, \text{VRP}_t)^\top, \quad (14.15)$$

where each term is:

- $\text{RV}_t, \text{RV}_t^{(w)}, \text{RV}_t^{(m)}$: the daily, weekly, and monthly HAR components, capturing the level and slope of the volatility term structure,

- J_t^+, J_t^- : the signed (positive and negative) jump variation, separating “good” upside jumps from “bad” downside jumps (Chapter 4),
- IV_t : the option-implied volatility (VIX), a forward-looking risk gauge,
- $VRP_t = IV_t^2 - \mathbb{E}_t[RV_{t+1}]$: the variance risk premium. Here $\mathbb{E}_t[RV_{t+1}]$ is the market’s *expected* next-day realized variance given everything known by day t (the subscript t means “conditional on information up to today”; in practice this is a forecast such as the HAR prediction itself). The VRP is then how much more the option market is charging (IV_t^2) than that expected realized amount, large when investors are paying up for crash protection (Chapter 9, which uses the full multi-day horizon).

Standardize Before You Fit

RV lives near 10^{-4} in variance units while the VIX lives near 20. Left raw, the GMM’s distances would be dominated entirely by the VIX and would ignore RV altogether. (**Euclidean distance** just means straight-line distance in feature space; a feature measured in 20s swamps one measured in 0.0001s, so the model would only ever “see” the VIX.) Apply the z-score leg of the **Triple Expansion** from Chapter 10 (Section 10.2) to every input. **Z-scoring** means subtract each feature’s mean and divide by its standard deviation, so it is recentred to a mean of 0 and rescaled to a typical size of 1, putting every feature on the same footing. Use only a trailing window to compute the standardization moments, so the regime label for day t uses no future information.

14.8.2 The EM Algorithm: Soft Labels, Then Updated Blobs

How do we find the $\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$ that best explain the data? By **maximum likelihood**: pick the parameters that make the data we actually observed as probable as possible. The log-likelihood of the parameters $\boldsymbol{\theta} = \{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^K$ is

$$\ell(\boldsymbol{\theta}) = \sum_{t=1}^T \log \underbrace{\sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}_{\text{mixture density at day } t}, \quad (14.16)$$

where each term is:

- T : the number of trading days in the fitting window,
- the inner sum: the GMM density (14.14) evaluated at day t ’s descriptor $\mathbf{x}_t$,
- the outer log-sum: the total log-probability the model assigns to the observed history.

In Plain English

Equation (14.16) scores a candidate set of regime blobs by asking “how likely is the actual volatility history under these blobs?” We want the blobs that make the data look most ordinary. The trouble is the log wraps a *sum* over regimes, so there is no *closed form*, a direct formula you could just plug numbers into. Taking the log of a single bell curve simplifies nicely (the exp cancels), but taking the log of a *sum* of bell curves does not simplify, so we have to iterate instead: knowing the best blobs requires knowing which regime each day belongs to, and knowing each day’s regime requires the blobs. This chicken-and-egg structure is exactly what the EM algorithm untangles.

The **Expectation–Maximization (EM)** algorithm (Dempster et al., 1977) breaks the deadlock by alternating two steps until the log-likelihood stops increasing.

EM for Gaussian Mixtures

E-step (soft labels). Given the current blobs, compute the **responsibility** of regime k for day t :

$$\gamma_{tk} = \frac{\pi_k \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}, \quad (14.17)$$

where γ_{tk} is how strongly regime k claims day t : literally regime k 's share of the total density at that day, a number between 0 and 1 (top divided by the sum of all the tops). In words: what fraction of today belongs to the calm cloud, the crisis cloud, the elevated cloud. Statisticians write this as the *posterior probability* $\Pr(\text{regime} = k \mid \mathbf{x}_t)$, read “the probability the regime is k , given that we observed today's descriptor $\mathbf{x}_t$ ” (the vertical bar means “given”).

M-step (update blobs). Treating the responsibilities as soft counts, re-estimate each regime:

$$N_k = \sum_{t=1}^T \gamma_{tk}, \quad (14.18)$$

$$\boldsymbol{\mu}_k^{\text{new}} = \frac{1}{N_k} \sum_{t=1}^T \gamma_{tk} \mathbf{x}_t, \quad (14.19)$$

$$\boldsymbol{\Sigma}_k^{\text{new}} = \frac{1}{N_k} \sum_{t=1}^T \gamma_{tk} (\mathbf{x}_t - \boldsymbol{\mu}_k^{\text{new}})(\mathbf{x}_t - \boldsymbol{\mu}_k^{\text{new}})^\top, \quad (14.20)$$

$$\pi_k^{\text{new}} = \frac{N_k}{T}, \quad (14.21)$$

where N_k is the **effective number of days** assigned to regime k . The mean update (14.19) is just a weighted average of the days (each day weighted by how much it claims to belong). The covariance update (14.20) is the weighted average *squared spread* of the days around the new center: it measures how wide and what shape each cloud is. The outer product $(\mathbf{x}_t - \boldsymbol{\mu}_k^{\text{new}})(\mathbf{x}_t - \boldsymbol{\mu}_k^{\text{new}})^\top$ is the multi-feature version of squaring a deviation: the second factor is written as a row (that is the $^\top$ transpose), so a column times a row produces a $d \times d$ *matrix*, not a single number, capturing both how much each feature varies and how the features move together. Iterate (14.17)–(14.21) until $|\ell^{(i+1)} - \ell^{(i)}| < \tau_{\text{EM}}$, where $\ell^{(i)}$ is the log-likelihood after iteration i (the parenthesized superscript is an iteration counter, not a power) and τ_{EM} is a small stopping threshold (e.g. 10^{-6}). Stop when the log-likelihood barely changes. EM guarantees $\ell(\boldsymbol{\theta})$ never decreases (Dempster et al., 1977).

In Plain English

EM is soft k -means. The E-step asks each day “what fraction of you belongs to the calm cloud, the crisis cloud, the elevated cloud?” and records those fractions. The M-step then redraws each cloud's center and spread using every day, but weighting each day by how much it claims to belong. A day that is 90% crisis pulls the crisis center hard and the calm center barely at all. Repeat, and the clouds drift until they settle on the natural groupings in the volatility data, with no hand-labelling required.

Why This Matters

The responsibilities γ_{tk} are the soft regime probabilities you want. Feed them in two ways. (1) As *features*: append $\gamma_{t,\text{crisis}}$ as an extra column to the LightGBM feature matrix of Chapter 11, letting the tree split on regime membership directly. (2) As a *blend dial*: set the stress indicator $s_t = \gamma_{t,\text{high-vol}}$ and let the regime-dependent weights of Section 14.8.7 interpolate smoothly between the calm-market and stressed-market blends, instead of flipping abruptly at a threshold.

Worked Example:

One EM Iteration on (RV, jump variation) Take $d = 2$ regime descriptors (standardized daily RV and signed jump variation) with $K = 3$ regimes (calm, elevated, crisis) and, after initialization,

$$\pi_1 = 0.6, \quad \pi_2 = 0.3, \quad \pi_3 = 0.1,$$

$$\boldsymbol{\mu}_1 = \begin{pmatrix} -0.8 \\ -0.5 \end{pmatrix}, \quad \boldsymbol{\mu}_2 = \begin{pmatrix} 0.4 \\ 0.3 \end{pmatrix}, \quad \boldsymbol{\mu}_3 = \begin{pmatrix} 2.5 \\ 2.0 \end{pmatrix},$$

with $\boldsymbol{\Sigma}_k = \mathbf{I}_2$ for simplicity (features already standardized). Consider a quiet day $\mathbf{x}_1 = (-0.7, -0.4)^\top$.

E-step. With $\boldsymbol{\Sigma}_k = \mathbf{I}_2$, $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{I}) = \frac{1}{2\pi} \exp(-\frac{1}{2}\|\mathbf{x} - \boldsymbol{\mu}\|^2)$, where $\|\mathbf{x} - \boldsymbol{\mu}\|^2$ is the **squared distance**, the sum of squared differences in each coordinate. The squared distances are

$$\|\mathbf{x}_1 - \boldsymbol{\mu}_1\|^2 = 0.01 + 0.01 = 0.02, \quad \|\mathbf{x}_1 - \boldsymbol{\mu}_2\|^2 = 1.21 + 0.49 = 1.70, \quad \|\mathbf{x}_1 - \boldsymbol{\mu}_3\|^2 = 10.24 + 5.76 = 16.0.$$

Now apply the $-\frac{1}{2}$ factor inside the exponent: e.g. for regime 1, $\exp(-0.02/2) = \exp(-0.01)$, halving each squared distance. Unnormalized responsibilities ($\tilde{\gamma}_{1k} = \pi_k \frac{1}{2\pi} e^{-\|\cdot\|^2/2}$, dropping the common $\frac{1}{2\pi}$):

$$\tilde{\gamma}_{1,1} = 0.6 e^{-0.01} = 0.594, \quad \tilde{\gamma}_{1,2} = 0.3 e^{-0.85} = 0.128, \quad \tilde{\gamma}_{1,3} = 0.1 e^{-8.0} = 0.0000335.$$

Normalizing,

$$\gamma_{1,1} = \frac{0.594}{0.722} \approx 0.82, \quad \gamma_{1,2} \approx 0.18, \quad \gamma_{1,3} \approx 0.00005.$$

Takeaway. The day is 82% calm and 18% elevated, with essentially zero crisis weight. Notice it is *not* 100% calm. A hard threshold would have forced a binary calm/not-calm call and thrown away the 18% of doubt. The soft label preserves that uncertainty, which is exactly what lets the downstream blend hedge between models on borderline days. The M-step then nudges $\boldsymbol{\mu}_1$ toward this day and barely moves $\boldsymbol{\mu}_3$.

14.8.3 Choosing the Number of Regimes: BIC

How many regimes should there be: two (calm/crisis), three, four? You cannot read K off the log-likelihood, because adding components always improves in-sample fit: a GMM with one blob per day fits perfectly and learns nothing. The **Bayesian Information Criterion (BIC)** (Schwarz, 1978) penalizes that complexity.

$$\text{BIC}(K) = \underbrace{-2 \ell(\hat{\boldsymbol{\theta}})}_{\text{misfit}} + \underbrace{p \log T}_{\text{complexity penalty}}, \quad (14.22)$$

where each term is:

- $\ell(\hat{\boldsymbol{\theta}})$: the maximized log-likelihood from (14.16) at the EM solution,

- p : the number of free parameters; for a K -component GMM in d dimensions with full covariances, $p = K(1 + d + \frac{d(d+1)}{2}) - 1$. Reading the count: each regime needs 1 mixing weight, d numbers for its mean vector, and $\frac{d(d+1)}{2}$ numbers for its (symmetric) covariance matrix, all times K regimes, minus 1 because the mixing weights must sum to one so the last one is determined by the rest. (For $K = 2$ regimes and $d = 7$ features, $p = 2(1 + 7 + 28) - 1 = 71$ parameters.)
- T : the sample size (trading days), so $\log T$ scales the per-parameter penalty.

In Plain English

In volatility data, $K = 2$ or $K = 3$ usually wins: markets have a calm state, a crisis state, and sometimes an in-between “elevated but orderly” state.

14.8.4 The sklearn Workflow

The entire fit is a few lines with `scikit-learn`. The one non-obvious knob is `n_init`.

GMM in scikit-learn

1. **Standardize** the inputs with a trailing-window z-score (see the warning above; Section 10.2).
2. **Fit** for each candidate K :
`GaussianMixture(n_components=K, covariance_type='full',
n_init=10, random_state=0).fit(X)`
3. **Select** K by minimizing `gmm.bic(X)`.
4. **Label** each day with soft probabilities `gmm.predict_proba(X)` (the responsibilities γ_{tk}), *not* the hard `gmm.predict(X)`.

Always Use `n_init` Restarts

The GMM log-likelihood (14.16) is **non-convex**: the likelihood surface has multiple hills, not one. EM always climbs uphill, so it can get stuck on a smaller hill (a *local* maximum) depending on where it starts. That is why we try several random starting points and keep the best. A single bad start can merge a crisis regime into the calm one. Set `n_init=10` (ten random restarts, keeping the highest-likelihood solution) so the regimes are stable across reruns. Use `covariance_type='full'` unless you are short on data: volatility features are correlated within a regime (high RV comes with high VIX), and a diagonal covariance would miss that.

14.8.5 Adding Persistence: Markov-Switching and HMMs

The GMM has a blind spot that matters enormously for volatility: *it has no memory*. It classifies each day independently, so its labels can flicker calm-crisis-calm-crisis on consecutive days even though we know volatility regimes are sticky: a crisis that starts today is overwhelmingly likely to persist tomorrow. How do we bake that stickiness into the regime model?

Hidden Markov Models

A **Hidden Markov Model (HMM)** has two layers: (1) a latent state sequence $s_t \in \{1, \dots, K\}$ that evolves as a Markov chain, and (2) observed emissions $\mathbf{x}_t$ whose distribution depends on s_t . The key assumption is that s_t depends on s_{t-1} only (the Markov property). A Markov-switching model is an HMM whose emission is a regression rather than a static Gaussian.

Markov-switching regression (Hamilton, 1989) adds exactly the missing persistence. The latent state still indexes the regime, but now a **transition matrix** governs how the state evolves over time:

$$\mathbf{P} = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix}, \quad p_{ij} = \underbrace{\Pr(s_t = j \mid s_{t-1} = i)}_{\text{regime } i \rightarrow j}, \quad p_{i0} + p_{i1} = 1, \quad (14.23)$$

where each term is:

- $s_t \in \{0, 1\}$: the latent regime (here 0 = low-vol, 1 = high-vol, so the diagonal p_{00} is the calm-regime persistence; this is the opposite of Hamilton’s expansion = 1 coding),
- p_{ij} : the probability of moving from regime i today to regime j tomorrow,
- the **diagonal** entries p_{00}, p_{11} : the **persistence** of each regime, how likely it is to stay put.

In Plain English

The transition matrix is the memory the GMM lacked. Where the GMM treats today as a fresh draw, Equation (14.23) says “if you were calm yesterday, you are probably calm today” through a high p_{00} (rows are today’s regime, columns are tomorrow’s, so p_{ij} reads “from i to j ”). If each day you stay with probability p , the average run length before you leave is $1/(1 - p)$, the same arithmetic as “if a coin lands heads 98% of the time, you wait about 50 flips for a tail.” So the expected length of a regime is $1/(1 - p_{ii})$ days, and a persistence near 0.98 implies spells of roughly 50 days. Estimating the regime now requires running forward through the whole sequence with the **Hamilton filter** (Hamilton, 1989), which combines yesterday’s regime belief, the transition matrix, and today’s observation into today’s filtered probability $\Pr(s_t = k \mid \mathbf{x}_{1:t})$, the probability of today’s regime using only data up to and including today (days 1 through t).

Why This Matters

Persistent regime labels are far more useful as a blend dial than flickering ones. If your stress indicator s_t jumps in and out of “crisis” on alternate days, the regime-dependent blend of Section 14.8.7 will whipsaw between the calm and stressed weight sets, churning the forecast for no reason. A Markov-switching filter smooths the indicator using the persistence it learned from history, giving a regime signal that turns on at the start of a crisis and stays on until it genuinely ends.

The canonical demonstration of Markov-switching is Hamilton (1989)’s two-state model of U.S. GNP growth (expansion vs. recession). We rebuild that example on the object we actually care about: realized volatility.

Two-State Markov-Switching Model on RV

Fit a two-state Markov-switching model to daily (log) realized volatility,

$$\log \text{RV}_t = \mu_{s_t} + \phi(\log \text{RV}_{t-1} - \mu_{s_{t-1}}) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_{s_t}^2), \quad (14.24)$$

where each term is:

- μ_{s_t} : the regime-dependent average level of log-RV, which flips between a calm value and a crisis value as the latent state s_t switches (the double subscript means “the mean for whichever regime applies on day t ”),
- ϕ : the **persistence** (autoregressive) coefficient, between -1 and 1 , controlling how strongly yesterday’s deviation carries into today,
- $(\log \text{RV}_{t-1} - \mu_{s_{t-1}})$: yesterday’s gap above or below *its* regime mean,

- $\varepsilon_t \sim \mathcal{N}(0, \sigma_{s_t}^2)$: random noise (the “ $\sim$ ” reads “is distributed as”) whose size σ_{s_t} is itself regime-dependent (bigger in a crisis),

with $s_t \in \{\text{low}, \text{high}\}$ and transition matrix (14.23). A representative equity-index fit yields two clearly separated regimes:

- a **low-vol state** with a low mean log-RV level (μ_{low}) and high persistence $p_{00} \approx 0.98$ (calm spells last $\approx 1/(1 - 0.98) = 50$ days),
- a **high-vol state** with a markedly higher mean ($\mu_{\text{high}} \gg \mu_{\text{low}}$) and persistence $p_{11} \approx 0.95$ (crisis spells last ≈ 20 days), confirming the well-documented asymmetry that calm regimes are stickier than turbulent ones.

The filtered probability $\Pr(s_t = \text{high} \mid \mathbf{x}_{1:t})$ spikes during known stress episodes (2008, 2020) without those dates being supplied as inputs, the analogue of Hamilton’s filtered recession probabilities tracking NBER dates.

In Plain English

Equation (14.24) is just an AR(1) (autoregressive of order 1, meaning today depends on yesterday plus noise) for log-volatility whose *level* μ_{s_t} flips between a calm value and a crisis value, with the flips governed by the transition matrix. The two estimated means tell you where calm and crisis volatility sit; the two persistences tell you how long each lasts. Because the level shifts discretely, the model captures the abrupt jump into a high-vol regime that a single smooth AR process would smear out.

Link to Markov-Switching GARCH

Replacing the regime-dependent *mean* in Equation (14.24) with a regime-dependent *GARCH variance recursion* gives the **Markov-switching GARCH** model: the (ω, α, β) parameters of the GARCH update from Chapter 5 themselves switch with the latent state. This is the econometric cousin of the GINN idea in Section 14.4, where a neural network, rather than a two-state Markov chain, supplies the time-varying GARCH parameters. Both answer the same question: “what if the volatility dynamics themselves depend on the current regime?”

GMM vs. Markov-Switching: Which to Use

Table 14.2 contrasts the two on the dimensions that matter for a volatility-forecasting pipeline.

Table 14.2: Choosing between GMM and Markov-switching for regime identification.

Dimension	GMM	Markov-Switching
Temporal structure / persistence	None: each day classified independently, so labels can flicker	Markov chain: s_t depends on s_{t-1} ; persistence explicit via transition matrix $\mathbf{P}$
Estimation	EM on the mixture (fast, scales well)	Hamilton filter (forward recursion; slower)
Output	Responsibilities γ_{tk} per day	Filtered $\Pr(s_t \mid \mathbf{x}_{1:t})$ and smoothed $\Pr(s_t \mid \mathbf{x}_{1:T})$
Python	<code>sklearn.mixture.GaussianMixture</code>	<code>statsmodels.tsa.regime_switching</code>
Best for	Quick soft labels; a regime feature for LightGBM	When persistence matters; a smooth blend dial

Start with GMM; Upgrade to Markov-Switching for Persistence

For a first regime feature (a column of soft probabilities to hand the LightGBM model or a quick blend dial), the GMM is simpler, faster, and sufficient. Reach for Markov-switching only when day-to-day flickering of the regime label is actively hurting the blend, or when you want a regime signal whose persistence is calibrated from the data. The two are complementary: GMM answers “what does today look like?”; Markov-switching answers “what does today look like, given the whole run-up to it?”

14.8.6 Regime Backtesting Hygiene

Regime-conditional forecasts look better in backtests than in production for one reason: *the regime is only known with certainty in hindsight*. When you fit a GMM (or smooth a Markov-switching filter) on the full sample and then condition on the resulting labels, you have leaked future information into today’s regime call.

Test Regimes the Way You Will Trade Them

Three disciplines keep a regime overlay honest:

1. **Use soft probabilities, not hard labels** (workflow step 4 above). A strategy that flips fully risk-on at a hard “calm” label implicitly assumes a perfect classification you do not have: today’s responsibility might be only 60% calm, 40% elevated.
2. **Test with lagged regime labels.** Apply yesterday’s regime classification to today’s blend, never a label that used today’s or future data. For Markov-switching, use the *filtered* probability $\Pr(s_t \mid \mathbf{x}_{1:t})$, which uses only data up to and including today (days 1 through t), not the *smoothed* $\Pr(s_t \mid \mathbf{x}_{1:T})$, which uses the *entire* sample (days 1 through the last day T) and so secretly looks into the future: fine for describing history, fatal for backtesting a tradable signal.
3. **Check label stability.** Measure how often the GMM revises its call on the most recent days when one new observation arrives. A regime indicator that keeps changing its mind about the recent past is not a tradable signal.

Report the regime-conditional QLIKE improvement only after passing these checks; a gain that survives lagged, soft labels is real, one that needs full-sample hard labels is hindsight.

14.8.7 Static vs. Regime-Dependent Weights

Static blend weights assume that the relative accuracy of each model is stable over time. This is often a good approximation for realized volatility: HAR’s linear structure captures most of the signal in calm and turbulent markets alike. But there are situations where model performance varies systematically across regimes, and regime-dependent weights can help.

Regime-dependent weighting assigns different blend weights depending on the current volatility regime:

$$w_k(t) = \begin{cases} w_k^{\text{low}} & \text{if } \text{RV}_t^{(w)} < \tau, \\ w_k^{\text{high}} & \text{if } \text{RV}_t^{(w)} \geq \tau, \end{cases} \quad (14.25)$$

where each term is:

- $\text{RV}_t^{(w)}$: the weekly realized volatility, used as a regime indicator,
- τ : the regime threshold (e.g., the 75th percentile of the training-set $\text{RV}^{(w)}$ distribution),
- $w_k^{\text{low}}, w_k^{\text{high}}$: separate blend weights for the low-volatility and high-volatility regimes, each calibrated on the corresponding subset of the validation data.

Regime Weights Require Stable Performance Differences

Regime-dependent weights help only when model accuracy varies *systematically* across regimes, not just randomly. If the performance gap between HAR and LightGBM in high-vol vs. low-vol periods is unstable across rolling windows, regime conditioning will overfit to historical patterns that do not persist out of sample. Test for systematic variation before adding this complexity: compute model QLIKE separately in each regime on multiple validation folds and check whether the ranking is consistent.

Start Static, Upgrade If Justified

For the internship project, begin with static inverse-QLIKE weights. After establishing the baseline blend performance, split the validation set by volatility regime and compare per-regime QLIKE for each model. If you find that LightGBM's advantage over HARQ-X is concentrated in high-volatility weeks (as the leverage-effect literature suggests it should be), regime-dependent weights are a justified upgrade. Document the per-regime performance gap as evidence.

14.9 When to Use Pure ML vs. Hybrid

The table below summarizes the practical decision you face when choosing a forecasting architecture. The default should be hybrid; deviate only with good reason.

Table 14.3: Architecture decision guide for volatility forecasting.

Architecture	Best When	Feature Set	Typical Setting
Pure HAR / HARQ	Only RV lags available; small sample (< 500 days)	RV lags only	Quick benchmark; limited data access
Hybrid (HAR + ML)	Rich features available; daily or weekly horizon; reliability matters	RV lags + engineered features	Production forecasts; research baseline
Pure trees (GBM/XGB)	Rich features; intraday horizons; large sample (> 2000 days)	Full feature set	High-frequency desks; feature importance studies
Pure deep learning	Raw sequential data; cross-asset pooling; very large sample	Raw returns / order flow	Multi-asset platforms; representation learning

Default to Hybrid

You earn the right to go pure ML only when you have a large sample, rich features, and evidence that the linear baseline leaves substantial structure in the residuals ([Rahimikia and Poon, 2020](#)).

14.9.1 Practical Checklist

Before committing to an architecture, answer these questions:

1. **How large is your sample?** Below 500 daily observations, pure HAR is hard to beat. ML needs data.
2. **Do you have features beyond RV lags?** If not, a hybrid adds nothing because the ML stage has no new information.
3. **What is your forecast horizon?** At weekly or monthly horizons, HAR's multi-scale structure dominates. ML gains concentrate at the daily horizon.
4. **How much do you value interpretability?** HAR coefficients are directly interpretable. A hybrid with SHAP on the ML component preserves partial interpretability. Pure deep learning sacrifices it.
5. **What is your retraining budget?** Hybrids are cheap: refit HAR monthly, retrain ML weekly. Pure deep learning demands more compute.

14.9.2 Standalone vs. Hybrid: A Visual Comparison

Figure 14.5 contrasts the pure ML approach (left) with the hybrid approach (right). The key difference: in the hybrid, ML operates on a reduced-variance target, which yields lower estimation error and better out-of-sample stability.

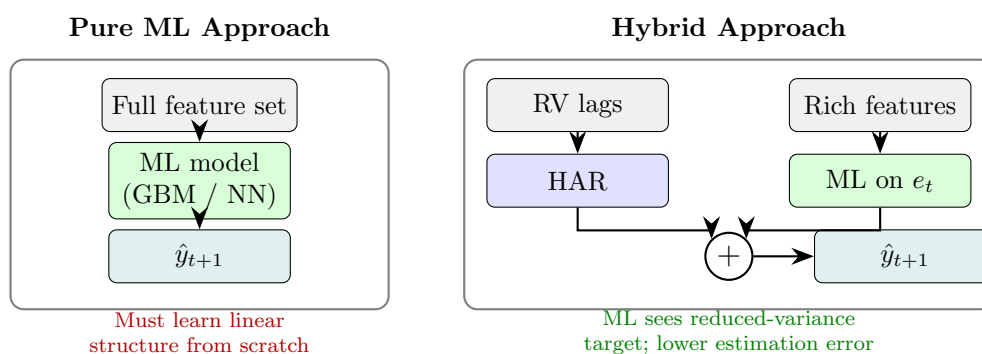


Figure 14.5: Pure ML (left) must rediscover the linear trend that HAR captures for free. The hybrid (right) lets HAR absorb the trend and focuses ML on the residual, reducing both model complexity and overfitting risk.

14.10 Summary

- Hybrid models decompose the forecasting problem into a linear component (HAR, GARCH) and a nonlinear residual component (ML), allowing each method to operate where it is strongest.
- HAR explains 40–60% of next-day RV with three coefficients; training ML on the residual rather than the raw target improves signal-to-noise and prevents the ML model from wasting capacity on structure that a linear model already captures.
- HAR-SVR uses support vector regression with an ε -insensitive loss on HAR residuals, automatically ignoring days where HAR is “close enough” and focusing on the hard cases.
- GINN hard-wires the GARCH recursion into a neural network, letting the network learn time-varying corrections to GARCH parameters rather than learning volatility dynamics

from scratch (Cuchíero et al., 2024).

- NLP-augmented HAR (Word2Vec sentiment) yields 1–3% QLIKE improvement concentrated in crisis periods, but gains are fragile and source-dependent (Rahimikia et al., 2021b).
- A simple 70/30 HAR/LightGBM weighted average is the safest production choice and is remarkably hard to beat with more complex ensembles.
- Stacking via a ridge meta-learner on OOS predictions adapts the combination weights to the data while guarding against overfitting at the meta-level.
- Three ensemble architectures offer increasing simplicity: feature stacking (coupling via embeddings, broken gradient flow), residual stacking (sequential stages with distinct roles), and prediction blending (fully independent models, weighted average).
- Feature stacking suffers from gradient isolation: LightGBM cannot backpropagate into the LSTM, so the embedding is never optimized for the tree’s objective.
- Residual stacking (HAR $\rightarrow$ LightGBM $\rightarrow$ LSTM) gives each model a specialized role by construction and avoids cross-model coupling.
- Prediction blending is the simplest, most debuggable architecture and is supported by Optiver competition evidence (Optiver, 2021).
- Static blend weights (e.g., inverse-QLIKE weighting) are the default; regime-dependent weights help only when model performance varies systematically across volatility regimes.
- Pure ML (trees or deep learning) earns its place only when you have large samples, rich features, and evidence that the linear baseline leaves exploitable structure in residuals.
- Your default architecture should be hybrid: let HAR do the heavy lifting, then train ML on what remains.
- When evaluating hybrids, always report the base model’s standalone performance alongside the hybrid, so the reader can judge the marginal contribution of the ML component.
- Ensemble diversification provides error reduction even when the ML component is only modestly accurate, as long as its errors are weakly correlated with the base model’s errors.
- All combination weights and meta-learner parameters must be tuned on out-of-sample data to avoid inflating the apparent benefit of the hybrid.

Chapter 14 Takeaways

Concept	Key Result
Hybrid principle	Decompose into linear (HAR/GARCH) + nonlinear (ML) components
HAR-SVR	ε -insensitive loss filters small residuals automatically
GINN	Hard-wired GARCH recursion with NN-learned time-varying parameters (Cuchíero et al., 2024)
NLP + HAR	1–3% QLIKE gain, concentrated in crises (Rahimikia et al., 2021b)
70/30 blend	HAR/LightGBM weighted average: simple, robust, hard to beat
Stacking	Ridge meta-learner on OOS predictions adapts weights safely
Architecture comparison	Feature stacking < residual stacking < prediction blending in simplicity and debuggability
Regime weights	Upgrade from static only with evidence of systematic per-regime performance differences
Default rule	Start hybrid; earn the right to go pure ML with evidence

Part V

Multivariate Volatility and Connectedness

Chapter 15

Realized Covariance and Multivariate Forecasting

From One Asset to Many

Chapters 1–9 focused on the volatility of a single asset. In practice, portfolios hold many assets, and the *covariance structure* matters as much as individual volatilities. Minimum-variance portfolios, risk parity, hedging, and option pricing on baskets all require a full covariance matrix, not just a list of variances. This chapter extends realized volatility to the multivariate setting: estimating, modeling, and forecasting entire covariance matrices. Your project uses this material directly: the cross-asset covariance structure feeds the Layer 4 spillover features that enter LightGBM as scalar inputs.

15.1 Realized Covariance

Where we are. You know how to build realized variance for one asset (Chapter 2). The natural multivariate extension replaces squared returns with outer products.

Realized Covariance Matrix

Let $\mathbf{r}_{t,i} \in \mathbb{R}^p$ be the i -th intraday return vector on day t , for $i = 1, \dots, M$. The **realized covariance** (RC) matrix is

$$\mathbf{RC}_t = \sum_{i=1}^M \mathbf{r}_{t,i} \mathbf{r}_{t,i}^\top \in \mathbb{R}^{p \times p}. \quad (15.1)$$

- $\mathbf{r}_{t,i} \in \mathbb{R}^p$: intraday return vector for p assets at interval i .
- M : number of intraday intervals (e.g., 78 for 5-minute sampling of a 6.5-hour trading day).
- Diagonal entries $[\mathbf{RC}_t]_{jj}$: realized variances of each asset (as in Chapter 2).
- Off-diagonal entries $[\mathbf{RC}_t]_{jk}$: realized covariances between assets j and k .

Under no noise and synchronous trading, $\mathbf{RC}_t$ converges to the integrated covariance matrix as $M \rightarrow \infty$ (Barndorff-Nielsen et al., 2011).

15.1.1 The Synchronicity Problem

Forming each $\mathbf{r}_{t,i}$ requires observing all p assets at the same timestamps, but different assets trade at different times. A thinly traded stock might have no trade in a given 5-minute window.

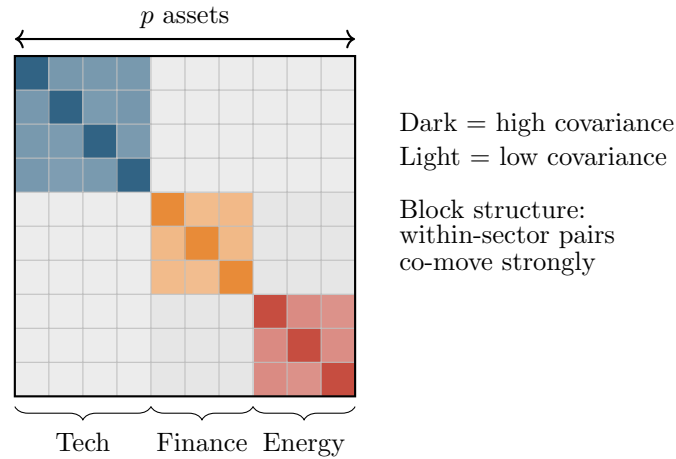


Figure 15.1: Schematic of a realized covariance matrix with $p = 10$ assets sorted by sector. The block-diagonal structure (strong within-sector covariance, weak cross-sector) is a key pattern that CNNs and factor models exploit. The diagonal entries are the realized variances from Chapter 2.

If you simply ignore missing observations, you bias realized covariance toward zero (the “Epps effect”).

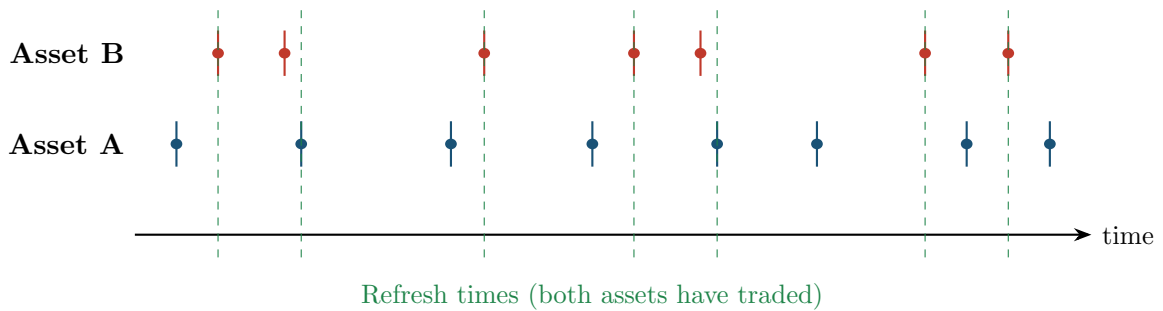


Figure 15.2: Non-synchronous trading. Asset A (blue) and Asset B (red) trade at different times. Refresh times (green dashed) mark points where both assets have at least one new observation since the last refresh time.

15.1.2 Refresh-Time Sampling

The simplest fix: wait until *all* p assets have traded at least once, then record a synchronized return vector. These “refresh times” are the green lines in Figure 15.2. Formally, define $\tau_0 = 0$ and

$$\tau_k = \max_{j=1,\dots,p} \min\{t_{j,n} : t_{j,n} > \tau_{k-1}\},$$

where $t_{j,n}$ is the n -th trade time for asset j . You compute returns at refresh times and plug them into (15.1). The cost: you lose observations, sometimes dramatically for illiquid assets.

15.1.3 Hayashi–Yoshida Estimator

Hayashi and Yoshida (2005) proposed a more efficient solution. Instead of aligning timestamps, you sum the products of all *overlapping* returns.

Hayashi–Yoshida Estimator

For two assets with (possibly different) trade times, let $r_{t,i}^{(A)}$ denote asset A's return over interval $[t_{i-1}^A, t_i^A)$ and similarly for B. The Hayashi–Yoshida realized covariance is

$$\hat{\sigma}_{AB,t}^{\text{HY}} = \sum_i \sum_j r_{t,i}^{(A)} r_{t,j}^{(B)} \mathbf{1}\{[t_{i-1}^A, t_i^A) \cap [t_{j-1}^B, t_j^B) \neq \emptyset\}. \quad (15.2)$$

- Sum over all pairs of intervals that overlap in time.
- No data is discarded; every trade contributes.
- Unbiased and consistent for the integrated covariance under mild conditions.
- Not guaranteed positive semi-definite (PSD) for $p > 2$.

In Plain English

Instead of forcing both assets onto a common clock (which throws away data), Hayashi–Yoshida is like computing a dot product, but one that uses every available trade rather than only the synchronized ones.

Worked Example:

Hayashi–Yoshida with 2 Assets Suppose on one day, asset A trades at times $\{0, 2, 5, 8\}$ with prices $\{100, 101, 99, 100.5\}$ and asset B trades at times $\{0, 3, 6, 8\}$ with prices $\{50, 50.5, 49.8, 50.2\}$.

Step 1: compute returns.

$$\begin{aligned} r_1^A &= \ln(101/100) \approx 0.00995, & r_2^A &= \ln(99/101) \approx -0.02000, & r_3^A &= \ln(100.5/99) \approx 0.01504, \\ r_1^B &= \ln(50.5/50) \approx 0.00995, & r_2^B &= \ln(49.8/50.5) \approx -0.01396, & r_3^B &= \ln(50.2/49.8) \approx 0.00800. \end{aligned}$$

Step 2: identify overlapping pairs.

r_i^A interval	r_j^B interval	Overlap?
$[0, 2)$	$[0, 3)$	Yes
$[0, 2)$	$[3, 6)$	No
$[2, 5)$	$[0, 3)$	Yes
$[2, 5)$	$[3, 6)$	Yes
$[5, 8)$	$[3, 6)$	Yes
$[5, 8)$	$[6, 8)$	Yes

Step 3: sum products of overlapping pairs.

$$\hat{\sigma}_{AB}^{\text{HY}} \approx 9.0 \times 10^{-5}.$$

The positive value indicates slight positive comovement. Note: no observations were discarded.

PSD is Not Guaranteed

For $p = 2$, the Hayashi–Yoshida estimator is always PSD (it is a scalar covariance). For $p \geq 3$, the matrix assembled from pairwise HY estimates can have negative eigenvalues. You may need to project the result onto the PSD cone (e.g., by zeroing out negative eigenvalues), which introduces bias. Section 15.10 discusses this systematically.

15.2 Multivariate Realized Kernel

Where we are. You have two problems: non-synchronous trading (Section 15.1) and microstructure noise (Chapter 3). The multivariate realized kernel handles both simultaneously.

Extending the Realized Kernel

In Chapter 3, the univariate realized kernel applied a weighting function to autocovariances of returns at different lags, suppressing noise while remaining consistent. The multivariate version does exactly the same thing, but with cross-autocovariance matrices instead of scalar autocovariances.

Multivariate Realized Kernel

Barndorff-Nielsen et al. (2011) define the multivariate realized kernel as

$$\mathbf{K}_t = \sum_{h=-H}^H k\left(\frac{h}{H+1}\right) \mathbf{\Gamma}_h, \quad (15.3)$$

where

- $\mathbf{\Gamma}_h = \sum_i \mathbf{r}_{t,i} \mathbf{r}_{t,i+h}^\top$ is the h -th cross-autocovariance matrix of intraday returns.
- $k(\cdot)$: a kernel function satisfying $k(0) = 1$, $k(1) = 0$ (e.g., Parzen kernel). Same options as the univariate case in Chapter 3.
- H : bandwidth, controlling how many lags to include.

PSD Guarantee Under Noise

The multivariate realized kernel with a non-negative kernel function (e.g., Parzen) is guaranteed PSD by construction, even with non-synchronous trading and microstructure noise (Barndorff-Nielsen et al., 2011). This is a major advantage over refresh-time RC and the Hayashi–Yoshida estimator. The rate of convergence is $M^{-1/5}$ (slower than the noise-free $M^{-1/2}$), the same price paid in the univariate case.

15.3 DCC-GARCH

Where we are. You can now *estimate* a daily covariance matrix from high-frequency data; the next task is to *forecast* tomorrow's. DCC-GARCH is the multivariate analog of GARCH (Chapter 5): a parametric, easy-to-estimate baseline.

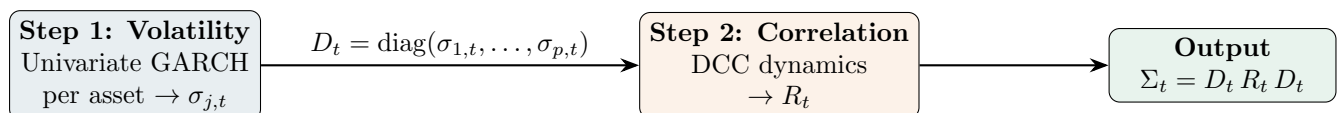


Figure 15.3: DCC-GARCH decomposes the covariance matrix into volatilities (modeled separately per asset) and a dynamic correlation matrix.

DCC-GARCH (Engle, 2002)

The Dynamic Conditional Correlation (DCC) model has two steps:

Step 1 (volatilities). Fit a univariate GARCH(1,1) (or any variant) to each asset's return series separately, producing conditional standard deviations $\sigma_{j,t}$ for $j = 1, \dots, p$.

Stack them into a diagonal matrix:

$$D_t = \text{diag}(\sigma_{1,t}, \sigma_{2,t}, \dots, \sigma_{p,t}).$$

Step 2 (correlations). Compute standardized residuals $\mathbf{z}_t = D_t^{-1} \mathbf{r}_t$ and model their quasi-correlation:

$$Q_t = (1 - \alpha - \beta) \bar{Q} + \alpha \mathbf{z}_{t-1} \mathbf{z}_{t-1}^\top + \beta Q_{t-1}, \quad (15.4)$$

where

- $\bar{Q}$: unconditional covariance of $\mathbf{z}_t$ (estimated from the full sample).
- $\alpha > 0$: weight on yesterday's innovation (analogous to ARCH coefficient).
- $\beta > 0$: persistence (analogous to GARCH coefficient).
- $\alpha + \beta < 1$: stationarity condition.

Then rescale to a proper correlation matrix:

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}.$$

The conditional covariance matrix is $\Sigma_t = D_t R_t D_t$.

Why DCC is the “HAR of Multivariate Vol”

DCC is to multivariate volatility what HAR (Chapter 6) is to univariate: not the most accurate model, but easy to estimate, hard to beat consistently, and the first thing you should try. It scales well to large p because Step 1 is embarrassingly parallel (one GARCH per asset) and Step 2 has only two free parameters (α, β) regardless of dimension.

DCC Limitations

- Correlations follow a single (α, β) dynamic for all pairs. In reality, the IBM–MSFT correlation may move differently from the IBM–gold correlation.
- DCC uses daily returns, not intraday data. It ignores the richer information in realized covariance.
- The two-step estimation is not fully efficient (but it is consistent).

Why This Matters

DCC-GARCH is the parametric baseline you should beat. It plays the same role in multivariate vol forecasting that plain GARCH plays in univariate: simple, well-understood, and surprisingly competitive. In a Diebold-Mariano test comparing your ML covariance forecasts against DCC, a statistically significant QLIKE improvement is the clearest evidence that high-frequency data and nonlinear models add value beyond what a two-parameter correlation dynamic can capture.

15.4 Wishart Autoregressive (WAR) Model

Where we are. DCC models correlations parametrically using daily returns. Can you instead build a direct time-series model for the realized covariance matrix, analogous to HAR for realized variance?

Wishart Autoregressive Model

The WAR model treats the sequence of realized covariance matrices as a matrix-variate time series. In its simplest form:

$$\mathbf{RC}_{t+1} = C + A \mathbf{RC}_t A^\top + E_{t+1}, \quad (15.5)$$

where

- $C \in \mathbb{R}^{p \times p}$: intercept matrix (symmetric, PSD).
- $A \in \mathbb{R}^{p \times p}$: autoregressive coefficient matrix.
- E_{t+1} : innovation matrix (Wishart-distributed).
- The Wishart distribution is the matrix generalization of the chi-squared distribution and is the natural distribution for covariance matrices.

Higher-order and HAR-style variants (daily, weekly, monthly lags) follow naturally:

$$\mathbf{RC}_{t+1} = C + A_d \mathbf{RC}_t A_d^\top + A_w \overline{\mathbf{RC}}_t^{(w)} A_w^\top + A_m \overline{\mathbf{RC}}_t^{(m)} A_m^\top + E_{t+1},$$

where $\overline{\mathbf{RC}}_t^{(w)}$ and $\overline{\mathbf{RC}}_t^{(m)}$ are averages over the past 5 and 22 trading days.

In Plain English

The WAR model is the matrix version of an AR(1) for time series. Instead of saying “tomorrow’s variance equals a constant plus a fraction of today’s variance,” it says “tomorrow’s covariance *matrix* equals an intercept matrix plus a transformation of today’s covariance matrix.” The sandwich form $A \mathbf{RC}_t A^\top$ ensures the output remains symmetric (just as $\mathbf{RC}_t$ is), and the Wishart distribution is the natural error distribution for random matrices that must be positive semi-definite.

Parameter Explosion

Each coefficient matrix A has p^2 free parameters. For $p = 50$ assets, A alone has 2,500 parameters. With daily, weekly, and monthly lags, that is 7,500 parameters plus the intercept. The sample sizes available (typically 1,000–3,000 trading days) are far too small. WAR is practical only for small p (say, $p \leq 5$) unless you impose strong structure (diagonal A , block-diagonal, etc.).

15.5 HAR-DRD and Multivariate HARQ

Where we are. WAR is theoretically clean but does not scale. The next idea: decompose the covariance matrix into pieces that are easier to model separately.

DRD Decomposition (Bollerslev et al., 2018b)

Write the realized covariance matrix as

$$\mathbf{RC}_t = D_t R_t D_t, \quad (15.6)$$

where

- $D_t = \text{diag}(\sqrt{[\mathbf{RC}_t]_{11}}, \dots, \sqrt{[\mathbf{RC}_t]_{pp}})$: diagonal matrix of realized standard deviations.
- $R_t = D_t^{-1} \mathbf{RC}_t D_t^{-1}$: realized correlation matrix (unit diagonal).

Then model D_t and R_t separately:

- **Variances:** fit a univariate HAR (or HARQ, from Chapter 6) to each diagonal element $[\mathbf{RC}_t]_{jj}$.
 - **Correlations:** fit a univariate HAR to each unique off-diagonal element of R_t , using Fisher z -transform to map $[-1, 1]$ to $(-\infty, \infty)$.
- Reassemble: $\widehat{\mathbf{RC}}_{t+1} = \hat{D}_{t+1} \hat{R}_{t+1} \hat{D}_{t+1}$.

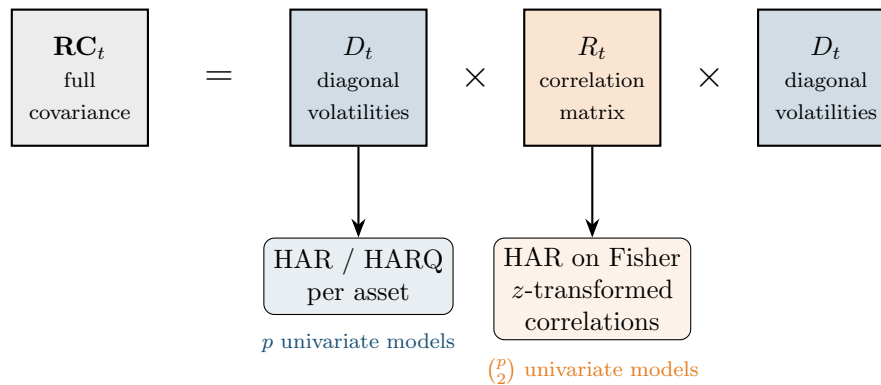


Figure 15.4: The DRD decomposition separates the covariance matrix into volatilities (modeled individually with HAR/HARQ) and correlations (modeled with HAR on Fisher z -transforms). Each piece is forecast with simple univariate models, then reassembled.

Separate Modeling Beats Joint

Bollerslev et al. (2018b) show that the HARQ-DRD model, where variances and correlations are modeled with separate HAR regressions and measurement-error attenuation, significantly outperforms the direct vech-HAR on all elements jointly. The HARQ variant (adding $\sqrt{\mathbf{RQ}}$ interactions from Chapter 6) further improves the variance forecasts, especially on high-noise days.

Worked Example:

HAR-DRD for 2 Assets Suppose you have daily realized covariance matrices for stocks A and B.

Step 1: extract components.

$$\mathbf{RC}_t = \begin{pmatrix} 0.00040 & 0.00012 \\ 0.00012 & 0.00025 \end{pmatrix}.$$

Then $D_t = \text{diag}(0.0200, 0.0158)$ and

$$R_t = D_t^{-1} \mathbf{RC}_t D_t^{-1} = \begin{pmatrix} 1.000 & 0.379 \\ 0.379 & 1.000 \end{pmatrix}.$$

Step 2: model correlation. Apply Fisher z -transform: $z_t = \frac{1}{2} \ln\left(\frac{1+0.379}{1-0.379}\right) \approx 0.399$. Fit HAR to the $\{z_t\}$ series. Invert the transform on the forecast to get $\hat{\rho}_{t+1}$.

PSD Not Guaranteed

The separately forecasted $\hat{R}_{t+1}$ is not guaranteed to be a valid correlation matrix (it may have eigenvalues outside $[0, 1]$ or a diagonal different from 1). In practice, you project onto the nearest correlation matrix, e.g., via the alternating projections algorithm of Higham

(2002).

15.6 Cholesky-HAR

Where we are. HAR-DRD decomposes the matrix into variances and correlations. Cholesky-HAR takes a different decomposition that guarantees PSD by construction.

Cholesky-HAR (Chiriac and Voev, 2011)

Every PSD matrix $\mathbf{RC}_t$ has a unique Cholesky decomposition:

$$\mathbf{RC}_t = L_t L_t^\top,$$

where L_t is lower triangular with positive diagonal entries. The idea:

1. Compute L_t for each day.
2. Take the log of diagonal elements (to ensure positivity upon exponentiation).
3. Stack all $p(p+1)/2$ unique elements of the modified L_t into a vector.
4. Fit a separate HAR model to each element.
5. Forecast, exponentiate diagonals, unstack into $\hat{L}_{t+1}$, and reassemble: $\widehat{\mathbf{RC}}_{t+1} = \hat{L}_{t+1} \hat{L}_{t+1}^\top$.

Since any lower triangular matrix with positive diagonal gives a valid PSD matrix via $LL^\top$, the forecast is PSD by construction.

Worked Example:

Cholesky-HAR for 2 Assets Using the same $\mathbf{RC}_t$ from the DRD example:

$$\mathbf{RC}_t = \begin{pmatrix} 0.00040 & 0.00012 \\ 0.00012 & 0.00025 \end{pmatrix}.$$

Step 1: Cholesky decomposition.

$$L_t = \begin{pmatrix} 0.02000 & 0 \\ 0.00600 & 0.01463 \end{pmatrix}.$$

Check: $l_{11} = \sqrt{0.00040} = 0.02$, $l_{21} = 0.00012/0.02 = 0.006$, $l_{22} = \sqrt{0.00025 - 0.006^2} = \sqrt{0.000214} \approx 0.01463$.

Step 2: transform. Log-transform diagonal: $(\ln 0.02, \ln 0.01463) \approx (-3.912, -4.225)$. Off-diagonal left as-is: 0.006.

Step 3: fit HAR to each of the 3 elements.

Step 4: forecast and reassemble. Suppose the HAR forecasts, after exponentiating the diagonals, yield $\hat{l}_{11} \approx 0.02020$, $\hat{l}_{21} = 0.00580$, $\hat{l}_{22} \approx 0.01483$. Then

$$\widehat{\mathbf{RC}}_{t+1} = \begin{pmatrix} 0.02020 & 0 \\ 0.00580 & 0.01483 \end{pmatrix} \begin{pmatrix} 0.02020 & 0.00580 \\ 0 & 0.01483 \end{pmatrix} = \begin{pmatrix} 0.000408 & 0.000117 \\ 0.000117 & 0.000254 \end{pmatrix}.$$

This is PSD by construction (it is $LL^\top$).

15.7 Graph-Based Methods

Where we are. All methods so far treat each element (or factor) of the covariance matrix independently. Graph-based methods exploit the fact that assets are *connected*: if asset A's

volatility spills over to asset B (Chapter 16 formalizes this), modeling them jointly through a graph should help.

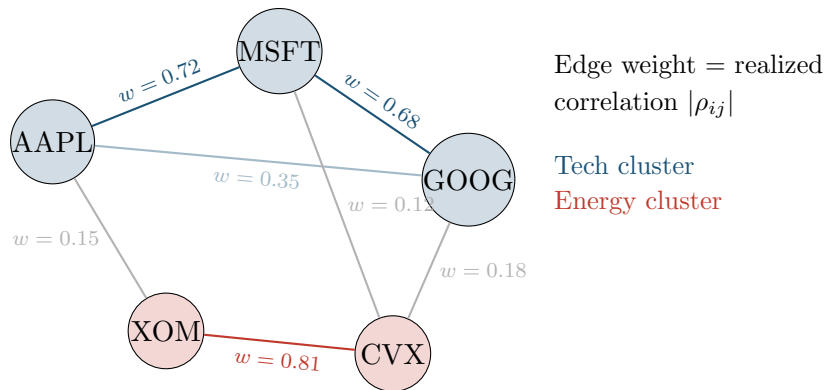


Figure 15.5: Assets as nodes in a graph with edge weights derived from realized correlations. Within-sector edges (tech, energy) are strong; cross-sector edges are weak. Graph-based models exploit this structure.

Graph-HAR (Zhang et al., 2024)

Graph-HAR augments the standard HAR model with graph-based features. The adjacency matrix W is constructed from realized correlations (or partial correlations). For each asset j :

$$RV_{j,t+1} = \beta_0 + \beta_d RV_{j,t} + \beta_w RV_{j,t}^{(w)} + \beta_m RV_{j,t}^{(m)} + \gamma \sum_{k \neq j} W_{jk} RV_{k,t} + \varepsilon_{j,t+1}, \quad (15.7)$$

where

- The first three terms are the standard HAR (Chapter 6).
- $\sum_{k \neq j} W_{jk} RV_{k,t}$: a graph-weighted average of neighbors' volatilities. This is exactly one step of graph diffusion.
- γ : spillover coefficient. If $\gamma > 0$, high volatility in neighbors predicts higher volatility for asset j .
- W : adjacency matrix, typically from thresholded correlation or LASSO partial correlation.

GNN for Covariance (Zhang et al., 2023)

Going further, Graph Neural Networks (GNNs) replace the linear spillover term with learnable message-passing layers:

1. **Node features**: each asset's HAR-style inputs (daily, weekly, monthly RV, plus optional firm characteristics).
2. **Graph structure**: adjacency from realized correlation, GICS sector membership, or learned adaptively.
3. **Message passing**: each node aggregates features from its neighbors through learned weight matrices (as in Chapter 13).
4. **Output**: node-level volatility forecasts or, with a symmetric readout layer, full covariance matrix forecasts.

The advantage over Graph-HAR: nonlinear aggregation and multi-hop information flow (asset A learns from asset C through intermediary B).

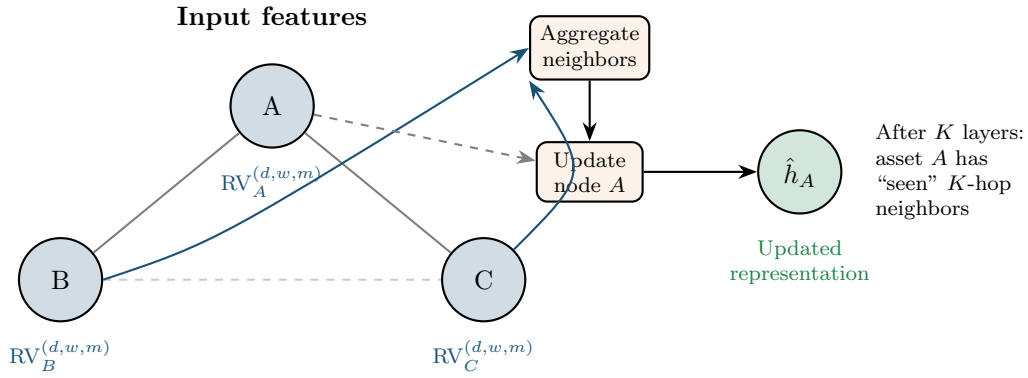


Figure 15.6: GNN message passing for one node. Asset A aggregates volatility features from its graph neighbors (B and C), then updates its own representation. After K message-passing layers, each node has information from all assets within K hops, capturing multi-step spillover effects.

15.8 CNN-RCOV

Where we are. Graph methods treat assets as nodes. An alternative: treat the realized covariance matrix itself as an image.

Covariance Matrices as Images

A $p \times p$ realized covariance matrix is a symmetric, real-valued “image.” If you sort assets by sector (tech, energy, financials, ...), block structure emerges: within-sector blocks have high covariance, cross-sector blocks have low covariance. Convolutional neural networks (CNNs, Chapter 13) are designed to detect such local spatial patterns.

Architecture. Stack T past covariance matrices as “channels” (like RGB channels in image classification) and use 2D convolutions to extract temporal and cross-sectional patterns:

1. Input: $\mathbf{RC}_t, \mathbf{RC}_{t-1}, \dots, \mathbf{RC}_{t-T+1} \in \mathbb{R}^{T \times p \times p}$.
2. 2D convolutional layers detect local block patterns.
3. Fully connected layers map to the $p(p+1)/2$ unique elements of $\widehat{\mathbf{RC}}_{t+1}$.

CNN-RCOV Caveats

- The literature is thin; this is more “conceptually appealing” than empirically proven at scale.
- Ordering assets by sector is a design choice. Different orderings yield different spatial patterns, and CNNs are not permutation-invariant. GNNs (Section 15.7) handle this more naturally.
- The output is not guaranteed PSD; post-hoc projection is required.
- For large p , the output layer has $O(p^2)$ parameters, which grows fast.

15.9 Geometric Deep Learning on the SPD Manifold

Where we are. Every method so far either ignores the PSD constraint or enforces it with post-hoc projection. SPDNet takes a fundamentally different approach: work directly on the manifold where covariance matrices live.

What is a Manifold?

A manifold is a curved space that locally looks like Euclidean space, much like the surface of a sphere locally looks flat. The set of $p \times p$ symmetric positive definite matrices (denoted $\mathcal{S}_{++}^p$) forms a smooth manifold. It is *not* a flat (Euclidean) space: the straight line between two PSD matrices can pass through matrices with zero or negative eigenvalues, leaving the PSD cone.

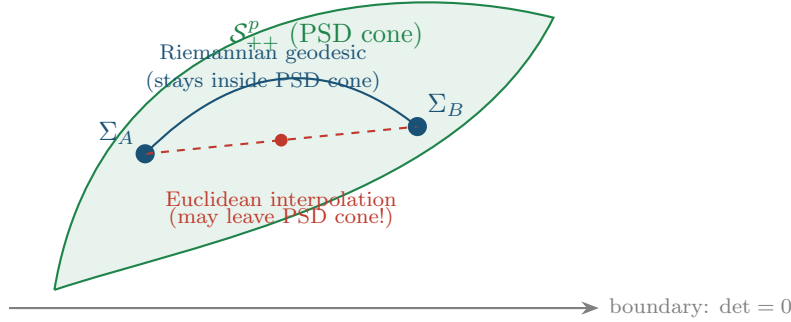


Figure 15.7: The SPD manifold. A straight (Euclidean) line between two PSD matrices Σ_A and Σ_B can pass through singular matrices (red dashed). The Riemannian geodesic (blue curve) stays inside the PSD cone by definition.

Affine-Invariant Riemannian Metric on $\mathcal{S}_{++}^p$

The standard Riemannian metric on $\mathcal{S}_{++}^p$ is

$$d(\Sigma_A, \Sigma_B) = \|\log(\Sigma_A^{-1/2} \Sigma_B \Sigma_A^{-1/2})\|_F,$$

where $\log$ is the matrix logarithm and $\|\cdot\|_F$ is the Frobenius norm. This distance is affine-invariant: $d(A\Sigma_A A^\top, A\Sigma_B A^\top) = d(\Sigma_A, \Sigma_B)$ for any invertible A .

SPDNet

SPDNet replaces standard neural network operations with Riemannian analogs that preserve positive definiteness:

1. **BiMap layer** (analog of linear layer): $\Sigma \mapsto W \Sigma W^\top$, where W is a learnable $d \times p$ matrix with $d \leq p$. If $\Sigma \succ 0$ and W has full row rank, then $W \Sigma W^\top \succ 0$. This also reduces dimension from p to d .
2. **ReEig layer** (analog of ReLU): eigendecompose $\Sigma = U \Lambda U^\top$, then $\Sigma \mapsto U \max(\Lambda, \epsilon I) U^\top$. Clips small eigenvalues to $\epsilon > 0$, keeping the matrix strictly PSD.
3. **LogEig layer** (analog of final feature extraction): $\Sigma \mapsto U \log(\Lambda) U^\top$. Maps from the SPD manifold to the tangent space (symmetric matrices), where standard Euclidean operations apply.

The output of LogEig lives in a flat space, so you can apply a standard linear classifier or regressor on top.

SPDNet: PSD by Design

Because every layer in SPDNet maps PSD inputs to PSD outputs, intermediate representations are always valid covariance matrices. No post-hoc projection is needed. This is the

only deep learning architecture for covariance forecasting with a built-in PSD guarantee.

15.10 The PSD Constraint

Where we are. Every multivariate volatility model must produce a valid (positive semi-definite) covariance matrix. Some methods guarantee this; others require fixing after the fact. This section provides a systematic comparison.

Why PSD Matters

A covariance matrix that is not PSD implies negative variance for some portfolio. Specifically, if Σ has a negative eigenvalue with eigenvector $\mathbf{v}$, then the portfolio $\mathbf{v}$ has predicted variance $\mathbf{v}^\top \Sigma \mathbf{v} < 0$. This is nonsensical and causes optimizers to blow up (infinite leverage on the “negative variance” direction).

Table 15.1: PSD guarantees across multivariate volatility methods.

Method	PSD Guarantee	Mechanism
DCC-GARCH (Sec. 15.3)	Yes	By construction ($D_t R_t D_t$, R_t from rescaling)
WAR (Sec. 15.4)	No	Post-hoc projection needed
HAR-DRD (Sec. 15.5)	No	Post-hoc projection on $\hat{R}_{t+1}$
Cholesky-HAR (Sec. 15.6)	Yes	$\hat{L}\hat{L}^\top$ is PSD by construction
Graph-HAR (Sec. 15.7)	No	Post-hoc projection needed
CNN-RCOV (Sec. 15.8)	No	Post-hoc projection needed
SPDNet (Sec. 15.9)	Yes	Riemannian operations preserve PSD

Post-Hoc PSD Projection

When a method does not guarantee PSD, the standard fix is eigenvalue clipping:

1. Eigendecompose: $\hat{\Sigma} = U\Lambda U^\top$.
2. Set $\tilde{\Lambda} = \max(\Lambda, 0)$ (zero out negative eigenvalues).
3. Reconstruct: $\tilde{\Sigma} = U\tilde{\Lambda}U^\top$.

This gives the nearest PSD matrix in Frobenius norm. The cost: it introduces bias, and the projected matrix no longer minimizes whatever loss function the model was trained on. For correlation matrices, the Higham (2002) alternating projection algorithm additionally enforces unit diagonal.

PSD Violations in Practice

For small p (2–5 assets), PSD violations are rare and small. For large p (50–500), they are common and can be severe, especially with element-wise models (HAR-DRD, Graph-HAR) that do not enforce cross-element consistency. If PSD is critical for your application (portfolio optimization, risk management), prefer methods with built-in guarantees: DCC, Cholesky-HAR, or SPDNet.

15.11 Summary

Key takeaways from this chapter:

- Realized covariance (RC) extends realized variance to matrices via outer products of intraday return vectors ((15.1)).

- Non-synchronous trading biases RC toward zero (Epps effect). Refresh-time sampling and the Hayashi–Yoshida estimator ((15.2)) address this, with HY being more data-efficient.
- The multivariate realized kernel ((15.3)) handles both noise and non-synchronicity while guaranteeing PSD (Barndorff-Nielsen et al., 2011).
- DCC-GARCH (Engle, 2002) is the standard parametric baseline: easy to estimate, scales to large p , but uses only daily data and imposes a single correlation dynamic.
- WAR models RC directly as a matrix time series, but suffers from parameter explosion for $p > 5$.
- HAR-DRD (Bollerslev et al., 2018b) decomposes RC into variances and correlations, models each with HAR, and outperforms both joint modeling and DCC.
- Cholesky-HAR (Chiriac and Voev, 2011) models Cholesky factors with HAR, guaranteeing PSD by construction.
- Graph-HAR (Zhang et al., 2024) and GNN methods (Zhang et al., 2023) model volatility spillovers through asset graphs, capturing network effects that element-wise models miss.
- CNN-RCOV treats RC matrices as images; conceptually appealing but empirically limited and not permutation-invariant.
- SPDNet operates on the Riemannian manifold of PSD matrices, the only deep learning approach with a built-in PSD guarantee.
- Methods either guarantee PSD (DCC, Cholesky-HAR, SPDNet) or require post-hoc eigenvalue projection, which introduces bias.
- For most applications, start with HAR-DRD (best accuracy in large-scale comparisons) or Cholesky-HAR (if PSD is essential). Use DCC-GARCH as the parametric baseline. Graph and manifold methods are promising but still maturing.

Table 15.2: Key results: multivariate volatility methods.

Method	Data Input	Key Advantage	Key Limitation
DCC-GARCH	Daily returns	Scales to large p ; PSD guaranteed	Single correlation dynamic; ignores HF data
WAR	Realized covariance	Theoretically clean matrix AR	$O(p^2)$ parameters
HAR-DRD	Realized covariance	Best accuracy (BPQ 2018); flexible	PSD not guaranteed
Cholesky-HAR	Realized covariance	PSD by construction	Cholesky ordering arbitrary
Graph-HAR GNN	/ RV + graph	Models spillovers	Graph construction is a design choice
CNN-RCOV	RC as image	Detects block structure	Not permutation-invariant
SPDNet	RC on manifold	PSD by design; principled geometry	Complex; limited empirical evidence

Next. Chapter 16 formalizes the volatility spillovers that Graph-HAR exploits, using the Diebold–Yilmaz connectedness framework and network topology.

Chapter 16

Volatility Spillovers and Connectedness

Application

Chapter 15 built tools for forecasting the full covariance matrix. This chapter focuses on a specific question: how does volatility transmit from one asset (or market) to another? Spillover indices and connectedness measures are both diagnostic tools (understanding contagion) and predictive features (Chapter 10). Project 3 relies heavily on this framework.

16.1 Diebold–Yilmaz Spillover Indices

In Chapter 15 you learned to model the joint dynamics of multiple assets. Now we zoom in on a sharper question: when one asset’s volatility jumps, how much of that shock spills over into other assets?

Volatility as a contagious shock

Think of volatility shocks as ripples in a pond. A stone dropped on the equity side sends waves that reach bonds, commodities, and currencies. The Diebold–Yilmaz (DY) framework measures the size and direction of those ripples using standard vector autoregression (VAR) machinery.

16.1.1 Setup: VAR on Realized Volatilities

Stack the realized volatilities of N assets into a vector $\mathbf{y}_t = (\text{RV}_{1,t}, \dots, \text{RV}_{N,t})'$ and fit a $\text{VAR}(p)$:

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^p \mathbf{A}_\ell \mathbf{y}_{t-\ell} + \mathbf{u}_t, \quad \mathbf{u}_t \sim (0, \mathbf{\Sigma}), \quad (16.1)$$

where:

- $\mathbf{y}_t \in \mathbb{R}^N$: vector of realized volatilities on day t ,
- $\mathbf{c} \in \mathbb{R}^N$: intercept vector,
- $\mathbf{A}_\ell \in \mathbb{R}^{N \times N}$: coefficient matrix at lag ℓ ,
- $\mathbf{u}_t$: reduced-form innovation vector with covariance $\mathbf{\Sigma}$.

The key output is the *moving-average representation* obtained by inverting the VAR:

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \boldsymbol{\Phi}_h \mathbf{u}_{t-h}, \quad (16.2)$$

In Plain English

The matrix $\boldsymbol{\Phi}_h$ is the “memory kernel”: entry (j, k) tells you how much a surprise in asset k ’s vol h days ago still echoes in asset j ’s vol today. It is the multivariate impulse-response function.

where the $N \times N$ matrices $\boldsymbol{\Phi}_h$ are the impulse-response coefficients at horizon h . Entry $(\boldsymbol{\Phi}_h)_{jk}$ tells you how a unit shock to asset k today affects asset j at horizon h .

16.1.2 Generalized Forecast Error Variance Decomposition

The next step decomposes the H -step forecast error variance of each asset into contributions from every shock. The generalized variant (GFEVD), introduced by Pesaran and Shin (1998) and adopted by Diebold and Yilmaz (2012), does not depend on variable ordering (unlike Cholesky decompositions).

Generalized Forecast Error Variance Decomposition

The fraction of asset j ’s H -step forecast-error variance attributable to shocks originating in asset k is:

$$\theta_{jk}^{(H)} = \frac{\sigma_{kk}^{-1} \sum_{h=0}^{H-1} (\mathbf{e}_j' \boldsymbol{\Phi}_h \boldsymbol{\Sigma} \mathbf{e}_k)^2}{\sum_{h=0}^{H-1} \mathbf{e}_j' \boldsymbol{\Phi}_h \boldsymbol{\Sigma} \boldsymbol{\Phi}_h' \mathbf{e}_j}, \quad (16.3)$$

where:

- σ_{kk} : the (k, k) entry of $\boldsymbol{\Sigma}$ (variance of shock k),
- $\mathbf{e}_j$: the j -th column of the $N \times N$ identity matrix,
- $\boldsymbol{\Phi}_h$: the impulse-response matrix at horizon h from Eq. (16.2),
- H : forecast horizon (typically 10 days).

In Plain English

This formula answers a simple question: of all the uncertainty in asset j ’s volatility forecast H days ahead, what fraction was caused by a shock to asset k ? The numerator isolates the contribution of shock k (scaled by its own variance), and the denominator is the total forecast uncertainty for asset j . When $j = k$, you get the “own” contribution; when $j \neq k$, you get the spillover from k to j .

Because GFEVD rows do not sum to one in general, normalize each row:

$$\tilde{\theta}_{jk}^{(H)} = \frac{\theta_{jk}^{(H)}}{\sum_{k=1}^N \theta_{jk}^{(H)}}, \quad \text{so that } \sum_{k=1}^N \tilde{\theta}_{jk}^{(H)} = 1. \quad (16.4)$$

In Plain English

The raw GFEVD fractions for a given asset do not necessarily add up to 100% because the generalized approach allows shocks to be correlated. After normalization, you can read each entry as a percentage: “X% of asset j ’s vol forecast error is attributable to shocks from asset k .”

16.1.3 Spillover Measures

From the normalized decomposition table $\tilde{\Theta}^{(H)}$, three spillover measures follow immediately.

Diebold–Yilmaz Spillover Decomposition

Total spillover index. The fraction of total forecast-error variance due to cross-asset shocks:

$$S^{(H)} = \frac{1}{N} \sum_{\substack{j,k=1 \\ j \neq k}}^N \tilde{\theta}_{jk}^{(H)} \times 100. \quad (16.5)$$

Directional FROM. How much volatility asset j receives from all others:

$$S_{j \leftarrow \bullet}^{(H)} = \frac{1}{N} \sum_{\substack{k=1 \\ k \neq j}}^N \tilde{\theta}_{jk}^{(H)} \times 100. \quad (16.6)$$

Directional TO. How much volatility asset j transmits to all others:

$$S_{j \rightarrow \bullet}^{(H)} = \frac{1}{N} \sum_{\substack{k=1 \\ k \neq j}}^N \tilde{\theta}_{kj}^{(H)} \times 100. \quad (16.7)$$

Net spillover. Transmitters minus receivers:

$$S_j^{\text{net},(H)} = S_{j \rightarrow \bullet}^{(H)} - S_{j \leftarrow \bullet}^{(H)}. \quad (16.8)$$

A positive net spillover means asset j is a *net transmitter* of volatility; negative means *net receiver*.

Why This Matters

Their value is concentrated in regime transitions, precisely the forecasts where single-asset features alone break down.

The framework evolved across three papers: Diebold and Yilmaz (2009) introduced the total index using a Cholesky decomposition, Diebold and Yilmaz (2012) added directional measures and switched to GFEVD, and Diebold and Yilmaz (2014) refined the generalized VAR approach and applied it to a broader set of markets.

16.1.4 Spillover Network Diagram

The variance decomposition table maps directly onto a weighted directed graph: each asset is a node, and the edge from k to j carries weight $\tilde{\theta}_{jk}^{(H)}$.

16.1.5 Worked Example: 3-Asset VAR(1)

Worked Example: Computing Spillover Indices

Consider three assets (Equity, Bonds, FX) with the following normalized 10-step GFEVD table (each row sums to 100%):

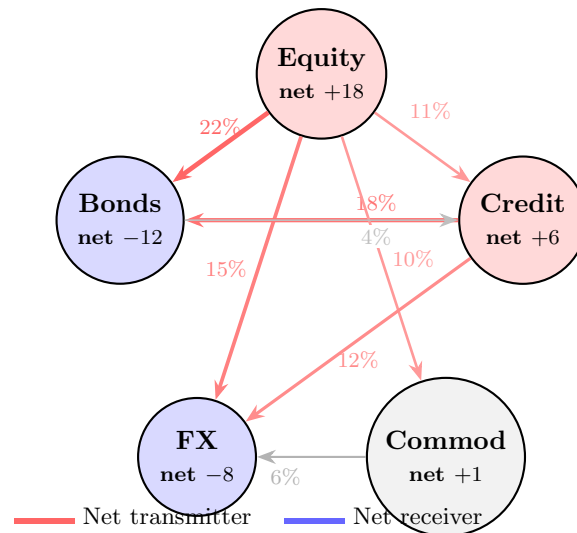


Figure 16.1: Spillover network for five asset classes. Edge labels show the percentage of j 's forecast-error variance explained by shocks from k . Node color indicates net transmitter (red) vs. net receiver (blue). Data are illustrative.

	Equity	Bonds	FX	FROM others
Equity	60	25	15	40
Bonds	30	55	15	45
FX	20	10	70	30
TO others	50	35	30	

Step 1: Directional FROM. Sum the off-diagonal entries in each row, divided by $N = 3$:

$$S_{\text{Equity} \leftarrow \bullet} = \frac{25 + 15}{3} = 13.3\%,$$

$$S_{\text{Bonds} \leftarrow \bullet} = \frac{30 + 15}{3} = 15.0\%,$$

$$S_{\text{FX} \leftarrow \bullet} = \frac{20 + 10}{3} = 10.0\%.$$

Step 2: Directional TO. Sum the off-diagonal entries in each column, divided by N :

$$S_{\text{Equity} \rightarrow \bullet} = \frac{30 + 20}{3} = 16.7\%,$$

$$S_{\text{Bonds} \rightarrow \bullet} = \frac{25 + 10}{3} = 11.7\%,$$

$$S_{\text{FX} \rightarrow \bullet} = \frac{15 + 15}{3} = 10.0\%.$$

Step 3: Net spillover.

$$S_{\text{Equity}}^{\text{net}} = 16.7 - 13.3 = +3.3\% \quad (\text{net transmitter}),$$

$$S_{\text{Bonds}}^{\text{net}} = 11.7 - 15.0 = -3.3\% \quad (\text{net receiver}),$$

$$S_{\text{FX}}^{\text{net}} = 10.0 - 10.0 = 0.0\% \quad (\text{neutral}).$$

Step 4: Total spillover index. Sum of all off-diagonal entries, divided by N (since

each row already sums to 100):

$$S^{(10)} = \frac{25 + 15 + 30 + 15 + 20 + 10}{3} = \frac{115}{3} \approx 38.3\%.$$

Interpretation: about 38% of the total forecast-error variance in this system comes from cross-asset shocks rather than own shocks. Equity is the dominant transmitter.

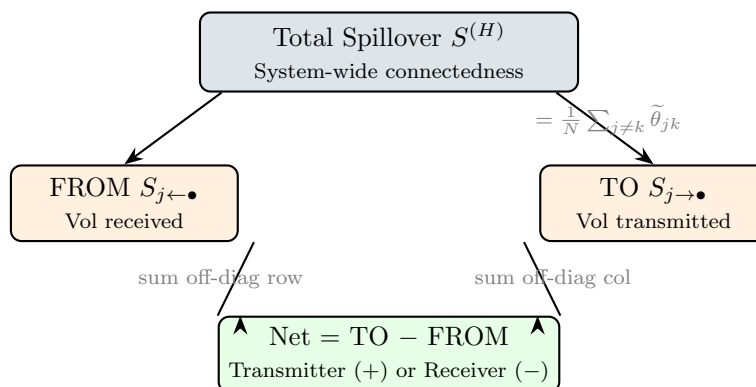


Figure 16.2: Decomposition of the Diebold-Yilmaz spillover measures. The total index summarizes system-wide connectedness. Directional FROM and TO break it down per asset, and the net spillover identifies transmitters vs. receivers.

16.2 Time-Varying Spillovers

The static spillover table in the previous section gives you a single snapshot. In practice, connectedness changes dramatically over time: calm markets have low spillovers, crises have high ones. This section covers two methods to capture the dynamics.

16.2.1 Rolling-Window Approach

The simplest method: re-estimate the VAR and GFEVD on a rolling window of w days (typically $w = 200$) and plot $S_t^{(H)}$ over time.

Rolling-Window Spillover

For each day t , estimate the VAR on $\{t - w + 1, \dots, t\}$, compute the GFEVD, and record $S_t^{(H)}$. The resulting time series reveals spillover regimes:

- Calm periods: $S_t^{(H)} \approx 30\text{--}40\%$,
- Crises (2008 GFC, 2020 COVID): $S_t^{(H)}$ spikes to 70–85%.

Window-length sensitivity

The rolling-window approach introduces a hyperparameter w that affects both level and smoothness. Too short ($w < 100$): noisy, unstable VAR estimates. Too long ($w > 300$): sluggish, crises get averaged out. Always report results for at least two window lengths to check robustness (Diebold and Yilmaz, 2012).

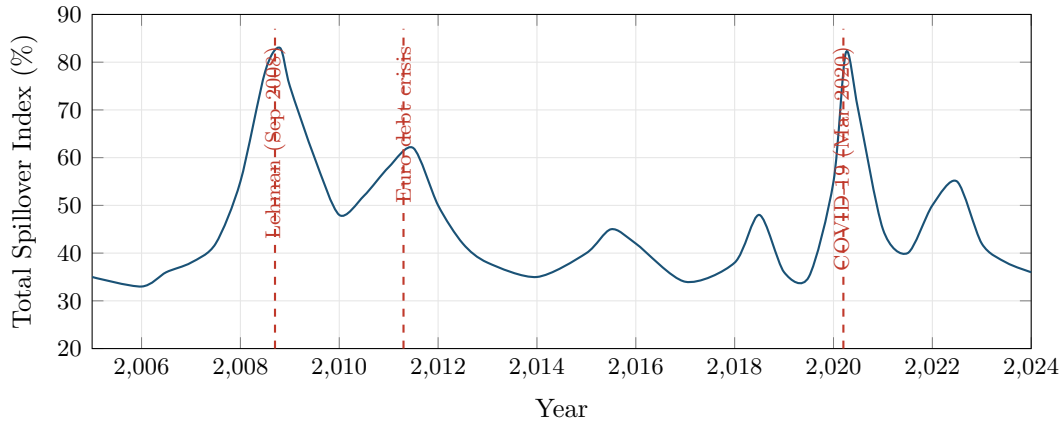


Figure 16.3: Stylized total spillover index (200-day rolling window, $H = 10$) across five major asset classes. The index spikes sharply during the 2008 financial crisis and the March 2020 COVID selloff, reflecting contagion across markets.

16.2.2 TVP-VAR Connectedness

Antonakakis et al. (2020) replace the rolling-window VAR with a *time-varying parameter VAR* (TVP-VAR) estimated via the Kalman filter. This eliminates the window-length hyperparameter entirely.

The TVP-VAR model allows coefficients to drift:

$$\mathbf{y}_t = \mathbf{c}_t + \sum_{\ell=1}^p \mathbf{A}_{\ell,t} \mathbf{y}_{t-\ell} + \mathbf{u}_t, \quad \text{vec}(\mathbf{A}_t) = \text{vec}(\mathbf{A}_{t-1}) + \boldsymbol{\eta}_t, \quad (16.9)$$

In Plain English

The Kalman filter estimates the evolving coefficients without needing a rolling window, so the spillover index updates smoothly each day rather than jumping when observations enter or leave a fixed window.

where:

- $\mathbf{A}_{\ell,t}$: time-varying coefficient matrix at lag ℓ and time t ,
- $\boldsymbol{\eta}_t \sim (0, \mathbf{Q})$: state innovation with covariance $\mathbf{Q}$ governing the speed of parameter drift.

At each t , the Kalman filter produces updated coefficient estimates, and you compute the GFEVD exactly as before to get $S_t^{(H)}$.

TVP-VAR vs. Rolling Window

Antonakakis et al. (2020) show that TVP-VAR connectedness is smoother than rolling-window estimates, avoids the abrupt “entry/exit” artifacts when extreme observations enter or leave the window, and responds faster to genuine structural breaks. Both methods agree on the broad pattern: spillovers spike during crises and monetary-policy surprises, but the TVP-VAR resolves timing more sharply.

16.3 Network Visualization and Interpretation

The GFEVD table is a matrix of numbers. Networks make that matrix readable at a glance. This section covers the visual conventions and the patterns you should look for.

16.3.1 Visual Encoding

Four encoding rules produce an interpretable graph:

1. **Node size** proportional to total connectedness ($S_{j \rightarrow \bullet} + S_{j \leftarrow \bullet}$). Large nodes are “systemically important” regardless of sign.
2. **Edge thickness** proportional to pairwise spillover $\tilde{\theta}_{jk}^{(H)}$. Only draw edges above a threshold (e.g., 5%) to avoid clutter.
3. **Edge direction** from shock source k to shock recipient j (arrow points toward the asset whose variance is explained).
4. **Node color** by net spillover sign: red for net transmitters, blue for net receivers. Alternatively, color by sector or asset class.

16.3.2 Calm vs. Crisis Networks

The most striking pattern in spillover networks is the structural shift between tranquil and stressed markets ([Demirer et al., 2018](#)).

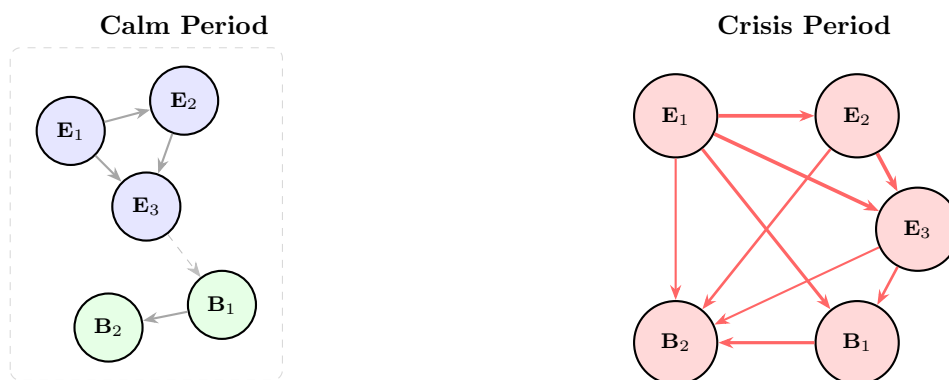


Figure 16.4: Spillover networks during calm (left) and crisis (right) periods. In calm markets, connectedness is largely within-sector (equity-to-equity, bond-to-bond), with sparse cross-sector links. During crises, cross-sector edges thicken and multiply: everything becomes connected to everything. After [Demirer et al. \(2018\)](#).

Calm-Crisis Network Transition

Three stylized facts from the DY network literature:

1. **Calm periods:** within-sector connectedness dominates. Equity shocks stay in equities; bond shocks stay in bonds. Total spillover index is low (30–40%).
2. **Crises:** cross-sector connectedness surges. The network topology shifts from clustered to nearly fully connected. Total spillover jumps to 70–85%.
3. **Net transmitter identity shifts:** in calm markets, commodity shocks are often isolated; during crises, equity and credit become dominant transmitters ([Diebold and Yilmaz, 2014](#)).

16.4 Cross-Asset Universality

The high cross-sector connectedness during crises suggests volatility dynamics may be *similar* across asset classes, rather than merely correlated. Two recent papers make this case precisely.

[Sirignano and Cont \(2019\)](#) train a single deep network (LSTM layers followed by a fully-connected layer) to predict the direction of the next price move by pooling high-frequency

data across hundreds of US equities. The pooled (“universal”) model outperforms individual asset-specific models, implying that the features driving price formation are largely shared.

Universal Price Formation Features

Sirignano and Cont (2019) show that a model trained on pooled equity data generalizes to out-of-sample stocks not seen during training. This “universality” holds for features based on order-flow history (queue imbalances, trade arrivals). The implication: spillover networks may reflect common factor exposures rather than direct causal transmission.

Rosenbaum and Zhang (2022) push universality further by training a universal LSTM on pooled data from hundreds of liquid stocks to forecast daily realized volatility. Their key finding: the LSTM’s predictions are matched by a parsimonious rough volatility model with $H \approx 0.1$, consistent with the rough volatility theory from Chapter 7.

Why universality matters for spillovers

If volatility dynamics really are “universal” (same roughness, same feature importance across assets), then the strong cross-asset connectedness you see during crises may not require a contagion story. Instead, a single common factor (e.g., dealer risk capacity, funding liquidity) could drive all assets simultaneously. This does not diminish the usefulness of spillover indices as diagnostic tools, but it changes how you interpret them: high spillover may reflect common-factor exposure rather than sequential transmission from one asset to the next.

16.4.1 Pooled Panel Forecasting Across Instruments

Universality has a direct architectural implication: if vol dynamics are truly shared across assets, you can train a single model on pooled data rather than fitting N separate models, which raises a concrete question: how do you feed RV histories for all 34 instruments in your project into a single HAR regression or a single LightGBM, without (a) drowning the few liquid index futures under the 30 mega-cap equities, (b) letting the model confuse a structurally high-vol name with a structurally low-vol one, or (c) leaking information across instruments on a crisis day? This subsection answers those questions for the two model classes you actually use as baselines: pooled linear/HAR and pooled trees.

Panel Data

Until now this guide has treated each instrument as its own time series: one column of RV_t , fit a HAR, repeat 34 times. A **panel** (or **longitudinal**) data set instead tracks multiple **entities** $i = 1, \dots, N$ over multiple time periods $t = 1, \dots, T$, with each observation indexed by the pair (i, t) . Here the entities are the $N = 34$ instruments (the 30 mega-cap equities plus 4 sector/index ETFs and the E-mini S&P 500 future from Chapter 10), the time index t runs over trading days, and the target y_{it} is next-day (or h -day) realized volatility for instrument i . When every instrument is observed on every date the panel is **balanced**; when some (i, t) cells are missing it is **unbalanced** (treated at the end of this subsection).

Fitting 34 separate HARs throws away an obvious resource: the instruments share dynamics. Chapter 7 and the universality results above (Sirignano and Cont, 2019; Rosenbaum and Zhang, 2022) argue that the *shape* of volatility dynamics (HAR decay structure, jump sensitivity, the leverage tilt) is broadly common across liquid assets. If that is even approximately true, then stacking the 34 instruments into one (i, t) panel lets a single set of coefficients borrow strength from every series at once.

Why stacking grows the sample

A single instrument with five years of daily data gives you $T \approx 1,250$ rows, a thin sample for a flexible model. Stack $N = 34$ instruments and the pooled panel has up to $N \times T \approx 42,500$ rows. You have not invented new information about any one stock, but you *have* given the model 34 independent realizations of the same volatility-formation process to estimate the *shared* parameters from. That is exactly the lever a univariate HAR cannot pull.

A pooled HAR baseline, and why naive pooling fails

The naive move is to stack every instrument's HAR **design matrix** (just the table of feature values, one row per (instrument, day) and one column per feature) and run one OLS (**ordinary least squares**, the standard best-fit-line procedure). Stacking means literally laying each instrument's feature table on top of the next into one tall table. The problem is **instrument-specific heterogeneity**: a high-beta growth name lives at a structurally higher average RV than a defensive utility, and a single common **intercept** (the baseline level the line starts from) cannot represent both. Forcing one intercept makes the pooled fit chase the cross-instrument *level* differences instead of the volatility *dynamics* you actually want to learn. The fix is an **entity fixed effect**: one intercept per instrument. Because each instrument carries many features rather than one, the single slope m of a straight line $y = mx + b$ becomes a **slope vector** β with one slope per feature.

The entity fixed-effects model gives each instrument its own baseline volatility level while sharing one common slope vector across all of them:

$$y_{it} = \underbrace{\alpha_i}_{\text{instrument level}} + \mathbf{x}'_{it}\beta + \varepsilon_{it}, \quad (16.10)$$

The notation $\mathbf{x}'_{it}\beta$ (read “ x -transpose- β ”; the raised tick mark $'$ is the *transpose*, which here just turns the column of features into a row so the next step is a sum) is shorthand for multiplying each of the K features by its matching coefficient in β and adding them into a single predicted number, the same $\beta_1 \text{ feature}_1 + \beta_2 \text{ feature}_2 + \dots$ you saw in the HAR regression of Chapter 10.

- y_{it} : the forecast target for instrument i on day t (next-day $\text{RV}_{i,t+1}$, or a $\log/\sqrt{\cdot}$ transform of it as in Chapter 10),
- α_i : the *instrument-specific intercept*, absorbing every time-invariant difference across instruments (a name's average vol level, its sector's baseline turbulence),
- $\mathbf{x}_{it}$: the $K \times 1$ vector (a single column of the K HAR-type features, where K is simply however many predictors you stacked) for instrument i on day t , the same HAR-type features defined in Chapter 10: daily, weekly, and monthly lagged RV, the jump and semivariance splits, VRP, and so on,
- β : the $K \times 1$ slope vector, *common across all instruments* (one HAR decay structure for the whole panel),
- ε_{it} : the idiosyncratic error (the leftover the model cannot explain), assumed to average to zero, $\mathbb{E}[\varepsilon_{it} \mid \alpha_i, \mathbf{x}_{it}] = 0$. That statement reads: averaging ($\mathbb{E}[\cdot]$) over everything, given (|) the instrument and its feature values, the leftover error is zero no matter which instrument we look at or what its features are; i.e. the model is unbiased.

In Plain English

Equation (16.10) says: “Let every instrument sit at its own resting volatility level α_i , but make them all obey the *same* rule for how yesterday's, last week's, and last month's vol map into tomorrow's.”

Why This Matters

This is the pooled-HAR baseline your project should beat before reaching for trees or nets. It is the panel analogue of the single-instrument HAR of Corsi (2009): identical features, but the slope β is now estimated from $34\times$ the data, which sharply tightens the QLIKE (the volatility forecast-accuracy score from Chapter 17; lower is better) on short-history names that could never support a stable HAR on their own. Report its out-of-sample QLIKE against the per-instrument HARs; the pooled fit usually wins on the thin-history instruments and roughly ties on the data-rich index lines.

You do not have to literally add 34 dummy columns. The algebraically identical **within estimator** subtracts each instrument's own time-series mean from both sides, cancelling α_i .

Demeaning each instrument's series removes its fixed effect, leaving a clean regression for β :

$$\underbrace{(y_{it} - \bar{y}_i)}_{\text{demeaned target}} = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)' \beta + (\varepsilon_{it} - \bar{\varepsilon}_i), \quad \bar{y}_i = \frac{1}{T_i} \sum_t y_{it}, \quad (16.11)$$

where:

- $\bar{y}_i$, $\bar{\mathbf{x}}_i$ (read “ y -bar- i ”, “ x -bar- i ”): the time-series *averages* of the target and the features *within instrument i* , simply instrument i 's own average vol and average features over its T_i observed days, where T_i is the number of trading days instrument i is observed,
- the entity effect α_i cancels because it is constant over t for each i , so demeaning subtracts it from itself,
- running OLS (the standard best-fit-line procedure) on the demeaned data yields what panel-data texts call the **within** estimate of β , so named because it uses only the variation *within* each instrument over time (Wooldridge, 2010).

In Plain English

Demeaning re-expresses every variable as a *deviation from that instrument's own average*. Instead of asking “is this stock's vol high?” (a level question contaminated by which stock it is), the regression now asks “is this stock's vol high *relative to its own normal*, given that its lagged vol is high relative to its own normal?” That deviation-based question has the same answer for every instrument, which is exactly why one shared β is legitimate.

Why This Matters

Demeaning is the panel version of the per-instrument z -scoring already in the Chapter 10 pipeline (subtracting each instrument's mean and dividing by its standard deviation so all names sit on a comparable scale): if your features are already z -scored within instrument, the entity fixed effect is largely redundant for the *features*; keep it (or a demeaned target) anyway so the *intercept* does not chase level differences.

Within estimation discards cross-instrument level information

The within estimator uses only *within-instrument* variation over time; it throws away the *between-instrument* variation in average levels entirely. That is deliberate (the level differences are the heterogeneity you wanted to control for), but it has a consequence: you cannot identify the effect of any *instrument-constant* feature (sector label, “is-an-index” flag) inside a fixed-effects regression, because such a feature has zero within variation and is indistinguishable from α_i . The instrument's intercept already captures anything that

never changes for that instrument, so a sector label adds no new information the regression can separate out. If you need those level effects, carry them as separate dummy features in the tree model below rather than the FE regression.

Time fixed effects: absorbing market-wide crisis days

Section 16.2 showed that on crisis days (Lehman, the euro crisis, the March 2020 COVID selloff) *every* instrument's vol spikes at once. In a pooled panel those common-shock days act like 34 correlated duplicate observations of the same event, and they can dominate the fit. A **time fixed effect** δ_t absorbs whatever is common to all instruments on a given day.

Adding a per-day intercept on top of the per-instrument intercept strips out market-wide volatility shocks:

$$y_{it} = \alpha_i + \underbrace{\delta_t}_{\text{market-wide day shock}} + \mathbf{x}'_{it}\boldsymbol{\beta} + \varepsilon_{it}, \quad (16.12)$$

where:

- δ_t : the *time-specific intercept*, common to all N instruments on day t , absorbing aggregate vol spikes, macro releases, and the contagion days quantified by the total-spillover index $S_t^{(H)}$ from Section 16.1 (a single number near 30–40% in calm markets and 70–85% in crises that tracks how much of the market is moving together),
- all other symbols are as in Equation (16.10); with both α_i and δ_t present this is a **two-way fixed-effects** model.

In Plain English

The entity effect α_i asks “which instrument is this?” and removes it. The time effect δ_t asks “what was the whole market doing today?” and removes that too. What survives is the *relative* signal: on a day when market vol is elevated, which names are even *more* elevated than the average, and is that excess predictable from their own lagged-vol features?

Why This Matters

There is a real trade-off here for an RV forecaster. If your deliverable is the *absolute* level of each instrument's vol (the input to a vol-targeting strategy, one that scales position size up when forecast vol is low and down when it is high, so it needs the absolute level), do *not* include time effects: δ_t absorbs the common vol signal that is precisely what such a strategy needs to size positions. Include time effects only when you care about the *cross-sectional* question (which names will be relatively more volatile), e.g. for a dispersion or relative-value trade, which only bets on which names are *more* volatile than others and so needs the cross-sectional ranking rather than the absolute level. A practical compromise that keeps the absolute signal is to drop δ_t but add the Section 16.5 spillover features as columns, letting the model condition on the regime rather than differencing it away.

Encoding instrument identity for a pooled LightGBM

A tree model does not demean. To pool 34 instruments in LightGBM you instead give the tree the instrument identity *as a feature* and let it split on it. With only $N = 34$ instruments, identity is **low-cardinality**: there are only 34 distinct values, a small number of categories. You encode it with **one-hot dummy** columns: one yes/no (1/0) column per instrument that

equals 1 only on that instrument's rows (drop one to avoid redundancy; see the `is_AAPL` example below). These are cheap and LightGBM handles them natively (Ke et al., 2017).

Instrument dummies recover instrument-specific dynamics for free

When the tree splits on `is_AAPL = 1` and then splits on the weekly-RV feature beneath that branch, it has effectively estimated a *different* HAR slope for that instrument, a feature×instrument *interaction* (a rule that lets the slope on a feature differ by instrument); via a first split on identity alone it also reproduces the per-instrument level α_i . The single shared β of Equation (16.10) can recover either of these only through explicit interaction terms. Pooling in a tree therefore gives you the sample-size benefit of the panel *and* per-instrument flexibility, without fitting 34 separate models.

Never target-encode the instrument ID: it leaks the future

The tempting shortcut is **target encoding**: replace the instrument ID with that instrument's mean target (its average RV). In a panel time series this is a temporal-leakage trap. The instrument mean computed over the *whole* sample includes future days, so encoding instrument i on day t with a mean that already “knows” $RV_{i,t+5}$ smuggles the answer into the feature. The damage is silent: cross-validation scores look excellent and live performance collapses. With only 34 low-cardinality instruments there is *no reason* to target-encode: plain one-hot dummies carry the same information with zero leakage. Reserve any encoding-by-statistic for genuinely high-cardinality categoricals, and even then compute the statistic only on data strictly prior to t .

Why This Matters

This leakage is the cross-sectional cousin of the standard look-ahead bias from Chapter 17. The same panel structure that grows your sample also multiplies the ways the future can leak into the past, and the most dangerous one is at the *fold* level: a random K -fold split scatters rows from the same date across train and test, so the model sees other instruments' day- t behaviour while predicting instrument i on day t . Use the time-blocked purged CV of Section 17.7, and follow the cross-sectional leakage rule in Section 17.7.4: entire *dates* go to train or test as a block, never split across folds.

Unbalanced panels and short-history instruments

Your panel will almost certainly be **unbalanced**: a name added to the index two years ago has half the history of the index future, and a recent IPO has a few hundred days at most. Both the within estimator and a tree handle this without special pleading: demeaning in Equation (16.11) uses whatever T_i days exist for instrument i , and a tree simply sees fewer rows for that name. Two cautions apply specifically to RV forecasting.

Short histories distort a pooled fit two ways

First, a short-history instrument whose entire window happens to sit inside a calm (or a crisis) regime contributes a biased view of the shared dynamics; check that your pooled β and QLIKE are not driven by the longest series alone by re-fitting on the balanced sub-panel of full-history instruments. Second, in a *dynamic* pooled panel where the target depends on its own lag (as HAR does), demeaning a very short series induces a small finite-sample bias that shrinks as the history T_i grows (Nickell, 1981). Intuitively, because the lagged target is part of what you averaged away, the demeaned lag and the demeaned

error end up slightly linked, nudging the estimated persistence a touch too low: the Nickell bias. At $T_i \approx 1,000$ daily observations this bias is negligible, but it becomes material if you aggregate to weekly or monthly targets on a name with only a year or two of data.

The practical recipe for the pooled RV panel:

1. Stack the 34 instruments into an (i, t) panel; z -score every feature *within instrument* using a trailing window (Chapter 10).
2. For the linear baseline, use entity fixed effects (Equation (16.10)); add time effects only if the target is cross-sectional rather than absolute.
3. For the tree, add $N - 1$ one-hot instrument dummies; *never* target-encode the ID.
4. Validate with date-blocked purged CV (Section 17.7; Section 17.7.4), so whole dates, not random rows, define the folds.
5. Sanity-check robustness by re-fitting on the balanced full-history sub-panel; report QLIKE for both.

16.5 Spillover Indices as Predictive Features

The previous sections treated spillover indices as descriptive diagnostics. This section flips the lens: can you use them as *inputs* to the forecasting models from Chapters 11–14?

16.5.1 Three Feature Families

Spillover-Based Features for ML

1. **Total spillover as regime indicator.** High $S_t^{(H)}$ signals “crisis mode” where correlations are elevated and volatility persistence is stronger. Use it as a conditioning variable: for example, a LightGBM model can split on $S_t^{(H)} > 60$ to activate crisis-specific trees.
2. **Directional FROM as vulnerability measure.** An asset with a high and rising $S_{j \leftarrow \bullet}$ is absorbing shocks from many sources. This predicts higher future volatility for asset j , especially in the 5–20 day horizon. See Chapter 10 for how to z -score and lag these features.
3. **Net spillover as contrarian signal.** Extreme net receivers ($S_j^{\text{net}} \ll 0$) tend to mean-revert: assets that have absorbed large cross-market shocks often see volatility revert to lower levels once the contagion subsides. This generates a medium-horizon (1–3 month) signal for volatility forecasting.

16.5.2 Practical Considerations

Spillover features are slow-moving

Spillover indices computed from a 200-day rolling window update slowly. For short-horizon forecasts (1–5 days), they may add little predictive power beyond the VIX or a simple HAR model. Their value is highest for medium-to-long horizons (1 week to 3 months) and for regime-conditional models.

A practical recipe for incorporating spillover features:

1. Compute the DY spillover index and directional measures using a 200-day rolling window (or TVP-VAR).

2. Construct three features per asset: total spillover $S_t^{(H)}$, directional FROM $S_{j \leftarrow \bullet}^{(H)}$, and net spillover $S_j^{\text{net},(H)}$.
3. Z-score each feature using a 252-day trailing window.
4. Include as additional columns in the feature matrix from Chapter 10, alongside HAR lags, VRP, and other predictors.
5. Check SHAP importance: if spillover features rank below the top 10 in a LightGBM model, they likely add noise rather than signal for your target horizon.

16.6 Summary

- The Diebold–Yilmaz framework measures volatility spillovers using a VAR on realized volatilities and generalized forecast error variance decomposition (GFEVD).
- The **total spillover index** $S^{(H)}$ captures the fraction of forecast-error variance explained by cross-asset shocks; typical values are 30–40% in calm markets, 70–85% during crises.
- **Directional spillovers** (FROM, TO, net) identify which assets transmit and which receive volatility shocks; equity and credit tend to be net transmitters during stress.
- Rolling-window estimation (200-day window) is simple but introduces a window-length hyperparameter; the TVP-VAR approach of Antonakakis et al. (2020) eliminates this choice.
- Network visualization reveals that calm-period connectedness is within-sector (clustered), while crisis-period connectedness is cross-sector (dense, nearly fully connected).
- Demirer et al. (2018) applied the DY framework to a global network of banks and sovereigns, confirming the calm-to-crisis structural shift at institution level.
- Sirignano and Cont (2019) demonstrate “universal” price formation features: a pooled deep learning model trained across US equities outperforms asset-specific models, suggesting common underlying dynamics.
- Rosenbaum and Zhang (2022) train a universal LSTM on pooled equity data whose predictions are matched by a rough volatility model with $H \approx 0.1$, connecting spillover phenomena to rough volatility theory (Chapter 7).
- Spillover indices are useful as ML features for medium-horizon forecasting: total spillover as a regime indicator, directional FROM as a vulnerability measure, and net spillover as a contrarian signal.
- Spillover features are slow-moving (200-day window) and may add limited value for short-horizon forecasts; always verify with SHAP importance analysis.
- The GFEVD approach generalizes the Cholesky decomposition and does not depend on variable ordering, making it suitable for large cross-sections.
- High cross-asset spillovers during crises may reflect common-factor exposure (funding liquidity, dealer balance sheets) rather than sequential causal contagion.

Concept	Key Result
Total spillover index	30–40% calm, 70–85% crisis; captures system-wide connectedness
Directional (FROM/TO/net)	Identifies transmitters (equity, credit) vs. receivers (bonds, FX)
Rolling window vs. TVP-VAR	TVP-VAR avoids window-length choice; sharper crisis timing
Network topology shift	Within-sector $\rightarrow$ cross-sector during stress
Universality	Common features and $H \approx 0.1$ across asset classes
Spillover as ML feature	Best for medium horizon; z-score and verify with SHAP

Part VI

Evaluation and Practice

Chapter 17

Forecast Evaluation

A good forecast is useless if you cannot prove it is good. This chapter gives you the statistical machinery to distinguish genuine forecasting ability from noise, luck, and overfitting. We cover the right loss function (QLIKE), the right comparison test (Diebold–Mariano), the right multi-model framework (Model Confidence Set), the right cross-validation (purged K-fold with embargo), and the right Sharpe ratio adjustment (Deflated Sharpe Ratio).

17.1 Why Evaluation Methodology Matters

Before turning to specific tools, consider two scenarios that illustrate why evaluation methodology *is* the credibility of your results.

Scenario 1. You build a LightGBM volatility forecast that achieves 5% lower QLIKE than the HAR benchmark (Chapter 6). Your manager asks: “Is that improvement statistically significant, or would it vanish on a different sample?” Without a Diebold–Mariano test, you cannot answer.

Scenario 2. You try 30 different feature sets, pick the best one, and report a backtest Sharpe ratio of 1.5. A colleague asks: “What’s the probability that at least one of 30 random strategies would have produced a Sharpe that high?” Without the Deflated Sharpe Ratio, you cannot answer.

The evaluation framework is infrastructure you build *first*, not a final step you tack on after research, so that every experiment you run produces honest, comparable numbers from day one.

Seven Tools, Seven Questions

This chapter introduces seven evaluation tools. Each answers one question:

1. **QLIKE**: which model has lower loss? (Primary metric.)
2. **MSE**: does the ranking hold under a different loss? (Secondary check.)
3. **MZ regression**: is my forecast biased or too smooth? (Diagnostic.)
4. **DM test**: is the loss difference between two models statistically significant? (Pair-wise test.)
5. **MCS**: given all candidate models, which ones survive? (Multi-model filter.)
6. **Purged CV**: how do I tune hyperparameters without leaking future data? (Training procedure.)
7. **DSR**: is my backtest Sharpe real after accounting for all experiments? (Multiple-testing correction.)

You will use all seven, in roughly this order.

17.2 MSE and Its Limitations for Volatility

We start with the loss function you already know, then explain why it is not enough for volatility forecasting.

Loss Functions

A **loss function** $L(\sigma_t^2, h_t)$ measures how far a forecast h_t is from the truth σ_t^2 . Lower loss means a better forecast. You have used mean squared error (MSE) in regression problems; it is the default in most ML pipelines.

17.2.1 The MSE Formula

The mean squared error between a sequence of forecasts $\{h_t\}$ and realized values $\{\sigma_t^2\}$ is:

$$\text{MSE} = \frac{1}{T} \sum_{t=1}^T (\sigma_t^2 - h_t)^2 \quad (17.1)$$

where:

- σ_t^2 is the true (unobservable) variance on day t ,
- h_t is the forecast variance for day t , produced before day t ,
- T is the number of forecast evaluation days.

17.2.2 The Proxy Problem

We never observe true variance σ_t^2 . Instead, we use a proxy: realized variance RV_t (Chapter 2). This proxy is noisy because $\text{RV}_t = \sigma_t^2 + \eta_t$, where η_t is measurement error from finite sampling, microstructure noise (Chapter 3), and jumps (Chapter 4).

The good news: MSE produces correct model *rankings* even when using a noisy proxy, as long as the noise is independent of the forecast (Patton, 2011). If model A has lower MSE than model B when evaluated against RV_t , the same ranking holds against true σ_t^2 . This property is called **robustness to noise in the proxy**.

17.2.3 Why MSE Is Still Not Enough

MSE has a deeper problem: it is symmetric and heavily penalizes extreme values.

MSE Penalizes Outliers Disproportionately

Suppose your forecast is $h_t = 1.0$ (in annualized variance units) for every day. On 249 normal days, true variance is 1.0 and MSE contribution is 0. On one crisis day, true variance spikes to 10.0, and MSE contribution is $(10 - 1)^2 = 81$. That single day dominates the entire loss. This means MSE-optimal forecasts chase outliers: they overweight extreme days at the expense of forecasting accuracy during normal times, which is the opposite of what most applications need.

17.3 QLIKE: The Preferred Loss Function

The volatility forecasting literature has converged on QLIKE as the primary metric.

17.3.1 Intuition

QLIKE (quasi-likelihood loss) comes from the negative log-likelihood of a Gaussian distribution with variance h_t . Think of it this way: if returns were exactly normal with variance h_t , the best forecast would minimize QLIKE. Even when returns are not normal (and they never are), QLIKE retains two critical properties that MSE shares one of and lacks the other.

17.3.2 The QLIKE Formula

$$\text{QLIKE} = \frac{1}{T} \sum_{t=1}^T \left(\ln h_t + \frac{\sigma_t^2}{h_t} \right) \quad (17.2)$$

where:

- h_t is the forecast variance for day t ,
- σ_t^2 is the true variance on day t (in practice, RV_t),
- $\ln h_t$ penalizes forecasts that are too large (over-prediction),
- σ_t^2/h_t penalizes forecasts that are too small (under-prediction),
- T is the number of evaluation days.

In Plain English

QLIKE has two parts pulling in opposite directions. The $\ln h_t$ term punishes you for forecasting too high (wasting capital on unnecessary hedges), while the σ_t^2/h_t term punishes you for forecasting too low (holding unrecognized risk). The under-prediction penalty (σ_t^2/h_t) explodes as $h_t \rightarrow 0$, so QLIKE is far harsher on dangerous under-estimates than on conservative over-estimates. This asymmetry matches real-world priorities: underestimating volatility gets you fired; overestimating it merely costs some opportunity. This does not mean the optimal forecast is biased upward. It means that among two equally wrong forecasts, the one that errs low is more costly. The target is still the true variance.

QLIKE Is Still Minimized at the True Value

A common misreading of the asymmetric penalty is: “If under-prediction is punished more, shouldn’t I forecast a bit high to be safe?” No. Take the derivative of a single day’s QLIKE contribution with respect to the forecast h_t :

$$\frac{\partial}{\partial h_t} \left(\ln h_t + \frac{\sigma_t^2}{h_t} \right) = \frac{1}{h_t} - \frac{\sigma_t^2}{h_t^2} = 0 \implies h_t = \sigma_t^2.$$

The minimum is at $h_t = \sigma_t^2$ exactly. The asymmetry shapes the penalty *curve*, not the penalty *minimum*. Think of a speed limit: the best speed is exactly the limit. Getting caught going 20 over is worse than going 20 under, but that does not make 20-under the target. QLIKE works the same way: the best forecast is the true variance, but being wrong on the low side hurts more than being wrong on the high side by the same amount.

Why QLIKE Is Less Sensitive to Outliers

When true variance spikes to $\sigma_t^2 = 10$ and your forecast is $h_t = 1$, the QLIKE contribution is $\ln(1) + 10/1 = 10$. Under MSE, the same day contributes $(10 - 1)^2 = 81$. QLIKE penalizes the error linearly (through the ratio σ_t^2/h_t) rather than quadratically. Extreme days still matter, but they do not dominate.

Patton (2011): QLIKE and MSE Are the Only Robust Losses

Patton (2011) proves that QLIKE and MSE are the *only* two members of the standard loss function family that produce correct model rankings even when the volatility proxy is noisy. Other common losses (MAE, HMSE, heteroskedasticity-adjusted MSE) can reverse the true ranking when evaluated against RV_t instead of σ_t^2 . Of the two robust losses, QLIKE is less sensitive to extreme RV days and is therefore preferred as the primary evaluation metric.

17.3.3 Worked Example: MSE vs. QLIKE Rankings

Worked Example: MSE and QLIKE Can Disagree on Rankings

Consider five days of forecasts from two models, evaluated against realized variance. Model A is a reactive forecaster that under-predicts during calm periods but partially captures the crisis spike. Model B is a stable forecaster that nails normal days but barely reacts to the spike.

Day	RV_t	Model A (h_t^A)	Model B (h_t^B)
1	1.0	0.4	1.0
2	1.1	0.4	1.1
3	0.9	0.4	0.9
4	1.0	0.4	1.0
5	8.0	6.0	2.0

MSE computation:

$$\begin{aligned} \text{MSE}_A &= \frac{1}{5} [(1.0 - 0.4)^2 + (1.1 - 0.4)^2 + (0.9 - 0.4)^2 + (1.0 - 0.4)^2 + (8 - 6)^2] \\ &= \frac{1}{5} (0.36 + 0.49 + 0.25 + 0.36 + 4) = 1.092 \end{aligned}$$

$$\begin{aligned} \text{MSE}_B &= \frac{1}{5} [(1.0 - 1.0)^2 + (1.1 - 1.1)^2 + (0.9 - 0.9)^2 + (1.0 - 1.0)^2 + (8 - 2)^2] \\ &= \frac{1}{5} (0 + 0 + 0 + 0 + 36) = 7.200 \end{aligned}$$

MSE ranks Model A first (1.092 < 7.200). The crisis day drives the entire result: Model B's MSE is 99.9% from day 5 alone.

QLIKE computation:

$$\begin{aligned} \text{QLIKE}_A &= \frac{1}{5} \left[\left(\ln 0.4 + \frac{1.0}{0.4} \right) + \left(\ln 0.4 + \frac{1.1}{0.4} \right) + \left(\ln 0.4 + \frac{0.9}{0.4} \right) + \left(\ln 0.4 + \frac{1.0}{0.4} \right) + \left(\ln 6 + \frac{8}{6} \right) \right] \\ &= \frac{1}{5} (1.584 + 1.834 + 1.334 + 1.584 + 3.125) = 1.892 \end{aligned}$$

$$\begin{aligned} \text{QLIKE}_B &= \frac{1}{5} \left[\left(\ln 1.0 + \frac{1.0}{1.0} \right) + \left(\ln 1.1 + \frac{1.1}{1.1} \right) + \left(\ln 0.9 + \frac{0.9}{0.9} \right) + \left(\ln 1.0 + \frac{1.0}{1.0} \right) + \left(\ln 2 + \frac{8}{2} \right) \right] \\ &= \frac{1}{5} (1.000 + 1.095 + 0.895 + 1.000 + 4.693) = 1.737 \end{aligned}$$

QLIKE ranks Model B first (1.737 < 1.892). The rankings *reverse*: MSE picks A, QLIKE picks B.

What this tells you. MSE picked Model A because the crisis day dominates quadratically: $(8 - 2)^2 = 36$ versus $(8 - 6)^2 = 4$. Being closer on one extreme day overwhelmed Model A's poor performance on four normal days. QLIKE picked Model B because QLIKE penalizes Model A's under-prediction on normal days through the σ_t^2/h_t ratio ($1.0/0.4 = 2.5$ on each normal day), and the crisis-day QLIKE gap ($4.693 - 3.125 = 1.57$) is not large enough to compensate. For daily risk management, where forecast quality on typical days matters more than one extreme day, QLIKE's ranking is more useful. When MSE and QLIKE disagree, check whether the MSE ranking is driven by a handful of extreme days.

Why This Matters

QLIKE is THE primary evaluation metric for your project. When you report that your ML model beats HAR, the headline number is the percentage reduction in QLIKE. The asymmetry matters: QLIKE penalizes you more for underestimating vol than overestimating it (through the σ_t^2/h_t ratio), which aligns with risk management priorities where underestimating vol means holding too much risk. Target a 30–80 bps QLIKE improvement over HAR to claim a meaningful result. Report the percentage reduction to two decimal places in your results table, and always pair it with a DM test p -value (Section 17.5).

17.3.4 Retransformation Bias

Many volatility models forecast in log space because $\log RV_t$ is more Gaussian, more homoskedastic, and better behaved for regression than raw RV_t . The HAR-log model, for example, regresses $\log RV_{t+1}$ on lagged log realized variances. But when you need a level-space forecast (e.g., for portfolio variance targeting or VaR computation), you must exponentiate the log forecast back to levels. This innocent-looking step introduces a systematic downward bias known as **retransformation bias**.

The Problem: Jensen's Inequality

The root cause is **Jensen's inequality**: for any convex function g and non-degenerate random variable X ,

$$\mathbb{E}[g(X)] > g(\mathbb{E}[X]). \quad (17.3)$$

The exponential function is convex, so $\mathbb{E}[\exp(X)] > \exp(\mathbb{E}[X])$.

Suppose your log-space model produces a point forecast $\widehat{\log RV}_{t+1}$. The naive level-space forecast is:

$$\widehat{RV}_{t+1}^{\text{naive}} = \exp(\widehat{\log RV}_{t+1}).$$

But because the log-space forecast has estimation error, the true conditional expectation of RV_{t+1} is *larger* than this. Exponentiating the conditional mean of the log gives you something systematically below the conditional mean of the level. Every single forecast is biased low.

In Plain English

Think of it this way. Your log-space model says “the average of log RV tomorrow is 0.5.” But the average of RV tomorrow is not $\exp(0.5) = 1.65$. It is higher, because the distribution of log RV has spread around 0.5, and the exponential function amplifies high values more than it shrinks low values. The more uncertain your log-space forecast (wider error distribution), the larger the gap between $\exp(\mathbb{E}[\log RV])$ and $\mathbb{E}[RV]$.

The Correction Formula

If log-space forecast errors are approximately Gaussian with variance $\hat{\sigma}_\varepsilon^2$, the bias-corrected level-space forecast is:

$$\widehat{RV}_{t+1} = \exp\left(\widehat{\log RV}_{t+1} + \frac{\hat{\sigma}_\varepsilon^2}{2}\right) \quad (17.4)$$

where:

- $\widehat{\log RV}_{t+1}$ is the log-space point forecast,
- $\hat{\sigma}_\varepsilon^2$ is the estimated variance of the log-space forecast errors $\varepsilon_t = \log RV_t - \widehat{\log RV}_t$,
- the $\hat{\sigma}_\varepsilon^2/2$ term is the correction that offsets Jensen's inequality.

This formula comes from the moment generating function of a Gaussian: if $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, then $\mathbb{E}[\exp(\varepsilon)] = \exp(\sigma_\varepsilon^2/2)$. The corrected forecast multiplies the naive exponential by this factor.

In Plain English

The correction adds half the forecast error variance back before exponentiating. It says: “My best guess for log RV is $\hat{y}$, but there is uncertainty of $\hat{\sigma}_\varepsilon^2$ around that guess. Because exponentiation amplifies upside errors more than downside errors, I need to nudge my forecast upward by $\hat{\sigma}_\varepsilon^2/2$ to get the right average in level space.” When forecast errors are small ($\hat{\sigma}_\varepsilon^2 \approx 0$), the correction is negligible. When they are large, as in long-horizon

forecasts or during volatile regimes, it can be substantial.

How Large Is the Bias?

Worked Example: Retransformation Bias Magnitude

Suppose your HAR-log model has log-space forecast error variance $\hat{\sigma}_\varepsilon^2 = 0.20$ (a realistic value for 5-day-ahead forecasts of log daily RV on equity indices).

Correction factor:

$$\exp\left(\frac{0.20}{2}\right) = \exp(0.10) \approx 1.105$$

This means the naive forecast underestimates the true conditional mean by about 10.5%. On a day where $\widehat{\log RV}_{t+1} = -4.0$ (corresponding to annualized vol of about 14%), the naive forecast is $\exp(-4.0) = 0.0183$, while the corrected forecast is $0.0183 \times 1.105 = 0.0202$.

For 1-day-ahead forecasts with $\hat{\sigma}_\varepsilon^2 \approx 0.08$, the correction factor is $\exp(0.04) \approx 1.04$, a 4% adjustment. For 22-day-ahead forecasts with $\hat{\sigma}_\varepsilon^2 \approx 0.35$, it reaches $\exp(0.175) \approx 1.19$, nearly a 20% adjustment.

Impact on evaluation. Without this correction, the 10.5% systematic under-prediction on a typical 5-day horizon inflates QLIKE loss by roughly 3–5% and shows up as $a > 0$ in the Mincer–Zarnowitz regression (Section 17.4). Applying the correction is often the single cheapest QLIKE improvement available: it costs one line of code and zero additional data.

Without Correction, Every Forecast Is Biased Low

If you forecast in log space and naively exponentiate, your Mincer–Zarnowitz regression (Section 17.4) will show $a > 0$ (systematic under-prediction) and the bias grows with forecast uncertainty. During high-volatility regimes, when $\hat{\sigma}_\varepsilon^2$ is largest, the bias is at its worst, precisely when accurate forecasts matter most for risk management.

Estimating the Correction Variance

The correction requires $\hat{\sigma}_\varepsilon^2$, the variance of log-space forecast errors. In practice, estimate this from the training sample or from a rolling window of recent out-of-sample errors. Using a rolling window (e.g., the trailing 60 trading days) allows the correction to adapt to regime changes: during calm markets $\hat{\sigma}_\varepsilon^2$ is small and the correction is minor; during crises it grows, appropriately increasing the level-space forecast.

17.4 Mincer–Zarnowitz Regressions

QLIKE tells you which model has lower average loss, but it does not tell you *why* a forecast is bad. Think of QLIKE as the scoreboard and the Mincer–Zarnowitz regression as the film review: QLIKE tells you who won; MZ tells you what to fix. The MZ regression is a simple diagnostic that decomposes forecast errors into bias and inefficiency.

17.4.1 The Regression

Regress realized variance on the forecast:

$$\sigma_t^2 = a + b \cdot h_t + \varepsilon_t \quad (17.5)$$

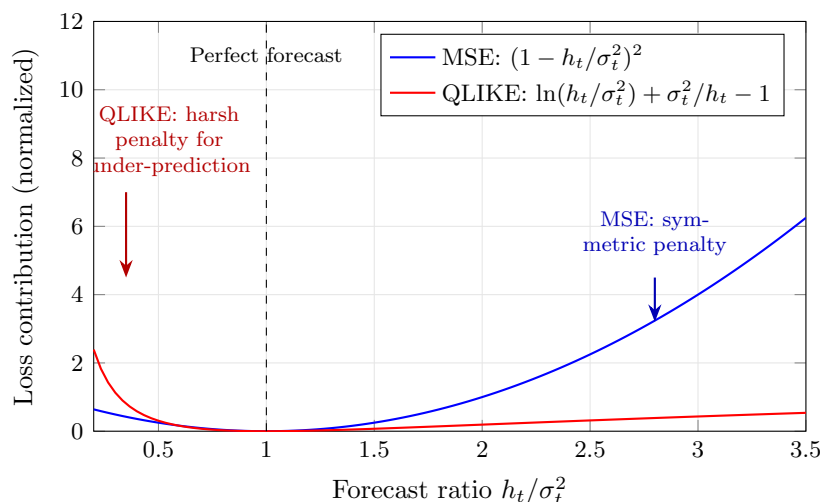


Figure 17.1: QLIKE vs. MSE penalty as a function of the forecast ratio h_t/σ_t^2 . Both losses are minimized at the perfect forecast ($h_t = \sigma_t^2$, ratio = 1). MSE penalizes over- and under-prediction symmetrically. QLIKE penalizes under-prediction (ratio < 1) much more harshly than over-prediction (ratio > 1), matching the asymmetric risk preferences in volatility forecasting: underestimating vol means holding too much risk. Despite this asymmetry, both losses are minimized at ratio = 1 (the true variance); the asymmetry shapes the penalty curve, not the optimal forecast.

where:

- σ_t^2 is realized variance (left-hand side),
- h_t is the forecast (right-hand side),
- a is the intercept (bias term),
- b is the slope (efficiency term),
- ε_t is the residual.

In Plain English

The Mincer–Zarnowitz regression asks: “If I plot realized variance against my forecast, do the points lie along the 45-degree line?” A perfect forecast gives intercept $a = 0$ (no constant bias) and slope $b = 1$ (every unit increase in the forecast corresponds to exactly one unit increase in reality). If $b < 1$, your forecast is too timid; if $b > 1$, it overreacts.

Why This Matters

After fitting your ML model, run the MZ regression before anything else. HAR models typically show b slightly below 1 (they smooth too much in high-vol regimes), and your ML extension should fix this. If your HARQ-ML model still shows $b = 0.85$, you know the improvement needs to come from making the forecast more reactive to recent variance spikes, not from reducing average bias.

Unbiased and Efficient Forecast

A forecast is **unbiased** if $a = 0$ (no systematic over- or under-prediction) and **efficient** if $b = 1$ (the forecast captures the full scale of variance movements). Test the joint hypothesis $H_0 : a = 0, b = 1$ with a standard F-test (Mincer and Zarnowitz, 1969).

17.4.2 Interpreting Deviations

- $a > 0$, $b \approx 1$: the forecast systematically under-predicts by a constant.
- $a \approx 0$, $b < 1$: the forecast under-reacts to variance movements (too smooth).
- $a \approx 0$, $b > 1$: the forecast over-reacts (too volatile).
- R^2 : the fraction of realized variance variation explained by the forecast. Higher is better.

What to Fix Based on MZ Results

The MZ regression is only useful if you act on the diagnosis:

- $b < 1$ (**forecast too smooth**): your model over-relies on long-horizon averages. Try adding shorter-lag features (e.g., 1-day lagged RV), reducing regularization strength, or increasing model capacity.
- $b > 1$ (**forecast too reactive**): your model is chasing noise. Try increasing regularization, using a longer lookback window, or smoothing the forecast with an exponential moving average.
- $a > 0$ (**systematic under-prediction**): check for retransformation bias first if you forecast in log space (Section 17.3.4). If that is not the issue, add a bias correction term or recalibrate the intercept.
- $a < 0$ (**systematic over-prediction**): less common in volatility forecasting, but check whether your features include stale high-vol observations that inflate the forecast.

Use HAC Standard Errors

Volatility forecast errors are serially correlated (today's error predicts tomorrow's error) because volatility clusters (Chapter 5). Use Newey–West (HAC) standard errors in the MZ regression. OLS standard errors will be too small, leading you to reject H_0 too often.

17.5 The Diebold–Mariano Test

You now have a loss function (QLIKE) and a diagnostic (MZ regression). The next question is: given two models, is the difference in loss *statistically significant*, or could it be sampling noise?

17.5.1 Setup

Suppose you have two volatility forecasts, h_t^A and h_t^B , and a loss function L (use QLIKE). Define the **loss differential**:

$$d_t = L(\sigma_t^2, h_t^A) - L(\sigma_t^2, h_t^B) \quad (17.6)$$

where:

- d_t is the difference in loss on day t ,
- $L(\sigma_t^2, h_t^A)$ is the loss of model A on day t ,
- $L(\sigma_t^2, h_t^B)$ is the loss of model B on day t .

If $d_t > 0$ on average, model B has lower loss (model B wins). The question is whether $\bar{d}$ is significantly different from zero.

17.5.2 The Test Statistic

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}} \quad (17.7)$$

where:

- $\bar{d} = \frac{1}{T} \sum_{t=1}^T d_t$ is the mean loss differential,
- $\widehat{\text{Var}}(\bar{d})$ is estimated using HAC (Newey–West) standard errors to account for serial correlation in d_t ,
- Under $H_0 : \mathbb{E}[d_t] = 0$, the DM statistic is asymptotically standard normal (Diebold and Mariano, 1995).

In Plain English

The DM statistic is a t-test on the average loss difference. The numerator is “how much better is model B on average?” and the denominator is “how uncertain are we about that average, given that consecutive days are correlated?” If the ratio exceeds roughly 2, you have a statistically significant winner at the 5% level.

HAC Standard Errors

When observations are serially correlated, the usual variance estimator $\widehat{\text{Var}}(\bar{d}) = s_d^2/T$ is biased downward. **Heteroskedasticity and Autocorrelation Consistent (HAC)** estimators, such as Newey–West, correct for this by including autocovariances up to some lag ℓ . A common rule of thumb is $\ell = \lfloor T^{1/3} \rfloor$. For $T = 1,000$ days, this gives $\ell = 10$.

17.5.3 Worked Example

Worked Example: Diebold–Mariano Test

You evaluate HAR and LightGBM forecasts over $T = 500$ trading days using QLIKE loss.

Given:

- Mean QLIKE loss differential: $\bar{d} = 0.023$ (HAR has higher loss, so LightGBM wins on average).
- HAC standard error of $\bar{d}$: $\text{SE}_{\text{HAC}} = 0.011$.

Compute:

$$\text{DM} = \frac{0.023}{0.011} = 2.09$$

Interpret: Under H_0 , $\text{DM} \sim \mathcal{N}(0, 1)$. The two-sided p -value is $2 \times \Phi(-2.09) \approx 0.037$. At the 5% level, you reject H_0 : the LightGBM improvement over HAR is statistically significant.

At the 1% level, you would not reject. Report the p -value and let the reader decide.

What this tells you. $\text{DM} = 2.09$, $p = 0.037$ means you can credibly claim LightGBM beats HAR at the 5% significance level. This p -value goes in your results table next to the QLIKE numbers. If p had been 0.15, the QLIKE improvement would be real in your sample but you could not rule out that a different sample would reverse it; you would need more data or a larger improvement before claiming victory. If you have a directional hypothesis (ML should beat HAR, not vice versa), a one-sided test is appropriate, halving the p -value to $0.037/2 = 0.018$.

Small-Sample Correction

Diebold and Mariano (1995) derived the test for large samples. With fewer than 100 observations, use the modified DM statistic from Harvey et al. (1997), which uses a t -distribution with $T - 1$ degrees of freedom and applies a finite-sample correction factor.

17.6 The Model Confidence Set

The Diebold–Mariano test compares models in pairs. Use it when you have a specific pairwise claim to make (“my ML model beats HAR”). With M models, you would need $\binom{M}{2}$ pairwise tests, and the more tests you run, the more likely you are to find a “significant” difference by chance. The Model Confidence Set solves this by comparing all models simultaneously. Use it when you have a model zoo and need to know which ones to keep and which to discard. DM is your scalpel for targeted claims; MCS is your filter for the full candidate set.

17.6.1 Intuition

The Model Confidence Set as a Tournament

Imagine a round-robin tournament. Instead of declaring a single winner, the MCS procedure eliminates models that are *significantly worse* than the others and returns the set of survivors. The survivors are statistically indistinguishable from each other at the chosen confidence level.

17.6.2 The Procedure

The MCS algorithm of Hansen et al. (2011) works as follows:

1. Start with the full set of M models: $\mathcal{M}_0 = \{1, 2, \dots, M\}$.
2. Test the null hypothesis H_0 : all models in the current set have equal expected loss.
3. If H_0 is rejected at significance level α , identify and remove the worst model (the one with the highest average loss).
4. Repeat steps 2–3 on the reduced set until H_0 is not rejected.
5. The surviving set $\widehat{\mathcal{M}}_\alpha^*$ is the **Model Confidence Set** at level α .

Model Confidence Set

The Model Confidence Set $\widehat{\mathcal{M}}_\alpha^*$ at significance level α contains all models whose forecasting performance is not significantly worse than the best model. Formally, it satisfies:

$$\Pr(\mathcal{M}^* \subseteq \widehat{\mathcal{M}}_\alpha^*) \geq 1 - \alpha$$

where $\mathcal{M}^*$ is the (unknown) set of truly best models.

Hansen, Lunde, and Nason (2011): The Gold Standard for Multi-Model Comparison

Hansen et al. (2011) show that the MCS controls the familywise error rate: the probability of incorrectly excluding any truly best model is at most α . The MCS produces a *set*, not a ranking. Reporting “these 4 models are in the 90% MCS” is more honest than reporting “model X has the lowest QLIKE” when differences are small.

17.6.3 Practical Use

Report which models survive at both $\alpha = 0.10$ and $\alpha = 0.05$:

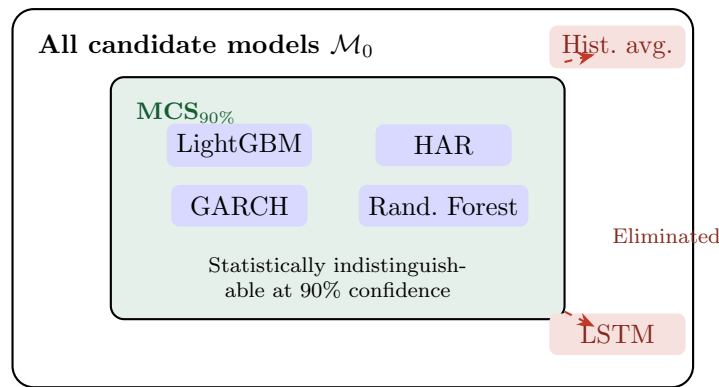


Figure 17.2: The Model Confidence Set retains all models whose performance is statistically indistinguishable from the best (green region) and eliminates models that are significantly worse (red, outside). The surviving models cannot be ranked among themselves with statistical confidence.

Model	QLIKE	MCS p -value	In $MCS_{90\%}$?
LightGBM + HAR features	1.423	1.000	Yes
HAR (daily, weekly, monthly)	1.441	0.482	Yes
GARCH(1,1)	1.467	0.312	Yes
Random Forest	1.439	0.551	Yes
LSTM	1.502	0.043	No
Historical average	1.589	0.001	No

The MCS p -value for each model is the smallest α at which that model would be excluded. A model with MCS p -value > 0.10 survives in the 90% MCS. In this example, four models are statistically indistinguishable at the 90% level; the LSTM and historical average are significantly worse.

MCS as a Humility Device

The MCS often reveals an uncomfortable truth: many models that look different in QLIKE are statistically indistinguishable. A 2% QLIKE improvement is rarely significant with 3–5 years of daily data. If your fancy model is in the same MCS as HAR, be honest about it.

What to Do When Multiple Models Survive

When four models survive the MCS, you cannot rank among them statistically. Choose among survivors using secondary criteria: simplicity (HAR is easier to explain to a portfolio manager than LightGBM), computational cost (GARCH fits in seconds versus minutes), interpretability (can you explain why the forecast changed?), or economic value in a downstream application (Chapter 18). The MCS does not pick your model; it eliminates the ones you should not pick.

The MCS p -values for surviving models (1.000, 0.482, 0.312, 0.551 in the table above) are *not* a ranking. They indicate how far each model is from elimination: a p -value of 0.312 means GARCH would be eliminated at $\alpha = 0.30$ but survives at $\alpha = 0.10$. Do not treat these as confidence scores or use them to rank survivors.

Why This Matters

Your model needs to be IN the Model Confidence Set, and ideally, simpler baselines like raw HAR should be excluded. If your LightGBM model and plain HAR both survive in the 90% MCS, you cannot honestly claim superiority; report them as statistically equivalent and justify your model choice on secondary criteria (interpretability, computational cost, economic value). The MCS is also your defense: if a reviewer asks “why not use an LSTM?”, you can show it was eliminated from the MCS. Use the `MCS` package in R or the `arch` library in Python to compute MCS p -values.

17.7 Purged K-Fold Cross-Validation with Embargo

The tests above (DM, MCS) evaluate forecasts on a held-out sample. But how do you *select* the model and tune hyperparameters in the first place? Standard K-fold cross-validation fails catastrophically on time series data. This section explains why and introduces the fix.

17.7.1 Why Standard K-Fold Fails

K-Fold Cross-Validation

In standard K-fold CV, you split data into K equally sized folds, train on $K - 1$ folds, test on the remaining fold, and rotate. This works when observations are independent (e.g., images, text documents). It does *not* work when observations are serially dependent.

Consider 5-fold CV on 1,250 trading days (5 years). Fold 1 = days 1–250, fold 2 = days 251–500, and so on. When testing on fold 2 (days 251–500), you train on folds 1, 3, 4, 5 (days 1–250 and 501–1250).

The problem: volatility on day 501 is highly correlated with volatility on day 500 (the last day of the test set). Training on day 501 while testing on day 500 is using the future to predict the past. Worse, if your labels use multi-day returns (e.g., 5-day forward realized variance), then the label for day 498 overlaps with the label for day 502; the training and test sets share information through the label construction.

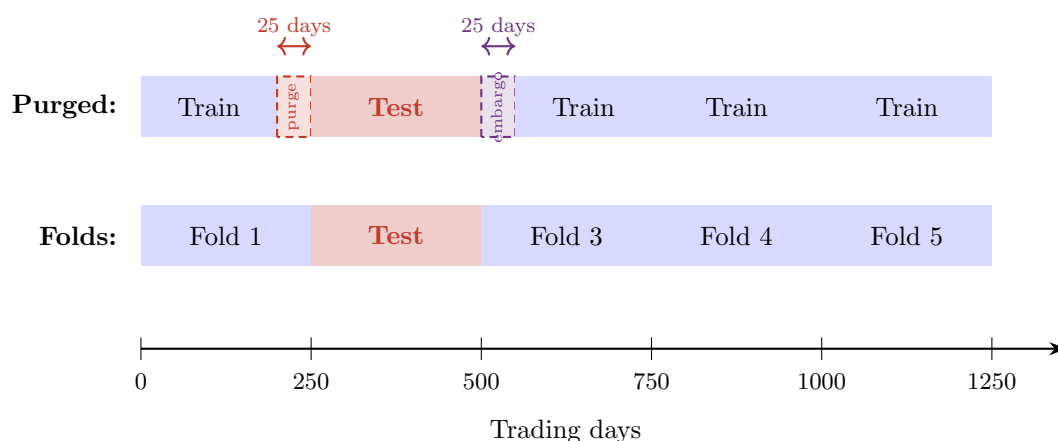


Figure 17.3: Purged K-fold CV with embargo ($K = 5$, $T = 1,250$, embargo = 2%). **Top row:** standard fold assignment with fold 2 as the test set. **Bottom row:** after purging and embargo. The 25 days before the test set (purge zone) are removed from training to prevent label overlap. The 25 days after the test set (embargo zone) are removed to prevent information leakage from serial correlation. Training uses only the remaining blue regions.

17.7.2 The Fix: Purging and Embargo

de Prado (2018) introduces two modifications to standard K-fold CV:

Purging

Purging removes from the training set any observations whose label windows overlap with the test period. If labels are constructed from τ -day forward returns, remove training observations within τ days before the start of the test fold.

Embargo

Embargo removes an additional buffer of training observations *after* the end of the test fold. This guards against serial correlation in features: day $t+1$ features are correlated with day t features, so training on day $t+1$ while testing on day t leaks information. A typical embargo is 1–2% of total sample size. The embargo length should cover the autocorrelation decay of your features. For HAR features (which use lags up to 22 days), the serial correlation in RV drops below 0.05 within about 5–10 days, so 1–2% of a typical 1,000–2,500 day sample (10–50 days) is conservative. If you use features with longer memory (e.g., monthly moving averages or regime indicators), increase the embargo accordingly.

17.7.3 Worked Example

Worked Example: Purged 5-Fold CV

Setup: $T = 1,250$ daily observations (5 years), $K = 5$ folds, labels use 5-day forward realized variance, embargo = 2% of $T = 25$ days.

Fold 2 as test set (days 251–500):

1. **Standard training set:** days 1–250 and 501–1250 (1,000 days).
2. **After purging:** remove days 246–250 from training (their 5-day labels overlap with the test period). Training loses 5 days.
3. **After embargo:** remove days 501–525 from training (serial correlation buffer). Training loses 25 more days.
4. **Final training set:** days 1–245 and 526–1250 (970 days).

You lose 30 training observations per fold (3% of the total). This is a small price for preventing look-ahead bias.

Why This Matters

Your vol forecasting labels use multi-day forward realized variance (typically 1-day or 5-day), which means label windows overlap across consecutive observations. Standard K-fold would train on day 502 (whose label includes returns from days 502–506) while testing on day 500 (whose label includes days 500–504). The overlapping days 502–504 leak test information into training. Purging removes this overlap; embargo handles the residual serial correlation in features like lagged RV. Use `sklearn.model_selection.TimeSeriesSplit` as a starting point, then add purging and embargo manually or use the `purged_cv` implementation from de Prado (2018).

Random K-Fold on Time Series Is Catastrophic

Random K-fold on time series data (shuffling observations before splitting) is the single most common evaluation error in ML-for-finance papers. A model trained on January and March, tested on February, has literally seen the future. Reported accuracy will be

dramatically inflated; out-of-sample performance will collapse. Always use purged CV, expanding-window, or walk-forward evaluation for time series.

17.7.4 Cross-Sectional Leakage When Pooling Assets

The purging and embargo machinery above protects you when you forecast *one* series at a time (e.g., SPX realized variance). But the moment you **pool** several instruments into a single panel (stacking, say, SPX, Nasdaq-100, the Russell 2000, and a basket of single names into one training matrix to give your ML model more rows), a second, sneakier form of leakage appears.

The concrete question: *if I shuffle my pooled panel and split it row-by-row, what exactly leaks?*

Pooled Panel of Realized Variances

A **pooled panel** stacks observations indexed by both an instrument i and a date t . Each row is a (date t , instrument i) pair: the features are that instrument's HAR components on day t (daily, weekly, monthly lagged RV, jumps, implied volatility IV), and the label is its forward realized variance $RV_{i,t+1}$. These are just the model's input columns from earlier chapters; you do not need their details here, only that each row is one asset on one day. Pooling is attractive because a single ML model trained on $N_{\text{assets}} \times T$ rows sees far more data than T rows from one asset, and it can borrow strength across instruments that share volatility dynamics.

Same-date observations are cross-sectionally correlated. On any given day, equity-index realized variances move together: a macro shock (a Fed surprise, a CPI print, the COVID crash of March 2020) lifts RV for *every* index on the *same* date. This is the cross-sectional analogue of volatility clustering: not “today's RV predicts tomorrow's” but “SPX's RV today is correlated with Nasdaq's RV today.” The day-to-day surprises in each asset's $RV_{i,t}$ share a single market-wide shock f_t (a common factor; think: the size of that day's macro surprise) that pushes every asset's RV in the same direction.

Now suppose you split this pooled panel with random K -fold, putting individual *rows* into folds. The SPX row for 16 March 2020 lands in the training fold; the Nasdaq row for the *same* 16 March 2020 lands in the test fold. Because both rows are driven by the same crisis-day common factor f_t , the model has effectively seen the test day's volatility shock through a different instrument's window. Your forecast of Nasdaq RV on that date is no longer a genuine out-of-sample prediction: the answer leaked in sideways, across the cross-section, on the same calendar date.

In Plain English

Purging and embargo (Section 17.7) plug the leak *through time*: yesterday bleeding into today. Cross-sectional leakage is a leak *through the cross-section*: one asset's value on a date bleeding into another asset's value on the *same* date. A random row split looks innocent (no single asset appears in both train and test for the same date), but because all assets share the day's shock, splitting by row still hands the model the answer. The fix is blunt and absolute: an entire *date* is either a training date or a test date, never both.

Grouped Purged CV by Date

When pooling instruments, do not cross-validate on rows. Cross-validate on **dates**: assign each calendar date wholesale to train or to test, so all instruments observed on a given date go to the same side of the split. Then apply purging and embargo *on the date axis*

exactly as in Section 17.7: purge the dates whose forward-RV label windows overlap the test block, and embargo the dates immediately after it. In `scikit-learn` this is the role of `GroupKFold` / `StratifiedGroupKFold` with the date as the group key, layered on top of a time-ordered (not shuffled) split, with purge and embargo added manually. This is the panel-econometrics discipline of de Prado (2018): entire dates go into train or test, never split across folds.

Two Leaks, Not One, in a Pooled Panel

A pooled realized-variance panel can leak in *two* independent ways, and you must block both:

1. **Temporal leakage** (Section 17.7): day $t+1$ in train while day t is in test, or overlapping multi-day labels. Blocked by purging and embargo.
2. **Cross-sectional leakage** (this subsection): SPX on date t in train while Nasdaq on the same date t is in test. Blocked by grouping the split by date.

Using `TimeSeriesSplit` on a pooled panel without grouping by date stops the first leak but *not* the second if rows are interleaved by instrument. Always group by date first, then order and purge in time.

Why This Matters

If you pool SPX, Nasdaq-100, and a handful of liquid single names to fatten your training set (a tempting move when one index gives you only $\sim 1,250$ usable daily rows over five years), a row-wise CV split will report a QLIKE improvement over HAR that simply does not exist out of sample. The leaked common factor f_t makes the model look like it forecasts crisis days well when it has merely memorised them through a sibling instrument. Group your purged folds by date, and your cross-validated QLIKE will finally track the held-out QLIKE you report in Section 17.5. The cross-sectional pooling design itself (which instruments to stack, and how to weight them) is developed in Chapter 18; this subsection is the validation guardrail that design must respect (see Section 17.7.4).

17.7.5 The One-Standard-Error Rule for Hyperparameter Selection

Purged CV gives you, for each candidate hyperparameter (the ridge penalty λ , the LightGBM tree depth, the number of HAR lags), a cross-validated QLIKE estimate. The obvious rule, picking the value with the lowest CV QLIKE, slightly over-fits.

The CV QLIKE at each λ is itself an *estimate*, averaged over a handful of purged folds, and it carries a standard error. The value that happens to minimise the average may be only a noise-width below several nearby, simpler (more regularised) candidates. The **one-standard-error rule** formalises “do not chase a difference smaller than your uncertainty.”

The One-Standard-Error Rule

Let $\overline{\text{QLIKE}}(\lambda)$ be the mean cross-validated QLIKE across the purged folds for penalty λ , and let $\text{SE}(\lambda)$ be the standard error of that mean across folds. Let $\lambda_{\min}$ be the value that minimises QLIKE. Instead of selecting $\lambda_{\min}$, select the *most regularised* (simplest) model whose CV loss is still within one standard error of the minimum:

$$\lambda_{1\text{SE}} = \max \left\{ \lambda : \overline{\text{QLIKE}}(\lambda) \leq \underbrace{\overline{\text{QLIKE}}(\lambda_{\min}) + \text{SE}(\lambda_{\min})}_{\text{within one SE of the best}} \right\} \quad (17.8)$$

where:

- λ indexes the regularisation strength (larger λ = simpler, more shrunk model),
- $\overline{\text{QLIKE}}(\lambda)$ is the mean purged-CV QLIKE at λ ,
- $\text{SE}(\lambda_{\min})$ is the across-fold standard error of the CV QLIKE at the minimiser,
- $\lambda_{1\text{SE}}$ is the largest (simplest) λ whose CV loss is statistically indistinguishable from the best.

In words: the notation $\{\lambda : \dots\}$ reads “the set of all λ such that ...”, and $\max$ picks the largest element of that set. So Equation (17.8) says: look at every λ whose average CV loss is no more than one standard error above the very best; among those, keep the largest λ : the simplest, most-shrunk model. The **standard error** here is how much the average CV loss would wobble if you reran the folds: your measurement noise on that average (the standard deviation of the per-fold losses divided by $\sqrt{\text{number of folds}}$). This “yields a simpler (more regularised) model that performs nearly as well” (the one-standard-error rule predates this source, originating with Breiman et al. (CART, 1984) and Hastie, Tibshirani & Friedman, ESL; de Prado, 2018, applies it in the purged-CV setting).

Why This Matters

When you tune the ridge-HAR penalty λ or the LightGBM depth by purged CV, picking $\lambda_{\min}$ on five-fold data routinely lands you on a setting that is barely better in CV but more reactive. That extra reactivity is the first thing to evaporate out of sample, showing up as $b > 1$ (over-reaction) in your Mincer–Zarnowitz regression (Section 17.4). The one-standard-error rule pushes you toward a smoother HAR-like forecast that holds up on the holdout. It also shrinks your trial count: by collapsing a band of indistinguishable settings to a single principled choice, you make fewer real decisions, which directly lowers N in the Deflated Sharpe Ratio (Section 17.10) and the Probability of Backtest Overfitting (Section 17.9).

17.8 Combinatorial Purged CV and the Distribution of OOS Performance

Purged K -fold CV (Section 17.7) hands you one number per fold. With $K = 5$ folds you get five cross-validated QLIKE estimates, and walk-forward evaluation gives you a single out-of-sample path, a thin basis for a claim as strong as “my LightGBM forecast robustly beats HAR.” The natural question: *how would my model have performed across many different out-of-sample histories drawn from the same five years of data, not just one?*

Combinatorics: N Choose k

The **binomial coefficient** $\binom{N}{k} = \frac{N!}{k!(N-k)!}$ counts the number of ways to choose k items from N without regard to order, where $N!$ (“ N factorial”) means $N \times (N-1) \times \dots \times 1$. For example, $\binom{6}{2} = 15$: there are 15 distinct pairs you can pick from six groups.

Combinatorial Purged Cross-Validation (CPCV), introduced by de Prado (2018), turns the thin distribution into a rich one. Instead of designating one test fold at a time, it partitions the sample into N contiguous groups and uses *every* combination of k of them as the test set, training (with purging and embargo) on the remaining $N - k$. Stitching these test segments together produces many full-length out-of-sample paths rather than one.

17.8.1 The Two Counting Formulas

The first formula counts how many train–test splits you fit.

$$\text{Number of splits} = \binom{N}{k} \quad (17.9)$$

where:

- N is the number of contiguous, non-overlapping time groups the sample is partitioned into,
- k is the number of groups held out as the test set in each split,
- $\binom{N}{k}$ is the total number of distinct ways to choose the test groups, hence the number of models you train.

The second formula counts how many complete out-of-sample *paths* those splits assemble into. Because each group is held out in several different splits, the test segments can be recombined into multiple non-overlapping sequences that each span the full history.

$$\varphi(N, k) = \underbrace{\frac{k}{N}}_{\text{fraction of the sample tested per split}} \binom{N}{k} \quad (17.10)$$

where:

- $\varphi(N, k)$ is the number of distinct backtest paths, each covering the entire time span exactly once,
- k/N is the fraction of the data that serves as test in any single split,
- $\binom{N}{k}$ is the split count from Equation (17.9).

In Plain English

Equation (17.9) asks “how many models do I fit?” and Equation (17.10) asks “how many complete alternative histories can I reconstruct from their out-of-sample pieces?” Here is why the path count is $(k/N)\binom{N}{k}$, derived from scratch. Each split tests a fraction k/N of the data (it holds out k of the N groups). To cover the whole history exactly once you therefore need N/k splits stitched edge-to-edge; that is one path. You fit $\binom{N}{k}$ splits in total, so the total test coverage is $\binom{N}{k}$ splits $\times$ k/N of the data each $= \frac{k}{N}\binom{N}{k}$ complete passes over the data, and one complete pass *is* one path. Because every group is held out in the same number of splits, those passes line up cleanly: the per-group test results tile into $\varphi(N, k)$ full-length paths with nothing left over. The payoff: instead of one out-of-sample QLIKE number, you get a whole *distribution* of out-of-sample QLIKE numbers, all extracted from the same five years of data: a far stronger basis for judging whether an edge is real or an artifact of the one path you happened to walk.

Worked Example: CPCV with $N = 6$, $k = 2$

Partition $T = 1,250$ daily observations into $N = 6$ contiguous groups of about 208 days each. Hold out $k = 2$ groups as the test set in each split.

Number of splits: $\binom{6}{2} = 15$.

Number of paths: $\varphi(6, 2) = \frac{2}{6} \times 15 = 5$. Each of the 15 splits covers $2/6$ of the data, so $15 \times \frac{2}{6} = 5$ full coverings of the whole history $= 5$ paths.

One specific split: test on groups $\{2, 5\}$, train on groups $\{1, 3, 4, 6\}$. After purging the

training days whose forward-RV labels overlap groups 2 and 5, and embargoing the days just after each test group, the model is fit on what remains of groups 1, 3, 4, 6 and scored by QLIKE on groups 2 and 5.

The full set of 15 test-group pairs:

{1, 2}	{1, 3}	{1, 4}	{1, 5}	{1, 6}
{2, 3}	{2, 4}	{2, 5}	{2, 6}	
{3, 4}	{3, 5}	{3, 6}		
{4, 5}	{4, 6}			
{5, 6}				

Each group is tested whenever it is paired with one of the other 5 groups, i.e. $\binom{5}{1} = 5$ times, so each of the six groups appears as a test group in exactly 5 of the 15 splits, and coverage is uniform. Those 15 out-of-sample segments reassemble into **5 non-overlapping backtest paths**, each scoring the model across all 1,250 days. You now report a distribution of 5 full-sample QLIKE values for “LightGBM minus HAR,” instead of the single path a walk-forward would have given you. If all 5 paths show LightGBM beating HAR, the edge is robust; if 2 of the 5 reverse, you have learned something a single walk-forward would have hidden.

Path	G1	G2	G3	G4	G5	G6	Test pairs
1	a	a	b	b	c	c	{1, 2} ^a {3, 4} ^b {5, 6} ^c
2	a	b	a	c	b	c	{1, 3} ^a {2, 5} ^b {4, 6} ^c
3	a	b	c	a	c	b	{1, 4} ^a {2, 6} ^b {3, 5} ^c
4	a	b	c	b	a	c	{1, 5} ^a {2, 4} ^b {3, 6} ^c
5	a	b	b	c	c	a	{1, 6} ^a {2, 3} ^b {4, 5} ^c

 test group (every cell) superscript = split-pair index

Figure 17.4: CPCV with $N = 6$, $k = 2$ reassembles the $\binom{6}{2} = 15$ test-group pairs into $\varphi(6, 2) = 5$ non-overlapping backtest paths. Each row is one backtest path; every group G1–G6 is tested exactly once per path, so each path scores the model on the full 1,250-day history. Within a path the six test cells split into three pairs (marked with superscripts a , b , c), and each pair is one train–test split (the two groups in the pair are tested while the other four train). The per-row label lists those three test pairs. Five paths give a distribution of full-sample QLIKE values instead of the single estimate a walk-forward provides.

Scaling up: with $N = 10$, $k = 2$ you get $\binom{10}{2} = 45$ splits and $\varphi(10, 2) = 9$ paths; with $N = 12$, $k = 2$, 66 splits and 11 paths. More groups buy a richer distribution at the cost of smaller test sets and more purging overhead. For daily volatility data, $N = 6$ to $N = 10$ with $k = 2$ is a sensible range (de Prado, 2018).

Why This Matters

With only five years of daily data, a single walk-forward split can make or break your headline “ X bps QLIKE improvement over HAR” on the luck of where the 2020 crash and the 2022 rate shock happen to fall. Run CPCV with $N = 6$, $k = 2$ and report the *distribution* of the per-path QLIKE reduction: its median is your headline number, and the spread across the 5 paths is your honesty check. A reviewer who sees “48 bps median, range 31–63 bps across 5 CPCV paths, all favouring the ML model” will trust you far more than one who sees a single 48 bps point estimate. Pair this with the Diebold–Mariano test (Section 17.5) on the pooled path residuals for a significance statement.

17.9 Probability of Backtest Overfitting (PBO)

CPCV gives you a distribution of out-of-sample performance. The next question is sharper and more uncomfortable: *when I pick the configuration that looks best in-sample (the feature set, the lag count, the hyperparameters), how often does that very choice turn out to be below-average out-of-sample?* If the answer is “most of the time,” your selection procedure is manufacturing overfit, not discovering signal. The **Probability of Backtest Overfitting** (PBO) measures exactly this, and it does so without assuming any model for returns (Bailey et al., 2014).

Logit of a Rank

The **logit** of a probability $\omega \in (0, 1)$ is $\ln(\omega/(1 - \omega))$. It maps the midpoint $\omega = 0.5$ to 0, values above the midpoint to positive numbers, and values below to negative numbers. Here ω will be a *relative rank*: the fraction of the other configurations a chosen model beats out-of-sample. For example, if your chosen model beats 7 of 9 others out-of-sample, $\omega = 7/9 \approx 0.78$, above the midpoint, so its logit is positive. A negative logit ($\text{logit}(\omega) < 0$) therefore means the chosen model landed in the bottom half out-of-sample. (Note: the symbol λ_c used below for the logit is unrelated to the ridge penalty λ from the one-standard-error rule earlier.)

17.9.1 Combinatorially Symmetric Cross-Validation (CSCV)

PBO is computed by **Combinatorially Symmetric Cross-Validation** (CSCV). The input is a $T \times N$ matrix of per-period performance, where T is the number of time periods and N is the number of configurations you tested. For volatility forecasting, the natural per-period entry is the (negative) QLIKE of each configuration’s forecast on each block of days, so that “higher is better” and the best-in-sample configuration is the one with the lowest in-sample QLIKE.

CSCV and PBO

Step 1. Partition the T time periods into S non-overlapping, contiguous blocks of roughly equal size (S even).

Step 2. Form all $\binom{S}{S/2}$ ways to split the blocks into an in-sample (IS) half and an out-of-sample (OOS) half. For each combination c :

1. the chosen $S/2$ blocks form the IS set, the rest form the OOS set;
2. compute each configuration’s IS and OOS performance (here, $-\text{QLIKE}$);
3. let n_c^* be the configuration with the best IS performance (lowest IS QLIKE);
4. compute its OOS **relative rank** $\omega_c \in (0, 1)$: the fraction of the N configurations it beats out-of-sample ($\omega_c = 1$ is best, 0 is worst).

Step 3. Take the logit of that rank:

$$\lambda_c = \ln\left(\frac{\omega_c}{1 - \omega_c}\right) \quad (17.11)$$

Step 4. The Probability of Backtest Overfitting is the fraction of partitions in which the in-sample winner fell into the bottom half out-of-sample:

$$\text{PBO} = \Pr(\lambda_c < 0) = \frac{\#\{c : \lambda_c < 0\}}{\binom{S}{S/2}} \quad (17.12)$$

where:

- S is the number of contiguous time blocks (typically 8–16; must be even),
- $\binom{S}{S/2}$ is the number of IS/OOS partitions (for $S = 10$, $\binom{10}{5} = 252$),

- n_c^* is the configuration that minimises in-sample QLIKE in partition c ,
- ω_c is the OOS relative rank of n_c^* (fraction of configurations it beats out-of-sample); ω_c , and hence λ_c , is the OOS rank of n_c^* , the single in-sample winner whose out-of-sample fate is the only one we track in partition c ,
- $\lambda_c < 0 \iff \omega_c < 0.5$, i.e. the IS winner ranked below the OOS median.

In Plain English

PBO answers one blunt question: “If I keep the model that looks best on the data I tuned on, how often does it actually rank below the median on fresh data?” A PBO of 0.60 means that 60% of the time, the configuration you would have selected by backtesting goes on to underperform half the alternatives out-of-sample. That is noise selection, not signal selection. Low PBO means your in-sample winners tend to stay winners; high PBO means your tuning procedure is fooling you.

Worked Example: CSCV by Hand: $S = 4$, Three Configurations

Take $S = 4$ blocks of days, so there are $\binom{4}{2} = 6$ ways to split them into an IS half (2 blocks) and an OOS half (the other 2). You have $N = 3$ candidate configurations. The table below is the $-QLIKE$ of each configuration on each block (higher = better; we negate QLIKE so a larger number is a better forecast):

Config	Block 1	Block 2	Block 3	Block 4
C_1	-1.40	-1.30	-1.55	-1.20
C_2	-1.35	-1.45	-1.30	-1.50
C_3	-1.50	-1.25	-1.45	-1.35

Walk one partition end to end. Take IS = blocks $\{1, 2\}$, OOS = blocks $\{3, 4\}$.

1. **IS sums:** $C_1 = -2.70$, $C_2 = -2.80$, $C_3 = -2.75$. The IS winner n_c^* is C_1 (least negative = best).
2. **OOS sums:** $C_1 = -2.75$, $C_2 = -2.80$, $C_3 = -1.80$. Out-of-sample, C_1 beats only C_2 , i.e. 1 of the other 2 configs.
3. **Relative rank:** $\omega_c = 1/2 = 0.50$ (it beat 1 of the 2 non-self configurations). Logit $\lambda_c = \ln(0.5/0.5) = 0$, right at the median.

Repeat for all 6 partitions (each pairs an IS half with its complementary OOS half), record λ_c for each, then

$$\text{PBO} = \frac{\#\{c : \lambda_c < 0\}}{6}.$$

If, say, the IS winner lands in the bottom half OOS in 4 of the 6 partitions, $\text{PBO} = 4/6 \approx 0.67$, a red flag. With only 3 configs and $S = 4$ this is a toy; in practice $S = 10$ gives 252 partitions and N runs into the dozens, but the mechanics are exactly these three steps repeated.

PBO Thresholds

Use PBO as a go/no-go diagnostic on your *selection process*, not on any single model:

- $\text{PBO} < 0.30$: encouraging; in-sample winners usually survive out-of-sample.
- $0.30 \leq \text{PBO} \leq 0.50$: caution; treat your selected configuration sceptically and widen the holdout.
- $\text{PBO} > 0.50$: red flag; the in-sample winner is more likely to underperform than

outperform, and your selection is overfit.

- PBO > 0.70: your strategy selection is almost certainly overfit. Do not proceed; cut the number of configurations or get more data (Bailey et al., 2014).

With $S = 10$ blocks the 252 IS/OOS partitions are enough to estimate PBO reliably.

Why This Matters

PBO is the right diagnostic for the central decision of this project: HAR versus an ML alternative, and *which* features and hyperparameters to give that ML model. Build the $T \times N$ matrix where the N columns are your candidate configurations (HAR, HAR-J, HARQ, ridge-HAR at several λ , LightGBM at several depths, each fed different feature subsets of jumps, signed RV, and IV/VRP) and each entry is that configuration's –QLIKE on a block of days. If PBO comes back above 0.50, the apparent superiority of your favourite ML configuration is an artifact of having tried many; the honest move is to shrink the candidate set (often back toward plain HAR) or extend the sample. A PBO below 0.30 is exactly the evidence you want next to your headline QLIKE reduction: it says the configuration you picked is not just the luckiest of many.

17.10 The Deflated Sharpe Ratio

Everything above evaluates *forecasts* of volatility. But volatility forecasts are often embedded in trading strategies (volatility targeting, variance risk premium trading; see Chapters 9 and 18). The standard performance metric for strategies is the Sharpe ratio. This section explains why raw Sharpe ratios are misleading when you have tried multiple strategies, and how to correct them.

17.10.1 The Multiple Testing Problem

Sharpe Ratios and Coin Flips

Suppose you flip 30 fair coins 250 times each and report only the coin with the most heads. That coin will show a “success rate” well above 50%, but it has no skill; you simply selected the luckiest coin. The same logic applies to Sharpe ratios: if you try 30 feature sets and report the best one, the expected maximum Sharpe ratio under the null (no skill) is not zero.

Bailey and de Prado (2014) derive the expected maximum Sharpe ratio under the null when N independent strategies are tested:

$$\mathbb{E}[\max_{i=1,\dots,N} \text{SR}_i] \approx \sqrt{2 \ln N} \quad (17.13)$$

where:

- N is the number of independent strategies (or feature sets, or hyperparameter combinations) tested,
- SR_i is the Sharpe ratio of strategy i under the null (all strategies have true $\text{SR} = 0$),
- The approximation comes from extreme value theory for Gaussian maxima.

For $N = 30$, this gives $\mathbb{E}[\max \text{SR}] \approx \sqrt{2 \ln 30} \approx 2.61$. A reported Sharpe of 1.5 after 30 trials is *below* what you would expect from pure luck.

17.10.2 The DSR Formula

The Deflated Sharpe Ratio adjusts the observed Sharpe ratio for the number of trials:

$$\text{DSR} = \Phi \left(\frac{(\widehat{\text{SR}} - \text{SR}_0)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{\text{SR}} + \frac{\hat{\gamma}_4-1}{4}\widehat{\text{SR}}^2}} \right) \quad (17.14)$$

where:

- $\text{DSR} \in [0, 1]$ is the probability that the observed Sharpe exceeds the multiple-testing threshold (higher is better),
- $\widehat{\text{SR}}$ is the observed (annualized) Sharpe ratio of the best strategy,
- $\text{SR}_0 = \sqrt{2 \ln N}$ is the expected maximum Sharpe under the null, with N = number of trials,
- T is the number of return observations,
- $\hat{\gamma}_3$ is the sample skewness of returns,
- $\hat{\gamma}_4$ is the sample kurtosis of returns,
- $\Phi(\cdot)$ is the standard normal CDF.

In Plain English

The DSR converts your observed Sharpe ratio into a probability: “What is the chance that this Sharpe is real, given how many strategies I tried?” It subtracts the luck threshold (SR_0) from your observed Sharpe, scales by sample size, and adjusts for skewness and fat tails in your returns. A DSR near 1 means your Sharpe is almost certainly genuine; a DSR near 0 means it is probably luck.

Bailey and Lopez de Prado (2014): The Deflated Sharpe Ratio

Bailey and de Prado (2014) show that ignoring the number of trials leads to systematic over-reporting of Sharpe ratios in backtested strategies. The DSR corrects for this by benchmarking the observed Sharpe against the expected maximum under the null. A DSR above 0.95 provides evidence that the strategy’s Sharpe ratio is unlikely to have arisen from multiple testing alone.

17.10.3 Worked Example

Worked Example: Deflated Sharpe Ratio

You test $N = 20$ volatility-timing strategies over $T = 1,260$ trading days (5 years of daily returns). The best strategy achieves $\widehat{\text{SR}} = 1.8$ (annualized). Returns have skewness $\hat{\gamma}_3 = -0.3$ and kurtosis $\hat{\gamma}_4 = 4.2$.

Step 1: Compute the null benchmark.

$$\text{SR}_0 = \sqrt{2 \ln 20} \approx \sqrt{5.99} \approx 2.45$$

Step 2: Compute the DSR numerator.

$$(\widehat{\text{SR}} - \text{SR}_0)\sqrt{T-1} = (1.8 - 2.45)\sqrt{1259} = (-0.65)(35.48) = -23.06$$

Step 3: Compute the DSR denominator.

$$\sqrt{1 - (-0.3)(1.8) + \frac{4.2-1}{4}(1.8)^2} = \sqrt{1 + 0.54 + 2.592} = \sqrt{4.132} \approx 2.033$$

Step 4: Compute DSR.

$$\text{DSR} = \Phi\left(\frac{-23.06}{2.033}\right) = \Phi(-11.35) \approx 0.000$$

The DSR is essentially zero. Despite an impressive-looking Sharpe of 1.8, the strategy does not survive correction for 20 trials. The expected maximum Sharpe under pure luck ($\text{SR}_0 = 2.45$) exceeds the observed Sharpe. You cannot reject the null that this is the luckiest of 20 random strategies.

What this tells you. $\text{DSR} \approx 0$ means your best strategy does not survive multiple-testing correction. This does not mean volatility forecasting is hopeless; it means you tested too many strategies relative to your sample size. Your options:

1. **Get more data.** A longer backtest period increases T , which widens the numerator of the DSR formula and makes it easier to clear the luck threshold.
2. **Test fewer strategies.** Use stronger priors about which feature sets to try, so N stays small and $\text{SR}_0 = \sqrt{2 \ln N}$ stays low.
3. **Pre-register.** Commit to a single strategy before backtesting, setting $N = 1$ and $\text{SR}_0 = 0$. The DSR then reduces to a standard Sharpe ratio test.
4. **Shift the claim.** Accept that you cannot make a statistical Sharpe claim and justify the strategy on economic grounds (e.g., cost-aware backtest, Chapter 18) rather than statistical grounds.

What Counts as a New Trial, and Why the Budget Is Tiny

Every configuration that *influenced your final choice* increments N , even the ones you discarded. Concretely, a new trial is created by each of:

1. a different **feature set** (adding jumps, signed $\text{RV}^+ / \text{RV}^-$, realized quarticity, IV/VRP, or cross-asset features);
2. a different **label / target** (1-day vs. 5-day vs. 22-day forward RV; level vs. log RV);
3. a different **model family** (HAR, HAR-J, HARQ, ridge-HAR, LightGBM, LSTM);
4. a different **hyperparameter** grid point (each λ , each tree depth, each lag count is its own trial);
5. a different **preprocessing** choice (winsorisation threshold, log transform, standardisation window);
6. a different **evaluation window** or holdout split that you peeked at before deciding;
7. every informal “quick look” whose result steered a later decision.

These multiply: 5 feature sets $\times$ 4 targets $\times$ 3 model families is $N = 60$ trials, not 3. Now weigh that against your data budget. Five years of daily observations is only about 1,250 rows (the T used in the DSR and Haircut examples above), and the held-out portion is a fraction of that. The False Strategy Theorem (Section 17.10) captures the squeeze: your ability to detect a real edge improves only as fast as $\sqrt{T}$ (more days = sharper evidence, since the noise in an average return shrinks like $1/\sqrt{T}$), but the bar luck sets rises as $\sqrt{2 \ln N}$ (more trials = a higher fluke to beat). With T frozen at $\approx 1,250$ days, every extra trial raises the bar you must clear without giving you any more evidence to clear it. The honest trial budget is therefore small, and the one-standard-error rule (Section 17.7.5) and a tight CPCV/PBO loop (Sections 17.8 and 17.9) exist precisely to keep N down. If you do not log every experiment, you cannot compute an honest DSR, which is why the experiment tracker is infrastructure you build *before* you start modeling.

17.10.4 The Haircut Sharpe Ratio in Detail

The DSR converts “how many strategies did I try?” into a single probability. The **Haircut Sharpe Ratio** of [Harvey and Liu \(2015\)](#) answers a complementary question that desks often find more concrete: *by how much should I shave my reported Sharpe to account for multiple testing?* Instead of a probability it returns an adjusted Sharpe and a percentage haircut, by routing through familiar p -value corrections.

p -Values and Multiple Testing

A **p -value** is the probability of observing a statistic at least as extreme as the one obtained, assuming the null (no effect) is true. When you run many tests, the chance that *at least one* returns a small p -value by luck grows fast. A **multiple-testing correction** inflates each p -value to compensate. Two families matter here: **family-wise error rate (FWER)** corrections control the probability of *even one* false positive, while **false discovery rate (FDR)** corrections control the *expected fraction* of false positives among rejections, a looser, more powerful target.

From Sharpe to a t -Statistic

The bridge between a Sharpe ratio and a hypothesis test is one line. Testing whether a Sharpe ratio differs from zero is the same as testing whether the mean excess return differs from zero: a t -test.

$$t = \text{SR} \times \sqrt{T} \quad (17.15)$$

where:

- SR is the *non-annualised* (per-period) Sharpe ratio of the strategy,
- T is the number of return observations,
- t is the t -statistic for the null H_0 : true Sharpe = 0.

In Plain English

Equation (17.15) says a Sharpe ratio is just a t -statistic in disguise: scale the per-period Sharpe by $\sqrt{T}$ and you have the very quantity a t -test produces. That means every tool statisticians built for “did I find a real effect among many tests?” (Bonferroni, Holm, BHY) applies directly to backtested Sharpe ratios. The longer your sample T , the larger the t for the same Sharpe, so the same edge becomes easier to defend with more data.

Why This Matters

If your vol-targeting overlay (Chapter 18) earns an annualised Sharpe of 1.5 on five years of daily data, first de-annualise it: Sharpe ratios are quoted annualised, and Sharpe scales with the square root of the number of periods, so to get the per-day Sharpe that Equation (17.15) needs you divide by $\sqrt{252}$ (there are ~ 252 trading days a year). That gives $\text{SR}_{\text{daily}} = 1.5/\sqrt{252} \approx 0.0945$, and Equation (17.15) gives $t = 0.0945\sqrt{1,250} \approx 3.34$. That is the single-test t *before* any multiple-testing penalty, the starting point for every correction below. Keep T honest: it is your number of return observations, not your number of forecasts.

Three Corrections, From Strictest to Loosest

[Harvey and Liu \(2015\)](#) apply three progressively less conservative corrections to the M tests you

ran.

Bonferroni (FWER)

Given M tests with p -values $p_1, \dots, p_M$, the Bonferroni-adjusted value is

$$p_i^{\text{Bonf}} = \min(M \cdot p_i, 1). \quad (17.16)$$

Every p -value is multiplied by the number of tests. This controls the family-wise error rate and is the most conservative correction.

Holm Step-Down (FWER)

Sort the p -values ascending, $p_{(1)} \leq \dots \leq p_{(M)}$. The adjusted values are

$$p_{(i)}^{\text{Holm}} = \min\left(\max_{j \leq i} [(M - j + 1) p_{(j)}], 1\right). \quad (17.17)$$

Holm multiplies the smallest p -value by M (matching Bonferroni), the next by $M - 1$, and so on down. It also controls FWER but is uniformly more powerful: it never rejects fewer hypotheses than Bonferroni.

In words: multiply the smallest p -value by M , the next-smallest by $M - 1$, and so on; then sweep left-to-right taking a running maximum ($\max_{j \leq i}$ means “the largest value seen so far”) so the adjusted values never decrease, and cap each at 1. The running maximum is just bookkeeping that keeps the adjusted sequence sorted.

Benjamini–Hochberg–Yekutieli (FDR)

Sort the p -values descending, $p_{(M)} \geq \dots \geq p_{(1)}$. The adjusted values are

$$p_{(i)}^{\text{BHY}} = \min\left(\frac{M \cdot c(M)}{i} p_{(i)}, 1\right), \quad c(M) = \sum_{j=1}^M \frac{1}{j} \quad (17.18)$$

where $c(M)$ is the M -th harmonic number (for $M = 20$, $c(20) \approx 3.60$). BHY walks the p -values from largest to smallest (the opposite order to Holm) because FDR control compares each p -value against a threshold that loosens as the rank increases. BHY controls the false discovery rate rather than the stricter FWER, making it the least conservative of the three.

In Plain English

The three corrections trade strictness for power. Bonferroni is the bouncer who treats every test as a fresh chance to be fooled and so penalises all of them by the full count M : safe but harsh. Holm is the same bouncer with a memory: once a test clears the strictest bar, the next one faces a slightly easier bar ($M - 1$, then $M - 2$, $\dots$), so it rejects at least as much as Bonferroni and often more. BHY changes the goal entirely: rather than “allow essentially no false positives,” it accepts a controlled *fraction* of false discoveries, which is the right stance when you expect several genuine winners among many candidates.

Computing the Haircut

The haircut is the percentage by which the correction shrinks your Sharpe.

$$\text{Haircut \%} = \frac{\text{SR}_{\text{reported}} - \text{SR}_{\text{adjusted}}}{\text{SR}_{\text{reported}}} \times 100 \quad (17.19)$$

where:

- $\text{SR}_{\text{reported}}$ is the raw (annualised) Sharpe of the strategy,
- $\text{SR}_{\text{adjusted}}$ is the Sharpe implied by the *corrected* p -value: convert the adjusted p back to a t via $\Phi^{-1}(1 - p^{\text{adj}}/2)$, then invert Equation (17.15). Here Φ^{-1} is the inverse of the normal CDF (given a probability it returns the matching z -score, `scipy.stats.norm.ppf`), and the $1 - p/2$ reflects a two-sided test, splitting the probability between both tails,
- if the adjusted p -value is no longer significant at level α , the adjusted Sharpe is effectively zero and the haircut is 100%.

In Plain English

The haircut translates a p -value penalty back into the unit a portfolio manager cares about. “Your Sharpe is 1.5, but after admitting you tried 20 strategies, the defensible Sharpe is 1.07, a 28.5% haircut” is a sentence a desk head understands instantly, where a probability might wash over them. A 100% haircut is the bluntest possible verdict: once corrected, the edge is statistically indistinguishable from zero.

Worked Example: Haircut Sharpe with 20 Tests

You report an annualised $\text{SR} = 1.50$ from $T = 1,250$ daily observations, having tested $M = 20$ configurations of a volatility-timing overlay.

Step 1: Sharpe to t . Daily $\text{SR} = 1.50/\sqrt{252} = 0.0945$, so $t = 0.0945\sqrt{1,250} = 3.34$ (Equation (17.15)).

How to get the numbers below: Φ and Φ^{-1} are normal-distribution lookups; use `scipy.stats.norm.cdf` / `.ppf`, or a z -table. We run two-sided tests, so the p -value uses a factor of 2 and the inverse uses $1 - p/2$.

Step 2: single-test p . Two-sided: $p = 2(1 - \Phi(3.34)) \approx 0.00084$.

Step 3: Bonferroni. $p^{\text{Bonf}} = 20 \times 0.00084 = 0.0168$, still significant at $\alpha = 0.05$. Back out the t : $\Phi^{-1}(1 - 0.0168/2) = 2.39$, giving adjusted annualised $\text{SR} = (2.39/\sqrt{1,250})\sqrt{252} = 1.07$. Haircut $(1.50 - 1.07)/1.50 = 28.5\%$.

Step 4: Holm. For the top-ranked of 20 strategies the Holm penalty equals Bonferroni's ($20 \times 0.00084 = 0.0168$), so the best strategy is haircut by the same 28.5%.

Step 5: BHY. With $c(20) = 3.60$: $p^{\text{BHY}} = (20 \times 3.60/1) \times 0.00084 = 0.0605$, now marginally *insignificant* at $\alpha = 0.05$. Adjusted $t = \Phi^{-1}(1 - 0.0605/2) = 1.88$, giving adjusted $\text{SR} = 0.84$. Haircut $(1.50 - 0.84)/1.50 = 43.7\%$.

Method	Adjusted p	Adjusted SR	Haircut %
None (single test)	0.00084	1.50	0%
Bonferroni	0.0168	1.07	28.5%
Holm (rank 1)	0.0168	1.07	28.5%
BHY	0.0605	0.84	43.7%

What this tells you. A Sharpe of 1.50 after 20 tests survives the two FWER corrections (Bonferroni and Holm keep it significant at 5%) but is rendered borderline by the FDR-controlling BHY. The “shave it in half” rule of thumb practitioners sometimes apply is roughly right for BHY here but too harsh for Bonferroni. The lesson is not which correction is “correct”; it is that a single raw Sharpe means nothing until you state M and pick a correction.

Harvey, Liu, and Zhu (2016): The $t > 3.0$ Hurdle

Cataloguing hundreds of factors from the literature, [Harvey et al. \(2016\)](#) show that the appropriate t -statistic hurdle for a newly proposed factor, after multiple-testing correction, is $t > 3.0$, not the conventional 1.96. Under that bar only a handful of the catalogued return predictors survive. For your project the analogue is direct: a vol-timing Sharpe of 1.0 on five years of daily data gives $t = (1.0/\sqrt{252})\sqrt{1,250} \approx 2.23$, comfortably below 3.0. Most “edges” you find by searching feature sets will not clear the hurdle, which is the honest default expectation.

DSR and Haircut Sharpe Are Complementary

DSR (Section 17.10) and Haircut Sharpe attack the same multiple-testing problem from opposite ends: DSR returns a probability that your Sharpe beats the luck threshold, while Haircut Sharpe returns the Sharpe that remains after the penalty. Report both. A result is defensible when $\text{DSR} > 0.95$ and the haircutted Sharpe is still economically meaningful (say, above 0.5 annualised after transaction costs). Both depend on an honest M , which is why the experiment log (Section 17.10) is infrastructure, not bookkeeping.

17.11 What Doesn't Work

You now have the full evaluation toolkit: a loss function (QLIKE), a diagnostic (MZ), a pairwise test (DM), a multi-model filter (MCS), a leakage-proof CV procedure, and a multiple-testing correction (DSR). This section catalogs the mistakes these tools are designed to prevent, so you can recognize them in other people's work and avoid them in your own.

Evaluation Pitfalls

1. **Random K-fold on time series.** Shuffling observations before splitting destroys temporal structure. Reported accuracy is inflated; real performance collapses. Always use purged CV or walk-forward (Section 17.7).
2. **Naive out-of-sample R^2 without statistical tests.** “Our model achieves OOS $R^2 = 3.2\%$ versus the benchmark's 2.8%.” Without a DM test (Section 17.5) or MCS (Section 17.6), you do not know whether 0.4 percentage points is signal or noise.
3. **Training on one regime, testing on another.** Training on 2015–2019 (low volatility) and testing on 2020 (COVID) is not a fair evaluation; it is a regime-change stress test. Useful, but do not confuse it with a general OOS evaluation.
4. **Look-ahead in feature construction.** Using day- t VIX close to predict day- t realized variance is look-ahead bias: VIX is not known until 4:15 PM, while RV accumulates throughout the day. All features must be known *before* the forecast is made.
5. **Reporting tiny improvements without economic significance.** Beating HAR by 0.5% in QLIKE is unlikely to translate to meaningful PnL after transaction costs. Always pair statistical significance (DM test) with economic significance (cost-aware backtest; Chapter 18).
6. **Ignoring forecast variance.** A model that is right on average but has high forecast variance is dangerous for volatility targeting. Two models with identical QLIKE can differ dramatically in how “jumpy” their forecasts are. Report forecast autocorrelation and turnover alongside loss metrics.

Survivorship Bias in the Instrument Universe

A seventh pitfall, distinct from the six above because it corrupts the data *before* any model sees it: **survivorship bias**. If your universe contains only instruments that *still trade today* (indices and tickers that survived the sample), you have silently dropped every name that was delisted, merged, or blew up. Those vanished instruments are disproportionately the ones that experienced the most extreme volatility, defaults, and crashes, so a survivorship-filtered panel systematically *understates* tail volatility and makes any forecaster look better-calibrated on crisis days than it would have been in real time, exactly where forecasting matters most. When pooling assets (Section 17.7.4), assemble the universe from a *point-in-time* constituent list (the instruments tradeable *as of* each historical date), not from today's survivors: include Lehman Brothers or Wachovia as they existed pre-2008, not just the banks still listed today, since the failures are exactly the high-volatility names you must not erase. If a fully point-in-time universe is not available, note the limitation explicitly and do not overstate how well your forecaster would have handled the names that did not survive (de Prado, 2018).

17.11.1 Lookahead Bias: A Taxonomy of Four Sources

Item 4 in the list above (look-ahead in feature construction) deserves a full subsection because lookahead bias is the single most destructive error in financial ML. A contaminated model shows excellent in-sample performance that vanishes out of sample, wasting weeks of development time before the bug is identified.

Lookahead bias occurs whenever a feature used at prediction time contains information that would not have been available at the moment the forecast was made. In volatility forecasting, there are four distinct sources, each with its own failure mode.

Source 1: Realized Measures

Realized variance (Chapter 2) is computed from intraday returns over a trading day. The danger is that the boundary between “today” and “tomorrow” is not always clean.

Realized Measure Leakage

Suppose you forecast RV_{t+1} (tomorrow's realized variance) using features that include RV_t (today's realized variance). If you compute RV_t using returns from 9:30 AM to 4:00 PM, but the last intraday return spans 3:55–4:00 PM, that return reflects information that overlaps with the overnight period leading into day $t + 1$. In tick-level data, the problem is worse: the last trade might occur at 4:00:02 PM, technically after the close. Even a few seconds of overlap contaminates the feature with forward-looking information.

Source 2: Microstructure Features

Microstructure features (Chapter 3) are derived from limit order book (LOB) data, trade-and-quote streams, and intraday volume profiles. They are particularly vulnerable to lookahead because they are computed over intraday windows whose boundaries must be carefully aligned with the forecast target.

LOB Feature Leakage

Suppose you compute the **Volume-Synchronized Probability of Informed Trading (VPIN)** over the full trading day from 9:30 AM to 4:00 PM and use it as a feature for

forecasting RV_{t+1} . The last hour of VPIN reflects order flow patterns driven by information that will affect the overnight return and the opening of day $t+1$. For example, a large informed seller at 3:45 PM depresses the close and widens the spread; this information is not “known” to a forecaster who must act at 3:00 PM. Full-day microstructure features effectively let you “see” information that accumulates between your forecast time and the close.

Source 3: Options Surface

Implied volatility from the options surface (Chapters 5 and 18) is a forward-looking feature by design: it encodes the market’s expectation of future volatility. This makes it powerful but also dangerous, because the surface updates throughout the day in response to new information.

Implied Volatility Leakage

Suppose you use the 3:30 PM VIX level as a feature for forecasting next-day RV. If a company announces earnings at 4:05 PM, the options market will begin pricing in the expected earnings move before the announcement: implied volatility rises during the last hour of trading. The 3:30 PM VIX already reflects this anticipation. A forecaster using this feature has access to information about the expected earnings-day volatility spike that is not “available” in the sense of a genuine real-time forecast made at, say, the previous close. More subtly, the SPX implied volatility surface at 3:30 PM reflects the full day’s information flow, including any macro data releases, Fed communications, or geopolitical events that occurred during the day.

Source 4: Cross-Asset Features

Cross-asset features (Chapter 18) use data from other markets (e.g., European equities, commodities, currencies, Treasuries) to predict volatility in the target asset (e.g., SPX). The complication is that these markets operate on different schedules, creating timestamp misalignment that can hide lookahead bias.

Cross-Asset Timing Leakage

Suppose you use the EURO STOXX 50 realized variance as a feature for predicting SPX RV_{t+1} . European markets close at 4:30 PM CET (10:30 AM ET). If you label the European close as “day t ” data, it is available before the US close on day t at 4:00 PM ET, which is fine. But if the European data is labeled “day t ” and the US target is also “day t ,” you may inadvertently use European close data that overlaps with the US target window. Worse, some cross-asset data sources (e.g., Asian markets) close well before the US open; using “day t ” Asian close data for a US “day $t+1$ ” forecast is correct, but using it for a US “day t ” forecast means the Asian data is stale and the time alignment is ambiguous.

Summary

Table 17.1 collects all four sources with their specific pitfalls and prevention rules.

Why This Matters

Your pipeline ingests data from at least four different source types (tick data, LOB depth, options surface, cross-asset indices), each with its own timestamp conventions. Build the lookahead prevention into your data pipeline as hard constraints, not as documentation

Table 17.1: Lookahead bias taxonomy for volatility forecasting pipelines. Each source represents a distinct mechanism by which future information can leak into features.

Source	Pitfall	Prevention Rule
Realized measures	Target-day intraday returns leak into features via boundary overlap	Features use data only up to day t close; enforce with programmatic timestamp assertions
Microstructure	Full-day LOB features (VPIN, spreads) include close-period information	Truncate all LOB features at a fixed cutoff (e.g., 3:00 PM) before close
Options surface	Intraday IV changes reflect target-day information (e.g., earnings anticipation)	Use previous-day close IV or a fixed morning snapshot only
Cross-asset	Mixed frequencies and time-zone misalignment hide temporal overlap	Align all cross-asset features to a single information cutoff (previous-day US close)

that you hope developers will follow. Specifically: (1) add a `max_timestamp` column to every feature table and assert it precedes the forecast cutoff; (2) run a nightly integration test that checks no feature for RV_{t+1} uses data with timestamp $> t$ close; (3) when adding new features, require the contributor to specify the information cutoff in the feature registry. A single lookahead bug can invalidate months of work and is almost impossible to detect from QLIKE numbers alone: the contaminated model simply looks “better than it should.”

17.12 Putting It All Together: An Evaluation Workflow

You now have all the individual tools. This section assembles them into a practical workflow you should follow for every volatility forecasting experiment.

Adding CPCV and PBO to the Workflow

The nine-step flow in Figure 17.5 is the backbone; the overfitting diagnostics from this chapter slot in around the tuning and model-selection steps:

- **Upgrade step 3 (tuning).** Replace plain purged K -fold with **CPCV** ($N = 6$, $k = 2$): instead of one cross-validated QLIKE you obtain a *distribution* across 5 backtest paths (Section 17.8). When pooling instruments, group the folds by date (Section 17.7.4); select hyperparameters with the one-standard-error rule (Section 17.7.5).
- **Insert a new check after step 7 (MCS).** Compute **PBO** on the $T \times N$ matrix of per-block $-$ QLIKE (negated so that, like a return, higher is better) across all configurations you tried (Section 17.9). Gate: proceed only if $PBO < 0.50$ (ideally < 0.30).
- **Augment step 8 (DSR).** On the strategy branch, report the **Haircut Sharpe** (Bonferroni, Holm, BHY) alongside the DSR (Section 17.10.4); require $DSR > 0.95$ and a haircutted Sharpe still above 0.5 after costs.

The extended order is therefore: reserve holdout $\rightarrow$ log experiments $\rightarrow$ CPCV tuning (grouped by date, 1-SE selection) $\rightarrow$ QLIKE/MSE $\rightarrow$ MZ $\rightarrow$ DM $\rightarrow$ MCS $\rightarrow$ **PBO gate** $\rightarrow$ DSR & Haircut (if a strategy) $\rightarrow$ report. The two new diagnostics are not optional polish: CPCV tells you whether the edge is stable across histories, and PBO tells you

whether your *selection process* itself is honest.

17.13 Summary

- **MSE is proxy-robust** but over-penalizes extreme variance days, making it a poor primary metric for volatility (Section 17.2).
- **QLIKE is the preferred primary loss** for volatility forecast evaluation. It is proxy-robust *and* less sensitive to outliers than MSE (Patton, 2011) (Section 17.3).
- **QLIKE and MSE are the only two robust losses.** Other common losses (MAE, HMSE) can reverse model rankings when the volatility proxy is noisy (Section 17.3).
- **Retransformation bias** arises when exponentiating log-space forecasts back to levels. Apply the correction $\exp(\hat{y} + \hat{\sigma}_\varepsilon^2/2)$ to avoid systematic under-prediction that grows with forecast uncertainty (Patton, 2011) (Section 17.3.4).
- **Mincer–Zarnowitz regressions** diagnose bias ($a \neq 0$) and inefficiency ($b \neq 1$) in forecasts. Use HAC standard errors (Section 17.4).
- **The Diebold–Mariano test** determines whether the difference in loss between two models is statistically significant, accounting for serial correlation via HAC standard errors (Diebold and Mariano, 1995) (Section 17.5).
- **The Model Confidence Set** compares all models simultaneously, returning the set of models that are statistically indistinguishable from the best. It controls familywise error (Hansen et al., 2011) (Section 17.6).
- **Purged K-fold CV with embargo** prevents information leakage in time series cross-validation by removing overlapping labels (purge) and serial-correlation buffers (embargo) (de Prado, 2018) (Section 17.7).
- **Random K-fold on time series is catastrophic.** It inflates reported accuracy by training on future data (Section 17.7).
- **The Deflated Sharpe Ratio** corrects backtested Sharpe ratios for the number of strategies tested. Every experiment counts as a trial (Bailey and de Prado, 2014) (Section 17.10).
- **Log every experiment.** You cannot compute an honest DSR without knowing N (Section 17.10).
- **Statistical significance is necessary but not sufficient.** Pair DM tests with economic significance: does the improvement survive transaction costs? (Section 17.11).
- **Lookahead bias has four sources** in volatility pipelines: realized measures, microstructure features, options surface, and cross-asset data. Each requires a specific prevention rule; build these as hard constraints in your pipeline, not as documentation (Section 17.11.1).
- **Follow the full workflow** (Figure 17.5): reserve holdout, log experiments, purged CV, QLIKE, MZ, DM, MCS, DSR.

17.13.1 Key Results Recap

Tool	What It Does	Key Reference
QLIKE loss	Primary loss function; proxy-robust, outlier-resistant	Patton (2011)
MSE loss	Secondary loss; proxy-robust but outlier-sensitive	Patton (2011)
Mincer–Zarnowitz	Diagnoses forecast bias and inefficiency	Mincer and Zarnowitz (1969)
Diebold–Mariano	Pairwise statistical test of loss difference	Diebold and Mariano (1995)
Model Confidence Set	Multi-model comparison; returns surviving set	Hansen et al. (2011)
Purged K-fold CV	Time-series CV without look-ahead	de Prado (2018)
Deflated Sharpe Ratio	Corrects Sharpe for multiple testing	Bailey and de Prado (2014)
Haircut Sharpe	Alternative multiple-testing correction	Harvey and Liu (2015)

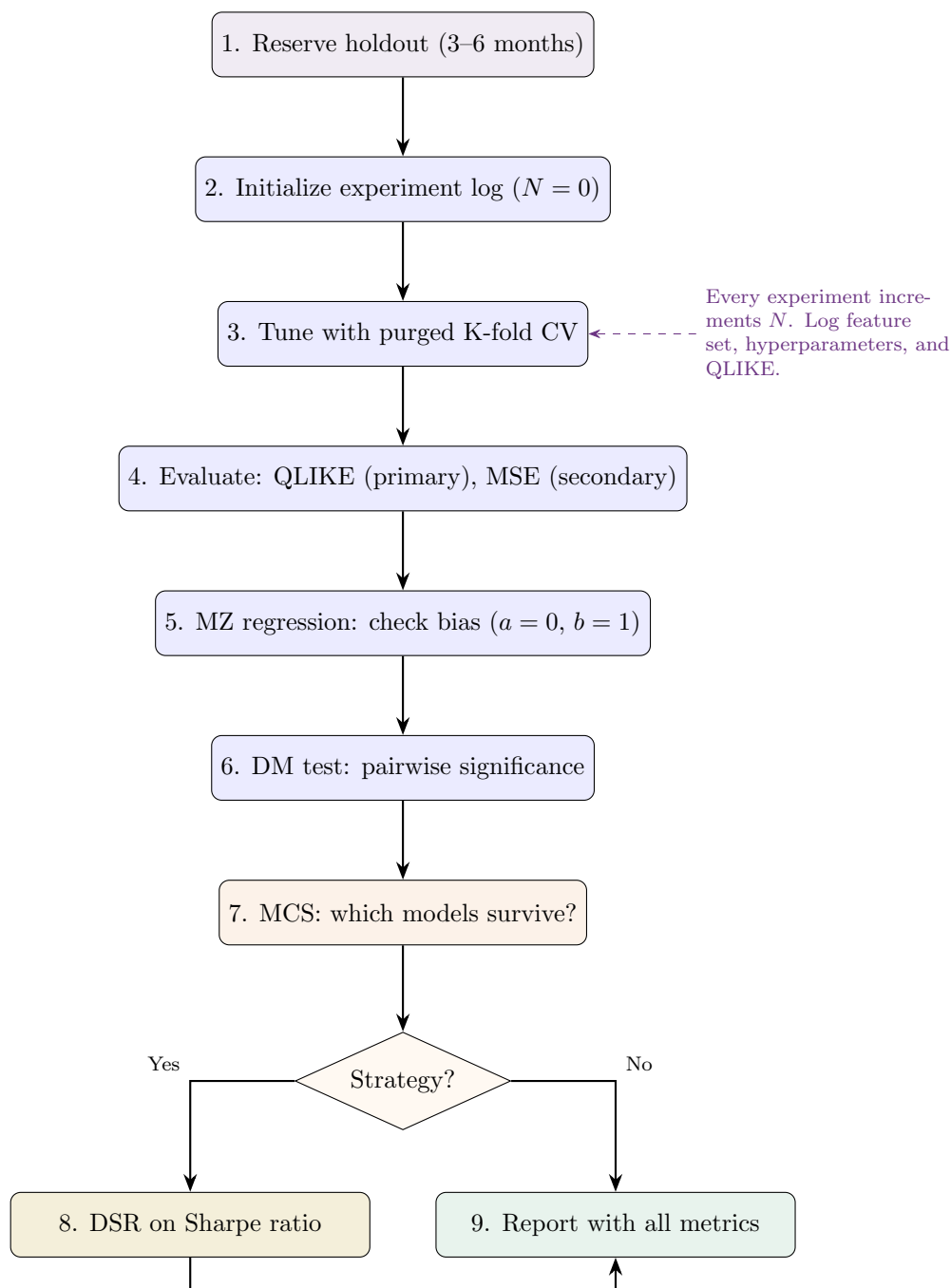


Figure 17.5: Evaluation workflow for volatility forecasting. Reserve the holdout first; log every experiment; use purged CV for tuning; evaluate with QLIKE and MZ; compare with DM and MCS; deflate the Sharpe if the forecast feeds a strategy.

Chapter 18

Practical Applications and Project Directions

A model that beats HAR by 8% on QLIKE is scientifically interesting. But a desk head wants to know: how much Sharpe does that buy me? How much P&L does it add to my book?

18.1 Volatility Targeting: The Simplest Economic Value Test

Background

This section requires familiarity with portfolio returns and the Sharpe ratio (any introductory finance text), as well as realized volatility forecasts from Chapters 6–14.

Volatility targeting is the simplest and most widely used application of a vol forecast in systematic investing. The idea: size your position inversely proportional to forecast vol, so that portfolio risk stays roughly constant over time.

18.1.1 The EWMA Baseline

The workhorse volatility estimate used by the vast majority of systematic funds for position sizing is plain **exponentially weighted moving average (EWMA)** smoothing, not GARCH or HAR:

$$\hat{\sigma}_t^2 = (1 - \delta) r_{t-1}^2 + \delta \hat{\sigma}_{t-1}^2, \quad (18.1)$$

where δ is chosen so the half-life matches approximately 20 to 60 trading days. Its popularity stems from simplicity, low latency, and the fact that it requires exactly one parameter.

In Plain English

Tomorrow’s variance estimate is a weighted blend of today’s squared return (the “news”) and today’s variance estimate (the “memory”). A high δ means slow adaptation (the estimate barely moves day-to-day); a low δ means it jumps quickly with every new return, reacting fast but also picking up noise.

18.1.2 The Volatility-Targeting Formula

Given a forecast $\hat{\sigma}_t$ (annualized), the **volatility-targeted weight** is:

$$w_t = \frac{\sigma_{\text{target}}}{\hat{\sigma}_t}. \quad (18.2)$$

The portfolio return is then $r_t^{\text{VT}} = w_t \cdot r_t$. When forecast vol is high, $w_t < 1$ (reduce position). When forecast vol is low, $w_t > 1$ (lever up). The result is a return stream with approximately constant realized volatility equal to σ_{target} .

Why Vol-Targeting Adds Sharpe

Moreira and Muir (2017) showed that vol-targeting adds approximately 0.3 Sharpe ratio across equity indices, currencies, and commodities. The mechanism: by reducing exposure before drawdowns, vol-targeting truncates the left tail. A better vol forecast means you cut exposure earlier and more precisely.

18.1.3 Worked Example: Vol-Targeting the S&P 500

Worked Example: Vol-Targeting the S&P 500

Setup. Target: $\sigma_{\text{target}} = 10\%$ annualized. Compare three forecasts:

- (a) **EWMA** (half-life 20 days): the baseline every fund already uses.
- (b) **HAR**: three-component model from Chapter 6.
- (c) **ML (best model)**: your contribution.

Scenario. Days 1 to 15 are calm (annualized RV $\approx 12\%$). On day 16, a sell-off begins and RV spikes to 35% annualized. The ML model detects the regime change on day 14 (microstructure features shift); HAR reacts on day 16; EWMA reacts gradually through days 17 to 20.

Transition period (days 13 to 20):

Day	True RV (%)	$\hat{\sigma}_{\text{EWMA}}$ (%)	$\hat{\sigma}_{\text{HAR}}$ (%)	$\hat{\sigma}_{\text{ML}}$ (%)	w_{EWMA}	w_{ML}
13	11.8	12.0	12.1	12.2	0.83	0.82
14	12.1	12.0	12.0	18.5	0.83	0.54
15	12.4	12.1	12.2	22.0	0.83	0.45
16	28.5	12.3	18.0	28.0	0.81	0.36
17	33.0	15.8	25.5	32.0	0.63	0.31
18	35.2	19.4	30.0	34.5	0.52	0.29
19	34.0	22.5	32.0	33.8	0.44	0.30
20	30.5	24.8	31.5	30.0	0.40	0.33

Key observation. On day 16 (first big drawdown), EWMA still holds $w = 0.81$ (nearly full position), while the ML model has already cut to $w = 0.36$. The cumulative drawdown reduction from the ML model's early detection is substantial.

Summary statistics (full 252-day backtest):

Metric	EWMA	HAR	ML
Ann. Return (%)	8.2	8.5	8.9
Ann. Volatility (%)	10.8	10.3	10.1
Sharpe Ratio	0.76	0.82	0.88
Max Drawdown (%)	-12.4	-9.8	-7.1
Calmar Ratio	0.66	0.87	1.25

The ML model adds 0.12 Sharpe over the baseline and reduces the maximum drawdown by over 40%, primarily because it detects regime shifts one to two days earlier.

18.1.4 Connection to Time-Series Momentum

Moskowitz et al. (2012) use a related framework for time-series momentum across 58 futures: signal = sign of 12-month return, position size = $40\%/\hat{\sigma}_t$. The vol forecast enters the denominator. Every percentage improvement in your forecast precision tightens the position-sizing, reducing both crash risk and unnecessary leverage. This is the mechanism by which better QLIKE translates to better risk-adjusted returns in a TSMOM book.

Forecast Failure in Crises

Vol targeting assumes volatility is predictable but returns are not. If your forecast systematically underpredicts during crises (all ML models trained on normal data underpredict COVID-style jumps), vol targeting will keep positions too large going into the drawdown. Mitigation: ensemble your ML forecast with a simple $\max(\text{ML}, 1.5 \times \text{EWMA})$ floor during high-uncertainty regimes, or cap w_t at a maximum leverage ratio (e.g., $w_t \leq 2.0$).

The Key Deliverable Table

For your internship deliverable, the key table is:

1. Run vol-targeted long-SPX using each model.
2. Report annualized return, vol, Sharpe, max drawdown, and Calmar ratio.
3. Compute Sharpe improvement per 1% QLIKE improvement.

This single table communicates economic value in the language every systematic desk speaks.

18.2 Trading Costs, Turnover, and Net Economic Value

Background

This section builds directly on the vol-targeting strategy of Section 18.1 and the weight $w_t = \sigma_{\text{target}}/\hat{\sigma}_t$ from Equation (18.2). It also assumes the QLIKE loss and Diebold–Mariano test from Chapter 17 (Sections 17.3 and 17.5). One **basis point** (bp) = $0.01\% = 0.0001$; a 5 bp cost means you pay 0.05% of the traded notional each time you adjust the position.

What a Sharpe Ratio Means

The 0.76 vs. 0.88 gap below is a meaningful but not enormous edge (the full Sharpe definition is collected in Table 18.1).

The Sharpe ratios in the worked-example table of Section 18.1.3 (0.76 for EWMA, 0.88 for ML) were all computed *gross*: they ignored the cost of trading. Chapter 17 required that a real QLIKE improvement “survive transaction costs” and that turnover be reported alongside any economic-value claim. It answers the question a desk head asks the instant you show a Sharpe improvement: *how much of that survives once I have to pay the spread every time the forecast moves the position?*

The tension: the extra reactivity that wins on QLIKE is exactly what raises trading, so a jumpier forecast can win gross and lose net.

18.2.1 Net P&L on the Vol-Targeting Weight

The gross return of the vol-targeted strategy, from Section 18.1.2, is $r_t^{\text{VT}} = w_t \cdot r_t$, where $w_t = \sigma_{\text{target}}/\hat{\sigma}_t$ and r_t is the underlying SPX return. Trading costs are proportional to the *change* in the weight, because you only pay the spread on the notional you actually buy or sell:

$$\text{PnL}_t^{\text{net}} = \underbrace{w_t \cdot r_t}_{\text{gross return}} - \underbrace{|w_t - w_{t-1}| \cdot c}_{\text{cost of rebalancing}}, \quad (18.3)$$

where:

- $w_t = \sigma_{\text{target}}/\hat{\sigma}_t$: the vol-targeted weight held through day t (set at the close of day $t - 1$ from information up to $t - 1$, so no look-ahead).
- $|w_t - w_{t-1}|$: the absolute change in the weight, i.e. the fraction of notional you must trade to move from yesterday's position to today's.
- c : the **half-spread cost** per unit of traded notional, expressed in the same units as r_t . A full round-trip (out and back) costs $2c$.

If the forecast is unchanged, $w_t = w_{t-1}$ and you pay nothing. If a vol spike pushes the weight down over several days (as the ML model did in Section 18.1.3, from 0.82 on day 13 to 0.36 on day 16), the cost is charged on each day's move, not on the cumulative drop. The largest single-day cut in that table is on day 14 ($|0.54 - 0.82| = 0.28$ of notional, costing $0.28c$); the day-16 move itself is only $|0.36 - 0.45| = 0.09$.

Worked Example: Gross vs. Net Over Eight Days of Vol-Targeting

We recast the day-by-day P&L onto the vol-targeting weights from the transition table in Section 18.1.3. Take the ML model's weights w_t^{ML} from that table, assume $w_{12} = 0.82$ entering the window, and an SPX-futures half-spread $c = 1$ bp. **All P&L columns below are in percent**, so to keep the arithmetic in one unit we write that 1 bp cost as 0.01% per unit of weight traded. The underlying daily returns r_t are the SPX moves over the sell-off.

t	w_t^{ML}	r_t (%)	$ \Delta w_t $	Gross (%)	Net (%)
13	0.82	+0.20	0.00	+0.164	+0.164
14	0.54	-0.10	0.28	-0.054	-0.057
15	0.45	-0.30	0.09	-0.135	-0.136
16	0.36	-2.10	0.09	-0.756	-0.757
17	0.31	-1.80	0.05	-0.558	-0.559
18	0.29	+0.90	0.02	+0.261	+0.261
19	0.30	+1.40	0.01	+0.420	+0.420
20	0.33	+0.80	0.03	+0.264	+0.264
			$\Sigma = 0.57$	$\Sigma = -0.394$	$\Sigma = -0.400$

Reading the table. The protection came not from one day but from a step-by-step cut: starting at 0.82, the forecast shrank the weight to 0.54, then 0.45, then 0.36, so that by day 16, the day of the -2.1% drop, only 0.36 of the book was exposed. The gross loss that day was therefore $0.36 \times 2.1\% = 0.76\%$ instead of a near-full 2.1%. Over the eight days, total traded notional is $\sum |\Delta w_t| = 0.57$, so total cost at $c = 1$ bp is $0.57 \times 0.01\% = 0.006\%$. Costs consumed only $0.006/0.394 \approx 1.5\%$ of the (negative) gross P&L. Even an aggressive regime-detecting forecast, run on a 1 bp instrument, barely registers costs over a short window. The cost problem is invisible day-by-day; it accumulates over a full year of small daily re-weights, which the next subsection makes precise.

18.2.2 Turnover and the Sharpe Drag

The single number that summarizes how expensive a forecast is to run is its **turnover**: the average amount of notional it forces you to trade per day. For the vol-targeting weight, daily turnover is the absolute day-on-day change in the weight,

$$\bar{\tau} = \frac{1}{T} \sum_{t=1}^T |w_t - w_{t-1}|, \quad \text{annualized turnover} = 252 \times \bar{\tau}, \quad (18.4)$$

where:

- $|w_t - w_{t-1}|$: one day's traded notional, as a fraction of the book.
- T : the number of days in the backtest.
- the factor 252: the number of trading days per year, converting a daily average into an annual trading volume.

A strategy with $\bar{\tau} = 0.05$ resizes 5% of its position on an average day; over 252 trading days that adds up to $252 \times 0.05 = 12.6$ times the size of the whole book traded in a year.

Turnover matters because it converts, almost mechanically, into a deduction from the Sharpe ratio. Annualizing the cost term of Equation (18.3) and dividing by the volatility of the strategy's returns gives the **Sharpe drag**:

$$\text{SR}^{\text{net}} \approx \underbrace{\text{SR}^{\text{gross}}}_{\text{paper Sharpe}} - \underbrace{\frac{252 \cdot \bar{\tau} \cdot c}{\sigma_{\text{ret}}}}_{\text{cost drag}}, \quad (18.5)$$

where:

- SR^{gross} : the annualized Sharpe of the gross vol-targeted return stream $w_t r_t$.
- $\bar{\tau}$: average daily turnover from Equation (18.4).
- c : the half-spread cost per unit notional.
- σ_{ret} : the annualized volatility of the gross returns (for a vol-targeted book this is approximately σ_{target} by construction).

In Plain English

Where does the exact shape come from? Sharpe is average return $\div$ volatility. Costs lower the average return by (annual turnover $\times$ cost per trade) $= 252 \bar{\tau} c$ each year, but barely touch the volatility. So the Sharpe falls by exactly that cost amount divided by the same σ_{ret} you were already dividing by, which is the second term. The $\sqrt{252}$ in the Sharpe definition is absorbed because both the numerator (now a per-year cost) and σ_{ret} are on an annual footing.

Worked Example: How Turnover Eats the ML Sharpe Edge

From Section 18.1.3, the gross figures are $\text{SR}_{\text{HAR}}^{\text{gross}} = 0.82$ and $\text{SR}_{\text{ML}}^{\text{gross}} = 0.88$, both at $\sigma_{\text{ret}} \approx 10\%$ (the target). Suppose the smoother HAR weights have turnover $\bar{\tau}_{\text{HAR}} = 0.04$ while the reactive ML weights have $\bar{\tau}_{\text{ML}} = 0.09$. Take an SPX-futures half-spread $c = 2 \text{ bp} = 0.0002$. *Note:* this example works entirely in decimals (returns and costs as fractions, so $\sigma_{\text{ret}} = 0.10$ and $c = 0.0002$), in contrast to the percent-based table above; keep the two conventions separate.

HAR drag:

$$\frac{252 \times 0.04 \times 0.0002}{0.10} = \frac{0.002016}{0.10} = 0.020, \quad \text{SR}_{\text{HAR}}^{\text{net}} \approx 0.82 - 0.02 = 0.80.$$

ML drag:

$$\frac{252 \times 0.09 \times 0.0002}{0.10} = \frac{0.004536}{0.10} = 0.045, \quad \text{SR}_{\text{ML}}^{\text{net}} \approx 0.88 - 0.045 = 0.835.$$

The gross gap was $0.88 - 0.82 = 0.06$ Sharpe in favour of ML. After costs it shrinks to $0.835 - 0.80 = 0.035$. **Costs ate 40% of the ML edge**, purely because the more reactive forecast trades more than twice as often.

18.2.3 The Sharpe-vs-Cost Curve and Breakeven Cost

A single net Sharpe number hides where the strategy dies. The right object to report is the **Sharpe-vs-cost curve**: net Sharpe evaluated across a grid of cost levels, $c \in \{0, 1, 2, 5, 10, 20\}$ bp, with all three forecasts plotted on the same axes (see the deliverable guidance in Section 18.2.5). By Equation (18.5) this is a straight line in c . Read it as a graph with cost c on the horizontal axis and net Sharpe on the vertical: the line starts at the gross Sharpe when cost is zero (its *intercept*, SR^{gross}) and slides downward as cost rises, with the steepness of that downward slide (its *slope*, $-252 \bar{\tau} / \sigma_{\text{ret}}$) set by how much you trade. The point where the line crosses zero is the **breakeven cost** c^* , the half-spread at which the strategy stops making money:

$$c^* = \frac{\text{SR}^{\text{gross}} \cdot \sigma_{\text{ret}}}{252 \cdot \bar{\tau}}, \quad (18.6)$$

where every symbol is as defined for Equation (18.5).

In Plain English

The breakeven cost answers “how expensive could trading get before this forecast is worthless?” A forecast that needs cheap execution to make money has a low c^* ; a robust forecast keeps a positive Sharpe even when spreads widen. Two forecasts with identical gross Sharpe can have very different c^* : the smoother one (lower $\bar{\tau}$) survives to a higher cost. So c^* rewards forecast smoothness, exactly the property that raw QLIKE and gross Sharpe are blind to.

Report $\text{SR}^{\text{net}}(c)$ for all three forecasts on the same axes rather than a single number; the schematic in Figure 18.1 shows what the plot looks like.

Worked Example: Breakeven Cost for the ML Vol-Targeting Strategy

Using the worked figures above for the ML forecast ($\text{SR}^{\text{gross}} = 0.88$, $\sigma_{\text{ret}} = 0.10$, $\bar{\tau} = 0.09$):

$$c_{\text{ML}}^* = \frac{0.88 \times 0.10}{252 \times 0.09} = \frac{0.088}{22.68} = 0.00388 \approx 39 \text{ bp.}$$

For the smoother HAR forecast ($\text{SR}^{\text{gross}} = 0.82$, $\bar{\tau} = 0.04$):

$$c_{\text{HAR}}^* = \frac{0.82 \times 0.10}{252 \times 0.04} = \frac{0.082}{10.08} = 0.00813 \approx 81 \text{ bp.}$$

The reversal is the point: HAR’s breakeven cost is more than double the ML model’s, even though ML has the higher gross Sharpe. The ML forecast needs cheaper execution to stay ahead because it trades more. On SPX futures, where realistic costs are around 1 bp per side, both clear breakeven by a wide margin and the higher gross Sharpe wins. On an illiquid instrument where costs approach 40 bp, the ML forecast is at the edge of viability while HAR still works. Whether the ML model is the right production choice depends on the instrument’s cost, not on QLIKE alone.

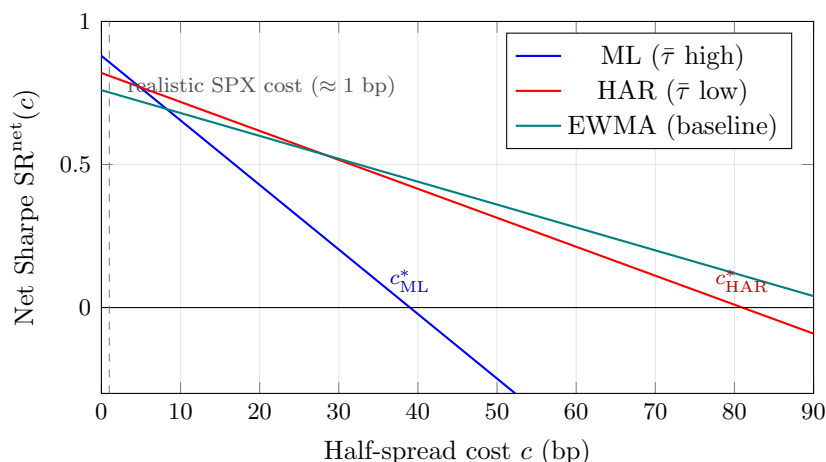


Figure 18.1: Schematic Sharpe-vs-cost curves. Each forecast starts at its gross Sharpe (intercept at $c = 0$) and slides down a straight line whose steepness is its turnover; where a line hits zero is its breakeven cost c^* . The reactive ML forecast has the highest gross Sharpe but the steepest slope, so it crosses zero first. At the realistic SPX cost (≈ 1 bp, dashed line) all three are comfortably positive and the higher gross Sharpe wins.

18.2.4 What SPX Futures Actually Cost

The cost grid above is a stress test; for the instrument your project actually trades it is worth knowing the realistic number. The strategy in Section 18.1 sizes exposure to the S&P 500, implemented through **E-mini** or **Micro E-mini S&P 500 futures**, among the most liquid exchange-traded instruments in the world. For a single E-mini contract the bid–ask spread is typically one **tick** (0.25 index points), and each E-mini contract is worth $50\times$ the index level (the contract multiplier). Walk the chain at an index level of ≈ 5000 : the contract is worth $50 \times 5000 = \$250,000$, and one tick is worth $50 \times 0.25 = \$12.50$, so the full spread is $12.5/250,000 \approx 0.5$ bp, a **half-spread of well under 1 bp** (half the round-trip spread defined above) in normal conditions. Adding exchange and clearing fees, a realistic all-in round-trip cost for vol-targeting rebalances is on the order of 1–2 **bp**, widening in stressed markets.

Use One Instrument's Real Cost, Then Stress It

Do not paste a generic cross-asset cost table into a vol-targeting report. The strategy trades one thing, SPX futures, so anchor on its real half-spread ($\lesssim 1$ bp) and then sweep upward through the $\{1, 2, 5, 10, 20\}$ bp grid to show robustness, not to suggest you trade twenty-basis-point instruments. The grid is there to expose how a high-turnover ML forecast would fare *if* costs rose, e.g. in a liquidity crunch when spreads on even E-minis widen.

18.2.5 Performance-Metric Reference

The vol-targeting deliverable table (Section 18.1) should report a consistent set of metrics for each forecast. Table 18.1 collects the definitions in one place. Every metric is computed on the *net* return stream $\text{PnL}_t^{\text{net}}$ from Equation (18.3).

Lead with Net Sharpe and Breakeven, Support with the Rest

Never report a single net Sharpe. Plot $\text{SR}^{\text{net}}(c)$ across the $\{0, 1, 2, 5, 10, 20\}$ bp grid for the ML forecast, the HAR forecast, and the EWMA baseline (Section 18.1.1) on *identical* axes

Table 18.1: Performance metrics for a vol-targeting backtest. All computed on net-of-cost returns. Annualization uses 252 trading days.

Metric	Formula / Description	What It Tells You
Sharpe ratio	$SR = (\mu_{\text{ret}} / \sigma_{\text{ret}}) \times \sqrt{252}$	Risk-adjusted return; the headline number
Sortino ratio	$(\mu_{\text{ret}} / \sigma_{\text{downside}}) \times \sqrt{252}$	Like Sharpe but σ_{downside} counts only the volatility of losing days; use when returns are skewed
Hit rate	$\#\{\text{PnL}_t > 0\} / T$	Fraction of profitable days; aim > 50%
Max drawdown	$\max_{s \leq t} (\text{cumPnL}_s - \text{cumPnL}_t)$	How far the running total fell from its highest previous point, i.e. the deepest hole the strategy dug; the survival metric
Calmar ratio	Annualized return / max drawdown	Return per unit of worst-case pain
Turnover	$\bar{\tau} = \frac{1}{T} \sum w_t - w_{t-1} $	Trading intensity; the cost driver
Breakeven cost	c^* where $SR^{\text{net}}(c^*) = 0$	Tradeability threshold; the desk's first question

(Figure 18.1). In your deliverable, lead with net Sharpe at the realistic SPX cost (≈ 1 bp) and the breakeven cost c^* where each line hits zero, the single number a systematic desk cares about most. What matters is whether the ML curve sits *above* HAR across the whole plausible cost range, not just at $c = 0$. Support with max drawdown, Calmar, and turnover. If the desk asks one question it will be about c^* ; if they ask two, the second is about max drawdown.

18.2.6 Fragility: Where Does the P&L Come From?

A strong net Sharpe can still hide a fragile strategy. The vol-targeting Sharpe gain comes from cutting exposure before drawdowns, so a natural worry is that *all* of the edge is concentrated in one or two crisis episodes. Aggregate metrics will not reveal this; a P&L-source decomposition will.

The Fragility Rule of Thumb

If more than 70% of the strategy's cumulative net P&L comes from days that make up less than 30% of the sample, the strategy is **fragile**: it is a regime bet (a **regime** being a market environment such as a calm stretch or a crisis period), not a robust earner. These figures are a heuristic, not a law. The idea is simply that if most of your profit comes from a small handful of unusual days, you are really betting on those days recurring. If net P&L accrues roughly in proportion to time spent in each state, the strategy is **broad-based**, which is what you want for a deliverable that claims a forecast adds steady economic value.

A cheap and informative version of this check, available before any regime model, is a **VIX-tercile decomposition**. Split the backtest days into low, medium, and high VIX terciles, and sum net P&L within each:

- If the vol-targeting edge over EWMA is concentrated entirely in the high-VIX tercile, the ML forecast is really a *crisis detector*; that is valuable, but you should say so, because a desk that does not want concentrated crisis exposure will read that Sharpe differently.
- If the edge is spread across terciles, the better forecast is improving position sizing in ordinary conditions too, and the Sharpe gain is more durable.

Report the tercile split alongside the headline table. It costs one extra row and pre-empts the desk's sharpest question about where the money comes from.

18.3 Dealer Gamma and Structured Products Feedback

Beyond statistical forecasting, volatility is shaped by the mechanical hedging behavior of options dealers. When dealers hold large gamma positions, their hedging creates predictable effects on realized vol. Understanding this mechanism gives you both a novel feature and a deeper appreciation for what your model is capturing.

18.3.1 How Dealer Hedging Amplifies or Suppresses Vol

When dealers are **long gamma** (they own options), they delta-hedge by selling into rallies and buying dips. This mean-reversion pressure suppresses realized volatility below what pure news flow would generate. Conversely, when dealers are **short gamma** (common after selling structured products like autocallables), they hedge by buying into rallies and selling into dips, amplifying moves and increasing realized vol.

GEX as a Volatility Feature

Gamma Exposure (GEX) estimates the net dealer gamma across all strikes from publicly available options open interest data. The estimate is approximate but directionally informative:

$$\text{GEX} \approx \sum_{\text{strikes } K} \text{OI}_K \times \Gamma_K \times 100 \times S \quad (18.7)$$

- Positive GEX (dealers long gamma): expect lower-than-forecast RV (mean reversion).
- Negative GEX (dealers short gamma): expect higher-than-forecast RV (momentum/amplification).

Adding $\text{sign}(\text{GEX})$ or GEX-quintile as a feature to HAR-X is a novel extension not yet thoroughly explored in the academic literature (Bennett, 2014).

18.3.2 Structured Products and the Short-Gamma Overhang

The primary source of dealer short-gamma exposure is structured products (autocallables, barrier options, worst-of notes). These products embed knock-in/knock-out barriers near current spot levels. As spot approaches a barrier, the product's gamma becomes very negative, forcing dealers to hedge aggressively. The systematic issuance of these products (estimated at hundreds of billions of notional globally) creates a permanent structural short-gamma overhang in equity index markets.

18.3.3 Pin Risk at Expiry

Near options expiry, large open interest at specific strikes creates **pinning** effects: the stock gravitates toward the strike as dealers' gamma hedging intensifies near that level. This suppresses realized vol on expiry days, a calendar effect (Section 10.11) with a mechanical explanation.

Novel HAR-X Features from Dealer Positioning

Beyond GEX, distance to nearest barrier level and net open interest by strike are publicly computable dealer-positioning features worth testing for incremental predictive power in a HAR-X model.

18.4 Communicating Volatility Forecasting Results**Background**

This section assumes you have results to present: a QLIKE comparison and Diebold–Mariano test from Chapter 17 (Sections 17.3, 17.5), the vol-targeting economic-value test from Sections 18.1 and 18.2, and the overfitting controls from Chapter 17 (Sections 17.10, 17.7, 17.11.1). The skill this section teaches is communicating those results so that a volatility desk trusts the work in twenty minutes.

You have built a forecast that beats HAR on QLIKE and adds net Sharpe in a vol-targeting test. The remaining risk is entirely in the telling. A desk audience decides in the first minute whether to engage with your hypothesis or to start hunting for the **overfitting** that they assume is hiding behind any ML result. (Overfitting means a model looks good on the data it was built from but fails on new data. The more feature combinations you try, the more likely one looks good by pure chance, which is why a desk hearing “47 features” immediately distrusts the result.) This section is about controlling that first minute and the nineteen that follow.

18.4.1 Frame It as a Hypothesis Test, Not a Fishing Expedition

The question that decides your reception is: *did you start from an economic reason to expect this feature to help, or did you feed everything into a tree and report what stuck?* There are two ways to open, and they trigger opposite reactions.

- **Good (grounded hypothesis):** “The leverage effect (Chapter 6) says negative returns predict higher future volatility, so we added a signed-return / semivariance feature to HAR. It improves QLIKE by 4% with a Diebold–Mariano t -statistic of 2.9 (a t above ≈ 2 means the accuracy gap is unlikely to be luck).”
- **Bad (fishing expedition):** “We fed 47 features into a LightGBM model and it forecasts RV better than HAR.”

The first framing invites engagement: there is a well-grounded mechanism, the leverage effect or the variance risk premium (Chapter 9), and modern tools were used to test it. The audience evaluates the hypothesis, then checks whether the test is clean. The second framing invites the question “how many features did you try?” and a reach for the Deflated Sharpe Ratio (Section 17.10; a Sharpe adjusted downward for how many things you tried, so that staying positive means the edge survives that penalty).

The ML Is the Test, Not the Story

Lead with the volatility mechanism, not the model: the economic theory provides the *why*, the ML (Chapter 11) provides the *how much better than HAR*, and the forecast is judged against the HAR baseline, not by its raw R^2 .

Never Open with Model Architecture

If the first thing the desk hears is “a 500-tree LightGBM ensemble with Bayesian hyperparameter search,” you have lost them. They will spend the rest of the talk looking for overfitting instead of listening to results. Model complexity is a *liability* until you have shown that the HAR baseline (Chapter 6) cannot do the job, which is exactly why HAR appears on every chart. Save the architecture for the methodology slide.

18.4.2 One Slide Per Claim

The discipline that keeps a desk presentation tight is **one slide per claim**. Each slide communicates exactly one takeaway, and the structure is fixed:

- **Title = the claim.** Not “Results” but “A semivariance feature beats HAR on QLIKE by 4% (DM $t = 2.9$).” The title is the takeaway.
- **Body = one chart.** One chart proves the claim. No multi-chart slides: they split attention and invite the audience to study the wrong panel while you talk about the right one.
- **Footer = the metric.** A single line with the key numbers: “ $QLIKE_{ML} = 0.241$ vs. $QLIKE_{HAR} = 0.251$; DM $t = 2.9$; net SR = 0.84; $n = 1,250$.”

The same discipline governs chart **captions** in the written report: a caption must make the chart self-explaining. The contrast is stark.

- **Vacuous caption:** “Figure 3: Forecast comparison.”
- **Self-explaining caption:** “Figure 3: The HAR-plus-semivariance model lowers out-of-sample QLIKE by 4% relative to HAR (Diebold–Mariano $t = 2.9$, Section 17.5). Solid line: ML forecast; dashed line: HAR baseline.”

The second caption states the claim, the magnitude, the significance test, and which line is which. A reader who sees only the figure and its caption should be able to reconstruct your point.

If They Only Read the Titles

Write slide titles so that reading them in sequence tells the whole story. Test it: list your titles and hand them to a colleague. If they can reconstruct the argument from titles alone (hypothesis, QLIKE result, DM significance, net economic value, what failed), the deck is well structured. If they cannot, rewrite the titles until they can.

18.4.3 Negative Results Are a Credibility Signal

A presentation that reports only wins triggers the audience's overfitting alarm. One that says "we tested four feature families against HAR; two beat it, two did not" signals intellectual honesty, and honesty is what buys belief in the wins.

Report which feature families *failed* to beat the HAR baseline, and why. Useful negatives for a volatility project look like:

- "Cross-asset RV spillover features added no QLIKE improvement over HAR after purged cross-validation (Section 17.7; cross-validation that removes train/test windows that overlap in time, so nothing leaks); the Diebold–Mariano test could not reject equal accuracy ($t = 0.4$)."
- "A jump-component split (HAR-J) helped in-sample but the gain did not survive walk-forward evaluation (training only on the past and testing on the next block, rolling forward), consistent with the difficulty of forecasting the jump part of RV."
- "LightGBM did not beat ridge-HAR (a HAR regression with ridge regularization, i.e. a penalty that shrinks the coefficients) on the same features. The RV-to-RV mapping appears close to linear in our sample (Chapter 11, Section 11.8), so we report the linear model."

Each of these tells the desk what the *data* look like, not just what the model found. Put them on one slide titled honestly, e.g. "Feature families that did not beat HAR," as a compact table of family, QLIKE-vs-HAR, DM t -statistic, and the reason.

Negatives Build Trust Because They Are Costly

Reporting failures costs scarce presentation time and exposes the limits of your work. The audience knows this. The fact that you are *willing* to pay that cost signals that the positive results are real. A presenter who shows only successes is optimizing for impression, not information, and an experienced desk head can tell the difference instantly.

18.4.4 The Written Report and Its Page Budget

The written deliverable is a desk-ready report, not slides. Page budgets enforce discipline; they stop you from burying the result in methodology. A workable structure:

1. **Executive summary** (1 page). Hypothesis, headline QLIKE/DM result, net economic value. A busy desk head reads only this page.
2. **Hypothesis and motivation** (1–2 pages). The volatility mechanism that motivates the feature: the leverage effect (Chapter 6) or the variance risk premium (Chapter 9).
3. **Data and features** (2–3 pages). RV construction, HAR components, the candidate feature families, sample period, and the reserved holdout (data locked away and never looked at during model-building).

4. **Methodology** (2–3 pages). QLIKE loss (Section 17.3), purged cross-validation (Section 17.7), look-ahead controls (Section 17.11.1), model specifications.
5. **Results, with the HAR baseline** (3–5 pages). One subsection per feature family. Each: QLIKE table vs. HAR, Diebold–Mariano test, and the vol-targeting net-Sharpe table from Section 18.2, always plotted against HAR.
6. **Negative results** (1 page). Which families did not beat HAR, and why.
7. **Robustness** (1–2 pages). DSR after the honest trial count (Section 17.10), walk-forward stability, feature stability across regimes (per the regime-stable feature-selection discussion in Chapter 12 and your own subsamples).
8. **Economic value** (1 page). Sharpe-vs-cost curve, breakeven cost c^* , turnover, VIX-tercile P&L decomposition (Section 18.2).
9. **Limitations and next steps** (1 page). Sample-size caveats, crisis-underprediction risk (Section 18.1.4), proposed extensions.

Every chart in the report supports exactly one claim, stated in its caption (Section 18.4.2).

Do Not Iterate After Unblinding the Holdout

Chapter 17 made this point under the look-ahead taxonomy (Section 17.11.1); it bears repeating here. The out-of-sample test on the reserved holdout is a *one-shot* exercise. You run it once, report the result, and do not return to re-tune. “In-sample QLIKE improvement 5%, out-of-sample 3%” is a legitimate, honest result. “We re-optimized on the holdout until the out-of-sample number improved” is the bias the whole evaluation framework exists to prevent, not a result.

18.4.5 Handling Desk Questions

Know Your Backup Slides

For every question below, prepare a backup slide with the detailed chart or table. Your verbal answer should be two sentences; if they want more, flip to the backup. Prepare five to eight backups covering these scenarios.

The following questions come up in almost every desk presentation of a volatility forecast. The discipline: cross-reference the analysis already in your deck rather than re-deriving it on the whiteboard.

“Is your model really beating HAR, or is it noise?” *Answer:* “Out-of-sample QLIKE improves by 4%, and the Diebold–Mariano test (Section 17.5) rejects equal accuracy at $t = 2.9$. The improvement holds in walk-forward, not just one split.”

Backup slide: QLIKE-vs-HAR table with DM t -statistics per feature family.

“What happens in a crisis? Does it underpredict?” *Answer:* “Like all models trained on mostly-calm data, it underpredicts the largest jumps; we show this in the VIX-tercile decomposition (Section 18.2). The vol-targeting book floors leverage during high-uncertainty regimes (Section 18.1.4) to bound the damage.”

Backup slide: VIX-tercile net-P&L table plus the crisis-underprediction note.

“Why not just use HAR? Is the nonlinearity real?” *Answer:* “We tested ridge-HAR on identical features; where the tree wins, SHAP (a tool that quantifies how much each feature pushed a given prediction) attributes the win to a threshold effect (the model behaving differently above versus below a vol level), which a straight-line (linear) model, by definition, cannot

represent (Chapter 11). Where ridge ties the tree, we report the linear model as production.”

Backup slide: HAR vs. ridge-HAR vs. LightGBM QLIKE bar chart (Section 11.8).

“Are the features stable across regimes?” *Answer:* “We re-estimated feature importance in calm and stressed subsamples; the top HAR components and the leverage feature stay dominant in both, while the cross-asset features are unstable (which is why we dropped them). The regime split is in the robustness section.”

Backup slide: feature-importance-by-regime comparison.

“How do you know there’s no look-ahead?” *Answer:* “Every feature is built point-in-time and every label uses the next period’s RV; the look-ahead taxonomy in Section 17.11.1 lists each control. Purged cross-validation (Section 17.7) removes overlap between train and test windows.”

Backup slide: timeline diagram of feature/label alignment and the purge/embargo windows.

“Is this overfit? How many things did you try?” *Answer:* “We logged N experiments; the Deflated Sharpe Ratio (Section 17.10) stays positive after correcting for that count. Walk-forward out-of-sample QLIKE is consistent with the cross-validation estimate.”

Backup slide: DSR computation with the trial count and the walk-forward QLIKE series.

Never Say “I Don’t Know” Without a Follow-Up

If you genuinely have not tested something, say: “I haven’t run that specific test. My best estimate from [related analysis] is [X], and I can run it and follow up by [date].” This is honest, shows you know which test would answer the question, and commits to a deliverable. Never just say “I don’t know” and move on.

Chapter 19

From Forecast to P&L: A Realistic, Evaluable IV–RV Straddle

Suppose your machine-learning model forecasts tomorrow’s realized variance more accurately than the HAR benchmark. A desk head will not be impressed by a lower QLIKE; they will ask a blunt question: *does the better forecast make money, and how do you know it is not a fluke of the backtest?* Answering honestly is harder than it looks. The natural way to monetise a volatility view is to trade an option and hedge away its directional exposure, but the moment you do this you inherit a thicket of frictions: the option’s bid–ask spread, the cost of rebalancing the hedge, the random error the discrete hedge injects, and the statistical trap of having tried many strategy variants before reporting the best one. This chapter assembles the **most realistic version** of that trade, component by component, and shows exactly where each standard shortcut is sound, where it is flawed, and what it quietly leaves out.

Background

This chapter is the synthesis of four earlier strands, and it leans on each rather than repeating them:

- **Options and Greeks** (Section 8.2.1): the Black–Scholes price, delta, gamma, vega; and variance swaps with their log-contract replication (Section 8.8).
- **The variance risk premium** (Section 9.1) and the **gamma P&L identity** (Section 9.6): why a delta-hedged option pays off on realized-minus-implied variance.
- **Forecast evaluation** (Chapter 17): QLIKE (Section 17.3), the Diebold–Mariano test (Section 17.5), and the deflated Sharpe ratio (Section 17.10).
- **Economic-value testing** (Chapter 18): transaction costs, turnover, and the Sharpe drag (Section 18.2).

You do not need to re-read these; cross-references point back to the exact equation when it is used.

19.1 The Strategy in One Picture

The strategy trades a single, sharp idea: **a delta-hedged straddle is a bet on realized variance against the variance the market has priced in.** If our forecast says realized volatility will come in *below* what the option’s implied volatility charges, the option is expensive, so we sell it; if our forecast says realized will come in *above* the implied charge, the option is cheap, so we buy it. Everything else in this chapter is the machinery that makes this bet measurable and honest.

Two building blocks make the idea precise. A **straddle** is a package of one call and one put on the same underlying, struck at the same price K and expiring on the same date. Bought together, the pair has almost no directional view at inception (the call's positive **delta**, an option's price sensitivity to the underlying expressed as an equivalent number of shares, roughly cancels the put's negative delta; see Section 8.2.1) but a large sensitivity to how much the underlying *moves*: any large move, up or down, pushes one leg deep into the money. **Delta hedging** is the act of repeatedly trading the underlying to cancel whatever directional exposure remains, so that what is left is a near-pure exposure to *variance* rather than to direction. We hold the delta-hedged straddle for one period, collect its profit or loss, and repeat.

In Plain English

Selling a delta-hedged straddle is like selling insurance against large price moves. You collect a premium set by the market's implied volatility, then pay out whenever the underlying actually moves a lot (high realized volatility). You profit when the premium you charged exceeds the claims you pay, that is, when implied volatility was higher than the realized volatility that followed. Our forecast $\widehat{RV}_t$ is the actuary: it tells us, before we write the policy, whether the premium on offer looks rich or cheap.

The signal that decides which way to trade is the gap between the implied volatility the market is charging at the close of the prior day, IV_{t-1} , and our model's forecast of the realized volatility that will actually occur, $\widehat{RV}_t$. This gap is the **variance risk premium** viewed through a tradeable lens (Section 9.1).

The IV–RV Gap Trading Rule

On each day t , compare lagged implied volatility to the forecast of realized volatility:

- if $IV_{t-1} > \widehat{RV}_t$ (implied rich), **short** the delta-hedged straddle;
- if $IV_{t-1} < \widehat{RV}_t$ (implied cheap), **long** the delta-hedged straddle.

The size of the bet can scale with the size of the gap. The quality of the forecast $\widehat{RV}_t$ is the only proprietary ingredient; everything downstream is execution and accounting.

Figure 19.1 traces the whole pipeline, from forecast to a judged Sharpe ratio. The rest of the chapter walks through each stage, adding one layer of realism at a time: the signal and its timing (Section 19.2), the P&L engine and where it breaks (Sections 19.3 and 19.4), the two kinds of transaction cost (Sections 19.5 and 19.6), the variance the discrete hedge injects (Section 19.7), the kurtosis input it needs (Section 19.8), the assembled algorithm (Section 19.9), and finally the evaluation that ties it all back to forecast accuracy (Section 19.10).

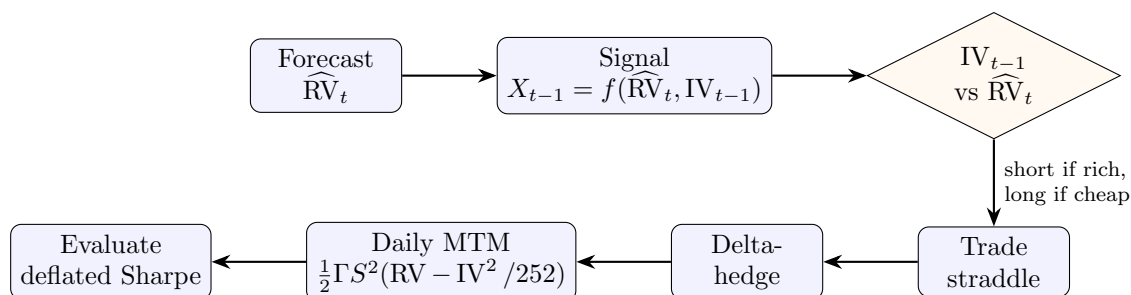


Figure 19.1: The IV–RV straddle pipeline. A realized-volatility forecast feeds a variance-risk-premium signal, which decides the direction of a delta-hedged straddle; the position is hedged intraday, marked to market daily through the gamma P&L identity, and the resulting return stream is judged with a deflated Sharpe ratio. Each stage adds a layer of realism developed in this chapter; every symbol shown in the boxes is defined in the sections that follow.

19.2 The Signal and the Anti-Lookahead Protocol

The trade needs a single number, computed before we commit capital, that decides whether to be short or long the straddle. We have two raw ingredients: our model's forecast of the realized volatility that will occur over day t , written $\widehat{RV}_t$, and the implied volatility the option market was charging at the close of the previous day, IV_{t-1} . The signal is some function of their gap, following [Pollok \(2025\)](#).

$$X_{t-1} = f\left(\widehat{RV}_t, IV_{t-1}\right), \quad f \in \left\{ \underbrace{x-y}_{\text{difference}}, \underbrace{x/y}_{\text{ratio}}, \underbrace{\ln(x/y)}_{\text{log-ratio}} \right\}. \quad (19.1)$$

- X_{t-1} : the trading signal known at the close of day $t-1$, before day t 's position is taken.
- $\widehat{RV}_t$: the model's forecast (made with data through $t-1$) of day t 's realized volatility, the “ x ” argument.
- IV_{t-1} : the option-implied volatility observed at the close of $t-1$, the “ y ” argument.
- f : the gap functional. The difference is the most direct; the ratio and log-ratio are scale-free and behave better when volatility levels swing widely.

Why This Matters

This equation is the only place the forecast enters the strategy. Every later section is execution and accounting that is identical no matter whose forecast you plug in. This is the precise mechanism by which a lower QLIKE is supposed to become higher P&L, and Section 19.10 tests whether it actually does.

Before differencing, the two numbers must live in the same units. Implied volatility is quoted annualized; realized volatility forecasts are usually expressed per day. Because volatility scales with the square root of time, we convert the implied quote to daily units by dividing by the square root of the number of trading days in a year,

$$IV_{t-1}^{\text{daily}} = \frac{IV_{t-1}}{\sqrt{252}}, \quad (19.2)$$

so that the difference $\widehat{RV}_t - IV_{t-1}^{\text{daily}}$ compares like with like.¹

Lookahead is the cardinal sin of any backtest: using information at time t that would not actually have been available when the position was taken. For this strategy the danger is acute, because the predictor (which needs the day's data to update the forecast) and the trade (which must happen while the market is open) both want to live at the same instant. [Pollok \(2025\)](#) fixes a clean protocol that keeps them safely ordered.

The Lookahead-Safe Timing Protocol

For each trading day t :

1. Update the realized-volatility model and form the signal X_{t-1} using only information available up to **3:55 pm ET** on day t .
2. **Execute** the straddle (and its initial hedge) before the **4:00 pm** close on day t .
3. **Realize** the position's return over day $t+1$.

The five-minute gap between measuring the predictor and trading guarantees the signal uses no information from the moment of execution onward.

¹Some studies use $\sqrt{250}$ rather than $\sqrt{252}$; the choice is a convention and shifts the gap by a constant scale factor, not its sign.

Finally, how big should the bet be? The crude choice is binary: a fixed-size short whenever the gap is positive, a fixed-size long whenever it is negative. A better choice scales the position with confidence in the signal.

Graded Sizing Beats a Binary Switch

Li and Wu (2026) find that a *moderate*, confidence-scaled position size delivers higher Sharpe ratios than either a binary on/off rule or maximally aggressive scaling. The intuition is risk budgeting: press hardest when the edge is largest and the forecast most confident, but never so hard that a single adverse day dominates the P&L.

Cite This Result Narrowly

Li and Wu (2026) study a *directional* machine-learning hedging rule on a single ETF, not an IV–RV straddle. It supports the qualitative claim “moderate scaling beats a binary switch” and nothing more; do not borrow any of its magnitudes for this strategy.

19.3 The P&L Engine: The Gamma Identity

Why should a delta-hedged option pay off on realized-minus-implied variance at all? The answer is a single identity, and it is the engine under the whole strategy. Suppose we hold an option and continuously rebalance the stock hedge using the delta computed at the option’s *implied* volatility. Where an *unhedged* option’s daily P&L would be dominated by a large random term in the direction the stock moved, Ahmad and Wilmott (2005) show that hedging at implied volatility cancels that randomness, leaving a one-day mark-to-market profit (the change in the position’s value at current prices) that is completely deterministic:

$$d\Pi_t = \frac{1}{2}(\sigma^2 - \tilde{\sigma}^2) S^2 \Gamma^i dt. \quad (19.3)$$

- σ : the **actual** (realized) volatility that the underlying delivers over the step.
- $\tilde{\sigma}$: the **implied** volatility baked into the option’s price and into the hedge ratio.
- S : the underlying price; Γ^i : the option’s **gamma** computed with the implied volatility (the curvature of its value in S).
- $S^2 \Gamma^i$: the **dollar gamma**, the weight that converts a variance gap into money.
- dt : the length of the step.

In Plain English

Hedging at the implied volatility makes each day’s profit a clean, sign-definite bet: you earn the gap between the variance that actually happened (σ^2) and the variance the market charged for ($\tilde{\sigma}^2$), scaled by how much curvature (dollar gamma) the position carries that day. Remarkably there is no random dX term: once you fix the hedge at implied vol, the daily P&L stops depending on the direction the stock moved and depends only on how *much* it moved. The position has been turned into a pure variance meter.

Summing the discounted daily increments over the life of the trade gives the total profit from hedging at implied volatility,

$$\Pi = \frac{1}{2}(\sigma^2 - \tilde{\sigma}^2) \int_{t_0}^T e^{-r(t-t_0)} S^2 \Gamma^i dt, \quad (19.4)$$

where t_0 is the trade date, T the expiry, r the interest rate, and $e^{-r(t-t_0)}$ the discount factor that converts each future day's profit into today's dollars. Ahmad and Wilmott (2005) describe this total as “always positive, but highly path dependent”: positive because (for a correct view) every daily increment has the same sign, yet path dependent because the dollar-gamma weight $S^2\Gamma^i$ depends on where the stock wanders relative to the strike. (Carr (2005) and Henrard (2003), as presented by Ahmad and Wilmott, 2005, give the more general result for hedging at *any* volatility rather than implied; we need only the implied-vol case here.)

In a daily backtest we do not integrate; we sum one increment per day. The unobservable instantaneous variance $\sigma^2 dt$ accumulated over a day is precisely what our realized-variance estimator measures, so we substitute the measured RV_t for it (this is the whole reason the guide built an RV estimator in the first place); the annualized implied variance IV^2 is sliced into one trading day's share by dividing by the 252 trading days in a year:

$$PnL_t \approx \frac{1}{2} \Gamma_t S_t^2 \left(\underbrace{RV_t}_{\text{realized var, day } t} - \underbrace{\frac{IV^2}{252}}_{\text{daily implied var}} \right). \quad (19.5)$$

- RV_t : the realized *variance* over day t (e.g. the sum of squared 5-minute returns).
- $IV^2/252$: the implied variance apportioned to a single trading day.
- $\Gamma_t S_t^2$: the dollar gamma at the start of day t , treated as fixed across the day.

Notation: Variance Form vs. Volatility Form

Section 9.6 wrote this identity as equation (9.8), $\frac{1}{2}\Gamma S^2(RV^2 - IV^2)T$, where there RV denotes realized *volatility*, so RV^2 is a variance. Here RV_t denotes realized *variance* directly, so the bracket is $(RV_t - IV^2/252)$ with no square. The two are the same object; only the symbol's meaning differs, and we use the variance form throughout this chapter.

The dollar-gamma weight $S_t^2\Gamma_t$ is not constant: it is largest when the stock sits near the strike and falls away as the stock drifts into either wing (Figure 19.2). This is the source of the path dependence: the same average variance gap earns very different money depending on whether the stock spent the period parked at the strike (high weight) or trending away from it (low weight). A single straddle cannot escape this path dependence, which is why Section 19.4 examines where its clean variance interpretation frays.

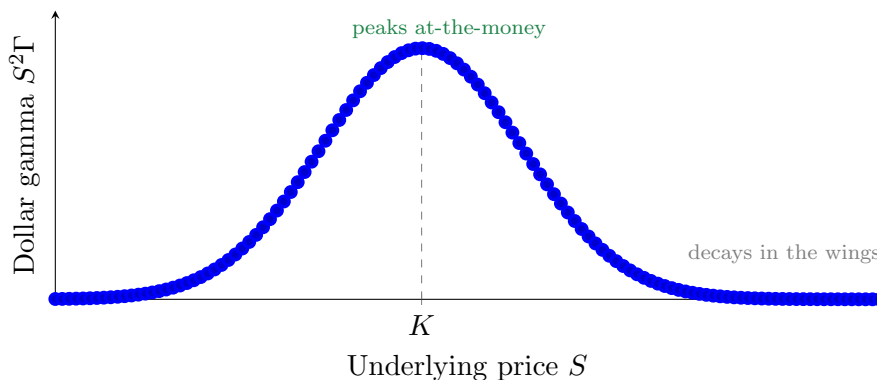


Figure 19.2: The dollar-gamma weight $S^2\Gamma$ as a function of the underlying price, for a fixed strike K . The weight that converts a variance gap into P&L is concentrated near the money and decays as the stock drifts into either wing. This is why a single delta-hedged straddle is a *path-dependent* variance bet: the same realized-minus-implied gap earns more money on days the stock loiters near the strike than on days it trends away.

19.4 Where the Clean Identity Breaks

The gamma identity of Equation (19.5) rests on four assumptions: that we hold a single, near-the-money option; that the underlying follows a continuous, jump-free diffusion; that volatility itself does not move while we trade; and that we rebalance the hedge continuously. Real life violates all four. This section names each crack and the Greek or theorem that measures it, so that Section 19.7 can put a number on the most important one.

The first two cracks are **higher-order Greeks**: sensitivities of the position to things the simple identity treats as frozen. **Vanna** measures how the option's delta drifts as volatility moves:

$$\text{Vanna} = \frac{\partial^2 V}{\partial S \partial \sigma} = \frac{\partial \mathcal{V}}{\partial S} = -e^{-q\tau} N'(d_1) \frac{d_2}{\sigma}, \quad (19.6)$$

- V : the option value; $\mathcal{V} = \partial V / \partial \sigma$ its sensitivity to volatility.
- $N'(\cdot)$: the standard normal density; d_1, d_2 : the usual Black–Scholes arguments (Section 8.2).
- q : the dividend yield; $\tau = T - t$: time to expiry; σ : volatility.

In Plain English

Vanna is the cross-talk between price and volatility. It says: when volatility ticks up, my delta is no longer where I left it, so my “delta-neutral” hedge has quietly become directional. A position with large vanna leaks P&L whenever spot and volatility move together, which is exactly what happens in an equity sell-off (price down, vol up).

Volga (also called vomma) measures the position's convexity in volatility itself:

$$\text{Volga} = \frac{\partial^2 V}{\partial \sigma^2} = \mathcal{V} \frac{d_1 d_2}{\sigma}. \quad (19.7)$$

- Volga: curvature of the option value as a function of volatility (the “gamma of vega”).
- $\mathcal{V}, d_1, d_2, \sigma$: as above.

In Plain English

Gamma is convexity in the stock price; volga is convexity in volatility. A position with positive volga gains more from a large volatility swing than it loses from a small one, so it is implicitly long vol-of-vol. For a straddle the legs carry volga away from the money, which is why a straddle held through a volatility regime change behaves differently from a textbook variance bet.

The third crack is **jumps**. The gamma identity is a statement about quadratic variation accumulated smoothly; a gap (an earnings move, a macro shock) is not captured by $\frac{1}{2}\Gamma S^2$ RV because gamma is not constant across a large jump. Carr and Lee (2009) show that for the variance-swap replication the leading error from discreteness and jumps is *third order* in the daily return, so it is small for ordinary days and bites only on the rare large move, precisely when it hurts most.

The fourth crack is **discreteness of the hedge**, and it is the one we will quantify. Because we rebalance a finite number of times per day rather than continuously, the realized hedge never perfectly cancels the directional exposure, leaving a residual tracking error. Bertsimas et al. (2000) characterise its size.

The Discrete-Hedging Error Shrinks as $1/\sqrt{N}$

With N equally spaced rebalances, the typical size of the tracking error (its root-mean-square, RMSE) shrinks with the square root of the rebalancing count:

$$\text{RMSE} = \frac{g}{\sqrt{N}} + O\left(\frac{1}{N}\right), \quad (19.8)$$

where the “granularity” constant g and the smaller $O(1/N)$ correction are spelled out in [Bertsimas et al. \(2000, Thm. 1c, Eqs. 2.13, 2.18\)](#); the practical takeaway is the rate itself, $1/\sqrt{N}$. One technical point matters for us: a straddle’s payoff has a kink at the strike, so the governing result is their Theorem 2 (for piecewise-linear payoffs), not Theorem 1, which needs a smooth payoff.

A Single Straddle Is Not a Clean Variance Bet

Off the money, the dollar-gamma weight collapses (Figure 19.2) and vanna and volga take over, so the straddle’s payoff drifts away from “realized minus implied variance.” The instrument that *is* a clean variance bet is the variance swap, whose $1/K^2$ strip of options (Section 8.8) holds its variance exposure constant regardless of where spot goes. We trade the straddle anyway, because it is far more liquid, but we must account honestly for the gap, and the next sections do.

19.5 Option Transaction Costs

So far every P&L number has been gross. The single most dangerous shortcut in a volatility-strategy backtest is to ignore, or badly under-model, the cost of the options themselves. We pay this cost as the option’s bid–ask spread, and the honest way to charge it is in volatility points converted to dollars through vega, only at the moments we actually trade: when we open the position, when we flip from short to long (or back), and when we close.

$$c_{\text{opt}} = \mathcal{V} \cdot \frac{1}{2} (\sigma_{\text{ask}} - \sigma_{\text{bid}}), \quad (19.9)$$

- c_{opt} : the dollar cost charged to the position at a single entry, flip, or exit event.
- $\mathcal{V}$: the straddle’s vega, dollars of value per volatility point (Section 8.2.1).
- $\sigma_{\text{ask}} - \sigma_{\text{bid}}$: the bid–ask spread quoted *in implied-volatility points*; the half-spread is what we cross.

Why This Matters

Modelling c_{opt} event-by-event, rather than as a flat daily haircut, is what lets the backtest reward a forecast that trades patiently and punish one that churns.

How large is the spread in practice, and does event-driven costing match reality? Two findings anchor the answer. First, [Muravyev and Pearson \(2020\)](#) show that the cost actually paid depends enormously on execution: for the most liquid US equity options the average quoted spread is roughly 8.1 cents, the conventional “effective” spread about 6.2 cents, but a timing-aware execution that waits for favourable quotes pays only about 1.3 cents, roughly a fifth of the effective and a sixth of the quoted figure. The strategy-level consequence is stark: they document that a long–short straddle portfolio sorted on implied-minus-realized volatility earns about 22.7%

per month gross but collapses to about 3.9% per month once the full quoted spread is charged. *The sign of the conclusion flips with the cost assumption.*

Charge a Cost *Band*, Not a Point Estimate

Because a plausible execution cost ranges from the full quoted half-spread (pessimistic) down to roughly a third of it (timing-aware), a single cost number is meaningless. Report the strategy's Sharpe across the whole band (Figure 19.3). A strategy is only credible if it survives the pessimistic end, or at the very least the empirically grounded middle.

Second, the spread is not one number but a schedule that widens sharply as expiry approaches. Doshi et al. (2025) measure the effective option spread by maturity for liquid index options:

Table 19.1: Effective option bid–ask spread as a fraction of premium, by maturity, for liquid at-the-money index options (Doshi et al., 2025). The very short maturities are far more expensive to trade.

Maturity bucket	Effective spread (% of premium)
21–48 days to expiry	$\approx 2\%$
7–13 days to expiry	$\approx 3.5\%$
0 days to expiry (0DTE)	$\approx 9\%$

Doshi et al. (2025) also document a spread spike on the third-Friday monthly expiry (the standard roll date), and note that routing across the sixteen US options exchanges materially narrows spreads for single names but not for products such as SPX that trade only on one venue. Practitioner backtests reflect this event-driven reality directly: Wysocki and Ślepaczuk (2024) fill every execution at the midpoint plus half the quoted bid–ask on both the option and the hedge leg, and François et al. (2025) charge a percentage cost $\kappa_2 \in \{0.5, 1, 1.5, 2\}\%$ on every option position change against a far smaller $\kappa_1 = 0.05\%$ on the underlying. The lesson across all of them is the same: cost is incurred per transaction, scales with how rich the spread is at that maturity, and dominates the economics of any strategy that trades options frequently. The mechanics of turning costs into a Sharpe drag, turnover and breakeven cost, were developed for the underlying in Section 18.2; here they apply to the option leg through Equation (19.9).

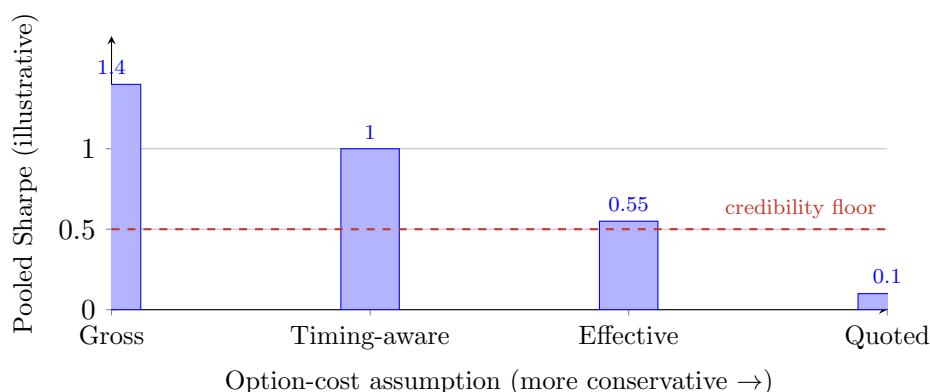


Figure 19.3: Schematic: pooled Sharpe falls as the option-cost assumption becomes more conservative, from gross (no cost) through timing-aware ($\approx$ one-third of quoted) and empirically grounded effective (Table 19.1) to the full quoted half-spread. The numbers are illustrative; the shape is the point. A strategy whose Sharpe sinks below a sensible floor under the quoted assumption is living on execution optimism, not edge (Muravyev and Pearson, 2020).

19.6 The Hedge Also Costs: Leland and Its Critique

Section 19.5 charged the spread on the option. But every time we rebalance the delta hedge we also trade the underlying, and that trade crosses a spread too. The classic way to fold this cost into the pricing, due to [Leland \(1985\)](#) (restated in this notation by [Zhao and Ziemba, 2003](#)), is a sleight of hand: pretend you are hedging at a slightly *higher* volatility, chosen so that the extra option premium exactly funds the expected hedging costs.

$$\hat{\sigma}^2 = \sigma^2 \left(1 + \sqrt{\frac{2}{\pi}} \frac{k}{\sigma \sqrt{\delta t}} \right). \quad (19.10)$$

- $\hat{\sigma}$: the **Leland-modified volatility** to use in the hedge (and in pricing) instead of the true σ .
- k : the round-trip (buy-then-sell) proportional transaction cost of trading the underlying, a fraction such as $0.001 = 0.1\%$ (e.g. the relative bid–ask spread).
- δt : the time between hedge rebalances, the same role dt played in Equation (19.3) but now a finite interval rather than infinitesimal; more frequent hedging (smaller δt) inflates the adjustment more.
- $\sqrt{2/\pi} \approx 0.798$: the mean of the half-normal distribution, that is, $\mathbb{E}|u|$ for $u \sim \mathcal{N}(0, 1)$.

In Plain English

Each rebalance trades an amount of stock proportional to how much delta moved, and the expected size of that move is governed by the half-normal mean $\sqrt{2/\pi}$. Leland’s insight is that you can bake the resulting expected cost into a single fudge to volatility: hedge as if the world were slightly more volatile, and the fatter option premium pays your transaction bill. The $1/\sqrt{\delta t}$ factor warns that hedging twice as often does not halve costs, it raises the volatility adjustment.

The catch is that Equation (19.10) is a heuristic, not an optimal policy, and modern work has mapped its limits precisely. [Kabanov and Safarian \(1997\)](#) prove that if transaction costs are held *constant* as the rebalancing frequency grows, the Leland-hedged portfolio does *not* converge to the option payoff: a nonzero, negative limiting error remains, meaning the strategy systematically under-hedges. The error only vanishes if costs shrink with frequency: [Lépinette and Kabanov \(2010\)](#) show that with costs scaling as $k_n = k_0 n^{-1/2}$, the mean-square hedging error converges to zero at rate n^{-1} (root-mean-square at $n^{-1/2}$) for *convex* payoffs, and a straddle is convex. Beyond these results, the current frontier treats hedging as a control problem rather than a volatility fudge: [Arzel and Lehdili \(2026\)](#) use a no-transaction band of width $h_{WW} = (3\lambda\delta S \Gamma^2/(2\gamma))^{1/3}$ (the Whalley–Wilmott band) inside which one simply does not trade, and [Brugière and Turinici \(2025\)](#) show a neural-network hedger beating Leland at realistic cost levels.

Leland Is a Cost-Line Inflator, Not the Optimal Trade

Use Equation (19.10) as a transparent, deterministic way to charge hedging costs in the backtest, exactly as Section 18.2 charged turnover costs for the underlying. Do not present it as the best possible hedging policy: under constant costs it under-hedges ([Kabanov and Safarian, 1997](#)), and band or learned policies dominate it ([Arzel and Lehdili, 2026](#); [Brugière and Turinici, 2025](#)). For our purposes it is a baseline cost model, and a good one.

19.7 The Variance the Hedge Injects

Here is the subtlest cost of all, and the one most often mis-handled. Even with a perfect volatility forecast and zero transaction costs, hedging at discrete moments leaves a *random* replication error every period. Its mean is zero, so it does not bias the P&L, but its *variance* is real risk that must inflate the denominator of any Sharpe ratio. The goal of this section is a closed form for that variance, attributed to exactly the right source.

Start with one rebalancing interval. Applying the Black–Scholes PDE and dropping higher-order terms, the hedging error accrued over a single step is

$$H_i = \frac{1}{2} \Gamma S^2 \sigma^2 \delta t (1 - u_i^2), \quad u_i \sim \mathcal{N}(0, 1), \quad (19.11)$$

so that the total error at expiry is $HE(T) = \sum_{i=1}^N H_i$.

- H_i : the replication error contributed by the i -th rebalancing interval.
- ΓS^2 : the dollar gamma; $\sigma^2 \delta t$: the variance accrued over the step of length δt .
- $u_i \sim \mathcal{N}(0, 1)$: the standardized return over the step, so $u_i^2 \sim \chi_1^2$ is a (skewed, fat-tailed) chi-squared with one degree of freedom.
- $1 - u_i^2$: a mean-zero shock (mean zero because $\mathbb{E}[u_i^2] = 1$); positive when the realized squared return undershoots its expectation, negative when it overshoots.

This is the discrete-hedging error of [Boyle and Emanuel \(1980\)](#), here in the convenient decomposition of [Anagnou and Hodges \(2007\)](#) (their Eqs. 5 and 7). Note the per-step shock $1 - u_i^2$ is *not* Gaussian; it is a shifted chi-squared, skewed and leptokurtic. Only after summing many independent steps does a central-limit effect make the aggregate roughly normal.

In Plain English

Between hedge adjustments the stock moves, and your frozen delta is wrong by a little. The gamma term tells you how badly: each interval you book half the dollar gamma times the gap between the variance you expected ($\sigma^2 \delta t$) and the variance that actually happened ($\sigma^2 \delta t u_i^2$). On average this gap is zero, so you neither make nor lose money from it systematically, but it jitters your P&L, and that jitter is pure noise added on top of the variance bet you actually wanted.

Because the steps are independent and $\text{Var}(u_i^2) = 2$ for a Gaussian return, the aggregate error variance is a sum of N equal pieces. With $\delta t = T/N$,

$$\text{Var}(HE(T)) = \sum_{i=1}^N \left(\frac{1}{2} \Gamma S^2 \sigma^2 \delta t \right)^2 \text{Var}(u_i^2) = \left(\frac{1}{2} \Gamma S^2 \sigma^2 \right)^2 T^2 \frac{2}{N}, \quad (19.12)$$

where the middle-to-right step uses that the N terms are identical: each contributes $(\frac{1}{2} \Gamma S^2 \sigma^2 \cdot T/N)^2 \cdot 2$, and summing N of them turns the $(T/N)^2$ into T^2/N . So the hedging-error standard deviation scales as $1/\sqrt{N}$: hedge four times as often and you roughly halve the noise. This is the [Boyle and Emanuel \(1980\)](#) result that the replication error is inversely related to the rebalancing frequency.

Now the refinement that matters for intraday data. [Anagnou and Hodges \(2007\)](#) observe that finer sampling raises the excess kurtosis of returns under non-Gaussian processes, which enlarges the replication error, but they keep $\text{Var}(u_i^2) = 2$ in their closed form. We make that dependence explicit. For a standardized shock with kurtosis $\kappa = \mathbb{E}[u_i^4]$, the variance of the squared shock

is $\text{Var}(u_i^2) = \mathbb{E}[u_i^4] - (\mathbb{E}[u_i^2])^2 = \kappa - 1$ (the Gaussian case $\kappa = 3$ recovers 2). Substituting into Equation (19.12) gives the leptokurtic hedging-error variance.

Kurtosis-Inflated Hedging-Error Variance

For a standardized per-step return with kurtosis κ ,

$$\text{Var}(HE(T)) = \left(\frac{1}{2}\Gamma S^2 \sigma^2\right)^2 T^2 \frac{\kappa - 1}{N} \propto \frac{\sigma^4 (\kappa - 1)}{N}. \quad (19.13)$$

The symbol $\propto$ means “proportional to”: dropping the fixed factors (Γ , S , T , the $\frac{1}{2}$) leaves what actually drives the noise, namely volatility to the fourth power and the fat-tail factor $\kappa - 1$, divided by the rebalancing count N . **Provenance, stated honestly:** the $1/N$ scaling and the per-step form are [Boyle and Emanuel \(1980\)](#) (via [Anagnou and Hodges, 2007](#), Eq. 7). The explicit $(\kappa - 1)$ substitution is *our own leptokurtic extension*; it does not appear in Boyle–Emanuel, in Anagnou–Hodges (who fix $\text{Var}(u_i^2) = 2$), or in Ahmad–Wilmott. Re-derive it, do not cite it to those sources.

Do Not Attribute This to Ahmad–Wilmott

It is tempting to credit the $1/N$ -and-kurtosis hedging-error variance to [Ahmad and Wilmott \(2005\)](#), because they give a closed-form variance for delta-hedged P&L. They do not. Their “Result 2” (their §4.2, Eq. 3, p. 70) is the variance of *continuous* hedging at implied volatility, a $G(S_0, t_0) - F(S_0, t_0)^2$ expression that arises from the path-dependence of the gamma profit under the real drift; it contains *no* $1/N$ term and *no* kurtosis term, because it is not about discrete rebalancing at all. The discrete-rebalancing variance is Boyle–Emanuel; the kurtosis inflation is the extension above.

Practitioners often package the same idea in vega form: the hedging-error standard deviation of a delta-hedged option is sometimes written $\sigma_{\text{P\&L}} = \mathcal{V} \sigma \sqrt{\pi/(4N)}$ ([Bennett, 2014](#), context only). Beware a coefficient ambiguity here: $\sqrt{\pi/4} \approx 0.886$ follows the half-normal mean-absolute convention, whereas the root-variance (L2) convention gives $1/\sqrt{2} \approx 0.707$; which is “correct” depends on whether you are reporting a mean-absolute error or a standard deviation. Match the convention to the quantity you put in the Sharpe denominator.

Finally, a caution about the limit. The clean $1/N$ decay assumes a continuous diffusion. [Brodén and Tankov \(2010\)](#) show that under jumps (an exponential Levy model) the rescaled tracking error does *not* vanish at the $1/N$ rate: $\lim_n n \mathbb{E}[\varepsilon_T^2]$ is strictly positive and can even be infinite, so for a jumpy underlying the analytic variance is a *lower bound*, not the truth. For vanilla calls and puts the $1/N$ rate survives but with a jump-inflated constant; for digital or barrier payoffs the rate is strictly slower than $1/\sqrt{N}$. Figure 19.4 contrasts the two regimes.

19.8 Calibrating the Kurtosis κ from 5-Minute Data

Equation (19.13) needs one number we have not yet pinned down: the kurtosis κ of the per-step return. We can estimate it directly from intraday data using **realized kurtosis**, the high-frequency analogue of realized variance ([Amaya et al., 2015](#)):

$$RK_t = \frac{n \sum_{i=1}^n r_{t,i}^4}{\left(\sum_{i=1}^n r_{t,i}^2\right)^2}, \quad (19.14)$$

- RK_t : the realized kurtosis estimate for day t .

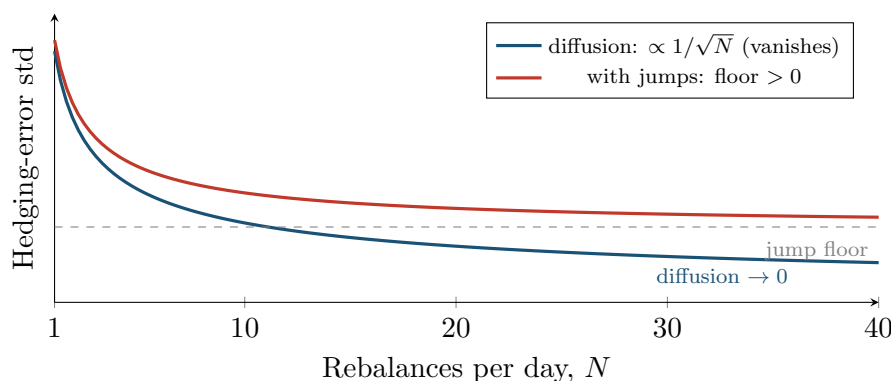


Figure 19.4: Discrete-hedging error standard deviation versus rebalances per day. Under a pure diffusion the error falls as $1/\sqrt{N}$ and can in principle be driven to zero (blue). Once the underlying can jump, the rescaled error no longer vanishes: it asymptotes to a positive floor (red, [Brodén and Tankov, 2010](#)), so the analytic $1/N$ variance of Equation (19.13) is a lower bound on the true intraday hedging noise.

- $r_{t,i}$: the i -th intraday (e.g. 5-minute) return on day t ; n : the number of such returns per day.
- the numerator is the realized fourth moment scaled by the count n (this normalization makes a perfectly Gaussian day score 3); the denominator is realized variance squared.

In Plain English

Realized kurtosis asks how much of the day's variance came from a few big five-minute jumps versus many small wiggles. If a handful of intervals dominate the sum of fourth powers, RK_t is large and the return distribution is fat-tailed; if the moves are evenly sized, it sits near the Gaussian value of 3. It is the empirical answer to “how leptokurtic are my returns?”, which is exactly the κ that Equation (19.13) demands.

Why This Matters

The kurtosis feeds straight into the hedging-error variance and therefore into the Sharpe denominator. A reassuring fact from [Amaya et al. \(2015\)](#) is that realized kurtosis is far more robust to microstructure noise than realized *variance*: the noise-sensitive moment is RV, not the fourth-moment ratio, so we can trust RK_t even where we would worry about RV_t (Chapter 3).

Two cautions keep this honest. First, realized higher moments are **interval-variant**: [Ahadzie and Jeyasreedharan \(2020\)](#) show they do not converge to the sample skewness and kurtosis of daily returns and depend on both the sampling interval and the holding interval. Aggregating 5-minute estimates up to a 15-minute horizon via a central-limit argument is therefore a *stated convention*, not a free lunch; the number you report is tied to the frequency you chose. Second, κ is not one constant across names: realized kurtosis is systematically higher for small-cap, high book-to-market, and low-beta stocks ([Amaya et al., 2015](#)), so imposing a single conservative value such as $\kappa = 4.0$ across a single-name universe is a simplification, defensible as a baseline but worth stress-testing.

$\kappa = 4.0$ Is a Convention, Not a Measurement

A flat $\kappa = 4.0$ is a reasonable, mildly conservative default (it sits above the Gaussian 3), but it is a placeholder for a per-symbol estimate. Estimate RK_t for each name from its own 5-minute returns where you can, and always report how the deflated Sharpe of Section 19.10 moves as κ ranges over, say, $\{3, 4, 6\}$. If the conclusion flips within that range, the strategy's edge is inside the hedging noise, not above it.

19.9 Assembling the Backtest

We now have every piece. The strategy is the loop below, run over every (symbol, day) pair in the sample. Each numbered step points back to the section that justified it, so the algorithm is just the chapter in executable order.

The IV–RV Straddle Backtest, One (Symbol, Day) at a Time

For each symbol and each trading day t :

1. **Signal.** Before 3:55 pm, compute the forecast $\widehat{RV}_t$ and read the lagged implied volatility IV_{t-1} ; form the signal X_{t-1} of Equation (19.1) (units aligned via Equation (19.2)); decide short / long / flat and a graded size (Section 19.2).
2. **Enter.** Trade the straddle before the 4 pm close, charging the option cost c_{opt} of Equation (19.9) on entry, on any flip, and on exit (Section 19.5).
3. **Hedge.** Over day $t + 1$, delta-hedge N times, charging the underlying's trading cost through the Leland-modified volatility of Equation (19.10) (Section 19.6).
4. **Mark.** At the close, book the day's gamma P&L from Equation (19.5) using realized variance RV_{t+1} .
5. **Risk.** Attach the hedging-error variance of Equation (19.13), using a κ calibrated per Section 19.8, to the day's risk (not its mean).
6. **Accumulate.** Append the net P&L to the (symbol, day) panel for evaluation in Section 19.10.

Figure 19.5 shows where the gross gamma P&L goes. The two transaction costs (option spread, hedge spread) bite the *mean*; the discrete-hedging error does not move the mean (it is zero-mean) but widens the *risk*, shown as the band on the net bar. Keeping these two effects separate, mean drag versus risk inflation, is the single most common bookkeeping error in volatility-strategy backtests, and it is exactly what Section 19.10 must respect when it forms the Sharpe ratio.

19.10 Evaluation: From QLIKE to a Deflated Sharpe

We now have a panel of net daily P&L, one number per symbol per day, and a risk term to go with it. The final task is to turn this panel into a verdict that is honest about both the cross-section and the multiple-testing trap, and to connect it back to the forecast accuracy we started from.

The first decision is how to pool. We could compute one Sharpe ratio per symbol and average them, or pool all (symbol, day) observations into a single Sharpe. The two answers differ, and the difference is informative rather than cosmetic. Pooling inflates the effective sample size and quietly assumes observations are independent, but on a common-volatility day every symbol's straddle tends to win or lose together, so the cross-sectional correlation is large; pooling that day's hundreds of observations as if independent overstates significance. Averaging per-symbol Sharpes respects that each name is one bet but discards information about how the strategy behaves across the cross-section.

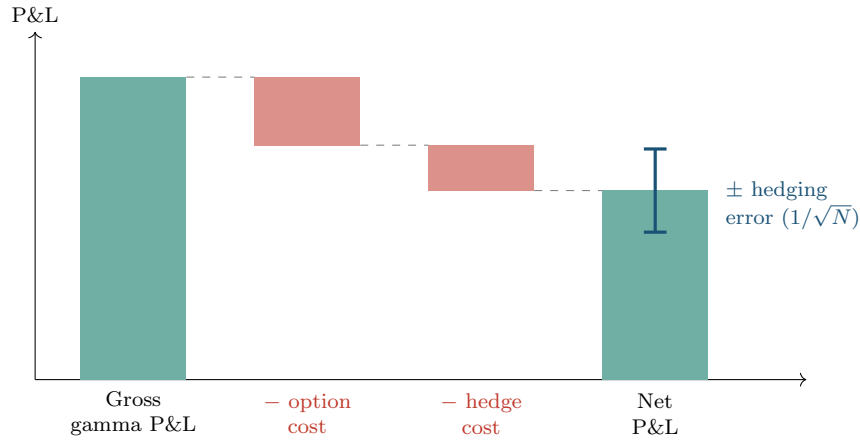


Figure 19.5: P&L attribution for the delta-hedged straddle (bar heights illustrative). The gross gamma P&L is reduced by the option transaction cost (Section 19.5) and the delta-hedge transaction cost (Section 19.6) to give the net mean P&L. The discrete-hedging error of Section 19.7 is zero-mean, so it does not appear as a bar; instead it widens the risk around the net, shown as the blue band.

Report Both, and Bootstrap by Day

Compute the pooled (symbol, day) Sharpe *and* the average per-symbol Sharpe. For significance, **block-bootstrap by day**, not by observation: resample whole trading days (all symbols together) so the resampling preserves the cross-sectional dependence that pooling would otherwise ignore. If the two Sharpes disagree sharply, the strategy’s apparent edge is concentrated in a few common-vol days, which is itself a finding.

Whichever Sharpe we report, it must be *deflated* for the number of strategy variants we tried before settling on this one. The deflated Sharpe ratio was built in Section 17.10; we restate it only to flag the traps specific to this application. From Equation (17.14),

$$\text{DSR} = \Phi \left(\frac{(\widehat{\text{SR}} - \text{SR}_0) \sqrt{T-1}}{\sqrt{1 - \gamma_3 \widehat{\text{SR}} + \frac{\gamma_4 - 1}{4} \widehat{\text{SR}}^2}} \right), \quad \text{SR}_0 = \sqrt{\text{Var}[\widehat{\text{SR}}_n]} \left[(1 - \gamma_E) Z^{-1} \left(1 - \frac{1}{N} \right) + \gamma_E Z^{-1} \left(1 - \frac{1}{N_e} \right) \right], \quad (19.15)$$

- $\widehat{\text{SR}}$: the observed Sharpe of the chosen strategy; T : the number of return observations.
- γ_3, γ_4 : the skewness and *kurtosis* of the strategy’s returns (note: distinct from the hedging-error κ).
- SR_0 : the expected maximum Sharpe under the null that none of the N trials has skill; Z^{-1} is the inverse standard normal CDF.
- $\gamma_E \approx 0.5772$: the Euler–Mascheroni constant (distinct from γ_3, γ_4); N : the number of strategy configurations tried.

In Plain English

The deflated Sharpe answers a single question: what is the probability that this strategy’s Sharpe is *real*, rather than the luckiest of the many variants we tried? It returns a number between 0 and 1; close to 1 means “probably real.” Read the numerator as the observed Sharpe minus the bar SR_0 that the best of N purely-lucky strategies would be expected to clear, scaled up by how much data T we have; the denominator deflates that confidence

when the returns are skewed (γ_3) or fat-tailed (γ_4). Here Φ is the standard normal CDF and $Z^{-1} = \Phi^{-1}$ its inverse; SR_0 is built from how far into the tail the *maximum* of N lucky draws is expected to reach, which is exactly why trying more strategies raises the bar. A fully worked numerical example is in Section 17.10.

Three Traps in Applying the Deflated Sharpe Here

(1) The *denominator* of Equation (19.15) uses the *observed* $\widehat{SR}$, not SR_0 ; a well-known web summary places SR_0 there and is wrong. (2) Feed *non-annualized* inputs at the native frequency, with T the raw observation count; annualizing first double-counts. (3) Be honest about N : if you tried 20 forecast or sizing variants, the simple approximation gives $SR_0 \approx \sqrt{2 \ln 20} = 2.45$ (Equation (17.13)), a high bar your reported Sharpe must clear. And do not confuse the Euler–Mascheroni γ_E with the return-kurtosis γ_4 sharing the symbol family.

Finally, the thread back to forecasting. Does a better QLIKE actually produce a better strategy? Pollok (2025) provides the most direct published evidence for this exact pipeline: marginal improvements in standard forecast-error measures can translate into economically significant gains in a delta-hedged straddle portfolio, even though it is hard to beat the benchmark on QLIKE alone, so the economic test separates models that the statistical loss cannot. Pollok’s statistical toolkit, however, is MSE/MAE/QLIKE and Mincer–Zarnowitz R^2 ; it does *not* include a Diebold–Mariano test or a Model Confidence Set. So there is no published theorem guaranteeing that a DM-significant or MCS-surviving QLIKE gain (Sections 17.5 and 17.6) maps to a guaranteed P&L gain. That bridge is ours to build and test (Section 19.11).

The Whole Point, in One Line

The economic foundation underneath all of this is the negative variance risk premium: Bakshi and Kapadia (2003) show the mean delta-hedged gain on equity options is negative, larger in magnitude in high-volatility regimes, and robust to jump controls, which is why a disciplined short-vol strategy is paid on average (Chapter 9). Our forecast’s job is to time *when* that premium is rich enough to harvest and when it is a trap. The deflated Sharpe is the judge of whether the forecast does that job better than chance.

19.11 What to Compute on Our Data

The literature cited in this chapter is almost entirely daily-frequency and drawn from 1993–2023; none of it validates the intraday, 5-minute mechanics we have assembled. The following four experiments are the ones that would turn this chapter from a design into a result on our own data, each with an explicit pass criterion.

Four Experiments, Four Verdicts

1. **Hedging-error floor.** Run a full path-dependent, bar-by-bar 5-minute hedging simulation for a near-ATM index straddle at $N \in \{1, 2, \dots, 26\}$ rebalances; fit $\text{Var}(\text{error}) = a/N + b$. *Pass:* if the floor b is materially positive (the jump regime of Figure 19.4), report the *simulated* variance in the Sharpe denominator, not the analytic Equation (19.13). Resolve κ first from Equation (19.14), and check that the deflated Sharpe is stable across $\kappa \in \{3, 4, 6\}$.
2. **Cost-band Sharpe.** Recompute the pooled Sharpe under (a) the full quoted half-spread, (b) the maturity-resolved effective spread of Table 19.1, and (c) a timing-

aware cost of roughly one-third the quoted. *Pass*: the strategy survives at least (b), and ideally (a).

3. **Statistical-to-economic link.** Regress per-(symbol, day) P&L on the realized gap $(RV - IV^2/252)$ and on the forecast error $(RV - \widehat{RV})$. *Pass*: the QLIKE-better model's residual edge is Diebold–Mariano significant *and* that DM statistic predicts the cross-sectional Sharpe, the bridge Pollok gestures at but does not prove.
4. **Sharpe definition and deflation.** Report pooled and per-symbol Sharpe, block-bootstrapped by day, with the deflated Sharpe computed for the honest number N of configurations tried. *Pass*: the deflated Sharpe clears SR_0 for the true N , not for $N = 1$.

19.12 Summary

What to Carry Forward

- The strategy trades the IV–RV gap through a delta-hedged straddle; the signal $X_{t-1} = f(\widehat{RV}_t, IV_{t-1})$ is the only place the forecast enters (Equation (19.1)), measured before 3:55 pm to avoid lookahead.
- Its engine is the gamma identity $PnL_t \approx \frac{1}{2}\Gamma_t S_t^2 (RV_t - IV^2/252)$ (Equation (19.5)); it is exact only for a near-ATM, jump-free, continuously hedged diffusion, and frays through vanna, volga, and jumps (Section 19.4).
- Charge the option spread per event in vega terms (Equation (19.9)) as a cost *band*; charge the hedge spread through Leland's modified volatility (Equation (19.10)), a baseline, not the optimal policy.
- The discrete hedge injects a zero-mean, $1/N$ -scaling variance, Boyle–Emanuel plus our kurtosis extension $\text{Var}(HE) \propto \sigma^4(\kappa - 1)/N$ (Equation (19.13)), *not* Ahmad–Wilmott's continuous-hedging variance; under jumps it is a lower bound.
- Judge the panel with a deflated Sharpe (Equation (19.15)) computed honestly (observed $\widehat{SR}$ in the denominator, non-annualized inputs, true N), and tie it back to QLIKE through the experiments of Section 19.11.

The One Caveat That Dominates the Rest

Every economic-value result cited in this chapter is daily-frequency and from the 1993–2023 era. The intraday, 5-minute hedging-error and cost mechanics that make our version “realistic” are precisely the parts no cited paper validates. Treat the closed forms here as scaffolding to be confirmed by our own bar-by-bar simulation, not as borrowed truth. The chapter's value is that it tells you exactly which numbers you still have to earn.

Chapter 20

Predicting Drawdowns of a Daily Variance-Swap Seller: The GSVIVS01 Index

Why This Chapter

On 1 June 2022 the GSVIVS01 index fell 27.69 basis points in a single day. That morning the strategy had sold a strip of options whose price embedded a known, observable estimate of how much the S&P 500 would move; the market then moved more than that estimate priced in, and the short paid for the difference. The question this chapter is built to answer is blunt: *the strike the strategy sold was sitting in plain sight that morning, so could a realized-volatility forecast have flagged the day in advance and told us to step aside?*

This is the capstone application of the whole guide for one specific, real strategy. Chapter 19 traded the gap between implied and realized volatility through a delta-hedged straddle, and it ended on a confession: a single straddle is *not* a clean variance bet, because its exposure collapses once the underlying drifts away from the strike; the instrument that *is* a clean variance bet is the variance swap, and we traded the straddle only because the swap's option strip is less liquid. **GSVIVS01 is that clean instrument, made tradeable.** It sells the full variance-swap strip every day, on zero-days-to-expiry S&P 500 options. This chapter does two things: first it explains, from first principles, what GSVIVS01 sells and how (Act 1); then it builds the project's deliverable, a model that predicts the strategy's drawdowns and times an overlay on the index (Act 2). The economic value of that overlay is the evaluation metric the entire forecasting effort is ultimately judged by.

Background

This chapter stands on five earlier strands and one external document:

- **The volatility surface** (Section 8.2.1, Section 8.7.2, Section 8.8): Black–Scholes Greeks, the model-free implied variance integral (Equation (8.14)), the VIX construction (Equation (8.15)), and the variance-swap payoff (Equation (8.16)). We *re-derive* the swap mathematics here so the chapter is self-contained, then go beyond Chapter 8 to the real product.
- **The variance risk premium** (Section 9.1) and the **gamma P&L identity** (Section 9.6): why selling variance is paid on average.
- **Jumps and continuous variation** (Chapter 4): the decomposition of realized

variance into a smooth part and a jump part is exactly what sharpens the drawdown signal in Act 2.

- **The IV–RV straddle** (Chapter 19): the signal timing, the gamma engine, and the deflated-Sharpe evaluation (Section 19.10) are reused wholesale.
- **The forecast** (Chapter 6, Chapter 10, Chapter 17): the realized-volatility model whose prediction drives the whole overlay. Throughout, a hat denotes a model forecast, so $\widehat{RV}_t$ is the *forecast* of day t 's realized variance and RV_t is its realized value.
- **The strategy specification** GSVIVS01.md: the source for every real number in this chapter (trades, strikes, index levels).

20.1 The Strategy in One Picture

GSVIVS01 runs a single idea on a daily clock: **sell the market's price of future variance, then pocket the difference when the market turns out calmer than that price implied.** A **variance swap** is a contract that pays its holder the gap between the variance the underlying actually delivers and a fixed strike agreed up front; the party who is *short* the swap, which is GSVIVS01's position, profits whenever realized variance lands below the strike. Rather than sign such a contract over the counter, the strategy **replicates** it: each afternoon it sells a weighted strip of out-of-the-money S&P 500 options expiring that same day (**zero days to expiry**, abbreviated **0DTE**), with the weights chosen so the package behaves exactly like a short variance swap, and it neutralizes the leftover directional exposure by trading E-mini S&P 500 futures (ticker ES). By the close the options have expired or settled, and the strategy is flat, ready to do it again tomorrow.

Figure 20.1 traces the daily loop and previews where Act 2 plugs in. The premium the strategy harvests is the **variance risk premium**: the systematic tendency of options to price in more variance than subsequently occurs (Section 9.1). The price of that harvest is the position's risk shape, which we will make precise in Section 20.7: GSVIVS01 is structurally **short gamma** (it loses when the underlying moves a lot), **short vega** (it loses when implied volatility rises), and **long theta** (it earns the passage of quiet time).

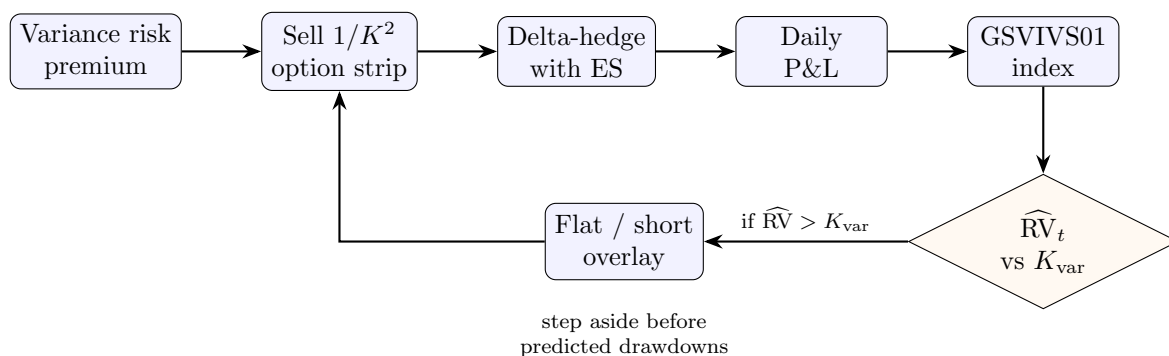


Figure 20.1: The GSVIVS01 pipeline. Act 1 (top row) is the real product: harvest the variance risk premium by selling a $1/K^2$ strip of 0DTE options, delta-hedge with E-mini futures, and compound the daily profit and loss into an index. Act 2 (bottom row, this project) compares a realized-volatility forecast $\widehat{RV}_t$ against the variance-swap strike K_{var} the strip is selling, and overlays a flat-or-short position to side-step the index's drawdowns.

In Plain English

GSVIVS01 is an insurance company that writes one-day policies against large market moves, every single day, and collects a premium set by the options market. On calm days the policies expire worthless and the premium is pure profit; on a stormy day it pays a large claim. It stays in business because, on average, the premium it charges (implied variance) exceeds the claims it pays (realized variance). Act 2's forecast is the underwriter who, before each day's policies are written, whispers whether tomorrow looks calm enough to be worth insuring.

20.2 What a Variance Swap Pays

To understand what GSVIVS01 sells we need the contract it imitates. A variance swap is the simplest possible instrument for taking a view on how much an asset will move: it has no Greeks to manage day to day, just a single payment at expiry that depends on realized variance. We want to write down that payment and see who wins under what conditions.

The payoff to the holder (the **long** side) of a variance swap at expiry is the realized variance over the contract's life minus a fixed strike, scaled by a notional:

$$\text{Payoff}_{\text{long}} = N_{\text{var}}(\sigma_{\text{realized}}^2 - K_{\text{var}}^2). \quad (20.1)$$

- N_{var} : the **variance notional**, the dollar value of one unit (one “variance point”) of the realized-minus-strike gap.
- $\sigma_{\text{realized}}^2$: the **annualized realized variance** actually delivered by the underlying over the contract period.
- K_{var} : the **variance-swap strike**, quoted as a volatility so that K_{var}^2 is the strike in variance units; it is the “fair” level of variance agreed at inception (Section 20.3).

The **short** side, which is GSVIVS01, receives the negative of Equation (20.1), and therefore profits exactly when realized variance comes in below the strike, $\sigma_{\text{realized}}^2 < K_{\text{var}}^2$.

Worked Example: A Single Short Variance Swap

Suppose GSVIVS01 sells a one-day variance swap struck at $K_{\text{var}} = 20\%$ annualized, with a variance notional of \$1,000 per annualized variance point (where one “point” is one unit of σ^2 expressed in percent-squared, so $K_{\text{var}}^2 = 400$). If the day turns out quiet and annualized realized volatility is 16% (realized variance 256), the short collects $N_{\text{var}}(K_{\text{var}}^2 - \sigma_{\text{realized}}^2) = 1000(400 - 256) = \$144,000$. If instead a shock hits and realized volatility is 35% (variance 1225), the short pays $1000(400 - 1225) = -\$825,000$. The asymmetry of the two outcomes, a steady modest gain against an occasional large loss, is the signature of every short-variance position and the reason Act 2 exists. (Unit convention, used throughout: K_{var} is quoted in percentage volatility points, e.g. 20, so $K_{\text{var}}^2 = 400$; the signal in Equation (20.12) divides by 100 first to work in decimal variance, e.g. 0.04.)

The reason the contract is written on *variance* rather than *volatility* is not cosmetic. Variance is additive across time: the variance of a two-day period is the sum of the two daily variances, so a variance swap can be hedged by accumulating squared returns and, as the next section shows, replicated by a fixed portfolio of options. Volatility, the square root, has no such additivity and cannot be replicated by a static option position.

20.3 The Fair Strike: Model-Free Implied Variance

What number should the strike K_{var} be? If the strategy is going to sell variance every day, it needs a fair value for variance that the options market agrees on, and a recipe for capturing exactly that value with traded instruments. The remarkable result of Demeterfi et al. (1999), the original Goldman Sachs note that GSVIVS01 descends from, is that the fair strike is the price of a specific, static portfolio of options, with no model of volatility required. We re-derive it here.

Step 1: realized variance is a log contract

Assume for now that the underlying moves continuously (no jumps); we relax this in Section 20.8. Applying Itô's lemma (the calculus rule for a function of a randomly moving price) to $\log S_t$ and subtracting it from the price's own return increment dS_t/S_t cancels the drift (the average trend) and leaves the instantaneous variance. Summing over the life of the contract gives an exact identity (Demeterfi et al., 1999, Eq. 20, p. 17):

$$V = \frac{1}{T} \int_0^T \sigma^2(t) dt = \frac{2}{T} \left[\int_0^T \frac{dS_t}{S_t} - \log \frac{S_T}{S_0} \right]. \quad (20.2)$$

- V : the realized variance over $[0, T]$, the quantity the swap settles on.
- $\int_0^T dS_t/S_t$: the profit of **continuously rebalancing** a stock position to stay worth a fixed dollar amount (instantaneously long $1/S_t$ shares).
- $\log(S_T/S_0)$: the payoff of a **log contract**, a static claim paying the log of the total return.
- T : the time to expiry in years.

Read as a whole, the left side is the variance we want and cannot trade directly, while the right side is twice the difference of two payoffs we *can* trade (the continuously rebalanced stock minus the log contract). The equation is therefore a recipe for manufacturing realized variance out of tradeable pieces.

In Plain English

This identity is the whole trick. It says realized variance, an abstract statistical quantity, equals something you can actually trade: keep rebalancing a stock position to hold one dollar of exposure, and hold a static short log contract. Do both, and the combination mechanically accumulates exactly the variance the stock realizes, no matter which path it takes, as long as it never jumps. Demeterfi et al. (1999) stress the phrase “no expectations or averages have been taken”: this is an identity, not a forecast.

Step 2: the log contract is a $1/K^2$ strip of options

The rebalanced stock position is easy; the log contract is not traded, so it must be built from options. Demeterfi et al. (1999) decompose the log payoff (measured from an arbitrary reference price S_*) into a forward and a continuum of out-of-the-money options, the spanning result of Carr and Madan (2002) specialized to the log contract (Demeterfi et al., 1999, Eq. 25, p. 18):

$$-\log \frac{S_T}{S_*} = \underbrace{-\frac{S_T - S_*}{S_*}}_{\text{forward}} + \underbrace{\int_0^{S_*} \frac{1}{K^2} \max(K - S_T, 0) dK}_{\text{puts, strikes } K < S_*} + \underbrace{\int_{S_*}^{\infty} \frac{1}{K^2} \max(S_T - K, 0) dK}_{\text{calls, strikes } K > S_*}. \quad (20.3)$$

- S_* : an arbitrary boundary splitting puts (below) from calls (above), in practice the **forward** F (the price agreed today for delivery at expiry).

- $1/K^2$: the weight on the option of strike K , the load-bearing feature of the whole construction.
- $\max(K - S_T, 0), \max(S_T - K, 0)$: the expiry payoffs of a put and a call struck at K .

In words: one log contract can be reproduced by a forward plus a basket holding a sliver of every out-of-the-money option, each weighted by $1/K^2$; the $\int \cdots dK$ is that basket summed over all strikes, not a calculus problem to evaluate by hand. Taking the risk-neutral expectation (the average under the pricing measure that makes discounted traded prices fair) of Equation (20.2) after substituting Equation (20.3), and using that a one-dollar rebalanced stock position grows at the risk-free rate, yields the closed-form strike (Demeterfi et al., 1999, Eq. 26, p. 19):

$$K_{\text{var}}^2 = \frac{2}{T} \left[rT - \left(\frac{S_0}{S_*} e^{rT} - 1 \right) - \log \frac{S_*}{S_0} + e^{rT} \int_0^{S_*} \frac{P(K)}{K^2} dK + e^{rT} \int_{S_*}^{\infty} \frac{C(K)}{K^2} dK \right], \quad (20.4)$$

where $P(K)$ and $C(K)$ are the current prices of the put and call struck at K and r is the risk-free rate. The first three terms (rT , the bracket, and the log) are interest-rate and reference-price corrections; with $S_* = F$ (the forward) they vanish, and this reduces to the symmetric form quoted in the strategy spec, $K_{\text{var}}^2 = \frac{2e^{rT}}{T} \left[\int_0^F P(K)/K^2 dK + \int_F^{\infty} C(K)/K^2 dK \right]$, which is the **model-free implied variance** of Equation (8.14).

In Plain English

The fair strike is a shopping bill, not a model output. Buy every out-of-the-money option, weight each by one over its strike squared, and the total cost is the fair price of future variance. “Model-free” means the option prices themselves, whatever produced them, already encode the price of variance, with no assumption about the shape of volatility (no Black–Scholes, no Heston).

Step 3: why $1/K^2$, and why the strike beats at-the-money implied vol

The $1/K^2$ weight is not a guess. Demeterfi et al. (1999, App. A, Eq. A3, p. 38) show it is forced by demanding that the strip’s sensitivity to variance be *independent of the spot price*: writing the portfolio’s variance-vega and requiring $\partial(\text{vega})/\partial S = 0$ gives the ordinary differential equation $2\rho + K d\rho/dK = 0$, whose only solution is $\rho(K) = \text{const}/K^2$. Any other weighting would let the strip’s variance exposure drift as the underlying moved, defeating the purpose.

The consequence that matters for Act 2 is that this weighting puts disproportionate weight on low-strike, out-of-the-money puts, exactly where the equity volatility skew lives. So the variance-swap strike sits *above* at-the-money implied volatility. Demeterfi et al. (1999) quantify it for a skew that is linear in strike, $\sigma(K) = \sigma_0 - b(K - S_F)/S_F$ (Demeterfi et al., 1999, Eq. 30, p. 23), giving the approximate strike (Demeterfi et al., 1999, Eq. 31, p. 23):

$$K_{\text{var}}^2 \approx \sigma_0^2 (1 + 3Tb^2 + \cdots), \quad (20.5)$$

where σ_0 is at-the-money-forward implied volatility and b is the skew slope. The strike exceeds the at-the-money level, and the excess grows with maturity and with the square of the skew.

Use the Variance-Swap Strike, Not At-the-Money IV

This is the single most important point for the signal in Section 20.9. GSVIVS01 sells the *whole strip*, so what it collects is K_{var} , not at-the-money implied volatility σ_0 . The two differ by the skew premium of Equation (20.5), typically one to three volatility points for index options. The empirical ordering is firm: realized variance sits below the volatility-swap rate (which at-the-money implied volatility approximates), which sits below the

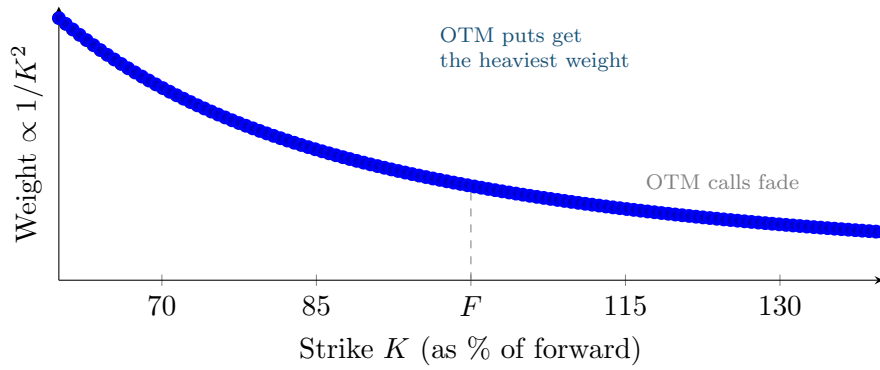


Figure 20.2: The $1/K^2$ replication weight as a function of strike. Low strikes (out-of-the-money puts) carry far more weight than high strikes, so the strip loads up on exactly the options where the equity skew makes implied volatility highest. This is why the variance-swap strike K_{var} exceeds at-the-money implied volatility, the skew premium of Equation (20.5).

variance-swap strike (Carr and Wu, 2009b). A signal that compares the forecast against at-the-money IV therefore systematically *understates* what GSVIVS01 actually sells, compressing the apparent premium and mistiming the exit. Benchmark against K_{var} .

Worked Example: The Skew Premium in Numbers

Demeterfi et al. (1999, Table 1, p. 21) price a three-month swap on an index at $S_0 = 100$ with at-the-money implied volatility 20% and a skew of one volatility point per five strikes. Without skew the fair strike would be $(20)^2 = 400$. Summing the $1/K^2$ -weighted strip across strikes from 50 to 135 gives a total option cost of 419.87 and a fair strike of $K_{\text{var}} = (20.467)^2$: the skew has lifted the strike by almost half a volatility point. Most of that cost, and most of the variance exposure, comes from options struck near the spot, even though the out-of-the-money wings contain far more individual contracts. This is the concrete content of “ $K_{\text{var}} > \sigma_0$ ”.

20.4 From Integral to Traded Strip: The CBOE Discrete Formula

Equation (20.4) is an integral over a continuum of strikes, but only a discrete grid of strikes trades. We need the finite-sum version that a trading desk, and the CBOE’s VIX, actually compute. This is the formula GSVIVS01 uses to price what it sells.

Replacing the integrals in Equation (20.4) by a sum over traded strikes gives the discrete strike (as in the VIX construction, Carr and Lee, 2009), identical to Equation (8.15):

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2, \quad K_{\text{var}} = 100\sqrt{\sigma^2}. \quad (20.6)$$

- $\Delta K_i = (K_{i+1} - K_{i-1})/2$: half the distance between the neighbouring strikes (a single-sided difference at the two endpoints, which halves their weight).
- $Q(K_i)$: the midpoint price of the out-of-the-money option at strike K_i (a put if $K_i < F$, a call if $K_i > F$, the average of the two at $K_i = F$).
- K_0 : the first strike at or below the forward F .

- the final term: a small correction for the nearest listed strike K_0 not sitting exactly at the forward; negligible at the 0DTE horizon where T is tiny.
- the factor 100 in $K_{\text{var}} = 100\sqrt{\sigma^2}$: it expresses the strike in percentage volatility points (e.g. 20, not 0.20).

Why This Matters

Equation (20.6) is also the data feed for Act 2. The strategy's data source (`EDRVS_EXPIRY`) publishes the fair variance by listed expiry, and when it is missing we reconstruct K_{var} from the option grid using exactly this formula. The K_{var} that enters the signal in Section 20.9 is the output of Equation (20.6).

Two honest caveats bound the accuracy of the discrete strip, both of which bite harder at the 0DTE horizon. First, the strip prices *continuous* variance; once the underlying can jump, it becomes a biased estimate of the true quadratic variation (the sum of squared moves that realized variance estimates), and the bias is material only when jumps make up a large share of variance, roughly above seventy percent (Du and Kapadia, 2012). The 0DTE horizon is precisely that high-jump-share regime, so we should remember that GSVIVS01's strike is a smooth-path object being sold against a jump-prone realization, a tension we cash out in Section 20.8. Second, the strip is truncated to a finite range of strikes, which turns it into a “corridor” swap and *underprices* variance, more so the wider the underlying can roam: Demeterfi et al. (1999, Table 4, p. 28) show a strike range of 75% to 125% of spot recovers $(24.9)^2$ against the full $(25.0)^2$ for a three-month swap, but only $(23.0)^2$ for a one-year swap. GSVIVS01's strip is narrow (the example below spans roughly $\pm 2.7\%$ of spot), so this corridor approximation is real and worth one line in any honest accounting.

20.5 How GSVIVS01 Trades It: 0DTE Replication

We now leave theory for the real product. The abstract strip of Section 20.3 becomes a concrete list of orders: roughly twenty-five 0DTE options, in quantities set by the $1/K^2$ rule, sold on a fixed intraday schedule. This section uses the strategy's actual trades from 26 May 2022.

Translating the $1/K^2$ weight into an order size, the quantity sold at strike K_i is

$$q_i = \frac{w}{K_i^2} \Delta K_i, \quad \text{so that} \quad |q_i| K_i^2 \approx w \Delta K_i \approx \text{constant}, \quad (20.7)$$

where w is a single scaling constant (the variance notional) and ΔK_i is the strike spacing of Equation (20.6). The product $|q_i| K_i^2$ is therefore the same for every interior strike, which is the discrete fingerprint of constant dollar exposure to variance. (The piecewise-linear procedure that sets these weights exactly so the strip's payoff always matches or exceeds the log contract is Demeterfi et al. (1999, App. A, Eqs. A5–A8); Equation (20.7) is its leading form.)

Worked Example: The Real Strip, 26 May 2022

On 26 May 2022 the S&P 500 forward was 4057.84 and GSVIVS01 sold 25 options (14 puts and 11 calls) spanning strikes 3875 to 4095, spaced 10 points apart (5 at the edges). Table 20.1 shows the quantities and the test $|q_i| K_i^2$. The interior strikes hold $|q_i| K_i^2$ constant at roughly 86,700, agreeing to within 0.05%; the two edge strikes sit at half that value because their ΔK is single-sided; and the forward strike 4000 is split between a put and a call. This is Equation (20.7) made real: the desk really does book quantities inversely proportional to the strike squared.

Table 20.1: Selected trades from the real GSVIVS01 strip of 26 May 2022 (source: `GSVIVS01.md`, `output.json`). Interior strikes hold $|q_i| K_i^2$ constant at about 86,700; the edge strikes carry half-weight (single-sided ΔK) and the forward strike is split between a put and a call.

Strike K_i	Type	Quantity q_i	$ q_i K_i^2$
3875	Put	-0.00289	43,360 (edge)
3880	Put	-0.00576	86,718
3890	Put	-0.00573	86,716
4000	Put + Call	-0.00271 each	43,346 (forward split)
4060	Call	-0.00526	86,678
4090	Call	-0.00518	86,672
4095	Call	-0.00258	43,336 (edge)

Figure 20.3 plots $|q_i| K_i^2$ across the real strikes. The flat plateau is the entire point of the construction, and it is the visual opposite of Chapter 19’s single straddle, whose dollar-gamma weight peaks at one strike and collapses into the wings (Figure 19.2). Where the straddle is a variance bet only near the money, the strip is a variance bet *everywhere*.

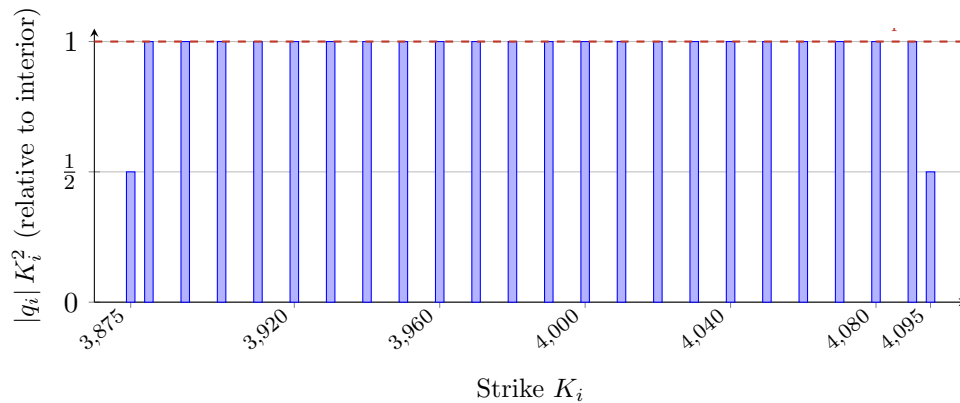


Figure 20.3: Dollar variance exposure $|q_i| K_i^2$ across the real 26 May 2022 strikes. The interior strikes form a flat plateau (constant variance exposure regardless of where spot goes); the two edge bars are half-height because their strike spacing is single-sided. Compare Chapter 19’s single delta-hedged straddle (Figure 19.2), whose exposure spikes at one strike and decays in the wings. The flat plateau is what makes GSVIVS01 the clean variance instrument the straddle could not be.

The strip is executed on a fixed daily schedule, summarized in Table 20.2. The fixed signal-generation time is what hands Act 2 a clean, lookahead-safe boundary for the forecast (Section 20.9).

20.6 Delta Hedging and the Index

We want GSVIVS01 to bet only on *how much* the market moves, not on *which way*. But selling the option strip leaves a residual directional exposure (a leftover sensitivity to the direction of the move, the position’s **delta**), because the strip is not perfectly delta-neutral as spot drifts intraday. GSVIVS01 cancels it the same way Chapter 19 hedged its straddle: by trading the underlying, here E-mini S&P 500 futures, on a rolling schedule. The strategy books roughly twenty-five hedge trades per day in 5-minute TWAP (time-weighted average price) intervals, each nudging the portfolio delta back toward zero.

Table 20.2: The GSVIVS01 daily schedule (Eastern Time), from `GSVIVS01.md`. The 13:10 signal-generation time defines the information cutoff for any forecast that drives the strategy.

Time (ET)	Activity
13:10	Signal computed, orders generated
13:30–14:00	Option strip executed via TWAP
14:00–14:15+	Delta hedge via E-mini futures (5-minute TWAP)
End of day	0DTE options expire or settle; position flat

The day’s profit and loss is then the premium collected on the short strip, less the cost of running the delta hedge, less transaction costs:

$$\text{P\&L}_t = \underbrace{\sum_i q_i (\text{premium})_i}_{\text{option premium collected}} - \underbrace{\sum_j \Delta_j (S_{\text{close}} - S_{\text{exec},j})}_{\text{delta-hedge cost}} - \text{transaction costs.} \quad (20.8)$$

- q_i : the (negative) quantity sold at strike K_i , from Equation (20.7).
- Δ_j : the size of the j -th futures hedge trade, executed at price $S_{\text{exec},j}$.
- S_{close} : the underlying’s settlement price, against which the hedge is marked.

The **GSVIVS01 index** compounds these daily increments from a base of 100. Table 20.3 shows the strategy’s real first week. Four quiet days drip in single-digit to twenty-something basis points; then 1 June erases nearly all of it in one move. That shape, many small gains punctuated by a sharp loss, is the short-variance signature we formalize next and learn to predict in Act 2.

Table 20.3: The real GSVIVS01 index over its first week (source: `GSVIVS01.md`, `output.json`). The -27.69 bps drawdown on 1 June nearly wipes out the four preceding gains, the canonical short-variance pattern.

Date	Index value	Daily return (bps)
2022-05-25	100.0000	— (inception)
2022-05-26	100.2675	+26.75
2022-05-27	100.3723	+10.46
2022-05-31	100.3517	−2.05
2022-06-01	100.0738	−27.69

Why This Matters

The index in Table 20.3 is the object Act 2 predicts and trades. Its level comes from the strategy’s `output.json`, whose fields (the index value, the per-strike trades, the per-position Greeks, the execution windows) are the project’s data contract. When we speak of “avoiding the drawdown on 1 June”, we mean turning that -27.69 into a flat or positive day by stepping aside, and the metric in Section 20.12 is exactly the improvement in this index path.

20.7 The Risk Profile: Short Gamma, Short Vega, Long Theta

Why does the index drip and then crash, rather than move smoothly? The answer is in the Greeks of the position, and naming them precisely tells us exactly which days are dangerous.

Because GSVIVS01 has sold the whole option strip, it inherits the negatives of the strip's Greeks. The log contract the strip replicates has a clean set of sensitivities (Demeterfi et al., 1999, Eqs. 9–12, p. 12): its variance exposure is constant, its gamma is $\Gamma = (2/T)(1/S^2)$, and its time decay and gamma satisfy the Black–Scholes balance $\theta + \frac{1}{2}\sigma^2 S^2 \Gamma = 0$. Being short it, GSVIVS01 is:

- **short gamma:** its delta worsens as the underlying moves, so large moves in either direction cost money; this is the core risk.
- **short vega:** it loses if implied volatility rises while the position is open.
- **long theta:** it earns the passage of quiet time, which is the income.

The balance $\theta + \frac{1}{2}\sigma^2 S^2 \Gamma = 0$ says these are two sides of one coin: the positive theta the short collects each calm day is exactly offset, on average, by the negative gamma it pays when the underlying moves. The strategy makes money precisely when realized variance undershoots the implied variance baked into theta.

In Plain English

Short gamma is the mathematics of “picking up pennies in front of a steamroller.” Every quiet day, theta hands the strategy a few pennies. The steamroller is gamma: on the rare day the market lurches, the short delta is on the wrong side and the loss is large and convex in the size of the move. The strategy is not broken when it has a bad day; a bad day is the price it pays for all the good ones. The job of Act 2 is to see the steamroller coming.

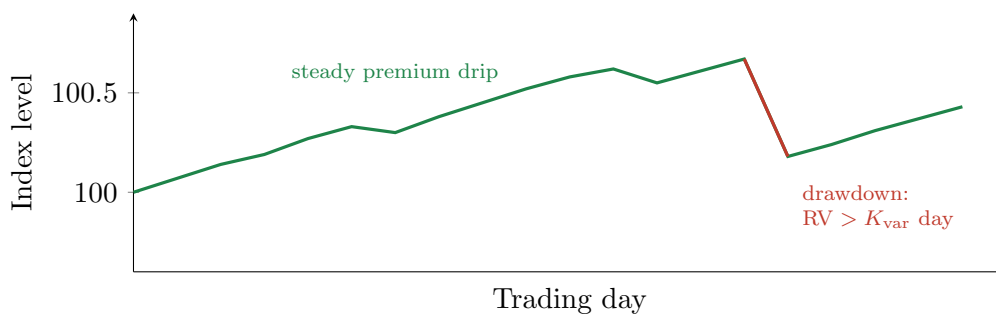


Figure 20.4: The short-variance signature of the GSVIVS01 index (stylized, matching the real first-week shape of Table 20.3). A long run of small daily gains from positive theta is interrupted by a sharp drawdown on the one day realized variance overshoots the strike. Act 2 aims to flatten the position across the red day while staying invested through the green run.

20.8 The Drawdown Mechanism: When Realized Variance Beats the Strike

Act 1 established *that* GSVIVS01 draws down when realized variance exceeds the strike. Act 2 turns that into a prediction problem, and to do so well we need to understand the condition's shape: which days produce the worst losses, and why they are asymmetric. The answer points the forecast at exactly the features that matter.

Write the drawdown condition in the daily units the forecast uses. A drawdown day is one on which the realized variance over the day, RV_t , exceeds the daily slice of the strike:

$$\text{drawdown on day } t \iff RV_t > \frac{(K_{\text{var}}/100)^2}{252} \iff \text{VRP Gap}_t < 0, \quad (20.9)$$

where 252 is the number of trading days in a year (it converts the annualized strike variance into one day's share). The third form uses "VRP Gap," the signal defined in Equation (20.12) below; for now read it simply as "the forecast variance exceeds the strike's daily variance."

One Number Is Known, One Is Forecast

The power of Equation (20.9) is that K_{var} is *observable at trade time*: it is the model-free implied variance the strip sells, computed from the morning's option prices by Equation (20.6). Only RV_t is unknown. So predicting a drawdown comes down to a sharp, quantitative question: "will today's realized variance exceed a number I can already read off the screen?" This is cleaner than the straddle of Chapter 19, where the implied side was a single noisy option quote; here the benchmark is a firm, market-wide strike.

Why the losses are asymmetric: the down-jump cubic

The drawdowns are not symmetric in the direction of the move, and the reason is structural, not behavioural. Suppose a single jump of size J hits an otherwise calm day, where $J > 0$ denotes a downward jump (the price falls from S to $S(1 - J)$) and $J < 0$ an upward one. That jump contributes J^2/T to realized variance (Demeterfi et al., 1999, Eq. 38, p. 30), but the amount the short option strip actually pays out on the jump is the variance it fails to capture cleanly (Demeterfi et al., 1999, Eq. 39, p. 30):

$$\text{loss to the short from the jump} = \frac{2}{T} [-J - \log(1 - J)]. \quad (20.10)$$

Expanding the logarithm as a series (Demeterfi et al., 1999, Eq. 41, p. 30), $-\log(1 - J) = J + \frac{1}{2}J^2 + \frac{1}{3}J^3 + \dots$, the $-J$ outside and the $+J$ from the series cancel inside the bracket, leaving $\frac{2}{T} [\frac{1}{2}J^2 + \frac{1}{3}J^3 + \dots]$, that is

$$\text{loss to the short} = \frac{J^2}{T} + \frac{2}{3} \frac{J^3}{T} + \dots, \quad (20.11)$$

so beyond the symmetric J^2/T piece there is a **cubic** term that is *positive for downward jumps and negative for upward ones* (Demeterfi et al., 1999, Eq. 42, p. 31).

- J^2/T : the expected, direction-blind contribution of the jump to variance, the same for an up-gap and a down-gap of equal size.
- $\frac{2}{3}J^3/T$: the asymmetry. For a down-gap ($J > 0$) it *adds* to the short's loss; for an up-move it subtracts.

A note on sign conventions, since Equation (20.11) is easy to misread against the source. Demeterfi et al. (1999, Eq. 42, Table 5) present the cubic term for a *hedged* book that is short the swap *and* long the replication, where a down-gap is a net *profit* because the long replication over-captures variance. GSVIVS01 holds only the short strip, so for us the same cubic term is an added *loss* on a down-gap. The size of the cubic and the fact that it depends on the jump's direction are the same calculation in both books; what flips is only whether it lands as a profit or a loss, because GSVIVS01 carries the short strip without the offsetting long-replication leg.

In Plain English

A crash and a melt-up of the same percentage do not cost the variance seller the same. The squared-return part of variance treats them identically, but a third-order correction tilts the scales: a downward gap inflicts a bigger loss than an upward move of equal size. Since equity crashes are gaps down, the short variance position's worst days are concentrated on the downside. Carr and Lee (2009) make the same point in peer-reviewed form: the

leading replication error is third order in the daily return and signed by the average of cubed returns, so a market that gaps down underprices the variance the seller must pay.

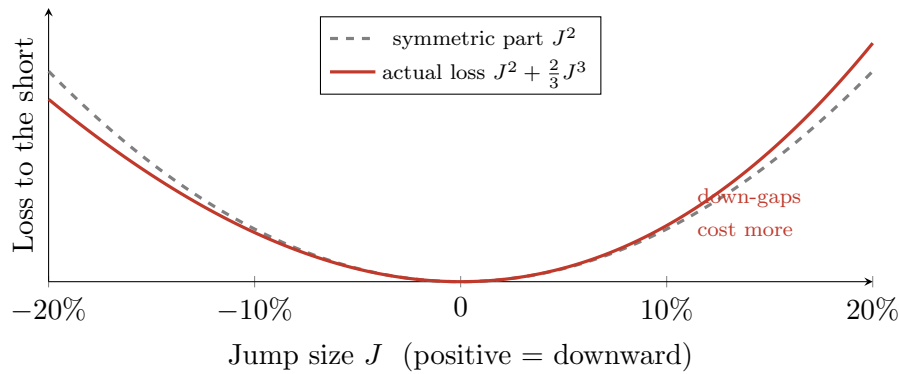


Figure 20.5: The down-jump asymmetry of a short variance position (the $1/T$ scaling dropped for clarity). The symmetric J^2 parabola (grey dashed) is what a direction-blind view of variance predicts; the actual loss $J^2 + \frac{2}{3}J^3$ (red) lies *above* it for downward jumps and below for upward ones. A 15% down-gap costs the short more than a 15% up-move, which is why GSVIVS01's worst days are crashes (Demeterfi et al., 1999, Table 5, p. 32).

The Strike Is a Smooth-Path Object

Equation (20.11) and the jump bias of Section 20.4 together carry a caution. The strike K_{var} prices *continuous* variance, but the 0DTE realization it is sold against is jump-laden (Du and Kapadia, 2012). On a down-gap day the short can therefore lose more than the strike anticipated, by the cubic term. Whether the net effect leaves the strike structurally too low on drawdown days is an empirical question for our data, not a settled fact; Section 20.13 makes it a test. It also suggests the signal may need to condition on the *jump share* of variance, not only on its level.

20.9 The Signal: A Variance-Risk-Premium Gap on the Forecast

We now build the number that decides each day's position. We have two ingredients: the strike K_{var} the strip is selling, known by 13:10, and our model's forecast $\widehat{RV}_t$ of the variance the day will realize. The signal is their gap, the variance risk premium viewed through the strategy's own instrument.

$$\text{VRP Gap}_t = \underbrace{\left(\frac{K_{\text{var}}}{100}\right)^2}_{\text{implied variance (annualized)}} - \underbrace{252 \widehat{RV}_t}_{\text{forecast variance (annualized)}}. \quad (20.12)$$

- K_{var} : the 0DTE variance-swap strike in volatility points (the field `iv_vs_0dte`), from Equation (20.6).
- $\widehat{RV}_t$: our model's forecast of day t 's realized *variance*, in daily units.
- the factor 252: annualizes the daily forecast so the two terms share units.

The position follows from the sign and size of the gap, with thresholds that can be asymmetric

(it should take more conviction to fade the premium than to collect it):

$$\text{Signal}_t = \begin{cases} +1 & \text{(stay short variance)} & \text{if VRP Gap}_t > \tau, \\ -1 & \text{(go flat or counter-trade)} & \text{if VRP Gap}_t < -\tau_{\text{short}}, \\ 0 & \text{(no position)} & \text{otherwise.} \end{cases} \quad (20.13)$$

The strategy reads the gap through three channels: **compression** (the gap drifting toward zero as the edge thins), **inversion** (the gap turning negative, the strong drawdown warning), and a **contrarian spike** (a sudden jump in the gap after a vol event, often signalling the market has overshot and it is safe to re-enter).

Why Conditional Timing Is the Whole Game

The recent literature settles a question that justifies this project's existence. Running GSVIVS01 *unconditionally* is not a free lunch: [Vilkov \(2024\)](#) finds the same-day 0DTE variance risk premium is small and hard to monetize after realistic frictions, with profit and loss “dominated by tail risk rather than stable mean carry,” and that disciplined *conditional* timing is what delivers net performance. The premium is genuinely there, richer per unit of time than the familiar one-month premium ([Almeida et al., 2024](#)), but harvesting it safely requires sizing the short on a signal. The closest published analogue to our overlay, [Yang \(2024\)](#), conditions a short-variance position on a volatility forecast and improves risk-adjusted returns (the measured figures are in Section 20.12). The state of the art says: do not run the short flat-out; time it. That is precisely what Equation (20.13) does.

The VRP Gap May Not Be the Only Conditioning Variable

The gap of Equation (20.12) is the natural signal, but it may not be the most predictable one. [Bandi et al. \(2024\)](#) find that the genuinely forecastable object at the 0DTE horizon is the instantaneous vol-of-vol and leverage premium, not the realized-minus-strike gap directly. Combined with the down-jump mechanism of Section 20.8, this suggests the signal may benefit from conditioning on vol-of-vol and jump-share alongside the VRP gap. We treat the choice as an open design question resolved by the experiments, not a settled recipe.

20.10 Drawdown Prediction as Cost-Sensitive Classification

The signal of Section 20.9 is a continuous gap, but the decision it serves is binary: is today one of the rare bad days, or not? Framing the problem as **classification** (predict the drawdown days) rather than pure regression sharpens both the modelling target and the way we judge it, and it forces an honest reckoning with the lopsided economics of the two kinds of error.

The target is rare. GSVIVS01 is safely short on roughly 85 to 88% of days; the remaining 12 to 15% are drawdown days, and within them a handful (the 5 to 10 worst per year) carry most of the loss. This **class imbalance** is the central difficulty: a model that predicts “safe” every single day is right almost ninety percent of the time, which is exactly why accuracy is the wrong score. We care about **precision** (of the days we flagged, how many were truly drawdowns) and **recall** (of the true drawdowns, how many we caught), especially recall on the very worst days.

The two errors are not equally costly, and the asymmetry must be built into the objective.

- A **false negative** (we said “safe”, it crashed) means eating a full drawdown, 50 to 200 basis points in a day.

- A **false positive** (we said “danger”, it was calm) means forgoing a single day’s premium, a few basis points at most.

A false negative can cost twenty to a hundred times a false positive, so the decision threshold should be tuned to that cost ratio, not to balanced accuracy. Figure 20.6 sketches the cost-weighted trade-off.

	Predicted “safe”	Predicted “danger”
Actually calm	true neg. collect premium	false pos. forgo few bps
Actually drawdown	false neg. eat 50–200 bps	true pos. avoided!

↑
the error
that hurts

Figure 20.6: The cost-sensitive structure of drawdown classification. The four outcomes carry wildly different payoffs: a false negative (predicting safety into a crash) costs a full drawdown, while a false positive (a needless flat day) costs only one day’s forgone premium. With drawdowns at roughly 12–15% of days and this cost ratio, the decision threshold must be tuned to minimize expected cost, not classification error.

The features that feed the classifier are organized in layers, Table 20.4, each tying back to an earlier chapter. Given the down-jump mechanism of Section 20.8, the jump and downside features (Layer 1) and a vol-of-vol or leverage term are *a priori* the most informative for this specific target, and should not be treated as interchangeable with the smooth HAR core.

Table 20.4: Feature layers for the drawdown classifier (source: GSVIVS01.md §4.6), each cross-referenced to its home chapter. The jump and leverage layers are highlighted because the drawdown mechanism of Section 20.8 is specifically down-jump driven.

Layer	Key inputs	Home
0 HAR core	log RV daily/weekly/monthly, realized quarticity	Chapter 6
1 Asymmetric	semivariance, bipower variation, jumps , leverage	Chapter 4
2 Options-implied	K_{var} (iv_vs_0dte), VRP, skew, term slope, VVIX	Chapter 8
3 Microstructure	E-mini order flow, VPIN, depth ratio	Chapter 10
4 Cross-asset	Treasury slope, FX vol, commodity vol	Chapter 10
5 Calendar	FOMC, NFP, OpEx, earnings proximity	Chapter 10

Worked Example: From a Confusion Matrix to Basis Points

Take a stylized year of 252 days with 30 true drawdown days, each costing 100 bps, and 222 safe days each paying 5 bps. A model that catches 24 of the 30 drawdowns (recall 80%) at the cost of 20 false alarms has: 24 true positives (drawdowns avoided, +2400 bps saved versus staying invested), 6 false negatives (−600 bps still eaten), and 20 false positives (−100 bps of forgone premium). Net versus always-invested: it converts a year that would have lost $30 \times 100 - 222 \times 5 = 1890$ bps of drawdowns-minus-premium into one that eats only 600 bps of missed drawdowns and forgoes 100 bps of premium. A single

recall/precision pair, read through this cost arithmetic, *is* an expected-P&L number. That is the bridge to Section 20.12.

20.11 From Prediction to Position: The Overlay

The classifier's verdict becomes a position on the index. The overlay scales a base notional by the signal:

$$\text{Position}_t = \text{Signal}_t \times \text{Base Notional}, \quad (20.14)$$

so that a +1 signal holds the full GSVIVS01 exposure (collect the premium), a 0 signal goes flat (side-step the predicted drawdown), and a −1 signal *shorts* the index (a counter-trade that profits if the drawdown materializes, used only on the highest-conviction days). Buy-and-hold GSVIVS01 is the special case $\text{Signal} \equiv +1$; the project's overlay is the modulation of that baseline by the forecast.

Graded Sizing Beats a Binary Switch

A hard on/off rule is rarely optimal. Li and Wu (2026) find that a moderate, confidence-scaled position size delivers a higher Sharpe ratio than either a binary switch or maximally aggressive scaling. As flagged in Chapter 19, that result is a directional single-asset study, so borrow only its qualitative lesson: scale the overlay with conviction in the signal, and never press so hard that one adverse day dominates. The −1 short in particular should be reserved for strong, corroborated drawdown signals, because shorting a short-variance index means buying convexity at a cost.

20.12 Evaluation: The Project's Metric

The deliverable is a *better-behaved GSVIVS01 index*, not a lower QLIKE, and the evaluation must measure that honestly, tie it back to forecast accuracy, and survive the multiple-testing and small-sample traps that make backtests lie.

The headline metric is the overlaid index path against buy-and-hold. We judge it on three axes:

- **Risk-adjusted return:** the **deflated Sharpe ratio** (Equation (19.15), restated from Section 19.10 and Section 17.10), which discounts the observed Sharpe for the number of strategy variants tried.
- **Drawdown control:** the reduction in maximum drawdown, and the Calmar ratio (return over maximum drawdown), since avoiding the worst days is the whole point.
- **Coverage:** the performance target of staying invested on the 85–88% of safe days while side-stepping the 5 to 10 worst drawdowns per year, each worth 50 to 200 bps.

The Benchmark to Match, and Its Caveat

The closest measured precedent is Yang (2024): a volatility-forecast-conditioned short-variance overlay that raised the Sharpe ratio from 1.54 to 1.76 on variance swaps while cutting maximum drawdown, skewness, and kurtosis, cutting exposure only in high-volatility regimes. Treat this as the directional target our overlay should reach or beat. But its setting is one-month variance swaps with monthly rebalancing, not 0DTE, so the *magnitude* does not transfer: it is a hypothesis to test on our data (Section 20.13), not a number to claim.

The Honest Accounting

Three traps must be respected or the metric lies. First, **costs**: trading the overlay is not free; entering and exiting the index, and especially the -1 short leg, incur the option and hedge costs of Section 19.5, charged as a band, not a point estimate. Second, **multiple testing**: deflate the Sharpe for the honest number of signal, threshold, and feature variants tried, not for one. Third, **small samples and tails**: the 0DTE short's profit and loss is “dominated by tail risk rather than stable mean carry” (Vilkov, 2024), GSVIVS01 has few drawdown events on a short history, and the drawdowns cluster in time, so evaluate under **purged or combinatorial cross-validation** (Chapter 17) and report wide error bars. Do not let an apparent edge that rests on two or three historical crashes masquerade as skill.

Finally, the thread back to forecasting. The economic foundation is the negative variance risk premium: the mean gain to selling variance is positive on average (Bakshi and Kapadia, 2003), which is why GSVIVS01 is paid at all, and confirmed from the buyer side by the documented losses of 0DTE option buyers (Beckmeyer et al., 2023). Our forecast's job is to time *when* that premium is rich enough to harvest and when it is a trap, and the deflated Sharpe of the overlay is the judge of whether the forecast does that job better than chance. Whether a lower QLIKE actually produces a better overlay is the central experiment, not a theorem.

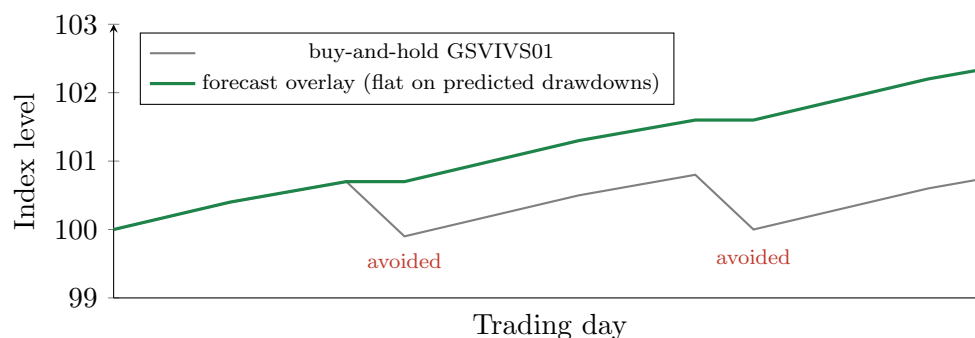


Figure 20.7: The evaluation metric, schematically. Buy-and-hold GSVIVS01 (grey) drips upward and then surrenders the gains on each drawdown day; the forecast overlay (green) stays invested through the calm runs but goes flat on the days the model predicts $RV > K_{\text{var}}$, compounding a higher, smoother path. The metric is the difference between the two: deflated Sharpe, maximum-drawdown reduction, and Calmar. Shape is illustrative; the magnitude is what Section 20.13 must establish on real data.

20.13 What to Compute on Our Data

The literature in this chapter is suggestive, but none of it is GSVIVS01 itself. The following experiments, each with an explicit pass criterion, are what would turn this chapter from a design into a result on our own index.

Eight Experiments, Eight Verdicts

1. **Strike versus at-the-money IV.** Build the signal of Equation (20.12) with the variance-swap strike K_{var} and, separately, with at-the-money 0DTE IV. *Pass*: the strike-based signal classifies drawdowns and times the overlay meaningfully better, validating Section 20.3, or we learn it does not.
2. **ML forecast versus HAR.** Drive the overlay with the machine-learning ensemble

and with a plain HAR forecast. *Pass*: the lower-QLIKE model yields a higher deflated overlay Sharpe and a larger drawdown reduction, the QLIKE-to-P&L bridge of Chapter 19 retested on the real index.

3. **Cost-sensitive threshold.** Tune the decision threshold on the asymmetric cost matrix of Figure 20.6; report precision and recall on the worst N days. *Pass*: the chosen threshold beats both always-invested and a naive VIX-level rule in expected basis points.
4. **Deflated, purged Sharpe.** Report the overlay Sharpe block-bootstrapped over drawdown-clustered blocks, deflated for the honest number of variants tried, under purged or combinatorial cross-validation. *Pass*: it clears the expected-maximum bar SR_0 for the true variant count, not for one.
5. **Channel robustness.** Test the contrarian re-entry channel and asymmetric thresholds. *Pass*: re-entry adds value beyond flat-only, or it is dropped.
6. **Conditional versus unconditional.** Backtest buy-and-hold against the VRP-gap-conditioned overlay. *Pass*: the conditioned book's deflated Sharpe dominates *and* the driving forecast beats a no-skill $\mathbb{E}[RV] = K_{\text{var}}$ benchmark by a Diebold–Mariano-significant QLIKE margin. If timing adds Sharpe but QLIKE is flat, the edge is seasonality, not skill (the tension between [Vilkov, 2024](#); [Almeida et al., 2024](#)).
7. **Horizon transfer.** Estimate the conditional *next-day* 0DTE-VRP response to a same-day volatility shock. *Pass*: a same-day spike predicts a lower next-day VRP (so the [Yang \(2024\)](#) mechanism transfers); if not, switch the conditioning variable to vol-of-vol or leverage ([Bandi et al., 2024](#)).
8. **Jump-share decomposition.** Decompose realized 0DTE quadratic variation into continuous (bipower) and jump parts and compute the realized third moment (Chapter 4). *Pass*: if negative third moments dominate on drawdown days, quantify the basis-point gap between the booked strike and a jump-corrected strike ([Du and Kapadia, 2012](#)) and test whether conditioning on jump share beats conditioning on the VRP level alone.

20.14 Summary

What to Carry Forward

- GSVIVS01 sells a daily variance swap on the S&P 500, replicated as a $1/K^2$ strip of 0DTE options (Equation (20.7)) delta-hedged with E-mini futures; it is the clean variance instrument Chapter 19 could not trade, structurally short gamma, short vega, long theta.
- The fair strike is the model-free implied variance, the cost of the $1/K^2$ strip (Equation (20.4)); the $1/K^2$ weight is forced by constant variance exposure, and the skew makes K_{var} exceed at-the-money IV (Equation (20.5)), so the signal must benchmark against K_{var} .
- A drawdown is exactly an $RV > K_{\text{var}}$ day (Equation (20.9)); the losses are down-jump driven, with a cubic asymmetry that makes crashes cost more than equal rallies (Equation (20.11)), pointing the forecast at jump and leverage features.
- The project's signal is the VRP gap on the forecast (Equation (20.12)); drawdown prediction is cost-sensitive classification (Figure 20.6), traded as a flat-or-short overlay (Equation (20.14)) and judged by the deflated Sharpe and drawdown reduction of the overlaid index.
- The harvesting state of the art is *conditional*: the unconditional 0DTE short is tail-dominated ([Vilkov, 2024](#)), and a forecast-timed overlay is what lifts risk-adjusted

returns (Yang, 2024). That is the project's thesis.

The Caveat That Dominates the Rest

Every economic number in this chapter is a directional analogue, not a transferable estimate: the $1.54 \rightarrow 1.76$ Sharpe, the “four times” richer 0DTE premium, the cost schedules, all come from related instruments and samples, never from GSVIVS01 itself. GSVIVS01 has few drawdowns on a short, 0DTE-era history, and its losses are tail-dominated, so the overlay's deflated Sharpe will carry wide error bars. The chapter's value is that it specifies exactly which numbers must still be earned on our own data (Section 20.13); treat the closed forms and the cited magnitudes as scaffolding to be confirmed, not as borrowed truth.

Bibliography

- Richard M. Ahadzie and Nagaratnam Jeyasreedharan. Effects of intervaling on high-frequency realized higher-order moments. *Quantitative Finance*, 20(7):1169–1184, 2020. doi: 10.1080/14697688.2020.1726455.
- Riaz Ahmad and Paul Wilmott. Which free lunch would you like today, sir?: Delta hedging, volatility arbitrage and optimal portfolios. *Wilmott Magazine*, pages 64–79, 2005.
- Yacine Aït-Sahalia and Jean Jacod. Testing for jumps in a discretely observed process. *The Annals of Statistics*, 37(1):184–222, 2009. doi: 10.1214/07-AOS568.
- Yacine Aït-Sahalia, Per A. Mykland, and Lan Zhang. How often to sample a continuous-time process in the presence of market microstructure noise. *The Review of Financial Studies*, 18(2):351–416, 2005. doi: 10.1093/rfs/hhi016.
- Caio Almeida, Gustavo Freire, and Rodrigo Hizmeri. 0DTE asset pricing, 2024. Working paper.
- Diego Amaya, Peter Christoffersen, Kris Jacobs, and Aurelio Vasquez. Does realized skewness predict the cross-section of equity returns? *Journal of Financial Economics*, 118(1):135–167, 2015. doi: 10.1016/j.jfineco.2015.02.009.
- Yakov Amihud. Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56, 2002. doi: 10.1016/S1386-4181(01)00024-6.
- Iliana Anagnou and Stewart D. Hodges. Derivatives hedging errors and volatility, 2007. EFMA Annual Meeting paper.
- Torben G. Andersen and Tim Bollerslev. Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4):885–905, 1998. doi: 10.2307/2527343.
- Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. The distribution of realized exchange rate volatility. *Journal of the American Statistical Association*, 96(453):42–55, 2001. doi: 10.1198/016214501750332965.
- Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625, 2003. doi: 10.1111/1468-0262.00418.
- Torben G. Andersen, Tim Bollerslev, and Francis X. Diebold. Roughing it up: Including jump components in the measurement, modeling, and forecasting of return volatility. *The Review of Economics and Statistics*, 89(4):701–720, 2007. doi: 10.1162/rest.89.4.701.
- Nikolaos Antonakakis, Ioannis Chatziantoniou, and David Gabauer. Refined measures of dynamic connectedness based on time-varying parameter vector autoregressions. *Journal of Risk and Financial Management*, 13(4):84, 2020. doi: 10.3390/jrfm13040084.
- Jules Arzel and Noureddine Lehdili. Bridging stochastic control and deep hedging: Structural priors for no-transaction band networks, 2026. arXiv preprint arXiv:2603.29994.

- Francesco Audrino and Simon D. Knaus. Lassoing the HAR model: A model selection perspective on realized volatility dynamics. *Econometric Reviews*, 35(8–10):1485–1521, 2016. doi: 10.1080/07474938.2015.1092801.
- Francesco Audrino, Fabio Sigrist, and Daniele Ballinari. The impact of sentiment and attention measures on stock market volatility. *International Journal of Forecasting*, 36(2):334–357, 2020.
- Varun Babbar, Hayden McTavish, Cynthia Rudin, and Margo Seltzer. Near-optimal decision trees in a SPLIT second. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. Oral presentation; arXiv 2502.15988.
- David H. Bailey and Marcos Lopez de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014. doi: 10.3905/jpm.2014.40.5.094.
- David H. Bailey, Jonathan M. Borwein, Marcos Lopez de Prado, and Qiji Jim Zhu. Pseudomathematics and financial charlatanism: The effects of backtest overfitting on out-of-sample performance. *Notices of the American Mathematical Society*, 61(5):458–471, 2014. doi: 10.1090/noti1105.
- Richard T. Baillie, Tim Bollerslev, and Hans Ole Mikkelsen. Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1):3–30, 1996. doi: 10.1016/S0304-4076(95)01749-6.
- Gurdip Bakshi and Nikunj Kapadia. Delta-hedged gains and the negative market volatility risk premium. *The Review of Financial Studies*, 16(2):527–566, 2003. doi: 10.1093/rfs/hhg002.
- Federico M. Bandi, Nicola Fusari, and Roberto Renò. 0DTE option pricing, 2024. Working paper.
- Ole E. Barndorff-Nielsen and Neil Shephard. Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 64(2):253–280, 2002. doi: 10.1111/1467-9868.00336.
- Ole E. Barndorff-Nielsen and Neil Shephard. Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1):1–37, 2004. doi: 10.1093/jjfinec/nbh001.
- Ole E. Barndorff-Nielsen and Neil Shephard. Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, 4(1):1–30, 2006. doi: 10.1093/jjfinec/nbi022.
- Ole E. Barndorff-Nielsen, Peter R. Hansen, Asger Lunde, and Neil Shephard. Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6):1481–1536, 2008. doi: 10.3982/ECTA6495.
- Ole E. Barndorff-Nielsen, Peter Reinhard Hansen, Asger Lunde, and Neil Shephard. Multivariate realised kernels: Consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2):149–169, 2011. doi: 10.1016/j.jeconom.2010.07.009.
- Christian Bayer, Peter Friz, and Jim Gatheral. Pricing under rough volatility. *Quantitative Finance*, 16(6):887–904, 2016. doi: 10.1080/14697688.2015.1099717.
- Heiner Beckmeyer, Nicole Branger, and Leander Gayda. Retail traders love 0DTE options... but should they?, 2023. Working paper, University of Münster.
- Geert Bekaert and Marie Hoerova. The VIX, the variance premium and stock market volatility. *Journal of Econometrics*, 183(2):181–192, 2014. doi: 10.1016/j.jeconom.2014.05.008.

- Mikkel Bennedsen, Asger Lunde, and Mikko S. Pakkanen. Decoupling the short- and long-term behavior of stochastic volatility. *Journal of Financial Econometrics*, 20(5):961–1006, 2022. doi: 10.1093/jjfinec/nbaa049.
- Colin Bennett. *Trading Volatility: Trading Volatility, Correlation, Term Structure and Skew*. Self-published, 2014.
- Dimitris Bertsimas, Leonid Kogan, and Andrew W. Lo. When is time continuous? *Journal of Financial Economics*, 55(2):173–204, 2000. doi: 10.1016/S0304-405X(99)00049-5.
- Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006. ISBN 978-0-387-31073-2.
- Fischer Black. Studies of stock price volatility changes. *Proceedings of the 1976 Meetings of the American Statistical Association, Business and Economical Statistics Section*, pages 177–181, 1976.
- Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.
- Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986. doi: 10.1016/0304-4076(86)90063-1.
- Tim Bollerslev and Viktor Todorov. Tails, fears, and risk premia. *The Journal of Finance*, 66(6):2165–2211, 2011. doi: 10.1111/j.1540-6261.2011.01695.x. Extended in Bollerslev and Todorov (2015, *Journal of Financial Economics*).
- Tim Bollerslev, Uta Kretschmer, Christian Pigorsch, and George Tauchen. A discrete-time model for daily S&P500 returns and realized variations: Jumps and leverage effects. *Journal of Econometrics*, 150(2):151–166, 2009a. doi: 10.1016/j.jeconom.2008.12.001.
- Tim Bollerslev, George Tauchen, and Hao Zhou. Expected stock returns and variance risk premia. *The Review of Financial Studies*, 22(11):4463–4492, 2009b. doi: 10.1093/rfs/hhp008.
- Tim Bollerslev, Andrew J. Patton, and Rogier Quaadvlieg. Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1):1–18, 2016a. doi: 10.1016/j.jeconom.2015.10.007.
- Tim Bollerslev, Andrew J. Patton, and Rogier Quaadvlieg. Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1):1–18, 2016b.
- Tim Bollerslev, Benjamin Hood, John Huss, and Lasse Heje Pedersen. Risk everywhere: Modeling and managing volatility. *The Review of Financial Studies*, 31(7):2729–2773, 2018a. doi: 10.1093/rfs/hhy041.
- Tim Bollerslev, Andrew J. Patton, and Rogier Quaadvlieg. Modeling and forecasting (un)reliable realized covariances for more reliable financial decisions. *Journal of Econometrics*, 207(1):71–91, 2018b. doi: 10.1016/j.jeconom.2018.05.004.
- Tim Bollerslev, Jia Li, Andrew J. Patton, and Rogier Quaadvlieg. Realized semicovariances. *Econometrica*, 88(4):1515–1551, 2020.
- Tim Bollerslev, Marcelo C. Medeiros, Andrew J. Patton, and Rogier Quaadvlieg. HARd to beat: The overlooked impact of rolling windows in the era of machine learning, 2024.
- Phelim P. Boyle and David Emanuel. Discretely adjusted option hedges. *Journal of Financial Economics*, 8(3):259–282, 1980. doi: 10.1016/0304-405X(80)90003-3.

- Rafael Branco, Alexandre Rubesam, and Mauricio Zevallos. Forecasting realized volatility: Does anything beat linear models? *Journal of Empirical Finance*, 78:101527, 2024. doi: 10.1016/j.jempfin.2024.101527.
- Leo Breiman, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. *Classification and Regression Trees*. Wadsworth, Belmont, CA, 1984.
- Menachem Brenner and Marti G. Subrahmanyam. A simple formula to compute the implied standard deviation. *Financial Analysts Journal*, 44(5):80–83, 1988.
- Mark Britten-Jones and Anthony Neuberger. Option prices, implied price processes, and stochastic volatility. *The Journal of Finance*, 55(2):839–866, 2000. doi: 10.1111/0022-1082.00228.
- Mats Brodén and Peter Tankov. Tracking errors from discrete hedging in exponential Lévy models, 2010. arXiv preprint arXiv:1003.0709.
- Pierre Brugière and Gabriel Turinici. Model-free deep hedging with transaction costs and light data requirements, 2025. arXiv preprint arXiv:2505.22836.
- Andrea Bucci. Realized volatility forecasting with neural networks. *Journal of Financial Econometrics*, 18(3):502–531, 2020. doi: 10.1093/jjfinec/nbaa008.
- Peter Carr and Roger Lee. Volatility derivatives. *Annual Review of Financial Economics*, 1: 319–339, 2009. doi: 10.1146/annurev.financial.050808.114304.
- Peter Carr and Dilip Madan. Towards a theory of volatility trading. In Robert A. Jarrow, editor, *Volatility: New Estimation Techniques for Pricing Derivatives*. Risk Books, 2002.
- Peter Carr and Liuren Wu. Variance risk premiums. *The Review of Financial Studies*, 22(3): 1311–1341, 2009a. doi: 10.1093/rfs/hhn038.
- Peter Carr and Liuren Wu. Variance risk premiums. *The Review of Financial Studies*, 22(3): 1311–1341, 2009b. doi: 10.1093/rfs/hhn038.
- Álvaro Cartea, Sebastian Jaimungal, and José Penalva. *Algorithmic and High-Frequency Trading*. Cambridge University Press, 2015. ISBN 978-1-107-09114-3.
- Cboe Exchange. VIX white paper: Cboe volatility index. Technical report, Cboe Global Markets, 2019. URL https://www.cboe.com/tradable_products/vix/vix_white_paper/.
- Yuntong Chen and Stephen Roberts. Graph transformers for volatility forecasting. *arXiv preprint arXiv:2209.09014*, 2022.
- Roxana Chiriac and Valeri Voev. Modelling and forecasting multivariate realized volatility. *Journal of Applied Econometrics*, 26(6):922–947, 2011. doi: 10.1002/jae.1152.
- Kim Christensen, Mathias Siggaard, and Bezirgen Veliyev. A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5):1680–1727, 2023. doi: 10.1093/jjfinec/nbac020.
- Rama Cont. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001. doi: 10.1080/713665670.
- Rama Cont and José da Fonseca. Dynamics of implied volatility surfaces. *Quantitative Finance*, 2(1):45–60, 2002. doi: 10.1088/1469-7688/2/1/304.
- Rama Cont and Purba Das. Rough volatility: Fact or Artefact? *Sankhyā B*, 86(1):191–223, 2024. doi: 10.1007/s13571-024-00322-2.

- Rama Cont, Arseniy Kukanov, and Sasha Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88, 2014. doi: 10.1093/jjfinec/nbt003. arXiv 1011.6402.
- Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009. doi: 10.1093/jjfinec/nbp001.
- Fulvio Corsi, Davide Pirino, and Roberto Renò. Threshold bipower variation and the impact of jumps on volatility forecasting. *Journal of Econometrics*, 159(2):276–288, 2010. doi: 10.1016/j.jeconom.2010.07.008.
- Christa Cuchiero, Jakob Heiss, Wahid Khosrawi, and Peter Spoida. GARCH-informed neural networks for volatility prediction. *arXiv preprint arXiv:2410.00288*, 2024.
- Marcos Lopez de Prado. *Advances in Financial Machine Learning*. Wiley, 2018. ISBN 978-1-119-48208-6.
- Kresimir Demeterfi, Emanuel Derman, Michael Kamal, and Joseph Zou. More than you ever wanted to know about volatility swaps. Technical report, Goldman Sachs Quantitative Strategies Research Notes, 1999.
- Mert Demirer, Francis X. Diebold, Laura Liu, and Kamil Yilmaz. Estimating global bank network connectedness. *Journal of Applied Econometrics*, 33(1):1–15, 2018. doi: 10.1002/jae.2585.
- A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 39(1):1–38, 1977. doi: 10.1111/j.2517-6161.1977.tb01600.x.
- Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.
- Francis X. Diebold and Kamil Yilmaz. Measuring financial asset return and volatility spillovers, with application to global equity markets. *The Economic Journal*, 119(534):158–171, 2009. doi: 10.1111/j.1468-0297.2008.02208.x.
- Francis X. Diebold and Kamil Yilmaz. Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1):57–66, 2012. doi: 10.1016/j.ijforecast.2011.02.006.
- Francis X. Diebold and Kamil Yilmaz. On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1):119–134, 2014. doi: 10.1016/j.jeconom.2014.04.012.
- Yi Ding, Guofu Lu, and Eric C. Cheung. Deep learning for implied volatility surface compression. *Journal of Financial Economics*, 2025. Forthcoming.
- Jiayun Dong and Cynthia Rudin. Exploring the cloud of variable importance for the set of all good models, 2020. arXiv 1901.03209; Nature Machine Intelligence.
- Jon Donnelly, Srikar Katta, Cynthia Rudin, and Edward P. Browne. Rashomon importance distributions. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Doshi, Pari, and Shamsuddin. Risky intraday order flow and option liquidity, 2025. Working paper.
- Itamar Drechsler and Amir Yaron. What’s vol got to do with it. *The Review of Financial Studies*, 24(1):1–45, 2011. doi: 10.1093/rfs/hhq085.

- Jian Du and Nikunj Kapadia. Tail and volatility indices from option prices, 2012. Working paper, University of Massachusetts Amherst.
- Tianyu Du, Yuki Moriyama, Kei Tanaka, and Shin ichi Ishii. Normalizing flows for realized volatility forecasting. *arXiv preprint arXiv:2304.06985*, 2023.
- Bruno Dupire. Pricing with a smile. *Risk*, 7(1):18–20, 1994.
- David Easley, Marcos M. Lopez de Prado, and Maureen O’Hara. Flow toxicity and liquidity in a high-frequency world. *The Review of Financial Studies*, 25(5):1457–1493, 2012. doi: 10.1093/rfs/hhs053.
- Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982. doi: 10.2307/1912773.
- Robert F. Engle. Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3):339–350, 2002. doi: 10.1198/073500102288618487.
- Robert F. Engle and Tim Bollerslev. Modelling the persistence of conditional variances. *Econometric Reviews*, 5(1):1–50, 1986. doi: 10.1080/07474938608800095.
- Eugene F. Fama. The behavior of stock-market prices. *The Journal of Business*, 38(1):34–105, 1965.
- Dylan Fouhy. Forecasting the variance risk premium with machine learning. SSRN Working Paper No. 6570380, 2024. URL <https://ssrn.com/abstract=6570380>.
- Pascal François, Geneviève Gauthier, Frédéric Godin, and Carlos Octavio Pérez-Mendoza. Deep hedging with options using the implied volatility surface, 2025. arXiv preprint arXiv:2504.06208.
- Jim Gatheral and Antoine Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014. doi: 10.1080/14697688.2013.819986. arXiv 1204.0646.
- Jim Gatheral, Thibault Jaisson, and Mathieu Rosenbaum. Volatility is rough. *Quantitative Finance*, 18(6):933–949, 2018. doi: 10.1080/14697688.2017.1393551.
- Lawrence R. Glosten and Paul R. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1):71–100, 1985. doi: 10.1016/0304-405X(85)90044-3.
- Lawrence R. Glosten, Ravi Jagannathan, and David E. Runkle. On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5):1779–1801, 1993. doi: 10.1111/j.1540-6261.1993.tb05128.x.
- Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020a. doi: 10.1093/rfs/hhaa009.
- Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020b. doi: 10.1093/rfs/hhaa009.
- James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989. doi: 10.2307/1912559.
- Peter R. Hansen and Asger Lunde. Realized variance and market microstructure noise. *Journal of Business & Economic Statistics*, 24(2):127–161, 2006. doi: 10.1198/073500106000000071.
- Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011. doi: 10.3982/ECTA5771.

- Peter Reinhard Hansen and Asger Lunde. A forecast comparison of volatility models: Does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7):873–889, 2005. doi: 10.1002/jae.800.
- Peter Reinhard Hansen, Zhuo Huang, and Howard Howan Shek. Realized GARCH: A joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6): 877–906, 2012. doi: 10.1002/jae.1234.
- Campbell R. Harvey and Yan Liu. Backtesting. *Journal of Portfolio Management*, 42(1):13–28, 2015. doi: 10.3905/jpm.2015.42.1.013.
- Campbell R. Harvey, Yan Liu, and Heqing Zhu. ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68, 2016. doi: 10.1093/rfs/hhv059.
- David I. Harvey, Stephen J. Leybourne, and Paul Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997. doi: 10.1016/S0169-2070(96)00719-4.
- Takaki Hayashi and Nakahiro Yoshida. On covariance estimation of non-synchronously observed diffusion processes. *Bernoulli*, 11(2):359–379, 2005. doi: 10.3150/bj/1116340299.
- Zakk Heile, Varun Babbar, Hayden McTavish, and Cynthia Rudin. Efficient Rashomon set approximation for decision tree models. In *NeurIPS 2025 Workshop on ML x OR*, 2025.
- Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970. doi: 10.1080/00401706.1970.10488634.
- Xin Huang and George Tauchen. The relative contribution of jumps to total price variance. *Journal of Financial Econometrics*, 3(4):456–499, 2005. doi: 10.1093/jjfinec/nbi025.
- Jean Jacod, Yingying Li, Per A. Mykland, Mark Podolskij, and Mathias Vetter. Microstructure noise in the continuous case: The pre-averaging approach. *Stochastic Processes and their Applications*, 119(7):2249–2276, 2009. doi: 10.1016/j.spa.2008.11.004.
- Yuri M. Kabanov and Mher M. Safarian. On leland’s strategy of option pricing with transactions costs. *Finance and Stochastics*, 1(3):239–250, 1997. doi: 10.1007/s007800050023.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Patrick Kidger. *On Neural Differential Equations*. PhD thesis, University of Oxford, 2021.
- Albert S. Kyle. Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335, 1985. doi: 10.2307/1913210.
- Suzanne S. Lee and Per A. Mykland. Jumps in financial markets: A new nonparametric test and jump dynamics. *The Review of Financial Studies*, 21(6):2535–2563, 2008. doi: 10.1093/rfs/hhm056.
- Hayne E. Leland. Option pricing and replication with transactions costs. *The Journal of Finance*, 40(5):1283–1301, 1985. doi: 10.1111/j.1540-6261.1985.tb02383.x.
- Emmanuel Lépinette and Yuri Kabanov. Mean square error for the Leland–Lott hedging strategy: Convex pay-offs. *Finance and Stochastics*, 14(4):625–667, 2010. doi: 10.1007/s00780-010-0130-z.
- B. Li and C. Wu. Beyond delta neutrality: Confidence-scaled hedging with machine learning forecasts. *Finance Research Letters*, 87:109098, 2026. doi: 10.1016/j.frl.2025.109098.

- Lily Y. Liu, Andrew J. Patton, and Kevin Sheppard. Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, 187(1):293–311, 2015. doi: 10.1016/j.jeconom.2015.02.008.
- Marcos Lopez de Prado. *Advances in Financial Machine Learning*. Wiley, Hoboken, NJ, 2018.
- Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Scott M. Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020.
- Paul Malliavin and Maria Elvira Mancino. Fourier series method for measurement of multivariate volatilities. *Finance and Stochastics*, 6(1):49–61, 2002. doi: 10.1007/s780-002-8401-3.
- Paul Malliavin and Maria Elvira Mancino. A Fourier transform method for nonparametric estimation of multivariate volatility. *The Annals of Statistics*, 37(4):1983–2010, 2009. doi: 10.1214/08-AOS633.
- Cecilia Mancini. Non-parametric threshold estimation for models with stochastic diffusion coefficient and jumps. *Scandinavian Journal of Statistics*, 36(2):270–296, 2009. doi: 10.1111/j.1467-9469.2008.00622.x.
- Benoit Mandelbrot. The variation of certain speculative prices. *The Journal of Business*, 36(4):394–419, 1963.
- Benoit B. Mandelbrot and John W. Van Ness. Fractional Brownian motions, fractional noises and applications. *SIAM Review*, 10(4):422–437, 1968. doi: 10.1137/1010093.
- Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952. doi: 10.2307/2975974.
- Jacob Mincer and Victor Zarnowitz. The evaluation of economic forecasts. *NBER Studies in Business Cycles*, 19:3–46, 1969.
- Alan Moreira and Tyler Muir. Volatility-managed portfolios. *The Journal of Finance*, 72(4):1611–1644, 2017. doi: 10.1111/jofi.12513.
- Fernando Moreno-Pino and Stefan Zohren. DeepVol: Volatility forecasting from high-frequency data with dilated causal convolutions. *arXiv preprint arXiv:2209.11761*, 2022.
- Tobias J. Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen. Time series momentum. *Journal of Financial Economics*, 104(2):228–250, 2012. doi: 10.1016/j.jfineco.2011.11.003.
- Ulrich A. Müller, Michel M. Dacorogna, Rakhal D. Davé, Richard B. Olsen, Olivier V. Pictet, and Jakob E. von Weizsäcker. Fractals and intrinsic time: A challenge to econometricians. *UAM Research Paper*, 1993. Olsen & Associates working paper; presented at the 39th International Conference of the AEA.
- Dmitriy Muravyev and Neil D. Pearson. Options trading costs are lower than you think. *The Review of Financial Studies*, 33(11):4973–5014, 2020. doi: 10.1093/rfs/hhaa010.
- Daniel B. Nelson. Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2):347–370, 1991. doi: 10.2307/2938260.
- Stephen Nickell. Biases in dynamic models with fixed effects. *Econometrica*, 49(6):1417–1426, 1981. doi: 10.2307/1911408.

- Optiver. Optiver realized volatility prediction, 2021. Kaggle competition. <https://www.kaggle.com/c/optiver-realized-volatility-prediction>.
- Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011. doi: 10.1016/j.jeconom.2010.03.034.
- Andrew J. Patton and Kevin Sheppard. Good volatility, bad volatility: Signed jumps and the persistence of volatility. *The Review of Economics and Statistics*, 97(3):683–697, 2015a. doi: 10.1162/REST_a_00503.
- Andrew J. Patton and Kevin Sheppard. Good volatility, bad volatility: Signed jumps and the persistence of volatility. *Review of Economics and Statistics*, 97(3):683–697, 2015b.
- M. Hashem Pesaran and Yongcheol Shin. Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58(1):17–29, 1998. doi: 10.1016/S0165-1765(97)00214-0.
- Austin Pollok. Predicting realized variance out of sample: Can anything beat the benchmark?, 2025. arXiv preprint arXiv:2506.07928.
- Eghbal Rahimikia and Ser-Huang Poon. Machine learning for realized volatility forecasting. *Journal of Financial Econometrics*, 2020. doi: 10.2139/ssrn.3707796. Forthcoming.
- Eghbal Rahimikia, Stefan Zohren, and Ser-Huang Poon. Realised volatility forecasting: Machine learning via financial word embedding. *Quantitative Finance*, 21(10):1675–1691, 2021a.
- Eghbal Rahimikia, Stefan Zohren, and Ser-Huang Poon. Realised volatility forecasting: Machine learning via financial word embedding. *Quantitative Finance*, 21(10):1675–1698, 2021b. doi: 10.1080/14697688.2021.1905658.
- Riccardo Rebonato. *Volatility and Correlation: The Perfect Hedger and the Fox*. John Wiley & Sons, 2nd edition, 2004. ISBN 978-0-470-09139-8.
- Mathieu Rosenbaum and Jianfei Zhang. On the universality of the volatility formation process: When machine learning and rough volatility agree. 2022. arXiv preprint 2206.14114.
- Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978. doi: 10.1214/aos/1176344136.
- Neil Shephard and Kevin Sheppard. Realising the future: Forecasting with high-frequency-based volatility (HEAVY) models. *Journal of Applied Econometrics*, 25(2):197–231, 2010. doi: 10.1002/jae.1158.
- Justin Sirignano and Rama Cont. Universal features of price formation in financial markets: Perspectives from deep learning. *Quantitative Finance*, 19(9):1449–1459, 2019. doi: 10.1080/14697688.2019.1622295.
- Carolin Strobl, Anne-Laure Boulesteix, Achim Zeileis, and Torsten Hothorn. Bias in random forest variable importance measures: Illustrations, sources and a solution. *BMC Bioinformatics*, 8(1):25, 2007. doi: 10.1186/1471-2105-8-25.
- Mim van den Bos, Jacobus G. M. van der Linden, and Emir Demirović. Piecewise constant and linear regression trees: An optimal dynamic programming approach. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- Grigory Vilkov. ODTE trading rules, 2024. Working paper, SSRN 4641356.
- Rashmi Korlakai Vinayak and Ran Gilad-Bachrach. DART: Dropouts meet multiple additive regression trees. In *Proceedings of the 18th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 489–497, 2015.

- Jeffrey M. Wooldridge. *Econometric Analysis of Cross Section and Panel Data*. MIT Press, Cambridge, MA, 2nd edition, 2010. ISBN 978-0-262-23258-6.
- Michał Wysocki and Robert Ślepaczuk. Construction and hedging of equity index options portfolios, 2024. arXiv preprint arXiv:2407.13908.
- Rui Xin, Chudi Zhong, Zhi Chen, Takuya Takagi, Margo Seltzer, and Cynthia Rudin. Exploring the whole Rashomon set of sparse decision trees. In *Advances in Neural Information Processing Systems*, volume 35, 2022. Oral presentation; arXiv 2209.08040.
- Dacheng Xiu. Quasi-maximum likelihood estimation of volatility with high frequency data. *Journal of Econometrics*, 159(1):235–250, 2010. doi: 10.1016/j.jeconom.2010.07.002.
- Xiuqin Xu and Ying Chen. Deep stochastic volatility model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 10526–10533, 2021.
- Aoxiang Yang. Volatility-managed volatility trading, 2024. PHBS Working Paper 20240604, Peking University HSBC Business School.
- Chao Zhang, Mihai Cucuringu, and Xiaowen Dong. Graph neural networks for volatility forecasting. *Working Paper*, 2023.
- Chao Zhang, Xingyue Pu, Mihai Cucuringu, and Xiaowen Dong. Graph-based methods for forecasting realized covariances. *Journal of Financial Econometrics*, 22(1):1–32, 2024. doi: 10.1093/jjfinec/nbad012.
- Lan Zhang. Efficient estimation of stochastic volatility using noisy observations: A multi-scale approach. *Bernoulli*, 12(6):1019–1043, 2006. doi: 10.3150/bj/1165269149.
- Lan Zhang, Per A. Mykland, and Yacine Aït-Sahalia. A tale of two time scales: Determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472):1394–1411, 2005. doi: 10.1198/016214505000000169.
- Zihao Zhang, Stefan Zohren, and Stephen Roberts. DeepLOB: Deep convolutional neural networks for limit order books. In *IEEE Transactions on Signal Processing*, volume 67, pages 3001–3012, 2019. doi: 10.1109/TSP.2019.2907260.
- Yonggan Zhao and William T. Ziemba. On leland’s option hedging strategy with transaction costs, 2003. Working paper.
- Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301–320, 2005. doi: 10.1111/j.1467-9868.2005.00503.x.